1. General

A. This pageblock contains MSG-3 task card data.

EWIS

TASK 05-42-02-100-801

2. Restore: Aft Passenger Compartment, Wet Areas, Sta 663.75 to Aft Pressure Bulkhead. (EZAP) (Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Procedure

NOTE: Galleys and lavs removed. Removal of sidewall panels, ceiling panels and return air grilles required in areas where galleys and lavs are located.

SUBTASK 05-42-02-160-001

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

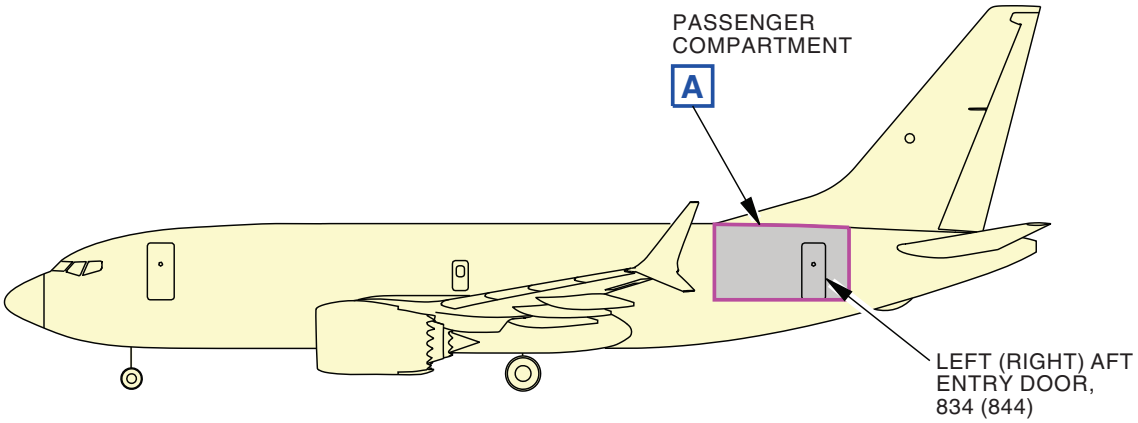
(2) Restore (Clean) areas above ceiling, and behind sidewalls and return air grilles in Aft Passenger Compartment, Wet Areas, Sta. 663.75 to Aft Pressure Bulkhead. (EZAP)

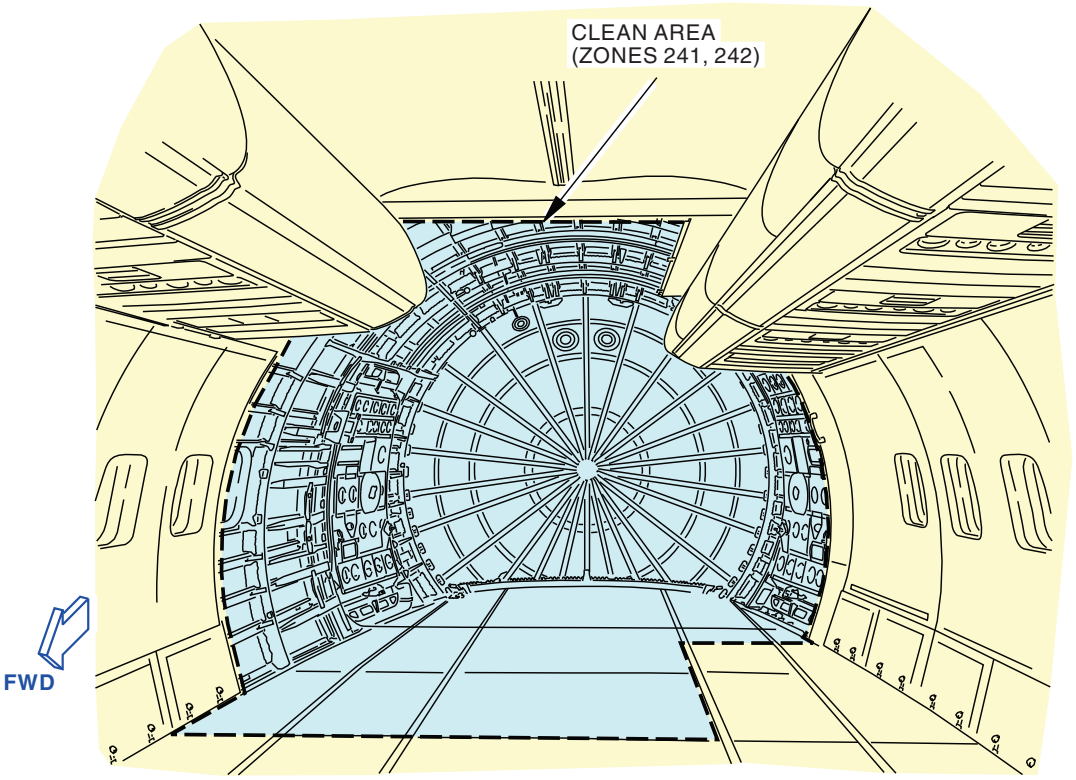
SUBTASK 05-42-02-910-001

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





PASSENGER COMPARTMENT
(LAVS, GALLEYS, CEILING, SIDEWALL PANELS, AND RETURN AIR GRILLES REMOVED)

A

2410370 S00061525398_V1

Passenger Compartment
Figure 201/05-42-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-100-802

3. **Restore: Aft Passenger Compartment, Dry Area, Sta 663.75 to Aft Pressure Bulkhead. (EZAP)**
(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Procedure

NOTE: Removal of sidewall panels, ceiling panels and return air grilles required.

SUBTASK 05-42-02-160-002

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

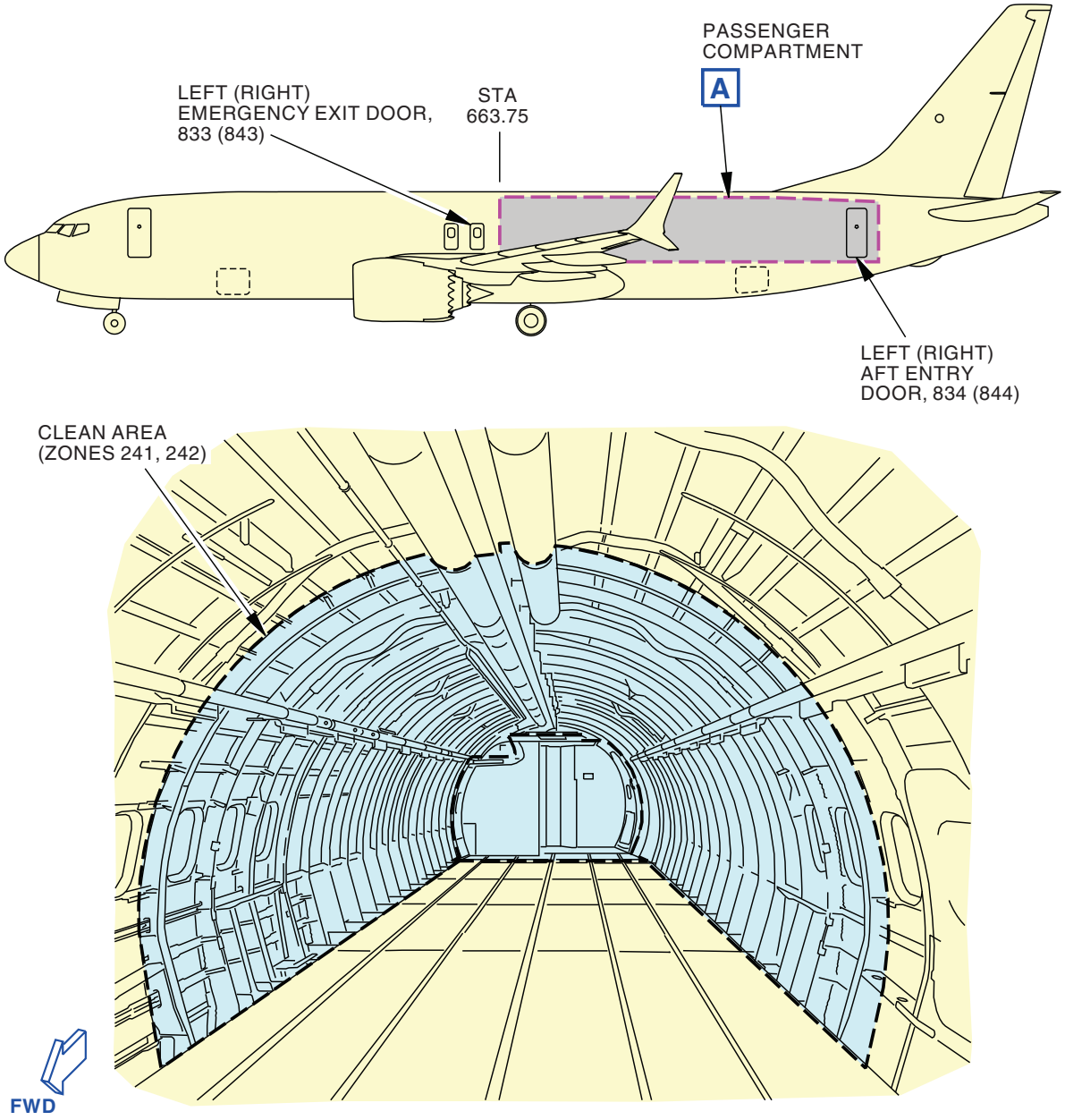
(2) Restore (Clean) areas above ceiling, and behind sidewalls and return air grilles in Aft Passenger Compartment, Dry Area, Sta. 663.75 to Aft Pressure Bulkhead. (EZAP)

SUBTASK 05-42-02-910-002

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PASSENGER COMPARTMENT
(SIDEWALL PANELS, CEILING PANELS AND RETURN AIR GRILLES REMOVED)

A

2410371 S00061525400_V1

Passenger Compartment
Figure 202/05-42-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-100-803

4. Restore: Forward Passenger Compartment, Wet Areas, Sta 360 to 663.75. (EZAP)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right

D. Procedure

NOTE: Galleys and lavs removed. Removal of sidewall panels, ceiling panels and return air grilles required in areas where galleys and lavs are located.

SUBTASK 05-42-02-160-003

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

(2) Restore (Clean) areas above ceiling, and behind sidewalls and return air grilles in Forward Passenger Compartment, Wet Areas, Sta. 360 to 663.75. (EZAP)

SUBTASK 05-42-02-910-003

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

EWIS

TASK 05-42-02-100-804

5. Restore: Forward Passenger Compartment, Dry Area, Sta 360 to 663.75. (EZAP)
(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Procedure

NOTE: Removal of sidewall panels, ceiling panels and return air grilles required.

SUBTASK 05-42-02-160-004

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

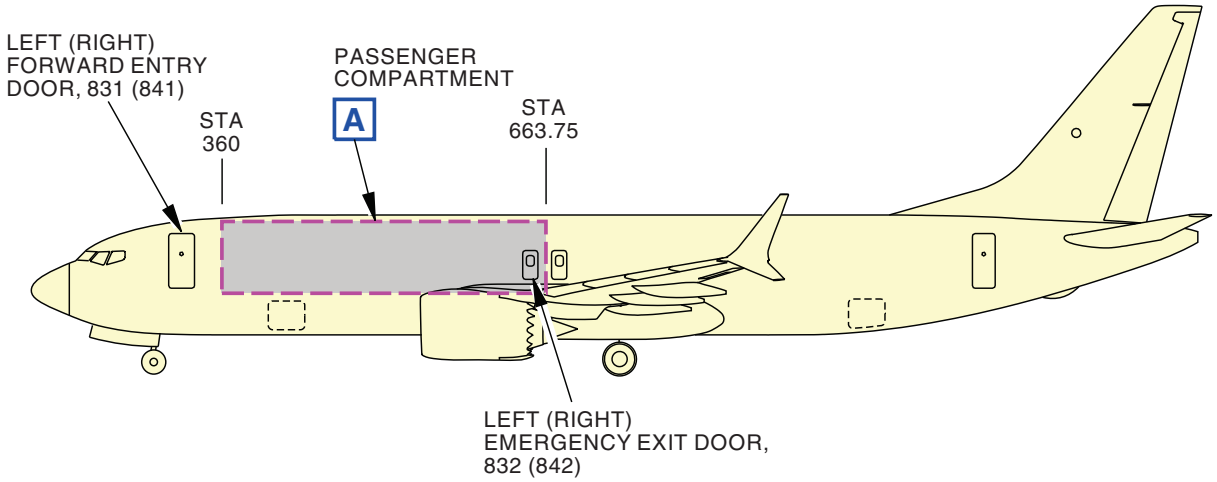
(2) Restore (Clean) areas above ceiling, and behind sidewalls and return air grilles in Forward Passenger Compartment, Dry Area, Sta. 360 to 663.75. (EZAP)

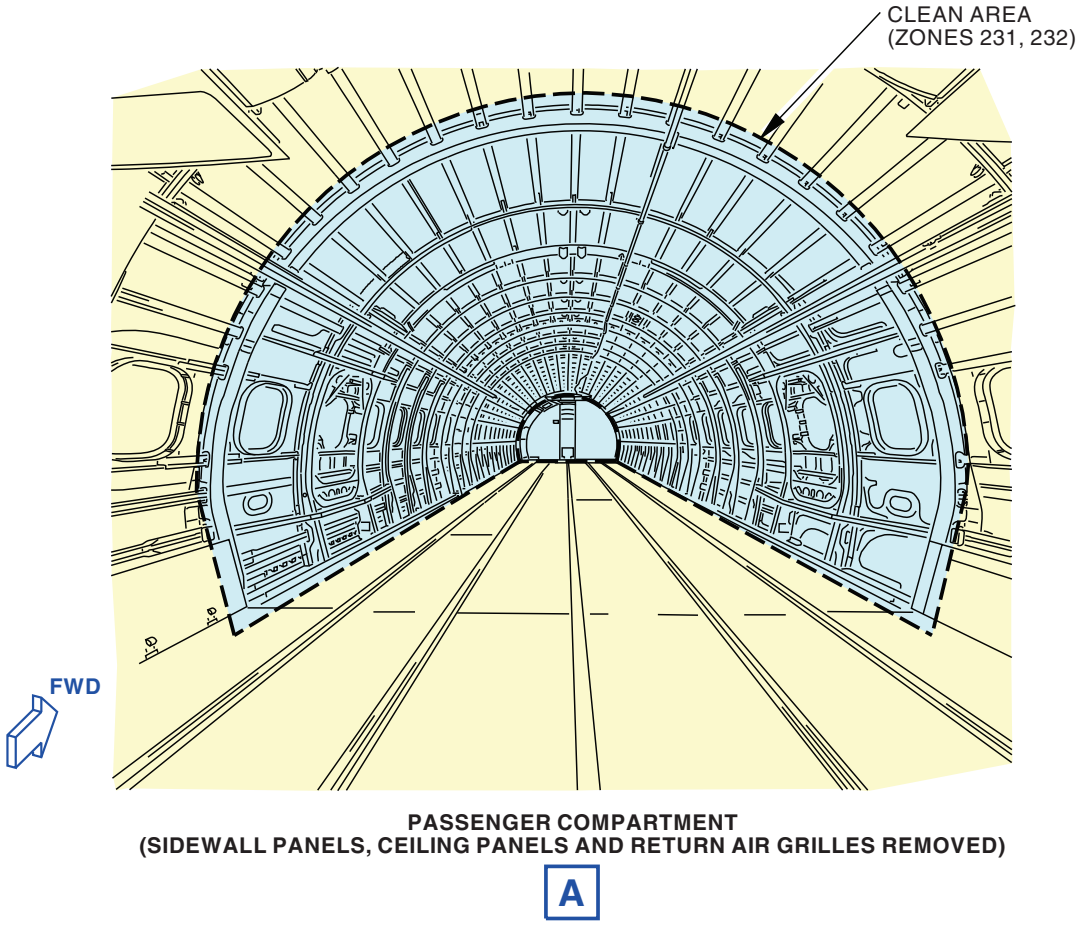
SUBTASK 05-42-02-910-004

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





PASSENGER COMPARTMENT
(SIDEWALL PANELS, CEILING PANELS AND RETURN AIR GRILLES REMOVED)

A

2410372 S00061525403_V1

Passenger Compartment
Figure 203/05-42-02-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-100-805

6. Restore: Areas Aft of Flight Compartment to Forward Entry Door. (EZAP)

(Figure 204)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
222	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Right

D. Procedure

NOTE: With access provided. Galleys and lavs removed. Ceiling and sidewall panels removal required.

SUBTASK 05-42-02-160-005

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

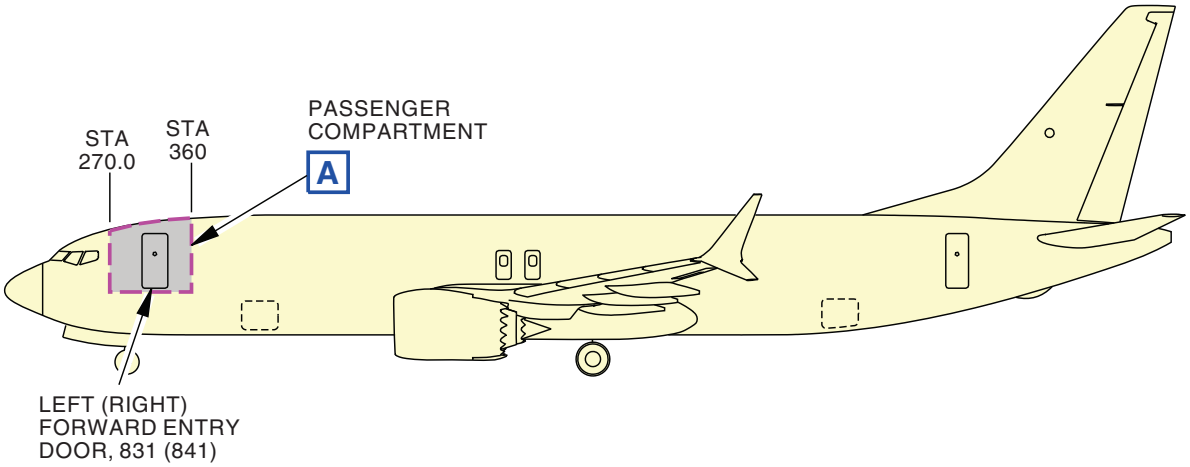
(2) Restore (Clean) areas above ceiling, behind sidewalls and under the raceway cover plates on both sides of the passenger compartment, aft of Flight Compartment to forward entry door. (EZAP)

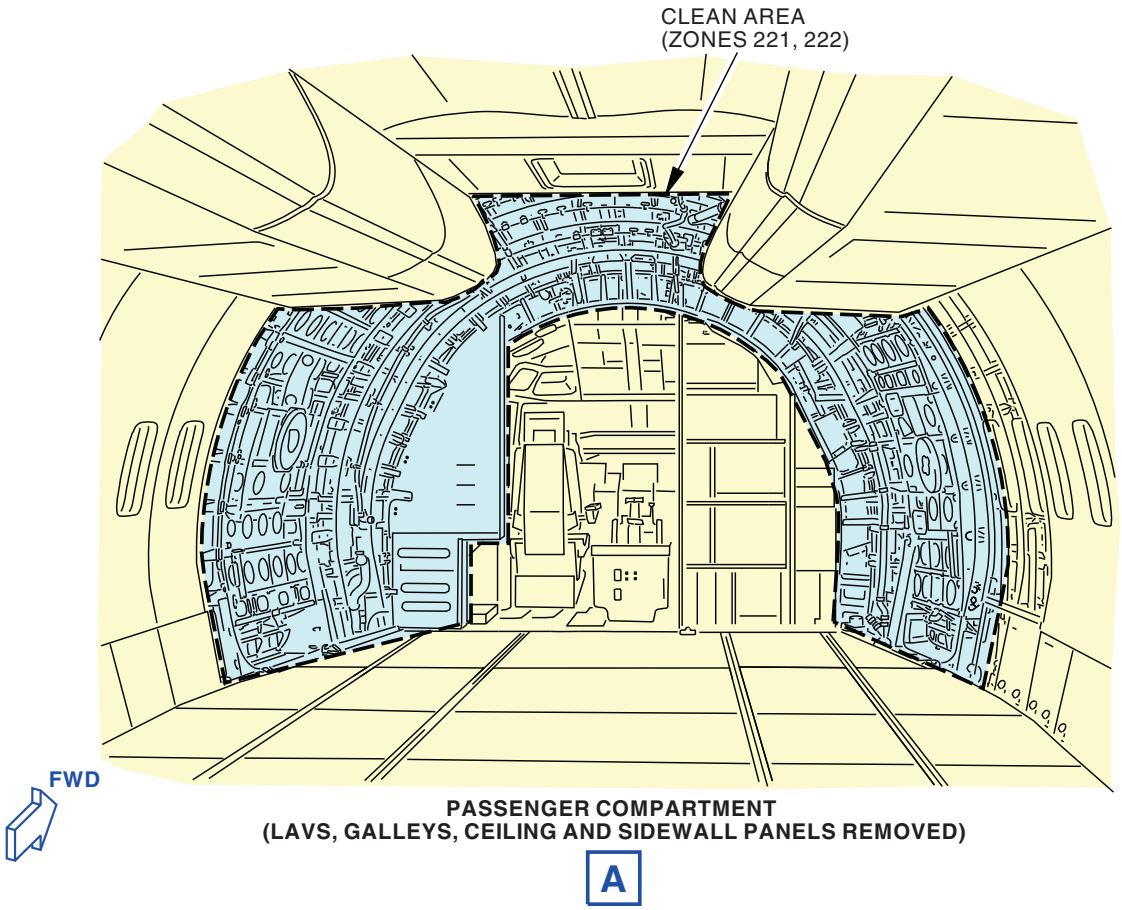
SUBTASK 05-42-02-910-005

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





Passenger Compartment
Figure 204/05-42-02-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-100-806

7. Restore: Area in Flight Compartment from Sta 186 to 211, WL 208 to 232 (EZAP)

(Figure 205)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-803	Cleaning to Remove Combustible Material (P/B 701)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

NOTE: Accessible areas forward and above the rudder pedals including areas under rudder pedal heelrest access panels.

SUBTASK 05-42-02-160-006

(1) Do this task: Cleaning to Remove Combustible Material, TASK 20-60-02-100-803.

SUBTASK 05-42-02-160-007

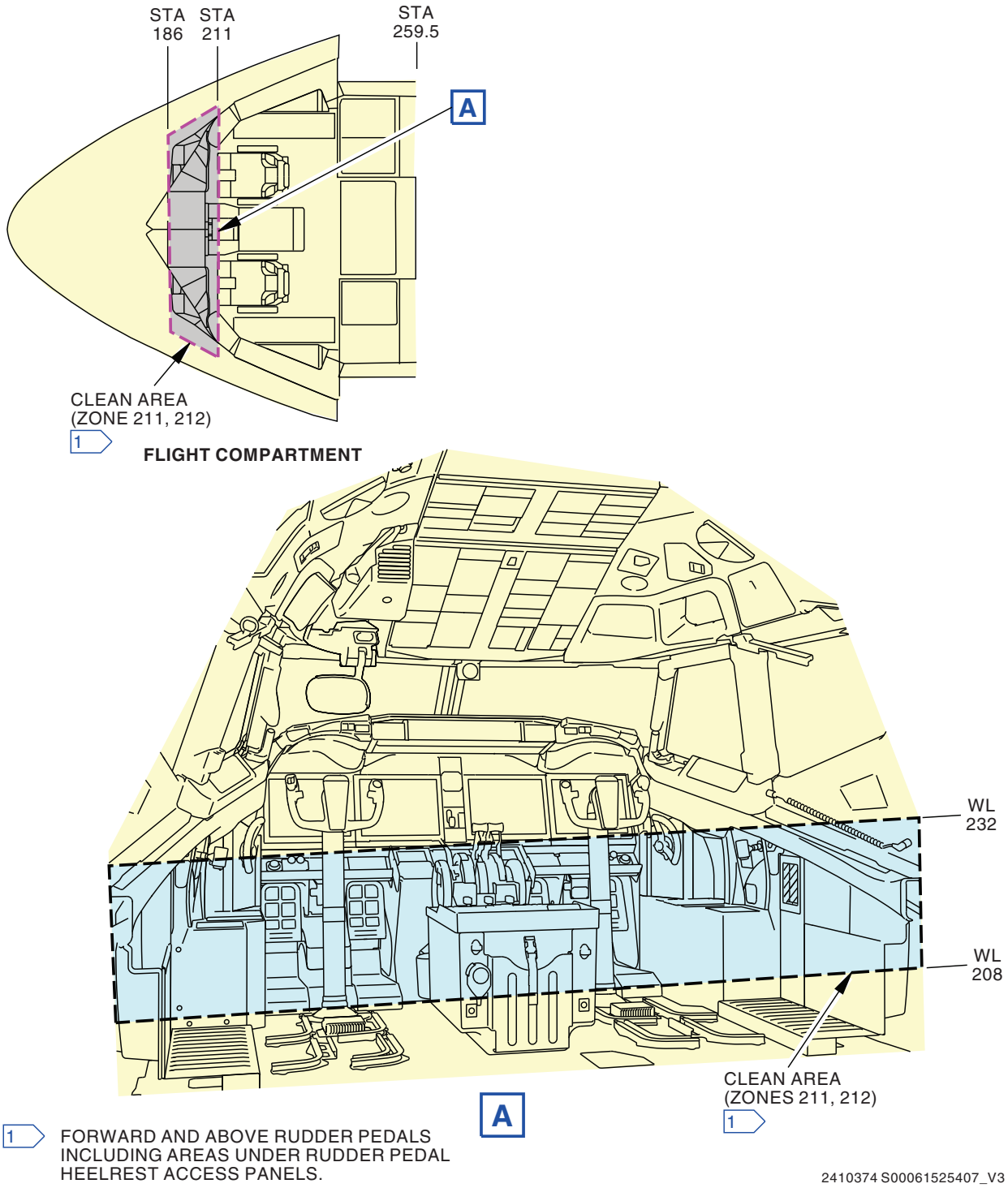
(2) Restore (clean) area in Flight Compartment from Sta 186 to 211, WL 208 to 232 (EZAP).

SUBTASK 05-42-02-910-006

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Compartment
Figure 205/05-42-02-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-210-801

8. General Visual Inspection: Exposed EWIS in the Flight Compartment excluding exposed EWIS in the P5, P6 and P18 panels. (EZAP)

(Figure 206)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-801	General Visual Inspection of EWIS (P/B 201)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Inspection

NOTE: With access provided. Seats removed. Control stand access panels, overhead & sidewall panels, glareshield, instruments/panels removal required.

SUBTASK 05-42-02-210-001

(1) Do this task: General Visual Inspection of EWIS, TASK 20-60-02-100-801.

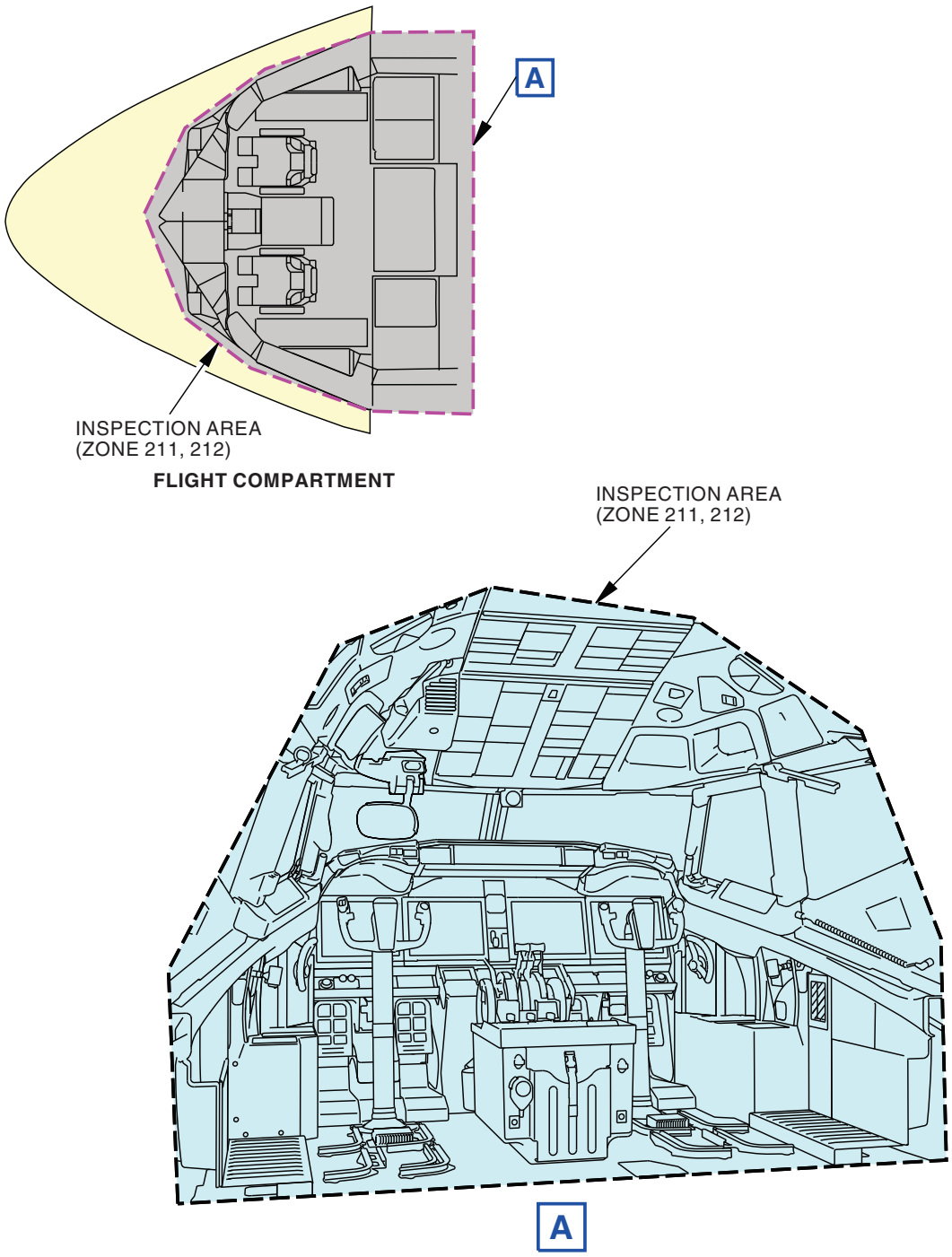
(2) Inspect (General Visual) all exposed EWIS in the Flight Compartment excluding exposed EWIS in the P5, P6 and P18 panels. (EZAP)

SUBTASK 05-42-02-910-007

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inspect (General Visual) - Exposed EWIS In Flight Compartment Excluding Exposed EWIS In the P3, P6, and P18 Panels
Figure 206/05-42-02-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-02-211-801

9. Detailed Inspection: Exposed EWIS in the P5, P6 and P18 panels. (EZAP)

(Figure 207)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Inspection

NOTE: Opening P5 (fwd/aft), P6 and P18 panels required.

SUBTASK 05-42-02-211-001

(1) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

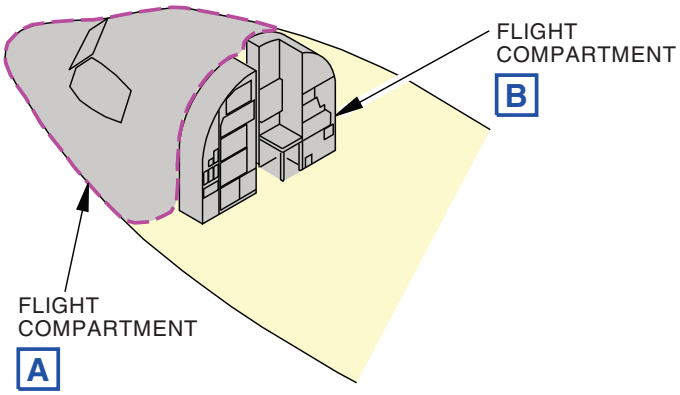
(2) Inspect (Detailed) all exposed EWIS in the P5 (fwd/aft), P6 and P18 panels. (EZAP)

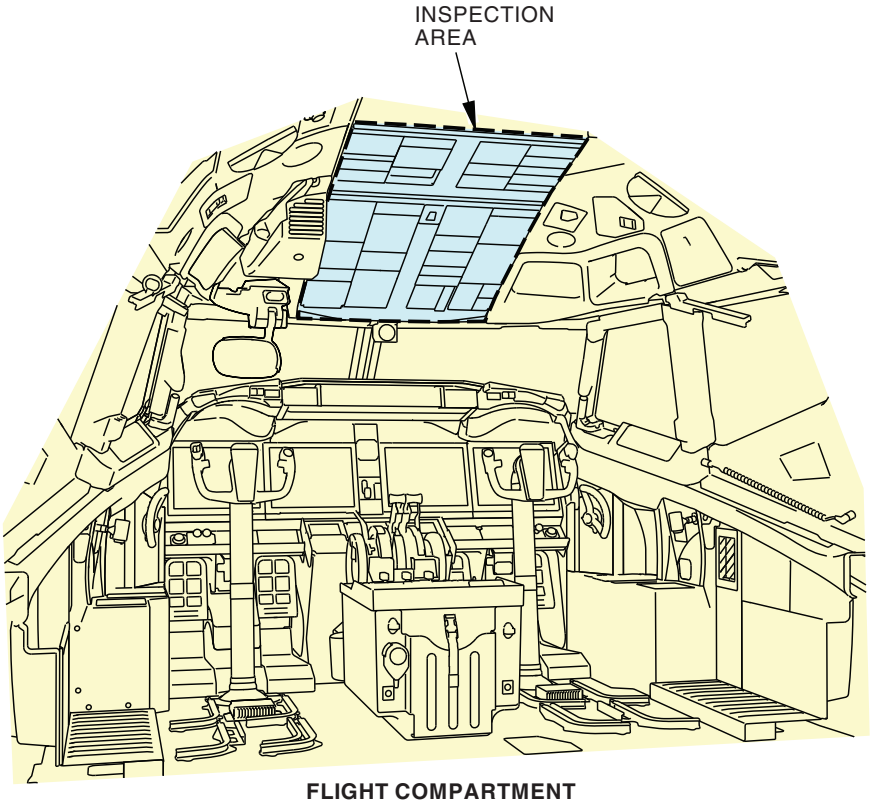
SUBTASK 05-42-02-910-008

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

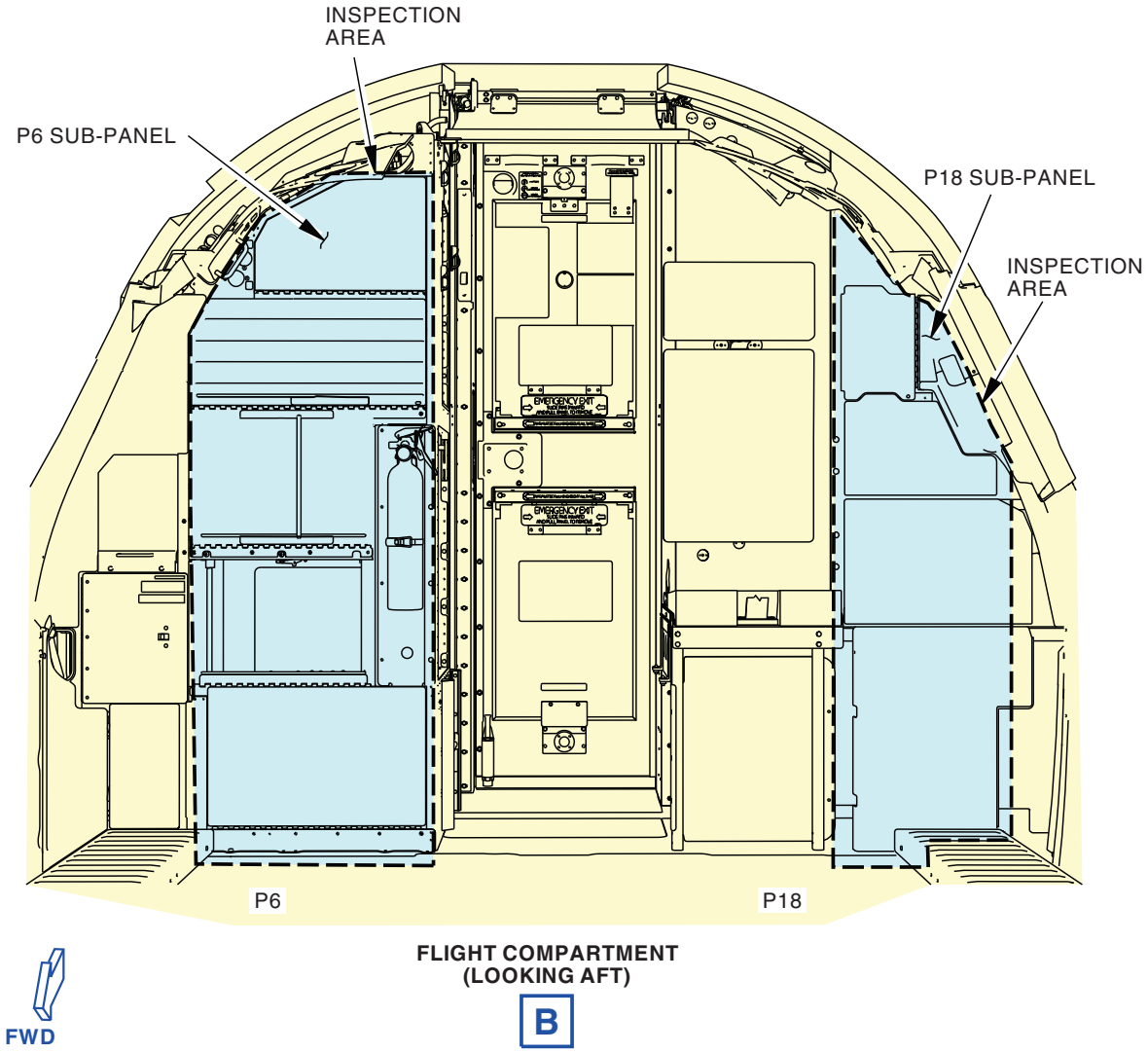




2410376 S00061525411_V1

Exposed EWIS in P5, P6, and P18 Panels
Figure 207/05-42-02-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exposed EWIS in P5, P6, and P18 Panels
Figure 207/05-42-02-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1. General

A. This pageblock contains MSG-3 task card data.

EWIS

TASK 05-42-03-210-801

2. General Visual Inspection: Exposed EWIS in the Tail Cone. (EZAP)

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-801	General Visual Inspection of EWIS (P/B 201)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

E. Inspection

SUBTASK 05-42-03-010-001

(1) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 05-42-03-210-001

(2) Do this task: General Visual Inspection of EWIS, TASK 20-60-02-100-801.

(3) Inspect (General Visual) all exposed EWIS in the Tail Cone. (EZAP)

SUBTASK 05-42-03-910-001

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

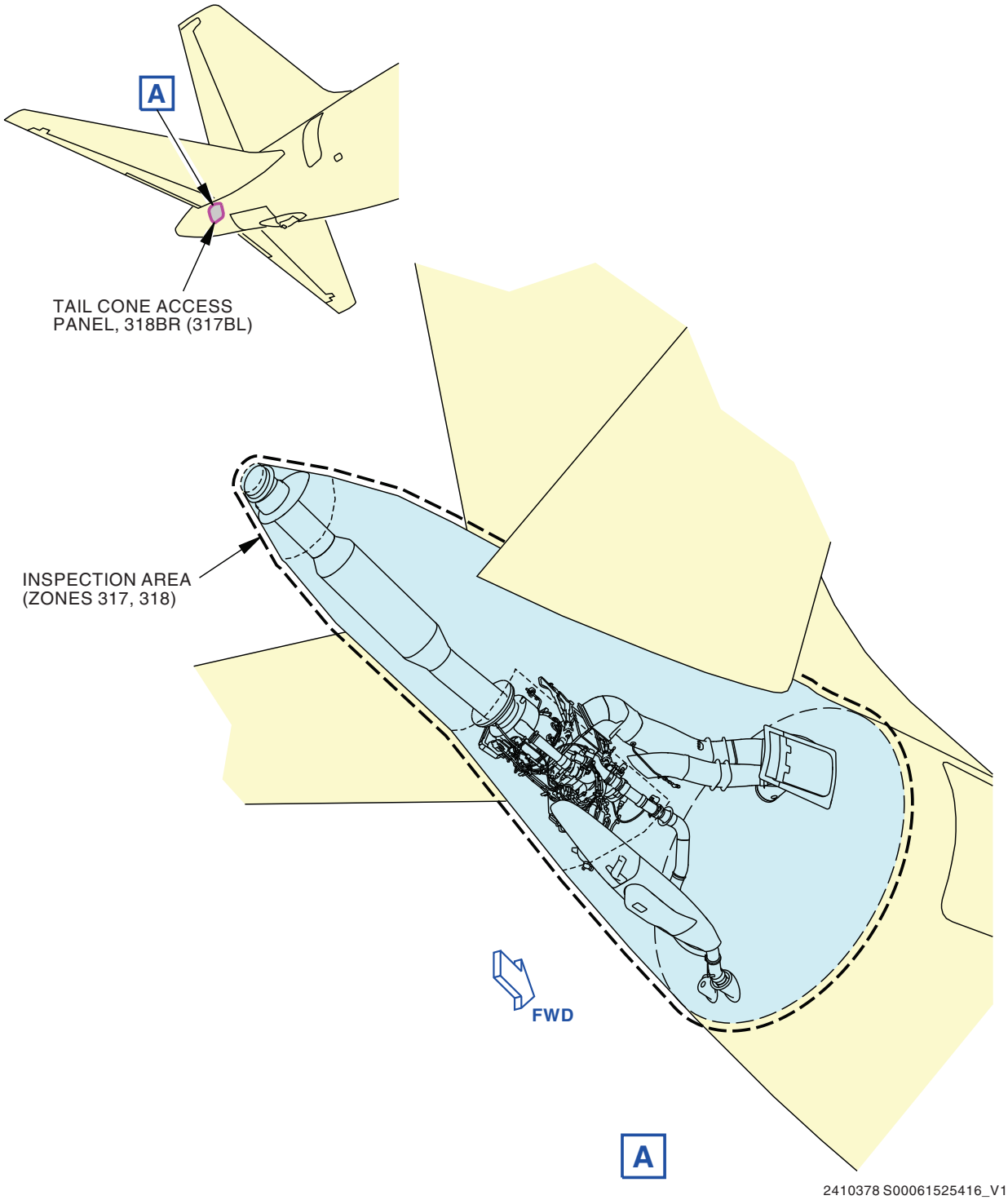
SUBTASK 05-42-03-410-001

(5) Close these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inspect (General Visual) - Exposed EWIS In Tail Cone
Figure 201/05-42-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-03-211-801

3. Detailed Inspection: APU Starter/Generator Power Feeder Wiring/Ignition Leads/ and connected EWIS in the APU compartment. (EZAP)

(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right

D. Access Panels

Number	Name/Location
315A	APU Cowl Door

E. Inspection

SUBTASK 05-42-03-010-002

(1) Open this access panel:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 05-42-03-211-001

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) the APU starter/generator power feeder wiring, ignition leads and connected EWIS in the APU Compartment. (EZAP)

SUBTASK 05-42-03-910-002

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

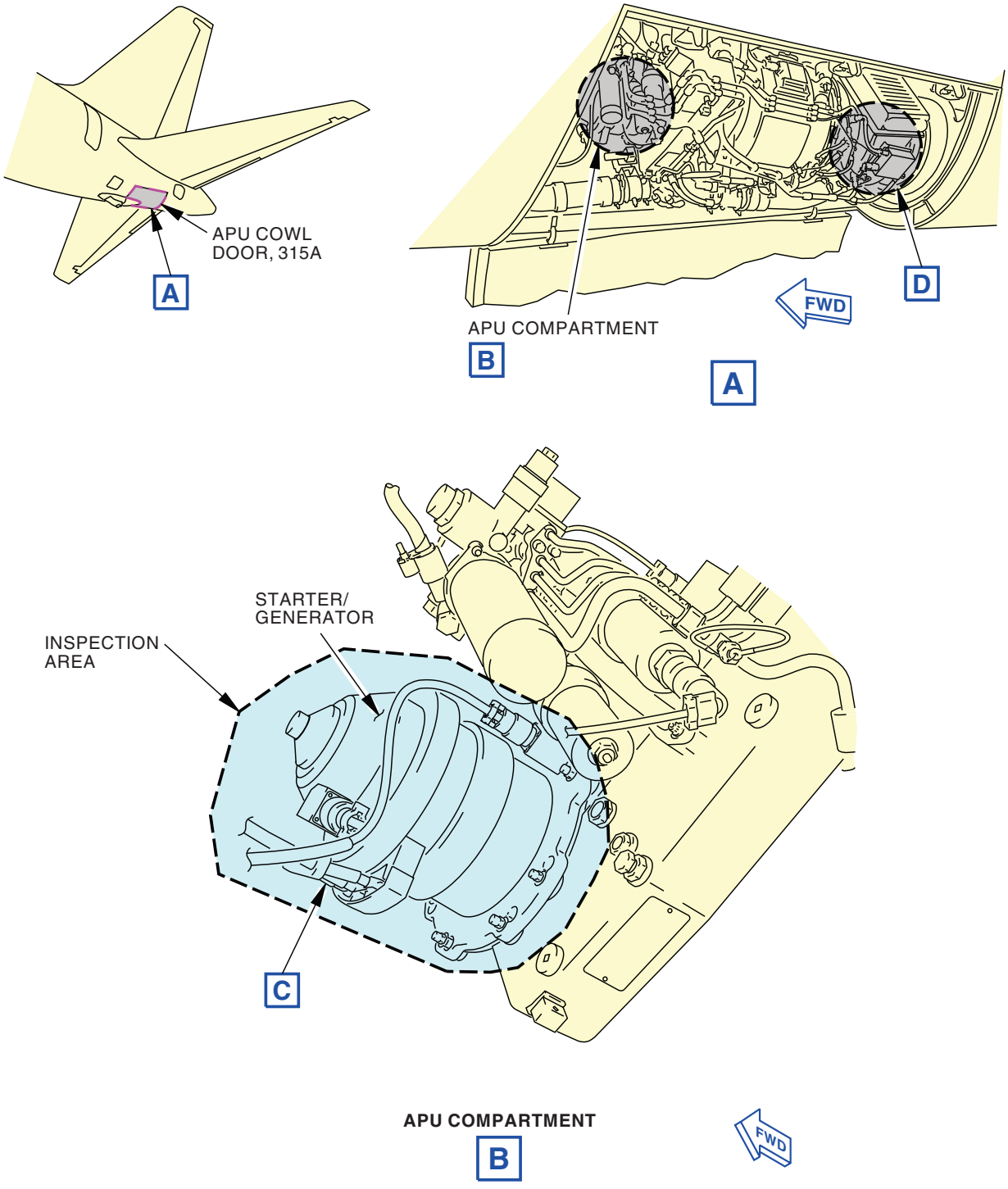
SUBTASK 05-42-03-410-002

(4) Close this access panel:

Number	Name/Location
315A	APU Cowl Door

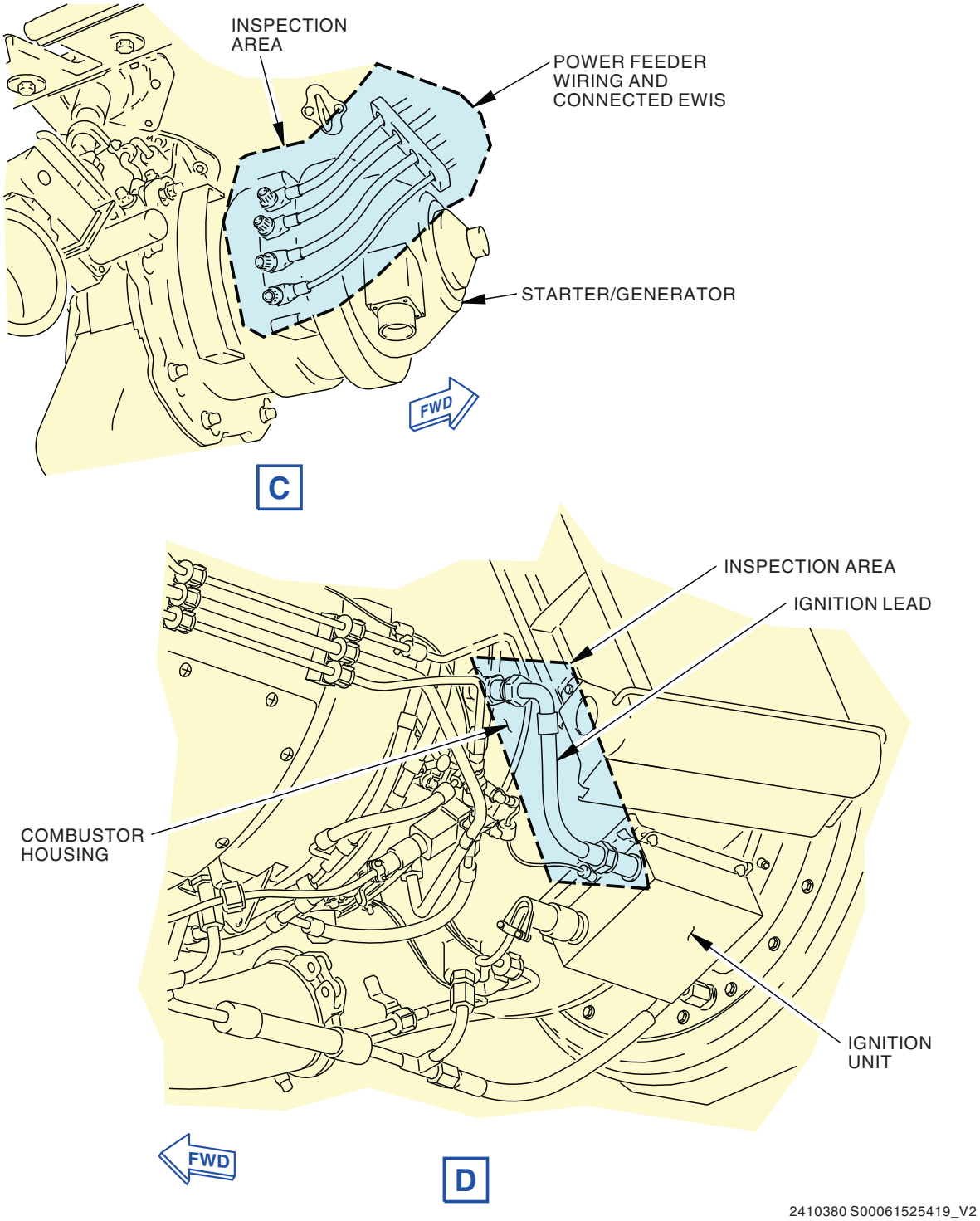
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Starter/Generator Power Feeder Wiring, Ignition Leads, and Connected EWIS
Figure 202/05-42-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Starter/Generator Power Feeder Wiring, Ignition Leads, and Connected EWIS
Figure 202/05-42-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-03-211-802

4. Detailed Inspection: APU Starter/Generator Power Feeder Wiring/EWIS in the area aft of Pressure Bulkhead. (EZAP)

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Inspection

SUBTASK 05-42-03-010-003

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 05-42-03-211-002

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) the APU starter/generator power feeder wiring and connected EWIS in the area aft of pressure bulkhead. (EZAP)

SUBTASK 05-42-03-910-003

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

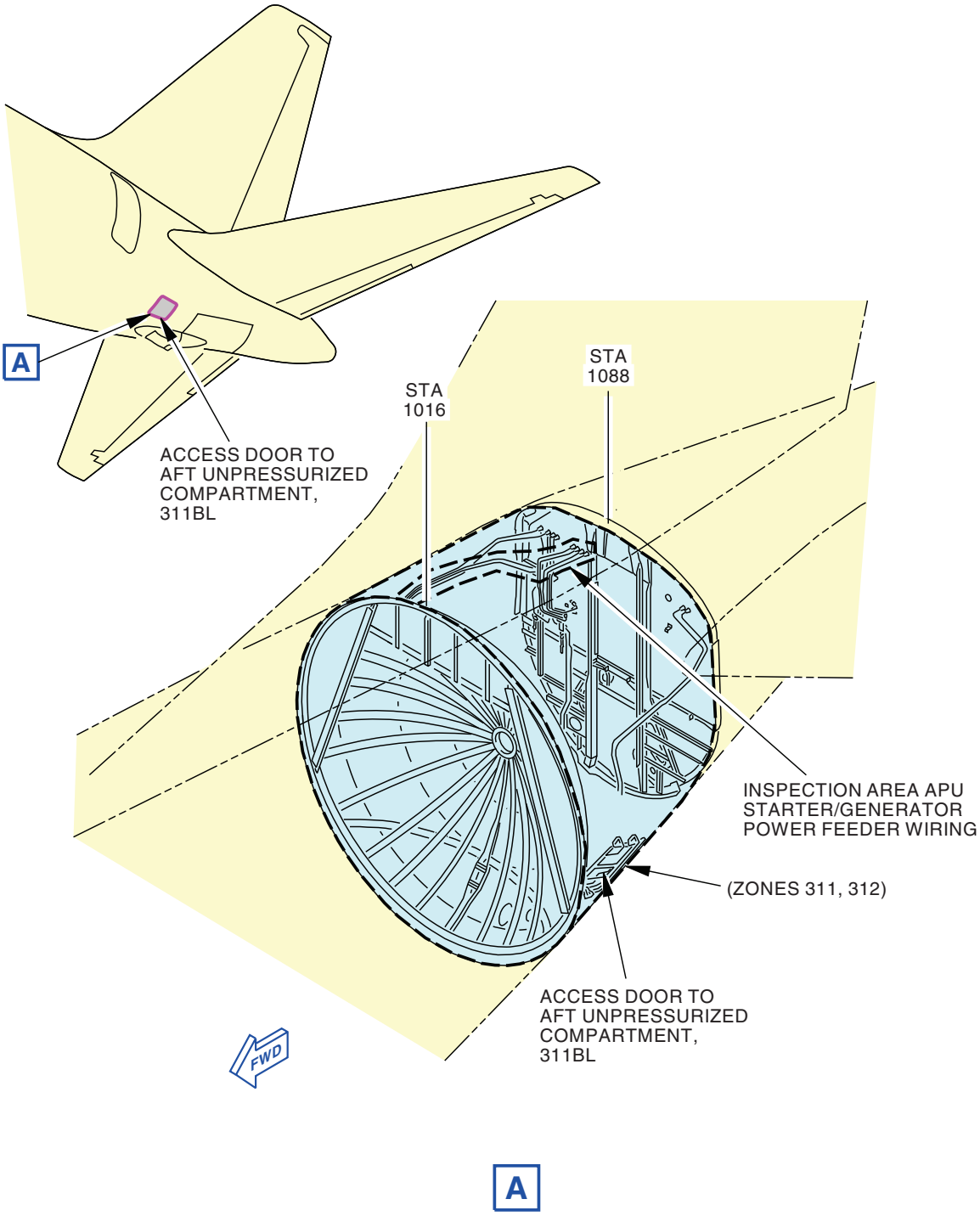
SUBTASK 05-42-03-410-003

(5) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410381 S00061525421_V1

Aft Pressure Bulkhead APU EWIS
Figure 203/05-42-03-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1. General

A. This pageblock contains MSG-3 task card data.

EWIS

TASK 05-42-04-211-801

2. Detailed Inspection: Exposed EWIS in Engine 1 (Engine fan and core sections). (EZAP)

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

E. Inspection

SUBTASK 05-42-04-010-001

(1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

SUBTASK 05-42-04-211-001

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in Engine 1 (Engine fan and core sections). (EZAP)

SUBTASK 05-42-04-910-001

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

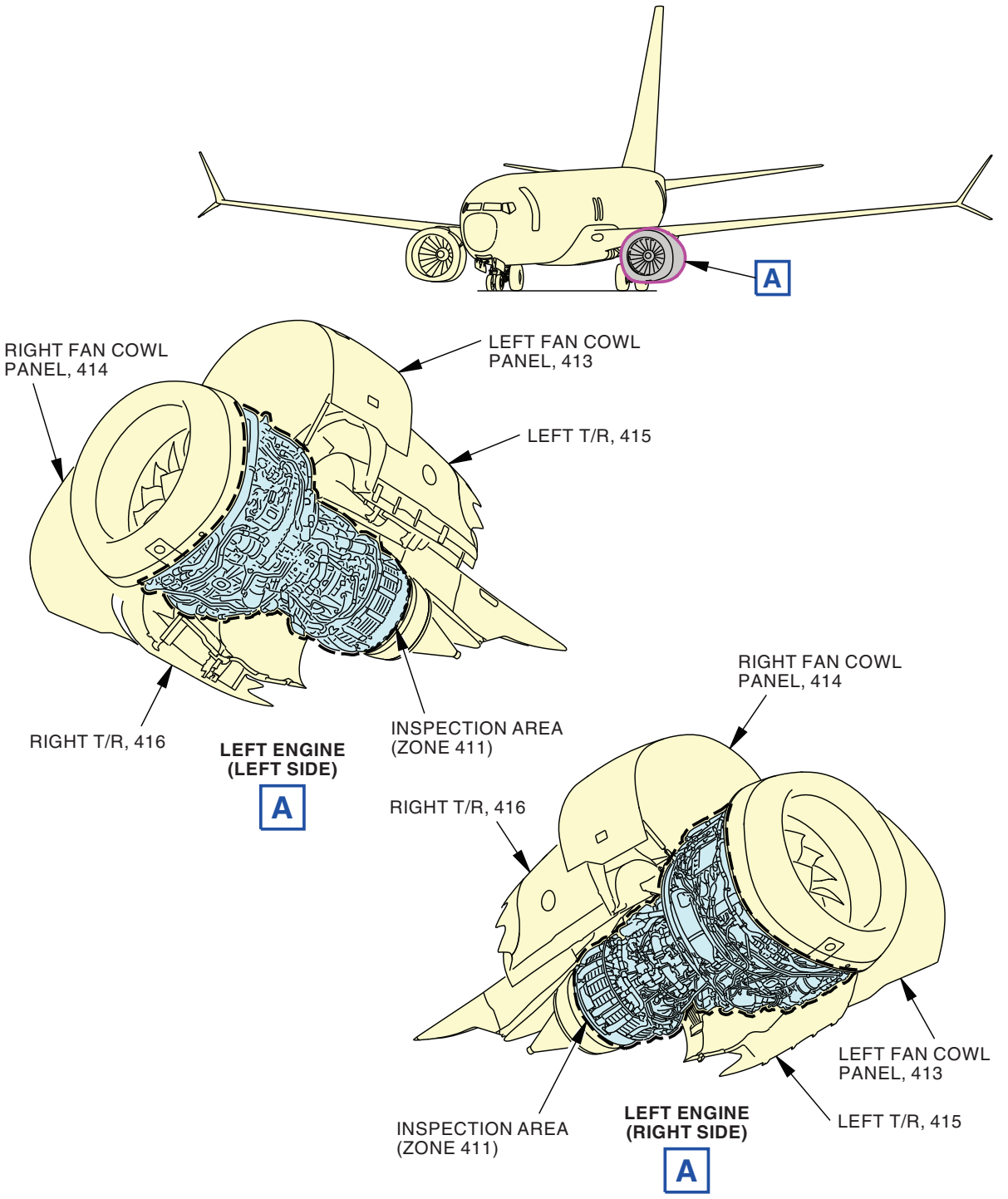
SUBTASK 05-42-04-410-001

(5) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



IDG Power Feeder and Connected EWIS
Figure 201/05-42-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-802

3. Detailed Inspection: Exposed EWIS in Engine 2 (Engine fan and core sections). (EZAP)

(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
421	Engine 2 - Engine

D. Access Panels

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Inspection

SUBTASK 05-42-04-010-002

(1) Open these access panels:

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 05-42-04-211-002

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in Engine 2 (Engine fan and core sections). (EZAP)

SUBTASK 05-42-04-910-002

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-04-410-002

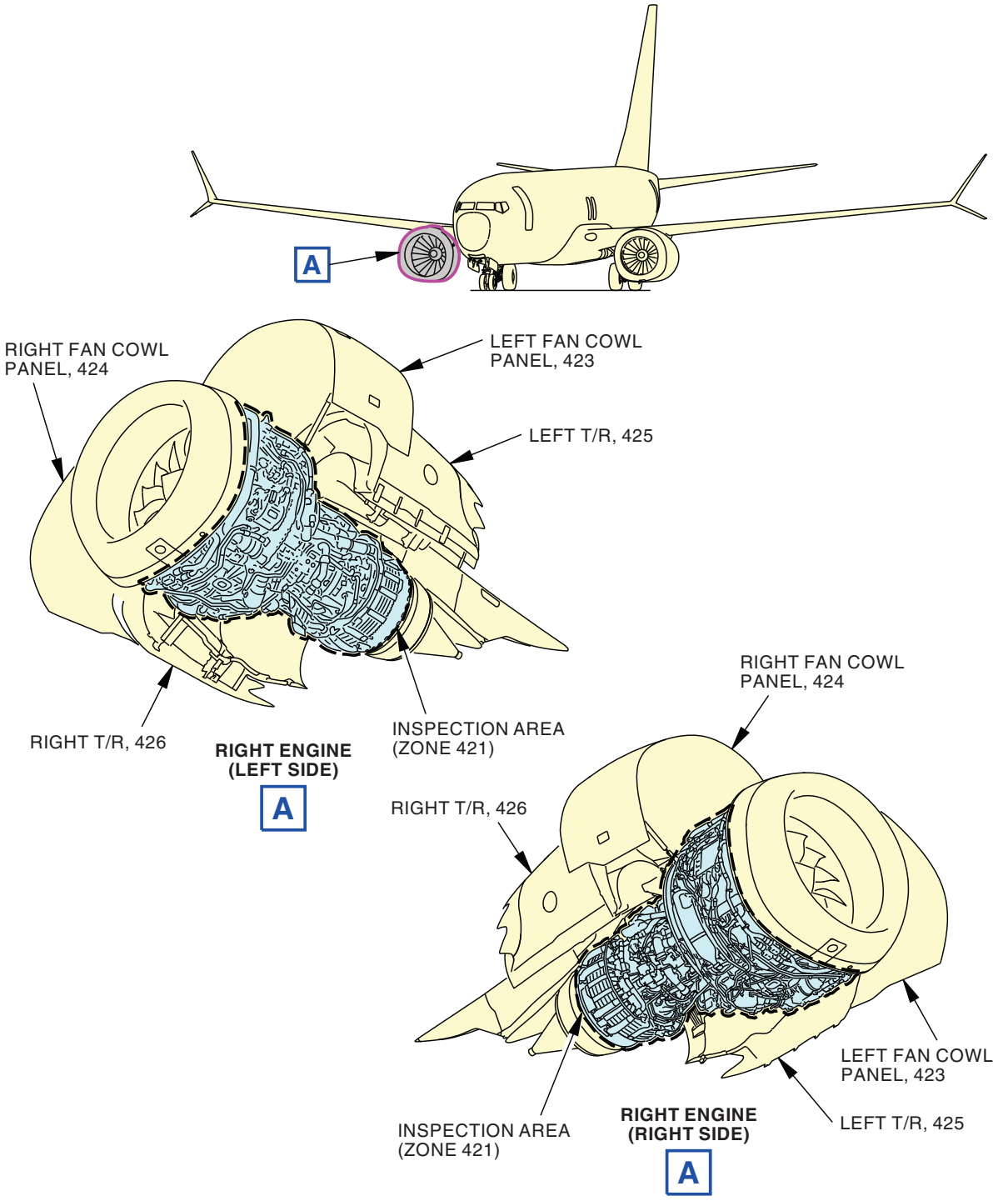
(5) Close these access panels:

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



IDG Power Feeder and Connected EWIS
Figure 202/05-42-04-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-803

4. Detailed Inspection: Exposed EWIS in the Forward Strut Fairing - Engine No. 1. (EZAP)

(Figure 203)

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
431	Engine 1 - Forward Strut Fairing

D. Access Panels

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 1
431DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 1

E. Detailed Inspection Exposed EWIS in the Forward Strut Fairing Engine No. 1. (EZAP)

SUBTASK 05-42-04-010-003

(1) Open these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 1
431DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 1

SUBTASK 05-42-04-211-003

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS in the Forward Strut Fairing - Engine No. 1. (EZAP).

SUBTASK 05-42-04-910-003

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

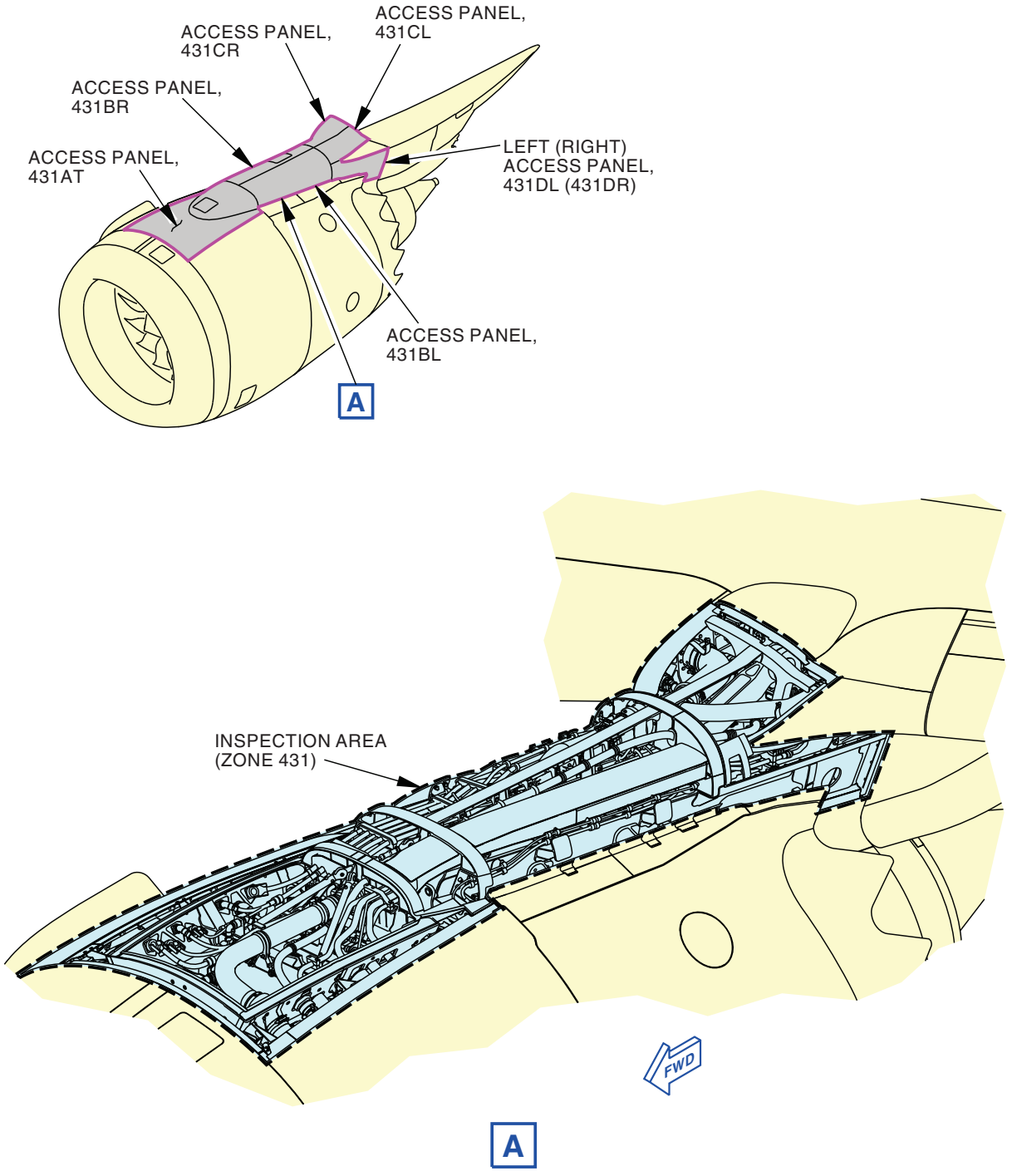
SUBTASK 05-42-04-410-003

(4) Close these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 1
431DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Strut Fairing - Engine 1
Figure 203/05-42-04-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-804

5. Detailed Inspection: Exposed EWIS in Fan Cowl Support Beam - Engine No. 1. (EZAP)

(Figure 204)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
432	Engine 1 - Fan Cowl Support Beam

D. Access Panels

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1

E. Inspection

SUBTASK 05-42-04-010-004

(1) Open this access panel:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1

SUBTASK 05-42-04-211-004

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS in Fan Cowl Support Beam - Engine No. 1. (EZAP).

SUBTASK 05-42-04-910-004

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

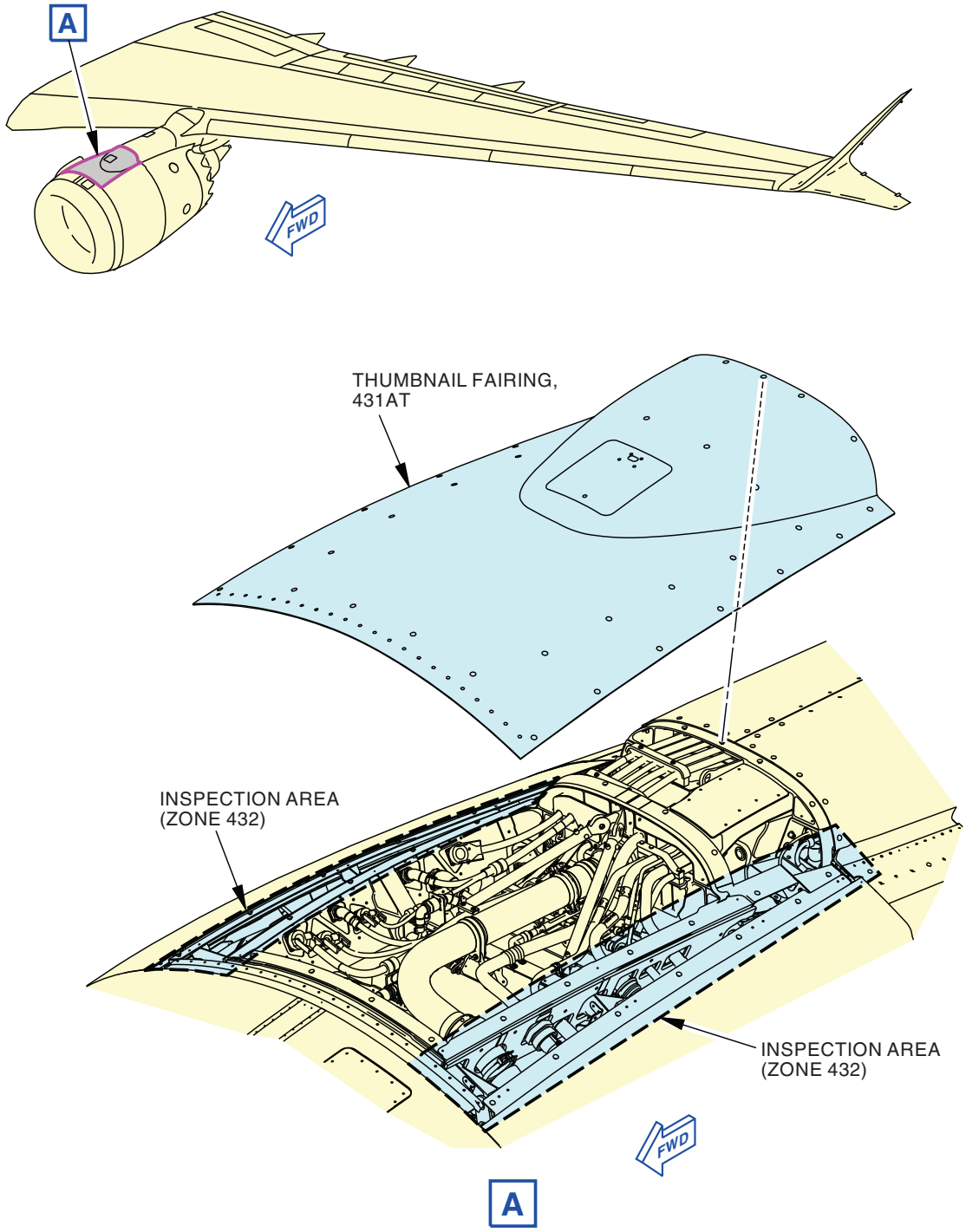
SUBTASK 05-42-04-410-004

(4) Close this access panel:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410384 S00061525430_V2

Fan Cowl Support Beam
Figure 204/05-42-04-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-805

6. Detailed Inspection: Exposed EWIS common to the Strut Torque Box - Engine No. 1. (EZAP)
(Figure 205)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
433	Engine 1 - Strut Torque Box

D. Access Panels

Number	Name/Location
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

E. Inspection

SUBTASK 05-42-04-010-005

(1) Open these access panels:

Number	Name/Location
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

SUBTASK 05-42-04-211-005

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS common to the Strut Torque Box - Engine No. 1. (EZAP).

SUBTASK 05-42-04-910-005

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

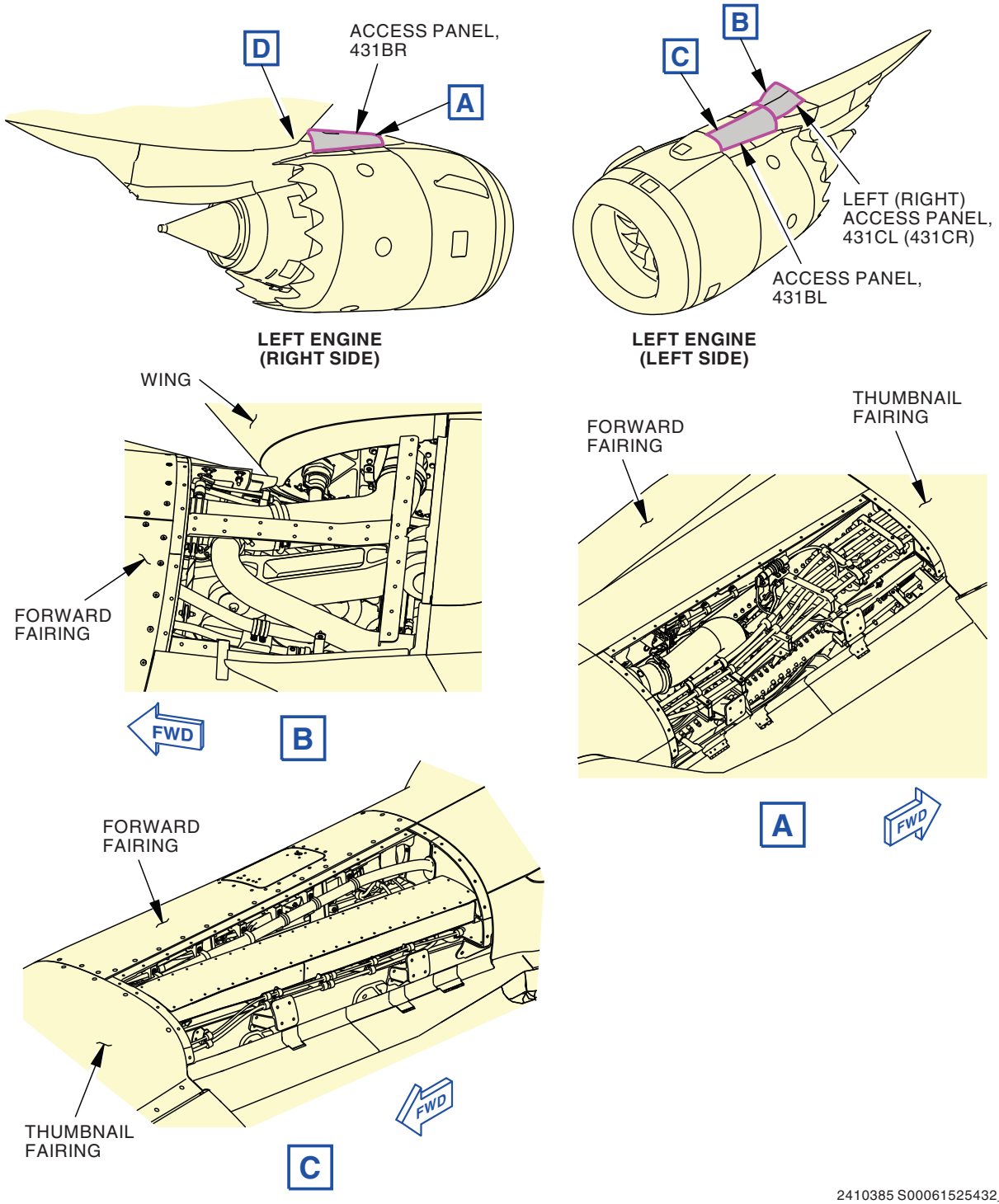
SUBTASK 05-42-04-410-005

(4) Close these access panels:

Number	Name/Location
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

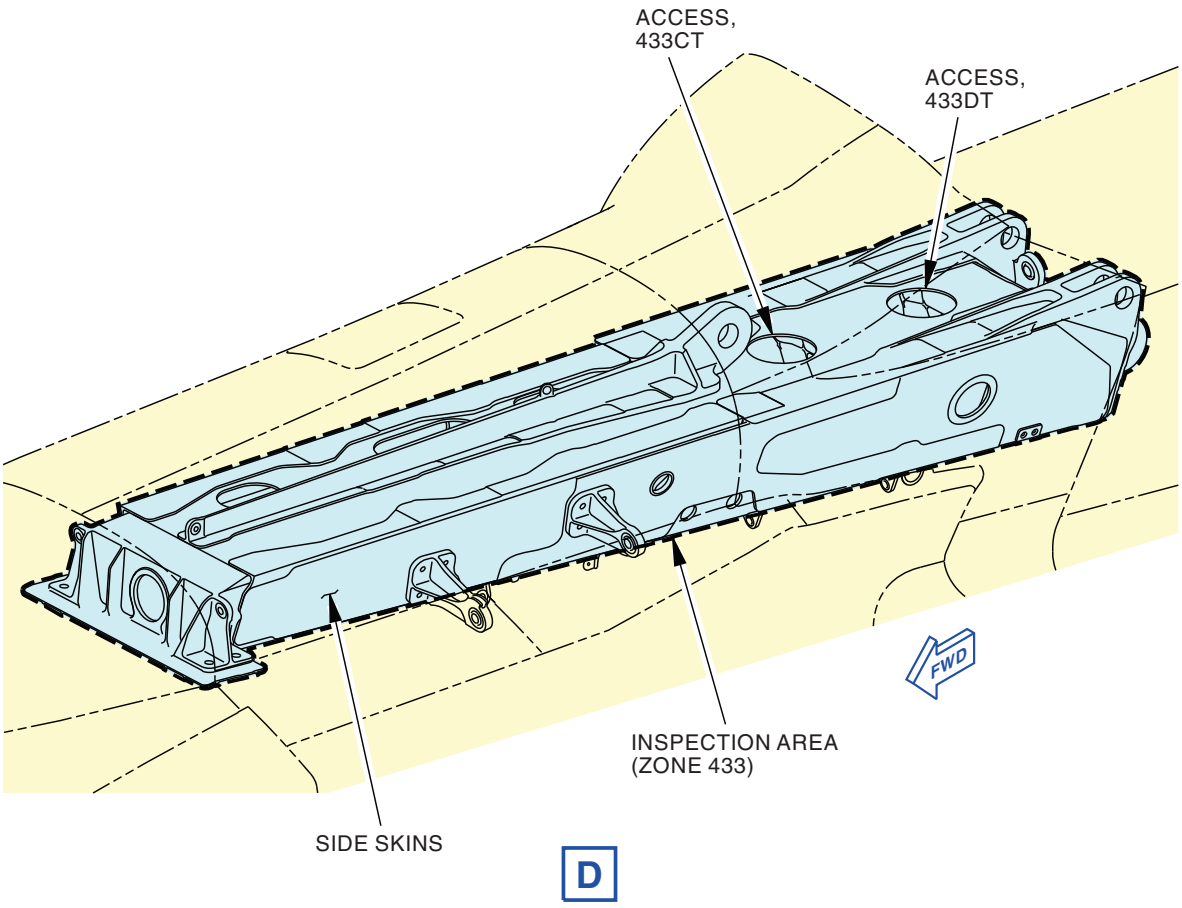
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exposed EWIS in Strut Torque Box - Engine 1
Figure 205/05-42-04-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410386 S00061525433_V3

Exposed EWIS in Strut Torque Box - Engine 1
Figure 205/05-42-04-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-806

7. Detailed Inspection: Exposed EWIS in the Forward Strut Fairing - Engine No. 2. (EZAP)

(Figure 206)

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
441	Engine 2 - Forward Strut Fairing

D. Access Panels

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 2
441DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 2

E. Detailed Inspection Exposed EWIS in the Forward Strut Fairing Engine No. 2. (EZAP)

SUBTASK 05-42-04-010-006

(1) Open these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 2
441DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 2

SUBTASK 05-42-04-211-006

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS in the Forward Strut Fairing - Engine No. 2. (EZAP).

SUBTASK 05-42-04-910-006

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

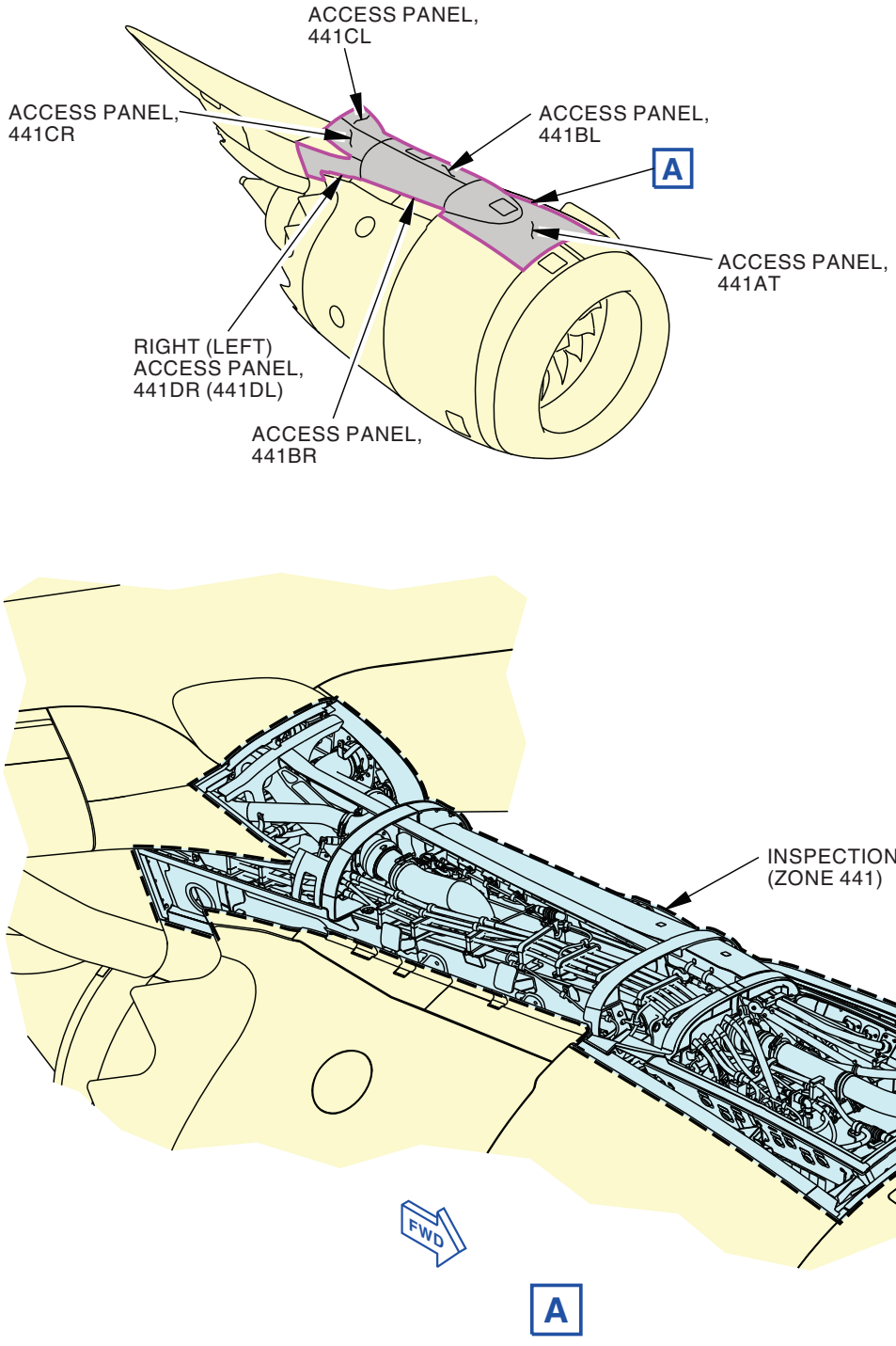
SUBTASK 05-42-04-410-006

(4) Close these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441DL	Forward Strut Fairing, Left Underwing (Cheek) Fairing, Strut 2
441DR	Forward Strut Fairing, Right Underwing (Cheek) Fairing, Strut 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Strut Fairing - Engine 2
Figure 206/05-42-04-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-807

8. Detailed Inspection: Exposed EWIS in Fan Cowl Support Beam - Engine No. 2. (EZAP)
(Figure 207)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
442	Engine 2 - Fan Cowl Support Beam

D. Access Panels

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2

E. Inspection

SUBTASK 05-42-04-010-007

(1) Open this access panel:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2

SUBTASK 05-42-04-211-007

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS in Fan Cowl Support Beam - Engine No. 2. (EZAP).

SUBTASK 05-42-04-910-007

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

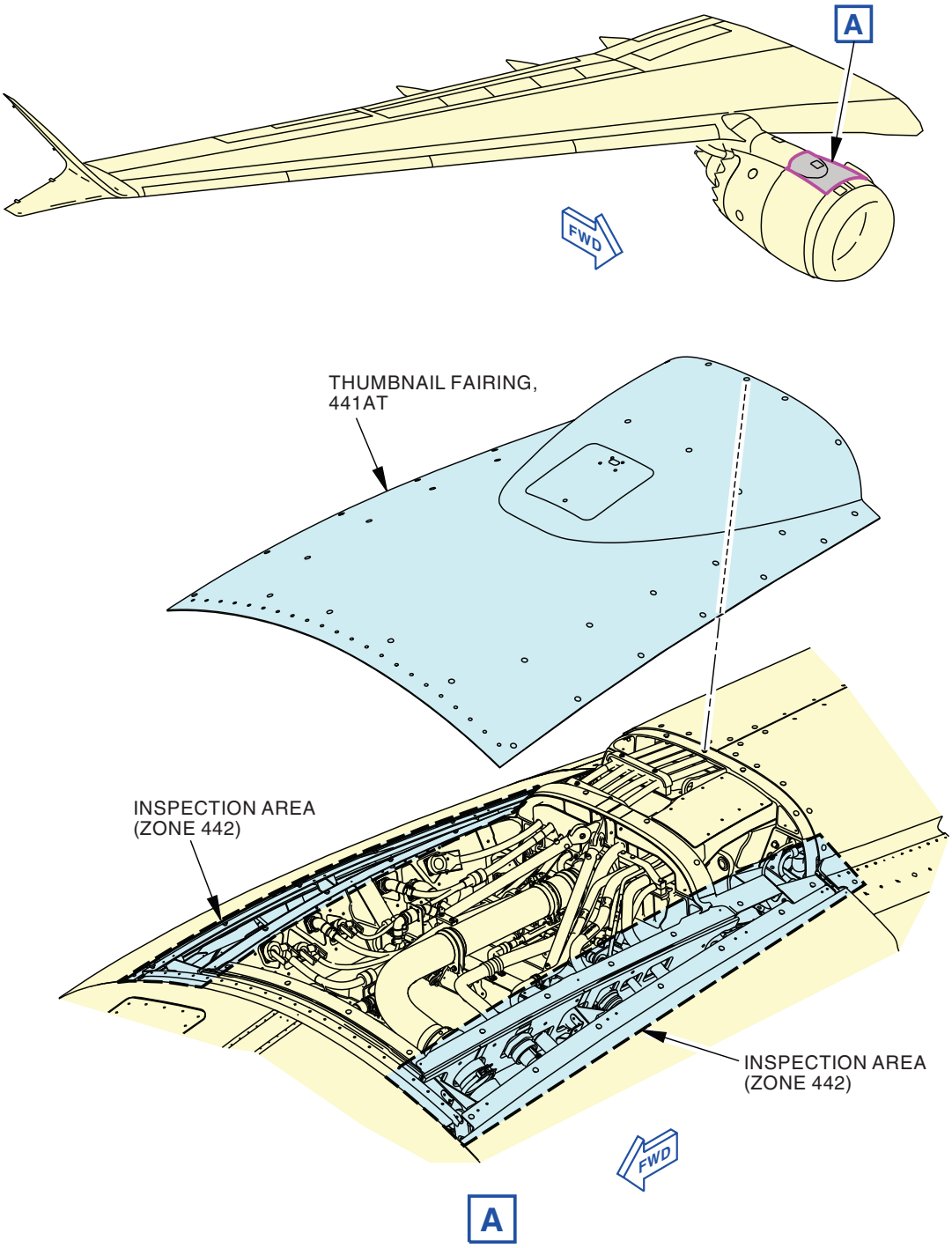
SUBTASK 05-42-04-410-007

(4) Close this access panel:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410388 S00061525437_V2

Fan Cowl Support Beam - Engine 2
Figure 207/05-42-04-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-04-211-808

9. Detailed Inspection: Exposed EWIS common to the Strut Torque Box - Engine No. 2. (EZAP)
(Figure 208)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
443	Engine 2 - Strut Torque Box

D. Access Panels

Number	Name/Location
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

E. Inspection

SUBTASK 05-42-04-010-008

(1) Open these access panels:

Number	Name/Location
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

SUBTASK 05-42-04-211-008

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(a) Inspect (Detailed) all exposed EWIS common to the Strut Torque Box - Engine No. 2. (EZAP).

SUBTASK 05-42-04-910-008

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

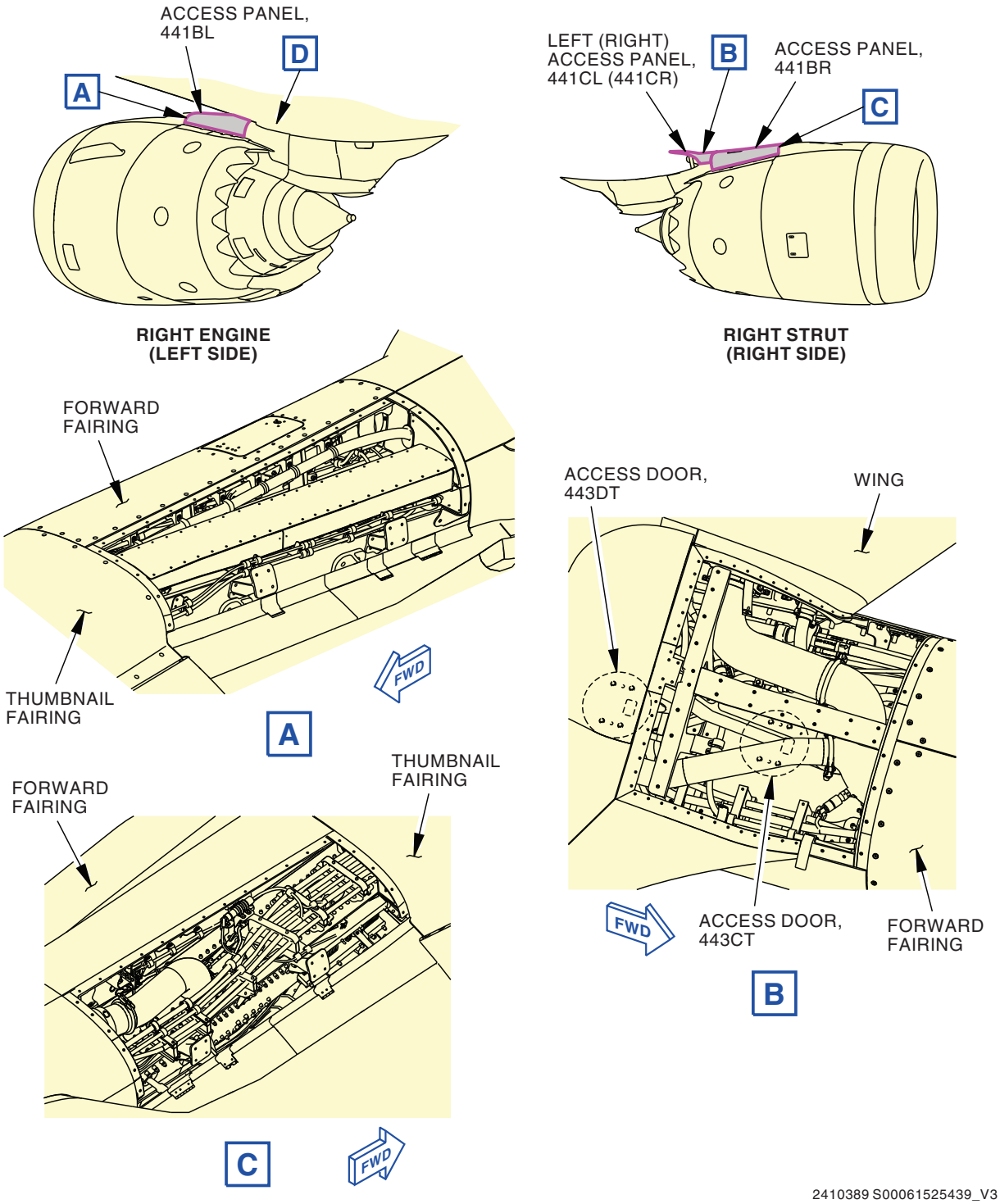
SUBTASK 05-42-04-410-008

(4) Close these access panels:

Number	Name/Location
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

——— END OF TASK ———

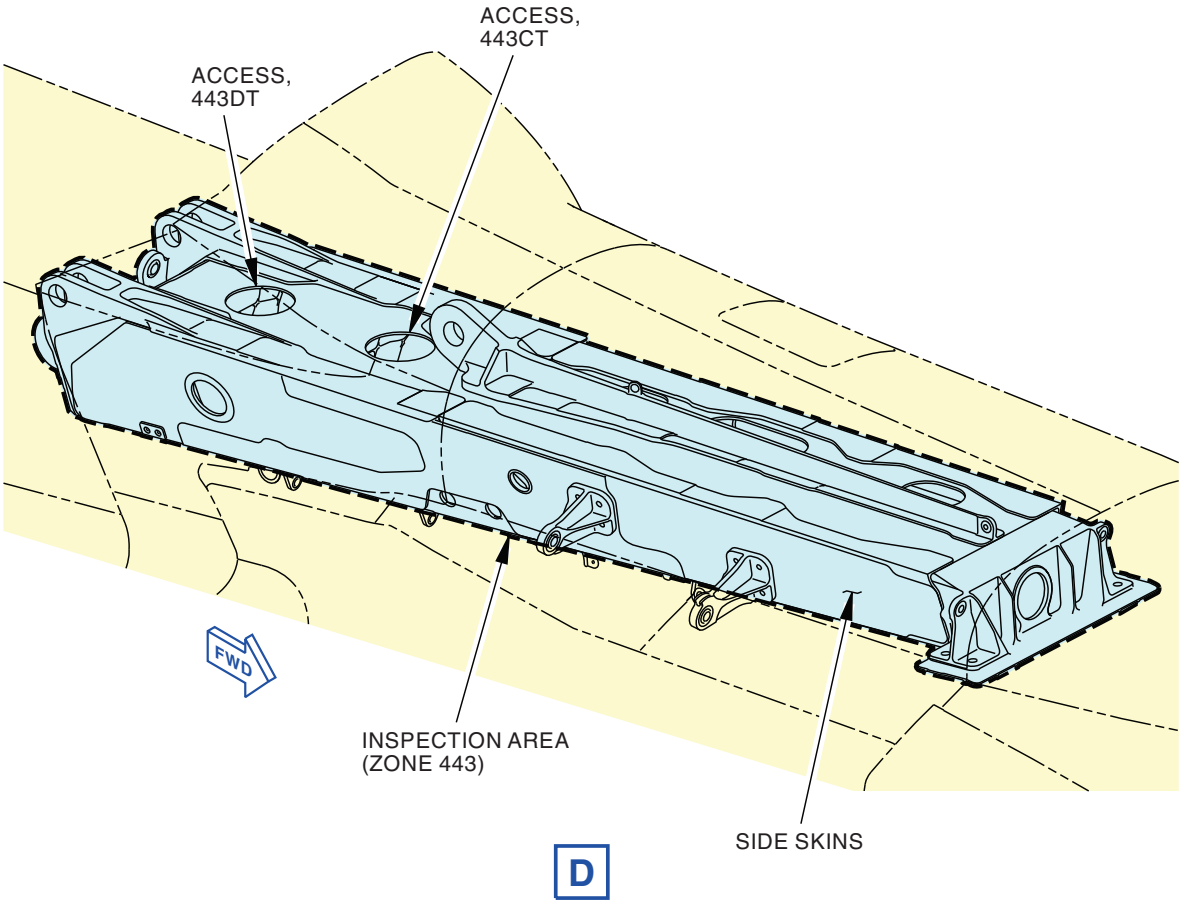
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410389 S00061525439_V3

Exposed EWIS in Strut Torque Box - Engine 2
Figure 208/05-42-04-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410390 S00061525440_V3

Exposed EWIS in Strut Torque Box - Engine 2
Figure 208/05-42-04-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ZONE 500 - LEFT WING - (EZAP)

1. General

A. This pageblock contains MSG-3 task card data.

EWIS

TASK 05-42-05-211-801

2. Detailed Inspection: Exposed EWIS in the area from Leading Edge to Front Spar - Inboard of Nacelle Strut - Left Wing. (EZAP)

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar

D. Access Panels

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
511BT	Fairing

E. Inspection

SUBTASK 05-42-05-010-001

(1) Open these access panels:

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
511BT	Fairing

NOTE: Leading edge flaps extended.

SUBTASK 05-42-05-211-001

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in the area from Leading Edge to Front Spar - Inboard of Nacelle Strut - Left Wing. (EZAP)

SUBTASK 05-42-05-910-001

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

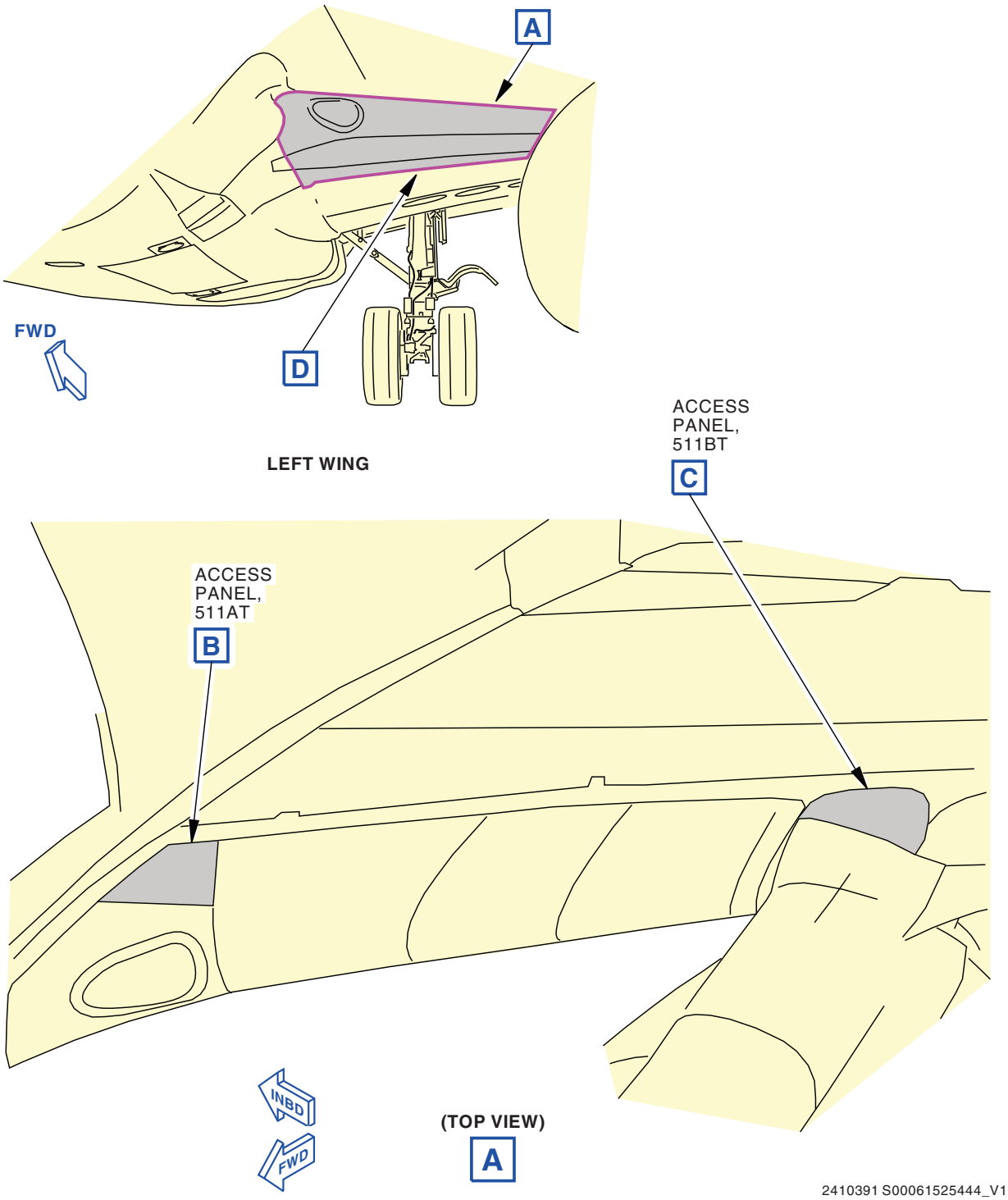
SUBTASK 05-42-05-410-001

(5) Close these access panels:

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
511BT	Fairing

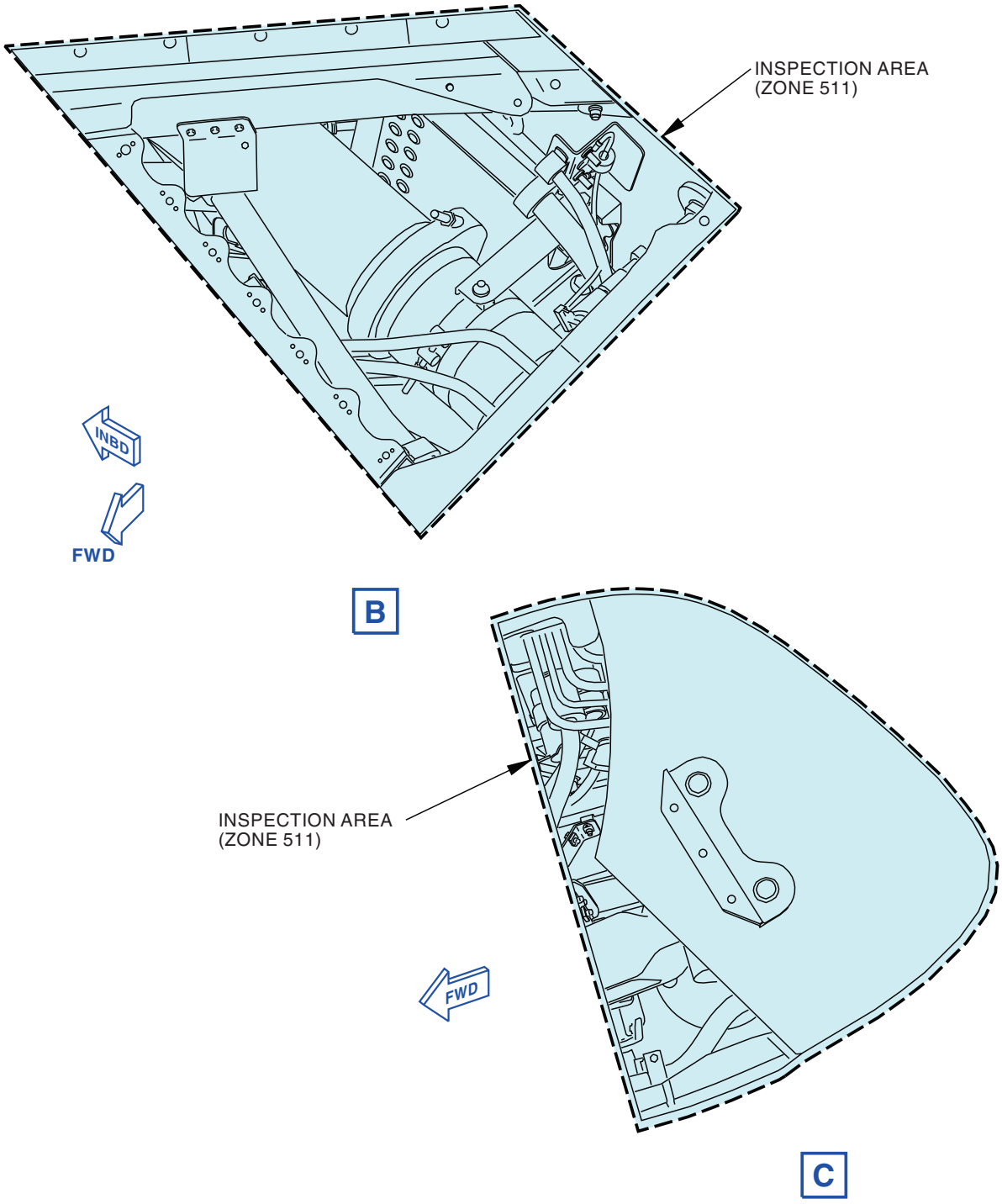
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



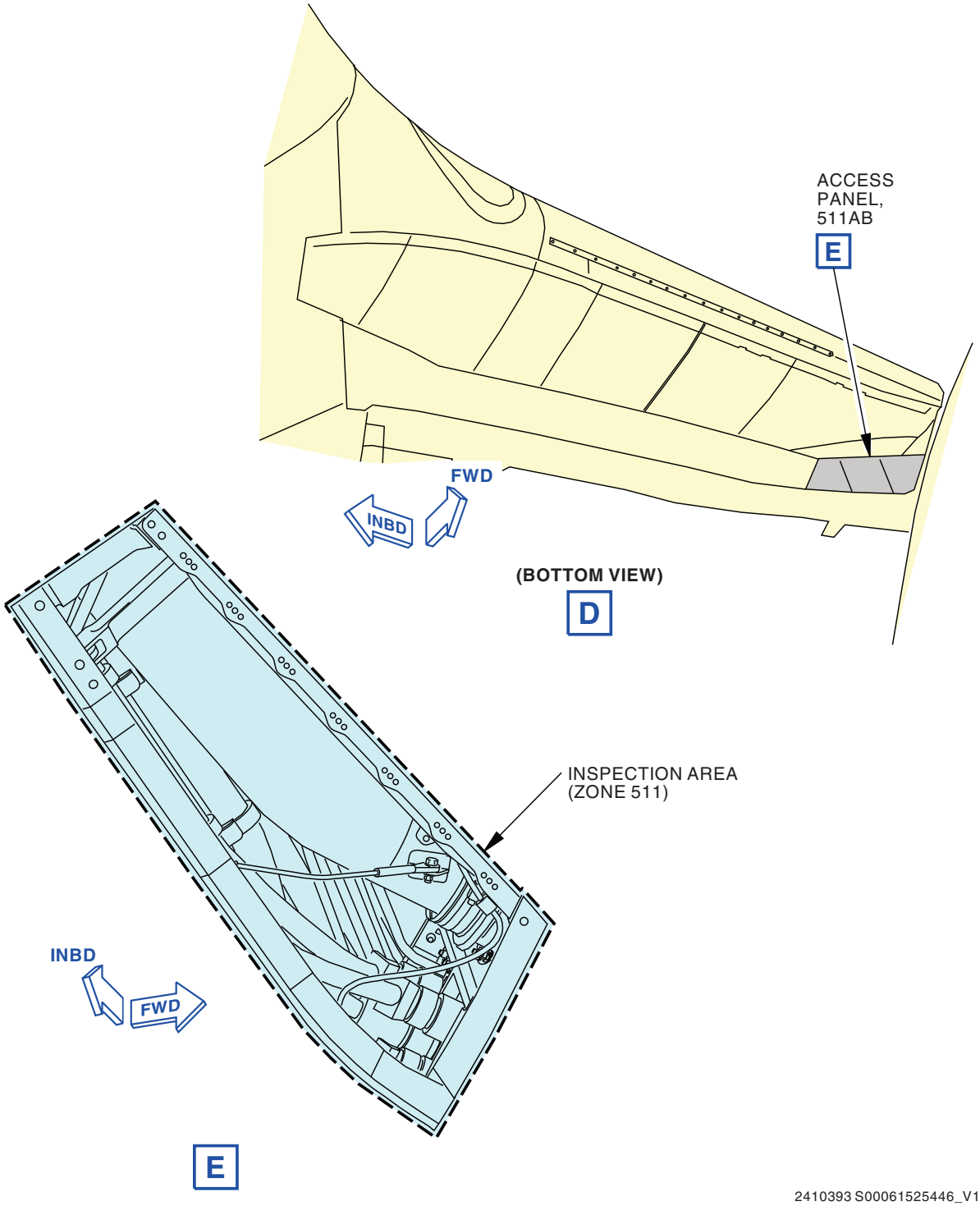
Leading Edge to Front Spar - Left Wing - Inboard
Figure 201/05-42-05-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Left Wing - Inboard
Figure 201/05-42-05-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Left Wing - Inboard
Figure 201/05-42-05-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-05-211-802

3. Detailed Inspection: Exposed EWIS in the area from Leading Edge to Front Spar - Outboard of Nacelle Strut - Left Wing. (EZAP)

(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

C. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar

D. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521AB	Outboard Leading Edge Blowout Door
521ABB	Lower Leading Edge Access Panel
521AT	Outbd Leading Edge - Gap Cover Access
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

E. Inspection

SUBTASK 05-42-05-010-002

(1) Open these access panels:

(TASK 57-41-02-000-801)

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521AB	Outboard Leading Edge Blowout Door
521ABB	Lower Leading Edge Access Panel
521AT	Outbd Leading Edge - Gap Cover Access
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

NOTE: Access is gained by extending slats, and through lower wing surface access panels.

SUBTASK 05-42-05-211-002

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

SUBTASK 05-42-05-211-009

(3) Inspect (Detailed) all exposed Electrical Wiring Interconnect Systems (EWIS) in the area from Leading Edge to Front Spar - Outboard of Nacelle Strut - Left Wing. (EZAP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-42-05-910-002

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-05-410-002

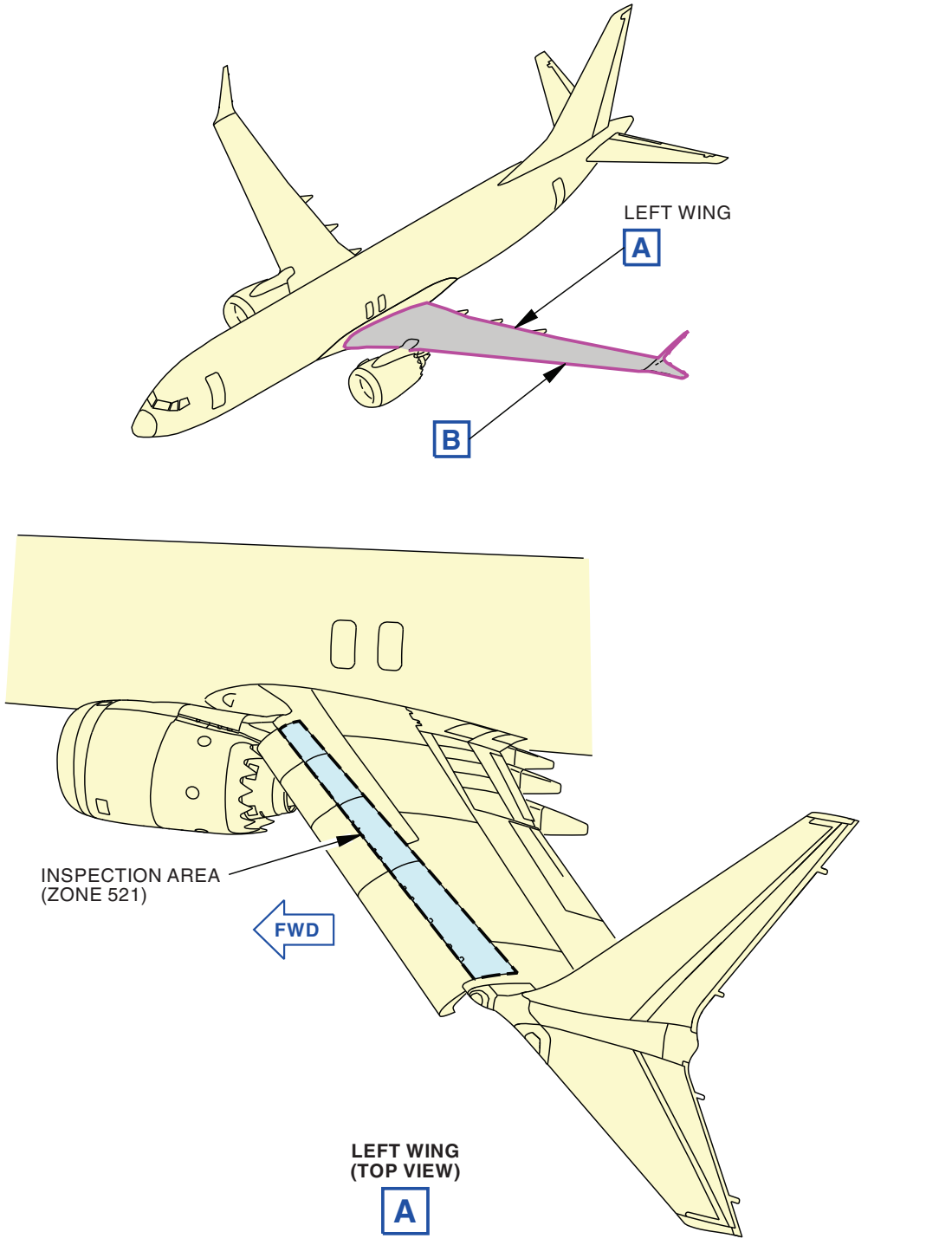
(5) Close these access panels:

(TASK 57-41-02-400-801)

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521AB	Outboard Leading Edge Blowout Door
521ABB	Lower Leading Edge Access Panel
521AT	Outbd Leading Edge - Gap Cover Access
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

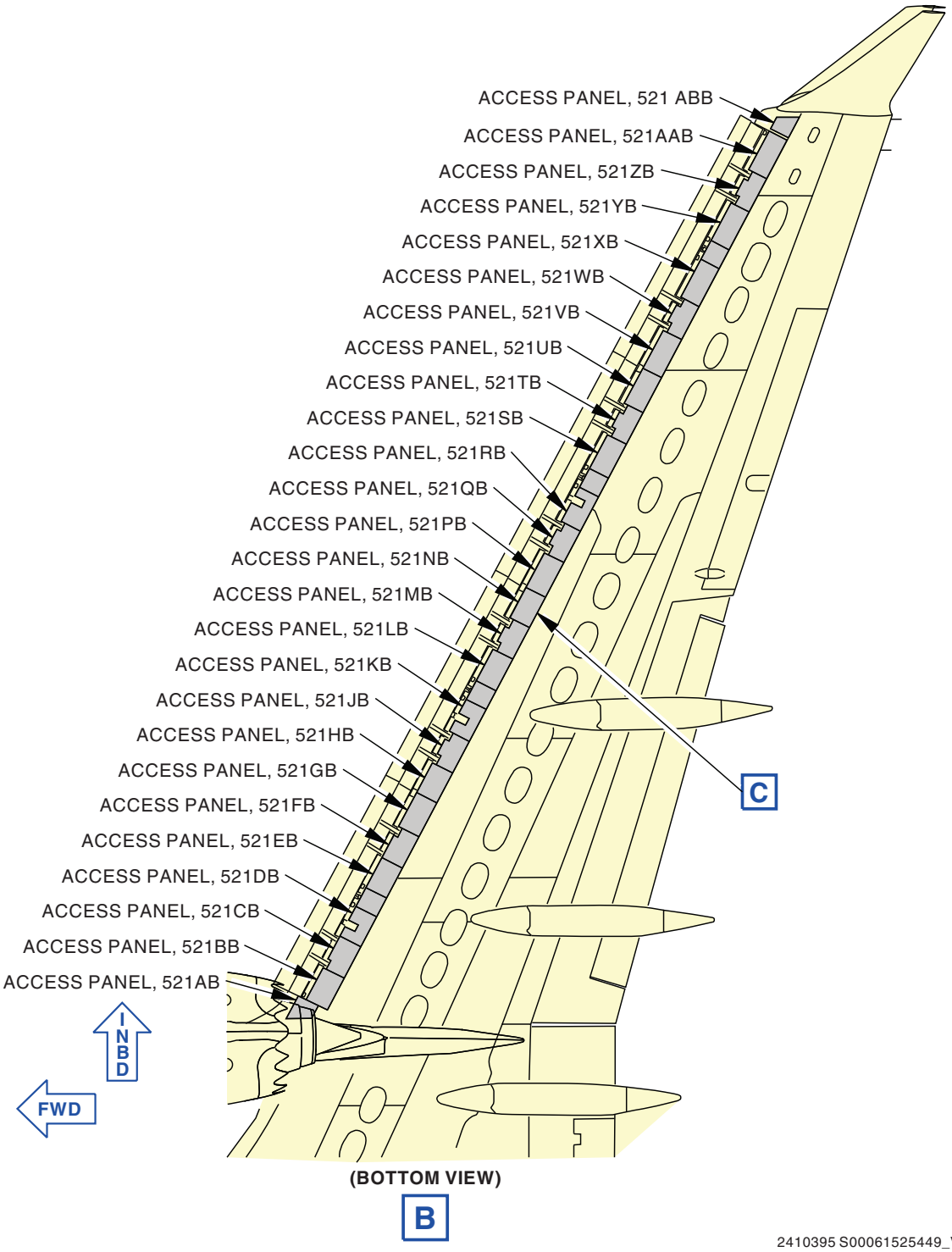
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



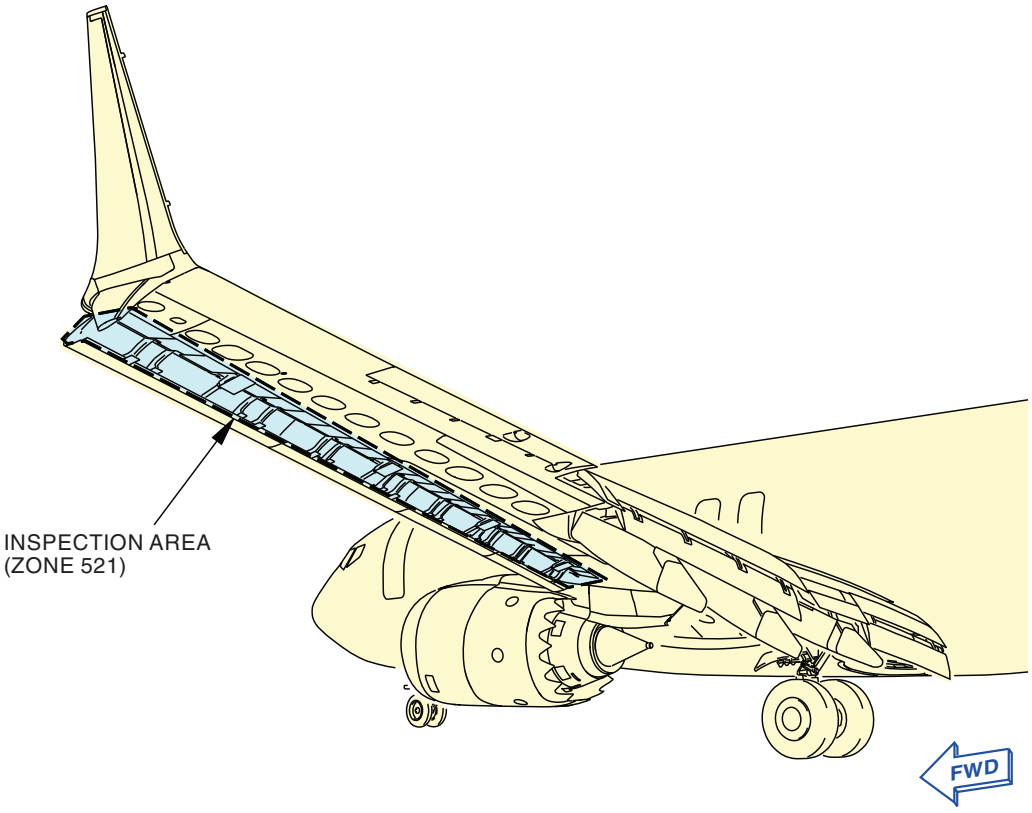
Leading Edge to Front Spar - Left Wing - Outboard
Figure 202/05-42-05-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Left Wing - Outboard
Figure 202/05-42-05-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(BOTTOM VIEW)

C

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Leading Edge to Front Spar - Left Wing - Outboard
Figure 202/05-42-05-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-05-211-803

4. Detailed Inspection: Exposed EWIS in the area from Rear Spar to Main Landing Gear Support Beam - Left Wing. (EZAP)

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam

D. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Inspection

SUBTASK 05-42-05-010-003

(1) Open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 05-42-05-211-003

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in the area from Rear Spar to Main Landing Gear Support Beam - Left Wing. (EZAP)

SUBTASK 05-42-05-910-003

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

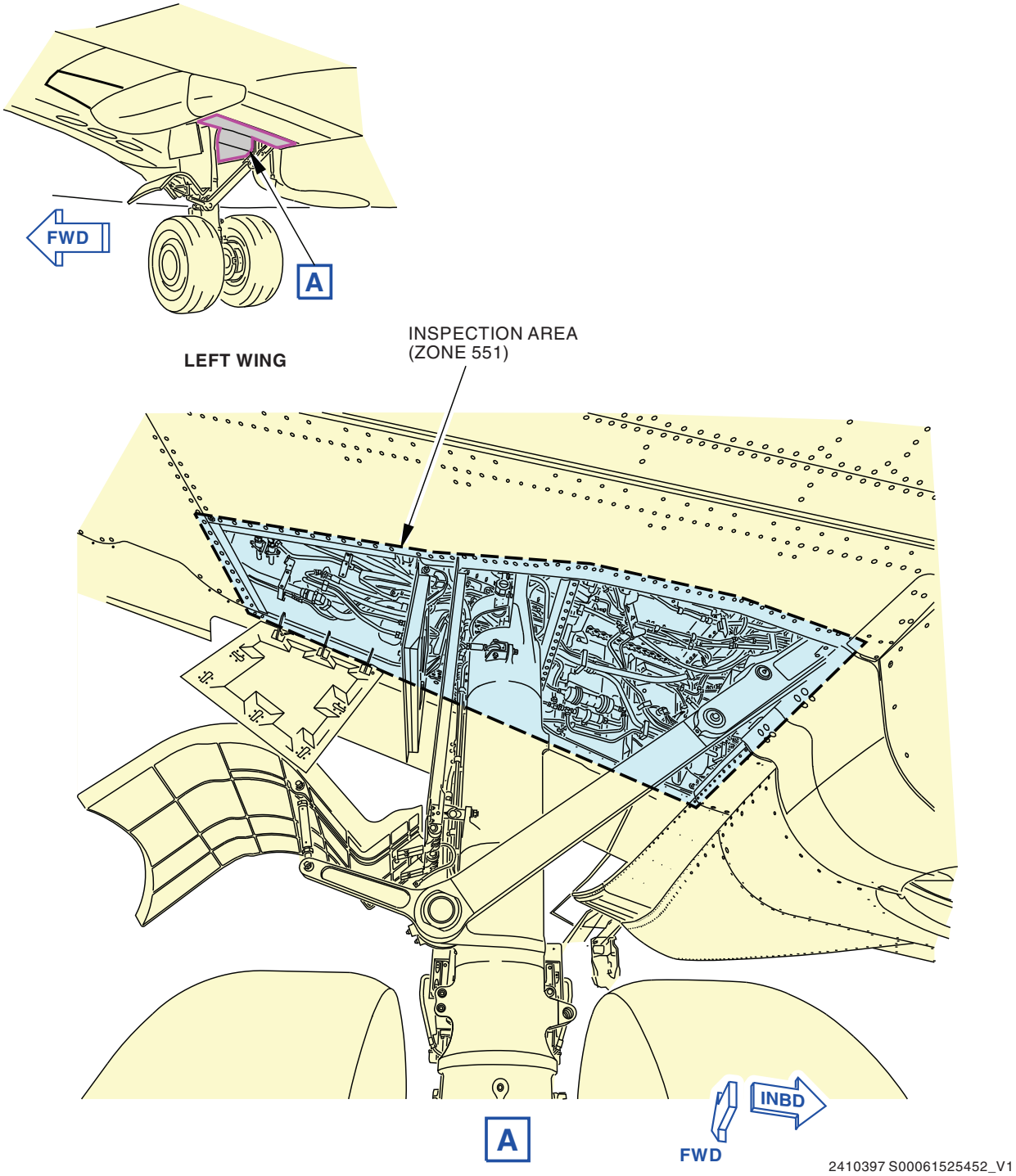
SUBTASK 05-42-05-410-003

(5) Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rear Spar to Main Landing Gear Support Beam - Left Wing
Figure 203/05-42-05-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-05-211-804

5. Detailed Inspection: Exposed EWIS in the area from Rear Spar to Trailing Edge - Outbd of Inbd Flap - Inbd of Fixed T.E. - Left Wing. (EZAP)

(Figure 204)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
561	Left Wing - Rear Spar to Trailing Edge, Outboard Of Inboard Flap, Inboard of Fixed Trailing Edge

D. Inspection

SUBTASK 05-42-05-211-004

(1) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

NOTE: Flaps extended, spoilers raised.

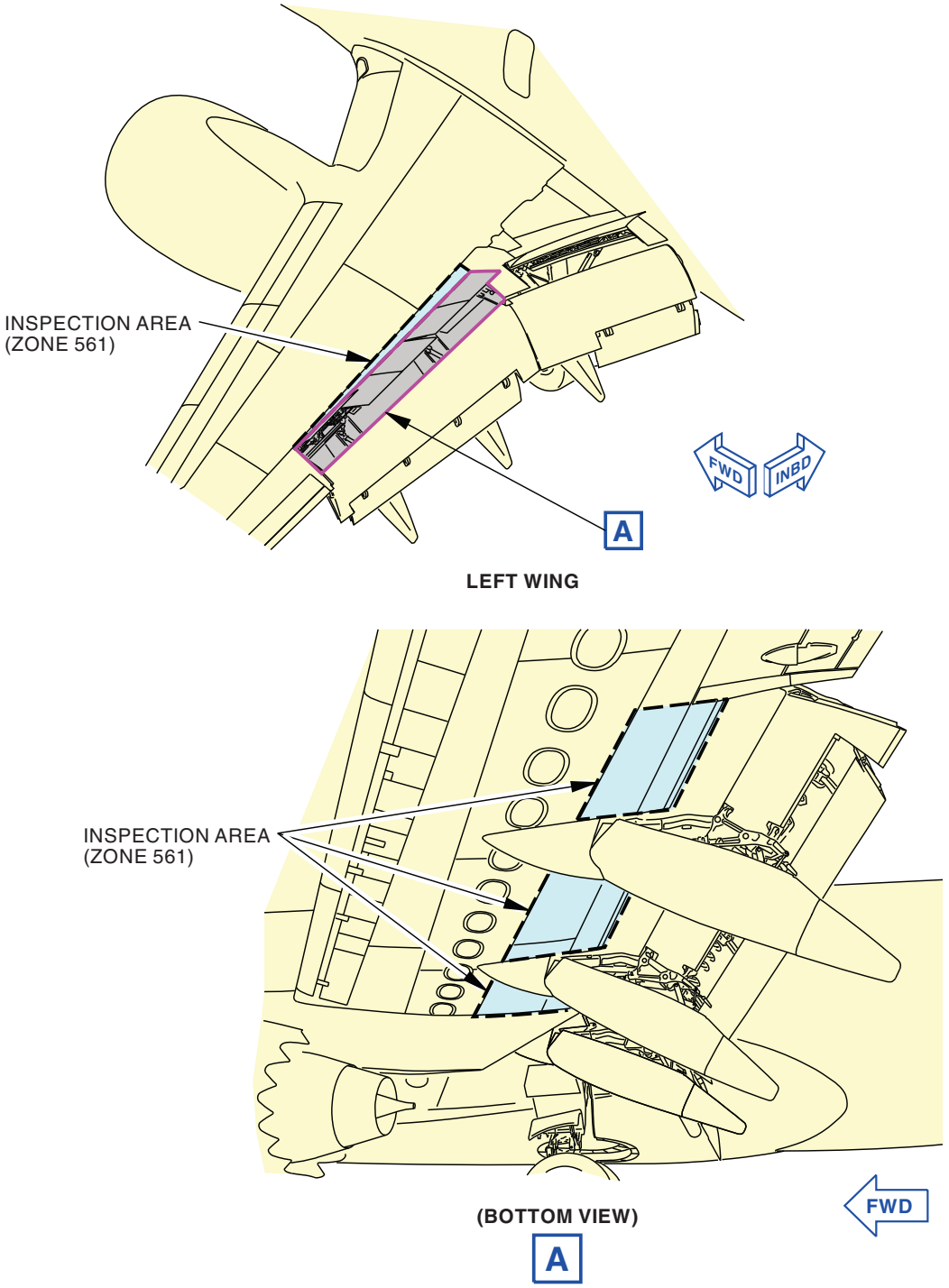
(2) Inspect (Detailed) all exposed EWIS in the area from Rear Spar to Trailing Edge - Outbd of Inbd Flap - Inbd of Fixed T.E. - Left Wing. (EZAP)

SUBTASK 05-42-05-910-004

(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rear Spar to Trailing Edge - Left Wing
Figure 204/05-42-05-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-05-211-805

6. Detailed Inspection: Exposed EWIS in the Center Fuel Tank - Left Wing (EZAP)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)

C. Location Zones

Zone	Area
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

E. Inspection

SUBTASK 05-42-05-010-004

(1) Open these access panels:

(TASK 28-11-31-000-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

SUBTASK 05-42-05-211-005

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in the Center Fuel Tank - Left Wing (EZAP).

SUBTASK 05-42-05-910-005

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801 for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-05-410-004

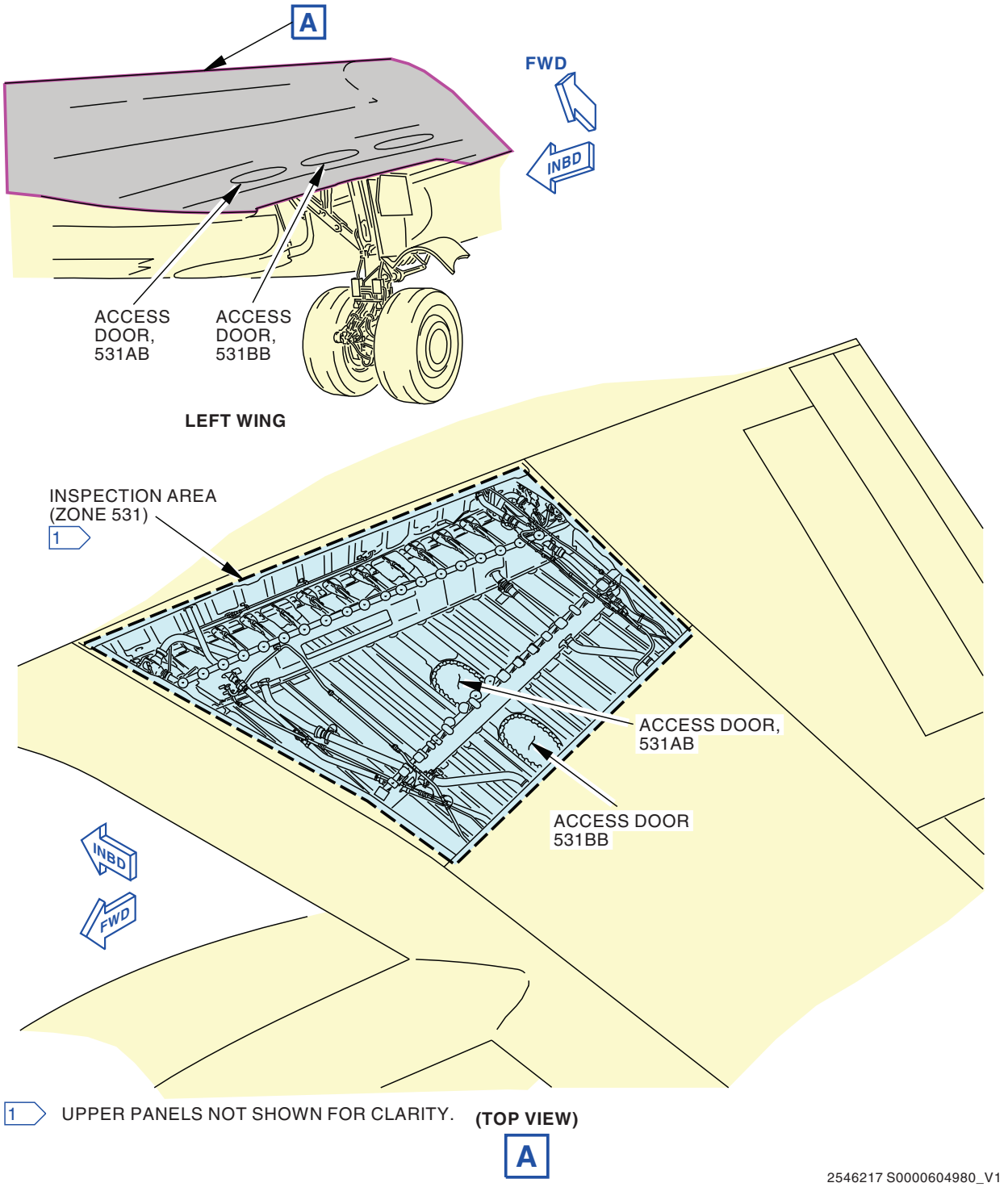
(5) Close these access panels:

(TASK 28-11-31-400-801)

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168
531BB	Center Tank Access Door - Wing Station 192

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Center Fuel Tank - Left Wing
Figure 205/05-42-05-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-05-211-806

7. Detailed Inspection: Exposed EWIS in the Main Fuel Tank - Left Wing

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)

C. Location Zones

Zone	Area
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50

D. Access Panels

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
532BB	Main Tank Access Door - Wing Station 265
532BZ	Main Tank Inner Access at Rib 6
532CB	Main Tank Access Door - Wing Station 290
532DB	Main Tank Access Door - Wing Station 313
532EB	Main Tank Access Door - Wing Station 337
532FB	Main Tank Access Door - Wing Station 367
532GB	Main Tank Access Door - Wing Station 390
532HB	Main Tank Access Door - Wing Station 417
532JB	Main Tank Access Door - Wing Station 443
532KB	Main Tank Access Door - Wing Station 470
532LB	Main Tank Access Door - Wing Station 496
532MB	Main Tank Access Door - Wing Station 523
532NB	Main Tank Access Door - Wing Station 549
532PB	Main Tank Access Door - Wing Station 576
532QB	Main Tank Access Door - Wing Station 602
532RB	Main Tank Access Door - Wing Station 629

E. Inspection

SUBTASK 05-42-05-010-005

(1) Open these access panels:

(TASK 28-11-11-000-801)

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
532BB	Main Tank Access Door - Wing Station 265
532BZ	Main Tank Inner Access at Rib 6
532CB	Main Tank Access Door - Wing Station 290
532DB	Main Tank Access Door - Wing Station 313
532EB	Main Tank Access Door - Wing Station 337
532FB	Main Tank Access Door - Wing Station 367
532GB	Main Tank Access Door - Wing Station 390
532HB	Main Tank Access Door - Wing Station 417
532JB	Main Tank Access Door - Wing Station 443
532KB	Main Tank Access Door - Wing Station 470
532LB	Main Tank Access Door - Wing Station 496
532MB	Main Tank Access Door - Wing Station 523
532NB	Main Tank Access Door - Wing Station 549
532PB	Main Tank Access Door - Wing Station 576
532QB	Main Tank Access Door - Wing Station 602
532RB	Main Tank Access Door - Wing Station 629

SUBTASK 05-42-05-211-006

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

(3) Inspect (Detailed) all exposed EWIS in the Main Fuel Tank - Left Wing (EZAP).

SUBTASK 05-42-05-910-006

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801 for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-05-410-005

(5) Close these access panels:

(TASK 28-11-11-400-801)

Number	Name/Location
434CL	Aft Strut Fairing, Left Access To Fuel Door, Strut 1
532AB	Main Tank Access Door - Wing Station 216
532AZ	Main Tank Inner Access at Rib 6
532BB	Main Tank Access Door - Wing Station 265
532BZ	Main Tank Inner Access at Rib 6
532CB	Main Tank Access Door - Wing Station 290
532DB	Main Tank Access Door - Wing Station 313
532EB	Main Tank Access Door - Wing Station 337
532FB	Main Tank Access Door - Wing Station 367
532GB	Main Tank Access Door - Wing Station 390
532HB	Main Tank Access Door - Wing Station 417
532JB	Main Tank Access Door - Wing Station 443
532KB	Main Tank Access Door - Wing Station 470
532LB	Main Tank Access Door - Wing Station 496
532MB	Main Tank Access Door - Wing Station 523

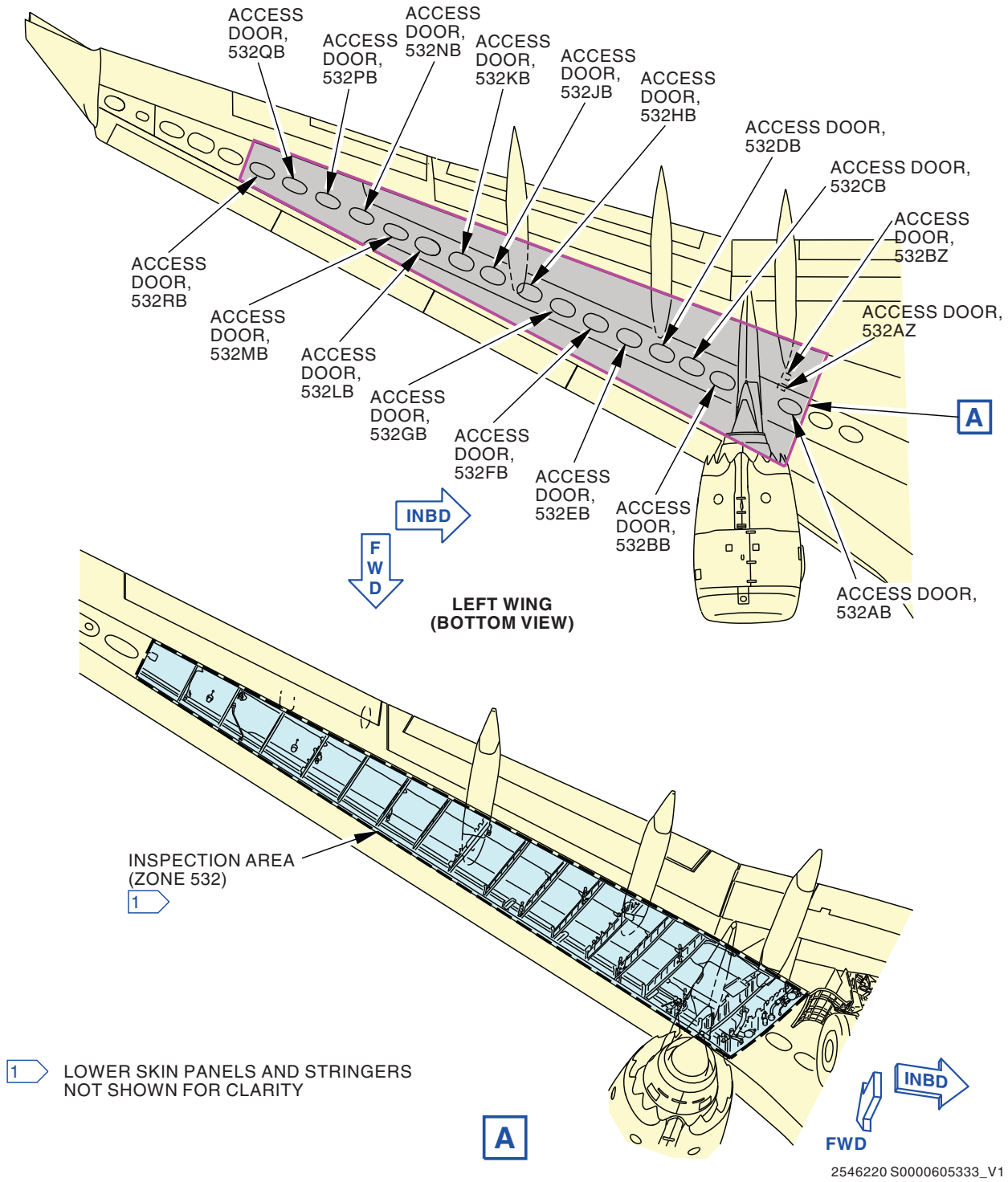
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
532NB	Main Tank Access Door - Wing Station 549
532PB	Main Tank Access Door - Wing Station 576
532QB	Main Tank Access Door - Wing Station 602
532RB	Main Tank Access Door - Wing Station 629

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EWIS in the Main Tank - Left Wing (EZAP)
Figure 206/05-42-05-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1. General

A. This pageblock contains MSG-3 task card data.

EWIS

TASK 05-42-06-211-801

2. Detailed Inspection: Exposed EWIS in the area from Leading Edge to Front Spar - Inboard of Nacelle Strut - Right Wing. (EZAP)

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
611	Right Wing - Leading Edge to Front Spar

D. Access Panels

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
611BT	Fairing

E. Inspection

SUBTASK 05-42-06-010-001

(1) Open these access panels:

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
611BT	Fairing

NOTE: Leading edge flaps extended.

SUBTASK 05-42-06-211-001

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

SUBTASK 05-42-06-200-001

(3) Inspect (Detailed) all exposed EWIS in the area from Leading Edge to Front Spar - Inboard of Nacelle Strut - Right Wing. (EZAP)

SUBTASK 05-42-06-910-001

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

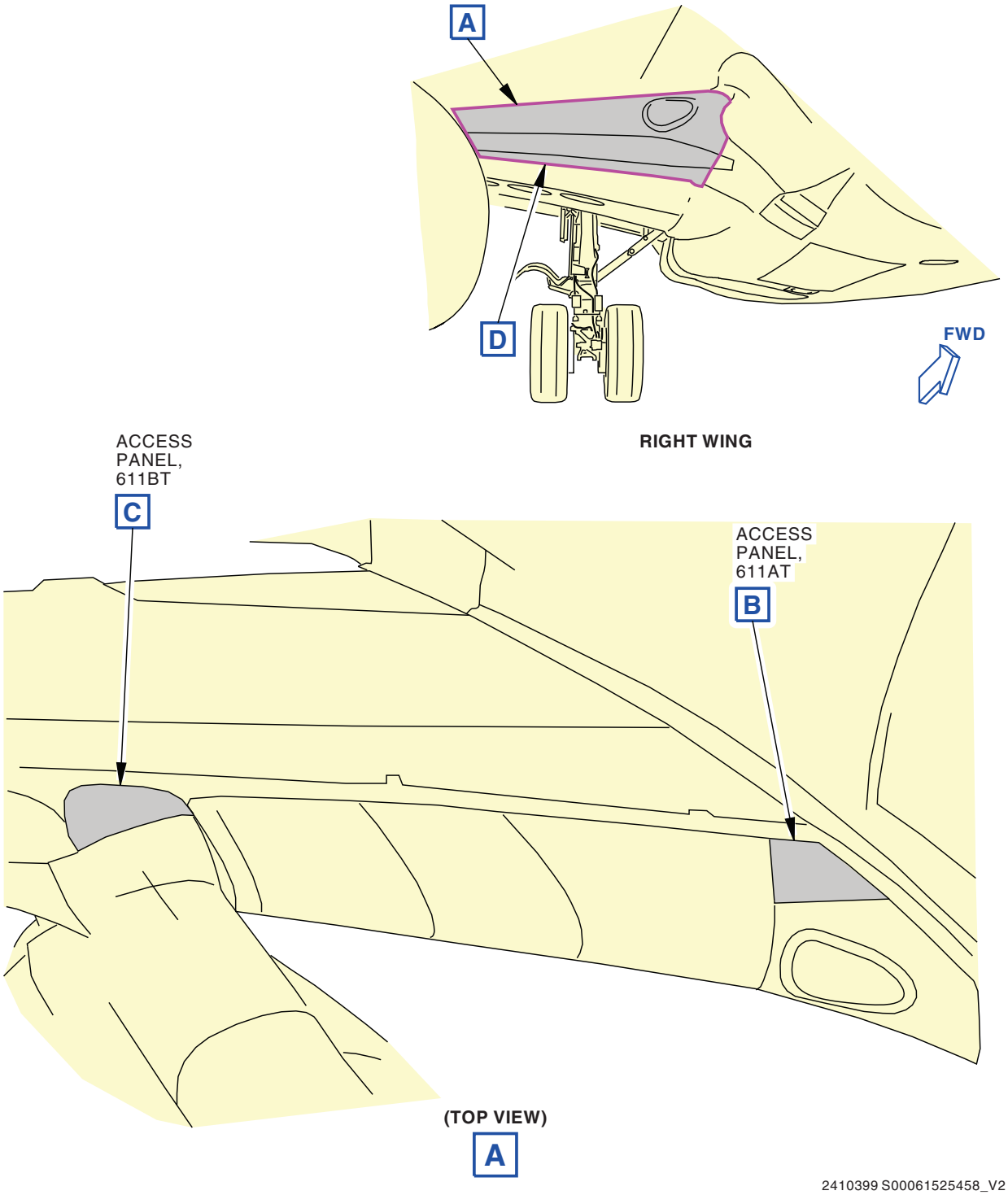
SUBTASK 05-42-06-410-001

(5) Close these access panels:

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
611BT	Fairing

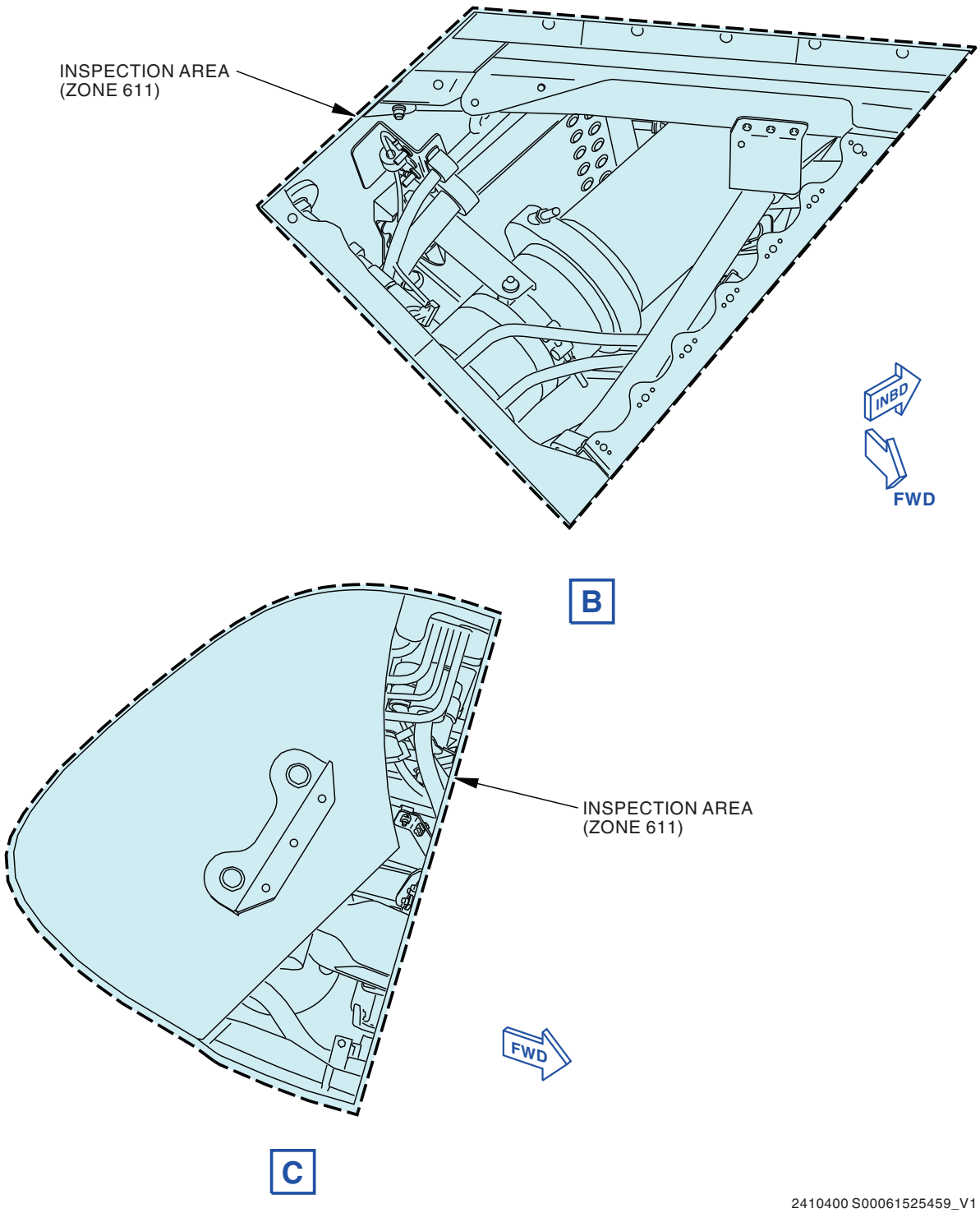
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



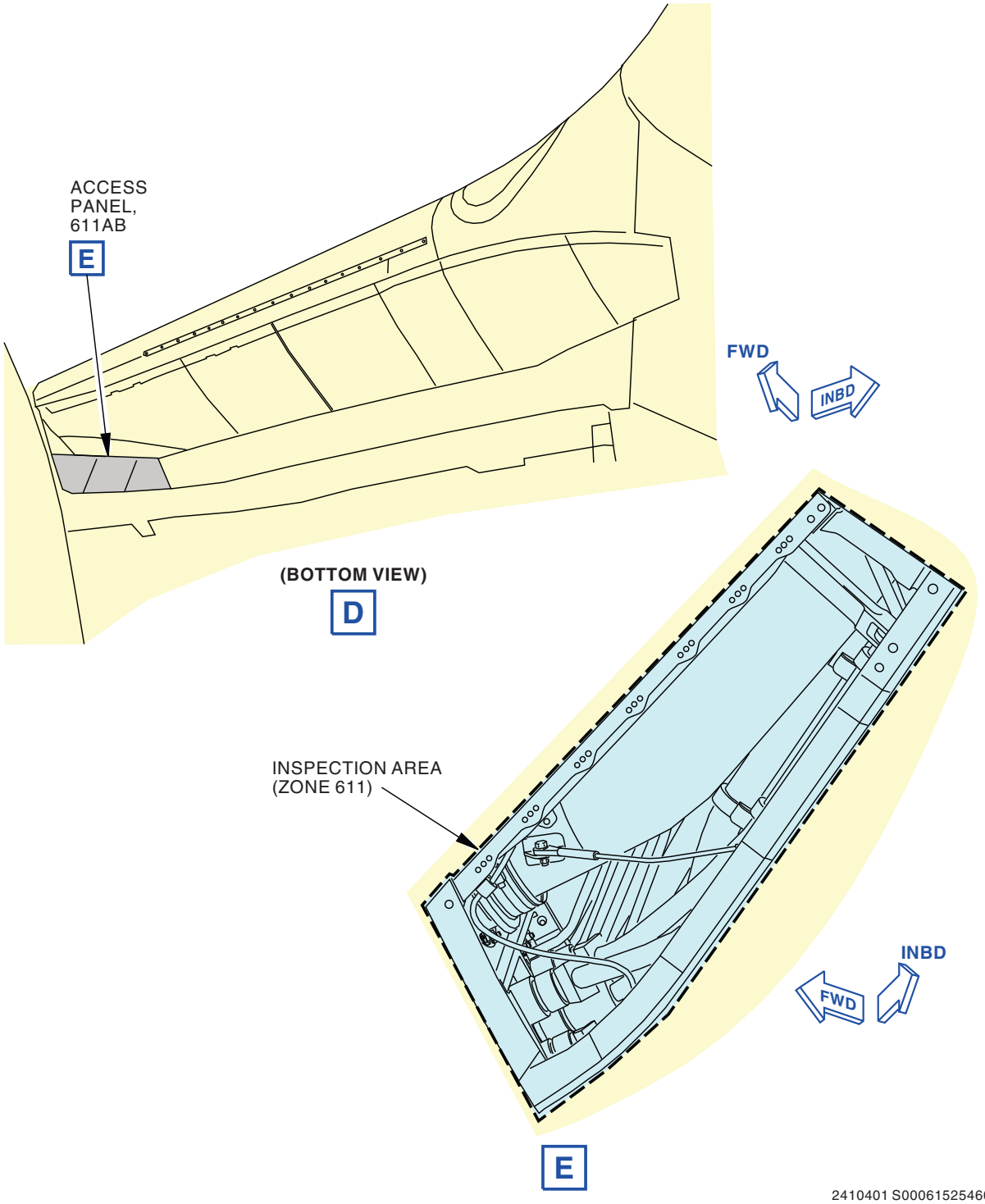
Leading Edge to Front Spar - Right Wing - Inboard
Figure 201/05-42-06-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Right Wing - Inboard
Figure 201/05-42-06-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Right Wing - Inboard
Figure 201/05-42-06-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-06-211-802

3. **Detailed Inspection: Exposed EWIS in the area from Leading Edge to Front Spar - Outboard of Nacelle Strut - Right Wing. (EZAP)**

(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

C. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

D. Access Panels

Number	Name/Location
621AAB	Lower Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621ABB	Lower Leading Edge Access Panel
621AT	Outbd Leading Edge - Gap Cover Access
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

E. Inspection

SUBTASK 05-42-06-010-002

(1) Open these access panels:

(TASK 57-41-02-000-801)

Number	Name/Location
621AAB	Lower Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621ABB	Lower Leading Edge Access Panel
621AT	Outbd Leading Edge - Gap Cover Access
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

SUBTASK 05-42-06-211-002

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

NOTE: Access is gained by extending slats, and through lower wing surface access panels.

SUBTASK 05-42-06-200-002

(3) Inspect (Detailed) all exposed Electrical Wiring Interconnect Systems (EWIS) in the area from Leading Edge to Front Spar - Outboard of Nacelle Strut - Right Wing. (EZAP)

SUBTASK 05-42-06-910-002

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-42-06-410-002

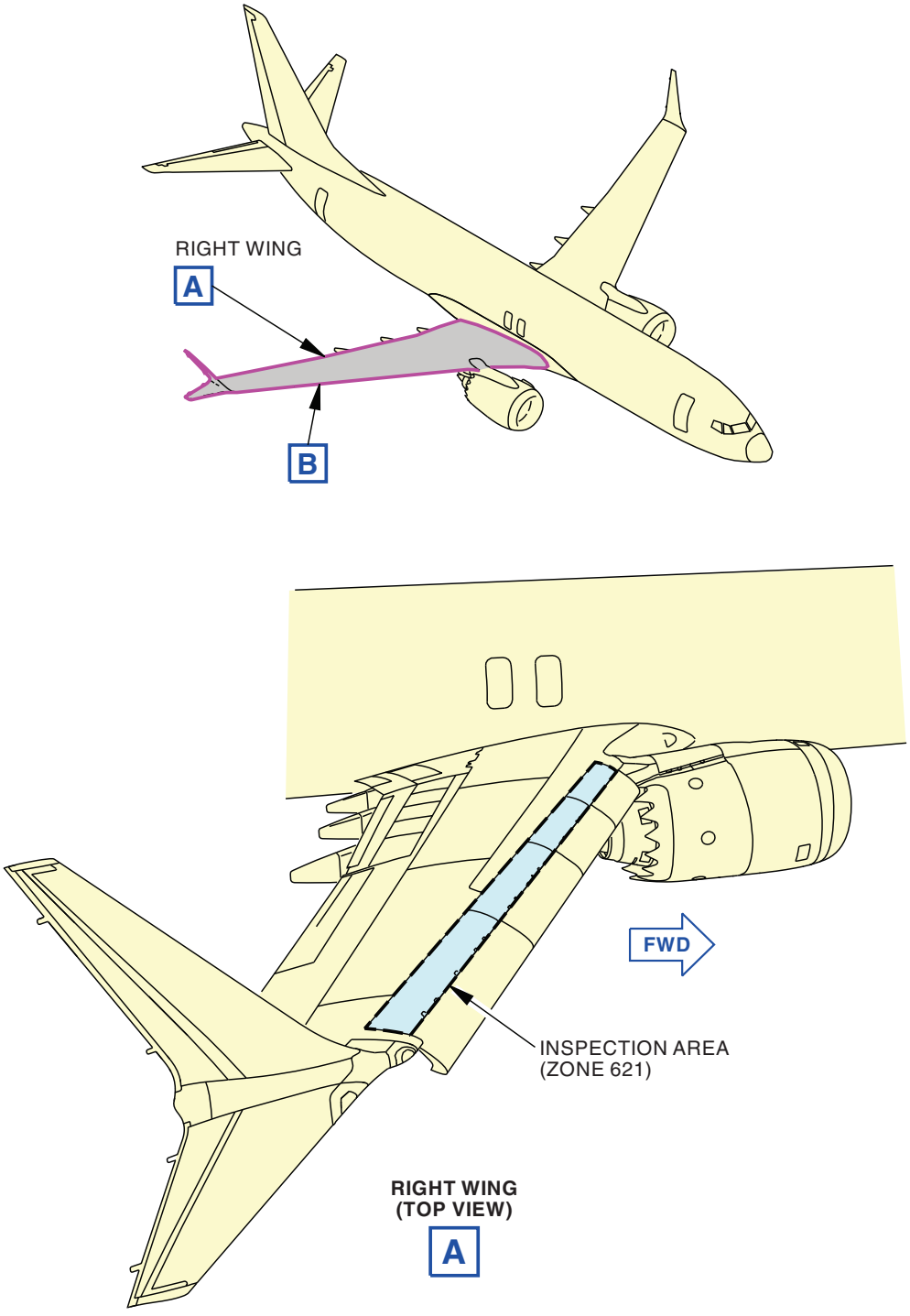
(5) Close these access panels:

(TASK 57-41-02-400-801)

Number	Name/Location
621AAB	Lower Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621ABB	Lower Leading Edge Access Panel
621AT	Outbd Leading Edge - Gap Cover Access
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



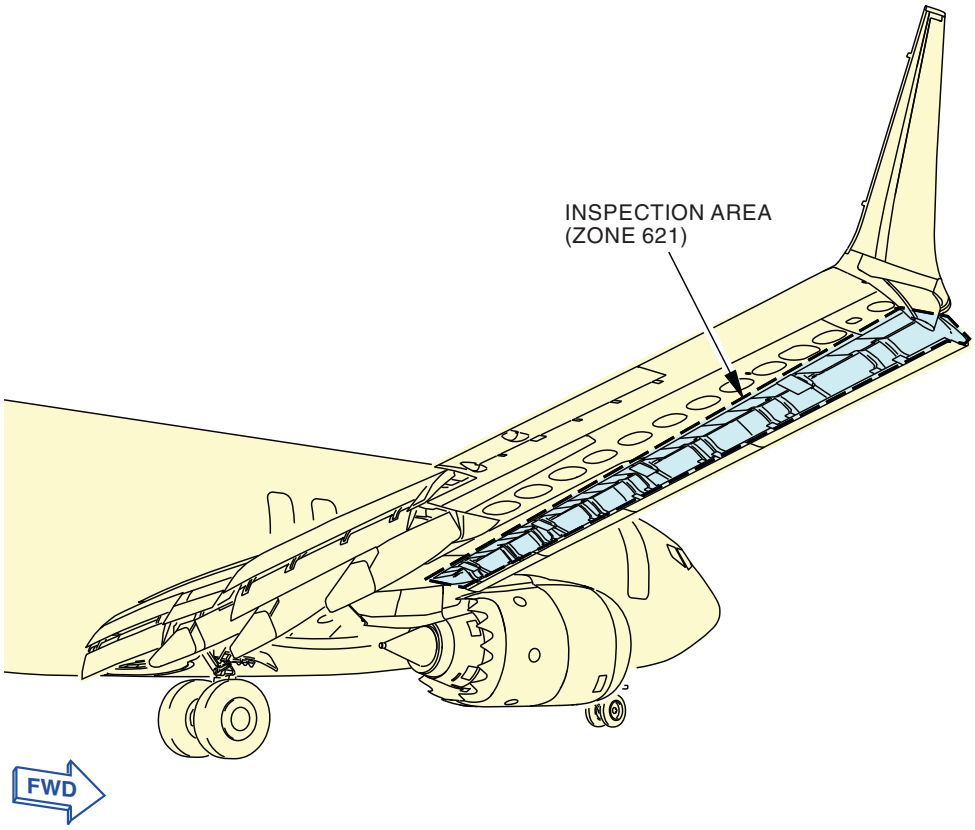
Leading Edge to Front Spar - Right Wing - Outboard
Figure 202/05-42-06-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge to Front Spar - Right Wing - Outboard
Figure 202/05-42-06-990-802 (Sheet 2 of 3)

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(BOTTOM VIEW)

C

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Leading Edge to Front Spar - Right Wing - Outboard
Figure 202/05-42-06-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-06-211-803

4. Detailed Inspection: Exposed EWIS in the area from Rear Spar to Main Landing Gear Support Beam - Right Wing. (EZAP)

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
651	Right Wing - Rear Spar to Landing Gear Support Beam

D. Access Panels

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Inspection

SUBTASK 05-42-06-010-003

(1) Open this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 05-42-06-211-003

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

SUBTASK 05-42-06-200-003

(3) Inspect (Detailed) all exposed EWIS in the area from Rear Spar to Main Landing Gear Support Beam - Right Wing. (EZAP)

SUBTASK 05-42-06-910-003

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

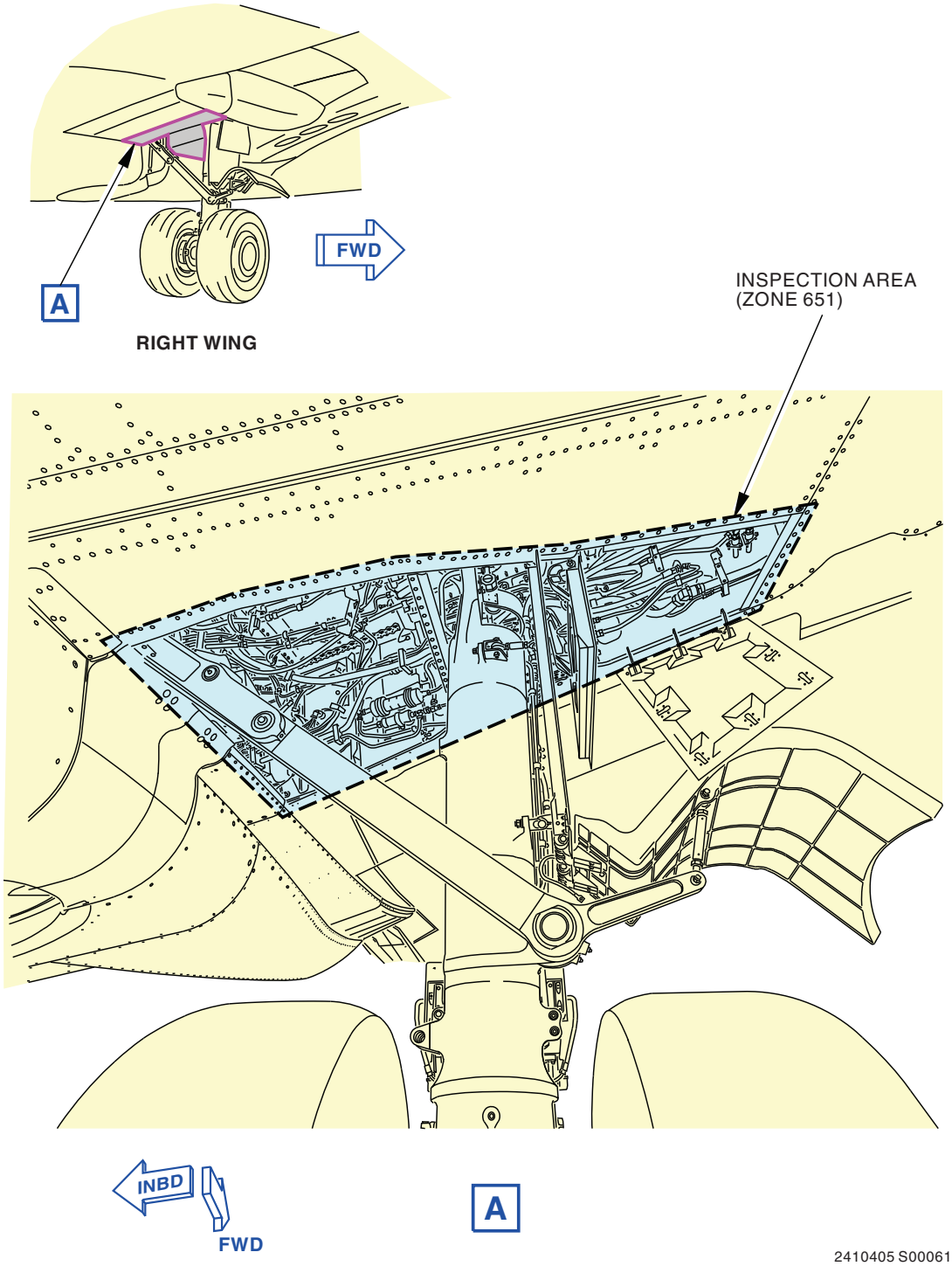
SUBTASK 05-42-06-410-003

(5) Close this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rear Spar to Main Landing Gear Support Beam - Right Wing
Figure 203/05-42-06-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-06-211-804

5. Detailed Inspection: Exposed EWIS in the area from Rear Spar to Trailing Edge - Outbd of Inbd Flap - Inbd of Fixed T.E. - Right Wing. (EZAP)

(Figure 204)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)

C. Location Zones

Zone	Area
661	Right Wing - Rear Spar to Trailing Edge, Outboard of Inboard Flap, Inboard of Fixed Trailing Edge

D. Inspection

SUBTASK 05-42-06-211-004

(1) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

NOTE: Flaps extended, spoilers raised.

SUBTASK 05-42-06-200-004

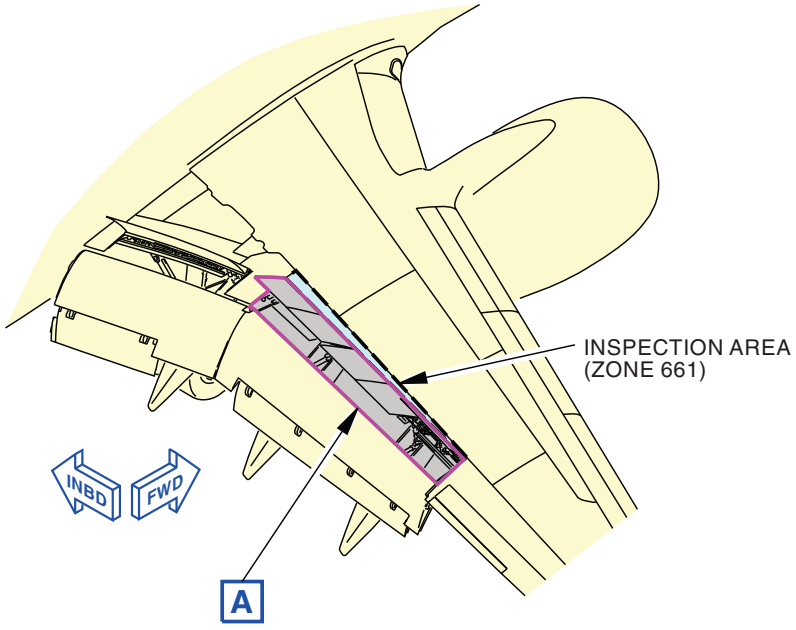
(2) Inspect (Detailed) all exposed EWIS in the area from Rear Spar to Trailing Edge - Outbd of Inbd Flap - Inbd of Fixed T.E. - Right Wing. (EZAP)

SUBTASK 05-42-06-910-004

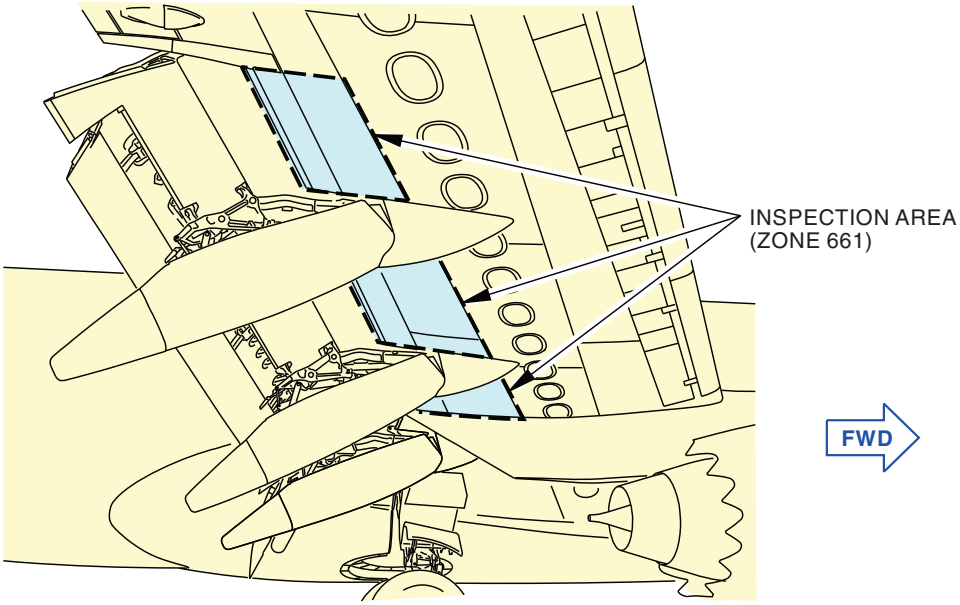
(3) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801, for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT WING



BOTTOM VIEW

A

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Rear Spar to Trailing Edge - Right Wing
Figure 204/05-42-06-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-06-211-805

6. Detailed Inspection: Exposed EWIS in the Center Tank - Right Wing

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
28-11-31-000-801	Center Tank Access Door Removal (P/B 401)
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)

C. Location Zones

Zone	Area
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5

D. Access Panels

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168
631BB	Center Tank Access Door - Wing Station 192

E. Inspection

SUBTASK 05-42-06-010-004

(1) Open these access panels:

(TASK 28-11-31-000-801)

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168
631BB	Center Tank Access Door - Wing Station 192

SUBTASK 05-42-06-211-005

(2) Do the task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

SUBTASK 05-42-06-200-005

(3) Inspect (Detailed) all exposed EWIS in the Center Fuel Tank - Right Wing (EZAP).

SUBTASK 05-42-06-910-005

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801 for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-06-410-004

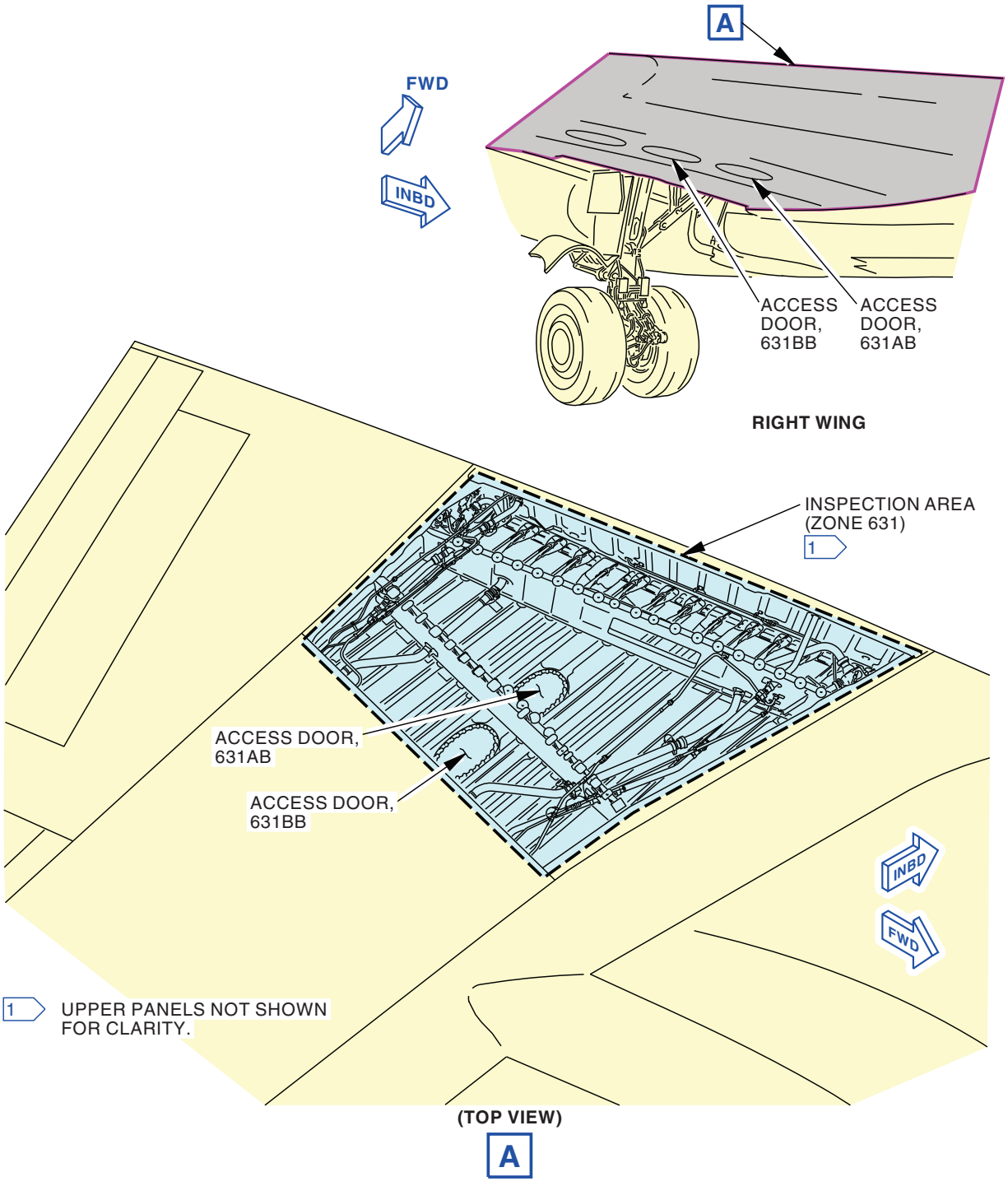
(5) Close these access panels:

(TASK 28-11-31-400-801)

Number	Name/Location
631AB	Center Tank Access Door - Wing Station 168
631BB	Center Tank Access Door - Wing Station 192

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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EWIS in the Center Fuel Tank - Right Wing (EZAP)
Figure 205/05-42-06-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EWIS

TASK 05-42-06-211-806

7. Detailed Inspection: Exposed EWIS in the Main Tank - Right Wing

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is an Enhanced Zonal Analysis Procedure (EZAP) task.

B. References

Reference	Title
20-60-02-100-802	Detailed Inspection of EWIS (P/B 601)
20-60-07-910-801	Protection of the EWIS During Maintenance (P/B 201)
28-11-11-000-801	Main Tank Access Door Removal (P/B 401)
28-11-11-400-801	Main Tank Access Door Installation (P/B 401)

C. Location Zones

Zone	Area
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Access Panels

Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6
632BB	Main Tank Access Door - Wing Station 265
632BZ	Main Tank Inner Access at Rib 6
632CB	Main Tank Access Door - Wing Station 290
632DB	Main Tank Access Door - Wing Station 313
632EB	Main Tank Access Door - Wing Station 337
632FB	Main Tank Access Door - Wing Station 367
632GB	Main Tank Access Door - Wing Station 390
632HB	Main Tank Access Door - Wing Station 417
632JB	Main Tank Access Door - Wing Station 443
632KB	Main Tank Access Door - Wing Station 470
632LB	Main Tank Access Door - Wing Station 496
632MB	Main Tank Access Door - Wing Station 523
632NB	Main Tank Access Door - Wing Station 549
632PB	Main Tank Access Door - Wing Station 576
632QB	Main Tank Access Door - Wing Station 602
632RB	Main Tank Access Door - Wing Station 629

E. Inspection

SUBTASK 05-42-06-010-005

(1) Open these access panels:

(TASK 28-11-11-000-801)

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Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6
632BB	Main Tank Access Door - Wing Station 265
632BZ	Main Tank Inner Access at Rib 6
632CB	Main Tank Access Door - Wing Station 290
632DB	Main Tank Access Door - Wing Station 313
632EB	Main Tank Access Door - Wing Station 337
632FB	Main Tank Access Door - Wing Station 367
632GB	Main Tank Access Door - Wing Station 390
632HB	Main Tank Access Door - Wing Station 417
632JB	Main Tank Access Door - Wing Station 443
632KB	Main Tank Access Door - Wing Station 470
632LB	Main Tank Access Door - Wing Station 496
632MB	Main Tank Access Door - Wing Station 523
632NB	Main Tank Access Door - Wing Station 549
632PB	Main Tank Access Door - Wing Station 576
632QB	Main Tank Access Door - Wing Station 602
632RB	Main Tank Access Door - Wing Station 629

SUBTASK 05-42-06-211-006

(2) Do this task: Detailed Inspection of EWIS, TASK 20-60-02-100-802.

SUBTASK 05-42-06-200-006

(3) Inspect (Detailed) all exposed EWIS in the Main Fuel Tank - Right Wing (EZAP).

SUBTASK 05-42-06-910-006

(4) Refer to Protection of the EWIS During Maintenance, TASK 20-60-07-910-801 for the protection and caution information that will minimize contamination and accidental damage to EWIS during maintenance.

SUBTASK 05-42-06-410-005

(5) Close these access panels:

(TASK 28-11-11-400-801)

Number	Name/Location
444CR	Aft Strut Fairing, Right Access To Fuel Door, Strut 2
632AB	Main Tank Access Door - Wing Station 216
632AZ	Main Tank Inner Access at Rib 6
632BB	Main Tank Access Door - Wing Station 265
632BZ	Main Tank Inner Access at Rib 6
632CB	Main Tank Access Door - Wing Station 290
632DB	Main Tank Access Door - Wing Station 313
632EB	Main Tank Access Door - Wing Station 337
632FB	Main Tank Access Door - Wing Station 367
632GB	Main Tank Access Door - Wing Station 390
632HB	Main Tank Access Door - Wing Station 417
632JB	Main Tank Access Door - Wing Station 443
632KB	Main Tank Access Door - Wing Station 470
632LB	Main Tank Access Door - Wing Station 496

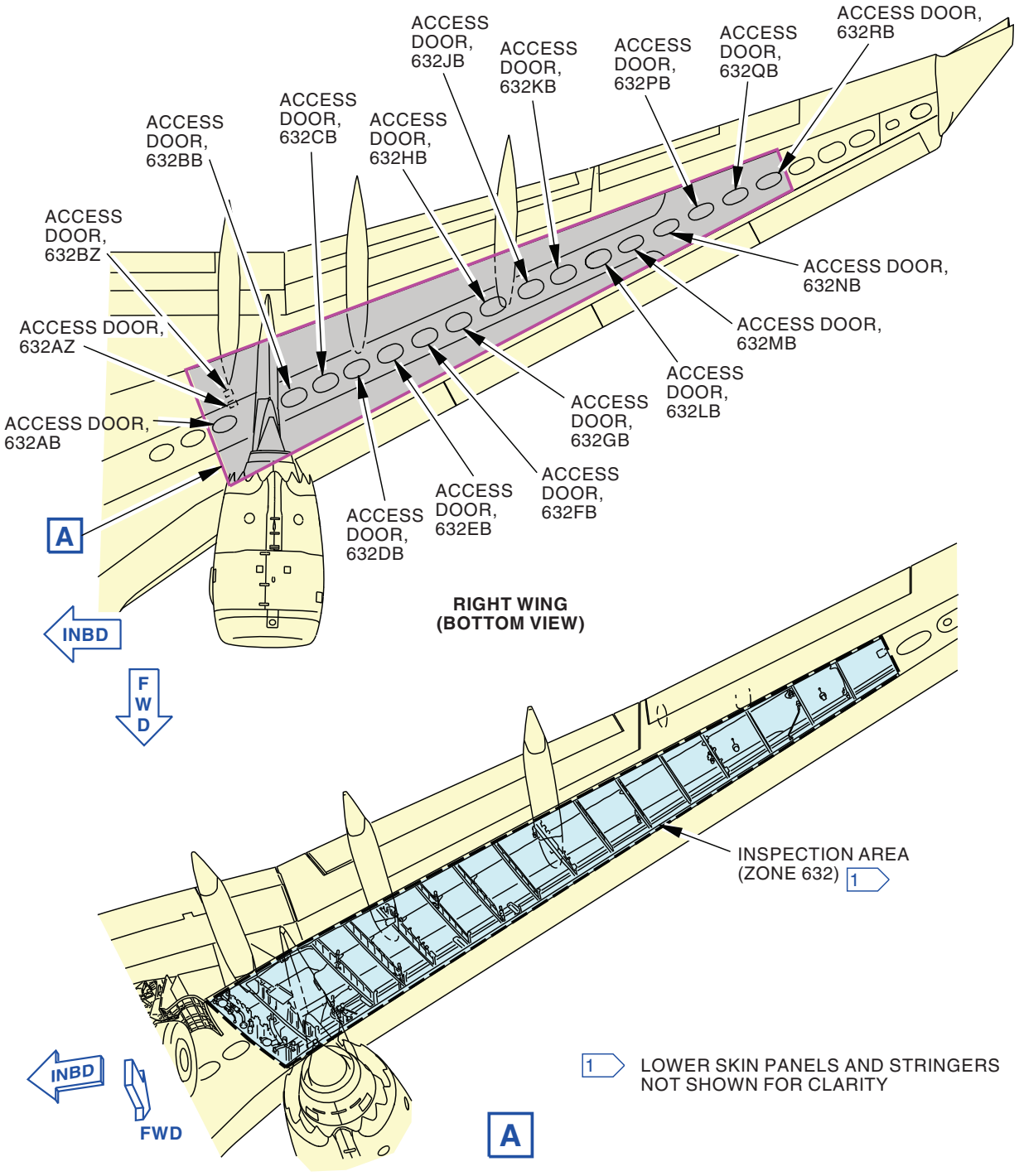
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
632MB	Main Tank Access Door - Wing Station 523
632NB	Main Tank Access Door - Wing Station 549
632PB	Main Tank Access Door - Wing Station 576
632QB	Main Tank Access Door - Wing Station 602
632RB	Main Tank Access Door - Wing Station 629

——— END OF TASK ———

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Exposed EWIS in the Main Tank - Right Wing (EZAP)
Figure 206/05-42-06-990-806

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HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection is for hard landings, or overweight/hard landings, or high drag or high side load conditions. It defines the inspections for areas other than the nose landing gear and forward fuselage for those conditions. The Phase I inspections of the nose landing gear and forward fuselage structure for these conditions are covered in procedure SUBJECT 05-51-37.

B. The Inspection

(1) The inspection is divided into two phases (Phase I and Phase II).

(a) The Phase I inspections in this procedure are for areas other than the nose gear. The nose gear inspections are contained in a separate procedure NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-37/201.

(b) The Phase I inspection is divided into two phases (Phase IA and Phase IB).

1) Phase IA is a visual inspection.

2) Phase IB contains inspections that require special tools or access to the structure. It is completed after Phase IA under the conditions described in this procedure.

(c) The Phase II inspection is a detailed structural inspection required if damage is found during the Phase I inspection.

NOTE: All inspections, especially visual inspections, are made with sufficient lighting conditions such as daylight, hangar lighting, flashlight or drop-light.

C. High Drag or High Side Load Condition

(1) A High Drag or High Side Load condition occurs if one or more of the following conditions are met:

(a) The flight crew reported a “high drag load” or “high side load” event.

(b) The airplane ran off the prepared surface.

(c) The airplane landed short of the prepared surface.

(d) The airplane made a landing and one or more tires were blown.

(e) One or more of the landing gear hit an obstacle.

D. Hard Main Gear Landing

(1) A Hard Main Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard landing” without specifying if it was hard on the main or nose gear, or

2) “hard main gear landing”, or

3) “hard bounced landing”.

E. Hard Nose Gear Landing

(1) A Hard Nose Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard landing” without specifying if it was hard on the main or nose gear, or

2) “hard nose gear landing”, or

3) “hard bounced landing”, or

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4) a landing where the nose gear contacted the prepared surface before the main gears.

F. Overweight Landing

(1) For the purpose of this inspection, an Overweight Landing occurs when:

(a) The landing weight is greater than the Maximum Landing Weight plus 1000 lb (454 kg).

G. The Inspection Logic

NOTE: The inspection logic is also shown on the flowchart in Figure 201.

(1) If it was a “Hard Nose Gear Landing”, or a “High Drag or High Side Load Condition”, do the NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-37/201.

(2) Was it a High Drag or High Side Load condition or a Bounced Landing?

(a) Yes: Do both the Phase IA and Phase IB inspections before the next flight.

(b) No: Is data available from the Flight Data Recorder (FDR) or other equivalent recording device?

NOTE: Answer "No” to this question if the data recording sample rate for the CG vertical load factor (VERTICAL ACCELERATION) is less than 16 samples/sec.

1) No: Do both the Phase IA and Phase IB inspections before the next flight.

2) Yes: Was the CG vertical load factor higher than the threshold for the corresponding roll angle in Figure 202 or was the roll angle greater than 6 degrees?

a) Yes: Do both the Phase IA and Phase IB inspections before the next flight.

b) No: Do the Phase IA inspection before the next flight. If damage is found, also do the Phase IB inspections.

NOTE: The CG vertical load factor (VERTICAL ACCELERATION) and roll angle (ROLL ATTITUDE) are determined from the FDR or equivalent recording device. The load factor is the maximum value of VERTICAL ACCELERATION that occurs in the first one-second period after touchdown and prior to speedbrake deployment. The roll angle is the maximum absolute value of ROLL ATTITUDE in the one-second period prior to the maximum load factor.

NOTE: The threshold is shown graphically in Figure 202. The threshold curve to use depends on the landing weight.

NOTE: The CG vertical load factor threshold may only be used if the recording sample rate for VERTICAL ACCELERATION is at least 16 samples/sec, if the landing was not bounced and if the roll angle is less than or equal to 6 degrees.

(3) If damage is found during any inspection phase, then the remaining inspection phases must be performed and all repairs be completed before the airplane is returned to service.

(4) When damage is found, Boeing recommends downloading the flight recorded data and retaining it for future consultation with Boeing regarding the incident.

H. When the conditional inspection tells you to "examine" or inspect a component, look for these conditions (replace or repair components, if necessary).

(1) Cracks

(2) Pulled apart structure or unusual gaps

(3) Loose paint (paint flakes)

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(4) Discoloration or scuffing
(5) Twisted parts (distortion)
(6) Wrinkles or buckles in the structure
(7) Bent components
(8) Fastener holes that become larger or deformed
(9) Loose fasteners
(10) Fasteners that have pulled out or are gone
(11) Delamination
(12) Fiber breakage
(13) Misalignment
(14) Interference
(15) Nicks or gouges
(16) Cracked fillet seals on sealed assemblies
(17) Other signs of damage

I. Fuse Pin Inspection Guidelines

NOTE: The fuse pin locations are shown in Figure 204.

(1) When the conditional inspection tells you to “examine” a fuse pin use this inspection method:

NOTE: The bolts must be inspected one at a time.

(a) Fuse pin removal and inspection.
1) Remove the load from the joint that has the fuse pin to be inspected.
2) Remove the bolt that goes through the fuse pin and the end caps of the fuse pin that will be inspected.
3) Remove the fuse pin.

(b) If distortion or other damage was not found during the fuse pin inspection, put the fuse pin installation back to the initial condition.

TASK 05-51-01-210-804

2. Phase IA Inspection

(Figure 201, Figure 202, Figure 203, Figure 204)

A. References

Reference	Title
05-51-37 P/B 201	NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
05-51-37-200-804	Phase I Inspection (P/B 201)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

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(Continued)

Zone	Area
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Nose Landing Gear and Forward Fuselage Inspection

SUBTASK 05-51-01-210-064

(1) Perform the Nose Landing Gear (NLG) and forward fuselage inspection procedure as stated in PAGEBLOCK 05-51-37/201, if it was a “Hard Nose Gear Landing”, or a “High Drag or High Side Load Condition”.

D. Main Landing Gear Inspection

SUBTASK 05-51-01-210-065

(1) Examine the Main Landing Gear (MLG) areas as follows (Figure 203, Figure 204):

(a) The tires for damage (TASK 32-45-00-700-803).

(b) The wheels for damage; cracks, missing tie-bolts, heat shield (TASK 32-45-00-700-801).

(c) The shock struts at the top and bottom end for signs of fluid leakage.

NOTE: A small quantity of hydraulic fluid on the inner cylinder is satisfactory.

(d) The top end of the shock strut for cracks and bolt distortion.

(e) The outer door and linkage.

(f) The center door and linkage.

(g) The inner door and linkage.

(h) The trunnions and shock strut top end, including bushing edges/flanges and visible pin's external surface plating.

(i) The side strut and trunnion attach fittings of the main landing gear.

(j) The landing gear walking beam to wing structure attachment fittings.

(k) The landing gear trunnion to landing gear walking beam attachment and landing gear trunnion to wing structure.

(l) The landing gear beam.

(m) The main landing beam outboard support fitting.

(n) The forward trunnion support fitting.

(o) The main landing gear beam inboard support (hanger link fitting).

(p) If the airplane departs the prepared surface, do the inspections outlined in Phase IB.

E. Fuselage, Wing, Empennage and Engine Strut/Nacelle Inspection

SUBTASK 05-51-01-210-066

(1) Examine the fuselage areas that follow:

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(a) Examine the upper and lower fuselage skin panels forward and aft of the wing.

NOTE: Permanent wrinkles frequently occur on the lower side of the fuselage skins aft of STA 727.

(b) If you found wrinkles on the skin panels and you did not know they were there before, perform an internal inspection.

NOTE: If the wrinkles were there before, make a careful inspection of the wrinkles for cracks.

(c) Examine the chords, stiffeners, webs and splices on the keel beam at station 663 thru 727A.

NOTE: It is only necessary to examine these structures if they are on the keel beam.

SUBTASK 05-51-01-210-067

(2) Perform a general visual examination of the areas around the probes and static ports.

(a) If obvious visual damage is found, do an RVSM check: SRM SUBJECT 51-10-03.

(3) Examine the wing lower surface leading edge, inspar, and trailing edge for signs of damage.

F. Cabin Inspections

NOTE: If inspections of the forward cabin were already performed during Phase I Inspection, TASK 05-51-37-200-804, then they do not need to be repeated.

SUBTASK 05-51-01-210-068

(1) Do the following cabin inspections that follow to find possible damage:

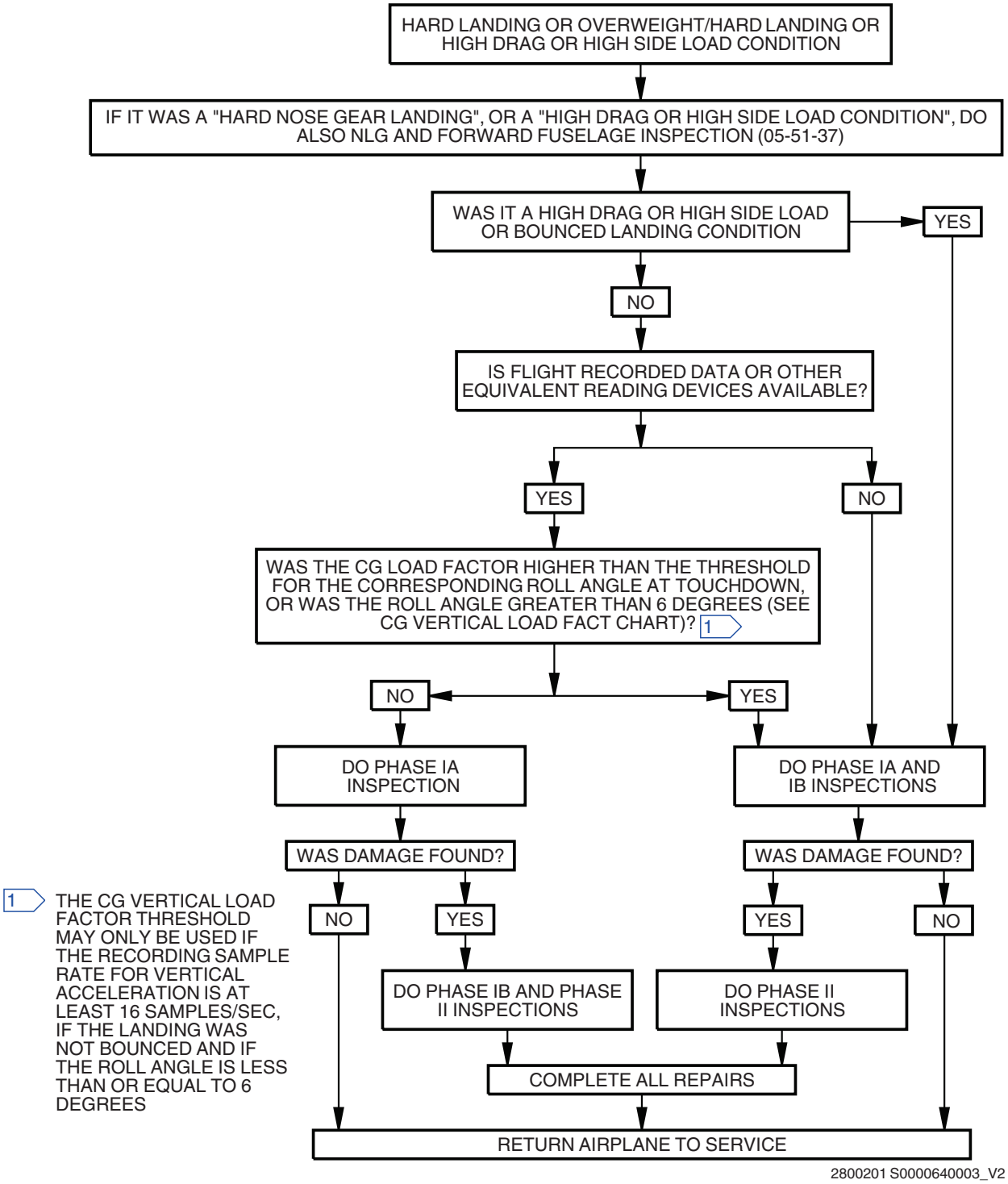
(a) Do a visual inspection of all ceiling panels (and video monitors, if applicable) for dislodging and evidence due to impact damage.

(b) Do a visual inspection of all overhead bins for evidence of looseness and impact damage.

(2) If any cabin damage was found in the Phase IA Cabin Inspections including the ceiling panels, overhead bins, or the video monitors then continue the Cabin Inspection in Phase IB and Phase II. If no damage was found, no further cabin inspections are required, and it is not necessary to continue the cabin inspections in Phase IB and Phase II.

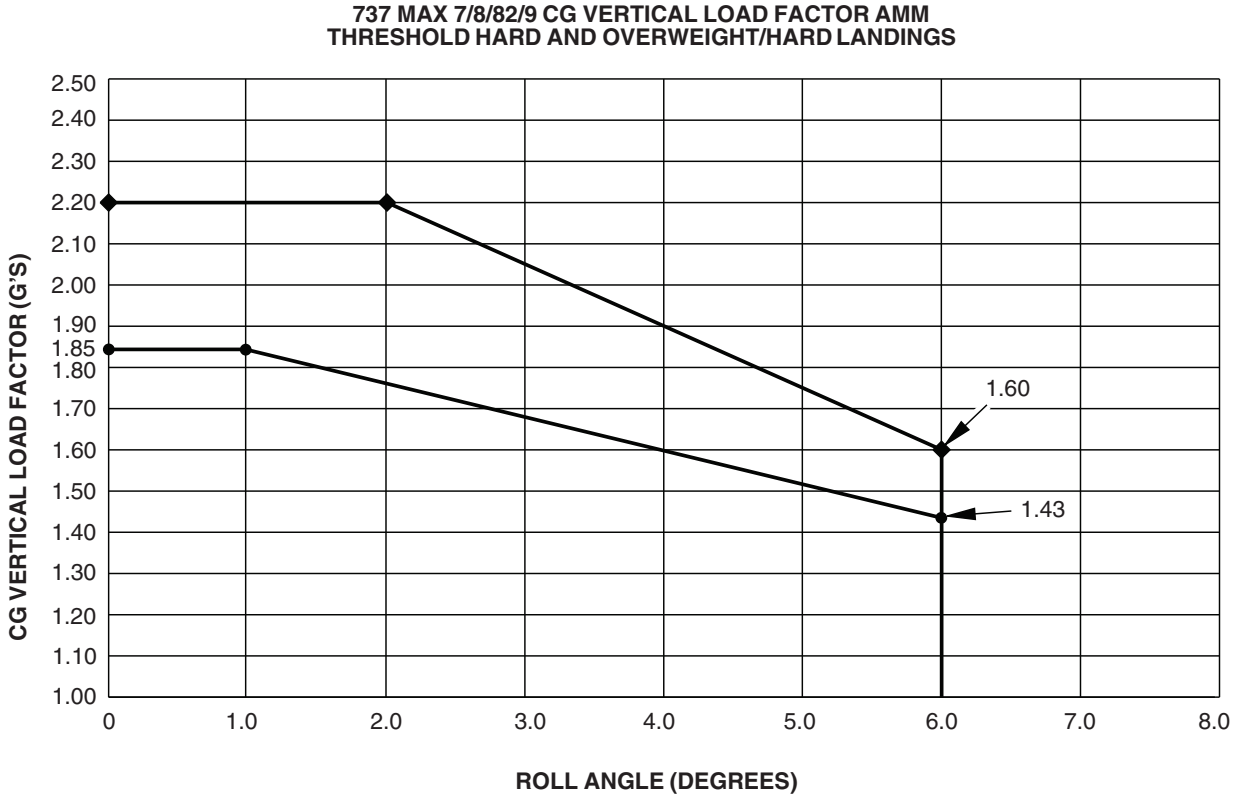
——— END OF TASK ———

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Hard or Overweight/Hard Landing or High Drag or High Side Load Condition Logic Flowchart
Figure 201/05-51-01-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

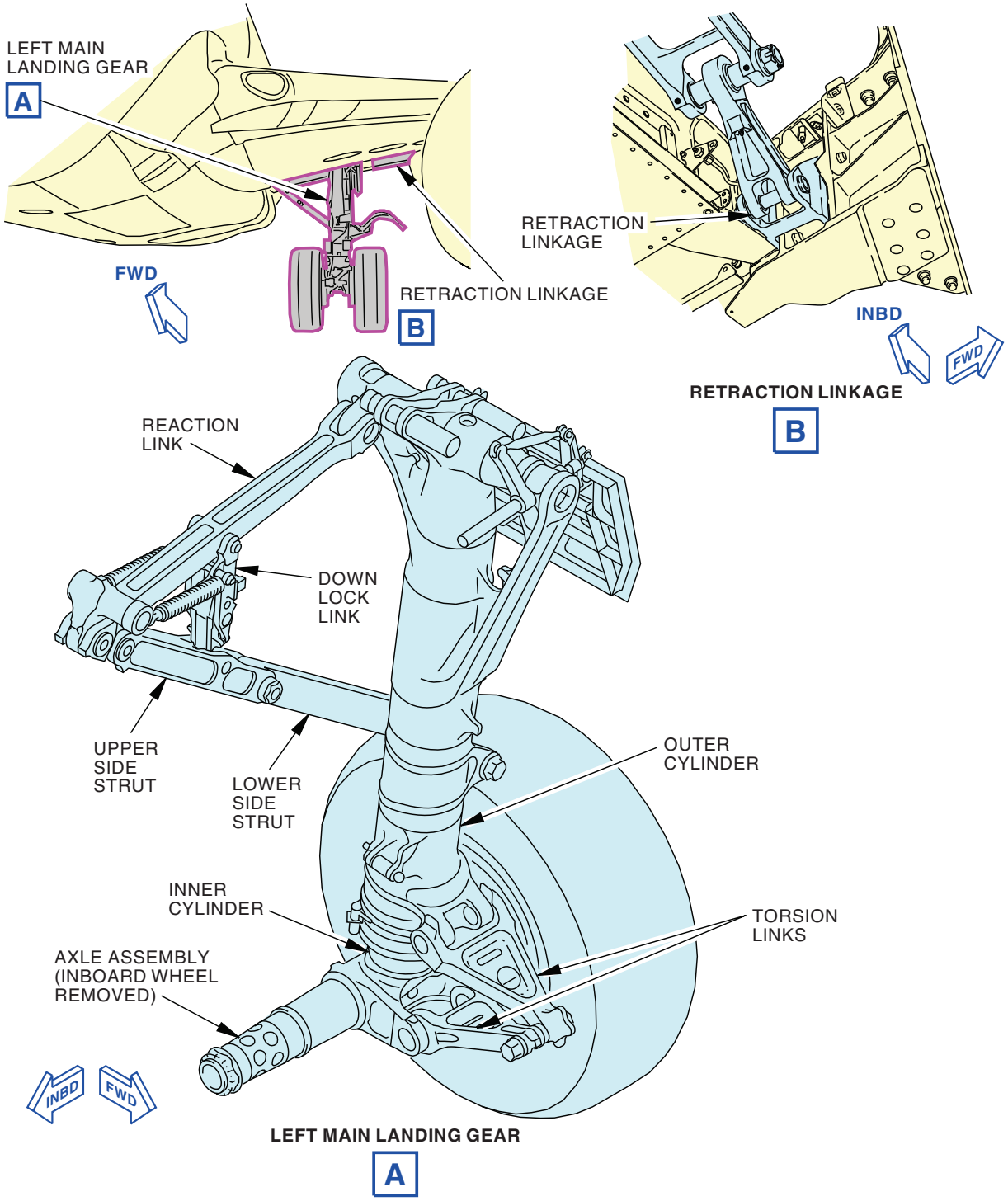
LANDING WEIGHT ≤ MLW + 1000 lbs (454 kg), 16+ SAMPLES/SEC

LANDING WEIGHT > MLW + 1000 lbs (454 kg), 16+ SAMPLES/SEC

2800209 S0000640004_V2

CG Vertical Load Factor Chart
Figure 202/05-51-01-990-810

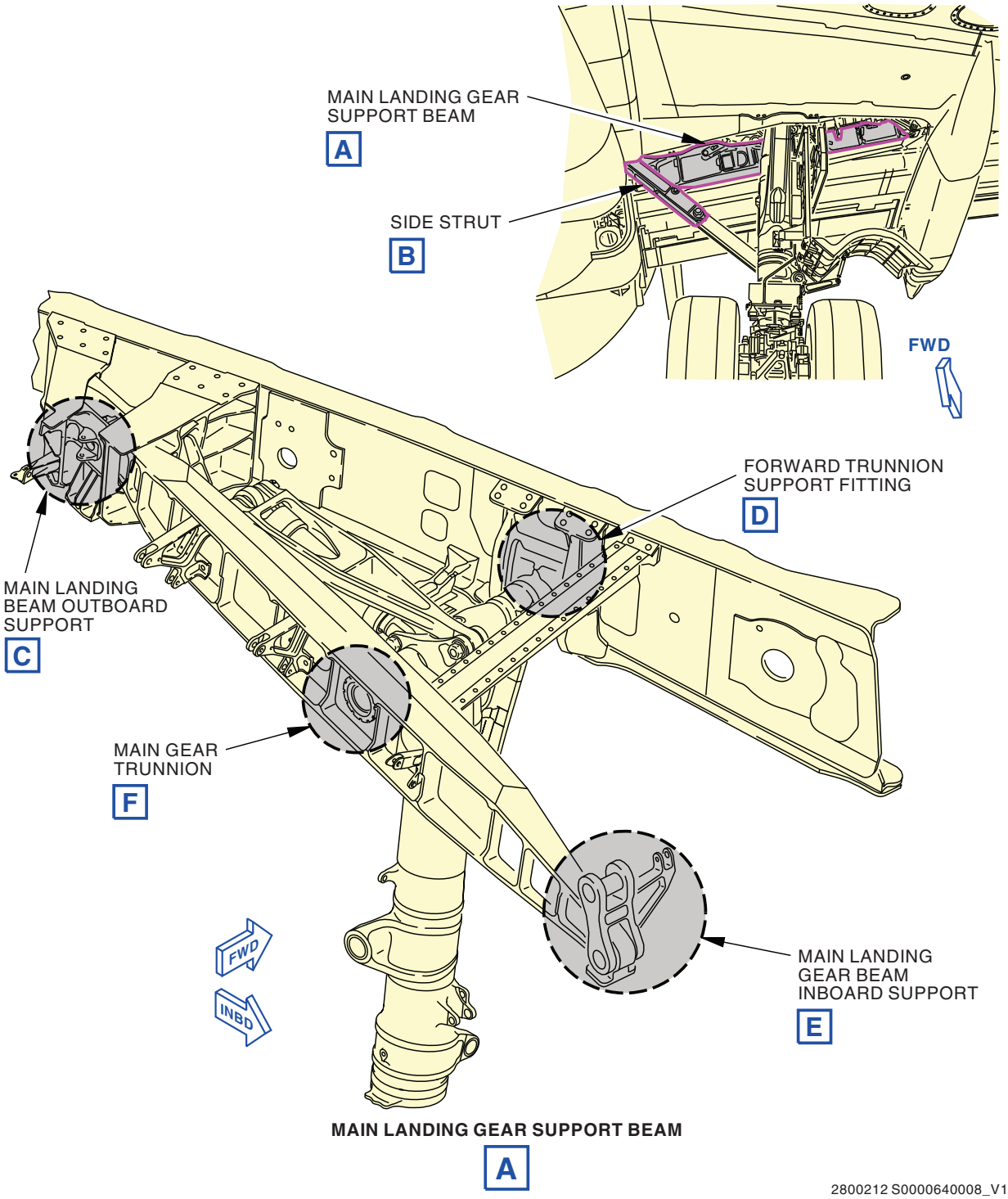
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2800211 S0000640007_V1

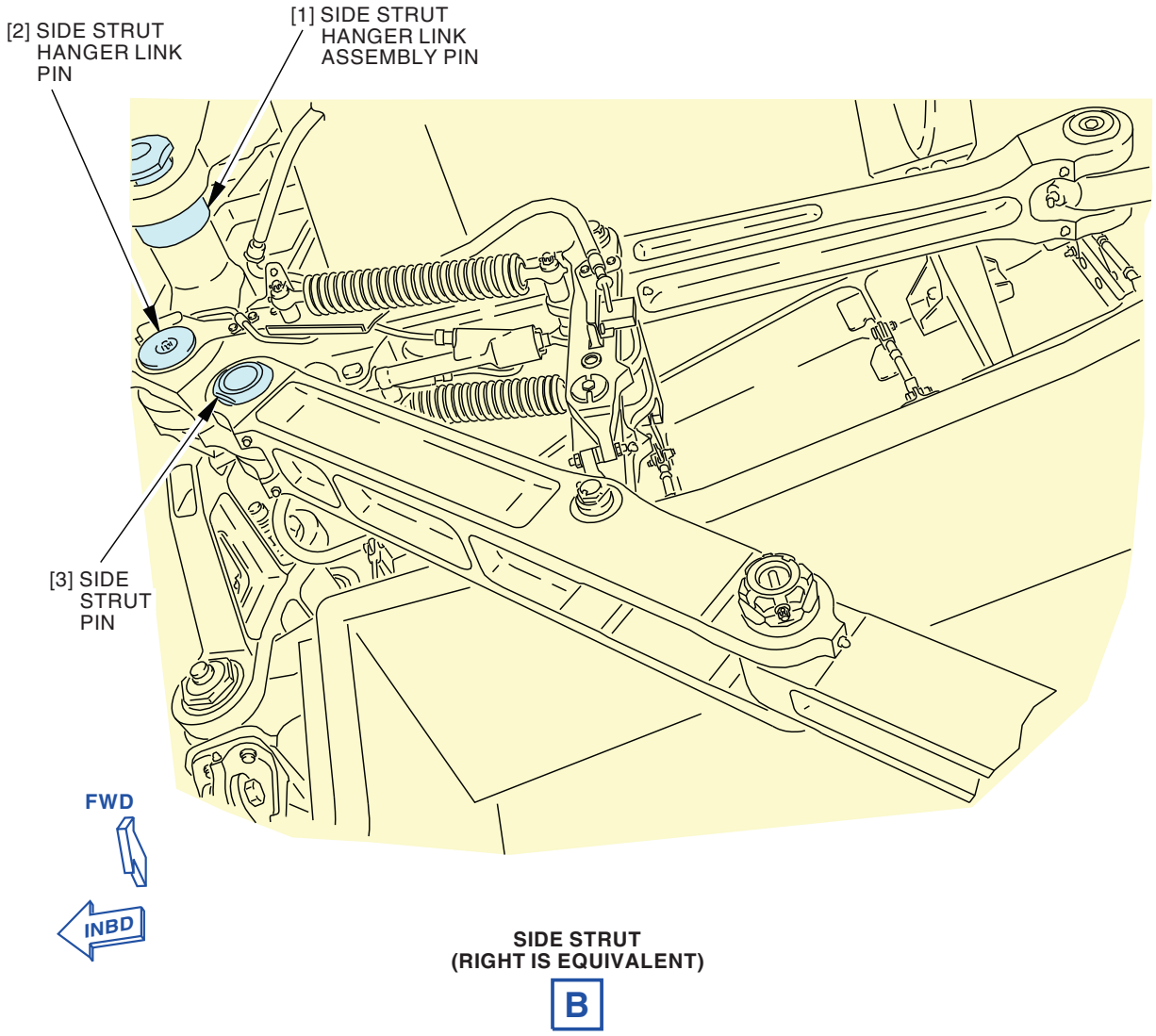
Main Landing Gear
Figure 203/05-51-01-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 1 of 6)

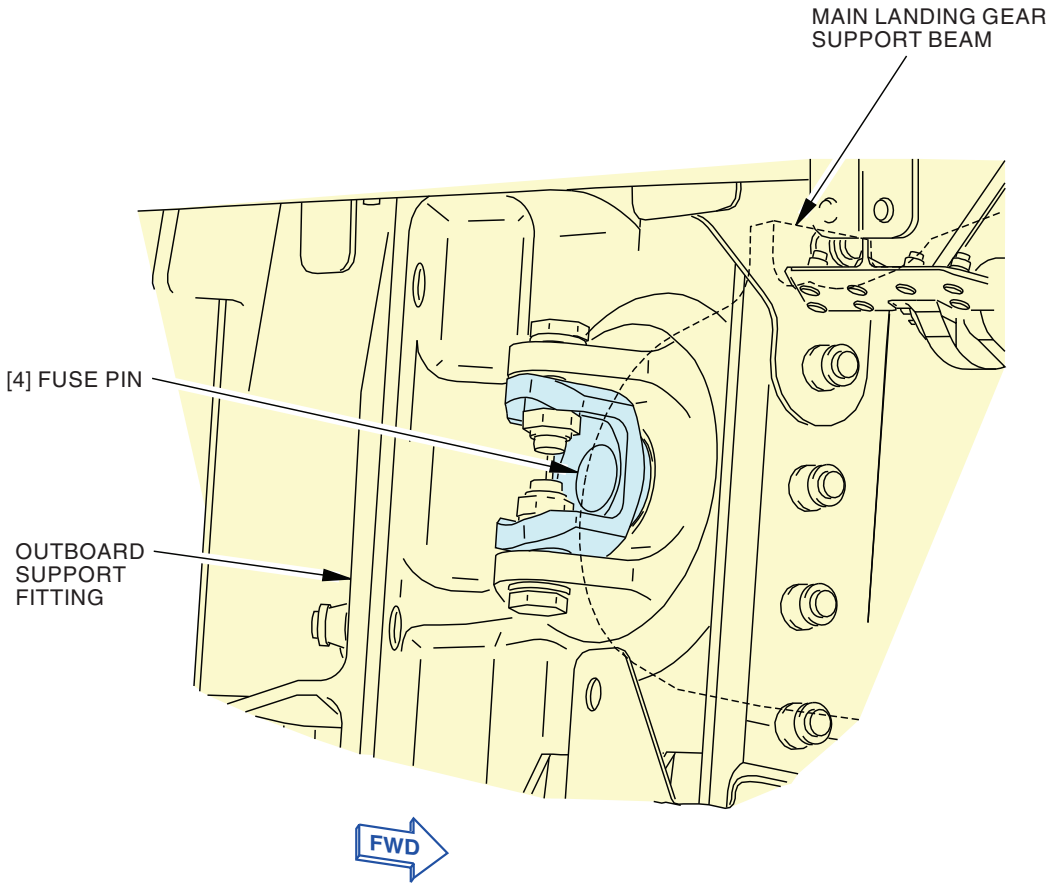
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2800213 S0000640009_V1

Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



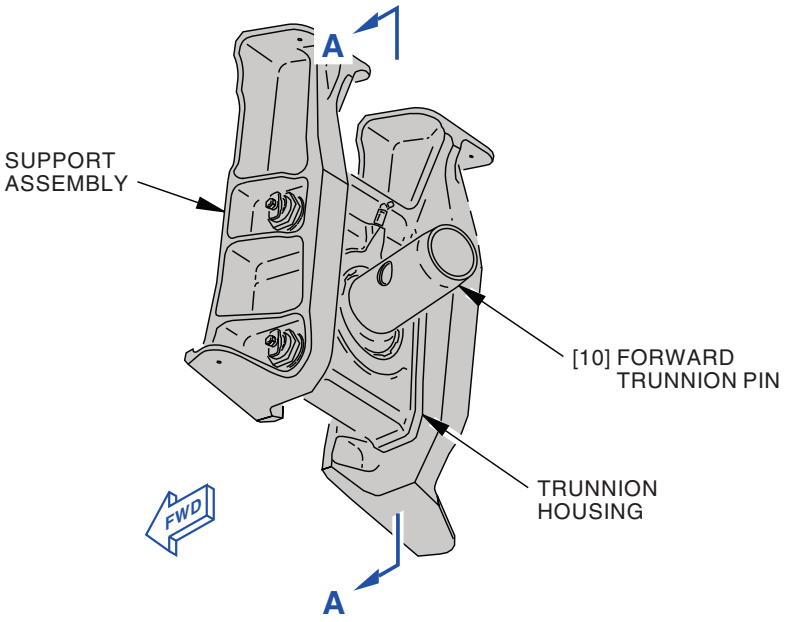
MAIN LANDING GEAR BEAM OUTBOARD SUPPORT

C

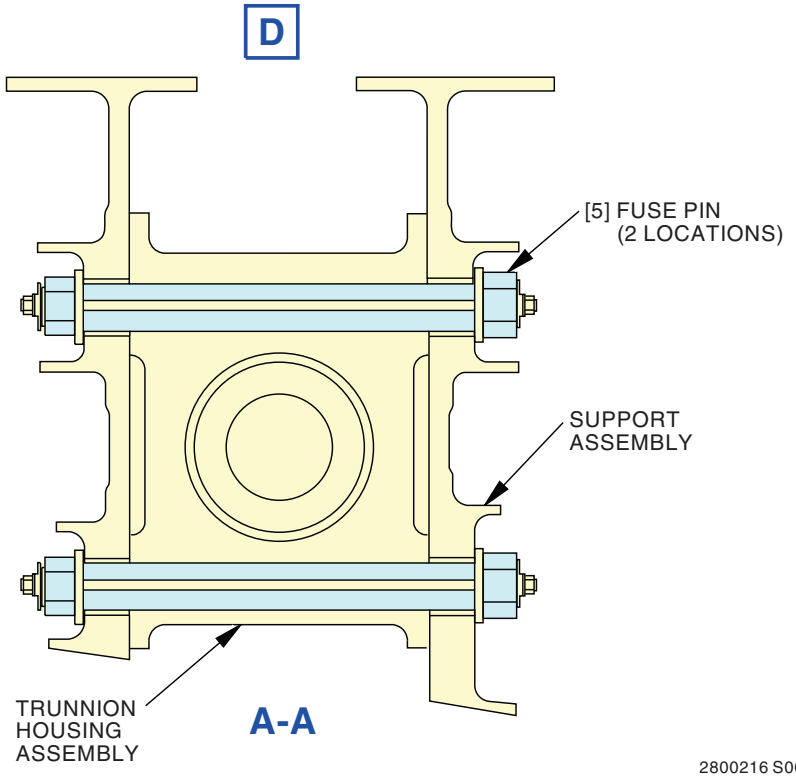
2800214 S0000640010_V1

Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



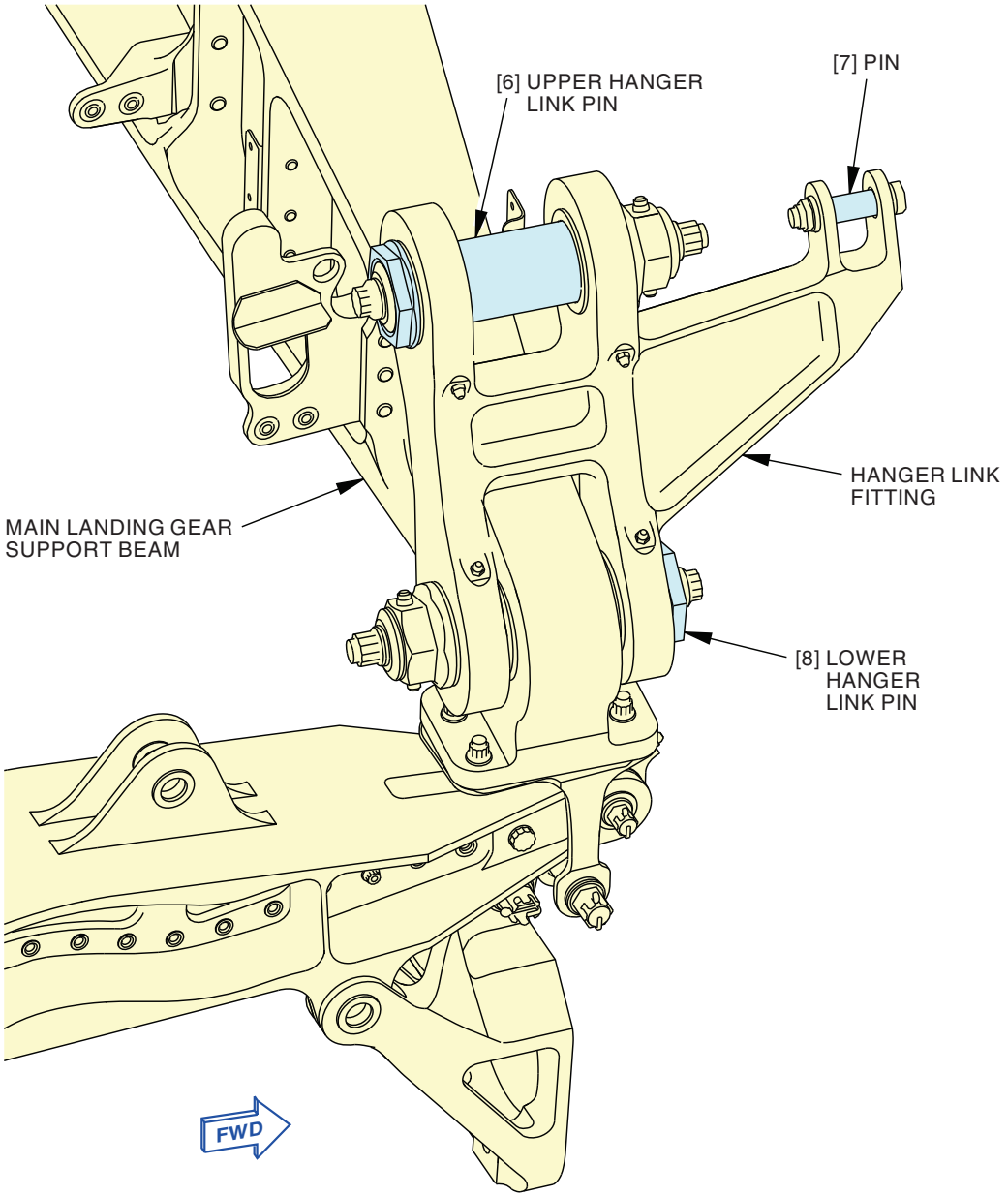
FORWARD TRUNNION SUPPORT FITTING



2800216 S0000640406_V2

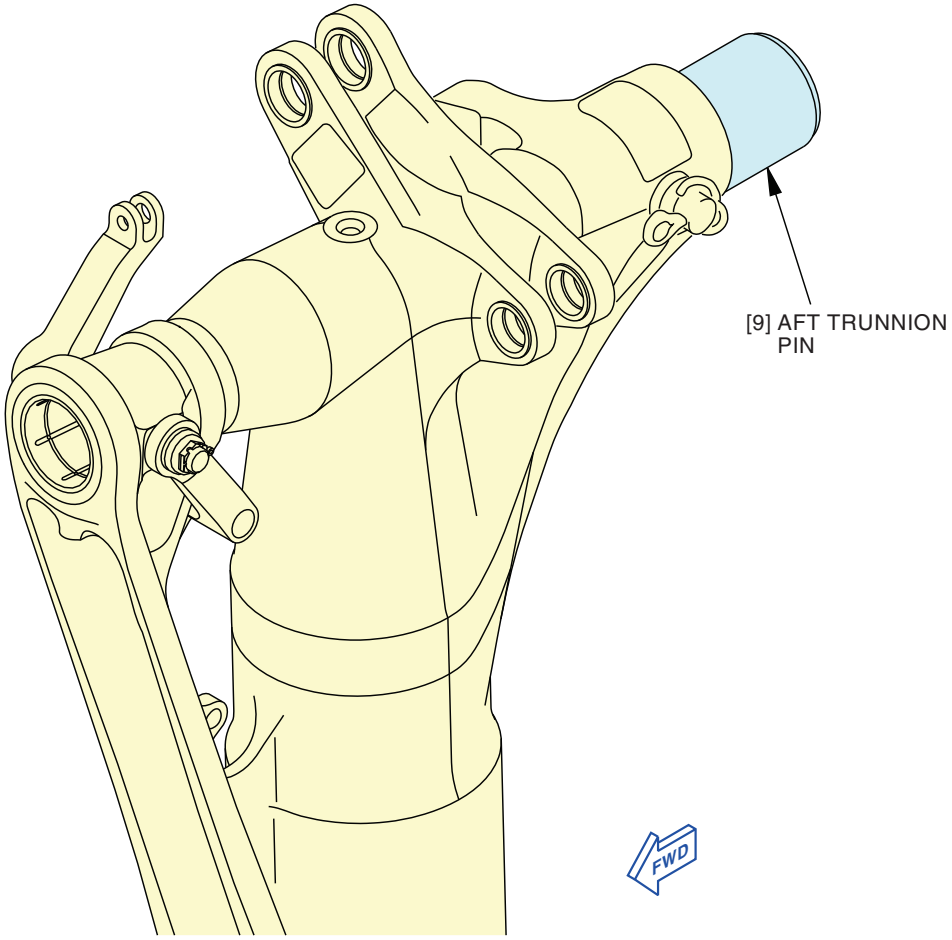
Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN GEAR TRUNNION

F

2800219 S0000640748_V2

Main Landing Gear Pins
Figure 204/05-51-01-990-812 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-51-01-210-805

3. Phase IB Inspection

A. References

Reference	Title
05-51-10-210-801	Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection (P/B 201)
05-51-37 P/B 201	NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
32-41-41 P/B 401	MAIN LANDING GEAR BRAKE - REMOVAL/INSTALLATION
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-700-802	Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-11 P/B 401	MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-1746	Pin - Lockout, L.E. Cruise Depressurization Valve Part #: C27038-6 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00
315	APU Compartment - Left
316	APU Compartment - Right
430	Subzone - Engine 1, Nacelle Strut
440	Subzone - Engine 2, Nacelle Strut
500	Left Wing
600	Right Wing
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

E. Main Landing Gear Inspection

SUBTASK 05-51-01-210-054

(1) Examine the main landing gear areas as follows:

(a) Examine the brake assemblies for damage (Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802).

(b) The doors and linkages of the main gear strut.

(c) Remove the forward trunnion fuse pin [5] and perform a detailed visual inspection if any of the following events occurred such that a significant drag loading on the main gear is expected:

1) The main landing gear hit an obstacle of large mass, or

2) The aircraft landed short of the prepared surface, or

3) The airplane ran off the prepared surface which produced a peak deceleration that was more severe than maximum effort braking.

NOTE: Inspection of the forward trunnion fuse pins may be deferred for non-revenue ferry flights only.

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SUBTASK 05-51-01-210-069

(2) If the airplane departs off the prepared surface, do the following:

(a) Examine the tires for damage (Landing Gear Tire Inspection, TASK 32-45-00-700-803).

(b) Examine the wheels for damage; cracks, missing tiebolts, heat shield (Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801).

1) After the examination, if the tires and wheels have not been damaged, do these steps:

a) Do the recovery and/or movement of the airplane to the prepared surface.

2) If the wheel heat shield(s) is/are bent and prevents the free rotation of the wheel and the tire assembly that contacts the brake:

a) Bend the heat shield(s) to allow free rotation of the wheel and tire assembly.

NOTE: The airplane can be towed to a location where further inspections and examinations will be performed.

3) After the examination, if the tires and wheels are determined to be damaged and prevents the recovery and/or movement of the airplane to the prepared surface, do these steps:

a) Replace the wheel and tire assemblies to allow recovery and movement of the airplane (MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-45-11/401).

4) Examine the brake assemblies for damage (Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802).

a) If a brake assembly is damaged and does not allow the rotation of the wheel and tire assembly, do these steps:

NOTE: If the wheel and tire assembly cannot rotate because the brake assembly is damaged, the brake assembly must be removed. After the brake assembly is removed and the wheel and tire assembly is installed, then the airplane can be towed safely.

<1> Remove the Wheel and Tire assembly (Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801).

<2> Remove the brake assembly (Main Landing Gear Brake Removal, TASK 32-41-41-000-801).

<3> Install the Wheel and Tire assembly (Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801).

NOTE: The wheel and tire assembly is installed to allow the airplane to be recovered.

5) After the airplane has been moved to a location for further examination, and before next flight, do these steps:

a) Replace the damaged wheel and tire assemblies (MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-45-11/401).

<1> If the wheel and tire assemblies were replaced before the airplane recovery. Clean the wheel and tire assemblies of unwanted material.

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<2> After the wheel and tire assemblies have been cleaned, examine the wheels and tires for damage (Landing Gear Tire Inspection, TASK 32-45-00-700-803, Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801).

<3> If no damage is found, the wheels and tires are serviceable.

b) Prior to flight, replace the brake assemblies (MAIN LANDING GEAR BRAKE - REMOVAL/INSTALLATION, PAGEBLOCK 32-41-41/401).

c) Remove the brake sleeves and examine the axles.

NOTE: There are occasions where the airplane lands and one MLG departs the prepared surface but then returns to the prepared surface and taxis to the gate. In this case, the wheel/tire- and brake-assemblies need to be cleaned of any debris and examined.

SUBTASK 05-51-01-210-070

(3) During a landing, if one main landing gear departs from the prepared surface and returns to the prepared surface and taxis to the gate, do these steps:

(a) Examine the tires for damage (Landing Gear Tire Inspection, TASK 32-45-00-700-803).

(b) Examine the wheels for damage; cracks, missing tiebolts, heat shield (Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801).

(c) Examine the brake assemblies for damage (Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802).

F. Nose Landing Gear Inspection

SUBTASK 05-51-01-210-055

(1) Perform the NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-37/201, if it was a "Hard Nose Gear Landing", or a "High Drag or High Side Load Condition".

G. Fuselage Inspection

NOTE: If inspections of the forward fuselage were already performed during Nose Gear and Forward Fuselage NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-37/201, then they do not need to be repeated.

SUBTASK 05-51-01-200-003

(1) Examine the aft side of the pressure bulkhead for signs of buckling, deformation or damage.

(a) Do a detailed visual inspection of the entire aft side of the BS 1016 aft pressure bulkhead. Give particular attention to the upper bulkhead structure between skin stringers S-10L and S-10R.

NOTE: Access to the interior unpressurized aft pressure bulkhead area is through the aft access hatch, 311BL.

H. Wing Area Inspection

SUBTASK 05-51-01-860-001

(1) Extend the Trailing Edge (TE) flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 05-51-01-860-002

(2) Extend the Leading Edge (LE) flaps and slats, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

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SUBTASK 05-51-01-040-008

WARNING DO THE DEACTIVATION PROCEDURE FOR THE TRAILING EDGE FLAP BEFORE YOU DO WORK ON THE FLAP SYSTEM. THE FLAPS MOVE QUICKLY. THEY CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-01-040-010

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) (Preferred Configuration) If the LE flaps and slat locks are available, do these steps to secure the area under the wing for inspection:

(a) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

(b) Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 05-51-01-040-011

WARNING DO THE ALTERNATIVE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) (Alternate Configuration) If the LE flaps and slat locks are not available, do these steps to secure the area under the wing for inspection:

NOTE: This alternative method is used to secure the areas under the wing LE and TE when no other maintenance action or structural inspection is required, scheduled or unscheduled.

(a) Do these steps to secure the hydraulic system:

1) Make sure that the hydraulic power is removed (TASK 29-11-00-860-805).

2) Make sure that the HYD PUMPS A ELEC 2 and the HYD PUMPS B ELEC 1 switches are in the OFF position.

3) Make sure that the HYD PUMPS ENG switch is in the ON position.

NOTE: The HYD PUMPS ENG switch should be left in the ON position. This will increase the life of the depressurization solenoid which becomes energized when the HYD PUMPS ENG switch is put in the OFF position.

4) Attach a DO NOT OPERATE tag, STD-858, on the HYD PUMPS switches.

(b) Make sure that the Proximity Sensor Electronics Unit (PSEU) is in the GROUND MODE (TASK 32-09-00-860-802).

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(d) Make sure that the ALTERNATE FLAP ARM switch is in the OFF position.

(e) Attach a DO NOT OPERATE tag, STD-858, on the flap control lever and the ALTERNATE FLAPS ARM switch.

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(f) Do these steps to secure the LE cruise depressurization valve (located in the main landing gear wheel well):

1) If the valve has a tab, cut the tab from the valve (SB 27-1242).

2) Remove the lockwire from the manual override lever (handle).

3) Move the manual override lever (handle) to the OFF position.

NOTE: Do not force movement of the lever with tools. The lever should move with very little force. If the lever is tight, you may need hydraulic pressure and the Flap/Slat Electronics Unit (FSEU) to relieve the pressure on the LE cruise depressurization valve.

4) Install the lockout pin, SPL-1746, in the manual override lever (handle) to keep the lever in the OFF position.

a) If the tool cannot be easily installed then it is necessary to supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.

<1> Use the FSEU procedure below but replace the selection of the LE Uncommanded Motion (UCM) shutoff valve for the LE DEPR valve.

NOTE: Two persons will be necessary when the FSEU is used to operate the LE cruise depressurization valve.

5) Attach a DO-NOT-REMOVE tag on the rig pin or lockout pin, SPL-1746.

(g) Do these steps to secure the LE UCM shutoff valve:

1) Open this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

2) Remove the lockwire from the manual override lever (handle).

3) Move the manual override lever on the LE UCM shutoff valve to the UP (OFF) position.

NOTE: The lever should move with very little force. If the lever is tight, you may need hydraulic pressure and the FSEU to release the pressure on the LE UCM shutoff valve.

a) If you cannot move the manual override lever, use the FSEU and hydraulic pressure to operate the LE UCM shutoff valve.

NOTE: Two persons will be necessary when the FSEU is used to release the pressure on the LE UCM shutoff valve.

<1> Supply system B hydraulic power (TASK 29-11-00-860-801).

<2> Use the FSEU to release the pressure on the LE UCM shutoff valve, as follows:

<a> Get access to the main equipment center, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

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<b> Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

<c> Push the ON/OFF button.

<d> EXISTING FAULTS..... press NO.

<e> Press NO until OTHER FUNCTIONS.

<f> OTHER FUNCTIONS..... press YES.

<g> SET OUTPUTS..... press YES.

<h> Press NO until LE UCM SHUTOFF VALVE.

<i> LE UCM SHUTOFF VALVE..... press YES.

WARNING

MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

<j> SET ON..... press YES.

<k> Remove system B hydraulic power (TASK 29-11-00-860-805).

<l> Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

b) Move the LE UCM shutoff valve up.

c) Install the lockout pin, SPL-1746, in the manual override lever (handle) to keep the lever in the OFF position.

NOTE: The tool will keep the handle in the UP (OFF) position.

d) Attach a DO-NOT-REMOVE tag on the lockout pin, SPL-1746.

(h) Do these steps to prevent LE devices extension or retraction by the standby hydraulic system:

1) Disconnect the electrical connector from the LE standby drive shutoff valve.

a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion (TASK 20-60-07-910-802).

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

b) If there was corrosion, correct the problem (SWPM 20-60-03).

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WARNING MAKE SURE THAT THE MANUAL OVERRIDE LEVER IS IN POSITION 1 TO PREVENT THE MOVEMENT OF THE LEADING EDGE SLATS. IF THE LEADING EDGE SLATS MOVE, DAMAGE TO AN OPEN THRUST REVERSER AND INJURIES TO PERSONNEL CAN OCCUR.

2) Make sure that the override lever is in the Position 1 (standby hydraulic power OFF).

NOTE: If the manual override lever at the LE standby drive shutoff valve is in Position 2 and the standby hydraulic system is then pressurized, the leading edge slats will move to the full extend position.

3) Install the lockout pin, SPL-1746.

NOTE: The lock, SPL-1675, can be used on the standby hydraulic system for the alternate deactivation (TASK 27-81-00-040-801).

4) Attach a DO NOT OPERATE tag, STD-858, on the LE standby drive shutoff valve.

(i) Do the LE flaps and slats tryout (TASK 27-81-00-040-801).

(j) Remove power from the airplane, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

(k) Make sure the BAT switch is in the OFF position.

(l) Attach a DO NOT OPERATE tag, STD-858, on the BAT switch.

SUBTASK 05-51-01-210-071

WARNING DO NOT GO INTO THE WING LEADING EDGE AND TRAILING EDGE AREAS. THESE FLIGHT CONTROL SURFACES CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Examine the wing area as follow:

NOTE: The wing areas must be examined from the ground. Unsatisfactory conditions or discrepancies can be visually identified from a safe distance. If damage is found in the extended wing areas, then it is necessary to install the LE flaps and slat locks before maintenance action.

(a) The extended wing TE for signs of damage.

(b) The extended wing LE for signs of damage.

(c) The strut-to-wing fairing panels on the upper and lower nacelle.

(d) If there was a dragged engine nacelle condition that occurred, do the following:

1) Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection, TASK 05-51-10-210-801.

I. Cabin Inspections

NOTE: If no damage was found in the Phase IA cabin inspections, then no further cabin inspections are required.

SUBTASK 05-51-01-210-056

(1) Do a detailed visual inspection of the latches, stops, potting inserts, and all fasteners on all ceiling panels equipped with video monitors.

(2) Continue the Cabin Inspection in the Phase II Inspection, TASK 05-51-01-210-806.

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J. Put the Airplane Back to it's Usual Condition

SUBTASK 05-51-01-410-009

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-01-080-005

(2) If installed, remove the LE flaps and slat locks.

(a) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 05-51-01-860-003

(3) If necessary, do these steps to restore electrical power to the airplane

(a) Remove the DO NOT OPERATE tag, STD-858, from the BAT switch.

(b) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 05-51-01-860-004

(4) If installed, remove the DO NOT OPERATE tag, STD-858, from the HYD PUMPS switches.

SUBTASK 05-51-01-440-011

(5) Do the activation procedure for the TE flap system.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-01-440-012

(6) Do the activation procedure for the LE flap and slat system.

(a) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

TASK 05-51-01-210-806

4. Phase II Inspection

(Figure 203, Figure 204)

A. References

Reference	Title
05-51-10-210-801	Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection (P/B 201)
05-51-37 P/B 201	NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
05-51-37-200-804	Phase I Inspection (P/B 201)
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
12-15-31-610-801	Main Landing Gear Shock Strut Fluid Check (P/B 301)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
32-00-10-211-801	Landing Gear Inner Cylinder Chrome Inspection (P/B 601)

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(Continued)

Reference	Title
32-11-00-000-802	Main Landing Gear Removal (P/B 401)
32-11-21-960-801	Active Seals with the Spare Seals Replacement (P/B 801)
32-21-00-200-801	Nose Landing Gear - Inspection (P/B 601)
32-21-00-700-801	Nose Landing Gear Torsional Freeplay Inspection (P/B 601)
32-21-24-000-801	NLG Inner Cylinder Removal (P/B 401)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-34-00-730-801	Main Gear Manual Extension System Test (Airplane on Jacks) (P/B 501)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-41-41-700-803	Main Landing Gear Brake Check (Wheel Removed from the Airplane) (P/B 601)
32-49-03-000-801	Nose Landing Gear Hubcap Assembly Removal (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
57-16-06-000-801	Main Landing Gear Support Beam Removal (P/B 401)
71-00-00-200-813-G00	Engine After a Hard Landing Inspection (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
120	Subzone - Body Station 396 to Body Station 540
140	Subzone - Body Station 727.00 to Body Station 1016.00
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Access Panels

Number	Name/Location
321A	Vertical Fin, Dorsal Fin

D. Main Landing Gear Inspection

SUBTASK 05-51-01-210-057

(1) Examine the Main Landing Gear (MLG) areas as follows (Figure 203, Figure 204).

(a) Examine the outer and inner cylinder lugs.

(b) Examine the trunnion fitting area of the outer cylinder including the bushing edges/flanges.

(c) Examine the torsion links.

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(d) Examine the actuator walking beam and linkage fittings.

(e) Examine the MLG beam fitting and aft trunnion support fitting on the shock strut for cracks.

(f) Examine these MLG pins and fuse pins: side strut hanger link assembly pin [1], side strut hanger link pin [2], side strut pin [3], fuse pin [4], fuse pin [5], upper hanger link pin [6], pin [7], lower hanger link pin [8], aft trunnion pin [9], and forward trunnion pin [10].

1) For vertical overload remove the MLG beam outboard attachment fuse pin and perform a detailed visual inspection.

a) Remove the MLG (TASK 32-11-00-000-802).

b) Remove the MLG support beam as required (TASK 57-16-06-000-801).

c) If damage is found remove the forward trunnion fuse pins and perform a detailed visual inspection.

2) For a high drag or high side load overload, remove the forward trunnion fuse pins and perform a detailed visual inspection.

a) Remove the MLG (TASK 32-11-00-000-802).

b) Remove the MLG support beam as required (TASK 57-16-06-000-801).

c) If damage is found remove MLG beam outboard attachment fuse pin and perform a detailed visual inspection.

(g) Make sure that the pressure and hydraulic fluid levels in the MLG shock struts are correct, do this task: Main Landing Gear Shock Strut Fluid Check, TASK 12-15-31-610-801.

1) If the fluid level is low; do these steps:

a) Do this task: Active Seals with the Spare Seals Replacement, TASK 32-11-21-960-801.

b) Remove the inner cylinder and examine it for distortions or cracks, do this task: Landing Gear Inner Cylinder Chrome Inspection, TASK 32-00-10-211-801.

c) Examine the orifice support tube, while it is on the airplane, for cracks.

(h) Examine the wheel well of the MLG for fuel or other fluid leaks.

(i) If a high drag or side load condition, and a hard landing occur at the same time, remove the MLG inner cylinders, do this task: Landing Gear Inner Cylinder Chrome Inspection, TASK 32-00-10-211-801.

1) Look for distortions or cracks in the inner cylinder.

(j) Examine the side strut and side strut attachments.

(k) If any MLG wheels are removed because of a blown tire, do the steps that follow:

1) Examine the wheel structure for damage.

2) Examine the brake assembly for damage (TASK 32-41-41-700-803).

E. Nose Landing Gear Inspection

NOTE: If no damage was found in the nose landing gear inspections of PAGEBLOCK 05-51-37/201, then no further nose landing gear inspections are required.

SUBTASK 05-51-01-210-058

(1) If not previously done during the inspection of TASK 05-51-37-200-804, do the following:

(a) Jack the nose landing gear (TASK 07-10-03-580-803 or TASK 07-10-21-580-801).

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SUBTASK 05-51-01-210-059

(2) Examine the nose landing gear areas that follows.

(a) Examine carefully the areas near the trunnion and the drag-strut support fittings.

(b) Do this task: Nose Landing Gear - Inspection, TASK 32-21-00-200-801.

(c) Do this task: Nose Landing Gear Torsional Freeplay Inspection, TASK 32-21-00-700-801.

(d) Examine the nose landing gear wheel well.

(e) Examine the drag strut support fittings.

(f) Examine the outer-cylinder-trunnion-fitting area and the drag brace linkage.

(g) Make sure that the pressure and hydraulic fluid level in the nose landing gear shock strut is correct, do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

1) If the pressure and hydraulic fluid level is not correct, remove the nose landing gear inner cylinder, do this task: NLG Inner Cylinder Removal, TASK 32-21-24-000-801.

a) Examine the inner cylinder for cracks and other types of damage.

b) Examine the orifice support tube for distortion and cracks.

(h) Examine the wheel well of the nose landing gear for buckling and other damage.

1) Make sure that there are no loose fasteners in the web of the nose wheel well near the trunnion support fittings.

2) Look for fluid leaks.

(i) Do a check of the rigging of the steering mechanism.

1) Do this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

(j) Examine the axle for bends.

1) Spin the wheels.

a) Make sure that there is no binding that may be caused by a bent axle.

<1> If it is necessary, do this task: Nose Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-801.

2) Place a level on the outboard wheel rims.

a) Make sure that both wheels are vertical.

(k) If it was a hard nose gear landing or a high drag or side load condition, do these steps:

1) Remove the nose landing gear inner cylinder, do this task: NLG Inner Cylinder Removal, TASK 32-21-24-000-801.

a) Examine the inner cylinder for damage.

F. Fuselage and Empennage Inspection

SUBTASK 05-51-01-040-009

(1) Examine the fuselage as follows:

(a) Examine the lower fuselage structure.

NOTE: Look carefully at the area below the body floor beams just aft of the wing center section.

(b) Examine the wing-to-fuselage joints and the wheel well bulkheads.

(c) Examine the upper fuselage structure above the primary wing spars.

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(d) Examine the area around leading edge dorsal fin:

Number	Name/Location
321A	Vertical Fin, Dorsal Fin

(e) Examine the external fuselage skin at BS 1016 for signs of buckling, deformation or damage.

1) Do a detailed visual inspection of the exterior fuselage skin 10.0 in. (25.4 cm) forward and aft of the BS 1016 skin butt joint between stringers S-10L and S-10R.

NOTE: Gain access to the exterior of the fuselage skin BS 1016 under the vertical stabilizer fairing, 323AL and 323AR.

(f) Examine the wheel well bulkheads of the main landing gear.

(g) Examine the side-of-body ribs for cracks in the area of the rear spar.

(h) Examine the rear spar for permanent buckles between the tank end and side of the body.

(i) Examine the nose landing gear wheel well aft bulkhead.

(j) Examine the fuselage structure immediately outboard of the nose wheel well.

(k) Examine the fuselage skin attachments above the landing gear beam.

(l) Look for signs of fuel leaks or other fluids leaking in the areas that follow:

1) The wing
2) The nacelles
3) The engine struts
4) The external surfaces of the fuselage
5) The nose landing gear wheel well
6) The main landing gear wheel well.

(m) Examine the lower fuselage (body sections 46 and 48) for signs that it touched the runway.

(n) If there are signs of damage, examine the areas that follow.

NOTE: Look for broken parts, bent parts, or other types of damage.

1) The Auxiliary Power Unit (APU) firewall
2) The APU drain
3) The APU doors
4) The drain mast
5) The lower fuselage skin, stringers, frames, and shears ties.

SUBTASK 05-51-01-210-060

(2) Make sure that the flight controls move freely.

G. Wing Area Inspection

SUBTASK 05-51-01-210-061

(1) Do an inspection of the wing areas that follow:

WARNING DO NOT GO INTO THE WING LEADING EDGE AND TRAILING EDGE AREAS BEFORE YOU INSTALL SAFETY LOCKS. IF YOU DO NOT OBEY, INJURY TO PERSONNEL FROM ACCIDENTAL FLAP/SLAT OPERATION CAN OCCUR.

(a) Install the leading edge flaps and slat locks.

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1) Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(b) Examine the wings and nacelles for fuel or other fluid leaks.

(c) Examine the wing ribs along the aft side of the rear spar, WSTA 228.25 and 253.00 for cracks.

(d) Examine the upper and lower trailing edge panels and the airplane structure for damage.

1) Make sure that you examine the area near the main landing gear beam carefully.

(e) Examine the flap areas that follow for sheared rivets and structure damage:

1) The inboard and outboard trailing edge flaps

2) The flap tracks

3) The drive screws

4) The linkages and fairings.

(f) Examine the inboard and outboard spoilers for sheared rivets and structure damage.

(g) Examine the flap track bolts on the outboard trailing edge flap for damage that can occur if the flaps touched the ground.

(h) Examine the flap track bolts on the inboard trailing edge flap for damage that can occur if the flaps touched the ground.

(i) Make sure that the flight controls move freely and cable tension is satisfactory.

(j) For airplanes with winglets:

1) Examine the winglet and attachment at rib 27 (WBL 658.17).

H. Do a Test of the Landing Gears

SUBTASK 05-51-01-580-003

(1) Do a check of the nose and main landing gear.

NOTE: Look for signs of interference, misalignment, or distortion.

(a) Lift the airplane on jacks, do these tasks: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

(b) Retract and extend the nose landing gear, do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

(c) Retract and extend the main landing gear, do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

(d) Retract and then extend the main landing gear with the manual system to make sure they operate correctly, do this task: Main Gear Manual Extension System Test (Airplane on Jacks), TASK 32-34-00-730-801.

(e) Retract and then extend the nose landing gear with the manual system to make sure it operates correctly, do this task: Nose Gear Manual Extension System Test - Airplane on Jacks, TASK 32-35-00-730-801.

I. Cabin Inspections

SUBTASK 05-51-01-210-062

(1) Continue the following inspection if any cabin damage was found in the above Cabin Inspection including the ceiling panels, central overhead bins, or the video monitors. If no damage was found, no further cabin inspection is required and it is not necessary to continue this Cabin Inspection.

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(a) If ceiling panels without video monitors are found dislodged or damaged: check for proper latching and installation. Repair as necessary.

(b) If central overhead stowage bins are found with impact damage or are loose, or if ceiling panels equipped with video monitors are found with broken latches: repair as necessary.

(c) Inspect all tie rods of affected zones for evidence of buckling or rupture.

1) If any defect is found, replace as necessary.

(d) Examine lavatory tie rod attachments for damage consisting of breakage, cracks, and deformation. Inspect the tie rods for evidence of buckling or rupture.

1) If any defect is found, replace as necessary.

(e) Examine the lavatory floor fittings for damage consisting of breakage, cracks, and deformation.

1) If any defect is found, replace as necessary.

(f) Examine galley tie rod attachments for damage consisting of breakage, cracks, and deformation. Inspect the tie rods for evidence of buckling or rupture.

1) If any defect is found, replace as necessary.

(g) Examine the galley floor fittings for damage consisting of breakage, cracks, and deformation.

1) If any defect is found, replace as necessary.

(h) Examine, closet tie rod attachments for damage consisting of breakage, cracks, and deformation.

1) If any defect is found, replace as necessary.

(i) Examine the closet floor fittings for damage consisting of breakage, cracks, and deformation.

1) If any defect is found, replace as necessary.

J. Engine Inspections

SUBTASK 05-51-01-210-063

(1) Do the steps that follow to examine the engine areas:

(a) Do an engine inspection after a hard landing.

1) Do this task: Engine After a Hard Landing Inspection, TASK 71-00-00-200-813-G00.

2) Examine the engine nacelle if it touched the ground.

(b) If damage was found on the strut to wing fairing, do this task: Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection, TASK 05-51-10-210-801.

K. Cargo Loading Inspection

SUBTASK 05-51-01-211-004

(1) Do a check of the cargo loading system to make sure it operates correctly with the containers and pallets.

(a) Look at the components that follow:

1) Make a check of the cargo restraint system in all cargo compartments.

2) Visually check the following:

a) The rollers

b) The side guides

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c) The seat tracks

d) The pallet locks.

3) Examine all components for breaks, cracks, and missing retaining lips.

a) Make sure that the cargo loading rollers (where installed) move freely.

L. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-01-410-010

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-01-410-011

(2) Close all of the cowling, panels, fairings, and doors.

SUBTASK 05-51-01-080-006

(3) Remove the leading edge flaps and slat locks.

(a) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 05-51-01-440-013

(4) Do the activation procedure for the trailing edge flap system.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-01-440-014

(5) Do the activation procedure for the leading edge flap and slat system.

(a) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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SEVERE OR UNUSUAL TURBULENCE, EXCESSIVE MANEUVER, BUFFET OR SPEEDS MORE THAN DESIGN LIMITS - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains a task that is an inspection of the airplane after one of the following occurs:

(1) Severe turbulence or unusual turbulence

(2) Excessive maneuver

(3) Buffets

(4) Airspeed more than its design limits.

TASK 05-51-04-210-801

2. Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection

A. General

(1) When the conditional inspection tells you to "examine" or inspect a component, look for these conditions (repair or replace components, if it is necessary):

(a) Cracks

(b) Pulled apart structure or unusual gaps

(c) Loose paint (paint flakes)

(d) Discoloration or scuffing

(e) Twisted parts (distortion)

(f) Wrinkles or buckles in the structure

(g) Bent components

(h) Loose fasteners

(i) Fasteners that have pulled out, deformed, or are gone

(j) Delaminations

(k) Fiber breakage

(l) Misalignment

(m) Interference

(n) Nicks or gouges

(o) Rotated (tilted) fastener heads

(p) Core damage

(q) Evidence of impact

(r) Other signs of damage.

B. Severe or Unusual Turbulence Condition

(1) Severe turbulence is identified as turbulence that causes large, abrupt changes in the altitude and/or attitude. The airplane could be out of control for short periods. It usually causes large variations in airspeed. Passengers and crew are moved violently against their seat belts and loose objects are moved around the airplane. Unusual turbulence could be vibration, flutter, or flight control related events that the flight crew might perceive as turbulence in a situation where it is not usually encountered.

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(2) An inspection is required when the aircraft has encountered severe or unusual turbulence. The pilot must report the suspected encounter of these events. If the pilot reports that the aircraft has encountered severe or unusual turbulence, refer to “Examine Airplane Structure and Wing Areas” and “Cabin Inspections” sections within this Task.

C. Excessive Maneuver Condition

(1) Excessive maneuvers are maneuvers that result in severe and abnormal aircraft response and/or attitude, and/or exceeding of the flight maneuvering vertical load factors, specified below. Excessive maneuvers could be due to rapid or large alternating control inputs i.e., control column, rudder pedals, and control wheel.

(a) The vertical acceleration load limits shown below are directly applicable to excessive maneuvers.

1) Flaps up . . . . . 2.5g to -1.0g

2) Flaps down . . . . . 2.0g to 0.0g

(2) An inspection is required when the aircraft has been involved in an excessive maneuver. The pilot must report the suspected encounter of this event. If the pilot reports that the airplane has been involved in an excessive maneuver, refer to "Examine Airplane Structure and Wing Areas" and "Cabin Inspections" in this section.

D. Maximum Operating Limit Speed

(1) The maximum design speed of the airplane for flight operations are as follows:

NOTE: The Maximum Operating Limit Speed is in Section 1, LIMITATIONS of the Airplane Flight Manual (AFM).

(a) The airplane Maximum Operating Speed (VMO) is 340 knots (630 km/h) when the airplane is at or below 25,968 ft (7915 m).

(b) Above 25,968 ft (7915 m), the airplane is limited to a Maximum Operating Mach (MMO) of 0.82.

(c) The overspeed clacker will activate when the airplane exceeds the VMO or MMO.

(2) The maximum indicated airspeed of the airplane under the Configuration Deviation List (CDL) 57-31-01 are as follows:

(a) The airplane VMO is 270 knots (500 km/h) when the airplane is at or below 25,968 ft (7915 m).

(b) Above 25,968 ft (7915 m), the airplane is limited to a MMO of 0.73.

E. References

Reference	Title
05-51-10-210-801	Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection (P/B 201)
27-11-00 P/B 501	AILERON - ADJUSTMENT/TEST
27-21-00 P/B 501	RUDDER AND RUDDER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST
27-31-00 P/B 501	ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST
27-41-00 P/B 501	HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST
27-51-00 P/B 501	TRAILING EDGE FLAP SYSTEM - ADJUSTMENT/TEST
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)

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(Continued)

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00 P/B 501	SPOILER CONTROL SYSTEM - ADJUSTMENT/TEST
27-62-00 P/B 501	SPEED BRAKE CONTROL SYSTEM - ADJUSTMENT/TEST
27-81-00 P/B 501	LEADING EDGE FLAP AND SLAT - ADJUSTMENT/TEST
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)

F. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors
800	Doors

G. Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection

SUBTASK 05-51-04-210-003

(1) If an unusual vibration occurs in-flight, do the "Examine Airplane Structure and Wing Areas" steps in this task: Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection, TASK 05-51-04-210-801.

SUBTASK 05-51-04-210-004

(2) If a stall occurs after the initial buffet or stick shaker condition, do the "Stall (After Initial Buffet or Stick Shaker) Structural Inspection" procedure.

SUBTASK 05-51-04-210-005

(3) If an unusual maneuver or a severe or unusual buffet condition occurs in flight, do the "Examine Airplane Structure and Wing Areas" procedure.

NOTE: Also do this examination if an unusual vibration occurs in flight.

SUBTASK 05-51-04-210-006

(4) Do the "Examine Airplane Structure and Wing Areas" procedure in the task if one or both conditions that follow occurs:

NOTE: When the overspeed event occurs at 25,968 ft (7915 m) or below, it should be viewed in knots. Overspeed events above 25,968 ft (7915 m) should be viewed in mach.

(a) The airplane speed exceeds the VMO by 20 knots (37 km/h) or more.

(b) The airplane speed exceeds the MMO by 0.02 Mach or more.

H. Examine Airplane Structure and Wing Areas

SUBTASK 05-51-04-860-001

(1) Extend the trailing edge flaps and the leading edge flaps and slats.

(a) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

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SUBTASK 05-51-04-040-001

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the deactivation procedure for the trailing edge flap system (TASK 27-51-00-040-801).

SUBTASK 05-51-04-040-002

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do the deactivation procedure for the leading edge flap and slat system with the flaps/slats in the fully extended position (TASK 27-81-00-040-801).

SUBTASK 05-51-04-480-001

(4) Install the lockset to prevent movement of the leading edge flaps/slats (TASK 27-81-00-480-801).

SUBTASK 05-51-04-210-008

(5) Do the inspections that follow:

(a) Examine the external surface of the fuselage.

1) Examine these areas carefully:

a) The keel beam between stations 540 and 727

b) The area above the wing aft of station 639

c) The bottom fuselage between stations 727 and 927

d) The fuselage between station 927 and 1088.

2) If you find external damage, examine all of the internal structure in the damaged area.

(b) Examine all of the external and internal structure of the fuselage, section 48 that you can get access to and/or is called out in this procedure for signs of buckling, deformation or damage.

NOTE: Look at the structure from the rear pressure bulkhead to the aft end of the airplane.

1) Examine these areas carefully:

a) Do a detailed visual inspection of the exterior fuselage skin 10 in. (25.4 cm) forward and aft of the Body Station (BSTA) 1016 skin butt joint between stringers S-10L and S-10R.

b) Do a detailed visual inspection of the entire aft side of the BSTA 1016 aft pressure bulkhead. Give particular attention to the upper bulkhead structure between skin stringers S-10L and S-10R.

NOTE: Access to the interior unpressurized aft pressure bulkhead area is through the aft access hatch, 311BL.

c) If you find damage in the previous inspection steps of the fuselage - section 48 - of the aft pressure bulkhead around BSTA 1016, and do a detailed visual inspection of the upper center web installation of the BSTA 1016 bulkhead for damage.

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<1> Remove the galley and lavatories as necessary to gain access to the upper center web installation.

NOTE: The upper center web installation of the BSTA 1016 bulkhead on the forward side of the pressure bulkhead.

d) Examine the exterior fuselage skin at BSTA 1016.

NOTE: Gain access to the exterior fuselage skin at BSTA 1016 behind the vertical stabilizer fairing (panels 323AL and 323AR).

2) Look at the jackscrew and hinges for signs of binding as follows:

a) Examine the stabilizer jackscrew-mechanism mount fittings and support structure.

(c) Examine the horizontal stabilizer external surfaces for signs of buckling as follows:

1) Make sure that you look at the skin splices.

2) If external damage to the horizontal stabilizer is found, do the steps that follow:

a) Examine the horizontal stabilizer rear spar.

b) Examine the horizontal stabilizer rear spar webs.

c) Examine the terminal fittings for the front and rear spar of the horizontal stabilizer.

d) Examine all the internal primary structure in the damaged area that has access.

(d) Examine the elevator external surfaces.

1) If you find external damage to the elevator, do these steps:

a) Examine the front spar web.

b) Examine the elevator actuator bearings for signs of binding.

c) Examine the horizontal stabilizer rear spar webs.

d) Move the elevator slowly by hand:

<1> Make sure that the elevator moves smoothly without binding.

<2> Make sure that there is no damage to the elevator hinge bearings.

NOTE: You can feel a slight vibration or racheting noise from the hinge bearings if there is damage.

(e) Examine the fin external surfaces for signs of buckling.

1) If you find external damage to the fin, do the steps that follow:

a) Examine the fin attach fittings.

b) Examine the fin rear spar web.

c) Examine all of the internal primary structure in the damaged area that has access.

(f) Examine the rudder for signs of buckling.

1) If you find external damage to the rudder, examine all of the internal primary structure in the damaged area that has access.

(g) Examine the external surfaces around the top and bottom wing-to-body attachment; this includes the wing-to-body fairing.

1) If you see external damage; do the steps that follow:

a) Examine the body-to-wing joints.

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b) Examine landing gear beam-to-body joints.

c) Examine the upper-wing skin splice.

(h) Examine the external surfaces of the wing at the skin splices.

1) Examine the external surface of the top of the wing trailing edge.

2) If there is external damage, do this step:

a) Examine all of the internal primary structure in the damaged area that has access.

(i) Examine the winglets.

(j) Examine the wing control surfaces as follows:

1) If you find external damage, do these steps:

a) Examine the spars.

b) Examine all of the internal primary structure in the damaged area that has access.

c) Examine the control surface attachments at the front and rear spars.

SUBTASK 05-51-04-720-001

(6) Do these steps to make sure that the flight controls move freely:

(a) Move the given systems through their usual range of movement:

- Flaps
- Slats
- Spoilers (includes Speedbrake)
- Ailerons
- Elevators
- Stabilizer
- Rudder.

(b) Make sure that there is full travel and the flight controls move freely.

NOTE: If the flight controls operate correctly, no more inspections are necessary.

(c) If unusual conditions are found for a given system, do these steps:

1) Do a check of all the flight control force specifications for that system.

2) Do a check of the cable tensions.

3) Do the tasks for the given system:

a) If it is necessary, do these tasks for the aileron: AILERON - ADJUSTMENT/TEST, PAGEBLOCK 27-11-00/501.

b) If it is necessary, do these tasks for the rudder: RUDDER AND RUDDER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-21-00/501.

c) If it is necessary, do these tasks for the elevators: ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-31-00/501.

d) If it is necessary, do these tasks for the horizontal stabilizer: HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-41-00/501.

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e) If it is necessary, do these tasks for the trailing edge flaps: TRAILING EDGE FLAP SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-51-00/501.

f) If it is necessary, do these tasks for the spoilers: SPOILER CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-61-00/501.

g) If it is necessary, do these tasks for the speedbrakes: SPEED BRAKE CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-62-00/501.

h) If it is necessary, do these tasks for the leading edge devices: LEADING EDGE FLAP AND SLAT - ADJUSTMENT/TEST, PAGEBLOCK 27-81-00/501.

SUBTASK 05-51-04-210-009

(7) Examine the engine strut panels, doors, and lower surface of the nacelle cowling.

(a) If you find unusual conditions, do this task: Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection, TASK 05-51-10-210-801.

SUBTASK 05-51-04-210-010

(8) Examine the areas that follow for signs of fuel leaks or other types of fluid leaks:

(a) The wing

(b) The engine nacelles

(c) The fuselage external surfaces

(d) All landing gear wheel wells.

SUBTASK 05-51-04-210-011

(9) If flight load accelerations are more than flight manual limits, and the airplane showed signs of much damage after the most recent severe turbulence event, check the airplane alignment.

NOTE: The alignment check procedure is in the Structural Repair Manual (SRM).

SUBTASK 05-51-04-210-012

(10) Examine the landing gear doors as follows:

(a) If the landing gear position lights were on during the turbulent part of the flight, do a check of the landing gear doors and uplocks.

SUBTASK 05-51-04-210-013

(11) Examine the inspection and blowout doors on the bottom surface of the wing and engine nacelles.

SUBTASK 05-51-04-210-018

(12) Examine the inspection and access doors on the bottom of the fuselage.

I. Stall (After Initial Buffet or Stick Shaker) Structural Inspection

SUBTASK 05-51-04-210-014

(1) Do the inspections that follow to find possible damage.

(a) Examine all of the internal structure of the fuselage, section 48 that has access as follows:

1) Look at the structure from the rear pressure bulkhead to the aft end of the airplane.

2) Look carefully at the areas that follow:

a) The aft fuselage bulkheads

b) The horizontal stabilizer center sections

c) The attach fittings for the horizontal stabilizer (Station 1088 and 1156)

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d) The terminal fittings for the front and rear spar of the horizontal stabilizer

e) The stabilizer hinge fittings

f) The stabilizer jackscrew-mechanism mount fittings and support structure.

3) Make sure that the stabilizer jackscrew and hinges do not bind.

SUBTASK 05-51-04-210-015

(2) Examine the horizontal stabilizer external surfaces as follows:

(a) Look for signs of buckling.

(b) Look at the skin splices for cracks.

(c) If you find external damage to the horizontal stabilizer, do these steps:

1) Examine the spars.

2) Examine all of the internal primary structure in the damaged area that has access.

SUBTASK 05-51-04-210-016

(3) Examine the elevator external surfaces.

(a) If you find external damage to the elevator, do these steps:

1) Examine the front spar web.

2) Examine the elevator actuator bearings for signs of binding.

3) Examine the elevator hinge bearing for signs of binding.

SUBTASK 05-51-04-720-002

(4) Do these steps to make sure that the flight controls move freely:

(a) Move the given systems through their usual range of movement:

- Flaps
- Slats
- Spoilers (includes Speedbrake)
- Ailerons
- Elevators
- Stabilizer
- Rudder.

(b) Make sure that there is full travel and the flight controls move freely.

NOTE: If the flight controls operate correctly, no more inspections are necessary.

(c) If unusual conditions are found for a given system, do these steps:

1) Do a check of all the flight control force specifications for that system.

2) Do a check of the cable tensions.

3) Do the tasks for the given system:

a) If it is necessary, do these tasks for the aileron: AILERON - ADJUSTMENT/TEST, PAGEBLOCK 27-11-00/501.

b) If it is necessary, do these tasks for the rudder: RUDDER AND RUDDER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-21-00/501.

c) If it is necessary, do these tasks for the elevators: ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-31-00/501.

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d) If it is necessary, do these tasks for the horizontal stabilizer: HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-41-00/501.

e) If it is necessary, do these tasks for the trailing edge flaps: TRAILING EDGE FLAP SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-51-00/501.

f) If it is necessary, do these tasks for the spoilers: SPOILER CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-61-00/501.

g) If it is necessary, do these tasks for the speedbrakes: SPEED BRAKE CONTROL SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 27-62-00/501.

h) If it is necessary, do these tasks for the leading edge devices: LEADING EDGE FLAP AND SLAT - ADJUSTMENT/TEST, PAGEBLOCK 27-81-00/501.

J. Cabin Inspection (After Severe, Unusual Turbulence, or Excessive Maneuver)

SUBTASK 05-51-04-210-017

(1) Do the inspections that follow to find possible damage:

(a) Do a visual inspection of all ceiling panels for dislodging and evidence do to impact damage.

(b) Do a visual inspection of all ceiling overhead bins for evidence of looseness and impact damage.

(c) Do a detailed visual inspection of all ceiling panels equipped with video monitors.

1) Inspect latches, stops, potting inserts and all fasteners.

(d) Continue the following inspection if you find cabin damage in the above cabin inspection including the ceiling panels, central overhead bins, or the video monitors.

1) If you do not find damage, further inspection is not necessary.

2) If you find ceiling panels without video monitors dislodged or damaged, do these steps:

a) Do a check for proper latching and installation.

b) Repair as necessary.

3) If you find impact damage on the central overhead stowage bins or they are loose, or if ceiling panels equipped with video monitors are found with broken latches, repair as necessary.

4) Inspect all tie rods of affected zones for evidence of buckling or rupture, replace as necessary.

5) Examine lavatory tie rod attachments for damage consisting of breakage, cracks, and deformation.

6) Examine the lavatory tie rods for evidence of buckling or rupture.

7) Examine the lavatory floor fittings for damage consisting of breakage, cracks, and deformation.

8) Examine galley tie rod attachments for damage consisting of breakage, cracks, and deformation.

9) Examine the galley tie rods for evidence of buckling or rupture.

10) Examine the galley floor fittings for damage consisting of breakage, cracks, and deformation.

11) Examine, closet tie rod attachments for damage consisting of breakage, cracks, and deformation.

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K. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-04-410-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-04-080-001

(2) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 05-51-04-440-001

(3) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-04-440-002

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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HIGH ENERGY STOP/HEAT DAMAGE CONDITION - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Inspect the landing gear wheels, brakes and axles after a high energy stop condition

(2) Heat Damage Inspection

(3) Tire Removal After Overspeed Landing.

TASK 05-51-07-210-801

2. High Energy Stop

A. General

(1) The level of the inspection depends on the level of absorbed kinetic energy.

(2) The heat absorbed by the airplane brake is slowly but not fully transmitted to the wheel, tire, axle, and the in-axle components.

(3) The wheels have fuse plugs which are designed to melt and release the tire pressure.

NOTE: The fuse plugs are design to melt at a specific temperature.

(a) This is to stop ruptures of the tires and wheels when the wheel temperature gets too high.

(4) If a brake fire occurs, refer to the section in this task titled, “High Energy Stop Condition Inspection - IN FUSE PLUG MELT range or Brake Fire.

(5) The brake cooling time starts once the airplane is stationary and no further brake applications are performed.

(6) Use the brake energy chart to approximate energy absorbed by a brake unit (Figure 201 or Figure 202).

(a) This chart will give only the approximate energy absorbed during the stop.

1) The factors that follow will also affect the energy absorbed:

a) The residual energy from the previous stop.

b) The runway slope.

c) The wind conditions.

d) The thrust reverser use.

(b) Listed below are example values to show you how to use the Brake Energy Chart to obtain result values (Figure 201 or Figure 202):

NOTE: The brake temperature monitoring system (BTMS) indications are also shown in the bottom of the figure. If brake cooling is determined from the BTMS, use the hottest brake indication 10 to 15 minutes after the airplane has come to a complete stop or in flight with gear retracted to determine recommended cooling schedule.

NOTE: The data values given below are for example only and must not be used to determine a specific airplane absorbed energy by a brake unit. The airplane operators must use actual values generated by the specific airplane high energy stop/heat event.

1) Example input values:

a) Gross Weight - 135,000 lb (61,235 kg)

b) Brakes on Speed - 120 knots (no wind)

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c) Pressure Altitude - 5000 ft (2 km)

d) Outside Air Temperature - 80°F (27°C)

e) Landing stop using maximum manual braking

f) No thrust reversers.

2) Example result values:

a) Resultant Brake Energy - 26,000,000 ft-lb (35,251,268 N·m)

b) Resultant is in the COOLING RECOMMENDED range.

(7) A high energy condition can be a refused takeoff.

(8) A high energy condition can also be any stop or sequence of stops that collect energy in the CAUTION or FUSE PLUG MELT range (Figure 201 or Figure 202).

(9) An indication that a high energy stop, or equivalent, has occurred is the release of a wheel fuse plug.

NOTE: If fuse plug releases do occur, the condition is a high energy stop within the fuse plug melt range.

(a) Energy levels are different between brakes after stops in the COOLING RECOMMENDED range.

(b) One or more brakes could have absorbed energy in the CAUTION range, and one or more wheel fuse plugs released.

(10) For main landing gear tires that have been subjected to unusual service events, such as high energy rejected takeoffs, high energy stops, or high energy overspeed landings where thermal fuse plugs have melted, these tires should be removed and discarded. If the main landing gear tires on the airplane remain inflated, remove the wheel-tire assembly and send the hardware to your local wheel shop for the inspection.

(a) Make sure to identify all tires of the unusual service event when returning the tires to the supplier.

(11) If the nose and main landing gear tires exceed their respective tire speed rating during the overspeed landing, remove the wheel-tire assembly and send the hardware to your local wheel shop for inspection.

(a) Make sure to identify all tires of the unusual service event when returning the tires to the supplier.

(12) If no high energy condition was determined, put the airplane back to its usual condition.

B. References

Reference	Title
05-51-95-100-801	Cleaning After Dry Chemical Fire Extinguisher Used (P/B 201)
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-700-802	Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-41-41-700-803	Main Landing Gear Brake Check (Wheel Removed from the Airplane) (P/B 601)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)

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(Continued)

Reference	Title
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. High Energy Stop Conditional Inspection - In CAUTION range

SUBTASK 05-51-07-210-001

(1) Do the steps that follow for the In caution range inspection:

(a) Move the airplane from the runway.

1) Do not use the brakes very much when you move the airplane.

2) Do not set the parking brake.

WARNING DO NOT GO NEAR THE MAIN LANDING GEAR FOR 1 HOUR AFTER THE AIRPLANE MAKES A HIGH-ENERGY STOP. INJURIES TO PERSONNEL CAN OCCUR.

(b) Let the brakes, tires, and wheels cool so you can touch them.

(c) Examine the tires (TASK 32-45-00-700-803).

NOTE: Low tire pressure compared to the other tires can indicate a melted fuse plug.

1) If a fuse plug has melted, do the High Energy Stop Conditional Inspection - In FUSE PLUG MELT range or Brake Fire.

(d) Examine the wheels (TASK 32-45-00-700-801).

(e) Examine the brakes (TASK 32-41-41-700-802).

(f) If the high energy condition was a refused takeoff, do this step:

1) Examine the thrust reverser blocker door cover assembly for cracking.

E. High Energy Stop Conditional Inspection - In FUSE PLUG MELT range or Brake Fire

SUBTASK 05-51-07-210-002

(1) Do the steps that follow for the In FUSE PLUG MELT range inspection:

(a) Move the airplane from the runway immediately because the tires will possibly deflate.

(b) Leave the parking brake off.

NOTE: Do not set the parking brake.

(c) Do one of these steps:

NOTE: Do not let the airplane move.

1) Keep the tow tug connected to the airplane.

2) Put the chocks on the nose landing gear tires (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

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WARNING DO NOT GO NEAR THE MAIN LANDING GEAR FOR 1 HOUR AFTER THE AIRPLANE MAKES A HIGH-ENERGY STOP. INJURIES TO PERSONNEL CAN OCCUR.

WARNING DO NOT APPLY EXTINGUISHER OR COOLANT DIRECTLY ON THE INFLATED TIRE OR THE WHEEL. AN EXPLOSION CAN BE CAUSED AND INJURY TO PERSONS CAN OCCUR.

(d) Do one of these steps to decrease the brake temperature:

NOTE: If a chemical agent is used to extinguish a brake-area fire, thoroughly rinse the extinguishing agent from the brakes and surrounding components once they have cooled. Use large amounts of low-pressure, clean water to rinse (Cleaning After Dry Chemical Fire Extinguisher Used, TASK 05-51-95-100-801).

NOTE: A different source of cooling can be an air conditioning cart or truck.

1) After 1 hour, use water mist or fog on the wheel or tire to decrease the temperature.
2) Wait 2 to 3 hours for the brakes, wheels and tires to cool so that they can be touched.

(e) Examine the tires on the axles where all tires are inflated (TASK 32-45-00-700-803).

(f) Examine the wheels on the axles where all tires are inflated (TASK 32-45-00-700-801).

(g) Examine the brakes on the axles where all tires are inflated (TASK 32-41-41-700-802).

(h) Remove all main gear tires and wheels on the axles where at least one tire was deflated (TASK 32-45-11-000-801):

NOTE: Do this step after the landing gear is cool and safe to go near.

(i) Discard the tires from the wheels with melted fuse plugs.

(j) For wheels with good fuse plugs do these steps:

1) Remove the wheel-tire assembly from service.

NOTE: Your preferred wheel shop or wheel repair facility will disassemble and examine the hardware to make sure it is serviceable.

2) Tag the wheel-tire assembly with the reason for removal.

(k) On the axles where at least one tire was deflated, look for seized or damaged brakes, do this task: Main Landing Gear Brake Check (Wheel Removed from the Airplane), TASK 32-41-41-700-803.

(l) Remove the antiskid wheel speed transducer on the axles where at least one tire was deflated, do this task: Antiskid Wheel Speed Transducer Removal, TASK 32-42-11-000-801.

(m) Remove damaged brakes from the axles (TASK 32-41-41-000-801).

(n) If a dry chemical fire extinguisher was used to extinguish a landing gear brake fire, do these steps:

1) Visually check for chemical fire extinguisher residue.
2) Remove the carbon brake assemblies that have chemical fire extinguisher.
3) Follow the specified and approved instructions for the disposition of the carbon brake assemblies.

(o) Examine the main landing gear axle as follows:

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1) Examine the Ti-Cad (titanium-cadmium) plating for heat damage.

NOTE: Heat damage can cause the steel substrate to become brittle. This condition is not always found by non-destructive inspection procedures.

NOTE: If you find an inorganic protective coating, the Ti-Cad plating could have been removed. The coating could have been applied during a previous inspection/repair.

a) Look for a white oxide material and blistered or melted plating.

b) If the plating shows signs of heat damage, remove the components.

c) Do this task: Heat Damage Inspection (Ammonium Persulfate Solution), TASK 05-51-07-280-801.

2) Examine the hydraulic fluid-resistant paint on the axles and the lower end of the shock strut inner cylinder as follows:

a) Look for a brown shade color that was caused by an overheat condition.

3) If the above two steps show that the coating has blistered and the parts have been overheated, do these steps:

a) Remove the parts for a hardness test.

b) Send the parts to the shop for approved repairs.

4) If the inspection in the first step above shows that the coating has melted, do this step:

a) Use the fluorescent magnetic particle inspection techniques to examine for evidence of embrittlement or cracking.

NOTE: Additional hand polishing and visual inspection techniques can be used if it is necessary.

WARNING MAKE SURE THAT THE POWER CABLE OF THE BORESCOPE IS IN GOOD CONDITION. IF THERE ARE CIRCUMFERENTIAL CUTS, FRAYED AREAS, OR RUPTURES TO THE EXTERNAL RUBBER COVER OF THE CABLE, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(p) Examine the paint (green primer or light gray enamel) on the inner surfaces (bore) of the axle with a borescope as follows:

NOTE: Use the borescope manufacturer instructions and look to a minimum depth of 16 inches (41 centimeters).

1) Look for discoloration or blistering.

NOTE: Green paint will change to a light brown or black color. Light gray paint will change to a yellow color.

2) If the plating shows signs of heat damage, do this task: Heat Damage Inspection (Ammonium Persulfate Solution), TASK 05-51-07-280-801.

3) If the paint shows only a minimum discoloration, you can wait for the Heat Damage Inspection (Ammonium Persulfate Solution).

a) When the airplane gets back to the primary base, do this task: Heat Damage Inspection (Ammonium Persulfate Solution), TASK 05-51-07-280-801.

NOTE: Do not land the airplane more than three times before you do this inspection.

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(q) If the high energy condition was due to a refused takeoff do this step:

1) Examine the thrust reverser blocker door cover assembly for cracking.

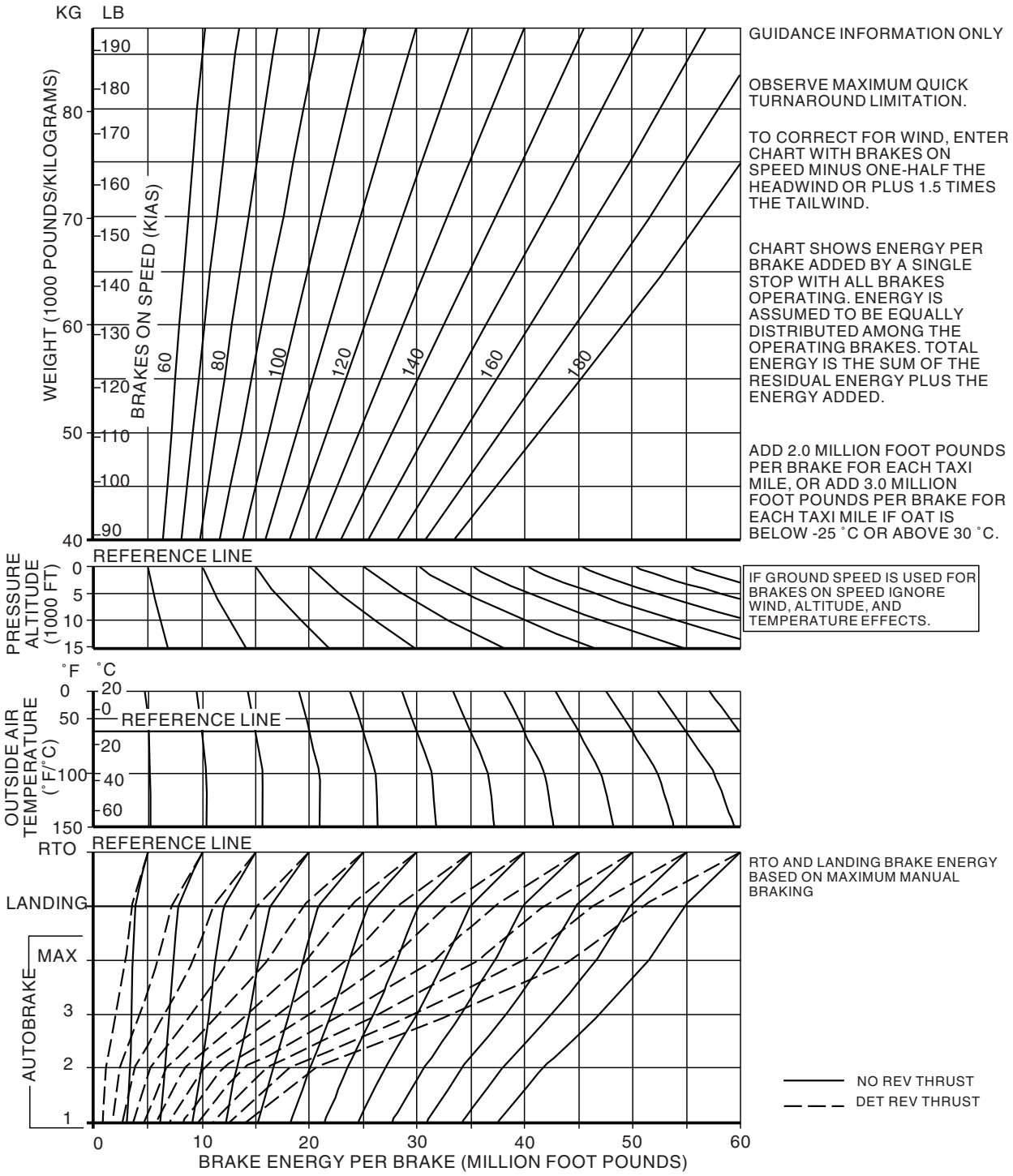
F. Put the Airplane Back to Its Usual Condition.

SUBTASK 05-51-07-420-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

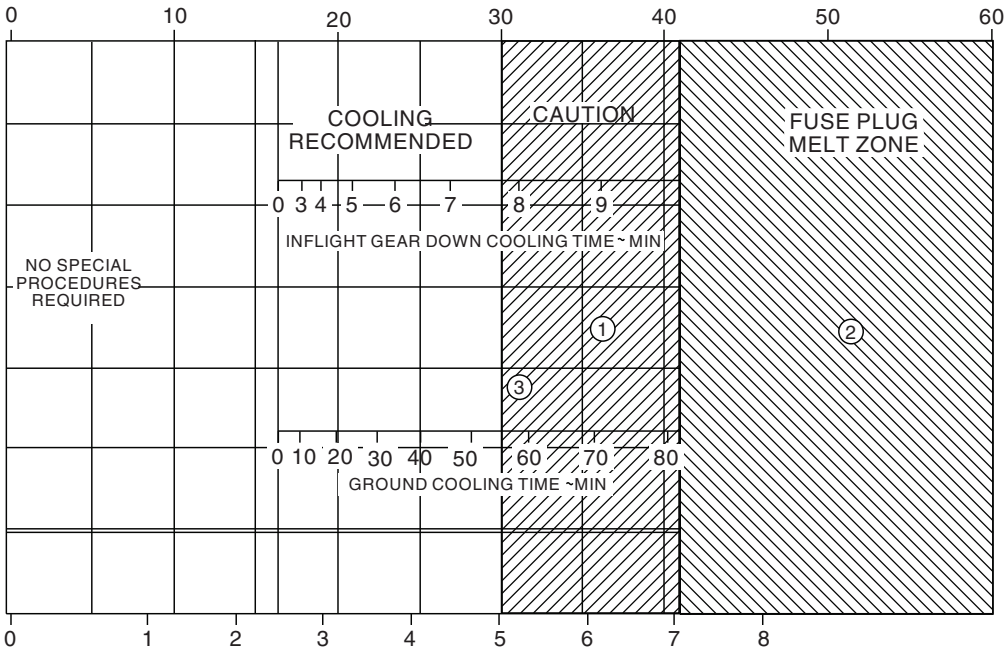
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Carbon Brake Cooling Schedule
Figure 201/05-51-07-990-805 (Sheet 1 of 2)

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OBSERVE MAXIMUM QUICK TURNAROUND LIMITATION

① WHEEL FUSE PLUGS MAY MELT. DELAY TAKEOFF AND INSPECT AFTER ONE HOUR. IF OVERHEAT OCCURS AFTER TAKEOFF, EXTEND GEAR SOON FOR AT LEAST 7 MINUTES.

② CLEAR RUNWAY IMMEDIATELY. UNLESS REQUIRED, DO NOT SET PARKING BRAKE. DO NOT APPROACH GEAR OR ATTEMPT TAXI FOR ONE HOUR. TIRE WHEEL AND BRAKE REPLACEMENT MAY BE REQUIRED. IF OVERHEAT OCCURS AFTER TAKEOFF, EXTEND GEAR SOON FOR AT LEAST 12 MINUTES.

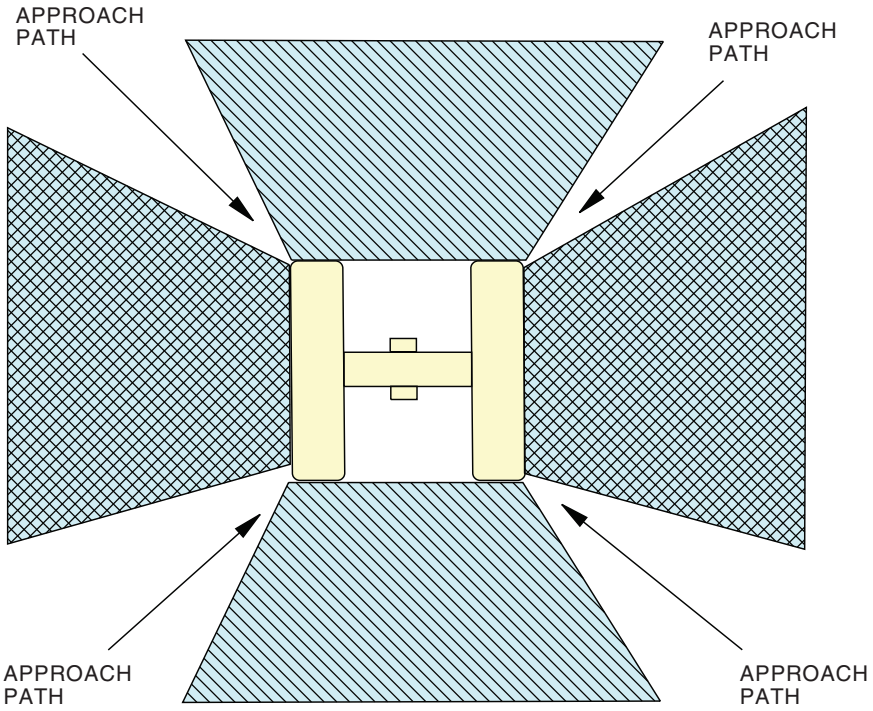
BRAKE TEMPERATURE MONITORING SYSTEM INDICATION MAY BE USED 10 TO 15 MINUTES AFTER AIRPLANE HAS COME TO A COMPLETE STOP OR INFLIGHT WITH GEAR RETRACTED TO DETERMINE RECOMMENDED COOLING SCHEDULE.

③ FOR AIRPLANES WITH TBMS (COMBINED TIRE PRESSURE AND BRAKE TEMPERATURE MONITORING SYSTEM), THE BEGINNING OF THE CAUTION RANGE CAN VARY FROM 5.0 TO 6.2 DEPENDING ON WHEEL TEMPERATURE.

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Carbon Brake Cooling Schedule
Figure 201/05-51-07-990-805 (Sheet 2 of 2)

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TIRE HAZARD AREA
RIM HAZARD AREA

NOTE:

DO NOT APPROACH MLG FOR INSPECTION OR INSTALLATION OF GEAR PINS UNTIL CLEARED

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Hot Brake and Tire Danger Areas
Figure 202/05-51-07-990-806

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TASK 05-51-07-280-801

3. Heat Damage Inspection (Ammonium Persulfate Solution)

A. General

(1) This inspection is used to find if the high strength steel that showed visible signs of heat damage, had changed.

(2) This inspection will examine for a change in the temper or heat treat properties.

(3) This inspection will usually occur on the components of the landing gear axles.

(4) The signs of discolored or blistered paint or plating shows that a general intense heat has changed the temper of the axle. When this happens, the ammonium persulfate procedure should be applied to the damaged part of the axle.

B. References

Reference	Title
32-11-85-000-801	Main Landing Gear Axle Removal (P/B 401)
32-11-85-420-801	Main Landing Gear Axle Installation (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-42-21-400-801	Antiskid/Autobrake Control Unit Installation (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)
51-21-21-370-801	Prepare the Surface to be Painted (P/B 701)
SOPM 20-10-02	Repair of Plated Steel Parts Heat Treated 200 to 305 KPSI

C. Consumable Materials

Reference	Description	Specification
B00081	Solvent - Trichloroethylene	ASTM D4080
B00102	Abrasive - Aluminum Oxide Coated Cloth	
B50262	Alcohol - Isopropyl, grade A (water content 0.1 percent max.)	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02412	Compound - Ammonium Persulfate	

D. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

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E. Heat Damage Inspection (Ammonium Persulfate Solution)

SUBTASK 05-51-07-120-001

WARNING

REFER TO THE APPLICABLE MANUFACTURER'S OR SUPPLIERS'S MSDS FOR:

- MORE PRECAUTIONARY DATA
- APPROVED SAFETY EQUIPMENT
- EMERGENCY MEDICAL AID.

TALK WITH THE LOCAL SAFETY DEPARTMENT OR AUTHORITIES FOR THE PROCEDURES TO DISCARD THESE HAZARDOUS AGENTS.

WARNING

THE ACTIVE INGREDIENTS OF ETCHANTS ARE POISONOUS AND CORROSIVE. MAKE SURE ALL PERSONS OBEY ALL OF THE PRECAUTIONS WHEN ETCHANTS ARE USED. - DO NOT USE IN AREAS WHERE THERE IS HIGH HEAT, SPARKS, OR FLAMES. - USE IN AN AREA OPEN TO THE AIR. - CLOSE THE CONTAINER WHEN NOT USED. - DO NOT GET ETCHANTS IN THE EYES, ON THE SKIN, OR ON YOUR CLOTHES. - DO NOT BREATHE THE GAS. - TALK WITH LOCAL SAFETY DEPARTMENT OR AUTHORITIES FOR: MORE PRECAUTIONARY DATA, APPROVED SAFETY EQUIPMENT, EMERGENCY MEDICAL AID, AND THE PROCEDURES TO DISCARD THESE HAZARDOUS AGENTS.

(1) Apply the solution, ammonium persulfate compound, G02412, as follows:

NOTE: 0.22 lb (100 g) of ammonium persulfate per 0.27 gal (1.02 l) of tap water will make the necessary solution.

(a) Remove the paint from the heat damage areas.

(b) Strip the cadmium surface with sandpaper, abrasive cloth, B00102, by hand.

(c) Strip a minimum area of 0.5 in. (12.7 mm) larger than the area that is to be tested.

(d) Clean the area to be etched, with an applicable solvent like solvent, B00081, or alcohol, B50262.

(e) Make sure that all the cadmium plating is removed by using the cadmium spot check solutions as follows:

1) Put a drop of Cadmium Spot Check Solution 1 (SOPM 20-10-02) on the test spot and let it stay for approximately 30 seconds.

2) Soak up the liquid drop onto a piece of filter paper.

3) Add a drop of Cadmium Spot Check Solution 2 (SOPM 20-10-02). If this becomes yellow, there is remaining cadmium on the surface of the part.

4) Repeat sanding plus solvent cleaning until the plating is completely removed.

(f) Apply the solution, ammonium persulfate compound, G02412, with a cheesecloth, cotton wiper, G00034, on the surface to be tested for 30 to 60 seconds.

(g) Rinse the surface with hot or cold water immediately or alcohol, B50262.

(h) Dry the surface with clean, dry compressed air, immediately.

SUBTASK 05-51-07-210-003

(2) Examine the etched areas as follows:

(a) Examine the parts with a bright light and without magnification for signs of heat damage.

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(b) If all of the etched area is the same shade of gray, there is no change to the temper of the steel.

(c) If the area has been too hot it will be a lighter or darker color than the normal color.

NOTE: Darker color shows over-tempered martensite.

NOTE: Darker color in a lighter circle shows untempered martensite surrounded by overtempered material.

(d) If the etched area is mottled (has spots or streaks of a different color), replace the axle (TASK 32-11-85-000-801 and TASK 32-11-85-420-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-07-120-002

(1) Remove the etch with applicable abrasive paper, abrasive cloth, B00102.

NOTE: Do this step if the part is serviceable.

NOTE: If it is not serviceable, install replacement parts.

SUBTASK 05-51-07-120-003

(2) Stylus the cadmium plate areas where the plating was removed.

SUBTASK 05-51-07-370-001

(3) Refinish the surface with the correct primer and paint (TASK 51-21-21-370-801).

SUBTASK 05-51-07-400-001

(4) Install the axle if it is applicable (TASK 32-11-85-420-801).

SUBTASK 05-51-07-400-002

(5) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

SUBTASK 05-51-07-400-003

(6) Do this task: Antiskid/Autobrake Control Unit Installation, TASK 32-42-21-400-801.

SUBTASK 05-51-07-400-004

(7) Do this task: Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801.

——— END OF TASK ———

TASK 05-51-07-000-801

4. Tire Removal After Overspeed Landing

A. General

(1) If the tires have been subjected to an overspeed landing above the rated tire speed, all tires on the airplane should be removed and discarded.

B. References

Reference	Title
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Tire Removal After Overspeed Landing

SUBTASK 05-51-07-020-001

(1) If the tires have been subjected to an overspeed landing above the rated tire speed, do these steps:

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(a) Do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

(b) Do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

(c) Discard all tires after removing them from the airplane.

——— END OF TASK ———

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FLAP/SLAT DOWN OVERSPEED OR ABOVE 20,000 FEET CONDITION - MAINTENANCE PRACTICES

1. General

A. Introduction

(1) If the trailing edge flaps, leading edge Krueger flaps, or leading edge slats (high lift devices) are extended at speeds more than permitted, or extended at altitudes above 20,000 ft (6.10 km), then inspect the affected components and structure with the criteria in this procedure.

(2) The conditional inspections are divided into two phases

(a) Phase I Inspection Phase I Inspection, TASK 05-51-08-210-801.

(b) Phase II Inspection Phase II Inspection, TASK 05-51-08-210-802.

(3) Load relief from the flap load relief system does not eliminate the need to do the inspections.

B. Determine Inspections (Figure 201 Condition Logic, Figure 202 Trailing Edge, and Figure 203 Leading Edge)

NOTE: The inspection logic is also shown on the flowcharts in the referenced figures.

(1) If the trailing edge flaps, leading edge Krueger flaps, or leading edge slats are extended at altitudes above 20,000 ft (6.10 km), use the following criteria:

(a) If speeds are within the permitted speeds, then do the Phase I Inspection on all high lift devices within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.

(b) If speeds are more than permitted, then contact Boeing for inspection requirements.

(c) Continue to the inspection requirements in Section E. Inspections.

(2) If the trailing edge flaps, leading edge Krueger flaps, or leading edge slats are extended at altitudes below 20,000 ft (6.10 km) at speeds more than permitted, refer to Section C. Trailing Edge and Section D. Leading Edge to determine inspection requirements.

NOTE: The inspection thresholds for the trailing edge flaps and leading edge devices are not the same and should be assessed independently.

(a) Look at Table 201 for a summary of the trailing edge flaps and leading edge device procedures.

(3) If Flight Data Recorder (FDR) data (or equivalent) is used to assess overspeed events, use the following guidelines. If data is not available, then use flight crew notes to complete the assessment.

(a) If the flaps are stable in a flap setting at the time of the suspected overspeed (not in transition), then it is acceptable to assess the overspeed using the detent position.

NOTE: Due to flap position indication system tolerances, the actual flap position as reported by the airplane data systems may be slightly higher or lower than the detent position.

(b) If flaps are in transition (moving between two flap settings) when the overspeed is suspected, use the following guidelines:

1) Use the actual flap position as reported by the airplane data systems to assess the overspeed.

2) Use the more restrictive (lower) speed from the next higher flap setting to determine inspection requirements.

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Table 201

Affected Devices Flight Hour/Cycle Limits for Inspections	
Condition	Task
The overspeed is greater than the Phase II speed limit in Table 202 (Trailing Edge) and/or Table 204 (Leading Edge).	Do the Phase I and Phase II Inspections on the affected devices before the next flight.
Vertical acceleration is within 0.7 to 1.3 G and the overspeed is less than or equal to 5 knots above the Phase I limits in Table 202 (Trailing Edge) and/or Table 204 (Leading Edge).	Do the Phase I Inspection on the affected devices within 100 flight hours of the overspeed event or within 25 flight cycles of the discovery of the overspeed event, whichever occurs later.
Vertical acceleration is within 0.7 to 1.3 G and the overspeed is greater than 5 knots above the Phase I limits and below the Phase II limits in Table 202 (Trailing Edge) and/or Table 204 (Leading Edge).	Do the Phase I Inspection on the affected devices within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.
Vertical acceleration is unknown, or outside 0.7 to 1.3 G but within 0.5 to 1.6 G, and the overspeed is above the Phase I limits and less than or equal to the Data Review Deferral limits in Table 203 (Trailing Edge) and/or Table 205 (Leading Edge).	Do the Phase I Inspection on the affected devices within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later. If vertical acceleration is unknown, then review FDR data or equivalent within 5 flight cycles of the overspeed event to determine vertical acceleration during the overspeed. • If the vertical acceleration during the overspeed was outside 0.5 to 1.6 G, contact Boeing for inspection requirements. • If the vertical acceleration data cannot be obtained, it is acceptable to do the Phase I and Phase II Inspections on the affected devices before the next flight.
Vertical acceleration is unknown, or outside 0.7 to 1.3 G but within 0.5 to 1.6 G, and the overspeed is above the Data Review Deferral limits in Table 203 (Trailing Edge) and/or Table 205 (Leading Edge).	Contact Boeing for inspection requirements or do the Phase I and Phase II Inspections on the affected devices before the next flight.
Vertical acceleration is known and outside 0.5 to 1.6 G.	Contact Boeing for inspection requirements or do the Phase I and Phase II Inspections on the affected devices before the next flight.

C. Trailing Edge (Figure 202)

NOTE: This procedure is summarized in Table 201 and the flowchart in the referenced figure.

(1) This procedure determines the trailing edge inspection requirements when the trailing edge flaps are extended at speeds more than permitted at altitudes below 20,000 ft (6.10 km).

(2) If the trailing edge flaps are extended at speeds above the Phase II limits in Table 202, do the Phase I and Phase II Inspections on the trailing edge before the next flight.

(a) Continue to the inspection requirements in Section E. Inspections.

(3) When vertical acceleration (Nz) is within 0.7 to 1.3 G, the following applies:

(a) If speed during the overspeed event is less than or equal to 5 knots above the Phase I limits in Table 202, do the Phase I Inspection on the trailing edge within 100 flight hours of the overspeed event or within 25 flight cycles of the discovery of the overspeed event, whichever occurs later.

(b) If speed during the overspeed event is greater than 5 knots above the Phase I limits and below the Phase II limits in Table 202, do the Phase I Inspection on the trailing edge within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.

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(c) Continue to the inspection requirements in Section E. Inspections.

(4) When vertical acceleration (Nz) is unknown, or outside 0.7 to 1.3 G but within 0.5 to 1.6 G, the following applies:

(a) If speed during the overspeed is above the Data Review Deferral limits in Table 203, contact Boeing for inspection requirements or do the Phase I and Phase II Inspections on the trailing edge before the next flight.

NOTE: When contacting Boeing for inspection requirements, having FDR data or equivalent can decrease response time. If the vertical acceleration during the overspeed is found to be within 0.7 to 1.3 G, refer to applicable inspection requirements.

I (b) If speed during the overspeed event is above the Phase I limits and less than or equal to the Data Review Deferral limits in Table 203, do the Phase I Inspection on the trailing edge within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.

(c) If vertical acceleration is unknown, then review the FDR data or equivalent within 5 flight cycles of the overspeed event to determine vertical acceleration during the overspeed.

1) If the vertical acceleration during the overspeed was outside 0.5 to 1.6 G, contact Boeing for inspection requirements.

2) If the vertical acceleration during the overspeed was within 0.5 to 1.6 G, no further action is necessary.

3) If the vertical acceleration data cannot be obtained, it is acceptable to also do the Phase II inspection before the next flight.

(d) Continue to the inspection requirements in Section E. Inspections.

(5) If vertical acceleration is known and outside 0.5 to 1.6 G, contact Boeing for inspection requirements on the trailing edge before the next flight.

NOTE: Phase I and Phase II Inspections can be done in lieu of contacting Boeing. Continue to the inspection requirements in Section E. Inspections.

Table 202

Trailing Edge Flap Inspection Speed Limits for Vertical Acceleration within 0.7 to 1.3 G						
Trailing Edge Flap Position	737–7/-8/-8200		737–9		737–10	
	Phase I Inspection Limit (KIAS)	Phase II Inspection Limit (KIAS)	Phase I Inspection Limit (KIAS)	Phase II Inspection Limit (KIAS)	Phase I Inspection Limit (KIAS)	Phase II Inspection Limit (KIAS)
1	250 *[1]	265 *[2]	250 *[1]	265 *[2]	250 *[1]	265 *[2]
2	250 *[1]	265 *[2]	250 *[1]	265 *[2]	250 *[1]	265 *[2]
5	250 *[1]	265 *[2]	250 *[1]	265 *[2]	250 *[1]	265 *[2]
10	210	231	210	229	210	229
15	200	229	200	227	200	227
25	190	227	195	211	195	211
30	175	191	180	196	180	196
40	166	182	171	190	171	192

*[1] 230 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

*[2] 245 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

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Table 203

Trailing Edge Flap Inspection Speed Limits for Vertical Acceleration unknown or within 0.5 to 1.6 G						
Trailing Edge Flap Position	737–7/-8/-8200		737–9		737–10	
	Phase I Inspection Limit (KIAS)	Data Review Deferral Limit (KIAS)	Phase I Inspection Limit (KIAS)	Data Review Deferral Limit (KIAS)	Phase I Inspection Limit (KIAS)	Data Review Deferral Limit (KIAS)
1	250 *[1]	255 *[2]	250 *[1]	255 *[2]	250 *[1]	255 *[2]
2	250 *[1]	255 *[2]	250 *[1]	255 *[2]	250 *[1]	255 *[2]
5	250 *[1]	255 *[2]	250 *[1]	255 *[2]	250 *[1]	255 *[2]
10	210	221	210	219	210	219
15	200	219	200	217	200	217
25	190	217	195	201	195	201
30	175	181	180	186	180	186
40	166	172	171	180	171	182

*[1] 230 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

*[2] 235 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

D. Leading Edge (Figure 203)

NOTE: This procedure is summarized in Table 201 and the flowchart in referenced figure.

(1) This procedure determines the leading edge inspection requirements when the leading edge Krueger flaps or leading edge slats are extended at speeds more than permitted at altitudes below 20,000 ft (6.10 km).

(2) If the leading edge Krueger flaps or leading edge slats are extended at speeds above the Phase II limits in Table 204, do the Phase I and Phase II Inspections on the leading edge before the next flight

(a) Continue to the inspection requirements in Section E. Inspections.

(3) When vertical acceleration (Nz) is within 0.7 to 1.3 G, the following applies:

(a) If speed during the overspeed event is less than or equal to 5 knots above the Phase I limits in Table 204, do the Phase I Inspection on the leading edge within 100 flight hours of the overspeed event or within 25 flight cycles of the discovery of the overspeed event, whichever occurs later.

(b) If speed during the overspeed event is greater than 5 knots above the Phase I limits and below the Phase II limits in Table 204, do the Phase I Inspection on the leading edge within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.

(c) Continue to the inspection requirements in Section E. Inspections.

(4) When vertical acceleration (Nz) is unknown, or outside 0.7 to 1.3 G but within 0.5 to 1.6 G, the following applies:

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(a) If speed during the overspeed event is greater than the Data Review Deferral limits in Table 205, contact Boeing for inspection requirements or do the Phase I and Phase II Inspections on the leading edge before the next flight.

NOTE: When contacting Boeing for inspection requirements, having FDR data or equivalent can decrease response time. If the vertical acceleration during the overspeed is found to be within 0.7 to 1.3 G, refer to applicable inspection requirements.

(b) If speed during the overspeed event is above the Phase I limits and below the Data Review Deferral limits in Table 205, do the Phase I Inspection on the leading edge within 100 flight hours of the overspeed event or within 5 flight cycles of the discovery of the overspeed event, whichever occurs later.

(c) If vertical acceleration is unknown, then review FDR data or equivalent within 5 flight cycles of the overspeed event to determine vertical acceleration during the overspeed.

1) If the vertical acceleration during the overspeed was outside 0.5 to 1.6 G, contact Boeing for inspection requirements.

2) If the vertical acceleration during the overspeed was within 0.5 to 1.6 G, no further action is necessary.

3) If the vertical acceleration data cannot be obtained, it is acceptable to also do the Phase II inspection before the next flight.

(d) Continue to the inspection requirements in Section E. Inspections.

(5) If vertical acceleration is known and outside 0.5 to 1.6 G, contact Boeing for inspection requirements on the leading edge before the next flight.

NOTE: Phase I and Phase II Inspections can be done in lieu of contacting Boeing. Continue to the inspection requirements in Section E. Inspections.

Table 204

Leading Edge Krueger Flap and Slat Inspection Speed Limits for Vertical Acceleration within 0.7 to 1.3 G		
Trailing Edge Flap Position	Phase I Inspection Limit (KIAS)	Phase II Inspection Limit (KIAS)
1	250 *[1]	265 *[2]
2	250 *[1]	265 *[2]
5	250 *[1]	265 *[2]
10	230	245
15	230	245
25	230	245
30	230	245
40	230	245

*[1] 230 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

*[2] 245 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

Table 205

Leading Edge Krueger Flap and Slat Inspection Speed Limits for Vertical Acceleration unknown or within 0.5 to 1.6 G		
Trailing Edge Flap Position	Phase I Inspection Limit (KIAS)	Data Review Deferral Limit (KIAS)

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Table 205 (Continued)

Leading Edge Krueger Flap and Slat Inspection Speed Limits for Vertical Acceleration unknown or within 0.5 to 1.6 G		
1	250 *[1]	253 *[2]
2	250 *[1]	253 *[2]
5	250 *[1]	253 *[2]
10	230	235
15	230	235
25	230	235
30	230	235
40	230	235

*[1] 230 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

*[2] 233 KIAS when Trailing Edge flaps are extended using the alternate (standby) system.

E. Inspections

(1) If a Phase II inspection is necessary and there are zero findings in the Phase I Inspection, this does not remove the requirement to do the Phase II Inspection.

(2) If overload damage is found during a Phase I Inspection, then do the Phase II Inspection before the next flight.

NOTE: It is only necessary to do the Phase II Inspection if you find the types of damage associated with an overload event as specified in the subsequent step.

(a) If the overload damage found during the Phase I Inspection is only common to the trailing edge flaps, then do the Phase II Inspection for the trailing edge structure only.

(b) If the overload damage found during the Phase I Inspection is only common to the leading edge Krueger flaps or leading edge slats, then do the Phase II Inspection for the leading edge structure only.

(c) If overload damage is found during the Phase I Inspection on one wing, then do the Phase II Inspections on both wings.

(3) When the conditional inspection tells you to examine a component, use the following guidance.

NOTE: All damage must be evaluated to determine what repairs or component replacements are necessary.

NOTE: If any damage found during a Phase I Inspection is related to an overload event, then a Phase II Inspection is necessary. Otherwise a Phase II Inspection is not necessary.

(a) Types of damage related to an overload event may include, but are not limited to, the following items:

1) Cracks.

2) Pulled apart structure.

3) Loose paint or paint flakes.

NOTE: If loose paint or paint flakes are found, the fastener holes should be inspected closely for damage related to an overload event. Loose paint or paint flakes with no other indication of an overload event do not require a Phase II Inspection to be performed.

4) Twisted or Bent parts.

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5) Wrinkles or buckles in the structure.

6) Fastener holes that became larger or longer.

7) Fasteners that have pulled through the skin.

8) Fasteners that are loose, missing, or deformed near critical joints.

NOTE: Critical trailing edge flap joints include the skin connections to carriages, carriage support ribs, and torque tube support ribs.

NOTE: Critical leading edge Krueger flap joints include the connections at the bullnoses and hinges.

NOTE: Critical leading edge slat joints include the skin connections to the actuator ribs, main track ribs, and auxiliary arm ribs.

NOTE: Corrosion around the fastener hole indicates the fastener has been loose or missing for an extended time and does not require a Phase II Inspection to be performed.

NOTE: Fasteners that are loose or missing away from critical joints do not require a Phase II Inspection to be performed.

9) Delaminations to solid laminate and sandwich structure that is not obviously caused by Foreign Object Debris (FOD), environmental conditions, etc.

10) Fiber breakouts.

11) Misalignment that is not due to rigging.

12) Interference or evidence of interference that has not been previously discovered.

13) Other signs of damage caused by overloading the structure.

(b) Types of damage not related to an overload event may include, but are not limited to, the following.

1) Corrosion.

2) Wear (tracks, wedges, seals, paint, etc.).

3) Loose paint or paint flakes with no other signs of an overload event.

4) Fasteners that are loose and have corrosion around the fastener hole.

5) Disbonds to metal-bond structure.

6) Other signs of damage caused by normal operating procedures.

TASK 05-51-08-210-801

2. Phase I Inspection

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

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B. Location Zones

Zone	Area
500	Left Wing
600	Right Wing

C. Examine the Trailing Edge Flaps

SUBTASK 05-51-08-010-001

(1) Extend the trailing edge flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 05-51-08-040-001

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the deactivation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-08-210-001

(3) Examine the trailing edge flap components as follows:

(a) Examine the external skin of all flaps.

(b) Examine the flaps adjacent to the support structure for openings, distortions, or split sealant.

NOTE: Corrosion of the internal structure can be caused by splits in the sealant beads.

D. Examine the Leading Edge Flaps and Slats, if applicable

SUBTASK 05-51-08-010-002

(1) Extend the leading edge flaps and slats, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 05-51-08-040-002

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do the deactivation procedure for the leading edge flap/slat system, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-51-08-210-002

(3) Examine the leading edge flaps and slat components as follows:

(a) Examine all of the flaps and slat skins.

(b) Examine the slats adjacent to main track and auxiliary arm attachments for openings, distortion, or split sealant.

NOTE: Corrosion of the internal structure can be caused by splits in the sealant beads.

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SUBTASK 05-51-08-211-001

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(4) Visually examine the area to make sure that there are no objects in the slat track housing assembly.

E. Put the Airplane Back to its Usual Condition if you will not do the Phase II Inspection

SUBTASK 05-51-08-440-001

(1) Do the activation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-08-440-002

(2) Do the activation procedure for the leading edge flap/slat system, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 05-51-08-410-001

(3) Retract the trailing edge flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 05-51-08-410-002

(4) Retract the leading edge flaps and slats, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

END OF TASK

TASK 05-51-08-210-802

3. Phase II Inspection

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-11 P/B 401	INBOARD TRAILING EDGE FLAP - REMOVAL/INSTALLATION
27-51-15 P/B 401	INBOARD FLAP TRACKS - REMOVAL/INSTALLATION
27-51-21 P/B 401	OUTBOARD TRAILING EDGE FLAP - REMOVAL/INSTALLATION
27-51-25 P/B 401	OUTBOARD FLAP TRACKS - REMOVAL/INSTALLATION
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
27-81-21 P/B 401	LEADING EDGE SLAT - REMOVAL/INSTALLATION
57-44-01-000-801	Outboard Leading Edge Slat Roller Removal (P/B 401)
57-44-01-400-801	Outboard Leading Edge Rollers Installation (P/B 401)

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B. Location Zones

Zone	Area
500	Left Wing
600	Right Wing

C. Trailing Edge Flaps Inspection, if applicable

SUBTASK 05-51-08-010-003

(1) Extend the trailing edge flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 05-51-08-040-003

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the deactivation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-08-210-003

(3) Examine the trailing edge flap components as follows:

(a) The flap tracks

(b) The flap linkages

(c) The track attachment points

(d) The flap track support fittings

(e) The wing in-spar surfaces near flap support fittings

(f) The flap carriages for cracks

1) The forward and aft main roller bolt assemblies including the bearings and mounting bolts (Figure 204, Figure 205).

a) To avoid re-rigging, remove, examine, and re-install the bolts one at a time.

<1> For the Inboard Flap:

<a> Tighten the nut to 160 in-lb (18.1 N·m) - 975 in-lb (110.2 N·m) (View A-A, View B-B, Figure 204).

<2> For the Outboard Flap Inboard Carriage:

<a> Tighten the nut to 240 in-lb (27.1 N·m) - 610 in-lb (68.9 N·m) (View A-A, View B-B, Figure 205).

<3> For the Outboard Flap Outboard Carriage:

<a> Tighten the nut to 235 in-lb (26.6 N·m) - 610 in-lb (68.9 N·m) (View A-A, View B-B, Figure 205).

b) If the cotter pin and nut cannot be removed, remove flaps to easier inspect the carriages and remove the main roller bolt assemblies (PAGEBLOCK 27-51-11/401 and PAGEBLOCK 27-51-21/401).

(g) The actuator support structure

(h) The actuator and drive mechanism.

SUBTASK 05-51-08-010-005

(4) Remove all of the trailing edge flap support-track fuse pins and bolts as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- INBOARD FLAP TRACKS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-15/401
- OUTBOARD FLAP TRACKS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-25/401.

(a) Remove the fuse pin or bolt one at a time.

1) Examine the fuse pins or bolts for cracks or distortion.

2) If it is necessary, replace removed fuse pins or bolts with new fuse pins or bolts.

(b) Make sure that all the fuse pins and bolts have been re-installed:

- INBOARD FLAP TRACKS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-15/401
- OUTBOARD FLAP TRACKS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-25/401.

D. Leading Edge Flaps and Slats Inspection, if applicable

SUBTASK 05-51-08-010-004

(1) Extend the leading edge flaps and slats, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 05-51-08-040-004

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do the deactivation procedure for the leading edge flap/slat system, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-51-08-210-004

(3) Examine the leading edge flap and slat components as follows:

(a) External skins of flaps and slats

(b) Leading edge flap hinge rib and linkage attachments

(c) Slat auxiliary and main tracks

(d) Leading edge flap and slat support ribs and adjacent skin panels

(e) All drive mechanisms and related support structure.

(f) Remove slat main track and auxiliary track roller bolts one at a time (PAGEBLOCK 27-81-21/401 and TASK 57-44-01-000-801).

NOTE: You can insert a retaining pin to hold the roller bearing or slat while the bolt is removed.

1) Make sure that the bolts are not cracked, bent, or show other signs of damage.

2) If it is necessary, replace removed bolts with new bolts.

(g) Make sure that all bolts have been re-installed (PAGEBLOCK 27-81-21/401 and TASK 57-44-01-400-801).

SUBTASK 05-51-08-211-002

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(4) Examine the area to make sure that there are no objects in the slat track housing assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-08-420-001

(1) If it is necessary, install the removed components or install replacement parts.

SUBTASK 05-51-08-440-003

(2) Do the activation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-08-440-004

(3) Do the activation procedure for the leading edge flap/slat system, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 05-51-08-410-003

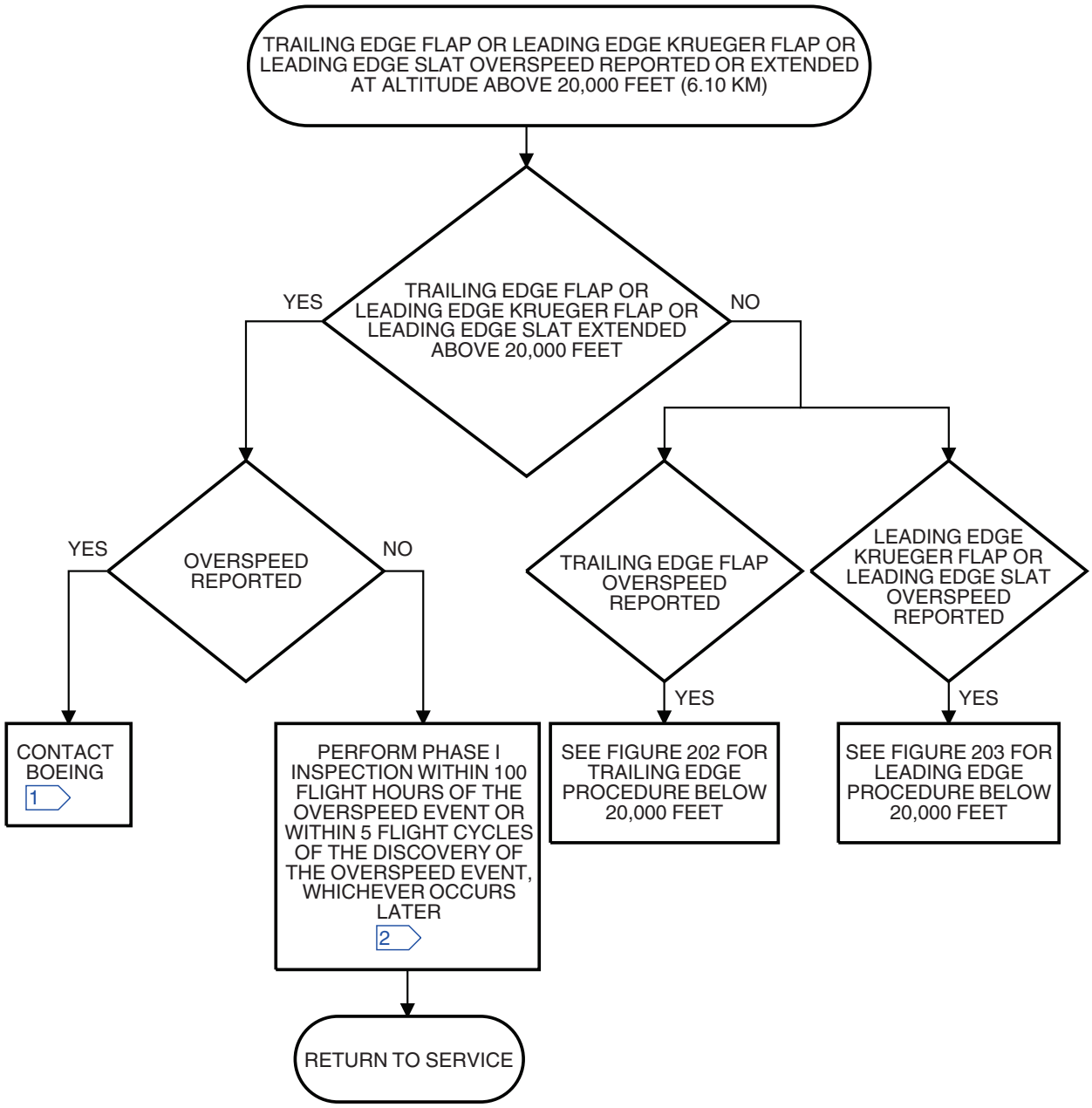
(4) Retract the trailing edge flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 05-51-08-410-004

(5) Retract the leading edge flaps and slats, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



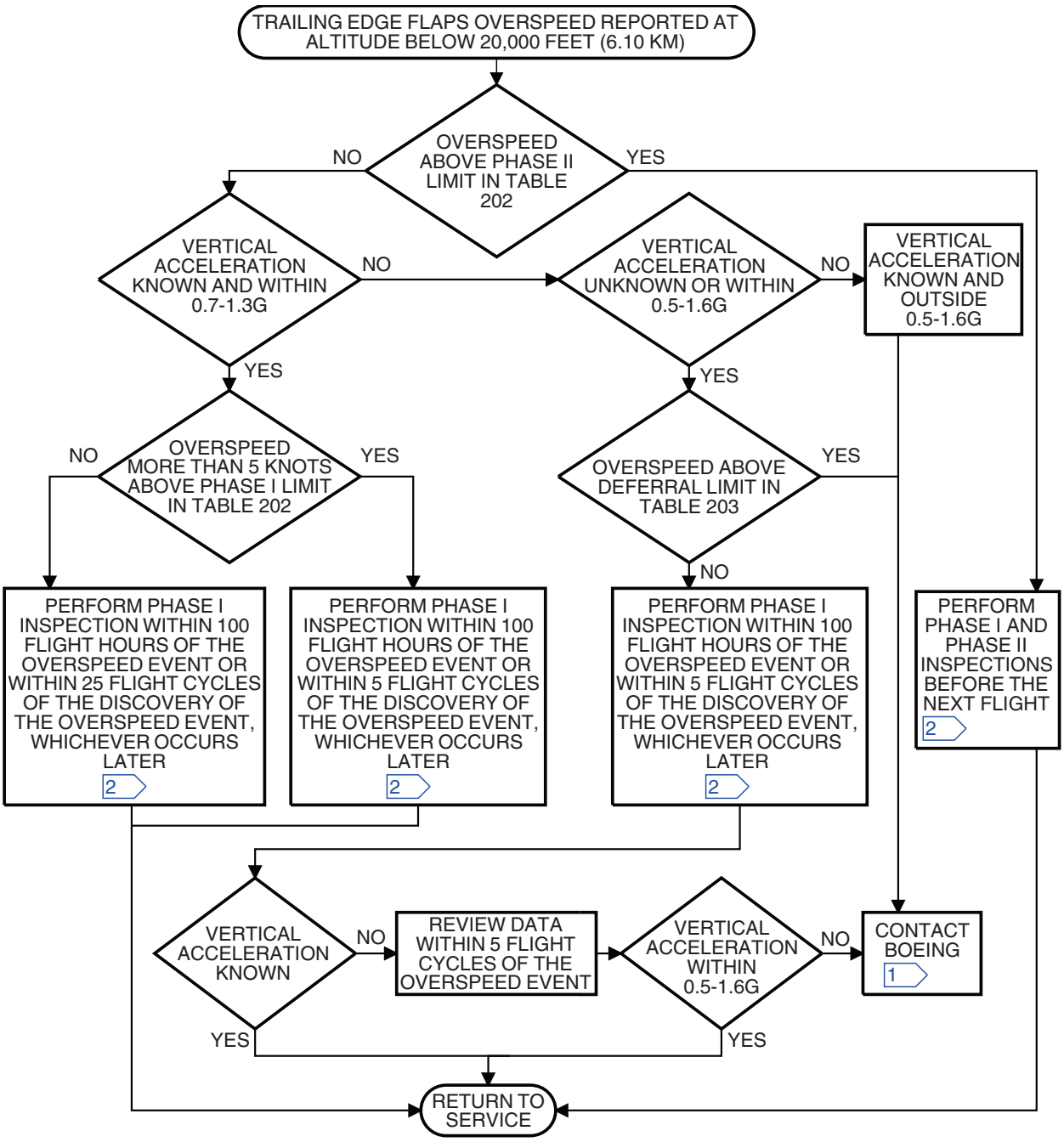
1 PHASE I AND PHASE II INSPECTIONS CAN BE DONE IN LIEU OF CONTACTING BOEING.

2 IF ANY DAMAGE RELATED TO AN OVERLOAD EVENT IS FOUND DURING THE PHASE I INSPECTION, THEN DO THE PHASE II INSPECTION BEFORE THE NEXT FLIGHT. REPAIR OR REPLACE COMPONENTS IF NECESSARY BEFORE NEXT FLIGHT.

3112382 S0000851045_V1

Flap/Slat Down Overspeed or Above 20,000 ft Condition Logic Flowchart
Figure 201/05-51-08-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



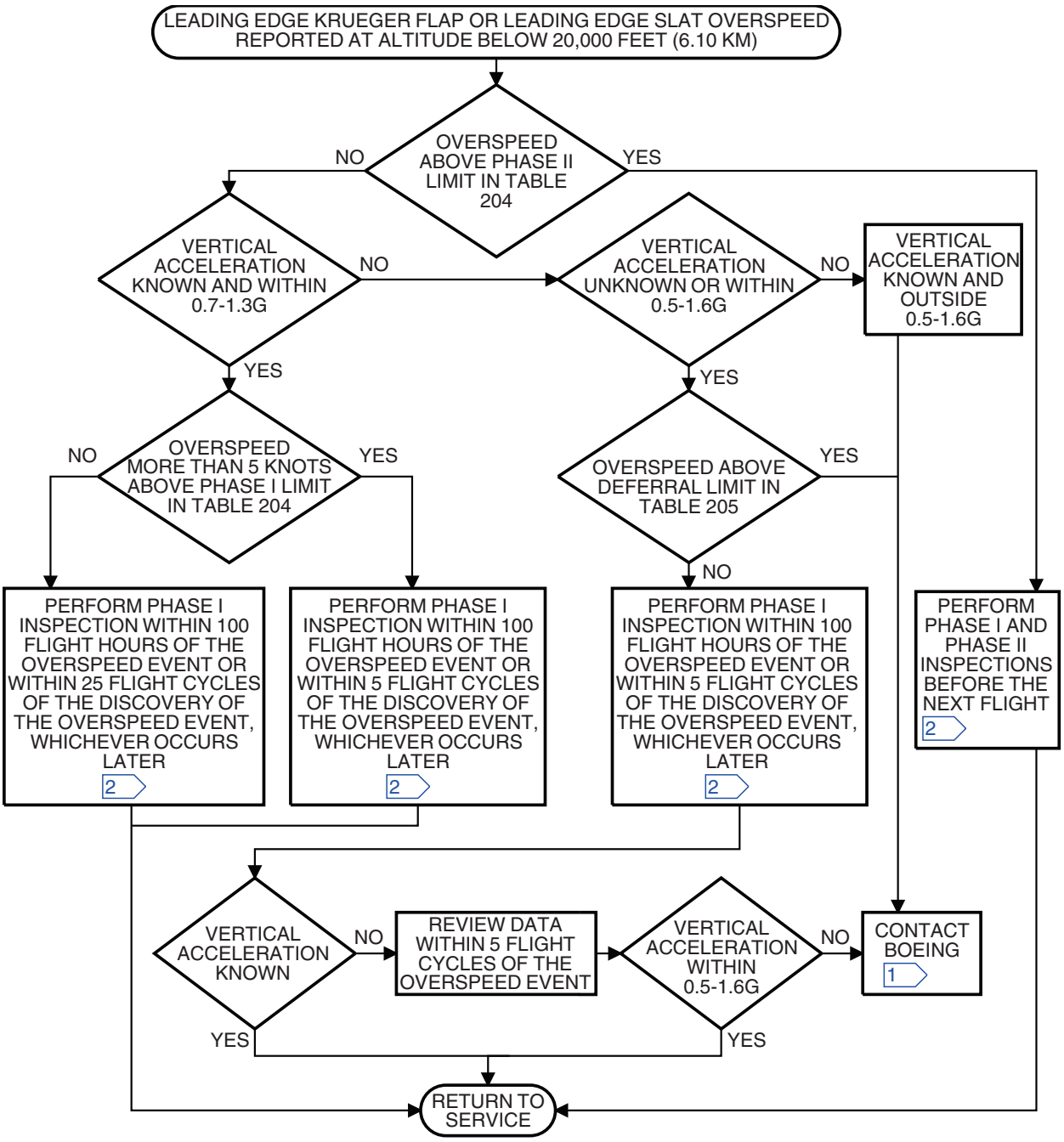
1 PHASE I AND PHASE II INSPECTIONS CAN BE DONE IN LIEU OF CONTACTING BOEING.

2 IF ANY DAMAGE RELATED TO AN OVERLOAD EVENT IS FOUND DURING THE PHASE I INSPECTION, THEN DO THE PHASE II INSPECTION BEFORE THE NEXT FLIGHT. REPAIR OR REPLACE COMPONENTS IF NECESSARY BEFORE NEXT FLIGHT.

3112383 S0000851047_V1

Flap/Slat Down Overspeed Condition – Trailing Edge
Figure 202/05-51-08-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details



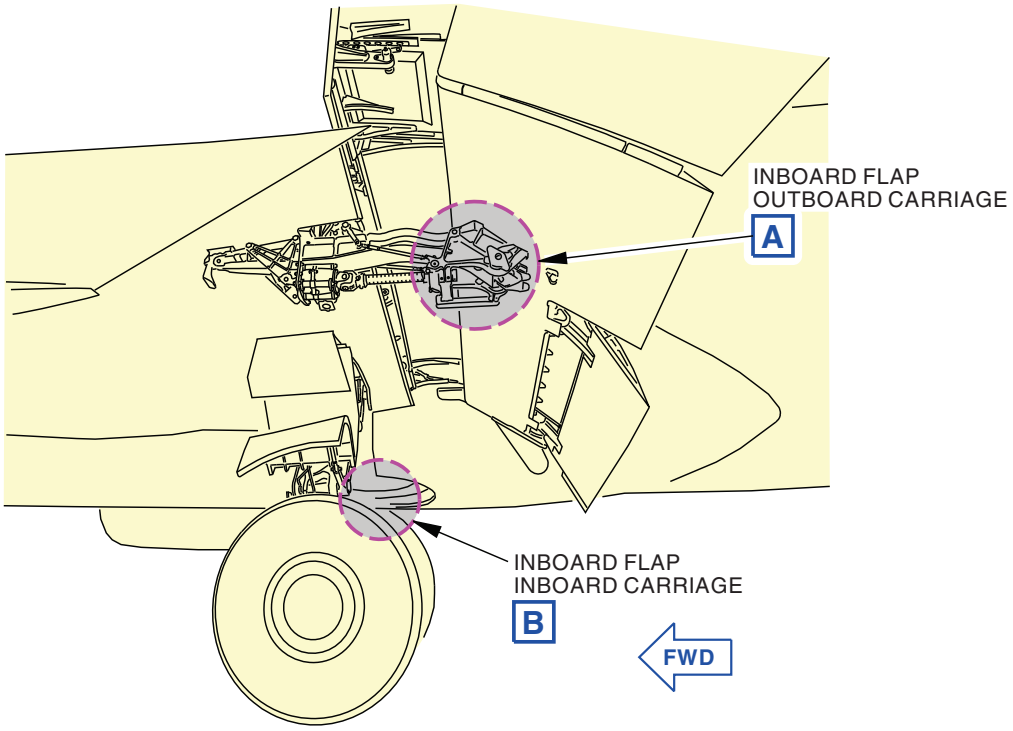
1 PHASE I AND PHASE II INSPECTIONS CAN BE DONE IN LIEU OF CONTACTING BOEING.

2 IF ANY DAMAGE RELATED TO AN OVERLOAD EVENT IS FOUND DURING THE PHASE I INSPECTION, THEN DO THE PHASE II INSPECTION BEFORE THE NEXT FLIGHT. REPAIR OR REPLACE COMPONENTS IF NECESSARY BEFORE NEXT FLIGHT.

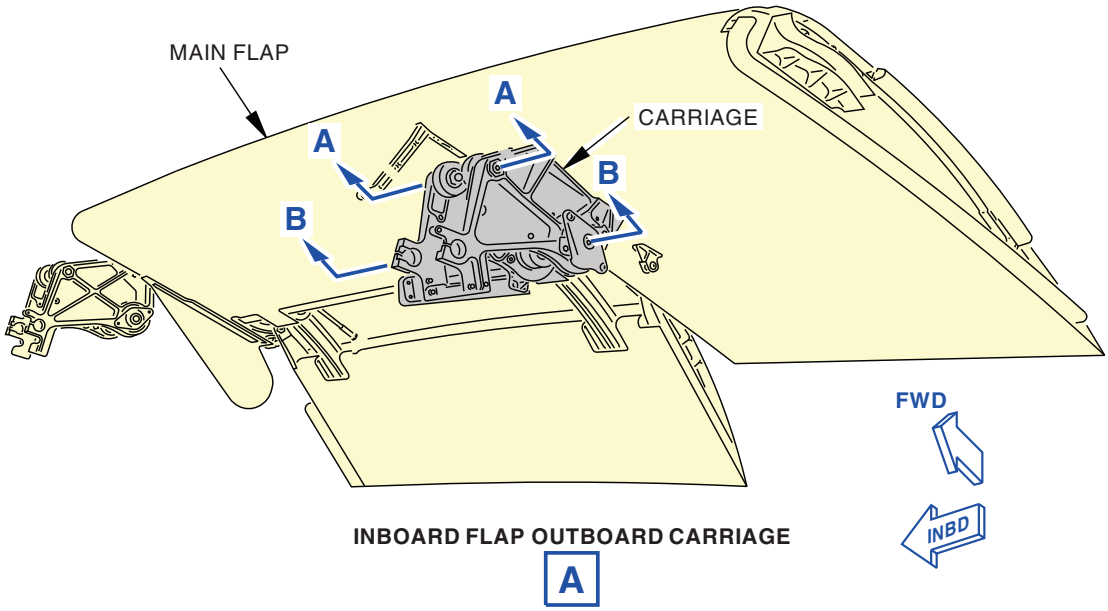
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Flap/Slat Down Overspeed Condition – Leading Edge
Figure 203/05-51-08-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)

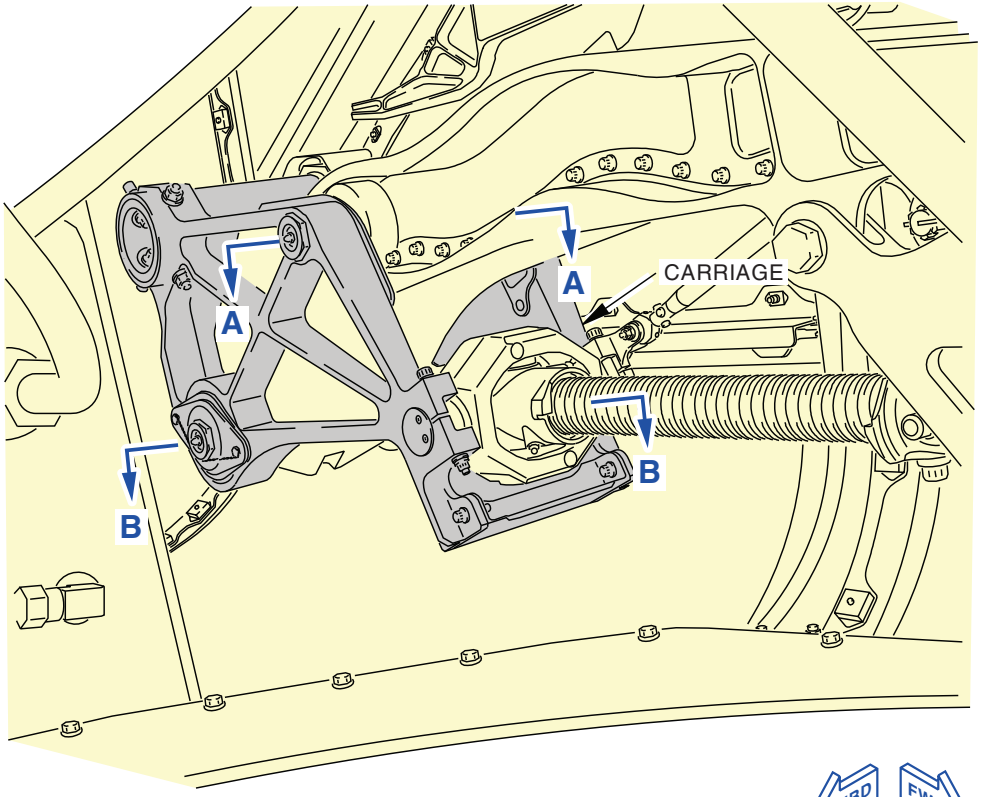


INBOARD FLAP OUTBOARD CARRIAGE
A

3009475 S0000783146_V1

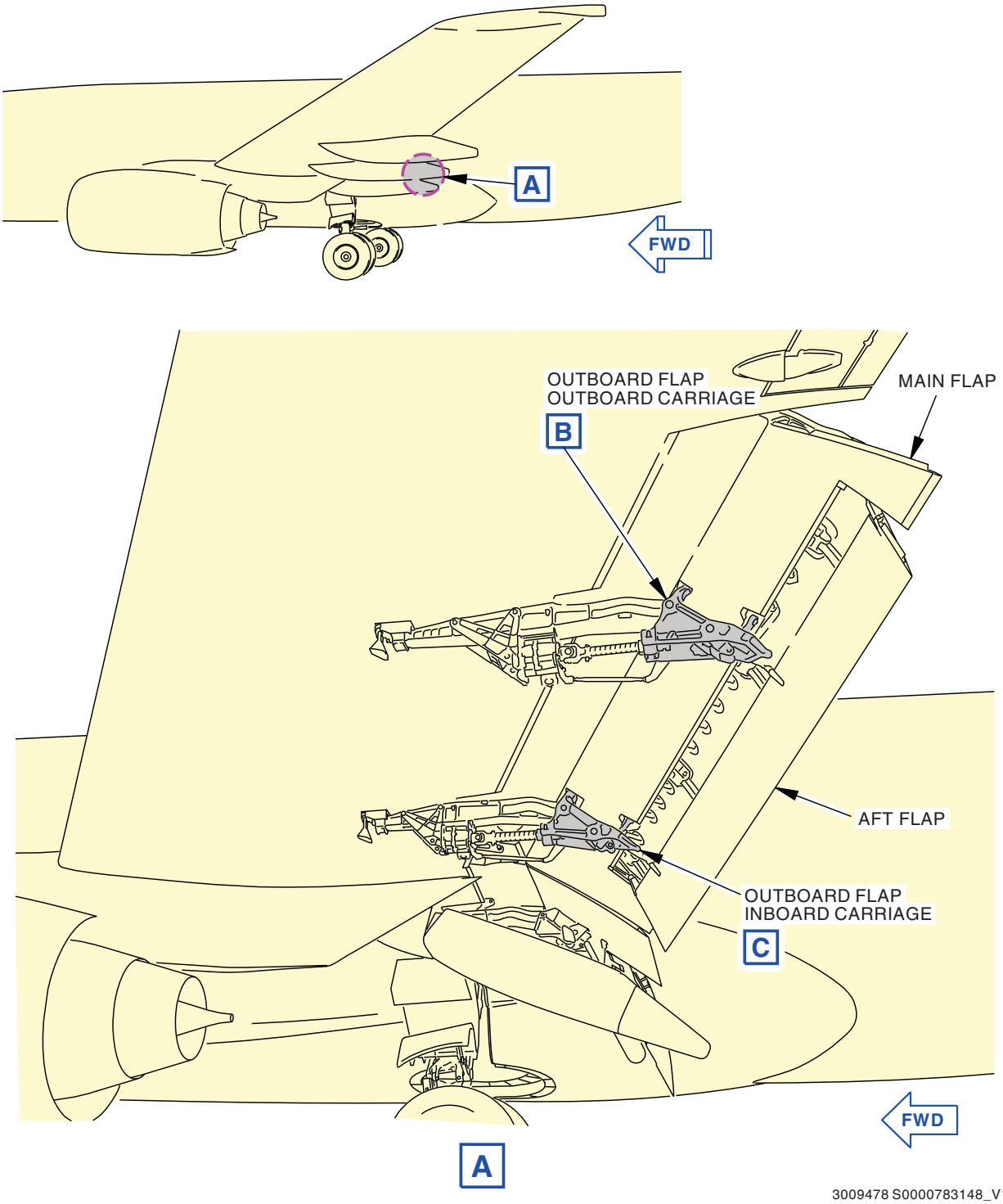
Inboard Flap Carriage Inspection
Figure 204/05-51-08-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



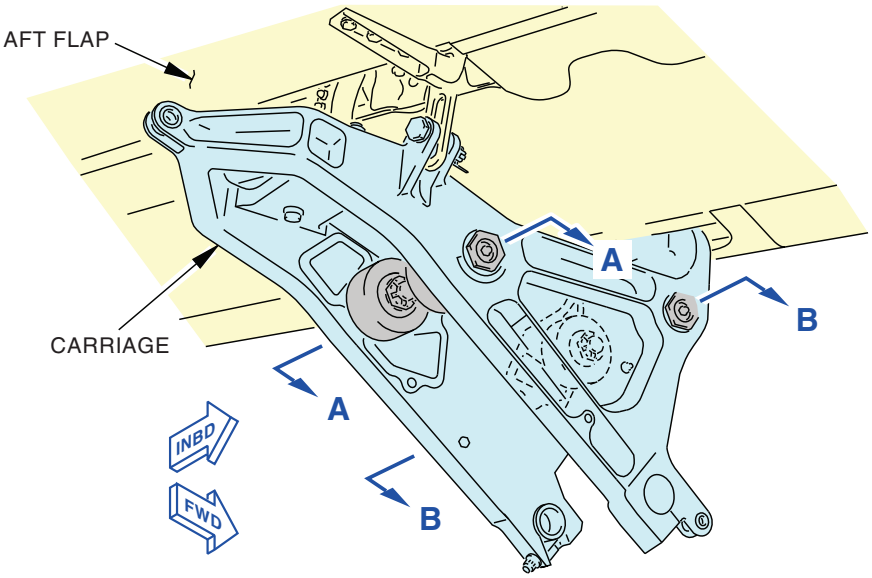
Inboard Flap Carriage Inspection
Figure 204/05-51-08-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



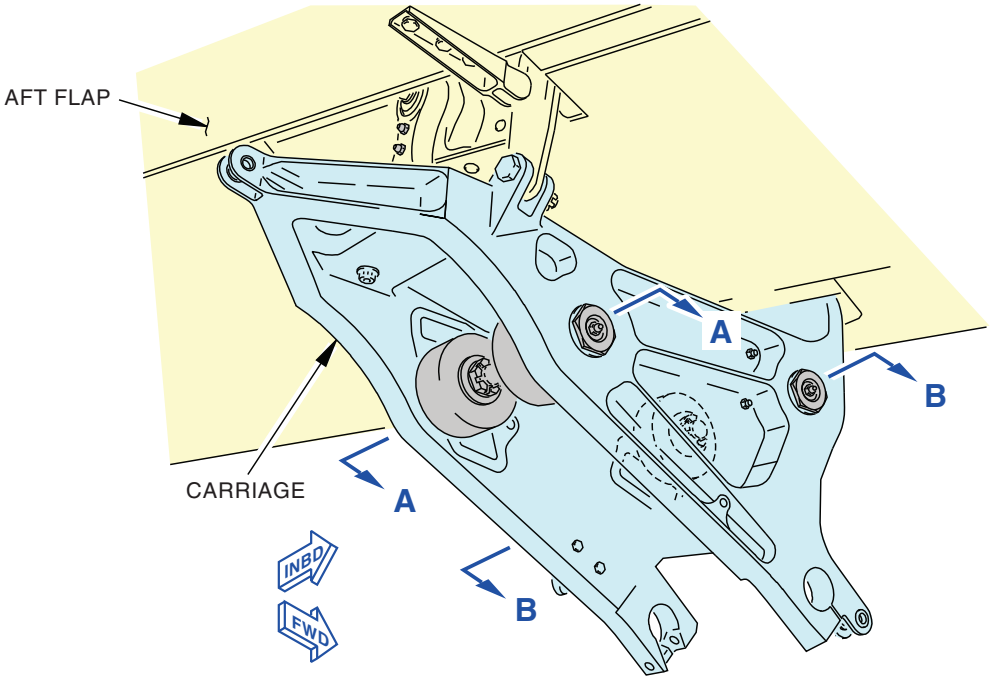
Outboard Flap Carriage Inspection
Figure 205/05-51-08-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD FLAP OUTBOARD CARRIAGE

B



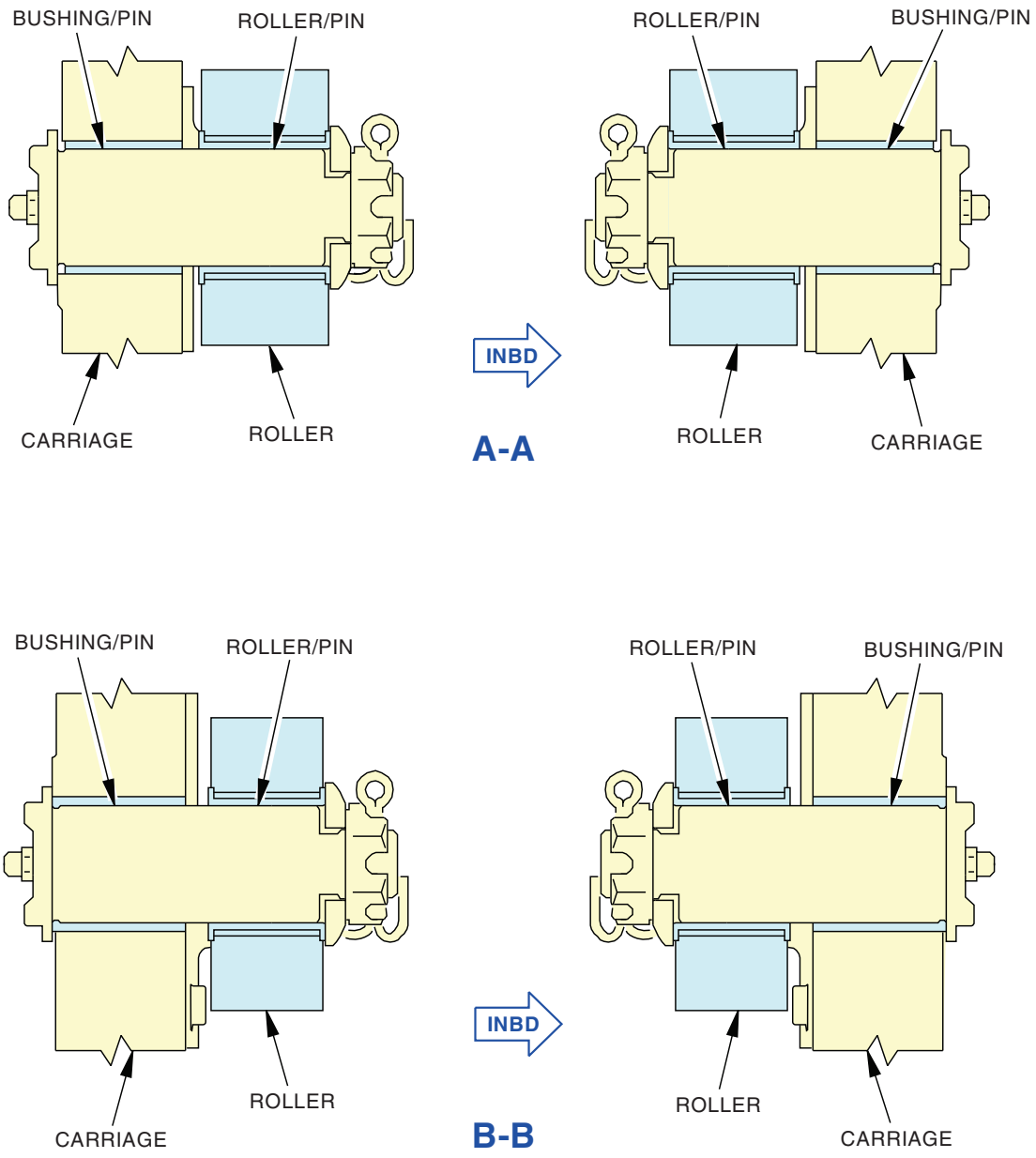
OUTBOARD FLAP INBOARD CARRIAGE

C

3009479 S0000783149_V1

Outboard Flap Carriage Inspection
Figure 205/05-51-08-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Flap Carriage Inspection
Figure 205/05-51-08-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OVERWEIGHT TAXI - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. Overweight Taxi is defined as taxiing at a weight that is more than the maximum-taxi weight (MTW) as specified in the Airplane Flight Manual (AFM).

(1) Before flight, you must decrease the airplane weight to that specified by the Airplane Flight Manual for takeoff.

B. Inspection Criteria; An immediate structural inspection is necessary if you:

(1) Taxi the airplane by more than 0.5% of the maximum-taxi-weight (MTW).

(2) Taxi the airplane overweight at any weight over the Maximum Design Taxi Weight (MTW) and have any of these conditions:

(a) High speed ground turn

(b) Sharp radius turn

(c) Heavy braking

(d) Taxi over rough pavement

(e) Pivoting (sharp radius turning with brakes on).

(3) If the criteria for the above paragraphs have not been met, no inspection is necessary.

(a) You must decrease the airplane weight to that specified by the Airplane Flight Manual before takeoff.

C. The Inspections

(1) The inspection is divided into Phase 1 and Phase 2.

(2) If the inspection for Phase 1 shows no signs of damage, the inspection is complete.

(3) If the Phase 1 inspection shows any sign of damage, the Phase 2 inspection must be done.

D. Inspections, Repairs, and Replacements

(1) When the conditional inspection tells you to "examine" or “inspect” a component, look for these conditions (repair or replace components, if it is necessary):

(a) Cracks

(b) Pulled apart structure or unusual gaps

(c) Loose paint (paint flakes)

(d) Twisted parts (distortion)

(e) Bent parts

(f) Wrinkles or buckles in structure

(g) Fastener holes that became larger or longer

(h) Loose fasteners

(i) Fasteners that have pulled out, deformed, or are gone

(j) Delaminations (a component with one or more layers pulled apart)

(k) Parts that are not aligned correctly

(l) Interference (clearance that is not sufficient between two parts)

(m) Discoloration (heat damage)

(n) Nicks or gouges

(o) Other signs of damage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Replace or repair the components that have one or more of the conditions given above.

TASK 05-51-09-212-801

2. Phase 1 Inspection

A. Airplane Inspection

SUBTASK 05-51-09-212-001

(1) Do the inspection of the main and nose landing gear.

(a) Examine all tires and wheels.

(b) Examine the support structure.

(c) Look for signs of fluid leakage at the top and bottom of the outer cylinder of the shock strut.

NOTE: A small quantity of hydraulic fluid on the surface of the inner cylinder of the shock strut is satisfactory.

SUBTASK 05-51-09-212-002

(2) Do the inspection of the landing gear, fuselage, and wing.

(a) Look for fuel leaks, and other fluid leaks, in the areas that follow:

1) All wheel well areas of the body, wing, and nose landing gear.

2) The lower external surface of the fuselage in the area of the wing-to-body fairing.

3) The wing.

——— END OF TASK ———

TASK 05-51-09-210-801

3. Phase 2 Inspection

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-31 P/B 301	MAIN LANDING GEAR SHOCK STRUT - SERVICING
12-15-41 P/B 301	NOSE LANDING GEAR SHOCK STRUT - SERVICING
32-11-00 P/B 401	MAIN LANDING GEAR - REMOVAL/INSTALLATION
32-21-00 P/B 401	NOSE LANDING GEAR - REMOVAL/INSTALLATION
32-32-00 P/B 501	MAIN GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST
32-33-00 P/B 501	NOSE GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST
32-51-00 P/B 501	NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST

B. Airplane Inspection

SUBTASK 05-51-09-212-003

(1) Do the inspection of the main landing gear and support structure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure the shock strut pressures are normal and the hydraulic fluids are at the correct levels (MAIN LANDING GEAR SHOCK STRUT - SERVICING, PAGEBLOCK 12-15-31/301).

(b) Lift the airplane with jacks, do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

(c) Examine the inner and outer cylinder lugs.

(d) Examine all structural components of the main landing gear and carefully examine the components that follow:

1) Shock strut
2) Drag strut
3) Torsion links
4) Side strut
5) Walking beam fitting of the shock strut
6) Walking beam for the main gear actuator
7) Linkage fittings of the walking beam and actuator
8) Strut doors and the mechanism that retracts and extends the doors

(e) Examine the support structure of the main landing gear and carefully examine the components that follow:

1) Landing gear beam
2) Support fittings for the landing gear beam:
   a) Inboard
   b) Outboard.
3) Trunnion support fittings and attachment:
   a) Forward
   b) Aft
4) Stabilizer link and fittings between the rear spar and landing gear beam
5) Body fitting for side strut and uplock attachment

(f) Examine all of the pin joints and fuse pin connections.

(g) If you found tire damage in Phase 1, do the steps that follow.

1) Remove and examine the wheel structure.
2) Remove and examine the brake assembly.
3) Examine the axles.

(h) If one or more of the conditions that follow occurred, remove, disassemble, and examine all parts of the shock strut (MAIN LANDING GEAR - REMOVAL/INSTALLATION, PAGEBLOCK 32-11-00/401).

1) The shock strut pressures were sufficiently low to cause damage.
2) The hydraulic fluid levels were sufficiently low to cause damage.
3) You found damage to one or more of the parts during your inspection of the landing gear.

   NOTE: Do not remove the orifice support tube to examine it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Make sure the main landing gear retracts and extends correctly (MAIN GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST, PAGEBLOCK 32-32-00/501).

(j) Lower the airplane from the jacks, do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 05-51-09-212-004

(2) Do the inspection of the nose landing gear and support structure.

(a) Make sure the shock strut pressures are normal and the hydraulic fluids are at the correct levels (NOSE LANDING GEAR SHOCK STRUT - SERVICING, PAGEBLOCK 12-15-41/301).

(b) Lift the nose of the airplane with jacks, do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

(c) If you found tire damage in Phase 1, do the steps that follow:

1) Remove and examine the wheel structure.

2) Examine the axle.

(d) Examine all structural components of the nose landing gear and carefully examine the components that follow:

1) Shock strut

2) Torsion links

3) Drag strut

4) Lock links

5) The attach nut of the orifice support tube (signs of fluid leakage).

(e) Examine the wheel well area and carefully examine the parts that follow:

1) Web (the left and right sidewalls)

2) Aft bulkhead

3) Trunnion attachments

4) Drag strut attachments.

(f) Examine the wheel well bulkheads and fuselage outboard of nose wheel well at (Body Stations 294.5 and 360).

NOTE: You can do this inspection on the forward side through the access holes in the sidewalls of the nose wheel well. Examine the aft side of the bulkhead at Body Station 294.5 from the electronics compartment.

(g) If one or more of the conditions that follow occurred, remove, disassemble, and examine all parts of the shock strut (NOSE LANDING GEAR - REMOVAL/INSTALLATION, PAGEBLOCK 32-21-00/401).

1) The shock strut pressures were sufficiently low to cause damage.

2) The hydraulic fluid levels were sufficiently low to cause damage.

3) You found damage to one or more of the parts during your inspection of the landing gear.

NOTE: Do not remove the orifice support tube to examine it.

(h) Make sure the nose landing gear steering system is adjusted and operates correctly (NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 32-51-00/501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Make sure the nose landing gear retracts and extends correctly (NOSE GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST, PAGEBLOCK 32-33-00/501).

(j) Lower the nose of the airplane from the jacks, do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 05-51-09-212-005

(3) Do the inspection of the fuselage.

NOTE: If you find external damage to the fuselage, always examine the adjacent internal structure.

(a) Examine the lower fuselage structure.

NOTE: Examine carefully in the area below the body crease from Body Station 727 to 100 in. (254 cm) aft.

(b) Examine carefully these components (Body Stations 540 through 727A):

1) Keel beam chords
2) Stiffeners
3) Webs and splices

(c) Also examine carefully the top of the fuselage between S-6 left and S-6 right (Body Stations 540, 664, 727).

(d) Examine the upper and lower fuselage skin panels forward and aft of the wing.

NOTE: Permanent wrinkles frequently occur on the lower side of the fuselage skins aft of Body Station 727. An internal inspection is necessary if you found wrinkles that were not there before the landing. Examine the wrinkles carefully for cracks.

(e) Examine the wing-to-fuselage joints (Body Stations 540 and 664).

(f) Examine the fuselage skin joints above the landing gear beam.

C. Put the Airplane Back to its Usual Condition

SUBTASK 05-51-09-430-001

(1) Install the components you removed or install replacement components.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DRAGGED ENGINE NACELLE/FAN BLADE OUT/ENGINE SEIZURE/THRUST REVERSER AND ATTACH POINTS DAMAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the following task:

(1) Dragged engine nacelle, fan blade out, engine seizure, engine, strut, and thrust reverser damaged conditional inspection.

B. This procedure gives an inspection for the conditions that follow:

(1) For the engine and strut after a dragged engine nacelle condition.

(2) After an engine seizure and other types of engine and strut damage.

(3) After an engine fan blade out and subsequent windmilling vibration.

(4) After a vibrational condition caused by loss of cowling or engine parts other than fan blades.

(5) After damage caused when hit by ground support equipment; except for minor, localized ground support equipment impact to the strut or cowling.

NOTE: These inspections are not necessary when the operator finds that small scratches/dents, etc, were not caused by the conditions in this procedure.

C. When the conditional inspection tells you to "examine" a component, look for these conditions (repair or replace components, if it is necessary):

(1) Cracks

(2) Pulled apart structure

(3) Excessive looseness or wear

(4) Twisted parts (distortion)

(5) Bent components

(6) Loose fasteners

(7) Fasteners holes that became larger or longer

(8) Fasteners that have pulled out or are missing

(9) Excessive loading

(10) Peening of pins and bushings

(11) Bushing/bearing damage

(12) Bushing bearing fusion

(13) Lug distortion or damage

(14) Bolt distortion

(15) Misalignment

(16) Interference

(17) Other obvious or unusual signs of damages.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-51-10-210-801

2. Dragged Engine Nacelle/Fan Blade Out/Engine Seizure/Engine and Strut Damage Conditional Inspection

A. References

Reference	Title
05-51-42-200-801	Engine Blade Out - EE Bay and Flight Deck Panels Inspection (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
54-52-06-000-801	Aft Fairing Access Panel Removal (P/B 401)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-06-200-801-G00	Blocker Door Visual Inspection (P/B 601)

B. Location Zones

Zone	Area
400	Powerplant and Nacelle Struts

C. Examine the Struts and Nacelles, and Adjacent Structure

SUBTASK 05-51-10-040-001

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do the steps that follow to deactivate the flaps and slats:

(a) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

(b) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-10-040-002

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 05-51-10-210-001

(3) Do the inspection of the struts and nacelles, and the adjacent structure.

(a) Visually examine the areas that follow:

1) The nacelle strut external surface.

2) The nacelle inlet, fan cowl, thrust reverser and exhaust (ground support equipment impacts).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) The lower surface of the nacelle cowlings (ground impact).

4) Engine fan blades (full or partial fan blade loss).

(b) If damage is found in the areas noted above (except for minor, localized ground support equipment impacts), do these steps:

1) Remove the nacelle strut access doors.

2) Remove the forward strut thrust reverser and overwing fairings.

3) Remove the forward strut fairing thumbnail fairing.

4) Remove the aft strut fairing panels, do this task: Aft Fairing Access Panel Removal, TASK 54-52-06-000-801.

5) Open the fan cowl and thrust reverser.

(c) Visually examine the following components for cracks, signs of offset deformation, or crank shafting of pins and/or bushings.

1) Strut fuse pin and link pin bores (perform examination in place - remove retention hardware).

2) Wing and strut fittings common to fuse and link pins.

3) Strut external upper spar and side skins.

4) Strut internal structure, including bulkheads and frames.

5) Forward engine mount assembly and aft mount assembly including thrust links.

SUBTASK 05-51-10-210-002

(4) Examine the thrust reverser, inlet and fan cowl areas that follow:

(a) Inspect the thrust reverser sleeves.

(b) Inspect the blocker door linkage for bent or broken links, do this task: Blocker Door Visual Inspection, TASK 78-31-06-200-801-G00.

(c) Inspect the inlet crushable spacers.

1) Look for loose attach bolts.

(d) Inspect the thrust reverser hinges and latches.

(e) Inspect the fan cowl hinges and latches.

SUBTASK 05-51-10-210-003

(5) If you think an engine seizure or internal engine damage has occurred, examine the engine.

(a) Do this task: Engine Operation Limits, TASK 71-00-00-800-802-G00.

(b) Remove the engine if it is necessary, do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

SUBTASK 05-51-10-210-004

(6) Examine the EEBay and flight deck, do this task: Engine Blade Out - EE Bay and Flight Deck Panels Inspection, TASK 05-51-42-200-801.

D. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-10-410-001

(1) Install all the components you removed if they are serviceable, or install replacement parts.

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SUBTASK 05-51-10-440-001

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 05-51-10-440-002

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do the steps that follow to activate the flaps and slats:

(a) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

(b) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GALLEY SPILLAGE - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) An inspection and the clean-up of the areas of contamination after a liquid is spilled in the galley.

B. If liquids are spilled inside the galley, you must remove all signs of liquid and fully dry the areas of contamination immediately.

C. If the floor area of the galley area is not sealed correctly, liquid contamination can result in corrosion of the floor structure.

NOTE: Moisture above the galley floor can go through small holes in the floor surface area and stay out of view. This could possibly cause unwanted damage to the airplane structure or to equipment in the area below the galley floor.

TASK 05-51-13-211-801

2. Galley Spill Conditional - Inspection

A. General

(1) This inspection procedure is to be done when liquid has been spilled in the galley area.

NOTE: When a small amount of liquid is spilled, an inspection is not usually necessary.

(2) In the galley areas, a water (liquid) barrier is installed to contain liquid when it is spilled. To protect the water barrier, vinyl mats cover the galley floor.

(3) In the adjacent entry and service door thresholds, waste water drains are installed to remove liquids (gray water) from the floor. These drain holes must be kept free from blockages.

(4) The integrity of the water barrier seal, in the forward galley areas, is most important because the location of the electronic equipment compartment is directly below the forward galley.

(5) A leak from the forward galley which has a damaged water barrier, into the compartment below, could possibly cause serious damage to the electronic equipment. The result could be a possible hazard to the flight safety of the airplane.

(6) In all areas where gray water contamination has to be removed and cleaned, you must use an approved disinfectant to kill all harmful micro-organisms.

(7) The flight safety of the airplane, together with the safety of health and the prevention of bad smells, must be your first consideration when you do this task.

B. References

Reference	Title
25-27-21 P/B 401	ENTRY AND SERVICE AREA FLOOR COVERING - REMOVAL/INSTALLATION
38-31-00 P/B 201	GRAY WATER/DRAIN SYSTEM - MAINTENANCE PRACTICES
53-21-00 P/B 401	PASSENGER CABIN FLOORS - REMOVAL/INSTALLATION
53-21-00 P/B 801	PASSENGER CABIN FLOORS - REPAIRS
53-21-11 P/B 801	WATER BARRIER - REPAIRS

C. Tools/Equipment

Reference	Description
STD-6812	Source - Dry Air, 30-35 inches of mercury (absolute)

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(Continued)

Reference	Description
STD-14360	Flashlight

D. Consumable Materials

Reference	Description	Specification
B01033	Cleaner - Disinfectant - Lysol	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Procedure

SUBTASK 05-51-13-160-001

(1) In the galley where the liquid was spilled, use a clean cloth or a sponge to remove all liquid from the floor of the galley area.

SUBTASK 05-51-13-010-001

(2) In the galley area where the liquid was spilled, at the next maintenance opportunity, remove the floor area covering to do an inspection of the waterseal. Do this task: ENTRY AND SERVICE AREA FLOOR COVERING - REMOVAL/INSTALLATION, PAGEBLOCK 25-27-21/401.

NOTE: The waterseal of the galley area can be damaged if care is not taken when you remove the vinyl floor mats.

SUBTASK 05-51-13-160-002

(3) Use Lysol cleaner, B01033 or equivalent cleaners when you clean areas where blue water contamination has occurred.

SUBTASK 05-51-13-160-003

(4) Make sure the door threshold drain, adjacent to the galley area, is clear and free from blockages (GRAY WATER/DRAIN SYSTEM - MAINTENANCE PRACTICES, PAGEBLOCK 38-31-00/201).

SUBTASK 05-51-13-212-001

(5) Inspect the fillet seal at the base of the galley for signs of damage.

SUBTASK 05-51-13-212-002

(6) Inspect the floor waterseal for signs of damage to materials and sealants.

SUBTASK 05-51-13-910-001

(7) If you find signs of damage to seals and/or sealing, refer to the approved repairs (PASSENGER CABIN FLOORS - REPAIRS, PAGEBLOCK 53-21-00/801, WATER BARRIER - REPAIRS, PAGEBLOCK 53-21-11/801).

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SUBTASK 05-51-13-010-002

(8) If the inspection is applicable to the forward galley, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 05-51-13-010-003

(9) If the inspection is applicable to the aft galley, do this task:

(a) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 05-51-13-212-003

(10) Use a flashlight, STD-14360 and do a close visual examination of the area below the floor of the galley.

SUBTASK 05-51-13-360-001

(11) If you find signs of moisture below the galley area, in electronic-equipment compartment or the aft cargo-compartment equipment-bay, do the steps that follow:

(a) Use a cotton wiper, G00034 to absorb and remove all moisture from the area.

(b) Do an examination of all the waste-water drain tubes and related clamps for signs of a leak.

(c) Tighten all clamps and/or replace damaged drain tubes as necessary.

(d) Examine the floor beams, intercostals and structure for signs of moisture and/or corrosion.

(e) Do an inspection of installed equipments, in the applicable equipment compartment for signs of contamination.

1) If you find signs of moisture on equipment electrical cables, use a dry air source, STD-6812 to remove all moisture from the contaminated areas.

NOTE: Electrical cable looms can hold moisture between the separate wires. If possible, open the wires in a contaminated cable when you dry the cable.

2) Do a check of contaminated connectors for moisture ingress and dry as necessary.

a) If you think a connector is contaminated with moisture, you must disconnect and fully examine the connector and the related equipment connector.

3) If you find signs of moisture, use a cotton wiper, G00034 or a dry air source, STD-6812 to remove all moisture and fully dry the contaminated components.

SUBTASK 05-51-13-942-001

(12) If you do not find signs of moisture and the water barrier is in a serviceable condition, do the close out procedure.

NOTE: The removal of the galley and repairs to the floor water barrier are only necessary when signs of damage and leaks in the galley floor are found.

SUBTASK 05-51-13-010-004

(13) If your inspection shows damage to the water barrier of the galley floor area and/or shows the moisture travel to go below the galley floor, remove the applicable galley.

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SUBTASK 05-51-13-010-005

(14) If necessary, remove the galley floor. Do this task: PASSENGER CABIN FLOORS - REMOVAL/INSTALLATION, PAGEBLOCK 53-21-00/401.

SUBTASK 05-51-13-160-004

(15) Fully clean the area and the floor panels and make sure you remove all the remaining sealant.

SUBTASK 05-51-13-350-001

(16) If corrosion is found on the airframe, refer to the Structural Repair Manual (SRM).

SUBTASK 05-51-13-420-001

(17) Install the galley floor. Do these tasks: PASSENGER CABIN FLOORS - REMOVAL/INSTALLATION, PAGEBLOCK 53-21-00/401 and PASSENGER CABIN FLOORS - REPAIRS, PAGEBLOCK 53-21-00/801.

SUBTASK 05-51-13-410-001

(18) Install the water barrier in the galley. Do this task: WATER BARRIER - REPAIRS, PAGEBLOCK 53-21-11/801.

SUBTASK 05-51-13-420-002

(19) Install the galley (as applicable).

SUBTASK 05-51-13-710-001

(20) If an equipment is found to have been contaminated with moisture, you must do an operational test of the equipment after all the moisture has been removed.

SUBTASK 05-51-13-940-001

(21) Do the close out procedure as follows:

(a) Install the galley floor covering. Do this task: ENTRY AND SERVICE AREA FLOOR COVERING - REMOVAL/INSTALLATION, PAGEBLOCK 25-27-21/401.

(b) Make sure that these access panels are closed:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MERCURY SPILLAGE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This section has a task to detect, isolate, examine, and clean the airplane after a mercury spill.

(1) Mercury spills that are not detected and cleaned immediately after a spill can spread and become very difficult to clean or cause extensive structural damage.

(2) Mercury spill containment and cleanup is recommended to be performed by personnel that have specialized training with mercury spills and hazardous material removal.

TASK 05-51-14-210-801

2. Mercury Spillage Conditional Inspection

(Figure 201)

A. References

Reference	Title
25-80-00-000-801	Insulation Blanket Removal (P/B 401)
25-80-00-400-801	Insulation Blanket Installation (P/B 401)

B. General

SUBTASK 05-51-14-160-001

(1) A mercury spill must be considered hazardous, and immediate action must be taken to isolate, inspect, and clean the aircraft.

(2) The corrosive action of mercury is very rapid in pitting and intergranular.

(3) Mercury amalgamates quickly with aluminum alloys at room temperature if the oxide film on the aluminum has been scratched or damaged.

(a) Once a small area of aluminum has been amalgamated, corrosion and liquid metal embrittlement occurs. This process is accelerated by moisture and particularly by salt water.

(4) The amalgamation of stressed aluminum structure may also result in rapid cracking similar to stress corrosion cracking.

(5) Mercury is not consumed in the amalgamation process. As the aluminum oxidizes, it separates from the amalgam and the mercury continues to attack fresh aluminum.

(6) The presence of organic finishes, greases or a thick continuous oxide layer retards the amalgamation process.

(7) When elemental mercury leaks on to an all-metallic structure, severe structural strength degradation can occur.

(a) The rate of the diffusion of the mercury into a metal is related to the type of metal touched by the mercury.

(b) The diffusion is also related to the type of finish protection on the metal.

(c) When the diffusion has started, it cannot be stopped.

(d) Structural degradation after a mercury spill is not always visually apparent until a load is applied to the structure.

(8) The inspection, removal, or repair must be made before the next flight.

(9) Structural repairs are necessary if there are signs of mercury contamination.

(10) Make sure there is no detectable mercury in the airplane.

(a) Use mercury sniffers and X-ray equipment (if available) to detect mercury.

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(b) The airplane is only clean when there is no detectable mercury.

C. Personal Protection

SUBTASK 05-51-14-160-002

(1) Throughout the decontamination process follow these steps:

(a) Monitor the mercury vapor levels near the spill area and protect personnel from exposure.

(b) Look for new free mercury deposits around the suspect spill area.

(c) Limit the number of persons around and on the airplane to the minimum required.

(d) Allow only persons trained in mercury spillage cleanup.

(2) Mercury vapor can cause effects in the central and peripheral nervous systems, lungs, kidneys, skin and eyes in humans. Acute exposure to high concentrations of mercury vapor causes severe respiratory damage, while chronic exposure to lower levels is primarily associated with central nervous system damage. Chronic exposure to mercury is also associated with behavioral changes and alterations in peripheral nervous system.

(a) Mercury Exposer Limits are based on local regulations. Refer to your local regulations for applicable exposure limits.

1) The following exposure levels are also available for your reference:

NOTE: Check the web sites for the latest information and exposure limits.

a) Per the US Environmental Protection Agency (EPA):

<1> For a 24 hours daily (continuous) exposer the safe limit for chronic inhalation of elementary mercury is 0.3 µg/m³ (microgram per cubic meter). This limit should be safe for 24 hour exposure of general population.

<2> The Lowest-Observed Adverse Effect Level (LOAEL) is 25 µg/m³ (converted to LOAEL [ADJ] of 9 µg/m³). The LOAEL is based on an 8-hour TWA occupational exposure. LOAEL [ADJ] limit is based on an 8-hour per day and 40 hours per week TWA occupational exposure. (Source http://www.epa.gov/iris/subst/0370.htm#refinhal).

b) Per the US Occupational Safety and Health Administration (OSHA):

<1> Permissible exposure limit (PEL) for mercury vapor is 100 µg/m³ of air as a ceiling limit. A worker's exposure to mercury vapor shall at no time exceed this ceiling level.

<2> OSHA recognizes the National Institute for Occupational Safety and Health (NIOSH) established recommended exposure limit (REL) for mercury vapor of 50 µg/m³ as a TWA for up to a 10-hour workday and a 40-hour workweek.

<3> OSHA also recognizes American Conference of Governmental Industrial Hygienists (ACGIH) assigned mercury vapor threshold limit value (TLV) of 25 µg/m³ as a TWA for a normal 8-hour workday and a 40-hour workweek. (Source http://www.osha.gov/SLTC/healthguidelines/mercuryvapor/recognition.html)

(3) Large amounts of mercury will vaporize at normal temperatures.

(a) Ventilate enclosed space because the air mass can become dangerous to personal health.

(4) Free mercury or amalgam must not be picked up by hand.

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(5) Clean tools with soap and hot water or steam bath.

(a) Discard drill bits after use on structure contaminated by mercury.

(6) Mercury can attach easily to clothing.

(a) Use protective mats or cardboard to prevent contamination of clothing and the spread of mercury to other areas.

(7) Mercury is highly toxic and spreads very easily from one surface to another by adhering to hands, shoes, clothes, tools, etc.

(a) The following precautions for mercury spills should be strictly followed:

1) Avoid contact with surfaces suspected of being contaminated. Use wood or fiber sheets to support body while working in the area.

2) Wear wing socks (shoe protectors) and protective (disposable) clothing in contaminated area to prevent scratching metal surfaces. Properly dispose of socks after use.

3) Do not wear clothing used in contaminated areas on jobs in uncontaminated areas. Dispose of wing socks and protective clothing into unused metal container outside of aircraft. Contact the Waste Management Service.

4) Wash shoes with soap and water. Clean all tools that have been used in contaminated area with steam or hot water and soap. Discard any drill bits used on mercury contaminated areas. Thoroughly clean vacuum cleaner, if used.

5) Always wash thoroughly with soap and water after contacting mercury. Keep hands away from mouth. Do not eat, smoke, or blow nose without first washing your hands thoroughly.

6) Appreciable amounts of mercury will vaporize at normal temperatures. A stagnant air will become dangerous to personal health.

7) Do not use cleaning aids such as solvents, solids, or polishes on contaminated areas. Such materials may promote corrosion.

8) If hands become contaminated with mercury while working with cleaning equipment, do not touch any exposed metal in surrounding area, as you may contaminate it.

D. Isolation of the Mercury

SUBTASK 05-51-14-160-003

(1) Isolate the mercury spill area to prevent contamination to other areas.

(2) Evaluate the spill for cleanup.

NOTE: Depending on the amount and extent of the spill, consider using a specialized cleanup company that has trained personnel for handling mercury spills and cleanup of hazardous materials.

(3) Make a record of the general quantity of the mercury spilled.

(4) Make a record of the location where the mercury was spilled.

(5) When mercury is found on floor, do not remove coverings, access/inspection plates, screws, rivets, bolts, etc., from floor until all free mercury is removed.

(6) Isolate the spilled mercury and avoid enlargement of the contaminated area. The following precautions are advised:

(a) If mercury is found on the floor, do not remove coverings, access/inspection plates, screws, rivets, bolts, etc., from the floor until all free mercury is removed.

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1) Any hole which is left open in the contaminated area of the floor may allow mercury to spread to structure underneath the floor.

(b) If hands become contaminated, do not touch any exposed metal in the surrounding area in case it becomes contaminated.

(c) Mercury spreads easily from one surface to another by adhering to hands, shoes, clothes, tools, etc. Keep traffic to a minimum in the contaminated area.

(d) Consider protecting uncontaminated areas by taping down protective material such as plastic sheets or toweling. Plastic sheeting or toweling can also be taped over a contaminated area to isolate it while equipment is being collected for the removal of mercury.

E. Detection and Cleanup

SUBTASK 05-51-14-160-004

(1) Because of the urgency involved in removing mercury from the airplane, most of these procedures utilize materials that are readily available locally. However, the use of some special equipment is beneficial in ensuring that all traces of mercury are removed, and the sources of supply for these items are as follows:

(a) Hg vacuum cleaner: Nilfisk of America Inc. (V57573).

(b) Sensing device (mercury sniffer): Beckman Industrial Corp. (V73138), or Sunshine Scientific Instruments Inc. (V06030).

(c) Nickel-plated carbon-fiber brushes: International Research and Development Co.

(d) Fine copper wire brush: Produced locally or commercially available.

(e) Mercury sponge: J.T. Baker Chemical Co. (V70829). Volume

(2) A grayish-white powder or fuzzy coating on aluminum surfaces indicates the presence of corrosion caused by spilled mercury.

(3) The presence of even small amounts of mercury can be detected by a sniffer.

NOTE: This is an electronic device that is sensitive to mercury vapor.

SUBTASK 05-51-14-160-005

(4) Inspect the metal floor, seat tracks, cargo rails, and adjacent structure for mercury and corrosion.

(a) Inspect skin and internal structure below point of spillage.

(b) Also inspect lowest point in fuselage below cargo compartment floor if mercury spill has occurred in cargo area.

NOTE: Mercury liquid will flow to lowest level.

(5) Inspect areas suspected of having mercury contamination using a 10X magnifying glass.

(a) If corrosion of aluminum has started there will be a light grayish “Christmas Tree” fuzz or gray powdery dust deposit, and severe structural damage may follow.

(6) If corrosion is evident and cleanup cannot be completed immediately, coat contaminated area with corrosion preventative compound or engine oil.

NOTE: This helps to slow down the corrosion rate and also helps to prevent spreading of mercury contamination.

(7) When mercury spills occur in lower cargo compartment, use a portable X-ray machine (if available) along the outside lower surface of the fuselage to check suspected hidden corrosion areas between the skin, stringers, and frames below the floor.

(a) Droplets of mercury will show on a radiograph as small white spots.

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(b) Corrosion and embrittlement will appear as tree-like forms completely penetrating a structural component.

(c) While the basic X-ray exposure and film density parameters are acceptable to Boeing, it is recommended to reverse the film and X-ray tube placement.

(d) X-Ray evaluation only needs to be performed in areas that are suspected to be contaminated.

1) Special attention must be given to the area along joints and crevices, e.g. skin/stringer joint, stringer clips.

WARNING

DO NOT GET MERCURY IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE MERCURY. WEAR PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU CLEAN A MERCURY SPILL. MERCURY IS POISONOUS AND CAN CAUSE INJURY TO PERSONS.

WARNING

FULLY REMOVE ALL MERCURY FROM THE AIRPLANE. MERCURY CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(8) Remove all free mercury without delay. The following methods of removal are suggested, the more primitive methods are less effective than those requiring special equipment, but any possible method should be attempted whereby speed is essential.

(a) Use paper or cardboard troughs to scoop up the mercury.

(b) Use a high-powered mercury vacuum cleaner with a trap, if available. For a suggested method of making a trap, see Figure 201.

1) Locate the vacuum outside the airplane in a well ventilated location to reduce the possibility of contamination to other areas.

2) The size of the pickup hose from the container should be about ¼ in. (6 mm) in diameter to increase the amount of suction applied to the mercury.

NOTE: Due to the weight of mercury, the container will catch the mercury before it can enter the vacuum cleaner hose.

(c) An all rubber storage battery water syringe, or a medicine dropper, may be used to remove mercury if the trap-type glass container and vacuum cleaner are not available.

(d) Cellulose tape may be used to pick up very tiny particles of mercury. General cleanup and inspection (using equipment available) should be made immediately after spillage occurs or is detected.

(e) A fine copper wire brush can also be used to pick up mercury. It is suggested that a suitable brush could be manufactured locally by using the fine wires used in domestic flexible electrical cable. The procedure for using the copper brush is as follows:

1) Use one of the special mercury pickup brushes that attract the mercury. Foam pads are also commercially available that will pick up mercury.

2) Dip the brush in nitric acid (HN03) to clean the wires.

3) Dip the brush into water to remove the acid.

4) Dip the brush into alcohol to remove the water.

5) Pick up the mercury with the brush. Mercury adheres to the copper wires by forming an amalgam. After the brush has collected a quantity of mercury it can be shaken off into a suitable container, and the picking up process continued.

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(9) Remove the insulation blankets in the spill area and discard them, do this task: Insulation Blanket Removal, TASK 25-80-00-000-801.

(a) Also remove all of the insulation blankets that are downstream of the spill.

NOTE: Make sure you remove all blankets that go to the nearest lateral fluid dams.

(10) Where there is evidence of the presence of mercury in joints, between faying surfaces, or trapped in any way between structural members, they must be disassembled as necessary to permit complete removal of the mercury.

(11) Inspect bronze/brass control cable turnbuckle barrels for mercury discoloration. Replace if slightest discoloration is detected.

SUBTASK 05-51-14-750-001

(12) Heat Cycling and Ventilation (optional):

(a) Boeing is aware of the use of a heat cycle to vaporize mercury at the microscopic level.

1) This process has been used by the industry in an effort to completely eliminate mercury from the airplane.

2) This process includes monitoring vapor levels at the elevated temperatures, and venting the aircraft to remove the trace amounts

3) This process is for mercury removal process when visual indication is not entirely possible.

(b) The heat and vent process is accomplished multiple times, reaching temperatures as high as 138°F (59°C).

NOTE: External sources are exclusively used for heating. Aircraft air conditioning packs and equipment cooling are not used to heat aircraft to ensure systems are isolated and vapors are not cycled throughout clean areas. Upon reactivation of aircraft, the aircraft air conditioning and the ECS are cycled to ensure vents are cleared and cabin is clean.

1) Heating to such levels is more than 2.5 times the vaporization temperature (mercury readily vaporizes at 55°F (13°C).

2) Six remote thermometers are used to monitor internal temperatures for 12-hour periods to ensure aircraft structure reaches the elevated temperatures.

3) Following the heat, the aircraft is vented for 12 hours to clear the vapor overboard.

4) Each cycle of heat and ventilation can give significant drops in vapor levels at high and low temperatures.

F. Put the Airplane Back to Its Usual Condition

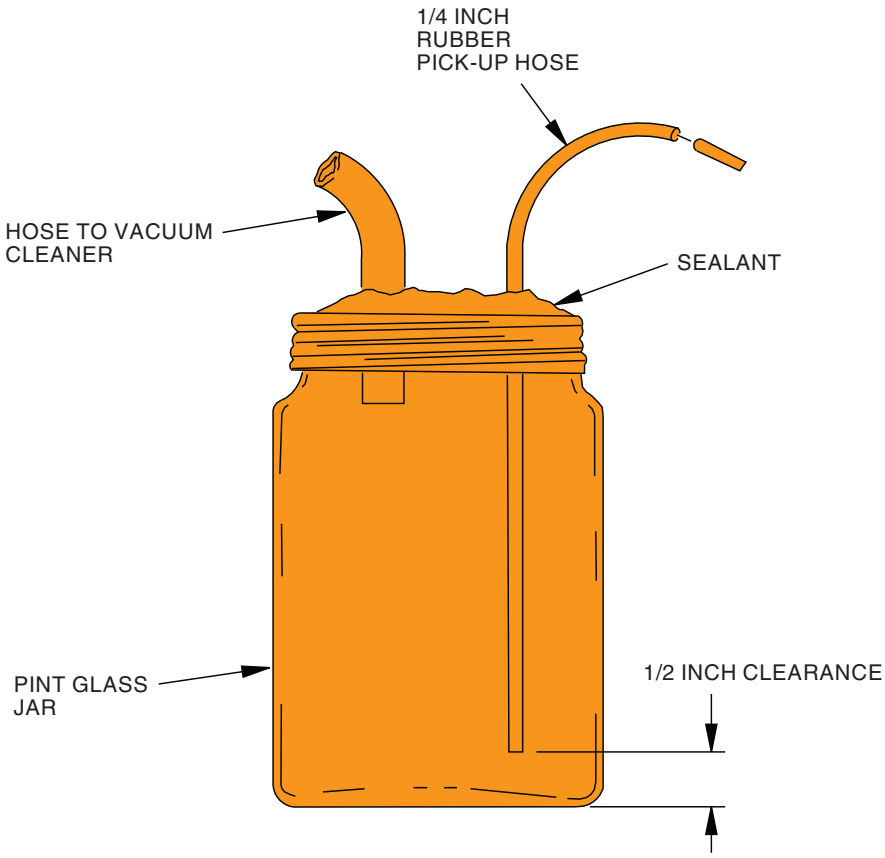
SUBTASK 05-51-14-410-001

(1) Install the components that you removed or install replaceable parts.

(a) Do this task: Insulation Blanket Installation, TASK 25-80-00-400-801.

——— END OF TASK ———

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NOTE:

MERCURY VAPOR MAY CONTAMINATE THE VACUUM CLEANER.
CLEAN THOROUGHLY AFTER USE.

2410412 S00061525504_V1

Vacuum Cleaner Mercury Trap
Figure 201/05-51-14-990-801

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BRAKE SEIZURE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to examine the wheel assemblies and brakes of the main landing gear after a brake seizure condition.

TASK 05-51-15-210-801

2. Brake Seizure Conditional Inspection

A. References

Reference	Title
05-51-07-210-801	High Energy Stop (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-700-803	Main Landing Gear Brake Check (Wheel Removed from the Airplane) (P/B 601)
32-45-00-700-802	Wheels Inspection (Wheel Removed from the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)

B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

C. Examine the Main Landing Gear Wheels and Brakes

SUBTASK 05-51-15-000-001

(1) Remove the wheel and tire assembly of the main landing gear, do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

SUBTASK 05-51-15-210-001

(2) Inspect the tires and wheels as follows:

(a) Do this task: Wheels Inspection (Wheel Removed from the Airplane), TASK 32-45-00-700-802.

(b) Do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

SUBTASK 05-51-15-000-002

(3) Remove the seized brake unit, do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

(a) Examine the brake unit grease O-ring seal(s), installed on the inner diameter of the brake, and make sure they are not damaged by heat or wear, do this task: Main Landing Gear Brake Check (Wheel Removed from the Airplane), TASK 32-41-41-700-803.

(b) Examine the wheel bearing seal adjacent to the brake and replace the seal if it has been damaged by heat.

SUBTASK 05-51-15-210-003

(4) Examine the brake sleeve and axle.

(a) If you find signs of an overheat condition, do this task: High Energy Stop, TASK 05-51-07-210-801.

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SUBTASK 05-51-15-210-004

(5) Examine the paint on the axles (High Energy Stop, TASK 05-51-07-210-801).

NOTE: Also look on the adjacent areas of the inner cylinder.

(a) Look for a brown shade color that is caused by an overheat condition.

(b) If the paint is a brown shade color, repair the damage.

D. Put the Airplane Back to its Usual Condition

SUBTASK 05-51-15-840-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLAT SPOTTED TIRES - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to inspect the wheels and brakes after the tires have become flat spotted.

(1) Flat spotted tires are from a locked wheel slide condition and not from a structural failure of the tire.

(2) You must do the inspections in the sequence given in this procedure to find the cause of the locked wheel condition.

B. The inspections are divided into two types.

(1) The first type of inspection is for conditions where only one tire is flat spotted.

(2) The second type of inspection is for conditions where multiple tires are flat spotted.

TASK 05-51-16-210-801

2. Flat Spotted Tires Conditional Inspection

A. References

Reference	Title
29-11-61-000-802	Return Filter Module Removal (P/B 401)
29-11-61-400-802	Return Filter Module Installation (P/B 401)
32-41-00-710-801	Normal and Alternate Brake Application Functional Test (P/B 501)
32-41-41-700-802	Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-42-00-720-801	Antiskid/Autobrake Control Unit Operational Test (P/B 501)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)
32-42-31-000-802	Antiskid Valve - Removal (P/B 401)
32-42-31-400-802	Antiskid Valve - Installation (P/B 401)
32-44-00-710-801	Parking Brake System Operational Test (P/B 501)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)

B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

C. One Flat Spotted Tire Condition

SUBTASK 05-51-16-210-001

(1) Do the following steps to examine a one flat spotted tire condition:

(a) Examine the tire, do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

(b) Examine the wheel, do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

(c) Examine the brake, do this task: Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802.

(d) Make sure that the brake operates correctly and is in a serviceable condition, do this task: Normal and Alternate Brake Application Functional Test, TASK 32-41-00-710-801.

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(e) Make sure that the routing of the hydraulic lines and wiring runs are correct.

(f) Do a functional test for the antiskid/autobrake system, do this task: Antiskid/Autobrake Control Unit Operational Test, TASK 32-42-00-720-801.

(g) Do the antiskid valve operational test, do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

(h) Do the transducer spin test and make sure that the correct brake releases when the transducer motion stops, do this task: Transducer Operational Test, TASK 32-42-00-710-803.

(i) If the tests above show that the system works correctly and inspections above do not show damage, replace the antiskid valve for the affected wheel, do the following steps:

1) Remove the antiskid valve, do this task: Antiskid Valve - Removal, TASK 32-42-31-000-802.

2) Install a new antiskid valve, do this task: Antiskid Valve - Installation, TASK 32-42-31-400-802.

D. Multiple Flat Spotted Tire Condition

SUBTASK 05-51-16-200-001

(1) Do the following steps to examine a multiple flat spotted tire condition:

(a) Examine the tires, do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

(b) Examine the wheels, do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

(c) Examine the brakes, do this task: Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802.

(d) Make sure that the brakes operate correctly and are in a serviceable condition, do this task: Normal and Alternate Brake Application Functional Test, TASK 32-41-00-710-801.

(e) Make sure that the routing of the hydraulic lines and wiring runs are correct.

(f) Do the antiskid valve operational test, do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

(g) Do the transducer spin test and make sure that the correct brakes release when the transducer motion stops, do this task: Transducer Operational Test, TASK 32-42-00-710-803.

(h) Make sure that the hydraulic return filter is not blocked.

1) Remove the hydraulic return filter module, do this task: Return Filter Module Removal, TASK 29-11-61-000-802.

2) Install the hydraulic return filter module, do this task: Return Filter Module Installation, TASK 29-11-61-400-802.

(i) Make sure that the parking brake valve operates correctly, do this task: Parking Brake System Operational Test, TASK 32-44-00-710-801.

1) Make sure that the valve is fully open when the parking brake releases.

(j) Do a BITE test for the antiskid/autobrake system, do this task: Antiskid/Autobrake Control Unit Operational Test, TASK 32-42-00-720-801.

1) Make sure that the antiskid/autobrake system works correctly.

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E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-16-410-001

(1) Install all the components that were removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WHEEL BEARING FAILURE/DAMAGE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection must be done before the next flight of the airplane.

B. This section has two tasks:

(1) Wheel Bearing Failure.

(2) Heat Damage Inspection of the Axle.

TASK 05-51-17-210-801

2. Wheel Bearing Failure

A. General

(1) This section gives an inspection procedure for landing gear wheels, brakes, axle sleeves, and axles after a wheel bearing failure.

(2) The degradation of wheel bearing components can occur as follows:

(a) Lubrication that is incorrect or not sufficient

(b) A decrease of the wheel bearing preload

(c) Contamination in the bearing

(d) Other damage that occurs.

(3) When you operate with damaged wheel bearings, it will cause heat in the adjacent parts:

(a) The axles

(b) The axle sleeves

(c) The spacers

(d) The wheel hubs.

(4) These parts can have scores if the bearing damage causes a seizure and then causes rotation that is relative to the adjacent parts.

(5) Do an inspection of the wheel bearings carefully during each tire change.

(6) The failure of a wheel bearing can be found by the symptoms or conditions that follow:

(a) Contamination of the wheel bearing grease with unwanted material.

(b) Metal particles or fragments in the area of the wheel and the brake

(c) The hubcap is damaged or is missing (main gear only)

(d) A wheel that is canted more than usual.

(e) The wheel and tire assembly are missing.

B. References

Reference	Title
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-41-41-700-803	Main Landing Gear Brake Check (Wheel Removed from the Airplane) (P/B 601)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)
32-42-21-400-801	Antiskid/Autobrake Control Unit Installation (P/B 401)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

D. Wheel Inspection

SUBTASK 05-51-17-210-001

(1) Do the wheel inspection. Do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

(a) When a main wheel shows signs of a wheel bearing failure, or other signs of damage, do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

NOTE: The wheel and tire assembly, with the bearings, must be sent to the shop for disassembly and a detailed inspection.

(b) When a nose wheel shows signs of a wheel bearing failure or other signs of damage, do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

NOTE: The wheel and tire assembly, and the bearings must be sent to the shop to be disassembled and inspected.

(c) Make sure the wheels that are removed have the correct wheel bearings installed.

E. Brake Inspection

SUBTASK 05-51-17-210-002

(1) Do an inspection of the brake assembly (main landing gear only). Do this task: Main Landing Gear Brake Check (Wheel Removed from the Airplane), TASK 32-41-41-700-803.

(a) Examine the parts that follow:

1) Rotors

2) Stators

3) Torque tube backing plate

4) Other parts of the brake assembly that can become damaged.

(b) When a brake assembly shows signs of damage or does not operate correctly, remove the brake assembly. Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: The brake assembly must be sent to the shop to be disassembled and inspected.

F. Axle Inspection

SUBTASK 05-51-17-210-003

(1) Do an inspection of the main gear axle and axle sleeve.

(a) Remove the components that follow if they have not been removed:

NOTE: Make sure you do a check for damage.

1) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

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2) Do this task: Antiskid Wheel Speed Transducer Removal, TASK 32-42-11-000-801.

(b) Remove the grease or corrosion inhibitive compound (CIC) from the axle if it is necessary.

(c) Visually examine the components that follow and look for paint or primer discoloration, blistering, and structural damage:

NOTE: Discoloration is green primer that has turned brown or black and/or enamel turned yellow, brown, or black.

NOTE: Damage is scoring, metal transfer, and/or displacement of material.

1) The outer diameter of the axle sleeve, check for scoring.

NOTE: Scoring will be caused when the wheel bearing turns on the axle sleeve.

a) Repair or replace the sleeve if necessary.

2) The brake sleeve, check for scoring on the end adjacent to the inner wheel bearing cone.

a) Repair or replace the sleeve if necessary.

(d) Visually examine the axle nut and washer for signs of scores or other damage.

1) Replace or repair the nut and washer if necessary.

(e) If there is no damage, discoloration, or blistering, you can leave the axle in service as it is.

NOTE: No more inspections to the axle are necessary.

(f) If the axle has signs of heat damage, do this task: Heat Damage Inspection of the Axle, TASK 05-51-17-210-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-17-410-001

(1) Install all of the components you removed if they are serviceable, or install replacement parts.

(a) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

(b) Do this task: Antiskid/Autobrake Control Unit Installation, TASK 32-42-21-400-801.

——— END OF TASK ———

TASK 05-51-17-210-802

3. Heat Damage Inspection of the Axle

A. References

Reference	Title
05-51-07-280-801	Heat Damage Inspection (Ammonium Persulfate Solution) (P/B 201)
32-11-85-000-801	Main Landing Gear Axle Removal (P/B 401)
51-21-98-300-801	Heat, Weather and Oil Resistant Inorganic Protective Coating Application (P/B 701)

B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

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C. Main Landing Gear (MLG) Axle

SUBTASK 05-51-17-210-004

(1) Examine the MLG axle for heat damage as follows:

NOTE: This step is necessary only when the axle sleeve show signs of scoring, discoloration, or other damage, and the axle sleeve has been removed.

(a) If it is necessary to reapply the heat-resistant protective coating, do this task: Heat, Weather and Oil Resistant Inorganic Protective Coating Application, TASK 51-21-98-300-801.

(b) Examine the light-gray hydraulic-fluid resistant paint on the axle.

1) Look for a brown color caused when the paint becomes too hot.

(c) Do a check for discoloration of the chrome plating on the axle lands where the sleeve touches.

(d) Examine the green primer on the inner surfaces (bore) of the axle with a borescope.

1) Use the borescope to a minimum depth of 16 in. (406 mm) and look for discoloration or blistered paint.

2) The green primer will turn to a light brown or black color when it becomes too hot.

NOTE: If the paint shows only a small discoloration, you can do the heat damage inspection when the airplane goes back to the primary base.

NOTE: Do not make more than three landings before you do this inspection.

SUBTASK 05-51-17-210-005

(2) If signs of heat damage to the main landing gear axle is found, do this task: Heat Damage Inspection (Ammonium Persulfate Solution), TASK 05-51-07-280-801.

(a) Remove the axle if it is necessary, do this task: Main Landing Gear Axle Removal, TASK 32-11-85-000-801.

SUBTASK 05-51-17-350-001

(3) If you find signs of damage in the areas that follow, repair or replace the damaged components:

NOTE: Heat damage to the cadmium plate can cause cadmium embrittlement of the steel substrate.

(a) The cadmium plate on the axle bore shows signs of heat damage.

(b) The chrome plate on the outer part of the axle shows signs of heat damage.

D. Nose Landing Gear (NLG) Axle

SUBTASK 05-51-17-210-006

(1) Do an inspection of the nose landing gear axle.

(a) Visually examine the parts that follow for scores or other damage:

1) The wheel bearing washer

2) The axle nut

3) The inner wheel bearing spacer.

(b) Replace or repair the components if necessary.

(c) Examine the NLG axle for damage.

1) Look for scores or discoloration of the chrome plated bearing lands on the axle.

2) Examine the light-gray, hydraulic-fluid-resistant, paint on the axle.

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a) The paint will become a brown color when it becomes too hot.

3) Examine the green primer on the inner surfaces (bore) of the axle with a borescope.

a) Use the borescope to a minimum depth of 7 in. (178 mm) and look for discoloration or blistering

b) The green primer will turn to a light brown or black color when it becomes too hot.

NOTE: If the paint shows only a small discoloration, you can do the heat damage inspection when the airplane goes back to the primary base.

NOTE: Do not make more than three landings before you do this inspection.

SUBTASK 05-51-17-210-007

(2) If signs of heat damage to the nose gear axle are found, do this task: Heat Damage Inspection (Ammonium Persulfate Solution), TASK 05-51-07-280-801.

SUBTASK 05-51-17-350-002

(3) If you find signs of damage in the areas that follow, repair or replace the damaged components:

NOTE: Heat damage to the cadmium plate can cause cadmium embrittlement of the steel substrate.

(a) The cadmium plate on the axle bore shows signs of heat damage

(b) The chrome plate on the outer part of the axle shows signs of heat damage.

E. Put the airplane back to its usual condition.

SUBTASK 05-51-17-400-001

(1) Install all of the components you removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BIRD/HAIL STRIKE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains these tasks:

(1) Bird strike and in-flight hail strike conditional inspection.

(2) Hail strike on the ground conditional inspection.

TASK 05-51-18-210-801

2. Bird Strike and In-flight Hail Strike Conditional Inspection

A. General

(1) All of the inspections are visual unless shown differently in the procedure.

(2) Examine the external surfaces of the airplane structure in the general area of the bird/in-flight hail strike.

(a) If the initial inspection shows structural damage, then the internal structure must be inspected.

(b) If the bird/in-flight hail struck the nose radome, the inside of the radome must be inspected even if no damage is found on the exterior of the nose radome.

1) If the outer skin-to-core of the radome does not compress when you apply strong finger pressure to impact area, then do the internal inspection at or before 3 flight cycles.

(c) Also inspect the hydraulic, pneumatic, and other systems in the area of the bird/in-flight hail strike for damage.

(3) When the conditional inspection tells you to "examine" a component, look for these conditions (repair or replace components, if it is necessary):

(a) Cracks

(b) Pulled apart structure

(c) Loose paint (paint flakes)

(d) Twisted parts (distortion)

(e) Bent components

(f) Ruptures

(g) Loose fasteners

(h) Fasteners holes that became larger or longer

(i) Fasteners that have pulled out or are gone

(j) Delamination

(k) Fiber breakouts

(l) Misalignment

(m) Interference

(n) Other signs of damage.

(4) If damage is found during these inspections, go to the related maintenance manual section for the repair.

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B. References

Reference	Title
12-40-04-100-801	Bird Strike Cleaning (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-710-801	Trailing Edge Flap System Operational Test (P/B 501)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-700-801	Leading Edge Flap and Slat Operational Test in Primary Mode (P/B 501)
53-54-00-211-801	Broadband Radome Inspection (P/B 601)
71-00-00-200-802-G00	Engine Foreign Object Damage and Birdstrike Inspection (P/B 601)

C. Location Zones

Zone	Area
111	Radome
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

D. Bird Strike and In-flight Hail Strike Conditional Inspection

SUBTASK 05-51-18-040-001

WARNING PUT ON EQUIPMENT FOR PROTECTION BEFORE YOU TOUCH THE BIRD CARCASS, BLOOD, GUTS, AND RESIDUE. THIS CAN CONTAIN BACTERIA AND VIRUSES THAT CAN CAUSE ILLNESSES, AND INJURIES TO PERSONNEL.

WARNING DO NOT LET THE BIRD CARCASS OR OTHER PIECES OF THE BIRD TOUCH YOUR SKIN. DISCARD THE BIRD PIECES IN A PLASTIC DISPOSAL BAG. THE BIRD PIECES CAN CONTAIN INFECTIOUS MATERIALS (BACTERIA AND VIRUSES). THEY CAN CAUSE ILLNESSES, AND INJURIES TO PERSONNEL.

(1) Before you touch any area of the airplane that may have been affected by the bird strike, do this task: Bird Strike Cleaning, TASK 12-40-04-100-801.

SUBTASK 05-51-18-040-004

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Extend the leading edge flap and slat system and the trailing edge flap system in the full extend position.

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(a) Do the deactivation procedure for the leading edge flap and slat system, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

(b) Do the deactivation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-18-210-001

(3) Examine the areas that follow, in particular, inspect the nearby components of these systems if evidence of bird remains or structural damage are found:

(a) The external areas of the radome for:

1) Cracks
2) Delamination
3) Soft spots
4) Core damage.

(b) The internal areas of the radome for:

1) Cracks
2) Delamination
3) Soft spots
4) Core damage.

(c) The pilots' windows for:

1) Delamination
2) Spalling
3) Cracks.

(d) The forward-body nose section "eyebrow" area above the windows and radome

(e) The wing

(f) The winglet

(g) The nacelle strut

(h) The leading edge fairing of the horizontal stabilizer

(i) The leading edge fairing of the vertical stabilizer

(j) Examine the strut panels, doors, and structure.

(k) Examine the wing leading edge and the wing trailing edge structure, panels, and doors.

1) Also examine the two sides of the honeycomb panels for:

a) Cracks
b) Delamination
c) Soft spots
d) Core damage.

(l) Examine the leading edge slat mechanism and track fairing links.

(m) Examine the trailing edge flap mechanism and track fairing links.

(n) Examine the flight control surfaces.

(o) The passengers windows for:

1) Displacement of the outer panel
2) Crazing

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3) Cracks

4) Other types of window damage.

(p) Examine the nose landing gear doors and linkage.

(q) Examine the main landing gear doors and linkage.

(r) If the landing gear was extended during the bird/in-flight hail strike, examine the landing gear.

SUBTASK 05-51-18-210-002

(4) Do a general visual inspection of each engine nose cowl and the engine intake for signs of bird strike or foreign object damage.

(a) If there is indication of a bird strike or foreign object damage, do this task: Engine Foreign Object Damage and Birdstrike Inspection, TASK 71-00-00-200-802-G00.

SUBTASK 05-51-18-210-003

(5) Examine the external fuselage structure.

(a) If you find external damage, get access to the internal areas and examine:

1) All of the frames

2) The fittings and the component mounts

3) All of the components in the area

4) The pipes

5) The pulleys

6) The mechanical linkage.

7) Refer to Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 for the analysis and continued service of airplanes with hail damage.

SUBTASK 05-51-18-211-001

(6) Inspect the broadband radome for damage (TASK 53-54-00-211-801).

SUBTASK 05-51-18-210-004

(7) Inspect the hydraulic, pneumatic, and other systems in the area of the bird/in-flight hail strike for damage.

SUBTASK 05-51-18-440-001

(8) Do the activation procedure for the leading edge flap and slat system, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 05-51-18-440-002

(9) Do the activation procedure for the trailing edge flap system, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-18-710-001

(10) Do the tests that follow if you worked on the system subsequent to the inspection for damage:

(a) Do this task: Leading Edge Flap and Slat Operational Test in Primary Mode, TASK 27-81-00-700-801.

(b) Do this task: Trailing Edge Flap System Operational Test, TASK 27-51-00-710-801.

E. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-18-400-001

(1) Install the components you removed or install replacement parts.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-51-18-212-801

3. Hail Strike on the Ground Conditional Inspection

A. General

(1) This inspection is required if hail reported as GR 0.25 in. (6.35 mm) or greater in size strikes the aircraft.

(a) If the GR hail struck the nose radome, the inside of the radome must be inspected even if no damage is found on the exterior of the nose radome.

1) If the outer skin-to-core of the radome does not compress when you apply strong finger pressure to impact area, then do the internal inspection at or before 3 flight cycles.

(b) For GR hail less than 0.25 in. (6.35 mm), no inspection is required.

(2) This inspection is not required if GS hail strikes the aircraft.

(a) GS hail is defined as white, opaque grains of ice. The pellets are round or sometimes conical and are less than 0.2 in. (5 mm) in diameter.

B. References

Reference	Title
27-51-00 P/B 201	TRAILING EDGE FLAP SYSTEM - MAINTENANCE PRACTICES
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-710-801	Trailing Edge Flap System Operational Test (P/B 501)
27-81-00 P/B 201	LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - MAINTENANCE PRACTICES
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-700-801	Leading Edge Flap and Slat Operational Test in Primary Mode (P/B 501)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors
800	Doors

D. Additional Information

NOTE: Additional information useful for this procedure may be found in these references:

- TRAILING EDGE FLAP SYSTEM - MAINTENANCE PRACTICES, PAGEBLOCK 27-51-00/201
- LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - MAINTENANCE PRACTICES, PAGEBLOCK 27-81-00/201.

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E. Hail Strike on the Ground Conditional Inspection

SUBTASK 05-51-18-040-002

> **WARNING** DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do the deactivation procedure for the leading edge flaps and slats system: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-51-18-040-003

> **WARNING** DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the deactivation procedure for the trailing edge flap system: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-18-212-001

(3) Do the visual examinations of the areas that follow:

NOTE: When you examine the honeycomb panels, look at the two sides for cracks, delamination, soft spots and core damage.

(a) The nose radome.

(b) The wings, for signs of hail damage to the areas that follow:

1) All horizontal surfaces
2) Leading edges
3) Trailing edges
4) Panels.

(c) The winglets.

(d) The flight control surfaces.

(e) The leading edge fairings on the horizontal stabilizer.

(f) The leading edge fairings on the vertical stabilizer.

(g) The external fuselage structure.

1) If signs of hail damage to the external fuselage structure is found, refer to Aerodynamic Smoothness Requirements/SRM PROCEDURE 51-10-01-0G-0 for investigation and clean-up of damage.

(h) The flight compartment windows, for signs of hail damage as follows:

1) Cracks
2) Displacement of the outer window
3) Other types of window damage.

(i) The forward fuselage section above the flight compartment windows and the radome.

(j) The wing tip fairings and navigation lights.

(k) The passenger and cargo doors.

(l) The nacelle strut.

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(m) The nose landing gear doors.

(n) The main landing gear doors.

(o) The passenger compartment windows for signs of hail damage as follows:

1) Cracks

2) Displacement of the outer window

3) Crazing

4) Other types of window damage.

(p) Examine the nose cowl and the engine for signs of foreign object damage.

(q) The broadband radome.

SUBTASK 05-51-18-212-002

(4) The allowable damage shown in the Structural Repair Manual (SRM) is to be used to define the limits of damage within each component.

SUBTASK 05-51-18-440-003

(5) Do the reactivation procedure for the leading edge flap and slat system: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 05-51-18-440-004

(6) Do the reactivation procedure for the trailing edge flap system: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-18-710-002

(7) Do the tests that follow if you worked on the system subsequent to the inspection for damage:

(a) Do this task: Leading Edge Flap and Slat Operational Test in Primary Mode, TASK 27-81-00-700-801.

(b) Do this task: Trailing Edge Flap System Operational Test, TASK 27-51-00-710-801.

——— END OF TASK ———

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LIGHTNING STRIKE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. After an airplane lightning strike report, an examination of the airplane is done to find possible damage.

B. The term "Examine" or "Examination" is used to describe the inspection actions intended to direct airline personnel to find lightning strike damage. These terms do not indicate the intensity, access or tools recommended for completion of the inspection action.

C. This procedure has these tasks:

(1) Inspection for Lightning Strike Damage

(a) Phase I Inspection

(b) Phase II Inspection

(2) Inspection and Operational Check of Radio and Navigation Systems

D. Lightning Strike Examination

(1) The lightning strike inspection is divided into two phases.

(a) Phase I - Inspection of metallic structures, non-metallic (composite) structures and system components.

(b) Phase II - General Visual Inspection (GVI) of all areas.

(2) Phase II Inspection may be deferred up to 50 flight cycles for the following conditions:

(a) No damage was found during Phase I Inspection.

(b) Damage was found and assessed/resolved accordingly during Phase I Inspection.

E. The metal fuselage and wing supply the basic protection from lightning for the fuel system and critical electronic systems.

(1) Necessary protection is also supplied for the non-metal structure by aluminum mesh (the rudder has a picture frame configuration).

(2) The external structure prevents fuel ignition and electrical/electronic system damage.

(3) Critical electronic systems also rely on wire shields and proper shield terminations for indirect effects protection.

F. Typical lightning strike behavior

(1) There will be a minimum of two attach points at different areas of the airplane surface, an entrance and an exit.

(2) When lightning initially strikes on the forward fuselage or engine nacelles, it will move rearward over the fuselage surface or over the wing surface.

(3) On the wing surface, the lightning will move aft of the nacelle or aft of the extended ends of the leading edge slats.

(4) When lightning initially strikes the aft area of the empennage or wing surface, it will stay there until the lightning current stops.

(5) Lightning strike attach points are typically found in Zone 1, but also occur in Zone 2 and 3.

(6) Lightning strikes in Zone 1 typically exit a different location in Zone 1. The external components most likely to be hit are listed below (Figure 202):

(a) wing leading edge

(b) wing trailing edge

(c) wingtips/winglets

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(d) engines

(e) vertical fin tip

(f) horizontal stabilizer tips

(g) elevators

(h) nose radome

(i) tailcone area

(j) broadband radome.

(7) Areas where initial lightning strikes typically do not occur are as follows:

(a) drain masts

(b) pitot probes

(c) blade antennas

(d) extended ends of leading edge slats

(e) trailing edge flap track fairing tips

(f) landing gear.

G. Direct Damage occurs when there is a visible effect that degrades the performance of a structure or components. Typical indications are as follows:

(1) Metallic surfaces appear burned, melted or show signs of metal distortion, and may have small circular melted marks of approximately ⅛ in. (3 mm) in diameter.

(2) Non-metallic surfaces (composite) appear burned, punctured or delaminated, and may have burns or black marks up to 4 in. (102 mm) in diameter.

(3) Marks appear along the larger area of a trailing edge surface.

(4) Holes appear in the skin, created from high intensity lightning.

(5) Skins and fasteners appear burned or discolored.

(6) Fluid leakage appears from damaged aluminum hydraulic tubing joints on the vertical stabilizer, or horizontal stabilizer, or empennage.

H. Indirect Damage can result from the large electrical transients caused by lightning. Typical indications are as follows:

(1) Failures in electrical/electronic systems equipment.

(2) Degraded or disturbed wire shielding and shield termination.

(3) Problems occur with components located external to the pressurized fuselage.

NOTE: While the electrical system is protected from a possibility of direct lightning strike to electrical/electronic wires and cables, damage can occur. A high intensity lightning strike can produce electromagnetic fields within the unpressurized areas that are large enough to cause possible damage to the electrical system components.

I. Static dischargers, located on the tips and trailing edges of airfoils, are frequently hit and damaged by lightning.

(1) The primary function of a static discharger is to dissipate static charges to prevent interference on communication and navigation equipment.

(2) Some static dischargers are put in areas to prevent damage to lights or other system components from a lightning strike.

(3) These devices do not prevent the lightning strikes on the airplane.

J. System Verification

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(1) If a system malfunction is reported by the flight crew then a system check is required.

(2) All installed systems required for flight must be operational.

TASK 05-51-19-210-802

2. Phase I Inspection

A. General

(1) Use of visual aids from the ground is acceptable to perform the following procedure.

(2) If a system malfunction is reported by the flight crew then a system check is required.

(3) Any requests for engineering analysis and repairs must contain the following information:

(a) Clearly defined damage:

1) Damage location defined with dimensioned on the production drawings and photos for reference.

2) Damage extent.

3) Cause of damage.

(b) Proposed repair Plan.

(c) Any limitations - equipment, materials, etc.

(d) Status of the aircraft and airplane Return to Service date.

(e) Desired outcome (Permanent, Time Limited/Temporary, or Ferry Flight).

(4) Phase II Inspection may be deferred up to 50 flight cycles for the following conditions:

(a) No damage was found during Phase I Inspection.

(b) Damage was found and assessed/resolved accordingly during Phase I Inspection.

B. References

Reference	Title
23-61-00-210-801	Static Discharger Inspection (P/B 601)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-807	Rudder Trim System Test (P/B 501)
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-710-801	Elevator and Elevator Trim Control System - Operational Test (P/B 501)
27-31-00-710-802	Elevator Control Column Override - Operational Test (P/B 501)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
28-41-00-720-801	Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check (P/B 501)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
33-51-00-720-801	Emergency Lights Operational Test (P/B 501)
34-11-01-200-801	Pitot Probe Detailed Inspection (P/B 601)
34-11-02-200-801	Static Port - Detailed Inspection (P/B 601)
34-21-04 P/B 401	AIR DATA MODULE - REMOVAL/INSTALLATION
34-21-05-200-801	Angle of Attack (AOA) Sensor - Inspection (P/B 601)

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(Continued)

Reference	Title
34-23-00-730-801	Standby Magnetic Compass System (Spot Check) - Inspection (P/B 201)
34-23-00-820-801	Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method (P/B 201)
34-23-00-820-802	Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method (P/B 201)
34-23-00-820-803	Standby Magnetic Compass - Degaussing (P/B 201)

C. **Location Zones**

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors
800	Doors

D. **Phase I - Metallic Surfaces Inspection**

SUBTASK 05-51-19-210-013

(1) Perform a ground level inspection of Zones 1, 2, and 3 visible external metallic surfaces for signs of lightning strike damage (Figure 202).

NOTE: Use of stands or ladders are not required.

(a) Look for visible signs of discolored paint, burn marks, melted areas, or holes in skin.

(b) If any damage was found, the damage must be assessed and/or repaired immediately per the Structural Repair Manual (SRM) or approved method.

SUBTASK 05-51-19-210-014

(2) Perform a ground level inspection around the fuel areas of the wing torque box (Figure 201).

NOTE: A wing torque box includes the outboard wing, the wing center section, and the side of the body structure. The outboard wing has upper and lower skin panels and front and rear spar with ribs found perpendicular to the rear spar. The center section contains an in-spar wing box and wing-to-body interface structure.

(a) If any damage was found, assess and/or repair the damage per the SRM or approved method.

SUBTASK 05-51-19-210-016

(3) Perform a ground level inspection of the wingtips or winglets metallic structure for lightning strike damage.

(a) Examine the fuel vent outlet and surge tanks for signs of damage.

(b) If any damage was found, address the damage per the SRM or approved method.

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E. Phase I - Composite Surfaces Inspection

SUBTASK 05-51-19-210-018

(1) Perform a ground level inspection of Zones 1, 2, and 3 external composite surfaces for signs of lightning strike damage(Figure 202).

NOTE: Use of stands or ladders are not required.

(a) Look for discolored paint, burns, punctures and delamination of composite honeycomb sandwich structure.

(b) Examine the external composite surfaces from the ground of the nose radome for burns, punctures, and pin holes in the composite honeycomb sandwich structure.

1) If damage is found, examine the internal surfaces and components under the nose radome.

(c) If lightning strike damage is found, examine all composite structure in perimeter of damaged area for signs of delamination.

1) Use instrumental or non-instrumental Non-Destructive Inspection (NDI) tap testing to find composite structure delamination you cannot see.

a) For a tap test, use a solid metal disk and tap the area adjacent to the damaged area lightly.

NOTE: If there is delamination, you will hear a sound that is different than the sound of a solid bonded area. Refer to the Nondestructive Test Manual.

2) If any damage was found, address the damage per the SRM or approved method.

SUBTASK 05-51-19-210-020

(2) Perform a ground level inspection of the wingtips or winglets composite structure.

(a) If any damage was found, address the damage per the SRM or approved method.

F. Phase I - System Components Inspection

SUBTASK 05-51-19-210-032

(1) Perform a ground level inspection of the Angle of Attack (AOA) sensors, the pitot probes, static ports, and the surrounding areas for damage.

(a) Look for damage to the AOA sensors.

1) If damage is found, do this task: Angle of Attack (AOA) Sensor - Inspection, TASK 34-21-05-200-801.

(b) Look for damage to the pitot probes.

1) If damage is found, do this task: Pitot Probe Detailed Inspection, TASK 34-11-01-200-801.

a) If the removal of the pitot probes is necessary, do this tasks: AIR DATA MODULE - REMOVAL/INSTALLATION, PAGEBLOCK 34-21-04/401.

(c) Look for damage to the static ports.

1) If damage is found, do this task: Static Port - Detailed Inspection, TASK 34-11-02-200-801.

a) If the removal of the static ports is necessary, do these tasks: AIR DATA MODULE - REMOVAL/INSTALLATION, PAGEBLOCK 34-21-04/401.

(d) If any damage is found around the AOA sensors, pitot probes and static ports.

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1) The damage must be assessed and/or repaired per the SRM and Reduced Vertical Separation Minimum (RVSM) Requirements/SRM PROCEDURE 51-10-03-0G-0 or approved method.

a) Look for burns, punctures, discolored paint, and general skin distortion.

SUBTASK 05-51-19-210-015

(2) Perform a ground level inspection of all external lights for lightning strike damage.

(a) Look for broken light assemblies, broken or cracked lenses, or other visible damage.

(b) Repair or replace damaged parts as necessary until the light functions properly.

(c) If damage was found on the Anti-Collision or Emergency Lights, do a test of the lights that follow:

1) Do this task: Anti-Collision Lights - Operational Test, TASK 33-44-00-710-801.

2) Do this task: Emergency Lights Operational Test, TASK 33-51-00-720-801.

SUBTASK 05-51-19-700-001

(3) Perform a ground level inspection of the flight control components for signs of lightning strike damage.

(a) If the flight control components show signs of lightning strike damage, perform a General Visual Inspection (GVI) of these items:

1) Surface hinges

2) Bearings

3) Bondings.

(b) If no damage is found, do a basic operational check of the flight controls as follows:

NOTE: If the flight controls operate correctly after you do these steps, no more inspections are necessary.

1) Move the rudder, control wheel, and control column in both directions.

2) Make sure that there is full travel and the flight controls move freely.

NOTE: Hold the nose wheel during the rudder check to prevent nose wheel movement. You can hold the nose wheel if you put in the steering lockout pin or remove the torsion link pin.

SUBTASK 05-51-19-720-001

(4) If the rudder system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the rudder system.

(a) Do this task: Rudder Trim System Test, TASK 27-21-00-700-807.

(b) Do this task: Rudder Pedal Forces Test, TASK 27-21-00-700-804.

(c) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

(d) Do this task: Rudder Centering Test, TASK 27-21-00-700-803.

SUBTASK 05-51-19-720-002

(5) If the elevator system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the elevator system.

(a) Do this task: Elevator Control Column Override - Operational Test, TASK 27-31-00-710-802.

(b) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

(c) Do this task: Elevator and Elevator Trim Control System - Operational Test, TASK 27-31-00-710-801.

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SUBTASK 05-51-19-720-003

(6) If the aileron system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the ailerons.

(a) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

SUBTASK 05-51-19-210-017

(7) Perform a ground level inspection of the Static Dischargers for lightning strike damage.

(a) If any Static Dischargers are damaged or missing, assess with: Static Discharger Inspection, TASK 23-61-00-210-801.

G. External Wire Shields Detailed Inspection

SUBTASK 05-51-19-211-001

(1) Examine the following:

NOTE: If the lightning strike attach points can not be identified, the external wire shield examination is not required.

NOTE: If there are no wire shield termination(s) in the area of the attach points, the external wire shield examination is not required.

(a) In the area of the lightning strike attach points, examine the wire shield terminations.

(b) If the flight crew has reported a problem with any system after the lightning strike, examine the wire shield terminations for that systems wiring.

(c) If damage is found at wire shield termination(s), check the full length of the wire bundle.

(d) Repair any damage found and do an operational check of the system that had the damage.

H. System Internal Test

SUBTASK 05-51-19-700-002

(1) If a lightning strike caused a system malfunction, do a full test of the defective system.

SUBTASK 05-51-19-700-003

(2) If a magnetic compass malfunction is reported in the maintenance log, do a check of the standby magnetic compass:

(a) Do this task: Standby Magnetic Compass System (Spot Check) - Inspection, TASK 34-23-00-730-801.

1) If the result is not in limits, do this task: Standby Magnetic Compass - Degaussing, TASK 34-23-00-820-803.

NOTE: It is not necessary to do one of these methods this time: Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method, TASK 34-23-00-820-802 or Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method, TASK 34-23-00-820-801.

(b) Do this task again to make sure the result is in the limits: Standby Magnetic Compass System (Spot Check) - Inspection, TASK 34-23-00-730-801.

1) If the result is not in limits, do one of these tasks: Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method, TASK 34-23-00-820-802 or Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method, TASK 34-23-00-820-801.

SUBTASK 05-51-19-710-022

(3) Make sure that the fuel quantity system is accurate.

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(a) Do the operational test for the Fuel Quantity Indicating System (FQIS) type on your airplane, do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

(b) If the FQIS test shows any problems, do this task: Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check, TASK 28-41-00-720-801.

SUBTASK 05-51-19-710-026

(4) If the rudder or vertical stabilizer had a lightning strike, do these steps:

(a) Look for leaks from the rudder hydraulic aluminum return tubing aft of the aft pressure bulkhead.

(b) If leaks are found, replace components that have leaks.

SUBTASK 05-51-19-710-027

(5) If the elevator or horizontal stabilizer had a lightning strike, do these steps:

(a) Look for leaks from the elevator hydraulic aluminum return tubing aft of the aft pressure bulkhead.

(b) If leaks are found, replace components that have leaks.

I. Inspection and Operational Check of Radio and Navigation Systems

SUBTASK 05-51-19-210-021

(1) Do the inspection of the radio and navigation systems.

NOTE: All installed systems required for flight must be operational.

(a) Refer to Figure 203 to determine if an operational check is necessary.

(b) If the radio and navigation systems do not operate correctly, do this task: Inspection and Operational Check of Radio and Navigation Systems, TASK 05-51-19-710-801.

——— END OF TASK ———

TASK 05-51-19-210-803

3. Phase II Inspection

A. General

(1) Any requests for engineering analysis and repairs must contain the following information:

(a) Clearly defined damage:

1) Damage location defined with dimensioned on the production drawings and photos for reference.

2) Damage extent.

3) Cause of damage.

(b) Proposed repair Plan.

(c) Any limitations - equipment, materials, etc.

(d) Status of the aircraft and airplane Return to Service date.

(e) Desired outcome (Permanent, Time Limited/Temporary, or Ferry Flight).

B. References

Reference	Title
23-61-00-760-801	Static Discharger Resistance Measurement (P/B 601)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)

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(Continued)

Reference	Title
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-807	Rudder Trim System Test (P/B 501)
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-710-801	Elevator and Elevator Trim Control System - Operational Test (P/B 501)
27-31-00-710-802	Elevator Control Column Override - Operational Test (P/B 501)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
28-41-00-720-801	Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check (P/B 501)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
33-51-00-720-801	Emergency Lights Operational Test (P/B 501)
34-23-00-730-801	Standby Magnetic Compass System (Spot Check) - Inspection (P/B 201)
34-23-00-820-801	Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method (P/B 201)
34-23-00-820-802	Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method (P/B 201)
34-23-00-820-803	Standby Magnetic Compass - Degaussing (P/B 201)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors
800	Doors

D. Phase II - Metallic Surfaces and Structures Inspection

SUBTASK 05-51-19-210-022

NOTE: If a 50 flight cycle deferral is provided, perform these steps on or before the 50 flight cycle limit.

(1) Perform a General Visual Inspection (GVI) of Zones 1, 2 and 3 metallic surface areas for signs of lightning strike damage (Figure 202).

(a) Utilize any available equipment at the time to examine required areas of the aircraft.

(b) Zone 1 includes these areas:

1) Wing Leading Edge (LE)

2) Wing Trailing Edge (TE)

3) Wingtips/Winglets

4) Engines

5) Vertical Fin Tip

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6) Horizontal Stabilizer Tips
7) Elevators
8) Nose Radome
9) Broadband Radome
10) Tailcone Area.

(c) Examine the fuel areas of the wing torque box (Figure 201).

NOTE: A wing torque box includes the outboard wing, the wing center section, and the side of the body structure. The outboard wing has upper and lower skin panels and front and rear spar with ribs found perpendicular to the rear spar. The center section contains an in-spar wing box and a wing-to-body interface structure.

(d) Examine all external metallic surfaces for any evidence of visible external lightning strike damage.

(e) Examine the external metallic surfaces to find the lightning strike attach points.

(f) Look in the areas where one surface stops and another surface starts (for example, butt joints and lap joints).

(g) Examine the metallic structure looking for signs of lightning strike damage such as: holes, pits, melting, burns, or discoloration.

CAUTION

SEAL OR REPAIR ALL DAMAGE. IF YOU DO NOT SEAL OR REPAIR ALL DAMAGE, MOISTURE CAN GET IN. MOISTURE CAN CAUSE MORE INTERNAL DAMAGE.

(h) Repair or seal the damaged areas per the Structural Repair Manual (SRM).

E. Phase II - Composite Surfaces and Structures Inspection

SUBTASK 05-51-19-210-024

NOTE: If a 50 flight cycle deferral is provided, perform these steps on or before the 50 flight cycle limit.

(1) Perform a GVI of Zones 1, 2 and 3 composite surface areas for signs of lightning strike damage (Figure 202).

(a) Utilize any available equipment at the time to examine required areas of the aircraft.

(b) Zone 1 includes these areas:

1) Wing LE
2) Wing TE
3) Wingtips/Winglets
4) Engines
5) Vertical Fin Tip
6) Horizontal Stabilizer Tips
7) Elevators
8) Nose Radome
9) Broadband Radome
10) Tailcone Area.

(c) Examine all composite structures for discolored paint, burned, or punctured skin.

(d) Examine the external composite surfaces to find the lightning strike attach points.

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(e) Look in the areas where one surface stops and another surface starts (for example, butt joints and lap joints).

(f) Examine the external composite surfaces of the nose radome for burns, punctures, and pin holes in the composite honeycomb sandwich structure.

1) If damage is found, examine the internal surfaces and components under the nose radome.

(g) Examine the external composite surfaces for discolored paint, burns, or punctures.

1) If lightning strike damage is found, examine all composite structure in perimeter of damaged area for signs of delamination.

NOTE: Types of damage you cannot see, like delamination, can extend to the areas beyond the lightning strike damage. You can find delamination by instrumental Non-Destructive Inspection (NDI) methods or by a tap test.

2) Use instrumental or non-instrumental NDI tap testing to find composite structure delamination you cannot see.

a) For a tap test, use a solid metal disk and tap the area adjacent to the damaged area lightly.

NOTE: If there is delamination, you will hear a sound that is different than the sound of a solid bonded area. Refer to the Nondestructive Test Manual.

CAUTION

SEAL OR REPAIR ALL DAMAGE. IF YOU DO NOT SEAL OR REPAIR ALL DAMAGE, MOISTURE CAN GET IN. MOISTURE CAN CAUSE MORE INTERNAL DAMAGE.

(h) Repair or seal the damaged areas per the SRM.

SUBTASK 05-51-19-210-028

(2) Perform a GVI of the wingtips or winglets composite structure.

(a) If the composite structure show signs of lightning strike damage, address the damage per the SRM or approved method.

F. Phase II - System Components Inspection

SUBTASK 05-51-19-210-023

(1) Perform a detailed inspection of the Angle of Attack (AOA) sensors.

(a) Do a movement check of the AOA sensors.

NOTE: The movement must be smooth and must not have any rough spots in the travel range during the movement check.

(b) Examine for melted sealant or damaged AOA sensors.

SUBTASK 05-51-19-210-033

(2) Examine the pitot probes, static ports, and the areas around them for damage and look for burns, punctures, discolored paint, and general skin distortion.

(a) If damage is found, the damage must be assessed and/or repaired per the SRM and Reduced Vertical Separation Minimum (RVSM) Requirements/SRM PROCEDURE 51-10-03-0G-0 or approved method.

SUBTASK 05-51-19-210-025

(3) Perform a GVI of all of the external lights.

(a) Look for these items:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Broken light assemblies.
2) Broken or cracked lenses.
3) Other visible damage.

(b) If you find damaged lights, do these steps:

1) Examine these items:
   a) The internal wires of the light fixture for damage.
   b) The first 3 ft (1 m) of wiring from the damaged light in the direction of the circuit breaker.
   c) The circuit breaker terminals and the first 1 in. (25 mm) of wires from the terminals.
2) Repair or replace damaged parts as necessary until the light functions properly.
3) If the circuit breaker did not trip, then no further examinations for the wiring of the damaged light are required.
4) If the circuit breaker did trip, then locate the source of the anomaly and repair as necessary.

(c) Do a test of the lights that follow:

1) Do this task: Anti-Collision Lights - Operational Test, TASK 33-44-00-710-801.
2) Do this task: Emergency Lights Operational Test, TASK 33-51-00-720-801.

SUBTASK 05-51-19-210-026

(4) If the flight control components show signs of lightning strike damage, perform a GVI of these items:

(a) Surface hinges.
(b) Bearings.
(c) Bondings.

SUBTASK 05-51-19-210-027

(5) If no damage is found, do a basic operational check of the flight controls as follows:

NOTE: If the flight controls operate correctly after you do these steps, no more inspections are necessary.

(a) Move the rudder, control wheel, and control column in both directions.
(b) Make sure that there is full travel and the flight controls move freely.

NOTE: Hold the nose wheel during the rudder check to prevent nose wheel movement. You can hold the nose wheel if you put in the steering lockout pin or remove the torsion link pin.

SUBTASK 05-51-19-720-007

(6) If the rudder system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the rudder system.

(a) Do this task: Rudder Trim System Test, TASK 27-21-00-700-807.
(b) Do this task: Rudder Pedal Forces Test, TASK 27-21-00-700-804.
(c) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.
(d) Do this task: Rudder Centering Test, TASK 27-21-00-700-803.

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SUBTASK 05-51-19-720-008

(7) If the elevator system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the elevator system.

(a) Do this task: Elevator Control Column Override - Operational Test, TASK 27-31-00-710-802.

(b) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

(c) Do this task: Elevator and Elevator Trim Control System - Operational Test, TASK 27-31-00-710-801.

SUBTASK 05-51-19-720-009

(8) If the aileron system did not operate correctly during the operational check of the flight controls steps above, do a functional test of the ailerons.

(a) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

G. Static Dischargers General Visual Inspection (GVI)

SUBTASK 05-51-19-210-029

(1) Perform a GVI of the static dischargers as follows:

(a) Make sure that the static dischargers are attached, installed correctly, and not broken.

(b) Examine the dischargers for the following:

1) Burns.

2) Rough coating.

3) Pitted metal discharger retainers.

(c) Examine the dischargers for the following:

NOTE: Bent pins must be made straight.

1) Broken pins.

2) Bent pins.

3) Blunted tungsten pins.

(d) Examine the discharger for the following:

1) Erosion of the discharger coating.

NOTE: The LE erosion of the discharger must not extend more than the 1/3 of the total length of the discharger.

2) Peeling of the tip cup.

SUBTASK 05-51-19-700-004

(2) Do a resistance test if there is a damaged static discharger, do this task: Static Discharger Resistance Measurement, TASK 23-61-00-760-801.

H. External Wire Shields Detailed Inspection

SUBTASK 05-51-19-211-002

(1) Examine the following:

NOTE: If the lightning strike attach points can not be identified, the external wire shield examination is not required.

NOTE: If there are no wire shield termination(s) in the area of the attach points, the external wire shield examination is not required.

(a) In the area of the lightning strike attach points, examine the wire shield terminations.

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(b) If the flight crew has reported a problem with any system after the lightning strike, examine the wire shield terminations for that systems wiring.

(c) If damage is found at wire shield termination(s), check the full length of the wire bundle.

(d) Repair any damage found and do an operational check of the system that had the damage.

I. System Internal Test

SUBTASK 05-51-19-700-005

(1) If a lightning strike caused a system malfunction, do a full test of the defective system.

SUBTASK 05-51-19-700-006

(2) If a magnetic compass malfunction is reported in the maintenance log, do a check of the standby magnetic compass:

(a) Do this task: Standby Magnetic Compass System (Spot Check) - Inspection, TASK 34-23-00-730-801.

1) If the result is not in limits, do this task: Standby Magnetic Compass - Degaussing, TASK 34-23-00-820-803.

NOTE: It is not necessary to do one of these methods this time: Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method, TASK 34-23-00-820-802 or Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method, TASK 34-23-00-820-801.

(b) Do this task again to make sure the result is in the limits: Standby Magnetic Compass System (Spot Check) - Inspection, TASK 34-23-00-730-801.

1) If the result is not in limits, do one of these tasks: Standby Magnetic Compass - Compass Rose or Inertial Reference System (IRS) Method, TASK 34-23-00-820-802 or Standby Magnetic Compass Swing - Standby Compass Calibrator (SCC) Method, TASK 34-23-00-820-801.

SUBTASK 05-51-19-710-024

(3) Make sure that the fuel quantity system is accurate.

(a) Do the operational test for the Fuel Quantity Indicating System (FQIS) type on your airplane, do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

(b) If the FQIS test shows any problems, do this task: Fuel Quantity Indicating System (FQIS) with the Fuel Measuring Sticks - Comparison Check, TASK 28-41-00-720-801.

SUBTASK 05-51-19-710-028

(4) If the rudder or vertical stabilizer had a lightning strike, do these steps:

(a) Look for leaks from the rudder hydraulic aluminum return tubing aft of the aft pressure bulkhead.

(b) If leaks are found, replace components that have leaks.

SUBTASK 05-51-19-710-029

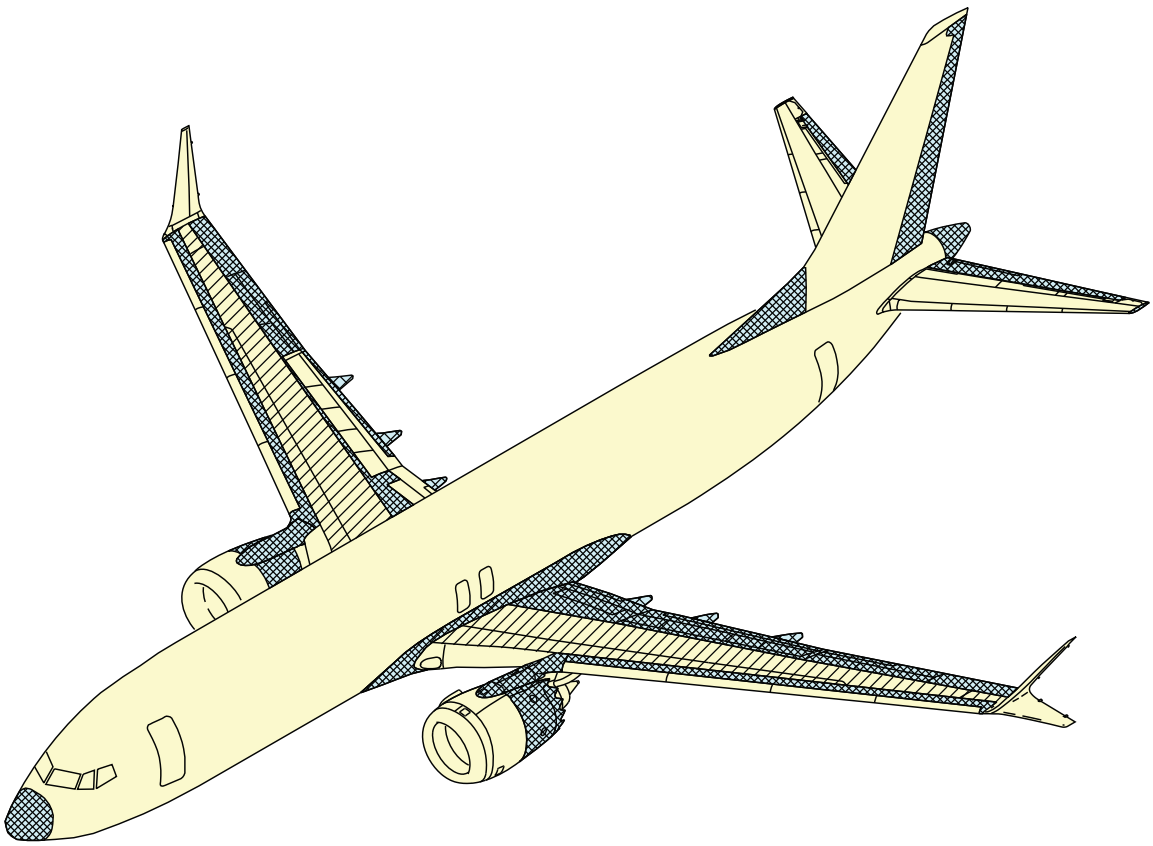
(5) If the elevator or horizontal stabilizer had a lightning strike, do these steps:

(a) Look for leaks from the elevator hydraulic aluminum return tubing aft of the aft pressure bulkhead.

(b) If leaks are found, replace components that have leaks.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



COMPOSITE APPLICATION
(SHADED AREAS CONTAIN COMPOSITE STRUCTURE, UNSHADED AREAS ARE METAL STRUCTURE)

WING TORQUE BOX

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Composite Application
Figure 201/05-51-19-990-802

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TASK 05-51-19-710-801

4. Inspection and Operational Check of Radio and Navigation Systems

A. General

(1) Refer to Figure 203 to determine if an operational check is necessary.

NOTE: It is not necessary to examine the coaxial cables and the connectors if one of the following conditions occurs:

- Radio system had no problems during and after the incident, and the system antenna was not damaged;
- Operational checks were done and no problems were found.

B. References

Reference	Title
20-10-72-210-801	Coaxial Cable Inspection with a Time Domain Reflectometer (P/B 201)
22-11-00-740-801	Digital Flight Control System (DFCS) - Operational Test (P/B 501)
23-12-00-730-801	VHF Communication System - System Test (P/B 501)
23-24-00-730-801	Emergency Locator Transmitter System Test (P/B 501)
23-28-00-700-801	SELCAL System - Operational Test (P/B 501)
31-65-00-710-801	MAX Display System - Operational Test (P/B 501)
34-31-00-710-801	Instrument Landing System - Operational Test (P/B 501)
34-32-00-710-801	Marker Beacon System - Operational Test (P/B 501)
34-33-00-710-801	Low Range Radio Altimeter (LRRA) System - Operational Test (P/B 501)
34-41-00-710-801	Weather Radar (WXR) Operational Test (P/B 501)
34-45-00-710-801	TCAS - Operational Test (P/B 501)
34-51-00-710-801	VOR System - Operational Test (P/B 501)
34-53-00-710-801	ATC System - Operational Test (P/B 501)
34-55-00-710-801	DME System - Operational Test (P/B 501)
34-57-00-730-801	Automatic Direction Finder System - System Test (P/B 501)
34-58-00-710-801	Global Positioning System - Operational Test (P/B 501)
53-52-00-200-801	Nose Radome - Inspection (P/B 601)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

D. Radio and Navigation Systems General Visual Inspection (GVI)

SUBTASK 05-51-19-210-010

(1) Examine the following systems for damage:

(a) High Frequency (HF)

(b) Very High Frequency (VHF)

(c) VHF Omnidirectional Range (VOR)

(d) Distance Measuring Equipment (DME)

(e) Air Traffic Control (ATC)

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(f) Instrument Landing System (ILS)

(g) Automatic Direction Finder (ADF)

(h) Radio Altimeter (RA)

(i) Marker Beacon (MB)

(j) Global Positioning System (GPS)

(k) Traffic Alert and Collision Avoidance System (TCAS)

(l) Emergency Locator Transmitter (ELT).

SUBTASK 05-51-19-210-030

(2) Examine the antennas for damage.

(a) If it is necessary, replace the damaged antennas.

NOTE: Replace the antennas if there is damage that lets water into the antenna, or as part of system malfunction troubleshooting. Antennas with lightly damaged finishes, small cracks, chips, peeling, or erosion are serviceable if the system operates correctly.

SUBTASK 05-51-19-210-031

(3) If it is necessary, examine the coaxial cables and connectors for damage.

(a) If one or more of the previous systems have problems with their operational checks, do these steps:

1) Examine the coaxial cables and connectors.

2) Do a test of the coaxial cables and connectors.

3) If it is necessary, replace the damaged coaxial cables.

SUBTASK 05-51-19-210-011

(4) Examine the nose radome as follows:

(a) Look for pin holes, punctures, and paint that has chipped.

1) If pinholes or punctures are found, check the radome for trapped moisture (TASK 53-52-00-200-801).

(b) Make sure that the radome bonding straps are attached correctly to the airframe.

(c) Repair or replace the lightning diverter strips if they are damaged.

(d) If the radome is damaged, examine the antenna and the waveguide for damage.

E. Radio and Navigation System Operational Checks

SUBTASK 05-51-19-710-007

(1) HF Systems, if it is installed

(a) Do an operational check on some frequencies of each HF System.

(b) Do a ground check of each system with a station that is a minimum of 100 miles away.

(c) Monitor some frequencies on each HF System.

1) Make sure that the HF receiver operates.

SUBTASK 05-51-19-710-008

(2) VHF System

(a) Do this task: VHF Communication System - System Test, TASK 23-12-00-730-801.

(b) Do a ground check of each VHF System with the local control tower facility.

1) Make sure that the VHF System works.

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(c) Monitor some frequencies on each VHF System.

1) Make sure that the VHF receiver works.

SUBTASK 05-51-19-710-009

(3) VOR System

(a) Do this task: VOR System - Operational Test, TASK 34-51-00-710-801.

SUBTASK 05-51-19-710-021

(4) ADF System

(a) Do this task: Automatic Direction Finder System - System Test, TASK 34-57-00-730-801.

(b) Select a local AM broadcast or non direction beacon (NDB) station.

(c) Make sure that you hear a loud and clear ADF audio signal at all audio select locations.

(d) Make sure that the ADF pointers point to the selected bearing.

SUBTASK 05-51-19-710-010

(5) Radar System

(a) Do this task: Weather Radar (WXR) Operational Test, TASK 34-41-00-710-801.

(b) Make sure that the system reads targets at applicable ranges.

SUBTASK 05-51-19-710-011

(6) RA System

(a) Do this task: Low Range Radio Altimeter (LRRA) System - Operational Test, TASK 34-33-00-710-801.

SUBTASK 05-51-19-710-012

(7) DME System

(a) Do this task: DME System - Operational Test, TASK 34-55-00-710-801.

SUBTASK 05-51-19-710-013

(8) ATC System

(a) Do this task: ATC System - Operational Test, TASK 34-53-00-710-801.

SUBTASK 05-51-19-710-014

(9) MB System

(a) Do this task: Marker Beacon System - Operational Test, TASK 34-32-00-710-801.

SUBTASK 05-51-19-710-015

(10) ILS

(a) Do this task: Instrument Landing System - Operational Test, TASK 34-31-00-710-801.

SUBTASK 05-51-19-210-012

(11) If it is necessary, examine and do a test of the coaxial cables and connectors.

(a) Do this task: Coaxial Cable Inspection with a Time Domain Reflectometer, TASK 20-10-72-210-801.

SUBTASK 05-51-19-710-016

(12) MAX Display System (MDS)

(a) Do this task: MAX Display System - Operational Test, TASK 31-65-00-710-801.

SUBTASK 05-51-19-710-017

(13) GPS

(a) Do this task: Global Positioning System - Operational Test, TASK 34-58-00-710-801.

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SUBTASK 05-51-19-710-018

(14) TCAS

(a) Do this task: TCAS - Operational Test, TASK 34-45-00-710-801.

SUBTASK 05-51-19-710-019

(15) ELT

(a) Do this task: Emergency Locator Transmitter System Test, TASK 23-24-00-730-801.

SUBTASK 05-51-19-710-020

(16) Digital Flight Control System (DFCS)

(a) Do this task: Digital Flight Control System (DFCS) - Operational Test, TASK 22-11-00-740-801.

SUBTASK 05-51-19-710-025

(17) Selective Call System (SELCAL)

(a) Do this task: SELCAL System - Operational Test, TASK 23-28-00-700-801.

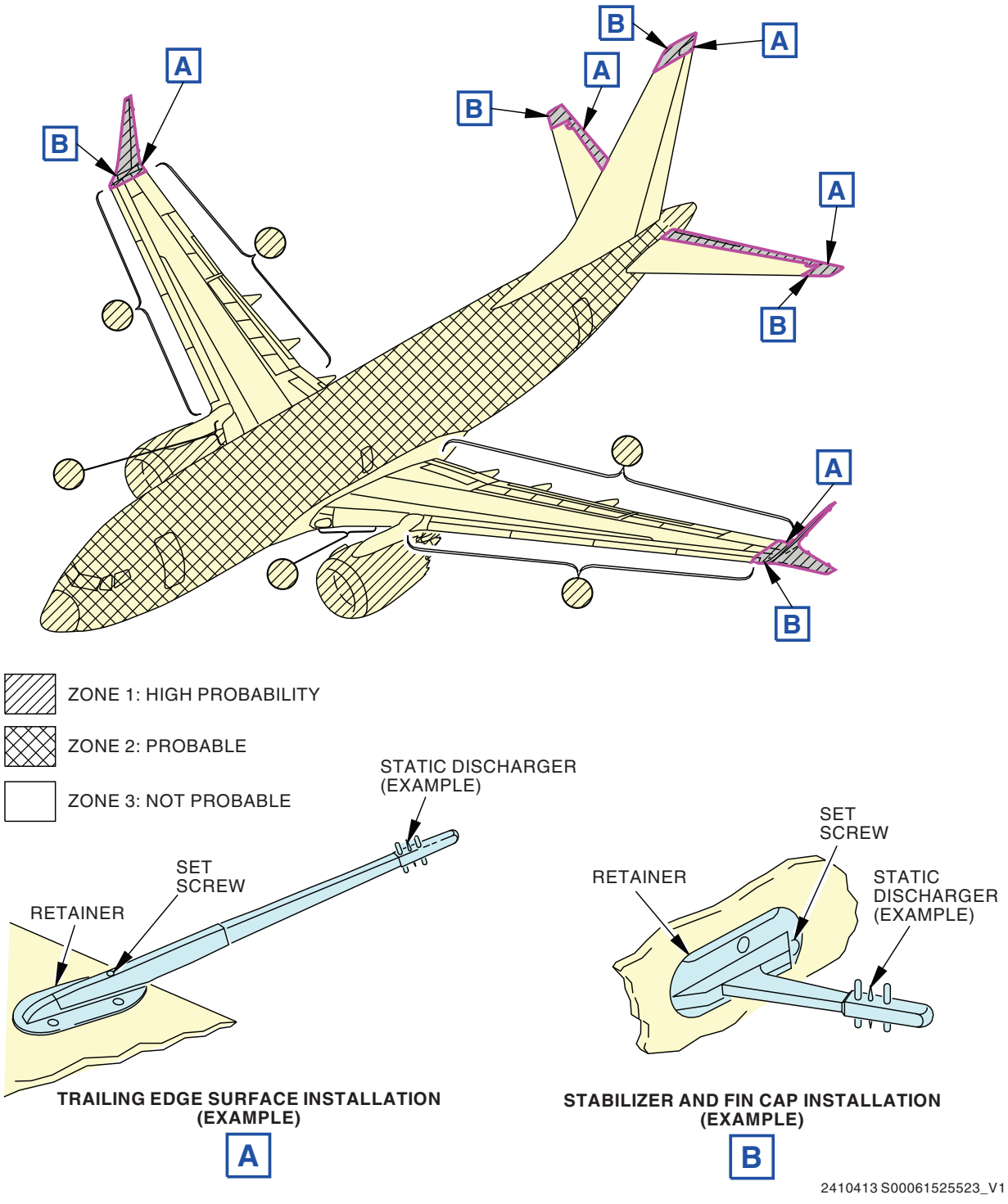
F. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-19-410-001

(1) Install the removed components if they are serviceable, or replacement parts.

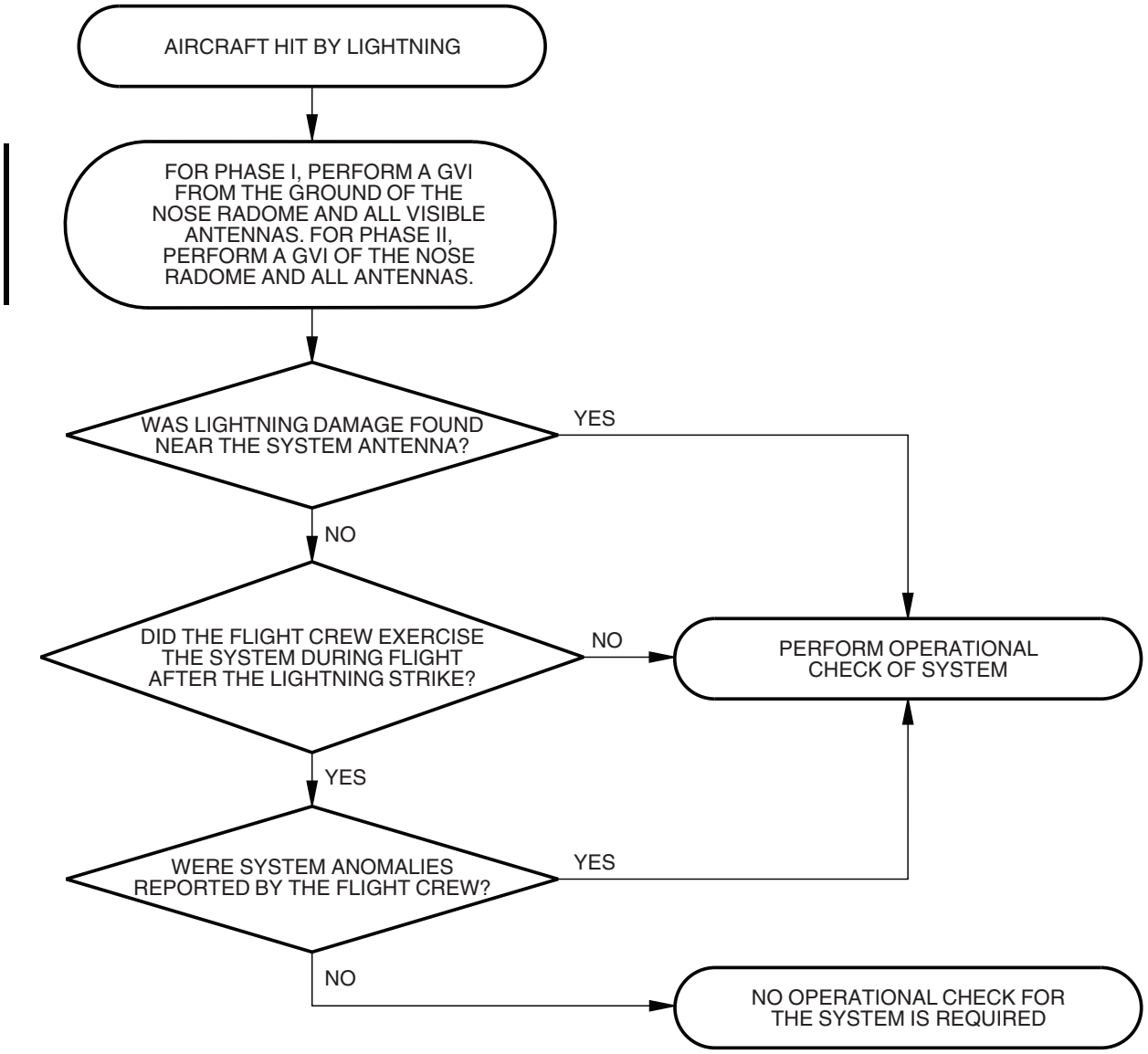
——— END OF TASK ———

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External Lightning Strike Areas
Figure 202/05-51-19-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Lightning Strike Inspections Flow Diagram
Figure 203/05-51-19-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FILIFORM CONDITION ON EXTERIOR DECORATIVE SURFACES (CONDITIONAL INSPECTION)

1. General

A. Filiform corrosion (FC) is a type of corrosion that may occur between the clad aluminum surface of the fuselage and the exterior decorative paint. FC appears as worm-like strands underneath the paint. Refer to CORROSION IDENTIFICATION - TYPES AND CAUSES, SUBJECT 51-00-50 for definition of FC.

B. There are two common initiation locations for FC on the fuselage. The first is common to the edges of parts and the second is common to fasteners. Moisture can ingress under the paint at these locations. FC usually does not penetrate through the clad layer of the aluminum.

TASK 05-51-21-210-801

2. Filiform Corrosion - (Conditional Inspection)

A. General

(1) The standard for the inspection is a General Visual Inspection.

NOTE: The Limited Return to Service Plan (LRTS) is only applicable to filiform corrosion.

(2) If FC is found, do a Limited Inspection in 30 days.

(a) The Limited Inspection is specified to areas prone to FC (Figure 201). These areas include:

- Butt joints
- Lap splices
- Cutouts
- Areas on and around fasteners common to the skin panels and lap splices
- Access doors

(b) All inspection areas are below the lower lap splice of the window belt at stringer S-14.

(3) Mild to moderate filiform corrosion (FC) that does not advance to severe must be removed within 3 years.

(4) The LRTS plan can be used if mild to moderate FC is found. All severe FC must be removed within 30 days with glass-bead blasting per Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 or CORROSION REMOVAL, SUBJECT 51-00-53.

(a) The LRTS plan has repeated inspections and surveys.

1) The repeat inspections are mandatory every 120 days after the discovery of the FC.

2) The LRTS plan is not applicable if all known areas of FC are removed from the aircraft. Subject aircraft can return to service.

B. References

Reference	Title
51-00-53	CORROSION REMOVAL

C. Tools/Equipment

Reference	Description
STD-4055	6 Inch Minimum Length Stainless Steel Pocket Ruler, 1/32" And 1/64" Increments, or Mm And Cm Increments
STD-6732	Camera - 35mm, Single Lens Reflex Type
STD-14360	Flashlight

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
100	Lower Half of Fuselage

E. Limited Inspection Procedure

SUBTASK 05-51-21-210-001

(1) Put the aircraft in an area that is free from rain and other conditions that can affect a visual inspection.

NOTE: Rain and moisture can prevent correct identification of FC. The inspection location must be dry.

SUBTASK 05-51-21-210-002

(2) Document each location of FC with a minimum of three pictures with a camera, STD-6732.

(a) A wide view picture to show the general location on the aircraft.

1) Include a monument in the picture. This will help to find the general area of the FC.

(b) A nearer picture, in the limits of the first picture, to show the location of the FC.

(c) A close-up picture of the FC.

1) Use a pocket ruler, STD-4055 to show the initial area dimension of the FC.

2) Identify the stations and stringer locations surrounding the FC.

3) A flashlight, STD-14360 can be used to help identify FC.

a) The flashlight angle can be varied to help maximize the contrast of the FC. Use a direction which shows the FC clearly.

NOTE: A recommended angle is 15 degrees or less.

(3) Compare the FC found with the images in Figure 202 and Figure 203.

(a) For mild to moderate FC, use one of the two options:

1) Option 1:

a) Utilize the LRTS plan, reference Figure 204.

2) Option 2:

a) Remove all found instances of FC with glass-bead blasting. Refer to Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 or CORROSION REMOVAL, SUBJECT 51-00-53. LRTS plan is not applicable and the aircraft can return to service

(b) For severe FC:

1) Severe FC must be removed from the aircraft within 30 days of initial discovery with glass-bead blasting. Refer to Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 or CORROSION REMOVAL, SUBJECT 51-00-53.

NOTE: The LRTS plan is not applicable to severe FC.

(c) If mild to moderate and severe FC is found, do one of the two options:

1) Option 1:

a) Utilize the LRTS plan for the mild to moderate FC.

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b) Severe FC must be removed from the aircraft within 30 days of initial discovery with glass-bead blasting. Refer to Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 or CORROSION REMOVAL, SUBJECT 51-00-53.

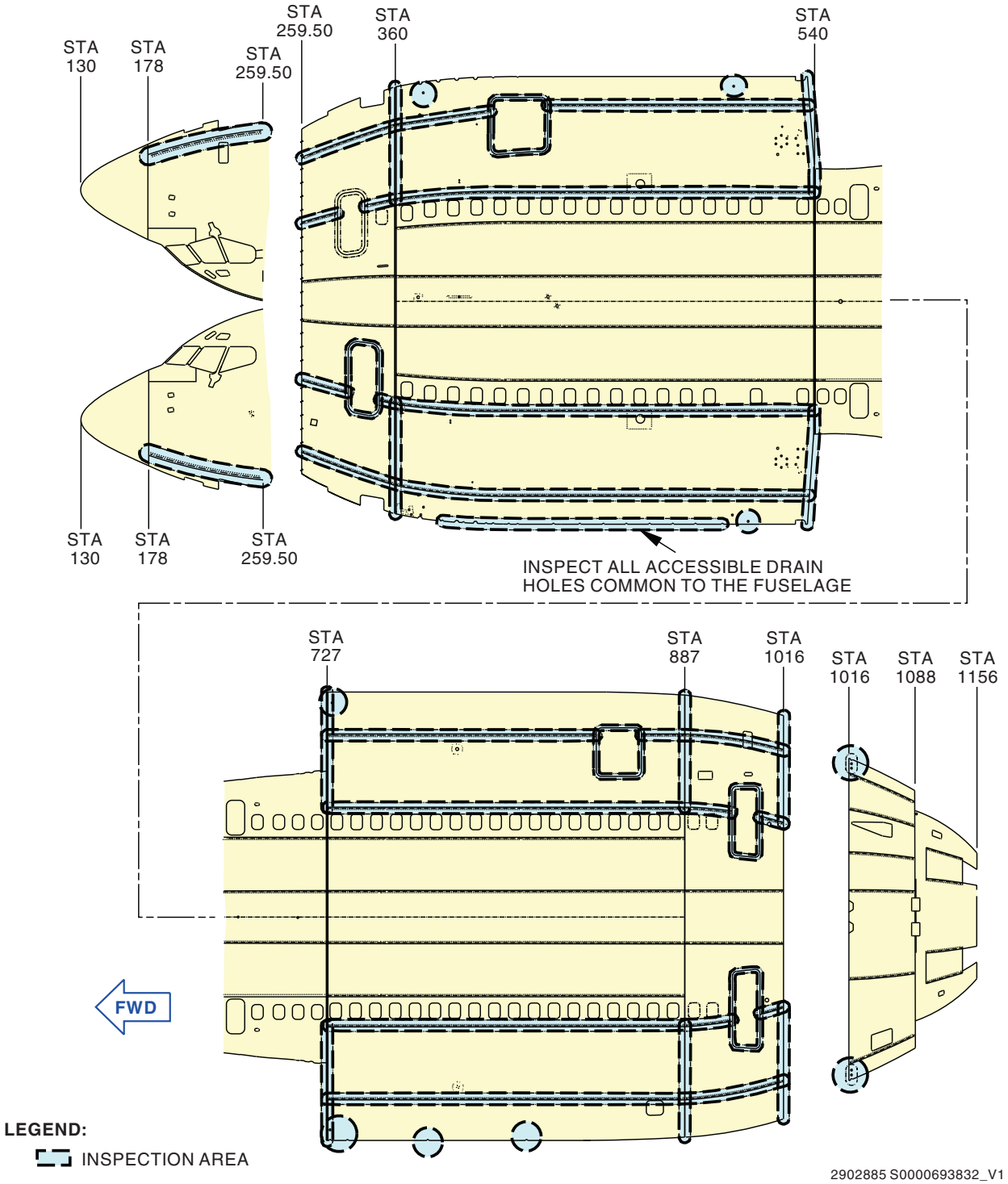
NOTE: The LRTS plan is not applicable to severe FC.

2) Option 2:

a) Remove all instances of FC from the aircraft with glass-bead blasting. Refer to Inspection and Removal of Damage/SRM PROCEDURE 51-10-02-0G-0 or CORROSION REMOVAL, SUBJECT 51-00-53. The LRTS plan is not applicable and the aircraft can return to service.

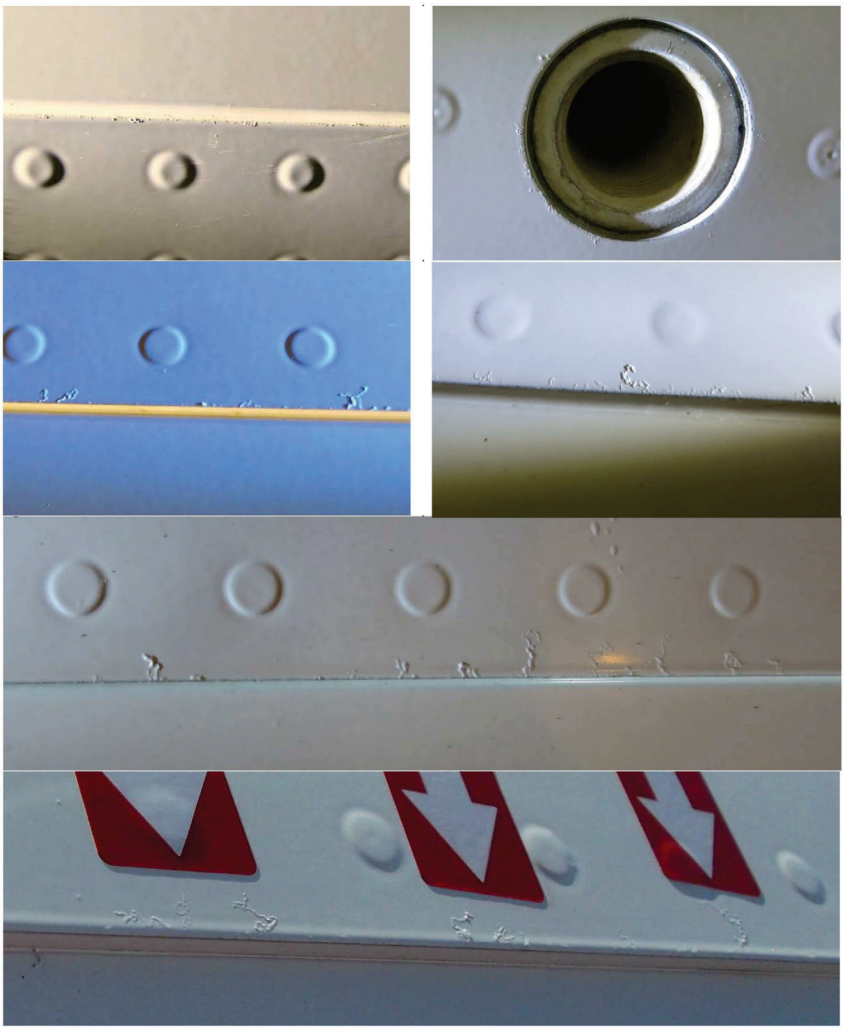
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Filiform Corrosion Inspection
Figure 201/05-51-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MILD FILIFORM CORROSION

NOTE:

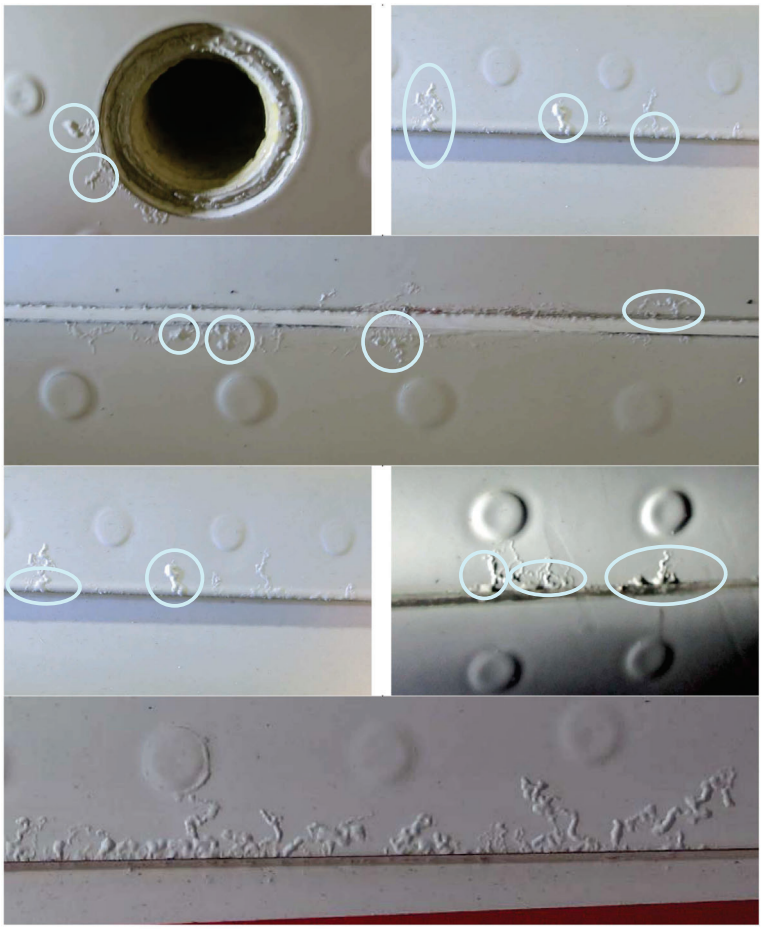
MILD FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- FILIFORM INITIATION POINTS VISIBLE ALONG EDGE.
- SHORT TO LONG SINGLE FILAMENTS.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.

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Filiform Corrosion - Edge Ratings
Figure 202/05-51-21-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MODERATE FILIFORM CORROSION

NOTE:

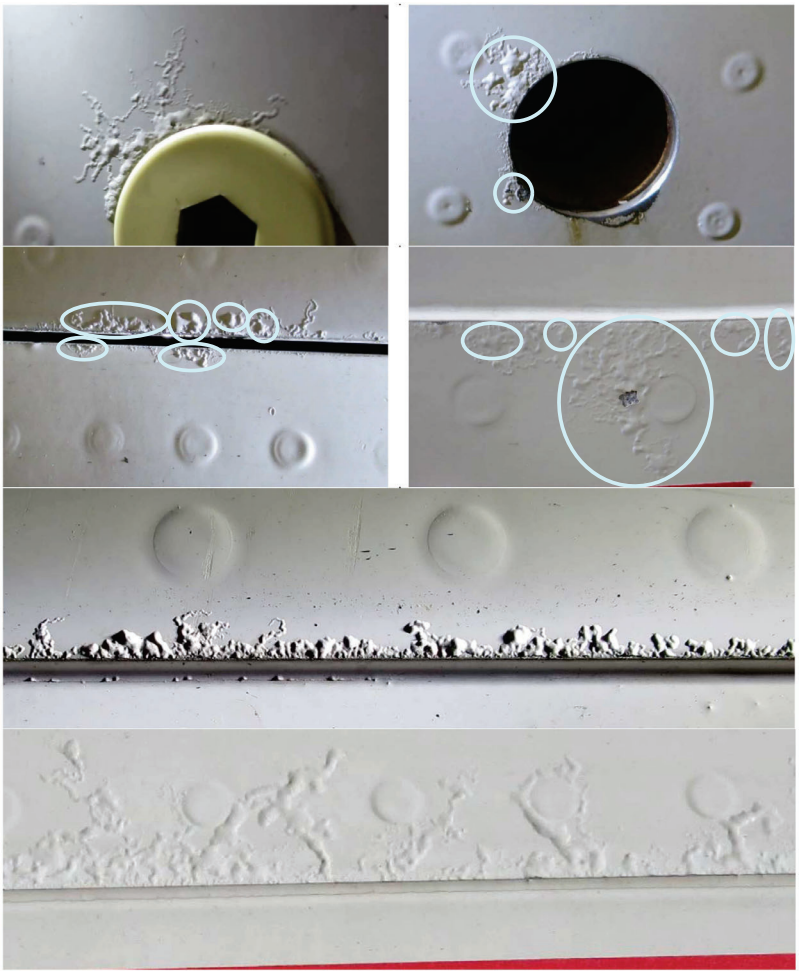
MODERATE FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- MULTIPLE SHORT AND LONG FILAMENTS, THICKER AND GREATER DENSITY COMPARED TO MILD FILIFORM.
- FILAMENTS BRANCHING OR MERGING.
- BLISTERS FORMING.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.
- IF AN OPERATOR OR MRO HAS DIFFICULTY DETERMINING IF THE RATING OF A SPECIFIC FILFORM CORROSION FINDING IS "MODERATE" OR "SEVERE". THEN THE FINDING SHOULD BE CONSIDER "SEVERE" AND ADDRESSED PER SECTION 2. ALTERNATIVELY, OPERATORS AND MROS MAY CONTACT BOEING WITH DETAILS OF THE FINDING(S) IN QUESTION IN ORDER TO OBTAIN SUPPORT WITH PROPERLY IDENTIFYING THE RATING.

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Filiform Corrosion - Edge Ratings
Figure 202/05-51-21-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SEVERE FILIFORM CORROSION

NOTE:

SEVERE FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- MULTIPLE BLISTERED FILAMENTS.
- FILAMENTS AND BLISTERS ARE INDISTINGUISHABLE.
- BLISTERS AND/OR FILIFORM PRESENT LENGTHWISE AND CONTINUOUS ALONG EDGE.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.
- IF AN OPERATOR OR MRO HAS DIFFICULTY DETERMINING IF THE RATING OF A SPECIFIC FILFORM CORROSION FINDING IS "MODERATE" OR "SEVERE". THEN THE FINDING SHOULD BE CONSIDER "SEVERE" AND ADDRESSED PER SECTION 2. ALTERNATIVELY, OPERATORS AND MROS MAY CONTACT BOEING WITH DETAILS OF THE FINDING(S) IN QUESTION IN ORDER TO OBTAIN SUPPORT WITH PROPERLY IDENTIFYING THE RATING.

2902889 S0000693836_V1

Filiform Corrosion - Edge Ratings
Figure 202/05-51-21-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MILD FILIFORM CORROSION

NOTE:

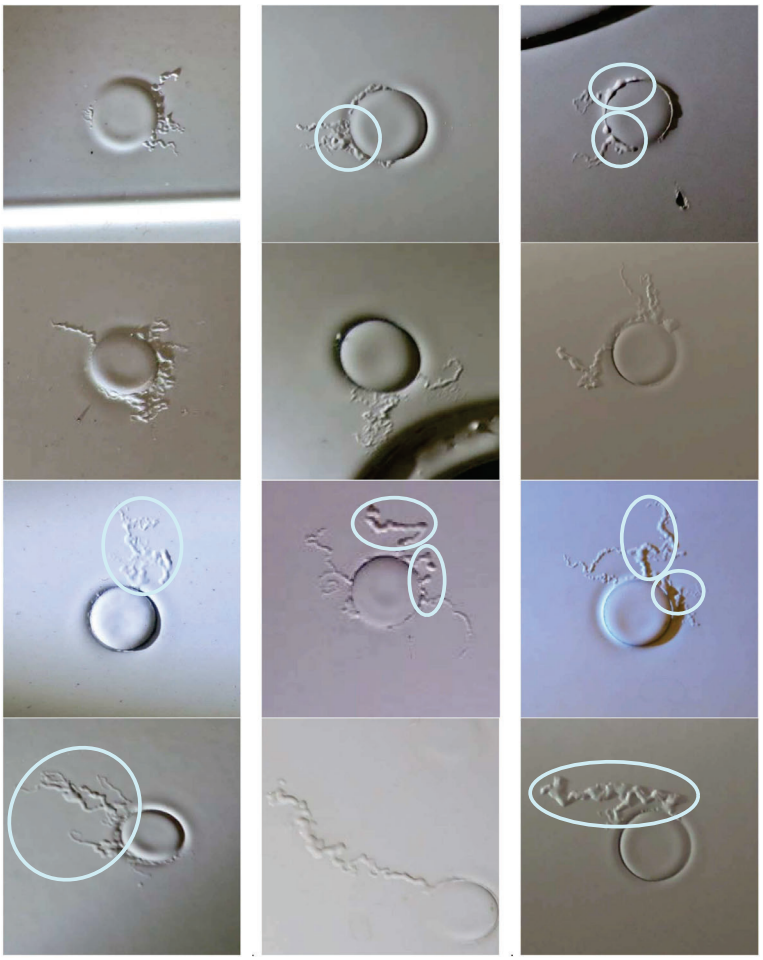
MILD FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- FILIFORM INITIATION POINTS VISIBLE ON OR AROUND FASTENER.
- SINGLE OR MULTIPLE THIN FILAMENTS.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.

2902892 S0000693837_V1

Filiform Corrosion - Fastener Ratings
Figure 203/05-51-21-990-803 (Sheet 1 of 3)

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MODERATE FILIFORM CORROSION

NOTE:

MODERATE FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- MULTIPLE SHORT AND LONG FILAMENTS, THICKER AND GREATER DENSITY COMPARED TO MILD FILIFORM.
- FILAMENTS BRANCHING OR MERGING.
- BLISTERS FORMING.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.
- IF AN OPERATOR OR MRO HAS DIFFICULTY DETERMINING IF THE RATING OF A SPECIFIC FILFORM CORROSION FINDING IS "MODERATE" OR "SEVERE". THEN THE FINDING SHOULD BE CONSIDER "SEVERE" AND ADDRESSED PER SECTION 2. ALTERNATIVELY, OPERATORS AND MROS MAY CONTACT BOEING WITH DETAILS OF THE FINDING(S) IN QUESTION IN ORDER TO OBTAIN SUPPORT WITH PROPERLY IDENTIFYING THE RATING.

2902893 S0000693838_V1

Filiform Corrosion - Fastener Ratings
Figure 203/05-51-21-990-803 (Sheet 2 of 3)

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SEVERE FILIFORM CORROSION

NOTE:

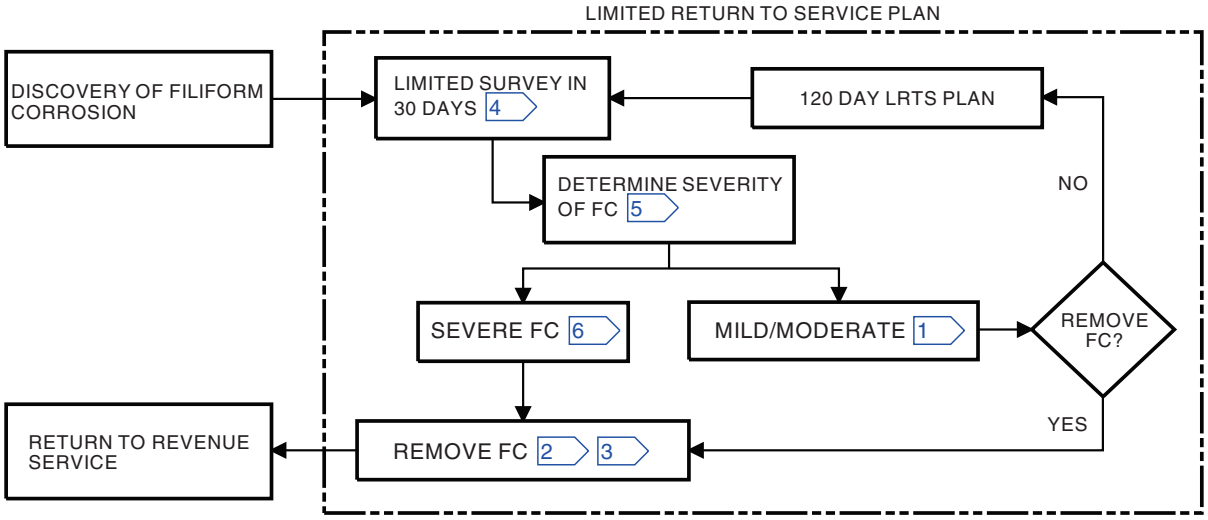
SEVERE FILIFORM CORROSION IS IDENTIFIED AS HAVING ANY OF THE FOLLOWING CONDITIONS:

- MULTIPLE BLISTERED FILAMENTS.
- FILAMENTS AND BLISTERS ARE INDISTINGUISHABLE.
- BLISTERS AND/OR FILIFORM CONTINUOUS AROUND THE FASTENER.
- THE DESCRIPTION AND IMAGES ARE GENERALIZED AND SHOULD NOT BE CONSIDERED TO BE ALL INCLUSIVE.
- IF AN OPERATOR OR MRO HAS DIFFICULTY DETERMINING IF THE RATING OF A SPECIFIC FILFORM CORROSION FINDING IS "MODERATE" OR "SEVERE". THEN THE FINDING SHOULD BE CONSIDER "SEVERE" AND ADDRESSED PER SECTION 2. ALTERNATIVELY, OPERATORS AND MROS MAY CONTACT BOEING WITH DETAILS OF THE FINDING(S) IN QUESTION IN ORDER TO OBTAIN SUPPORT WITH PROPERLY IDENTIFYING THE RATING.

2902894 S0000693840_V1

Filiform Corrosion - Fastener Ratings
Figure 203/05-51-21-990-803 (Sheet 3 of 3)

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1 MILD/MODERATE FILIFORFM CORROSION (FC) THAT DOES NOT ADVANCE TO SEVERE MUST BE REMOVED WITHIN 3 YEARS OF INITIAL DISCOVERY.

2 REMOVE FC WITH GLASS-BEAD BLASTING PER SRM 51-10-02 OR AMM 51-00-53.

3 ONCE ALL THE FC HAS BEEN REMOVED FROM THE SUBJECT AIRCRAFT, IT IS NO LONGER REQUIRED TO ABIDE BY THE INSPECTION CYCLES OF THE LRTS PLAN.

4 DOCUMENT FC AREAS AND PERFORM REPEAT INSPECTION AT DOCUMENTED FC AREAS IF THE AIRCRAFT IS UNDER THE LRTS PLAN.

5 IF SEVERE FC IS DISCOVERED, REMOVAL OF THE SEVERE FC IS REQUIRED WITHIN 30 DAYS TO ALLOW AIRCRAFT TO OPERATE UNDER THE LRTS PLAN.

6 SEVERE FC THAT IS DISCOVERED MUST BE REMOVED WITHIN 30 DAYS.

2902895 S0000693841_V2

Limited Return to Service Plan
Figure 204/05-51-21-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HYDRAULIC FLUID REACTION WITH TITANIUM - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure has this task:

(1) An inspection of the titanium parts contaminated with fire-resistant hydraulic fluid.

TASK 05-51-22-210-801

2. Titanium Parts Contaminated With Fire-Resistant Hydraulic Fluid Inspection

A. General

(1) This task gives instructions to examine bare, painted, or gold/green coated titanium parts.

(a) The inspection is necessary when the titanium touches fire resistant hydraulic fluid.

(b) If the temperature of the hydraulic fluid is less than 270°F (132°C), then the hydraulic fluid is considered low temperature for the purpose of this inspection.

(c) If the temperature of the hydraulic fluid is greater than 270°F (132°C), or if the hydraulic fluid leaks onto a high temperature duct, then the hydraulic fluid is considered high temperature for the purpose of this inspection. Additional cleaning steps are required in these cases.

NOTE: Hydraulic fluid above 270°F (132°C) will cause embrittlement of the titanium.

(d) Signs of hydraulic fluid contamination are a light glossy brown film, or dull black carbonaceous residue.

1) Also, look for a bare surface on painted titanium ducts.

B. Tools/Equipment

Reference	Description
STD-1064	Scraper - Phenolic, Hard Resin

C. Consumable Materials

Reference	Description	Specification
B00005	Cleaner - Alkaline - Cee Bee 280	BAC5744
B00008	Cleaner - Alkaline - Oakite 204	
B00015	Cleaner - Heavy Duty, Multipurpose - Calla 301A	BAC5744, AMS 1526-A, AMS 1550-A
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00094	Solvent - Toluene	A-A-59107
B00130	Alcohol - Isopropyl	TT-I-735
B00230	Cleaner - Metal - Altrex B	
B00402	Cleaner - Aerospace Equipment	MIL-PRF-87937
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B50085	Solvent - Skykleen 1000	BAC5750
C00018	Paint - High Temperature For Titanium - B2000	
C00314	Coating - Low Emissivity Gold Coating	BMS10-82 Type I

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Inspection of the Titanium Parts

SUBTASK 05-51-22-860-001

(1) Make sure that the black inorganic finish is not hydraulic fluid.

NOTE: Titanium parts found in hydraulic fluid areas have a flat, black inorganic finish, or gold coat for protection. The torque box interior structure has a green color finish.

SUBTASK 05-51-22-210-001

(2) Examine all of the titanium parts for contamination, metal deterioration, or paint deterioration in these areas:

NOTE: Look for contamination, metal deterioration, or paint deterioration.

(a) High temperature ducts:

1) The ducts in the wheel well areas
2) The engine fan case
3) The engine fire walls
4) The ducts in the aft fuselage between the pressure bulkhead and APU firewall
5) The pneumatic ducting along the leading edge of the wing.

(b) Other titanium parts:

1) Some strut attach fittings and spar webs
2) The torque box interior structure
3) The landing gear beam
4) The APU support structure
5) The stabilizer links.

E. Cleaning of the Contaminated Titanium Parts

SUBTASK 05-51-22-110-001

(1) To clean the titanium parts that were contaminated with low-temperature hydraulic fluid, do these steps:

NOTE: When hydraulic fluid leaks onto a high temperature duct, the hydraulic fluid should be considered high temperature.

(a) Clean the titanium with one of the manual cleaner:

1) solvent, B00094
2) Altrex B cleaner, B00230
3) Honey Bee 60 cleaner, B00638
4) solvent, B50085.

(b) Remove the hydraulic fluid residue with solvent, B50085, or solvent, B00062, and a hard resin phenolic scraper, STD-1064.

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SUBTASK 05-51-22-100-001

(2) To clean and treat titanium parts that were contaminated with high-temperature hydraulic fluid, do these steps:

CAUTION DO NOT USE METHYL ALCOHOL, OR CHLORINATED SOLVENTS ON TITANIUM OR TITANIUM ALLOY PARTS. THESE MATERIALS CAUSE DAMAGE TO TITANIUM PARTS.

(a) Clean the titanium to remove the oil and other foreign material with solvent, B00094.

(b) Remove the soft hydraulic fluid and remaining hard hydraulic fluid contamination (shows as a light glossy, brown film) with cleaner, B00402.

NOTE: Cee Bee 280 cleaner, B00005, Oakite 204 cleaner, B00008, and Calla 301A cleaner, B00015 are optional.

1) To remove the thick layers of the remaining hard hydraulic fluid contamination, let the cleaner set for approximately to 20 - 40 minutes.

2) After the contamination is soft, remove it with a hard resin phenolic scraper, STD-1064.

NOTE: Do not use a power brush or abrasive blast to remove the contamination.

a) Use a light abrasive to remove the remaining dried fluid.

(c) Replace all parts that show paint deterioration and/or metal that is etched.

(d) For gold-plated titanium ducts, do these steps:

1) Clean the gold coated titanium ducts with cotton wiper, G00034, and alcohol, B00130.

2) If the duct is clean, apply coating, C00314, or B2000 paint, C00018, to provide a protective coating for the duct.

NOTE: B2000 paint, C00018, may be applied over worn or scarred coating, C00314, or to bare titanium duct.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-22-400-001

(1) If the components are serviceable, install the removed parts.

(a) If the components are not serviceable, install replacement parts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN OVERPRESSURE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This task examines components which can be damaged because of overpressurization.

TASK 05-51-24-200-801

2. Cabin Overpressurization Conditional Inspection

A. General

(1) Equipment connected to pitot static systems will not be damaged by internal cabin pressure.

(2) Equipment temporarily disconnected from pitot static systems, or carried as spares, may be damaged by cabin pressure greater than 16 psi (110 kPa).

B. Precautions

(1) If personnel are in the airplane during a test, they must be in good physical condition. If any person experiences pain during pressure change, you must lower the rate of change of pressure or make the absolute pressure stable immediately until the person can either make the pressure equal in their ears, or be removed from the airplane.

NOTE: The ability to adjust to increasing or decreasing pressure is an individual characteristic, and one which varies from day to day depending upon a person's health. People with sinus or ear problems should not be exposed to pressure changes. Similarly a person experiencing headache, toothache, head cold, hay fever, etc., should be excluded from airplane during test. Individuals subjected to pressure changes should be trained in techniques for venting the ears.

C. References

Reference	Title
35-12-00-800-801	Bleed the Crew Oxygen System Prior to System Maintenance or Repair (P/B 201)

D. Procedure

SUBTASK 05-51-24-720-001

(1) If a cabin overpressurization occurs that is more than 16 psi (110 kPa), but less than 21 psi (145 kPa), do the step that follows:

(a) Do a check of all components with pitot static connections that were disconnected, or carried as spares when the overpressurization occurred.

SUBTASK 05-51-24-720-002

(2) If a cabin overpressurization occurs that is more than 21 psi (145 kPa), do the steps that follow:

(a) Do the above step for pressurization between 16 psi (110 kPa) and 21 psi (145 kPa).

(b) Do a check of the oxygen mask regulators from the oxygen system in the crew cabin, do this task: Bleed the Crew Oxygen System Prior to System Maintenance or Repair, TASK 35-12-00-800-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-24-410-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXTREME DUST OR SAND CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This section gives procedures to examine the dust and sand sensitive areas on the airplane when there are extreme dust or sand conditions.

B. This procedure is for extreme dust and sand conditions during the two conditions that follow:

(1) The airplane's flight path went through a cloud of extreme dust or sand.

(2) If the airplane was parked or taxiing during the extreme dust or sand condition.

TASK 05-51-27-210-801

2. Extreme Dust or Sand Conditional Inspection

A. References

Reference	Title
12-11-00-680-801	Fuel Tank Sumps - Fuel Sampling (P/B 301)
12-14-00-600-801	Potable Water System - Drain (P/B 301)
12-40-00-100-801	Clean (Wet Wash) the External Surfaces of the Airplane (P/B 201)
12-40-00-100-802	Clean (Waterless Wash) the External Surfaces of the Airplane (P/B 201)
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
21-32-01-000-801	Positive Pressure Relief Valve Removal (P/B 401)
21-32-02-000-801	Positive Pressure Relief Valve Filter Removal (P/B 401)
21-32-03-000-801	Negative Pressure Relief Valve Removal (P/B 201)
21-51-01-000-801	Left Flow Control and Shutoff Valve Removal (P/B 401)
21-51-01-000-802	Right Flow Control and Shutoff Valve Removal (P/B 401)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
21-51-08-000-801	High Pressure Water Separator Removal (P/B 401)
21-51-12-000-801	Condenser Removal (P/B 401)
21-51-21-000-801	Ram Air Inlet Actuator Removal (P/B 401)
21-51-22-000-801	Ram Air Inlet Deflector Door Removal (P/B 401)
21-51-22-000-802	Ram Air Inlet Deflector Door Shaft Assembly Removal (P/B 401)
21-51-23-000-801	Ram Air Inlet Modulation Panels Removal (P/B 401)
21-61-10-100-801	Cabin Temperature Sensor Cleaning (P/B 201)
28-11-11-000-802	Surge Tank Access Door Removal (P/B 401)
29-11-00-200-801	Hydraulic Fluid Check (P/B 601)
30-21-11-000-801	Engine Anti-Ice Valve Removal (P/B 401)
35-12-00-700-801	Crew Oxygen Stowage Box Test (Mask Stowed in Stowage Box) (P/B 501)
35-12-00-700-802	Crew Oxygen Mask-Regulator Test (P/B 501)
35-22-00-700-801	Passenger Oxygen System Automatic Actuation Functional Test (P/B 501)
36-11-02-000-801	Intermediate Pressure Check Valve (IPCV) Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-05-000-801	Intermediate Manifold Pressure (PI) Sensor Removal (P/B 401)
36-11-06-000-801	Duct Vent Valve Removal (P/B 401)

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(Continued)

Reference	Title
36-12-01-000-802	Bleed Air Precooler Removal with Engine Installed (P/B 401)
36-12-03-000-801	Fan Air Modulating Valve Removal (P/B 401)
36-13-03-000-801	Ground Pneumatic Connector Check Valve Removal (P/B 401)
36-13-04-000-801	Bleed Air Isolation Valve Removal (P/B 401)
36-14-02-000-801	APU Check Valve Removal (P/B 401)
36-21-01-000-801	Duct Pressure Transducer Removal (P/B 401)
38-11-03-000-801	Fill/Overflow Valve Removal (P/B 401)
38-14-01-000-801	Water Quantity Transmitter Removal (P/B 401)
38-42-10-960-801	Bleed Air Filter Element Replacement (P/B 401)
47-32-04-000-802	NGS Filter Element - Removal (P/B 401)
49-15-11-200-801	Air Inlet Seal Inspection (P/B 601)
49-15-15-200-801	APU Air Inlet Diffuser Ducts Inspection (P/B 601)
49-52-13-000-801	Bleed Air Duct Removal (P/B 401)
49-81-11-200-802	Exhaust Duct Muffler Inspection (P/B 601)
49-91-41-200-801	Oil Cooler Inspection (P/B 601)
71-00-00-200-806-G00	Heavy Sand/Dust Ingestion and Volcanic Ash Ingestion Inspection (P/B 601)
80-11-02-000-801-G00	Starter Air Valve Removal (P/B 401)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

C. Extreme Dust or Sand Conditional Inspection

SUBTASK 05-51-27-200-001

(1) Examine the windshield.

CAUTION DO NOT OPERATE THE WINDSHIELD WIPERS ON DRY GLASS. ALWAYS SUPPLY WATER TO THE WINDSHIELDS BEFORE YOU OPERATE THE WIPER. THE WIPER WILL CAUSE DAMAGE TO A DRY WINDSHIELD.

(a) Make sure that the windshield is free from dust and sand.

SUBTASK 05-51-27-200-002

(2) Examine the surfaces that follow to make sure that they are free from sand and dust.

(a) Fuselage.

(b) Upper surfaces of the wings and other control surfaces.

SUBTASK 05-51-27-200-003

(3) Examine the areas that follow to make sure that they are free from sand and dust.

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(a) Probes.

(b) Sensors.

(c) Ports.

1) Including static ports and pitot tubes.

(d) Vents.

(e) Drains.

(f) Left and right ram air inlets.

(g) Cabin pressure outflow valve.

(h) Positive pressure relief valves.

SUBTASK 05-51-27-200-004

(4) Examine the leading edge flaps.

(a) Make sure that they are not damaged.

SUBTASK 05-51-27-200-005

(5) Examine the power plant areas to make sure that they are free from sand and dust.

(a) Engine inlet and cowling.

(b) Indicator probes.

(c) Fan and blades.

1) Make sure that the fan blades are free to turn and are not damaged.

(d) Do the inspection that follows before the next flight if there is dust or sand in the power plant areas (TASK 71-00-00-200-806-G00).

SUBTASK 05-51-27-200-006

(6) Examine the fuel tank vents.

(a) Make sure that they are free from sand and dust.

SUBTASK 05-51-27-200-007

(7) Examine the landing gear areas that follow to make sure that they are free from sand and dust.

(a) Landing gear shock struts.

(b) Landing gear actuator pushrods.

(c) Landing gear uplock actuator pushrods.

(d) Landing gear doors.

SUBTASK 05-51-27-200-008

(8) Examine the control surfaces that follow to make sure that they are free from sand and dust.

(a) Vertical stabilizer.

(b) Horizontal stabilizer.

(c) Stabilizer trim mechanisms.

(d) Flap screw mechanisms.

(e) Leading edges.

1) Make sure that all leading edges are not damaged.

SUBTASK 05-51-27-211-001

(9) Visually examine all the exposed portions of the flight control cable.

(a) Make sure that they are free from sand and dust.

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SUBTASK 05-51-27-200-009

(10) Examine the Auxiliary Power Unit (APU) areas that follow to make sure that they are free from dust or sand.

(a) Do the next check to be sure that there is no damage to the APU:

1) Operate the APU on the ground and monitor the pneumatic performance in the modes that follow:

a) The Main Engine Start (MES) operating mode.

b) The Environmental Control System (ECS) operating mode.

NOTE: Use the applicable APU manufacturer recommendations for engine inspection for flights through areas of low level dust or sand concentrations.

(b) Examine the inlet and exhaust ducts for dust or sand contamination.

NOTE: Do not perform a borescope inspection on the APU.

1) If there is dust or sand in the APU inlet or exhaust ducts, look for dust or sand in the components that follow:

a) Do this task: Oil Cooler Inspection, TASK 49-91-41-200-801.

b) Do this task: Air Inlet Seal Inspection, TASK 49-15-11-200-801.

c) Do this task: APU Air Inlet Diffuser Ducts Inspection, TASK 49-15-15-200-801.

d) Do this task: Exhaust Duct Muffler Inspection, TASK 49-81-11-200-802.

e) Do this task: Bleed Air Duct Removal, TASK 49-52-13-000-801.

SUBTASK 05-51-27-200-010

(11) Examine the air conditioning areas to make sure that they are free from dust and sand.

(a) Do the steps that follow to examine the air conditioning system:

1) Examine the aft outflow valve, do this task:Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

a) If you find dust or sand in the valve, clean or replace the valve.

2) Remove and examine the negative pressure relief doors, do this task: Negative Pressure Relief Valve Removal, TASK 21-32-03-000-801.

a) If you find dust or sand in the door area, clean or replace the door and linkage.

3) Remove and examine the positive pressure relief valves, do this task: Positive Pressure Relief Valve Removal, TASK 21-32-01-000-801.

a) If you find dust or sand in the valves, clean or replace the valves.

4) Remove and examine the positive pressure relief valve filters, do this task: Positive Pressure Relief Valve Filter Removal, TASK 21-32-02-000-801.

a) If you find dust or sand in the filters, clean or replace the filters.

5) Examine the ram air inlet deflector door, do this task: Ram Air Inlet Deflector Door Removal, TASK 21-51-22-000-801.

a) If you find dust or sand in the deflectors, clean the deflectors and movable joints.

6) Examine the ram air inlet deflector door shaft assemblies, do this task: Ram Air Inlet Deflector Door Shaft Assembly Removal, TASK 21-51-22-000-802.

a) If you find dust or sand in the shaft assemblies, clean the shaft assemblies.

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7) Examine the ram air inlet modulation panels, do this task: Ram Air Inlet Modulation Panels Removal, TASK 21-51-23-000-801.
   a) If you find dust or sand in the modulation panels, clean the panels and movable joints.
8) Examine the ram air inlet modulation panel shaft assemblies.
   a) If you find dust or sand in the shaft assemblies, clean the shaft assemblies.
9) Examine the ram air exhaust modulation louver assemblies.
   a) If you find dust or sand in the louver assemblies, clean the louvers, bearings and movable joints.
10) Examine the ram air inlet actuators, do this task: Ram Air Inlet Actuator Removal, TASK 21-51-21-000-801.
   a) If you find dust or sand in the actuators, clean or replace the actuators.
11) Remove and examine the Air Cycle Machine (ACM), do this task: Air Cycle Machine Removal, TASK 21-51-04-000-801.
   a) Check for dust or sand in the ACM valves.
      <1> If you find dust or sand in the ACM valves, clean or replace the ACM.
   b) Remove the high pressure water separator, do this task: High Pressure Water Separator Removal, TASK 21-51-08-000-801.
   c) Examine the water for contamination.
   d) If there is contamination in the water of the high pressure water separator, do the steps that follow:
      <1> Remove the ACM, do this task: Air Cycle Machine Removal, TASK 21-51-04-000-801.
      <2> Inspect the ACM for torus erosion.
         NOTE: Torus erosion is caused because atmospheric contamination in the bleed air is trapped in the ACM torus and moves around the torus, which erodes the inner surface of the torus.
12) Remove and examine the air cycle cooling pack primary and secondary heat exchangers.
   a) If you find dust or sand in the heat exchangers, clean or replace the heat exchangers.
13) Make sure that the heat exchanger passes the Pressure Flow Tests to verify that the unit is fully cleaned and not clogged.
   a) Do this every 2000-3000 hours.
14) Examine the water extract nozzles and drain lines.
   a) Clean the water extract nozzles and drain lines if dust or sand is found.
15) Remove and examine the condensers, do this task: Condenser Removal, TASK 21-51-12-000-801.
   a) If you find dust or sand in the condensers, clean or replace the condensers.
16) Remove and examine the flow control pack valves, do these tasks: Left Flow Control and Shutoff Valve Removal, TASK 21-51-01-000-801, Right Flow Control and Shutoff Valve Removal, TASK 21-51-01-000-802.
   a) If you find dust or sand in the pack valves, clean or replace the pack valves.

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17) Remove and examine the recirculating air fans.
    a) If you find dust or sand in the fans, clean or replace the fans.
18) Remove and examine one of the trim air nozzles.
    a) If you find dust or sand in one of the trim nozzles, remove and clean all of the trim nozzles.
19) Remove and examine the recirculation air filters.
    a) If you find dust or sand in the filters, clean or replace the filters.
20) Remove and examine the recirculation air check valves.
    a) If you find dust or sand in the valves, clean or replace the valves.
21) Remove and examine the equipment cooling automatic flow control valve.
    a) If you find dust or sand in the valve, remove or replace the valve.
22) Remove and examine the exhaust fans and check valves.
    a) If you find dust or sand in the fans or valves, clean or replace the fans and valves.
23) Remove and examine the pack temperature control valve.
    a) If you find dust or sand in the control valves, clean or replace the valves.
24) Do a visual examination of the temperature sensors and connectors in both air conditioning pack bays.
    a) Make sure that the temperature sensors and connectors are not damaged and are installed correctly.
25) Make sure that the flight compartment and cabin temperature sensors are not clogged.
    a) If you find dust or sand in the flight compartment or cabin temperature sensors, clean or replace the sensors, do this task: Cabin Temperature Sensor Cleaning, TASK 21-61-10-100-801.
26) Remove and examine the trim air valve.
    a) If you find dust or sand in the valves, clean or replace the valves.

SUBTASK 05-51-27-200-012

(12) Examine the Nitrogen Generation System (NGS).
    (a) Examine the NGS Filter (TASK 47-32-04-000-802).
        1) If you find dust or sand in the NGS filter, clean or replace the filter, if necessary.

SUBTASK 05-51-27-200-013

(13) Examine the Pneumatic System.
    (a) Do this task: Bleed Air Precooler Removal with Engine Installed, TASK 36-12-01-000-802.
    (b) If you find dust or sand in the bleed air precooler, remove the components that follow and look for dust or sand.
        1) Do this task: Duct Vent Valve Removal, TASK 36-11-06-000-801.
        2) Do this task: PRSOV Removal, TASK 36-11-04-000-801.
        3) Do this task: Intermediate Pressure Check Valve (IPCV) Removal, TASK 36-11-02-000-801.
        4) Do this task: Starter Air Valve Removal, TASK 80-11-02-000-801-G00.

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5) Do this task: Bleed Air Isolation Valve Removal, TASK 36-13-04-000-801.

6) Do this task: APU Check Valve Removal, TASK 36-14-02-000-801.

7) Do this task: Ground Pneumatic Connector Check Valve Removal, TASK 36-13-03-000-801.

8) Do this task: Engine Anti-Ice Valve Removal, TASK 30-21-11-000-801.

9) Do this task: Duct Pressure Transducer Removal, TASK 36-21-01-000-801.

10) Do this task: Intermediate Manifold Pressure (PI) Sensor Removal, TASK 36-11-05-000-801.

11) Do this task: Fan Air Modulating Valve Removal, TASK 36-12-03-000-801.

SUBTASK 05-51-27-200-014

(14) Examine the Fuel System.

(a) Look for dust or sand in a fuel sample from each fuel tank sump, do this task: Fuel Tank Sumps - Fuel Sampling, TASK 12-11-00-680-801.

(b) Look for dust or sand in a fuel sample from each surge tank sump (if there is fuel in the surge tanks).

(c) Visually inspect for dust or sand in each surge tank, do this task: Surge Tank Access Door Removal, TASK 28-11-11-000-802.

SUBTASK 05-51-27-200-015

(15) Examine the Oxygen System.

(a) If you find dust or sand in the passenger and control cabin, examine the oxygen system.

1) Remove the stowage boxes for the mask and regulator and look for dust or sand, do this task: Crew Oxygen Stowage Box Test (Mask Stowed in Stowage Box), TASK 35-12-00-700-801.

2) Examine portable oxygen masks for dust or sand, do this task: Crew Oxygen Mask-Regulator Test, TASK 35-12-00-700-802.

3) Examine the passenger oxygen masks for dust or sand, do this task: Passenger Oxygen System Automatic Actuation Functional Test, TASK 35-22-00-700-801.

SUBTASK 05-51-27-200-016

(16) Examine the Potable Water System.

(a) Remove the bleed air filter and look for dust or sand contamination, do this task: Bleed Air Filter Element Replacement, TASK 38-42-10-960-801.

(b) If there is dust or sand in the air filter, do these steps:

1) Visually inspect for dust or sand in a sample from the potable water tanks, do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

2) Remove the fill/overflow valves and look for dust or sand contamination, do this task: Fill/Overflow Valve Removal, TASK 38-11-03-000-801.

3) Remove the water quantity transmitters and look for dust or sand contamination, do this task: Water Quantity Transmitter Removal, TASK 38-14-01-000-801.

SUBTASK 05-51-27-200-017

(17) Examine the Hydraulic System.

(a) Visually inspect for dust or sand in a sample of the hydraulic fluid, do this task: Hydraulic Fluid Check, TASK 29-11-00-200-801.

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D. Clean the Dust or Sand Contamination

SUBTASK 05-51-27-100-001

(1) If you find dust in the areas you examined, remove the dust.

(a) Do this task: Clean (Wet Wash) the External Surfaces of the Airplane, TASK 12-40-00-100-801 or Clean (Waterless Wash) the External Surfaces of the Airplane, TASK 12-40-00-100-802.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-27-400-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-27-200-011

(2) Operate the air conditioning packs and check for the conditions that follow.

(a) Unusual noise or vibration.

NOTE: If you find unusual noise or vibration, use the Honeywell CMM 21-50-78 for ACM maintenance.

(b) Leaking on all ducting in the two air conditioning pack bays.

——— END OF TASK ———

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ICE OR SNOW CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This section gives the task to examine the airplane surface when there are ice or snow or slush conditions.

(1) When there is ice, snow, frost, and/or slush conditions before a flight, examine the surfaces of the airplane.

TASK 05-51-28-210-801

2. Ice or Snow Condition Maintenance Practices

A. References

Reference	Title
12-33-01-600-801	Cold Weather Maintenance Procedure (P/B 301)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
500	Left Wing
600	Right Wing

C. Procedure

SUBTASK 05-51-28-210-001

(1) When there are ice or snow conditions, examine the airplane.

(a) Look at the areas that follow for ice or snow:

1) Fuselage.
2) Wings.
3) TE flaps.
4) LE flaps.
5) LE slats.
6) Ailerons.
7) Stabilizers.
8) Elevators.
9) Rudder.
10) Accessible areas of the balance panels.
11) Accessible areas of control surface hinge points and hinges.
12) Gaps between wing structure and aileron nose.
13) Gaps between fin and rudder nose.
14) Gaps between horizontal stabilizer and elevator nose.
15) Engine fan blades.
16) Broadband radome.

(b) The areas that follow must not have any snow or ice on them before takeoff:

1) The total area of wing upper surface.

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2) The wing leading edge.

3) All control surfaces.

4) All tab surfaces.

5) Balance panel cavities.

6) The engine fan blades.

(c) Look in the engine inlet cowl for ice and snow.

1) Make sure the fan turns freely.

2) If there is ice on the engine fan blades, do this task: Cold Weather Maintenance Procedure, TASK 12-33-01-600-801.

SUBTASK 05-51-28-610-001

(2) If there is ice, snow, slush, and/or frost, do this task: Cold Weather Maintenance Procedure, TASK 12-33-01-600-801.

——— END OF TASK ———

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EXCEEDING MAXIMUM NOSE LANDING GEAR TOWING ANGLE OR MAXIMUM TOWING LOAD (CONDITIONAL INSPECTION) - MAINTENANCE PRACTICES

1. General

A. The procedure is necessary for an accidental turn of the nose landing gear for these conditions:

(1) The steering angle is larger than the maximum steering angle with torsion links attached.

(2) Steering from the flight deck with the tow bar installed and the steering system connected.

(3) Towing load more than the maximum towing load.

NOTE: Oversteer is when the steering angle is more than the maximum steering angle with torsion links attached.

B. This inspection procedure is divided into two phases (Phase I and Phase II).

(1) If the inspection during Phase I does not show that damage has occurred, no more inspections are necessary.

(2) If the Phase I inspection shows that damage has occurred, the Phase II inspection must be done.

(3) The Phase I inspection should be completed prior to the Phase II inspection.

C. When the conditional inspection tells you to "examine" a component, look for these conditions (replace or repair components, if it is necessary).

(1) Cracks

(2) Pulled apart structure

(3) Loose paint (paint flakes)

(4) Twisted parts (distortion)

(5) Leakage

(6) Bent components

(7) Fasteners holes that become larger or longer

(8) Loose fasteners

(9) Fasteners that have pulled out or are gone

(10) Delamination

(11) Misalignment

(12) Interference

(13) Other signs of damage.

TASK 05-51-29-200-801

2. Phase I Inspection

A. References

Reference	Title
09-10-00-990-803	Figure: Airplane Towing Loads and Turning Radius (P/B 201)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-802	Wheels Inspection (Wheel Removed from the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-51-31-220-801	Nose Gear Steering System Cables Inspection (P/B 601)

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B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

C. Nose Landing Gear Area, Exceeding Maximum Steering Angle, Maximum Towing Load.

(Figure 09-10-00-990-803)

SUBTASK 05-51-29-200-001

(1) Examine the nose landing gear areas that follow:

(a) The tires, do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

(b) The wheels, do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

1) If the examination finds any wheels damaged, do this task: Wheels Inspection (Wheel Removed from the Airplane), TASK 32-45-00-700-802.

(c) If damage occurred during towbarless towing, do these steps:

1) Verify the axle is not bent or deformed.

2) Examine the axle threads for excessive damage.

(d) Make sure all of the fasteners are installed in the correct positions.

(e) Make sure there are no loose fasteners in the web of the nose wheel well near the trunnion support.

(f) Examine the steering cable control system for signs of cable overload, do this task: Nose Gear Steering System Cables Inspection, TASK 32-51-31-220-801.

1) Verify the pulley brackets are not deformed.

2) Verify the proper cable rigging.

(g) Examine the nose gear steering actuator rods for signs of necking or contact with the steering collar.

1) Verify the actuator rods are not bent.

(h) Examine the actuator rod end attachments to the steering collar for signs of deformation and excessive freeplay.

(i) If damage occurred during towbarless towing examine the following pins for signs of crank shafting, cracks or any other gross defects.

1) Torsion Link Pin, upper

2) Torsion Link Pin, lower

3) Torsion Link Pin, apex

4) Steering Collar Pins

(j) Examine the actuator trunnion attachments to the steering plates for signs of deformation and excessive freeplay.

(k) Examine the nose gear torque links for signs of deformation and excessive freeplay.

(l) Operationally check the steering system by commanding slow tiller inputs in both directions.

1) Verify no system binding occurs and there is no leakage from steering actuator trunnions and rod seals and steering metering valve.

(m) Examine the upper and lower ends of the shock strut of the nose gear for fluid leakage.

(n) Examine the outer cylinder of the nose landing gear.

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(o) Examine the nose landing gear inner cylinder at tow fittings.

(p) Examine the nose landing gear inner cylinder torsion links attach lugs.

(q) Examine the nose landing gear trunnions for signs of damage.

(r) Examine the nose landing gear trunnion attachment areas.

1) Examine bolted connections (attaching upper end of outer cylinder and upper end of upper drag brace).

(s) Examine both upper and lower drag brace segments, attach lugs, and pins.

(t) Examine the doors, hinges and retraction mechanism of the nose landing gear and the nose landing gear taxi light bracket for signs of damage.

——— END OF TASK ———

TASK 05-51-29-200-802

3. Phase II Inspection

A. References

Reference	Title
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
32-21-00-200-801	Nose Landing Gear - Inspection (P/B 601)
32-21-00-700-801	Nose Landing Gear Torsional Freeplay Inspection (P/B 601)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

C. Nose Landing Gear Operational Check

SUBTASK 05-51-29-200-002

(1) Retract and then extend the nose landing gear to make sure it operates correctly. Do one of the steps that follow:

(a) Do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

(b) Do this task: Nose Gear Manual Extension System Test - Airplane on Jacks, TASK 32-35-00-730-801.

D. Nose Landing Gear Inspections

SUBTASK 05-51-29-200-003

(1) Examine the nose landing gear areas that follow:

(a) Do this task: Nose Landing Gear - Inspection, TASK 32-21-00-200-801.

(b) Do this task: Nose Landing Gear Torsional Freeplay Inspection, TASK 32-21-00-700-801.

(c) Examine the shock strut of the nose landing gear for fluid leakage.

(d) Examine the shock strut of the nose landing gear to make sure the pressure is satisfactory.

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1) Do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

(e) Remove the inner cylinder if:

1) There is visible damage to any of the following structural components:
   a) Nose landing gear inner cylinder.
   b) Nose landing gear outer cylinder.
   c) Drag strut.
2) There are signs of a shock strut fluid leakage.
3) It is known the maximum towing load has been exceeded.

(f) If you remove the inner cylinder, examine it as follows:

1) Perform a visual examination looking for distortion and cracks.
2) Do a dimensional inspection.
3) Do a NDT inspection, if appropriate.

(g) Do a check of the rigging of the steering mechanism.

1) Do this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

——— END OF TASK ———

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VOLCANIC ASH CONDITION - MAINTENANCE PRACTICES

1. General

A. This procedure has two tasks:

(1) Volcanic ash operational encounter conditional inspection

(2) Volcanic ash ground fallout conditional inspection.

TASK 05-51-31-210-801

2. Volcanic Ash Operational Encounter Conditional Inspection

A. General

(1) This inspection is divided into three phases (Phase I, Phase II and Phase III).

(a) Phase I (IA and IB) is a general visual inspection for one or more of the following:

1) The flight crew reported the airplane flight path was through a cloud of visual volcanic ash.

2) The airplane lands or takes off in volcanic ash fallout conditions.

NOTE: Fallout is identified as ash drifting down from the air, or is blown about by the wind. Fallout is also caused by the ash being blown or moved about because of the movement of airplanes and ramp vehicles.

3) An airplane encounters volcanic ash while operating on the ground during volcanic ash fallout conditions.

(b) Phase II is applicable when any of the Phase IB inspections show positive results.

(c) Phase III is applicable when any of the Phase II inspections show positive results.

(2) Volcanic ash has the qualities that follow:

(a) It is hard and highly abrasive.

(b) It usually contains trace chemicals which can be acidic (pH between 2 and neutral).

(c) It can cause erosion/abrasion, etching or shot peening on exposed surfaces and remove protective finishes.

(d) The dimensions of most ash particles are less than 5 microns with trace amounts that are more than 50 microns.

(e) It resembles talcum powder.

(f) It stays on exposed lubricated surfaces.

(g) Able to enter small orifices and pass by seals.

(h) Can cause unusual wear and contamination of many airplane systems and filter elements.

WARNING

DO NOT GET VOLCANIC ASH IN YOUR MOUTH, EYES, OR ON YOUR SKIN. PUT ON A RESPIRATOR MASK THAT REMOVES VOLCANIC ASH PARTICLES SUFFICIENTLY. PUT ON CLOTHES AND GOGGLES (OR OTHER APPROVED EYE PROTECTION) THAT KEEP VOLCANIC ASH OFF YOU. VOLCANIC ASH CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Volcanic ash can cause discomfort and injury to persons during fallout conditions, and during inspection and cleaning.

(a) Precautions must be followed when you work in a volcanic ash environment. This will prevent the entry of volcanic ash into your eyes and your respiratory (breathing) system.

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(4) Guidance for flight through areas of low level volcanic ash concentrations.

(a) Boeing guidance is to always avoid operation in visible ash.

(b) For operations in low volcanic ash concentrations (not visible), the Airframe and Systems can handle these operations without requiring unique or special inspections outside the normal operating and maintenance experience of individual airlines.

(c) Airframe and systems degradation can occur at different rates within many operating environments. As this normal degradation occurs and varies between operating environments, airlines typically adjust their maintenance practices to ensure airplane degradation (e.g. plugged heat exchangers, filters, abrasions, etc.) doesn't cause unscheduled airplane downtime. This same maintenance philosophy applies to operations in low levels of ash concentrations. A continuous operation in areas of low level volcanic ash concentrations is of interest to Boeing in collecting operator observations and experiences. If significant observations are discovered beyond normal variations currently known, Boeing will share these observations and any improved recommendations for maintenance practices. The diagram in Figure 201 illustrates general areas where Boeing recommends increased observation during normal hanger visits, and potential temporary adjustments for maintenance programs based on the observations, to prevent unscheduled maintenance. Observations should include signs of unusual or accelerated abrasions and / or ash accumulation.

(5) Volcanic ash encounters can show these properties inside the engine:

(a) As the ash goes through the combustor, it can change from ash to a glass material in a plastic flow or a molten condition.

(b) In the molten condition, the ash will have very high adhesive qualities.

1) The ash will bond to the nozzle guide vanes of the high pressure turbine.

2) The ash will also bond to the rotor blades of the high pressure turbine.

a) When the ash bonds to these components, it causes a decrease in the turbine flow area.

b) The ash also causes a decrease of nozzle guide vane and turbine blade cooling air.

(c) It may obstruct fuel injectors and cause engine shutdown.

(d) Ash usually accumulates in the turbine section and can continue to cause damage unless fully cleaned.

(e) Ash can cause erosion of the High Pressure Compressor (HPC) airfoils.

(6) Indicators for the flight crew that an airplane has encountered volcanic ash in flight are related to odor, visibility, haze, changing engine conditions, airspeed, pressurization, and static discharges.

(a) Odor - When encountering a volcanic ash cloud, flight crews usually notice a smoky or acrid odor that can smell like electrical smoke, burned dust, or sulfur.

1) Smoke or dust appearing in the flight deck which may cause the flight crew to don oxygen masks.

(b) Visibility - Flight deck windshields become opaque with limited or no view.

(c) Haze - Flight crews, as well as cabin crews and passengers, can see a haze develop within the airplane. Dust can settle on surfaces.

1) Sharp distinct shadows cast by landing lights as compared to the diffused shadows observed in clouds.

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(d) Changing engine conditions - Surging, torching from the tailpipe, and flameouts can occur. Engine temperatures can change unexpectedly and a white or orange glow can appear at the engine inlets.

1) Multiple engine malfunctions, such as stalls, increasing Exhaust Gas Temperature (EGT), torching, flameout, and loss of thrust causing immediate departure from assigned altitude.

2) On engine restart attempts, engines may accelerate to idle very slowly, especially at high altitudes.

(e) Airspeed - If volcanic ash fouls the pitot tube, the indicated airspeed can decrease or fluctuate erratically.

(f) Pressurization - Cabin pressure can change, including possible loss of cabin pressurization.

(g) Static discharges - An electrostatic discharge/St. Elmo's Fire or glow can occur. In these instances, blue-colored sparks can appear to flow up the outside of the windshield or a white glow can appear at the leading edges of the wings or at the front of the engine inlets.

1) An electrostatic discharge/St. Elmo's Fire around the windshields.

NOTE: This electrical discharge is often referred as St. Elmo's Fire, which is an electrical weather phenomenon where luminous plasma is created by a coronal discharge originating from a grounded object in an atmospheric electric field (such as those generated by thunderstorms or thunderstorms created by a volcanic explosion).

2) A bright glow in the engine inlets along with subsequent engine shutdown.

(7) The following is a guide to help identify positive volcanic ash results versus false positive results. Minimum experience inspecting for volcanic ash may induce a number of probable results which have not been recognized previously, but are well known results caused by normal operation.

(a) Results caused by in-flight volcanic ash encounters are usually symmetrically on both sides of the airplane.

NOTE: Positive results on one side of the airplane can be from single engine taxi ground operations or operation through ash that is not uniform thickness.

(b) If there are signs of volcanic ash on one engine, the other engine(s) should show the same signs.

1) For example, results in one pack system should be similar on all packs.

2) For example, an engine spinner or inlet lip skin shows abrasion, the other engine(s) should show similar abrasion.

(c) Volcanic ash typically has a light grey color and looks like talcum powder.

(d) Black or dark debris on the aircraft skin, engine and fan blades are normal debris usually caused by industrial dust.

(e) Yellow or green debris inside of pitot tubes and/or similar sensors are typical of plant pollen.

(f) White debris on acoustic panels of the engine inlet, fan blades, outlet guide vanes etc. can be residuals of water.

(g) White signs on the fuselage and wings can be caused by aircraft dry cleaning.

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(h) A check of the maintenance history in the area of your results may indicate residual material from recent maintenance.

1) For example, a relubrication of the fan blades may also cause unusual debris on spinner, blades and engine inlet.

B. References

Reference	Title
12-11-00-680-801	Fuel Tank Sumps - Fuel Sampling (P/B 301)
12-14-00-600-801	Potable Water System - Drain (P/B 301)
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
21-32-01-000-801	Positive Pressure Relief Valve Removal (P/B 401)
21-32-02-000-801	Positive Pressure Relief Valve Filter Removal (P/B 401)
21-32-03-000-801	Negative Pressure Relief Valve Removal (P/B 201)
21-51-01-000-801	Left Flow Control and Shutoff Valve Removal (P/B 401)
21-51-01-000-802	Right Flow Control and Shutoff Valve Removal (P/B 401)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
21-51-08-000-801	High Pressure Water Separator Removal (P/B 401)
21-51-12-000-801	Condenser Removal (P/B 401)
21-51-21-000-801	Ram Air Inlet Actuator Removal (P/B 401)
21-51-22-000-801	Ram Air Inlet Deflector Door Removal (P/B 401)
21-51-22-000-802	Ram Air Inlet Deflector Door Shaft Assembly Removal (P/B 401)
21-51-23-000-801	Ram Air Inlet Modulation Panels Removal (P/B 401)
21-61-10-100-801	Cabin Temperature Sensor Cleaning (P/B 201)
24-11-13 P/B 401	INTEGRATED DRIVE GENERATOR AIR/OIL HEAT EXCHANGER - REMOVAL/INSTALLATION
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
28-11-11-000-802	Surge Tank Access Door Removal (P/B 401)
29-11-00-200-801	Hydraulic Fluid Check (P/B 601)
30-11-11-000-801	Wing Thermal Anti-Ice Valve Removal (P/B 401)
30-21-11-000-801	Engine Anti-Ice Valve Removal (P/B 401)
32-11-00-200-801	Main Landing Gear Inspection (P/B 601)
32-21-00-200-801	Nose Landing Gear - Inspection (P/B 601)
32-31-51-200-801	Landing Gear Selector Valve - Inspection (P/B 601)
32-41-31-200-801	Brake Metering Valve Detailed Inspection (P/B 601)
32-41-41-700-802	Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane) (P/B 601)
33-41-00-960-803	Wing Illumination Light - Light Assembly Replacement (P/B 201)
33-42-21-960-801	Fixed Landing Light - Main Array Assembly Replacement (P/B 201)
33-42-21-960-803	Runway Turnoff Light - Light Assembly Replacement (P/B 201)
33-43-11-000-801	Forward Position Light Removal (P/B 401)
33-43-11-400-801	Forward Position Light Installation (P/B 401)
33-43-12-000-801	Aft Marker Light Removal (P/B 401)

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(Continued)

Reference	Title
33-43-12-400-801	Aft Marker Light Installation (P/B 401)
33-43-13-960-801	Tailcone Position Lights Replacement (P/B 201)
33-44-01-960-801	Upper Anti-Collision Light - Light Assembly Replacement (P/B 201)
33-44-02-960-802	Lower Anti-Collision Light - Light Assembly Replacement (P/B 201)
33-44-13-000-801	Winglet Anti-Collision Light Removal (P/B 201)
33-44-13-400-801	Winglet Anti-Collision Light Installation (P/B 201)
33-49-00-960-802	Logo Light - Light Assembly Replacement (P/B 201)
35-12-00-700-801	Crew Oxygen Stowage Box Test (Mask Stowed in Stowage Box) (P/B 501)
35-12-00-700-802	Crew Oxygen Mask-Regulator Test (P/B 501)
35-22-00-700-801	Passenger Oxygen System Automatic Actuation Functional Test (P/B 501)
36-11-02-000-801	Intermediate Pressure Check Valve (IPCV) Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-05-000-801	Intermediate Manifold Pressure (PI) Sensor Removal (P/B 401)
36-11-06-000-801	Duct Vent Valve Removal (P/B 401)
36-12-01-000-801	Bleed Air Precooler Removal with Engine Removed (P/B 401)
36-12-03-000-801	Fan Air Modulating Valve Removal (P/B 401)
36-13-03-000-801	Ground Pneumatic Connector Check Valve Removal (P/B 401)
36-13-04-000-801	Bleed Air Isolation Valve Removal (P/B 401)
36-14-02-000-801	APU Check Valve Removal (P/B 401)
36-21-01-000-801	Duct Pressure Transducer Removal (P/B 401)
38-11-03-000-801	Fill/Overflow Valve Removal (P/B 401)
38-14-01-000-801	Water Quantity Transmitter Removal (P/B 401)
38-42-10-960-801	Bleed Air Filter Element Replacement (P/B 401)
47-32-04-000-802	NGS Filter Element - Removal (P/B 401)
49-15-11-200-801	Air Inlet Seal Inspection (P/B 601)
49-15-15-200-801	APU Air Inlet Diffuser Ducts Inspection (P/B 601)
49-52-13-000-801	Bleed Air Duct Removal (P/B 401)
49-81-11-200-802	Exhaust Duct Muffler Inspection (P/B 601)
49-91-41-200-801	Oil Cooler Inspection (P/B 601)
71-00-00-200-806-G00	Heavy Sand/Dust Ingestion and Volcanic Ash Ingestion Inspection (P/B 601)
80-11-02-000-801-G00	Starter Air Valve Removal (P/B 401)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
300	Empennage
400	Powerplant and Nacelle Struts
500	Left Wing

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(Continued)

Zone	Area
600	Right Wing
700	Landing Gear and Landing Gear Doors
800	Doors

D. Phase IA General Visual Inspection

SUBTASK 05-51-31-210-001

(1) Do a visual inspection of the airframe for abrasion damage:

(a) Radome for unusual abrasion

(b) Wing leading edges, landing lights and navlight glazings for unusual abrasion or crazing.

(c) Cockpit windows for unusual abrasion, cracking, crazing, discoloration or decreased visibility.

(d) Landing gear painted surfaces for abrasion.

(e) Pack ram air inlets for unusual abrasion.

(f) Broadband radome for unusual abrasion.

SUBTASK 05-51-31-210-002

(2) Do a visual inspection of the sensors and probes for abrasion damage:

(a) Angle of Attack (AOA) sensors for unusual abrasion and free movement

(b) TAT probes for unusual abrasion

(c) Pitot tubes for unusual abrasion

(d) Static ports for unusual abrasion.

SUBTASK 05-51-31-210-003

(3) Do a visual inspection of the following areas for ash contamination:

(a) Engine inlet lip skins, fan blades, and spinners

(b) Pitot probes and static ports

(c) Drain holes (bilge drains and engine drains)

(d) Lubricated surfaces (actuator rams, landing gear inner cylinders)

(e) Pack ram air inlets.

SUBTASK 05-51-31-210-004

(4) Do a visual inspection of the Auxiliary Power Unit (APU) for damage:

NOTE: Inspection can be performed from ground level.

NOTE: Use the applicable APU manufacturer recommendations for engine inspection for flights through areas of low level volcanic ash concentrations.

(a) APU inlet for unusual abrasion

(b) APU exhaust for unusual abrasion

(c) APU drains for abnormal conditions such as excessive or unexpected fuel or oil drainage.

SUBTASK 05-51-31-210-005

(5) Do a visual inspection of each engine for damage:

NOTE: Use the applicable engine manufacturer recommendations for engine inspection and sampling for flights through areas of low level volcanic ash concentrations.

(a) Engine nose cowl inlet lips for unusual abrasion

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(b) Engine spinner for unusual abrasion

(c) Fan Outlet Guide Vane (OGV)s leading edge for unusual abrasion (through fan blades).

SUBTASK 05-51-31-210-006

(6) If any of the Phase IA inspections show positive results, then the Phase IB inspection must be completed before the next flight.

NOTE: Typically, damage from volcanic ash encounters occur symmetrically on both sides of the aircraft. You may also extend the visual inspections to leading edges of stabilizers, antennae etc., as far as visible from ground, to confirm the inspection results.

E. Phase IB General Visual Inspection

SUBTASK 05-51-31-290-001

(1) Borescope first stages of HPC and High Pressure Turbine (HPT) blades for unusual abrasion or ash deposit on one engine.

(a) If the one engine does not show signs of unusual abrasion or grey dust accumulation, the other engine(s) do not need to be inspected.

SUBTASK 05-51-31-210-007

(2) If any of the Phase IB inspections show positive results, then the Phase II inspection must be completed before the next flight.

F. Phase II Inspection

SUBTASK 05-51-31-210-008

(1) Examine the engines.

(a) Do the steps that follow to examine each engine.

1) Examine each engine for damage, do this task: Heavy Sand/Dust Ingestion and Volcanic Ash Ingestion Inspection, TASK 71-00-00-200-806-G00.

2) Examine the engine control mechanisms and linkages for volcanic ash.

a) Examine the Variable Stator Vane (VSV) and Variable Bleed Valve (VBV) system for volcanic ash.

NOTE: These are mechanically controlled by a series of linkages to adjust the airflow through the engine.

b) Examine the High Pressure Turbine Active Clearance Control (HPTACC) and the Low Pressure Turbine Active Clearance Control (LPTACC) valves and tubing for volcanic ash.

(b) If any of the Phase II inspections show positive results, then the Phase III inspection must be completed before the next flight.

G. Phase III

SUBTASK 05-51-31-210-009

(1) Air Conditioning System

(a) Do the steps that follow to examine the air conditioning system:

1) Examine the aft outflow valve, do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

a) If you find volcanic ash in the valve, clean or replace the valve.

2) Remove and examine the negative pressure relief doors, do this task: Negative Pressure Relief Valve Removal, TASK 21-32-03-000-801.

a) If you find volcanic ash in the door area, clean or replace the door and linkage.

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3) Remove and examine the positive pressure relief valves, do this task: Positive Pressure Relief Valve Removal, TASK 21-32-01-000-801.
   a) If you find volcanic ash in the valves, clean or replace the valves.
4) Remove and examine the positive pressure relief valve filters, do this task: Positive Pressure Relief Valve Filter Removal, TASK 21-32-02-000-801.
   a) If you find volcanic ash in the filters, clean or replace the filters.
5) Examine the ram air inlet deflector door, do this task: Ram Air Inlet Deflector Door Removal, TASK 21-51-22-000-801.
   a) If you find volcanic ash in the deflectors, clean the deflectors and movable joints.
6) Examine the ram air inlet deflector door shaft assemblies, do this task: Ram Air Inlet Deflector Door Shaft Assembly Removal, TASK 21-51-22-000-802.
   a) If you find volcanic ash in the shaft assemblies, clean the shaft assemblies.
7) Examine the ram air inlet modulation panels, do this task: Ram Air Inlet Modulation Panels Removal, TASK 21-51-23-000-801.
   a) If you find volcanic ash in the modulation panels, clean the panels and movable joints.
8) Examine the ram air inlet modulation panel shaft assemblies.
   a) If you find volcanic ash in the shaft assemblies, clean the shaft assemblies.
9) Examine the ram air exhaust modulation louver assemblies.
   a) If you find volcanic ash in the louver assemblies, clean the louvers, bearings and movable joints.
10) Examine the ram air inlet actuators, do this task: Ram Air Inlet Actuator Removal, TASK 21-51-21-000-801.
   a) If you find volcanic ash in the actuators, clean or replace the actuators.
11) Remove and examine the Air Cycle Machine (ACM), do this task: Air Cycle Machine Removal, TASK 21-51-04-000-801.
   a) Check for volcanic ash in the ACM valves.
      <1> If you find volcanic ash in the ACM valves, clean or replace the ACM.
   b) Remove the high pressure water separator, do this task: High Pressure Water Separator Removal, TASK 21-51-08-000-801.
   c) Examine the water for contamination.
   d) If there is contamination in the water of the high pressure water separator, do the steps that follow:
      <1> Remove the ACM, do this task: Air Cycle Machine Removal, TASK 21-51-04-000-801.
      <2> Inspect the ACM for torus erosion.
         NOTE: Torus erosion is caused because atmospheric contamination in the bleed air is trapped in the ACM torus and moves around the torus, which erodes the inner surface of the torus.
12) Remove and examine the air cycle cooling pack primary and secondary heat exchangers.

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a) If you find volcanic ash in the heat exchangers, clean or replace the heat exchangers.

NOTE: For more information, refer to the Honeywell CMM 21-50-14.

13) Make sure that the heat exchanger passes the Pressure Flow Tests to verify that the unit is fully cleaned and not clogged.

a) Do this every 2000-3000 hours.

14) Examine the water extract nozzles and drain lines.

a) Clean the water extract nozzles and drain lines if volcanic ash is found.

15) Remove and examine the condensers, do this task: Condenser Removal, TASK 21-51-12-000-801.

a) If you find volcanic ash in the condensers, clean or replace the condensers.

16) Remove and examine the flow control pack valves, do these tasks: Left Flow Control and Shutoff Valve Removal, TASK 21-51-01-000-801, Right Flow Control and Shutoff Valve Removal, TASK 21-51-01-000-802.

a) If you find volcanic ash in the pack valves, clean or replace the pack valves.

17) Remove and examine the recirculating air fans.

a) If you find volcanic ash in the fans, clean or replace the fans.

18) Remove and examine one of the trim air nozzles.

a) If you find volcanic ash in one of the trim nozzles, remove and clean all of the trim nozzles.

19) Remove and examine the recirculation air filters.

a) If you find volcanic ash in the filters, clean or replace the filters.

20) Remove and examine the recirculation air check valves.

a) If you find volcanic ash in the valves, clean or replace the valves.

21) Remove and examine the equipment cooling automatic flow control valve.

a) If you find volcanic ash in the valve, remove or replace the valve.

22) Remove and examine the exhaust fans and check valves.

a) If you find volcanic ash in the fans or valves, clean or replace the fans and valves.

23) Remove and examine the pack temperature control valve.

a) If you find volcanic ash in the control valves, clean or replace the valves.

24) Do a visual examination of the temperature sensors and connectors in both air conditioning pack bays.

a) Make sure that the temperature sensors and connectors are not damaged and are installed correctly.

25) Make sure that the flight compartment and cabin temperature sensors are not clogged.

a) If you find volcanic ash in the flight compartment or cabin temperature sensors, clean or replace the sensors, do this task: Cabin Temperature Sensor Cleaning, TASK 21-61-10-100-801.

26) Remove and examine the trim air valve.

a) If you find volcanic ash in the valves, clean or replace the valves.

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SUBTASK 05-51-31-210-010

(2) Nitrogen Generation System (NGS)

(a) Examine the NGS Filter (NGS Filter Element - Removal, TASK 47-32-04-000-802).

1) If you find volcanic ash in the NGS filter, clean or replace the filter, if necessary.

SUBTASK 05-51-31-210-011

(3) Pneumatic System

(a) Examine the Pneumatic System.

1) Do this task: Bleed Air Precooler Removal with Engine Removed, TASK 36-12-01-000-801.

2) If you find ash in the bleed air precooler, remove the components that follow and look for volcanic ash.

a) Do this task: Duct Vent Valve Removal, TASK 36-11-06-000-801.

b) Do this task: PRSOV Removal, TASK 36-11-04-000-801.

c) Do this task: Intermediate Pressure Check Valve (IPCV) Removal, TASK 36-11-02-000-801.

d) Do this task: Starter Air Valve Removal, TASK 80-11-02-000-801-G00.

e) Do this task: Bleed Air Isolation Valve Removal, TASK 36-13-04-000-801.

f) Do this task: APU Check Valve Removal, TASK 36-14-02-000-801.

g) Do this task: Ground Pneumatic Connector Check Valve Removal, TASK 36-13-03-000-801.

h) Do this task: Engine Anti-Ice Valve Removal, TASK 30-21-11-000-801.

i) Do this task: Duct Pressure Transducer Removal, TASK 36-21-01-000-801.

j) Do this task: Intermediate Manifold Pressure (PI) Sensor Removal, TASK 36-11-05-000-801.

k) Do this task: Fan Air Modulating Valve Removal, TASK 36-12-03-000-801.

SUBTASK 05-51-31-210-012

(4) External Light Lens

(a) Examine all of the external lights lenses for abrasions and damage.

NOTE: If the lights or lenses are frosted or damaged, replace then when it is possible.

1) Do this task: Wing Illumination Light - Light Assembly Replacement, TASK 33-41-00-960-803.

2) Do this task: Fixed Landing Light - Main Array Assembly Replacement, TASK 33-42-21-960-801.

3) Do this task: Runway Turnoff Light - Light Assembly Replacement, TASK 33-42-21-960-803.

4) Do this task: Upper Anti-Collision Light - Light Assembly Replacement, TASK 33-44-01-960-801.

5) Do this task: Lower Anti-Collision Light - Light Assembly Replacement, TASK 33-44-02-960-802.

6) Do this task: Logo Light - Light Assembly Replacement, TASK 33-49-00-960-802.

7) Do this task: Tailcone Position Lights Replacement, TASK 33-43-13-960-801.

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8) Do these tasks: Forward Position Light Removal, TASK 33-43-11-000-801, Forward Position Light Installation, TASK 33-43-11-400-801.

9) Do these tasks: Aft Marker Light Removal, TASK 33-43-12-000-801, Aft Marker Light Installation, TASK 33-43-12-400-801.

10) Do these tasks: Winglet Anti-Collision Light Removal, TASK 33-44-13-000-801, Winglet Anti-Collision Light Installation, TASK 33-44-13-400-801.

SUBTASK 05-51-31-210-013

(5) Fuel System

(a) Look for ash in a fuel sample from each fuel tank sump, do this task: (Fuel Tank Sumps - Fuel Sampling, TASK 12-11-00-680-801).

(b) Look for ash in a fuel sample from each surge tank sump (if there is fuel in the surge tanks).

(c) Visually inspect for ash in each surge tank, do this task: Surge Tank Access Door Removal, TASK 28-11-11-000-802.

SUBTASK 05-51-31-210-014

(6) Oxygen System

(a) If you find volcanic ash in the passenger and control cabin, examine the oxygen system.

1) Remove the stowage boxes for the mask and regulator and look for ash, do this task: Crew Oxygen Stowage Box Test (Mask Stowed in Stowage Box), TASK 35-12-00-700-801.

2) Examine portable oxygen masks for ash, do this task: Crew Oxygen Mask-Regulator Test, TASK 35-12-00-700-802.

3) Examine the passenger oxygen masks for ash, do this task: Passenger Oxygen System Automatic Actuation Functional Test, TASK 35-22-00-700-801.

SUBTASK 05-51-31-211-001

(7) APU

(a) Operate the APU on the ground and monitor APU pneumatic performance and operability in both Main Engine Start (MES) and Environmental Control System (ECS) operating modes to confirm no significant damage has occurred to the APU.

NOTE: Use the applicable APU manufacturer recommendations for engine inspection for flights through areas of low level volcanic ash concentrations.

(b) Examine the APU for volcanic ash contamination.

NOTE: Do not perform a borescope inspection of the APU.

1) Examine the inlet and exhaust ducts for ash contamination.

2) If there is ash in the APU inlet or exhaust ducts, look for volcanic ash in the components that follow:

a) Do this task: Oil Cooler Inspection, TASK 49-91-41-200-801.

b) Do this task: Air Inlet Seal Inspection, TASK 49-15-11-200-801.

c) Do this task: APU Air Inlet Diffuser Ducts Inspection, TASK 49-15-15-200-801.

d) Do this task: Exhaust Duct Muffler Inspection, TASK 49-81-11-200-802.

e) Do this task: Bleed Air Duct Removal, TASK 49-52-13-000-801.

SUBTASK 05-51-31-210-015

(8) Potable Water System

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(a) Examine the potable water system for volcanic ash contamination.

1) Remove the bleed air filter and look for ash contamination, do this task: Bleed Air Filter Element Replacement, TASK 38-42-10-960-801.

2) If there is ash in the air filter, do these steps:

a) Visually inspect for ash in a sample from the potable water tanks, do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

b) Remove the fill/overflow valves and look for ash contamination, do this task: Fill/Overflow Valve Removal, TASK 38-11-03-000-801.

c) Remove the water quantity transmitters and look for ash contamination, do this task: Water Quantity Transmitter Removal, TASK 38-14-01-000-801.

SUBTASK 05-51-31-210-016

(9) Smoke Detectors

(a) Examine all of the cargo compartment smoke detectors for volcanic ash contamination.

SUBTASK 05-51-31-210-017

(10) Fire Extinguisher Bottle Nozzles

(a) Examine the nozzles of the fire extinguisher bottles that follow for volcanic ash contamination:

1) The engine fire extinguisher bottles

2) The APU fire extinguisher bottle.

SUBTASK 05-51-31-210-018

(11) Engine Accessories

(a) Examine the engine accessories that follow for volcanic ash contamination:

1) Examine the integrated drive generator air and oil cooler for ash, do this task: INTEGRATED DRIVE GENERATOR AIR/OIL HEAT EXCHANGER - REMOVAL/INSTALLATION, PAGEBLOCK 24-11-13/401.

SUBTASK 05-51-31-210-026

(12) Nacelle

(a) Examine for abrasion damage.

1) Examine all streamwise transverse interfaces between nacelle component to include:

a) Inlet lip skin to inlet barrel skin

b) Inlet to fan cowl interfaces

c) Fan cowl to thrust reverser interfaces

d) Thrust reverser to Core Cowl for those installations with a separate core cowl.

2) Make sure that you inspected all the external surfaces and the internal duct surfaces of the nacelle components.

(b) Examine for abrasion damage on all streamwise transverse edges of access panels and latch access doors.

(c) Examine all latch mechanisms for debris contamination.

(d) Examine all hinge mechanisms for debris contamination.

(e) Examine the drag links for abrasion damage and drag link attach ends for debris contamination.

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(f) Examine the TR blocker door hinges for debris contamination and streamwise transverse edges for erosion damage.

(g) Examine the nacelle seals for debris contamination.

1) If contamination is present, clean the seals and lands of the particulate contamination.

(h) Examine the Thrust Reverser Actuation System (TRAS) for particulate contamination.

SUBTASK 05-51-31-210-019

(13) Landing Gear

(a) Examine the landing gear components that follow for volcanic ash contamination:

NOTE: If you find volcanic ash, remove the ash.

1) The bare inner cylinders of the nose and main landing gear shocks struts.

2) All bare piston rods on all hydraulic actuators.

3) All attachment points for the landing gear components.

4) All landing gear:

a) door hinges

b) lock mechanisms

c) Do this task: Main Landing Gear Inspection, TASK 32-11-00-200-801.

d) Do this task: Nose Landing Gear - Inspection, TASK 32-21-00-200-801.

e) all proximity switches.

5) Landing gear brakes, do this task: Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane), TASK 32-41-41-700-802.

a) The selector valves, do this task: Landing Gear Selector Valve - Inspection, TASK 32-31-51-200-801.

b) The brake metering valves, do this task: Brake Metering Valve Detailed Inspection, TASK 32-41-31-200-801.

SUBTASK 05-51-31-210-020

(14) Hydraulic Systems

(a) Examine the hydraulic system for signs of volcanic ash contamination.

1) Visually inspect for ash in a sample of the hydraulic fluid, do this task: Hydraulic Fluid Check, TASK 29-11-00-200-801.

SUBTASK 05-51-31-210-021

(15) Flight Controls

(a) Examine the flight controls for signs of volcanic ash contamination:

NOTE: The flight control include the ailerons, elevator, rudder, leading edge flaps and slats, and spoilers.

WARNING DO NOT GO INTO THE WING LEADING EDGE AND TRAILING EDGE AREAS BEFORE YOU INSTALL SAFETY LOCKS. IF YOU DO NOT OBEY, INJURY TO PERSONNEL FROM ACCIDENTAL FLAP/SLAT OPERATION CAN OCCUR.

1) Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

2) Examine the bare actuator piston rods of the components that follow:

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a) the ailerons
b) the elevator
c) the rudder
d) the leading edge flaps and slats
e) the spoilers.

3) the control cables in unpressurized areas
4) the stabilizer jackscrew
5) the flight control surface hinges
6) the trailing edge flap tracks
7) the flap ball screw surfaces
8) the flap mechanisms
9) the trailing edge flap track fairings and linkages
10) the leading edge flap and slat mechanisms
11) the aileron and elevator balance panels.

SUBTASK 05-51-31-210-022

(16) Wing and Nacelle Anti-Ice System

(a) Examine the anti-ice system for signs of volcanic ash contamination.

1) Remove the anti-ice valves for the wing and look for volcanic ash contamination, do this task: Wing Thermal Anti-Ice Valve Removal, TASK 30-11-11-000-801.

2) Remove the anti-ice valves for the inlet cowl from the engine and look for volcanic ash contamination, do this task: Engine Anti-Ice Valve Removal, TASK 30-21-11-000-801.

SUBTASK 05-51-31-210-023

(17) Internal Areas of the Airplane

(a) Examine the internal areas of the airplane that follow for signs of volcanic ash contamination.

1) Visually inspect ash on these components in the upper and main compartments:

a) the flight compartment
b) the powered crew seats
c) the flight instruments
d) the electrical/electronic control panels
e) the passenger compartment
f) the closets
g) the passenger seats
h) the lavatories
i) the lavatory components
j) the galleys
k) the galley components
l) the floor coverings
m) refrigeration/chiller units (if installed)

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2) baggage and cargo systems
3) electrical and electronic compartments
4) Electronic Equipment Racks
5) control cables and mechanisms.

SUBTASK 05-51-31-211-002

(18) Examine the broadband radome for volcanic ash.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-31-420-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-31-860-001

(2) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 05-51-31-860-002

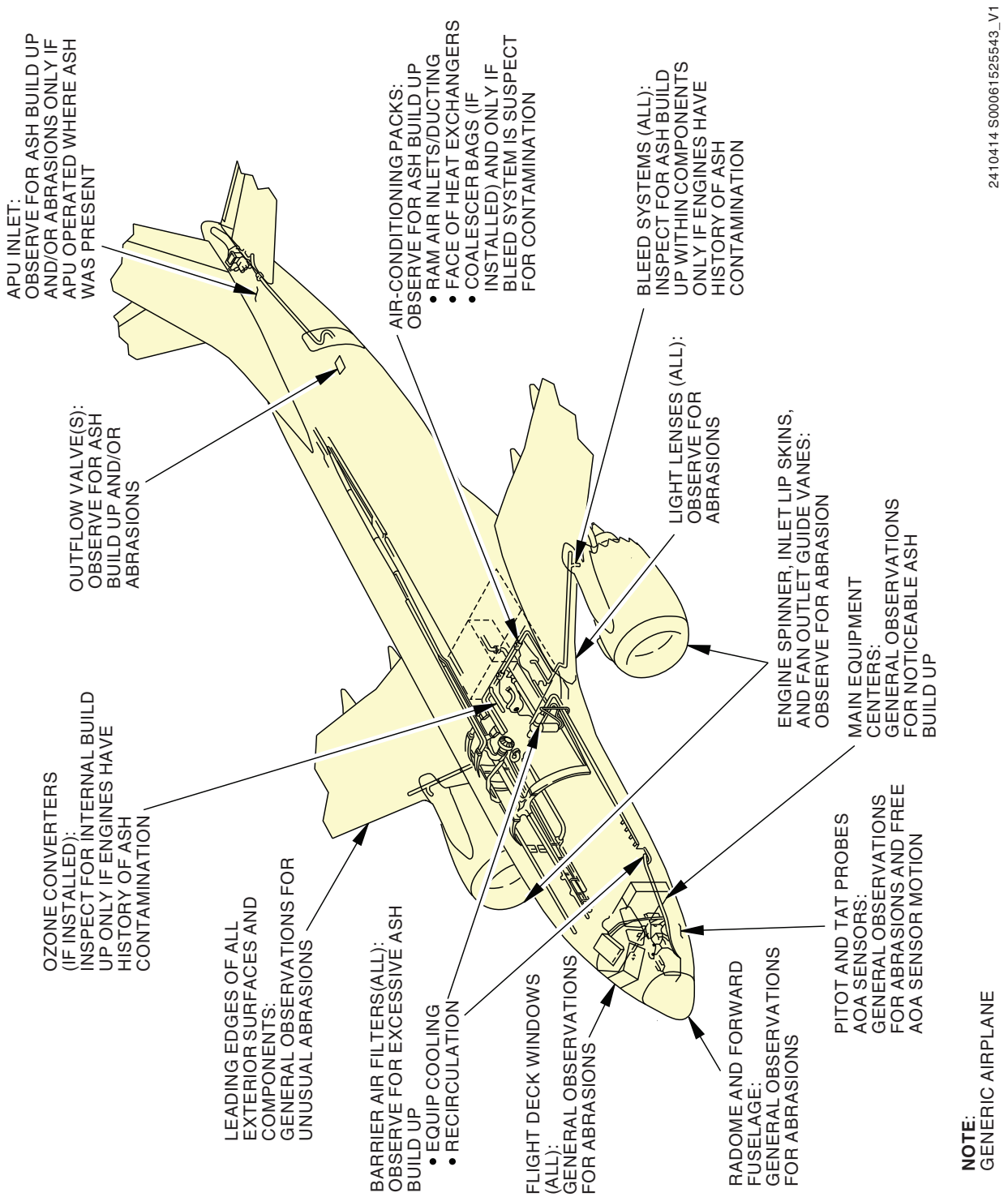
(3) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-31-860-003

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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Airframe and Systems Observation Areas
Figure 201/05-51-31-990-801

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TASK 05-51-31-200-801

3. Volcanic Ash Ground Fallout Conditional Inspection

A. General

(1) This task has information to examine a parked airplane for contamination in volcanic ash fallout conditions.

(2) Fallout is identified as ash drifting down from the air, or is blown about by the wind. Fallout is also caused by the ash being blown or moved about because of the movement of airplanes and ramp vehicles.

(3) For airplanes parked in fallout conditions, protect all inlets of systems, probes, static ports, engines, gaps, etc. with suitable covers.

(a) Any covered areas should have appropriate streamers or markings, to be removed before operation.

(4) Volcanic ash has the qualities that follow:

(a) It is hard and highly abrasive.

(b) It usually contains trace chemicals which can be acidic (pH between 2 and neutral).

(c) It can cause erosion/abrasion, etching or shot peening on exposed surfaces and remove protective finishes.

(d) The dimensions of most ash particles are less than 5 microns with trace amounts that are more than 50 microns.

(e) It resembles talcum powder.

(f) It stays on exposed lubricated surfaces.

(g) Able to enter small orifices and pass by seals.

(h) Can cause unusual wear and contamination of many airplane systems and filter elements.

WARNING

DO NOT GET VOLCANIC ASH IN YOUR MOUTH, EYES, OR ON YOUR SKIN. PUT ON A RESPIRATOR MASK THAT REMOVES VOLCANIC ASH PARTICLES SUFFICIENTLY. PUT ON CLOTHES AND GOGGLES (OR OTHER APPROVED EYE PROTECTION) THAT KEEP VOLCANIC ASH OFF YOU. VOLCANIC ASH CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Volcanic ash can cause discomfort and injury to persons during fallout conditions, and during inspection and cleaning.

(a) Precautions must be followed when you work in a volcanic ash environment. This will prevent the entry of volcanic ash into your eyes and your respiratory (breathing) system.

(6) The following is a guide to help identify positive volcanic ash findings versus false positive findings. Minimum experience inspecting for volcanic ash may induce a number of probable findings which have not been recognized previously, but are well known findings caused by normal operation.

(a) Black or dark debris on the aircraft skin, engine and fan blades are normal debris usually caused by industrial dust.

(b) Yellow or green debris inside of pitot tubes and/or similar sensors are typical of plant pollen.

(c) White debris on acoustic panels of the engine inlet, fan blades, outlet guide vanes etc. can be residuals of water.

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(d) White signs on the fuselage and wings can be caused by aircraft dry cleaning.

(e) A check of the maintenance history in the area of your findings may indicate residual material from recent maintenance.

1) For example, a relubrication of the fan blades may also cause unusual debris on spinner, blades and engine inlet.

B. References

Reference	Title
12-40-00-100-804	Clean the Exterior Surface of Volcanic and Fire Ash (P/B 201)

C. Procedure

SUBTASK 05-51-31-210-024

(1) Examine the airplane exterior surfaces and the area (ground, service equipment, buildings) around the airplane for any indication of ash.

(a) If no ash is found, no more inspection or cleaning is necessary.

SUBTASK 05-51-31-160-001

(2) If any ash is found, remove all traces of the ash, including areas such as the fuselage crown, horizontal surfaces, inlets, wheel wells, and exposed chrome common to landing gear.

(a) do this task:, Clean the Exterior Surface of Volcanic and Fire Ash, TASK 12-40-00-100-804.

SUBTASK 05-51-31-210-025

(3) Make sure the area around the airplane is free from ash.

NOTE: Coordinate with the local airport authority to make sure the ramps, taxiways, and runways are clear from ash contamination.

——— END OF TASK ———

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TAIL STRIKE/SKID CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection procedure must be done if an airplane tail strike (tail dragged or touched the runway) occurred when the airplane landed or during a takeoff.

(1) If the green part of the warning decal on the aft side of the tail skid cartridge is not visible, a tail strike may have occurred.

B. The tail skid assembly located on the keel line consists of a drag lever, with a drag shoe.

(1) The tail skid assembly is designed to drag the drag shoe along the ground and also to compress the crushable cartridge in the shock absorber.

C. Damage can occur to the fuselage when the tail touches the runway.

(1) If you find damage, refer to the applicable Structural Repair Manual (SRM) procedures.

D. This procedure has the following task(s):

(1) Tail Strike Inspection.

(2) Tail Skid Inspection.

TASK 05-51-32-210-801

2. Tail Strike Inspection

A. References

Reference	Title
27-41-00-700-801	Stabilizer Manual Trim and Trim Indicator - Test (P/B 501)
27-41-11-000-801	Horizontal Stabilizer Removal (P/B 401)
27-41-11-400-801	Horizontal Stabilizer Installation (P/B 401)
27-41-41-210-801	Stabilizer Control Mechanism Detail Visual Inspection (P/B 601)
51-21-99 P/B 701	DECORATIVE EXTERIOR PAINT SYSTEM - CLEANING/PAINTING

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
144	Area Below Aft Cargo Compartment - Right
145	Aft Cargo Compartment Equipment Bay - Left
310	Fuselage - Body Station 1016.00 to Body Station 1260.00

C. Tail Strike Inspection Procedure

SUBTASK 05-51-32-210-001

(1) Examine the tail skid assembly drag shoe for tail strike damage as follows:

NOTE: The cartridge assembly warning placard must be examined immediately after a tail strike. The skirt fairing can continue to extend from gravity. This will cause an error when reading the warning placard/decal.

(a) Do this task: Tail Skid Inspection, TASK 05-51-32-210-802.

SUBTASK 05-51-32-210-002

(2) Inspect the external skin surfaces of the lower aft fuselage.

(a) Look from station 847 to the APU compartment.

1) Look for signs of:

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a) scrapes
b) holes
c) cracked skin
d) burns
e) buckled belly skin
f) loose fasteners
g) missing fasteners.

(b) Look at the following areas for broken, bent (distorted), missing parts or other damage:
1) The APU firewall
2) The APU drain
3) The doors of the APU compartment
4) The drain mast.

(c) Inspect the upper fuselage structure aft of the wing rear spar for buckled structure, cracks and paint that has flaked.

SUBTASK 05-51-32-200-001

(3) Examine all of the external and internal structure of the fuselage, section 48, Body Station (BSTA) 1016, that you can get access to and/or is called out in this procedure for signs of buckling, deformation or damage.

NOTE: Look at the structure from the rear pressure bulkhead to the aft end of the airplane.

(a) Do a detailed visual inspection of the exterior fuselage skin 10 in. (25.4 cm) forward and aft of the BSTA 1016 skin butt joint between stringers S-10L and S-10R.

NOTE: Gain access to the exterior of the fuselage skin at BS 1016 under the vertical stabilizer fairing, 323AL and 323AR.

(b) Do a detailed visual inspection of the entire aft side of the BSTA 1016 aft pressure bulkhead.
1) Give particular attention to the upper bulkhead structure between skin stringers S-10L and S-10R.

NOTE: Access to the interior unpressurized aft pressure bulkhead area is through the aft access hatch, 311BL.

SUBTASK 05-51-32-210-003

(4) When you find external damage, inspect the fuselage internally as follows:

(a) In the area of the tail strike:
1) Inspect for bent or cracked stringers, frames, or clips.
2) Inspect for buckled or wrinkled webs.
3) Inspect for loose or missing fasteners and paint that has flaked.
4) Inspect the lower area of the aft pressure bulkhead for damage.
5) Inspect for pulled apart structure or unusual gaps.

(b) Aft pressure bulkhead at BSTA 1016:
1) Examine at BSTA 1016 the upper center web installation on the forward side of the pressure bulkhead for any damage.

NOTE: Gain access to inspect the upper center web installation by removing the galley and lavatories as necessary.

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SUBTASK 05-51-32-800-001

(5) If damage is found in the above paragraphs, use the applicable BSTA and do the repairs.

D. Stabilizer Position Indicator System Procedure

SUBTASK 05-51-32-210-004

(1) Inspect the stabilizer position indication system components, do this task: Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801.

(a) Inspect for broken cables.

(b) Inspect for cables that are out of the pulley groove.

(c) When there is no internal damage to the fuselage structure, do the operational test for the stabilizer position indication system.

1) Do this task: Stabilizer Control Mechanism Detail Visual Inspection, TASK 27-41-41-210-801.

2) Do this task: Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801.

(d) When there is internal damage to the fuselage structure, do the system test for the stabilizer position indicating system.

SUBTASK 05-51-32-210-005

(2) If the stabilizer position indicating system components are damaged:

(a) Do this task: Horizontal Stabilizer Removal, TASK 27-41-11-000-801

(b) Do this task: Horizontal Stabilizer Installation, TASK 27-41-11-400-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-32-410-001

(1) Install all of the components you removed if the components are serviceable, or install replacement parts.

SUBTASK 05-51-32-370-001

(2) If the tail skid drag shoe is within tolerance and the paint finish on the lower surface of the tail skid drag shoe has been gouged or scratched by contact with the runway, paint the skid drag shoe in accordance with the Decorative Exterior Finishes procedure (DECORATIVE EXTERIOR PAINT SYSTEM - CLEANING/PAINTING, PAGEBLOCK 51-21-99/701).

NOTE: If the tail skid drag shoe scratch or gouge damage is not painted over, a person may interpret the damage as new and request an unnecessary inspection. The addition of paint can help to determine when new tail strikes have occurred.

SUBTASK 05-51-32-200-003

(3) Make sure that pressurization mode selector is in the AUTO position.

——— END OF TASK ———

TASK 05-51-32-210-802

3. Tail Skid Inspection

A. References

Reference	Title
32-71-00-210-801	Tailskid Cartridge Assembly Inspection/Check (P/B 601)
32-71-00-211-801	Tailskid Shoe Inspection (P/B 601)

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B. Location Zones

Zone	Area
100	Lower Half of Fuselage
144	Area Below Aft Cargo Compartment - Right
145	Aft Cargo Compartment Equipment Bay - Left
310	Fuselage - Body Station 1016.00 to Body Station 1260.00

C. Tail Skid Inspection

SUBTASK 05-51-32-210-006

(1) Inspect the external tail drag surfaces.

(a) Visually inspect the tail skid and skin surfaces of the lower aft fuselage (TASK 32-71-00-211-801).

1) Make sure that the tail drag shoe is serviceable and within tolerance limits.

(b) Inspect the tail skid cartridge assembly (TASK 32-71-00-210-801).

NOTE: The cartridge assembly warning placard must be examined immediately after a tail strike.

1) Inspect the tail skid fuse pin for signs of shearing.

a) Replace the fuse pin if damaged.

(c) Inspect the tail skid drag assembly for cracks or fractures.

(d) Inspect from station 847 to the Auxiliary Power Unit (APU) compartment.

NOTE: This inspection is the lower area of Section 47 and 48.

1) Look for signs of:

a) scrapes

b) holes

c) cracked skin

d) burns

e) buckled belly skin

f) loose fasteners

g) missing fasteners.

SUBTASK 05-51-32-210-007

(2) Inspect the internal tail drag surfaces.

(a) When you find external damage, examine the fuselage internally in the area of the tail drag.

(b) Look for bent, cracked, pulled apart structure or unusual gaps:

1) stringers

2) frames

3) clips

4) wrinkled webs

5) look for loose or missing fasteners.

(c) Examine the lower area of the aft pressure bulkhead for damage.

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SUBTASK 05-51-32-800-002

(3) If you find internal or external damage, use the applicable Structural Repair Manual (SRM) for the repairs.

SUBTASK 05-51-32-200-002

(4) If you found damage to the bottom of the fuselage, or you have to replace the crushable cartridge, do this task: Tail Strike Inspection, TASK 05-51-32-210-801.

NOTE: It is not necessary to repeat inspection steps that have already been performed as part of the tail skid inspection procedure.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-32-410-002

(1) Install all of the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-32-200-004

(2) Make sure that pressurization mode selector is in the AUTO position.

——— END OF TASK ———

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HOT AIR DUCT RUPTURE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This section has two tasks to examine the airplane after a hot air duct rupture.

B. The first task examines the airplane structure for possible heat damage caused by a ruptured duct.

C. The second task examines the airplane nacelle structure for possible structural damage caused by a ruptured duct that has occurred below the strut for which there was a subsequent deployment of the affected thrust reverser.

TASK 05-51-34-210-801

2. Airplane Structure Hot Air Duct Rupture Conditional Inspection

A. Location Zones

Zone	Area
100	Lower Half of Fuselage
400	Powerplant and Nacelle Struts
500	Left Wing
600	Right Wing

B. Procedure

SUBTASK 05-51-34-210-001

(1) Examine the area of the airplane structure which has had a high heat condition caused by a ruptured pneumatic duct.

(a) If you find damage, refer to Structures/SRM PROCEDURE 51-00-00-0G-0 for the repairs.

C. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-34-410-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

END OF TASK

TASK 05-51-34-200-801

3. Nacelle Structure Hot Air Duct Rupture Conditional Inspection

A. Location Zones

Zone	Area
400	Powerplant and Nacelle Struts

B. Procedure

SUBTASK 05-51-34-210-002

(1) Contact Boeing for specifics of information related to the inspection of the inner wall for both location and the method of inspection.

C. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-34-410-002

(1) Install the components you removed if they are serviceable, or install replacement parts.

END OF TASK

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OVERWEIGHT LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection is only required for overweight landings that have no flight crew comments regarding if the landing was “hard” or “not hard”.

(1) Inspections for hard overweight landings are covered in procedure HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

(2) Inspections for overweight landings that are explicitly called “not hard” by the flight crew log are not necessary because overweight landings at normal sink rates do not cause structural damage.

B. The Inspection

(1) The inspection is a visual inspection of the landing gear.

(a) If damage is found during this inspection then the Hard Landing inspection in procedure HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201 must be performed.

C. High Drag or High Side Load Condition

(1) A High Drag or High Side Load condition occurs if one or more of the following conditions is met:

(a) The flight crew reported a “high drag load” or “high side load” event.

(b) The airplane ran off the prepared surface.

(c) The airplane landed short of the prepared surface.

(d) The airplane made a landing and one or more tires were blown.

(e) One or more of the landing gear hit an obstacle.

D. Hard Main Gear Landing

(1) A Hard Main Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard landing” without specifying if it was hard on the main or nose gear, or

2) “hard main gear landing”, or

3) “hard bounced landing”.

E. Hard Nose Gear Landing

(1) A Hard Nose Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard landing” without specifying if it was hard on the main or nose gear, or

2) “hard nose gear landing”, or

3) “hard bounced landing”, or

4) a landing where the nose gear contacted the prepared surface before the main gears.

F. Overweight Landing

(1) For the purpose of this inspection, an Overweight Landing occurs when:

(a) The landing weight is greater than the Maximum Landing Weight (MLW) plus 1000 lb (454 kg).

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G. The Inspection Logic

NOTE: The Flight Crew Training Manual states the following for overweight landings: “if the overweight landing was not a hard landing the flight crew should record that the landing was not a hard landing”.

(1) If the flight crew recorded that the overweight landing was “hard”, do the HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

NOTE: The Overweight Landing inspection is not required when the landing was both overweight and hard because the inspection steps contained within this procedure are duplicated in the Hard Landing Inspection.

(2) If the flight crew recorded that the overweight landing was “not hard”, then do the applicable Landing Gear portion of the Overweight Landing inspection (Overweight Landing Inspection, TASK 05-51-35-210-803).

(3) Alternatively, the main landing gear inspection can be waived if all of the following are true:

(a) It was not a bounced landing, and

(b) The Center of Gravity (CG) vertical load factor from the flight recorded data (or equivalent recording device) was less than or equal to the threshold for the corresponding roll angle of Figure 201 (Sheet 1).

NOTE: The CG vertical load factor threshold may only be used if the recording sample rate for VERTICAL ACCELERATION is at least 8 samples/sec, if the landing was not bounced and if the roll angle is less than or equal to 6 degrees.

NOTE: The CG vertical load factor (VERTICAL ACCELERATION) and roll angle (ROLL ATTITUDE) are determined from the Flight Data Recorder (FDR) or equivalent recording device. The load factor is the maximum value of VERTICAL ACCELERATION that occurs in the first one-second period after touchdown and prior to speedbrake deployment. The roll angle is the maximum absolute value of ROLL ATTITUDE in the one second period prior to the maximum load factor.

(4) If damage is found during any of the necessary inspections from this section, this task must be performed before the next flight: HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

NOTE: There may not be any applicable inspections.

H. When the conditional inspection tells you to "examine" a component, look for these conditions (replace or repair components, if it is necessary).

(1) Cracks

(2) Pulled apart structure or unusual gaps

(3) Loose paint (paint flakes)

(4) Discoloration

(5) Twisted parts (distortion)

(6) Wrinkles or buckles in the structure

(7) Bent components

(8) Loose fasteners

(9) Fasteners that have pulled out or are gone

(10) Delamination

(11) Fiber breakage

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(12) Misalignment

(13) Interference

(14) Nicks or gouges

(15) Cracked fillet seals on sealed assemblies

(16) Other signs of damage.

TASK 05-51-35-210-803

2. Overweight Landing Inspection

(Figure 201)

A. References

Reference	Title
32-11-00-200-801	Main Landing Gear Inspection (P/B 601)
32-21-00-200-801	Nose Landing Gear - Inspection (P/B 601)

B. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

C. Overweight Landing Inspection

SUBTASK 05-51-35-210-018

(1) If the logic requires to inspect the main landing gear, examine the areas of the main landing gear that follow:

(a) Examine the main landing gear tires and wheels for signs of damage.

(b) Examine the upper and lower ends of the main landing gear shock strut for signs of hydraulic fluid leakage, (Main Landing Gear Inspection, TASK 32-11-00-200-801).

NOTE: A small quantity of hydraulic fluid on the inner cylinder is satisfactory.

SUBTASK 05-51-35-210-012

(2) If the logic requires to inspect the nose landing gear, examine the areas of the nose landing gear that follow:

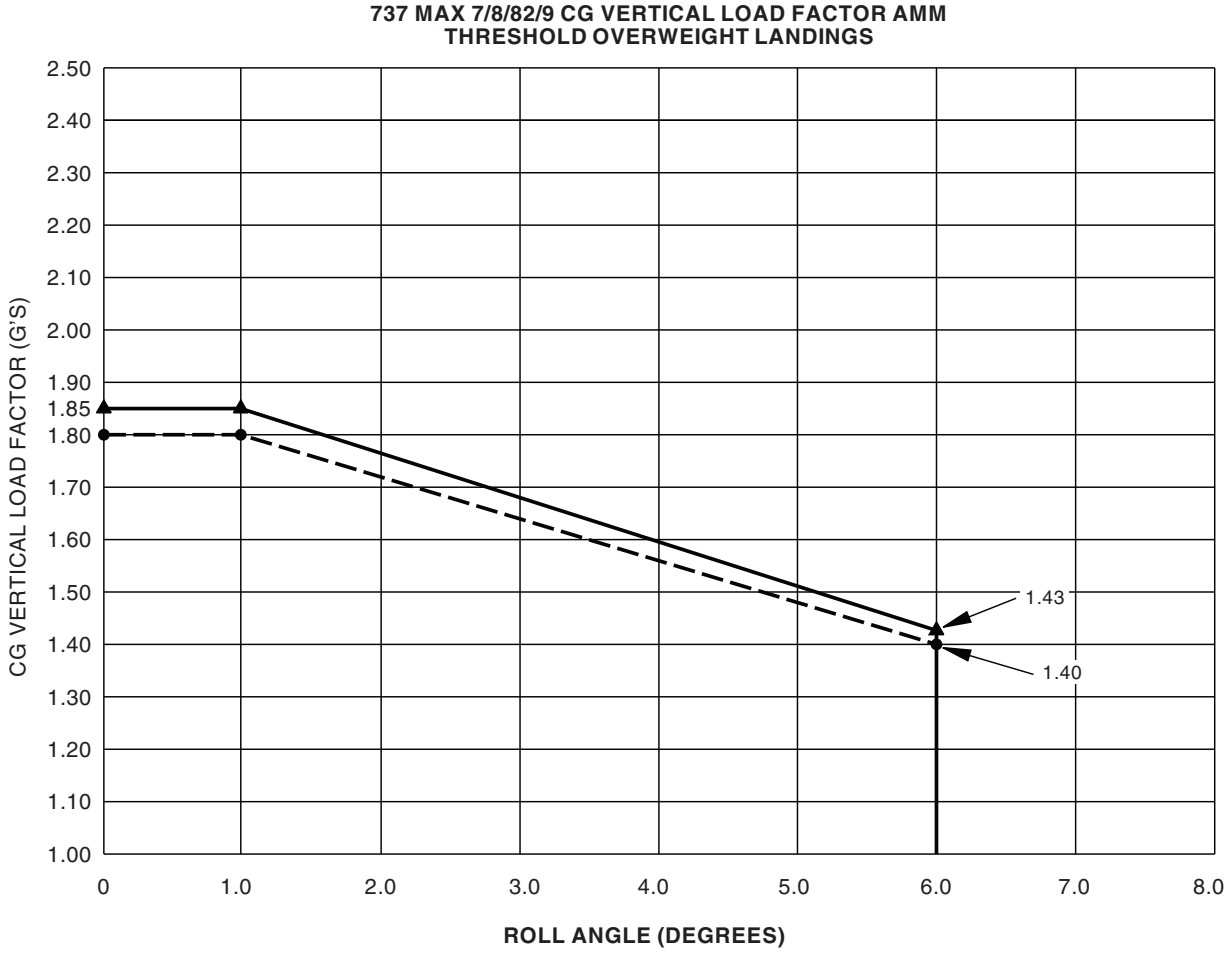
(a) Examine the tires and wheels of the nose landing gear for signs of damage.

(b) Examine the upper and lower ends of the shock strut of the nose landing gear for signs of hydraulic fluid leakage, (Nose Landing Gear - Inspection, TASK 32-21-00-200-801).

NOTE: A small quantity of hydraulic fluid on the inner cylinder is satisfactory.

——— END OF TASK ———

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CG Load Factor Chart
Figure 201/05-51-35-990-802

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WINGLET STRIKE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection procedure must be done if an airplane winglet strike occurred when the airplane was parked, taxiing, or towing.

B. The winglet is located on the tip of the wing and is attached to the wing closure rib.

C. If a winglet is damaged, there can also be damage to the wing.

(1) If you find damage to the wing, refer to the Structural Repair Manual (SRM) procedures.

D. This procedure has the following tasks:

(1) Winglet Strike Inspection Phase I

(2) Winglet Strike Inspection Phase II.

TASK 05-51-36-210-801

2. Winglet Strike Inspection Phase I

A. General

(1) This task has the steps to inspect the winglet if there is a winglet lower blade strike condition.

(2) If there is a winglet upper blade strike condition, do this task: Winglet Strike Inspection Phase II, TASK 05-51-36-210-802.

B. References

Reference	Title
57-31-21-000-802	Winglet Lower Blade Removal (P/B 401)
57-31-21-400-802	Winglet Lower Blade Installation (P/B 401)
57-31-21-480-801	Winglet Cover Plate Installation (P/B 201)

C. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
534	Left Wing - Dry Bay
600	Right Wing
627	Right Winglet
634	Right Wing - Dry Bay

D. Winglet Strike Inspection

SUBTASK 05-51-36-211-001

(1) Remove the applicable damaged winglet lower blade (TASK 57-31-21-000-802).

SUBTASK 05-51-36-211-005

(2) If the lower winglet blade was damaged, do these steps:

(a) Do a detailed visual inspection of the winglet lower blade to winglet upper blade attachment.

(b) Do a detailed visual inspection of the exposed attach fittings of the winglet blade ribs for evidence of cracks, buckling, elongated holes, or other damage.

(c) Do a detailed visual inspection of the winglet upper blade to wing surface.

1) Inspect the aero-sealed and unsealed portions for evidence of binding, interference, or other damage.

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(d) Do a detailed visual inspection of the internal and external surfaces at the wing lower skin between Rib 16 and Rib 27.

1) Examine the surfaces for wrinkles, loose fasteners, cracks, or other damage.

2) For the internal surface of the wing, do a detailed visual inspection the fasteners common to the ribs shear tie and tension clip.

NOTE: The detailed visual inspection of the internal surfaces at the wing lower skin between Rib 16 through 27 may be deferred up to 10 flight cycles if no damage is found on the external surface.

(e) If no damage was found on the winglet upper blade or on the lower blade attachment fittings or fasteners, do one of the steps that follow:

1) Install the serviceable winglet lower blade (TASK 57-31-21-400-802).

2) Install the winglet cover plate (TASK 57-31-21-480-801).

(f) If any damage is found, do this task: Winglet Strike Inspection Phase II, TASK 05-51-36-210-802.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-36-410-001

(1) If there is damage to structure (attached to winglet), use the applicable Structural Repair Manual and do repairs.

NOTE: This structure (attached to winglet) includes the closure rib, front spar, rear spar, upper, and lower inspar skins.

——— END OF TASK ———

TASK 05-51-36-210-802

3. Winglet Strike Inspection Phase II

A. General

(1) This task is used when one of these conditions occurred:

- Winglet upper blade strike condition, or
- Damage is found during the Wing Strike Inspection Phase I.

B. References

Reference	Title
57-31-21-000-801	Winglet Removal (P/B 401)
57-31-21-400-801	Winglet Installation (P/B 401)
737 NDT Part 6, 51-00-00, Procedure 16	Aluminum Part Fastener Hole Inspection (Rotary Scanner)
737 NDT Part 6, 51-00-00, Procedure 23	Aluminum Part Surface Inspection (Impedance Plane Display)

C. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
534	Left Wing - Dry Bay
600	Right Wing
627	Right Winglet

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(Continued)

Zone	Area
634	Right Wing - Dry Bay

D. Phase II

SUBTASK 05-51-36-020-001

(1) Do this task: Winglet Removal, TASK 57-31-21-000-801.

SUBTASK 05-51-36-211-006

(2) Do a detailed visual inspection of the closure rib for evidence of cracks, buckling, elongated holes or other damage.

NOTE: Use 10x magnification around the winglet attach fastener holes.

SUBTASK 05-51-36-211-007

(3) Do a detailed visual inspection to these areas, for evidence of cracking, buckling or other damage:

(a) Closure rib forward and aft attach fitting.

(b) Adjacent areas of the front and rear spar including chords and web from both the forward and aft sides.

SUBTASK 05-51-36-211-008

(4) Do a detailed visual inspection of the upper and lower inspar skins adjacent to the closure rib for evidence of cracking or buckling.

SUBTASK 05-51-36-250-007

(5) Do a high frequency eddy current inspection of each winglet attach bolt hole in the closure rib and bolt holes common to the forward and aft rib attach fittings (737 NDT Part 6, 51-00-00, Procedure 16).

(a) Make sure that no cracks exist.

SUBTASK 05-51-36-250-009

(6) Do a detailed visual inspection of each winglet attach bolts common to the closure rib, forward, and aft rib attach fittings to ensure no galling, wear, gouge, crack, and corrosion.

SUBTASK 05-51-36-250-008

(7) Do a high frequency eddy current inspection to the locations that follow (737 NDT Part 6, 51-00-00, Procedure 23):

- aft surface of the forward closure rib attach fitting around each of the fasteners that attach the fitting to the front spar
- forward surface of the aft closure rib attach fitting around each of the fasteners that attach the fitting to the rear spar
- upper and lower inspar skins around each of the fasteners that attach the closure rib to the skin
- upper and lower inspar skins around each of the fasteners that attach the front spar to the skin within 8 in. (203 mm) of the end rib
- upper and lower inspar skins around each of the fasteners that attach the rear spar to the skin within 8 in. (203 mm) of the end rib.

(a) Make sure that no cracks exist.

SUBTASK 05-51-36-210-002

(8) Do a general visual inspection of the upper and lower wing surfaces for any additional damage.

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SUBTASK 05-51-36-211-009

(9) Do a detailed visual inspection of the internal and external surfaces at the wing lower skin between Rib 16 and Rib 27.

(a) Examine the surfaces for wrinkles, loose fasteners, cracks, or other damage.

(b) For the internal surface of the wing, do a detailed visual inspection the fasteners common to the ribs shear tie and tension clip.

NOTE: The detailed visual inspection of the internal surfaces at the wing lower skin between Rib 16 through 27 may be deferred up to 10 flight cycles if no damage is found on the external surface.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-36-350-001

(1) If there is damage to the structure (attached to winglet), use the applicable Structural Repair Manual and do the repairs.

NOTE: This structure (attached to the winglet) includes the closure rib, front spar, rear spar, and upper and lower inspar skins.

SUBTASK 05-51-36-420-001

(2) Do this task: Winglet Installation, TASK 57-31-21-400-801.

——— END OF TASK ———

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NOSE LANDING GEAR AND FORWARD FUSELAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection is for hard landings, or overweight/hard landings, or high drag or high side load conditions. It defines the nose landing gear and forward fuselage inspections for those conditions. Inspections of other structure for these same conditions are covered in procedures HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

B. The Inspection

(1) The inspection contains one phase (Phase I).

(a) After Phase I inspection is complete, return to the inspection: HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201 and complete the inspection.

C. High Drag or High Side Load Condition

(1) A High Drag or High Side Load condition occurs if one or more of the following conditions is met:

(a) The flight crew reported a “high drag load” or “high side load” event.

(b) The airplane ran off the prepared surface.

(c) The airplane landed short of the prepared surface.

(d) The airplane made a landing and one or more tires were blown.

(e) One or more of the landing gear hit an obstacle.

D. Hard Main Gear Landing

(1) A Hard Main Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard landing” without specifying if it was hard on the main or nose gear, or

2) “hard main gear landing”, or

3) “hard bounced landing”.

E. Hard Nose Gear Landing

(1) A Hard Nose Gear Landing occurs if one of the following conditions is met:

(a) The landing is reported by the flight crew as:

1) “hard nose gear landing”, or

2) “hard landing” without specifying if it was hard on the main or nose gear, or

3) “hard bounced landing”, or

4) a landing where the nose gear contacted the prepared surface before the main gears.

F. Overweight Landing

(1) For the purpose of this inspection, an Overweight Landing occurs when:

(a) The landing weight is greater than the Maximum Landing Weight (MLW) plus 1000 lb (454 kg).

G. The Inspection Logic

(1) Do the following inspections when the flight crew reports a hard landing or high drag or high side load condition.

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(2) Did the flight crew state either of the following: It was not a Hard Nose Gear Landing, or it was hard on the main gear only.

(a) Yes: The inspections in this procedure are not required.

(b) No: Do the Phase I inspection before the next flight.

(3) If damage is found in the Phase I inspection, the Hard Landing Phase II (Phase II Inspection, TASK 05-51-01-210-806) inspections must be completed before the next flight.

(4) If directed to this procedure from the Hard Landing inspection, then at the completion of this procedure, return to the inspection where you came from and complete that inspection.

H. When the conditional inspection tells you to "examine" a component, look for these conditions (replace or repair components, if it is necessary).

(1) Cracks

(2) Pulled apart structure or unusual gaps

(3) Loose paint (paint flakes)

(4) Discoloration

(5) Twisted parts (distortion)

(6) Wrinkles or buckles in the structure

(7) Bent components

(8) Fastener holes that become larger or deformed

(9) Loose fasteners

(10) Fasteners that have pulled out or are gone

(11) Delamination

(12) Fiber breakage

(13) Misalignment

(14) Interference

(15) Nicks or gouges

(16) Cracked fillet seals on sealed assemblies

(17) Other signs of damage.

TASK 05-51-37-200-804

2. Phase I Inspection

A. References

Reference	Title
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-802	Wheels Inspection (Wheel Removed from the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-21 P/B 401	NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)

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B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Nose Landing Gear Area

SUBTASK 05-51-37-210-022

(1) Examine the nose landing gear tires for damage (Landing Gear Tire Inspection, TASK 32-45-00-700-803).

(2) Examine the wheels for damage; cracks, loose and/or missing tie-bolts (Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801).

(3) Examine the upper and lower ends of the nose landing gear shock strut for signs of hydraulic fluid leakage.

NOTE: A small quantity of hydraulic fluid on the inner cylinder is satisfactory.

(4) Examine the outer cylinder of the nose landing gear.

(5) Examine the nose landing gear areas that follow:

(a) Place a level on the outboard wheel rim.

1) Make sure both wheels are vertical.

(b) The nose wheel well.

1) Make sure all of the fasteners are installed in the correct positions.

2) Make sure there are no loose fasteners in the web of the nose wheel well near the trunnion support.

3) Examine the doors, hinges and retraction mechanism of the nose landing gear.

(c) If the airplane departed the prepared surface, do these steps:

1) After examination of the wheels and tires, do this step if the wheels and tires are not damaged:

a) Do the recovery and/or movement of the airplane to the prepared surface.

NOTE: The airplane can be towed to a location where further inspections and examinations will be performed.

2) After examination of the wheels and tires, do this step if the damage to the wheels and tires prevents the recovery and/or movement of the airplane to the prepared surface:

a) Replace the wheel and tire assemblies to allow recovery and movement of the airplane (NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-45-21/401).

3) After the airplane has been moved to a location for further examination, and before next flight, do these steps:

a) Do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

b) Examine the axle for damage.

c) Do this task: (Wheels Inspection (Wheel Removed from the Airplane), TASK 32-45-00-700-802).

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d) Do this task: Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

e) If the wheel and tire assemblies were replaced during the airplane recovery, clean the wheel and tire assemblies of unwanted material.

f) After the wheel and tire assemblies have been cleaned, examine the wheels and tires for damage (Landing Gear Tire Inspection, TASK 32-45-00-700-803, Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801).

g) If no damage is found, the wheels and tires are serviceable.

4) Do an operational check of the nose landing gear steering system.

D. Fuselage Inspections

SUBTASK 05-51-37-210-023

(1) Perform a walk around General Visual Examination of the forward fuselage from the ground and examine the structure for damage in the upper or lower fuselage.

(a) Examine the fuselage for buckles/wrinkles or other damage.

NOTE: If any damage is detected in this step or the following steps, inspections to determine the full extent of the damage must be done with sufficient lighting and from within touching distance.

(2) Examine the Reduced Vertical Separation Minimum (RVSM) areas around probes and static ports.

(a) If obvious visual damage is found, then perform an RVSM check using this task:SRM SUBJECT 51-10-03

E. Cabin Inspections

SUBTASK 05-51-37-210-024

(1) Do the following cabin inspections to find possible damage, if not already performed in a previous task:

(a) Do a visual inspection of all ceiling panels (and video monitors, if applicable) for dislodging and evidence due to impact damage.

(b) Do a visual inspection of all overhead bins for evidence of looseness or impact damage.

SUBTASK 05-51-37-211-003

(2) Do a detailed visual inspection of the latches, stops, potting inserts, and all fasteners on all ceiling panels equipped with video monitors.

(a) If any defect is found, repair or replace as necessary.

(b) If ceiling panels without video monitors are found displaced or damaged: Check for correct latching and installation. Repair if necessary.

SUBTASK 05-51-37-210-025

(3) If overhead stowage bins have impact damage or are loose: repair if necessary.

END OF TASK

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TASK 05-51-37-200-805

3. Phase II Inspection

A. References

Reference	Title
05-51-01-210-806	Phase II Inspection (P/B 201)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Nose Landing Gear Areas

SUBTASK 05-51-37-210-026

(1) Do the Nose Landing Gear Area inspection of the Hard Landing Phase II Inspection, TASK 05-51-01-210-806 to check for any additional damage.

D. Forward Fuselage Areas

SUBTASK 05-51-37-210-027

(1) Do the Fuselage and Wing Area inspections of the Hard Landing Phase II Inspection, TASK 05-51-01-210-806 to check for any additional damage.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-37-860-002

(1) Install the components you removed or install replacement components.

——— END OF TASK ———

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ENGINE BLADE OUT - EE BAY AND FLIGHT DECK PANELS- MAINTENANCE PRACTICES

1. General

A. This subject will address the maintenance practices as appropriate following a significant blade loss event and prior to returning the airplane to revenue service. Significant blade loss events will be considered those which result in excess of 0.75 blade equivalent unbalance and/or have reports of abnormally high vibration levels reported during flight.

B. The following are the minimum maintenance inspection checks:

(1) General Visual Check

(2) Electrical Equipment Rack (E1, E2, E3, E4, E5, and E8) Inspection.

TASK 05-51-42-210-801

2. Engine Blade Out - General Visual Check

A. General Visual Check

SUBTASK 05-51-42-211-001

(1) When you do the general visual check, look for the conditions that follow.

NOTE: Repair all damage found per the standard SRM procedure.

(a) Cracks
(b) Pulled apart structure
(c) Loose paint (paint flakes)
(d) Twisted parts (distortion)
(e) Bent components
(f) Fastener holes that become larger or longer
(g) Loose fasteners
(h) Fasteners that have pulled out or are gone
(i) Delaminations
(j) Misalignment
(k) Interference
(l) Excessive looseness or wear
(m) Excessive loading
(n) Peening of pins and bushings
(o) Bushing/bearing damage
(p) Bushing bearing fusion
(q) Bolt distortion
(r) Lug distortion or damage
(s) Other signs of damage

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-51-42-200-801

3. Engine Blade Out - EE Bay and Flight Deck Panels Inspection

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-10-07-400-801	E/E Box Installation (P/B 401)

B. Electrical Equipment Rack Inspection

SUBTASK 05-51-42-200-001

(1) Inspect for damage in the areas that follow:

NOTE: Repair all damage found per the standard SRM procedure.

(a) E1 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, do this task: E/E Box Removal, TASK 20-10-07-000-801.

(b) E2 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, do this task: E/E Box Removal, TASK 20-10-07-000-801.

(c) E3 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, do this task: E/E Box Removal, TASK 20-10-07-000-801.

(d) E4 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, do this task: E/E Box Removal, TASK 20-10-07-000-801.

(e) E5 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, do this task: E/E Box Removal, TASK 20-10-07-000-801.

(f) E8 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, if installed, do this task: E/E Box Removal, TASK 20-10-07-000-801.

C. Flight Deck Panel Inspection

SUBTASK 05-51-42-210-001

(1) Inspect for damage in the flight deck panels that follow:

NOTE: Repair all damage found per the standard SRM procedure.

(a) P1 Captain instrument Panel

(b) P2 Center Instrument Panel

(c) P3 First Officer Instrument Panel

(d) P5 Forward and Aft Overhead Panels

(e) P7 Glareshield Panel

(f) P8 Aft Electronic Panel

(g) P9 Forward Electronic Panel

(h) P18 Component Circuit Breaker Panel

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-42-410-001

(1) Re-install the E/E boxes removed for inspection access, do this task: E/E Box Installation, TASK 20-10-07-400-801.

(2) Re-install all flight deck panels removed for inspection access.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NACELLE/STRUT PRESSURE RELIEF DOORS OPEN CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains two tasks:

(1) Examine the Nacelle/Strut Structure

(2) Examine the Pneumatic Ducts

B. The pressure relief doors are made to open to prevent too much pressure in the nacelle/strut structure.

(1) There can be too much pressure in the nacelle/strut when a pneumatic duct breaks.

(2) Also, the pressure relief doors can open if a condition occurs which can cause temporary or permanent distortion.

(a) Example conditions are as follows, but others can also occur:

1) hard Landings

2) severe turbulence

3) engine Vibration

TASK 05-51-44-200-801

2. Examine the Nacelle/Strut Structure

A. Location Zones

Zone	Area
400	Powerplant and Nacelle Struts

B. Procedure

SUBTASK 05-51-44-210-001

(1) Examine the nacelle/strut inner and outer surfaces for:

NOTE: Carefully examine the surfaces for discoloration and/or paint that is flaked. This could be a sign of a broken pneumatic duct.

(a) distortion

(b) cracks

(c) pulled fasteners

(d) fasteners which are missing

(e) discolored paint

(f) paint that is flaked

——— END OF TASK ———

TASK 05-51-44-210-801

3. Examine the Pneumatic Ducts

A. Location Zones

Zone	Area
400	Powerplant and Nacelle Struts

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B. Procedure

SUBTASK 05-51-44-210-002

(1) Examine the components that follow:

(a) hard ducts

(b) flexible sense lines for cracks and tears

(c) clamps for tightness

(d) nuts for tightness

(e) fasteners for tightness

C. Put the Airplane Back to It's Initial Condition

SUBTASK 05-51-44-410-001

(1) Install the components you removed if they are serviceable.

(a) If the components are not serviceable, install replacement parts.

——— END OF TASK ———

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LANDING GEAR OPERATION ABOVE DESIGN SPEED CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to inspect the airplane when the landing gear has been operated in flight at speeds more than the permitted placard speed.

TASK 05-51-47-210-801

2. Landing Gear Operation Above Design Speed Condition, Conditional Inspection

A. Location Zones

Zone	Area
700	Landing Gear and Landing Gear Doors

B. Examine the Main Gear Door Areas

SUBTASK 05-51-47-210-001

(1) Examine main landing gear areas that follow:

(a) The doors.

(b) The hinges.

(c) The linkage and linkage support structure.

(d) The fairing panels.

1) Look for:

a) distortion

b) cracks

c) misalignment

d) displacement

e) fastener holes that are elongated or torn

f) fasteners that have pulled out

g) fastener locations where fasteners are not there

h) other signs of damage.

SUBTASK 05-51-47-210-002

(2) Examine the systems that are installed in the wheel wells of the main landing gear for damage or possible malfunction.

C. Examine the Nose Gear Door Area

SUBTASK 05-51-47-210-003

(1) Examine the nose landing gear areas that follow:

(a) The doors.

(b) The hinges.

(c) The linkage and linkage support structure.

(d) The fairing panels.

1) Look for:

a) distortion

b) cracks

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c) misalignment

d) displacement

e) fastener holes that are elongated or torn

f) fasteners that have pulled out

g) fastener locations where fasteners are not there

h) other signs of damage.

SUBTASK 05-51-47-210-004

(2) Examine the systems that are installed in the nose wheel well for damage or possible malfunction.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-47-410-001

(1) Install the components that were removed, or install replacement parts.

——— END OF TASK ———

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FUEL IMBALANCE EXCEEDANCE CONDITIONAL INSPECTION - MAINTENANCE PRACTICE

1. General

A. This procedure contains the task for maintenance actions or structural inspections related to exceeding the Airplane Flight Manual (AFM) fuel imbalance limits.

TASK 05-51-48-211-801

2. Fuel Imbalance Exceedance - Conditional Inspection

A. General

(1) The Airplane Flight Manual (AFM) for the 737 models identifies that the lateral fuel imbalance between main wing tanks must be scheduled to be zero. Random fuel imbalance between main wing tanks for towing, taxi, takeoff, flight, or landing is limited to 1000 lb (454 kg).

Operators experiencing fuel imbalances greater than 1000 lb (454 kg) should review the below criteria for evaluation of what maintenance actions or structural inspections are required.

B. References

Reference	Title
05-51-01 P/B 201	HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
05-51-01-210-804	Phase IA Inspection (P/B 201)
05-51-04-210-801	Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection (P/B 201)

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
531	Left Wing - Center Fuel Tank, Rib 1 to Rib 5
532	Left Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing BL 643.50
631	Right Wing - Center Fuel Tank, Rib 1 to Rib 5
632	Right Wing - Main Tank, Rib 5 to Rib 22, Wing Station 204.25 to Wing Station 643.50

D. Evaluation

NOTE: Boeing developed the following questions that operators can use to do an evaluation for wing tank lateral fuel imbalances exceeding 1000 lb (454 kg).

SUBTASK 05-51-48-210-001

(1) If the imbalance exceedance occurred during flight (climb/cruise/descent), then the following questions apply:

(a) Was the amount of imbalance greater than or equal to 3000 lb (1361 kg)?

1) If yes, do this task: Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection, TASK 05-51-04-210-801.

(b) Was the amount of imbalance greater than 1000 lb (454 kg) but less than 3000 lb (1361 kg)?

1) If yes, the following questions apply:

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a) Did the airplane encounter any moderate turbulence (gusts) during the imbalance?

<1> If yes, do this task: Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection, TASK 05-51-04-210-801.

b) Did the airplane experience significant load factors or perform rapid or large alternating control inputs during the imbalance?

<1> If yes, do this task: Severe or Unusual Turbulence, Excessive Maneuver, Buffet, or Speeds More than the Design Limits Conditional Inspection, TASK 05-51-04-210-801.

SUBTASK 05-51-48-210-002

(2) If the imbalance exceedance occurred during landing, then the following questions apply:

(a) Was the amount of imbalance greater than or equal to 3000 lb (1361 kg)?

1) If yes, do the visual inspection portion of Phase IA Inspection, TASK 05-51-01-210-804.

(b) Was the amount of imbalance greater than 1000 lb (454 kg) but less than 3000 lb (1361 kg)?

1) If yes, the following questions apply:

a) Was there a large roll angle during landing?

<1> If yes, do the visual inspection portion of Phase IA Inspection, TASK 05-51-01-210-804.

SUBTASK 05-51-48-210-003

(3) If the imbalance exceedance occurred during takeoff, taxi, or towing, then the following questions apply:

(a) Was the amount of imbalance greater than or equal to 3000 lb (1361 kg)?

1) If yes, do the high drag/side load inspection (HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201).

(b) Was the amount of imbalance greater than 1000 lb (454 kg) but less than 3000 lb (1361 kg)?

1) If yes, the following questions apply:

a) Was noticeable runway roughness encountered during takeoff, taxi, or towing?

<1> If yes, do the high drag/side load inspection (HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201).

b) Did the airplane experience a hard ground turn or a significant side load event during taxi or towing?

<1> If yes, do the high drag/side load inspection (HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201).

c) Was hard braking applied during taxi or towing?

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<1> If yes, do the high drag/side load inspection (HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201).

NOTE: If a lateral imbalance exceedance has persisted for an unknown number of flights for a particular airplane, Boeing requires that all the inspections noted above be performed. This is due to the increased probability that the aircraft would have experienced loads above those typically encountered in normal operation.

NOTE: If none of the inspections above are applicable to the particular event, the airplane may return to revenue service. If inspections are determined necessary based on the operator's answers to the questions above, all inspections must be performed before the airplane may return to revenue service.

——— END OF TASK ———

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Extreme High Wind Event - Conditional Inspection

TASK 05-51-51-200-801

1. Extreme High Wind Event - Conditional Inspection

A. General

(1) This procedure has information for an aircraft that has had an extreme high wind event while parked or moored.

(a) Extreme high wind events are tornadoes and microbursts, or when wind speeds exceed 75 mph (65 knots).

(b) Evidence of an extreme high wind event are if the airplane moved over the chocks or if there are tire skid marks on the ground where the airplane was parked. This may indicate that the airplane has moved from the position in which it was originally parked.

1) If the airplane has moved over wheel chocks, it can cause damage to the tires, wheels, and landing gear.

(2) Extreme high wind events are variable and unpredictable.

(3) Extreme high winds from the side and aft direction while parked can cause damage to the aircraft flight control surfaces and flight control mechanisms internal to the wings, horizontal stabilizer, and vertical stabilizer. It is possible that this damage will not be seen during a walk-around or during pre-flight checks. The integrity of the flight controls is very important to the safe flight of the airplane.

(4) When the conditional inspection tells you to “examine” a component, look for these damage conditions:

- Cracks
- Twisted parts (distortion)
- Misalignment
- Wrinkles
- Buckles
- Pulled apart structure
- Loose paint (paint flakes)
- Loose fasteners
- Fasteners that have pulled out or are missing
- Fastener holes that become larger or longer
- Delaminations
- Interference
- Other signs of damage

(5) When the conditional inspection tells you to do a “detailed visual inspection” of a component, look for these damage conditions:

- Cracks
- Deformation
- Loose components
- Sheared components
- Bent components

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- Missing attachments
- Distortion

(6) Boeing recommends that operators give Boeing the results of this procedure (if you find damage or if you do not find damage).

NOTE: Boeing recommends that when the operator gives Boeing their results they title the message “Extreme High Wind Event Inspection Results.”

(a) Boeing recommends that the operator give the data that follows when contacting Boeing.

1) The results of the inspections listed below for the flight controls, doors, and landing gear.
2) The results of cycling the flight controls.
3) If the airplane has moved from the location where it was parked.
   a) If the airplane moved over the chocks.
      <1> If the main landing gear tires or nose landing gear tires have been replaced.
   b) If there are tire skid marks on the ground.
4) Wind characteristics:
   a) Wind direction and airplane parking direction (angle between the two directions)
   b) Average wind speed
   c) Maximum wind gust speed
   d) Type of wind event
5) Gross Weight
6) Center of Gravity (CG)
7) Airplane Restraints
   a) chocks
   b) mooring configuration
   c) brakes
8) Surface Conditions
   a) icy
   b) wet
   c) dry
   d) clean
   e) dirty/sandy
   f) ground friction test
9) Ambient Air Temperature (C°) at time of event

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)

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(Continued)

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
32-13-11 P/B 401	MAIN LANDING GEAR SHOCK STRUT DOORS - REMOVAL/INSTALLATION
32-45-11 P/B 401	MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION
32-45-21 P/B 401	NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION

C. Extreme High Wind Event Conditional Inspection

SUBTASK 05-51-51-200-001

(1) Flight Controls

(a) Examine the external surfaces of the flight control surfaces that follow for damage.

1) Flaps
2) Slats
3) Ailerons
4) Spoilers
5) Elevators
6) Rudder
7) If there is damage, do a detailed visual inspection of the surface(s) that have damage.
   a) Do a detailed visual inspection of the surface(s) that have damage.
   b) Inspect the surface hinge line for surface over-travel.
8) If there is structural damage to the control surface, refer to the applicable Structural Repair Manual (SRM) procedure.

(b) Examine the actuator attach points and support structure.

(c) Examine the hinge attach structure of the flaps, the flaps at the rear spar, slats, ailerons, spoilers, elevators, and rudder for damaged and loose parts.

1) Make sure that the flaps are in the fully extended position (TASK 27-51-00-860-803, TASK 27-81-00-860-803).
2) If there is damage or any loose parts, do the steps that follow.
   a) Do a detailed visual inspection of the applicable actuator, control surface, attach fittings, and support structure.
   b) If there is structural damage to the component, refer to the applicable Structural Repair Manual (SRM) procedures.

(d) Move the flaps, slats, spoilers, ailerons, elevators, and rudder through their usual range of movement.

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1) Make sure that the surfaces move freely and that there is no interference with the structure or components around them.
   a) If the surface does not move freely or if there is interference with the structure or components around it, refer to the applicable Structural Repair Manual (SRM) procedures.
2) Make sure that there is a ground observer during the flight control surface movement.
   a) Make sure that there is communication set up between the ground observer and the personnel in the flight deck.
   b) Make sure that the ground observer can see all flight control surfaces.
   c) Operate each flight control system through its full range of movement.
   d) Make sure that the flight control positions are correct as monitored by the ground observer.

SUBTASK 05-51-51-200-002

(2) Doors

WARNING DO NOT OPERATE A DOOR THAT WAS OPEN DURING A VERY HIGH WIND EVENT UNTIL IT IS INSPECTED FOR DAMAGE. LOSS OF LIFT POWER, ENERGIZING OF EMERGENCY EQUIPMENT, OR FAILURE OF HOLD-OPEN SYSTEMS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Examine each external door of the airplane that was open at the time of the high wind event.

NOTE: For example: doors for stairs, fuselage doors, maintenance access panels, engine cowling, and landing gear doors (excluding main gear shock strut doors).

NOTE: If a door was closed and latched during the extreme high wind event, no inspection of that door is necessary.

1) Examine the door hinge and door open/close mechanism for evidence of damage.

NOTE: Examples of damage evidence are: looseness, cracking, chipped or cracked coatings, and distortion.

a) If evidence of damage is found, do the steps that follow.

<1> Do a detailed visual inspection of the structure around the door, the door structure, the mechanical parts of the door, and the mechanical parts attached to the surrounding structure.

<a> If there is damage to the door or surrounding structure, refer to the applicable Structural Repair Manual (SRM) procedures.

NOTE: Structural damage may be hidden behind attached components. If visible damage is identified, additional inspections may be necessary.

<b> If there is damage to the mechanical parts, repair or replace the damaged components, as necessary.

2) Examine the exterior skin surface of the door for evidence of damage.

NOTE: Examples of damage evidence are: cracks, buckles, chipped or cracked coatings, and loose or missing attach parts.

a) If evidence of damage is found, do the steps that follow.

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<1> Do a detailed visual inspection of the exterior skin surface of the door.

<a> If there is damage to the exterior skin surface, refer to the applicable Structural Repair Manual (SRM) procedures.

3) Operate the door through the normal range of motion (from open to close).

a) Make sure that there is no interference or binding due to deformation of the door or fuselage structure, or fuselage supports.

NOTE: Damage may be hidden behind attached components. If visible or operational damage is identified, additional inspections may be necessary.

<1> If there is interference or binding, do the steps that follow.

<a> Do a detailed visual inspection of the fuselage supports.

<b> If there is damage to the fuselage supports, repair or replace the components, as necessary.

4) Operate the door latching and locking mechanism, if installed.

a) Make sure that there is no interference or binding due to deformation of the mechanism.

NOTE: Damage may be hidden behind attached components. If visible or operational damage is identified, additional inspections may be necessary.

<1> If there is interference or binding, do the steps that follow.

<a> Do a detailed visual inspection of the fuselage mechanisms.

<b> If there is damage to the mechanism, repair or replace the components, as necessary.

SUBTASK 05-51-51-200-003

(3) Landing Gear

(a) Jack the main landing gear, do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801, or Lift the Airplane on Axle Jacks-Handling, TASK 07-10-03-580-801.

(b) Examine the main landing gear areas as follows:

1) The tires

a) If the airplane has moved over the main landing gear wheel chocks, replace the tire(s) that moved over the chock(s), do this task: MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-45-11/401.

2) The wheels

3) The shock strut of the main gear for fluid leakage

4) The top end of the shock strut for cracks and bolt distortion

5) The landing gear beam

6) If moored, inspect for damage at and near the mooring equipment interface locations.

7) If the airplane departed the prepared surface, do the steps that follow:

a) Remove the wheels and examine the wheel/brake interface.

<1> Remove any dirt, debris, or foreign objects.

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b) Remove and examine the brake units.

<1> Remove any dirt, debris, or foreign objects.

c) Remove the brake sleeves and examine the axles.

<1> Remove any dirt, debris, or foreign objects.

8) The main landing gear axles

9) The main landing gear tow fittings

10) The skins, attach points, and linkages of the main gear shock strut doors

a) If evidence of damage is found, do the steps that follow:

<1> Do a detailed visual inspection of the structure around the door, the door structure, the mechanical parts of the door, and the mechanical parts attached to the surrounding structure.

<2> If there is damage to the surrounding structure, refer to the applicable Structural Repair Manual (SRM) procedures.

NOTE: Structural damage may be hidden behind attached components. If visible damage is identified, additional inspections may be necessary.

<3> If there is damage to the door structure, mechanical parts, repair or replace the damaged components, as necessary. Do the applicable task: MAIN LANDING GEAR SHOCK STRUT DOORS - REMOVAL/INSTALLATION, PAGEBLOCK 32-13-11/401.

(c) Jack the nose landing gear, do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801, or Lift the Nose Landing Gear Axle With Axle Jacks-Handling, TASK 07-10-03-580-803.

(d) Examine the nose landing gear areas as follows:

1) The tires

a) If the airplane has moved over the nose landing gear wheel chocks, replace the nose landing gear tires, do this task: NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-45-21/401.

2) The wheels

3) The nose wheel well.

a) Make sure that all of the fasteners are installed in the correct positions.

b) Make sure that there are no loose fasteners in the web of the nose wheel well near the trunnion support.

c) Examine the upper and lower ends of the shock strut of the nose gear for fluid leakage.

d) Examine the outer cylinder of the nose landing gear.

e) Examine the doors, hinges, and retraction mechanism of the nose landing gear.

4) If moored, inspect for damage at and near the mooring equipment interface locations.

5) If the airplane departed the prepared surface, do the following:

a) Remove and examine the wheels.

<1> Remove any dirt, debris, or foreign objects.

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b) Do an operational check of the nose landing gear steering system.

6) The nose landing gear axle

7) The nose landing gear tow fitting

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-51-580-001

(1) Lower the main and nose landing gear, do these tasks: Lower the Airplane off Jacks, TASK 07-10-01-580-802 and Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 05-51-51-400-001

(2) Install the components that were opened for checks.

SUBTASK 05-51-51-080-001

(3) Remove all tools and equipment from the work area.

——— END OF TASK ———

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Heavy Water Spray from Tires During Takeoff and Landing - Conditional Inspection

TASK 05-51-52-200-801

1. Heavy Water Spray from Tires During Takeoff and Landing - Inspection/Check

A. General

(1) This procedure has one task:

(a) This task is to inspect the aircraft exterior when there is possible water damage.

B. Procedure

SUBTASK 05-51-52-211-001

(1) Make sure that the exterior components are attached correctly and are not damaged.

(a) Examine the areas that follow.

1) Wing leading edge structure between the fuselage and the engine.
2) Lower surfaces of the fuselage.
3) Antenna and water drain masts.
4) Landing gear doors.
5) Inboard trailing edge flap panels.
6) Landing gear hydraulic plumbing.
   a) Make sure that the plumbing is attached correctly.
   b) Look closely at the lower surface of the track.
7) Electrical conduits.
8) Switches, switch targets, and mounting brackets.
9) Temperature sensors and electrical connectors.

(b) If damage to any of the exterior components is found, see the SRM for repairs.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INTERIOR ICE CONDITION - MAINTENANCE PRACTICES

1. General

A. In the usual operation of the airplane, air comes through the airplane structure. During flight, the water vapor in the air can condense and freeze in the airplane. When you continuously operate the airplane in cold weather, ground temperatures below the freezing point will not let the ice melt. Too much ice will lower the performance of the airplane below the permitted values.

TASK 05-51-53-210-801

2. Interior Ice Removal

(Figure 201)

A. General

(1) This task is for the removal of interior ice for these conditions:

NOTE: This procedure will remove the ice build-up on the inside of the airplane structure.

(a) Airplanes that have continuous operation in cold weather

(b) Ice is found on interior equipment, furnishings, electrical components, wiring, control cables, pulleys, door mechanisms, or structure.

(2) In the usual operation of the airplane, air comes through the airplane structure. During flight, the water vapor in the air can condense and freeze in the airplane. When you continuously operate the airplane in cold weather, ground temperatures below the freezing point will not let the ice melt. Too much ice will lower the performance of the airplane below the permitted values.

B. References

Reference	Title
53-31-01-200-801	External Drainage Inspection/Check (P/B 601)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

D. Upper Lobe Ice Inspection

SUBTASK 05-51-53-010-001

(1) Get access to the main deck overhead above WL 270 in the front, middle and aft stations.

(a) To get access on passenger airplanes, you must also remove some of the main deck ceiling panels.

SUBTASK 05-51-53-010-002

(2) Remove or lift the insulation blankets to do a check for ice.

(a) Look for ice collection in the main deck ceiling areas.

1) If you do not find ice, do the Lower Lobe Ice Inspection.

2) If you do find ice, do the Overall Airplane Ice Removal.

E. Lower Lobe Ice Inspection

NOTE: Do this procedure to find if ice has collected in the lower lobes (below WL 210).

SUBTASK 05-51-53-010-003

(1) Remove the floor panels over the aisle in the main deck at approximately STA 660.

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SUBTASK 05-51-53-210-001

(2) Look for ice collection between the longitudinal floor beams over the wing box and wheel well.

SUBTASK 05-51-53-010-004

(3) Remove the sidewall liners in the aft lower cargo compartment at approximately STA 747.

SUBTASK 05-51-53-210-002

(4) Look for ice collection in the bilge area.

(a) If you do find ice, do the Overall Airplane Ice Removal.

(b) If you do not find ice, no further action is necessary.

F. Overall Airplane Ice Removal

NOTE: Do this procedure to remove all ice that has collected throughout the airplane structure.

SUBTASK 05-51-53-580-001

(1) Move the airplane to a hangar where the air temperature is above 32°F (0°C).

SUBTASK 05-51-53-010-005

(2) Open the doors on the main deck to let the air flow freely.

SUBTASK 05-51-53-010-006

(3) Remove all the main deck seats and floor panels to get access to the ice.

SUBTASK 05-51-53-010-007

(4) Remove the insulation blankets from the sidewall and below the floor in the forward and aft lower lobe cargo compartment where it is possible.

NOTE: This will let the water flow freely to the structural drains and will make sure the blankets do not get wet.

SUBTASK 05-51-53-010-008

(5) Remove the insulation blankets in the nose wheel well area (forward of the EE bay).

SUBTASK 05-51-53-660-001

WARNING

DO NOT PUT HOT AIR DIRECTLY ON HALON OR OXYGEN BOTTLES. IF YOU PUT HOT AIR ON THE BOTTLES, YOU CAN CAUSE AN EXPLOSION.

CAUTION

THE TEMPERATURE OF THE HOT AIR MUST NOT BE ABOVE 150 DEGREE F (70 DEGREE C). HOT AIR CAN CAUSE DAMAGE TO AIRPLANE SYSTEMS.

(6) Use a conditioned ground cart to blow hot air on the open airplane structure in the lower lobes and main deck (if it is applicable) to melt the ice that has collected.

SUBTASK 05-51-53-610-001

(7) Make sure the airplanes structural drains in the bilge are open to let the water drain.

(a) Do this task: External Drainage Inspection/Check, TASK 53-31-01-200-801.

SUBTASK 05-51-53-660-002

(8) Use Figure 201 to find the time needed to melt the ice.

NOTE: You can use a conditioned ground cart to heat the inside of the airplane. This will shorten the time needed to melt the ice.

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SUBTASK 05-51-53-020-001

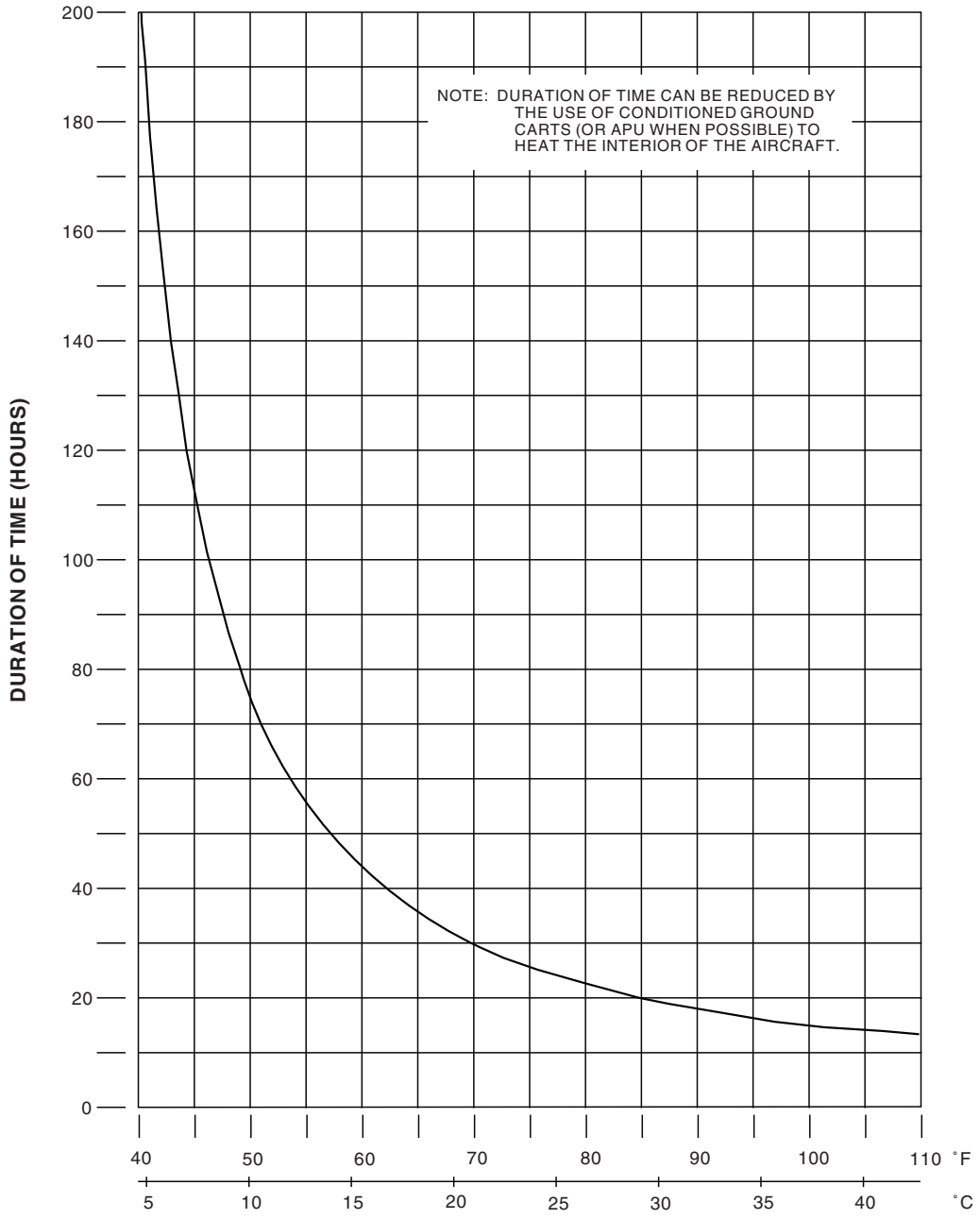
(9) You must remove or dry all wet insulation blankets before you install them again.

SUBTASK 05-51-53-840-001

(10) After the water drains, put the airplane back to its usual condition.

——— END OF TASK ———

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Air Temperature Outside Airplane Time Required to Melt Interior Ice
Figure 201/05-51-53-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRE TREAD LOSS OR TIRE BURST - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to inspect the airplane after a tire has lost it's tread, or has burst.

B. When a landing gear tire loses it's tread or a tire bursts, a careful inspection of some areas is necessary.

(1) These inspections look for damage and/or blockage caused by pieces of the tire.

C. Damage caused by the tire pieces can be seen as pieces of rubber and/or bitumen marks on the surface of the structure.

TASK 05-51-54-210-801

2. Tire Tread Loss or Tire Burst Conditional Inspection

A. References

Reference	Title
05-51-16-210-801	Flat Spotted Tires Conditional Inspection (P/B 201)
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-700-804	Control Wheel Centering Check (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-807	Control Wheel Travel Stop Test (P/B 501)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-720-801	Aileron Power Control Unit (PCU) Pogos - Functional Test (P/B 501)
27-11-00-730-801	Aileron Trim Response Test (P/B 501)
27-11-00-820-805	Aileron Feel and Centering Unit Adjustment (P/B 501)
27-51-00-210-801	Trailing Edge Flap Drive System Inspection (P/B 601)
27-51-00-720-803	Alternate Flap Control System Functional Test (P/B 501)
27-51-00-730-801	Trailing Edge Flap System Test (P/B 501)
27-59-00-730-801	Flap Skew Detection System Functional Test (P/B 501)
27-61-00-710-801	Spoiler Control System Operational Test (P/B 501)
27-62-00-710-801	Speed Brake Control System Operational Test (P/B 501)
27-62-00-800-803	Speed Brake Lever Friction Test (P/B 501)
27-81-00-860-801	Leading Edge Flap and Slat System Operation With Primary Control (P/B 201)
27-81-00-860-802	Leading Edge Flap and Slat System Operation With Alternate Control (P/B 201)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
71-00-00-200-802-G00	Engine Foreign Object Damage and Birdstrike Inspection (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
300	Empennage
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

SUBTASK 05-51-54-210-001

(1) Do the steps that follow to inspect for damage after a tire tread loss or tire burst:

(a) Examine the areas that follow for damage, delamination, or local separation, and soft spots because of core crush:

NOTE: To find delamination or local separation areas, tap the surface with a coin. To find areas of core crush, push on the surface.

1) The wing upper inboard fixed trailing edge panel
2) The wing lower inboard fixed trailing edge panel
3) The inboard trailing edge flaps
4) The inboard trailing edge fairings
5) The Very High Frequency (VHF) No. 2 antenna
6) The lower fuselage wing-to-body fuselage panels.

(b) Examine the inboard trailing edge flaps for tire fragments lodged in linkages or in flap carriages (main and foreflap sequence carriages).

(c) Examine the components that follow for damage and blockage because of pieces of the tire.

1) The trailing edge flap jackscrews
2) The actuating linkages
3) The actuating rods.

(d) Examine the landing gear wheel well areas as follows for damage, distortion, and blockage from pieces of the tire:

1) The wheel well doors
2) The wheel well door actuating rods
3) The hydraulic plumbing
4) The flap drive torque tubes
5) The control cables and mechanisms
6) The frangible hydraulic fittings located in the wheel well fairing.

(e) Examine the other tires for possible damage. Do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

NOTE: Carefully inspect the other tires for slash and/or wire penetration damage of the tread or the sidewall.

1) If the damaged tire is deflated, or is more than 20% below the specified pressure, replace the wheel and tire assembly that is on the same axle.

(f) If the tire burst was because of a tire skid, do this task: Flat Spotted Tires Conditional Inspection, TASK 05-51-16-210-801.

SUBTASK 05-51-54-210-002

(2) Examine the engine nacelles and inlets for damage when the nose landing gear or left/right main landing gear tire/tires tread is lost or the tire/tires have burst, do this task: Engine Foreign Object Damage and Birdstrike Inspection, TASK 71-00-00-200-802-G00.

SUBTASK 05-51-54-210-003

(3) If it is determined that the tire burst occurred when the landing gear was retracted, you must look for damage in the wheel well areas.

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(a) If the damage is visible on the forward bulkhead, do these tasks:

- Aileron Feel and Centering Unit Adjustment, TASK 27-11-00-820-805
- Manual Reversion Control Friction Test, TASK 27-11-00-700-803
- Control Wheel Centering Check, TASK 27-11-00-700-804
- Aileron Control Wheel Force Test, TASK 27-11-00-700-805
- Control Wheel Travel Stop Test, TASK 27-11-00-700-807
- Aileron Travel Test, TASK 27-11-00-700-810
- Aileron Power Control Unit (PCU) Pogos - Functional Test, TASK 27-11-00-720-801
- Aileron Trim Response Test, TASK 27-11-00-730-801
- Spoiler Control System Operational Test, TASK 27-61-00-710-801
- Speed Brake Control System Operational Test, TASK 27-62-00-710-801
- Speed Brake Lever Friction Test, TASK 27-62-00-800-803.

(b) If the damage is visible on the aft bulkhead, do these tasks:

- Trailing Edge Flap System Test, TASK 27-51-00-730-801
- Alternate Flap Control System Functional Test, TASK 27-51-00-720-803
- Trailing Edge Flap Drive System Inspection, TASK 27-51-00-210-801
- Flap Skew Detection System Functional Test, TASK 27-59-00-730-801
- Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801
- Leading Edge Flap and Slat System Operation With Alternate Control, TASK 27-81-00-860-802

(c) If there is no visible damage on either bulkhead, do the tasks assigned above for both damage on the forward and aft bulkheads.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-54-410-001

(1) Install all the components you removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SMOKE OR FUMES IN CABIN - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the information to perform an inspection to determine the cause of smoke or fumes entering the cabin during operation.

TASK 05-51-56-200-801

2. Smoke or Fumes in Cabin (Conditional Inspection)

A. Examine the Cabin

SUBTASK 05-51-56-280-001

(1) If smoke or fumes are detected in the cabin during airplane operation, refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACID SPILLAGE CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure has one task:

(1) A conditional inspection for acid spillage.

B. This task has an inspection and clean up procedures for areas where acid has touched the airplane.

C. The primary source of acid spillage is in the battery compartments where acid electrolytes may overflow during charging or spill during battery servicing.

D. Acid based corrosion removal compounds and airplane cleaners are used quite extensively during routine maintenance repair. Spills do occur at times and thorough neutralizing and/or rinsing is necessary to preclude corrosion damage.

E. Containers of acid concentrates or acid based chemicals may be part of a cargo and may be broken during loading or unloading. Spillage from such sources are usually larger in scale than spills previously mentioned. It is, therefore, advisable that the acid spillage be neutralized as soon as possible.

F. Operators should also be cognizant of the fact that acids may deteriorate nonmetallic materials such as fabrics, wood, leather, etc.

TASK 05-51-57-000-802

2. Corrosion Removal After Acid Spills

A. References

Reference	Title
51-00-58	STANDARD TREATMENT METHODS
51-00-59	STANDARD PREVENTIVE MAINTENANCE PROCEDURES

B. Tools/Equipment

Reference	Description
STD-122	Brush
STD-419	Gloves - Rubber, Elbow Length
STD-421	Goggles - Eye Protection
STD-774	Respirator - Full-Face, Air Purifying (with filters/cartridges)
STD-815	Sponge
STD-1392	Shield - Facial Protection
STD-1393	Aprons
STD-1394	Boots
STD-1395	Head Gear
STD-3904	Goggles - Safety
STD-6391	Mop - long handle (4 - 5 foot), or long wooden stick with shop cloth attached to one end
STD-8874	Brush - Soft

C. Consumable Materials

Reference	Description	Specification
B00095	Compound - Sodium Bicarbonate	ASTM D928
B00334	Cleaner - Sodium Bicarbonate	ASTM D928, or commercially available product

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(Continued)

Reference	Description	Specification
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23
G50071	Compound - Corrosion Inhibiting, Heavy Duty	BMS3-35
G50412	Paper - Litmus (Used to find the acidity or alkalinity of a liquid solution)	
G51294	Cloth - 100% Synthetic or Blended Synthetic, Cotton or Cellulose Material	AMS3819 Class 2 Grade A

D. Corrosion Removal After Acid Spills

SUBTASK 05-51-57-910-001

WARNING DO NOT LET THE ACIDS TOUCH YOU. YOU MUST WEAR CLOTHING THAT SUPPLIES PROTECTION. YOU MUST WEAR GLOVES AND A FACE SHIELD. THE ACIDS ARE POISONOUS AND CAN CAUSE INJURIES TO PERSONS. IF THE ACID GETS INTO YOUR EYES, FLUSH YOUR EYES WITH COOL WATER. GET MEDICAL AID IMMEDIATELY.

(1) Take these precautions when you work with an acid spill or vacuum waste system spills/leaks condition:

(a) Adequate protective clothing (rubber or plastic gloves, goggles, face shields, aprons, boots, head gear, etc.) should be worn when handling or working in acid contaminated areas.

1) elbow length rubber gloves, STD-419
2) safety goggles, STD-3904 or eye protection goggles, STD-421
3) face shield, STD-1392 or full-face respirator, STD-774
4) aprons, STD-1393
5) boots, STD-1394
6) head gear, STD-1395.

(b) Wash hands after using acid neutralizing treatment solution and/or materials before eating or smoking.

(c) Waste materials, solvents, chemical solutions, wiping rags, masking materials, etc., shall be collected and disposed of safely.

SUBTASK 05-51-57-940-001

(2) Do these steps to isolate the contaminated area:

(a) Do not allow acid spills to spread from areas of contamination.

(b) Using plastic sheets is advised for protection of equipment beneath battery areas. If equipment is operating, venting requirements should be maintained.

(c) Consider protecting uncontaminated areas by taping down protective material such as plastic sheets.

SUBTASK 05-51-57-110-001

(3) Do these steps to clean the area of an acid spillage:

(a) If equipment is adjacent to the treatment area use plastic sheets to cover the equipment to prevent inadvertent splashing of acids or treatment fluids.

(b) Wipe up excess fluids with cloth and discard cloth into plastic container for disposition.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Mix sodium bicarbonate cleaner, B00334 or compound, B00095 with water to create a solution that will neutralize the acid.

(d) Apply the solution with a cloth, G51294, mop, STD-6391, brush, STD-122, or sponge, STD-815.

1) Make sure that the solution goes into the contaminated faying surface joints.

NOTE: A pressure application of the solution can be necessary to flush the faying surface joints and some access areas fully.

(e) Apply the solution until bubbles cease in the acid/solution.

NOTE: When the bubbles ease, the acid has become neutral.

(f) Let the solution stay on the surface for 5 minutes after the bubbles stop.

(g) Remove the solution with mop, STD-6391 or sponge, STD-815.

(h) Flush the area with large quantities of clean water. Pressure application may be required to flush the joint thoroughly.

1) Rub the surface with a soft brush, STD-8874.

(i) Do a test of the cleaned area with litmus paper, G50412.

NOTE: The litmus paper should read between 7 and 8 when the area is fully cleaned.

(j) Wipe dry with clean cloth, G51294.

SUBTASK 05-51-57-410-001

(4) Do these steps to put the airplane back to its usual condition:

(a) After thoroughly dry, repair or replace damaged finishes if it is necessary. Refer to SUBJECT 51-00-58 and SUBJECT 51-00-59 for protective finish systems.

(b) Apply water displacing, corrosion inhibiting compound, G50071 or corrosion inhibiting compound, G00009 over entire area.

NOTE: For details of how to apply the corrosion inhibiting compound, refer to SUBJECT 51-00-59.

NOTE: If insulation blankets have become contaminated with vacuum waste, they must be replaced to prevent reintroduction of the acidic substance on the structure.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND HANDLING EQUIPMENT HITS AIRPLANE OR PART DEPARTS AIRPLANE (CONDITIONAL INSPECTION) - MAINTENANCE PRACTICES

1. General

A. This procedure has steps to do an inspection after ground handling equipment hits the airplane or parts depart the airplane.

B. An impact incident can be more than allowable damage limits.

C. Visually examine the external surfaces of the airplane in the general area of the impact which includes the nearest support structure.

(1) If you find damage at the support structure, internal NDI inspections and disassembly is possible.

D. When the conditional inspection tells you to examine a component, look for these conditions:

(1) Cracks
(2) Gaps
(3) Damaged paint
(4) Discoloration
(5) Scuffing
(6) Twisted or distorted parts
(7) Dents, gouges, scratches
(8) Elongated or distorted fastener holes
(9) Loose or missing fasteners
(10) Delamination
(11) Fiber damage
(12) Soft spots
(13) Core damage
(14) Misalignment
(15) Interference
(16) Other signs of damage.

E. Description of Low Velocity, High Energy Impact Events

(1) You can think of an impact incident as a Low Velocity, High Energy Blunt Impact when the type, force, or cause is more than the structure was first designed for without the result of damage you can see. Examples of events which can be considered Low Velocity, High Energy Blunt Impacts are (but are not limited to):

(a) An airport jetway that hits the fuselage at more than normal operational speeds or angles.
(b) Ground Support Equipment that hits the structure.
(c) Ground Support Equipment that does not have the correct protection that hits the airplane structure.
(d) Impact by a blunt, high-mass object at low speed.
(e) Impact by parts that depart the airplane in flight.

(2) If any damage is found that is within the applicable allowable damage limits in the Structural Repair Manual (SRM) for that structure, this inspection is not required.

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TASK 05-51-59-200-801

2. Ground Handling Equipment Hits the Airplane or Part Departs the Airplane Inspection

A. Examine the Airplane Structure

SUBTASK 05-51-59-210-001

(1) Visually examine the external structure adjacent to the 12 in. (305 mm) of the impact location.

(a) If you find signs of damage, refer to the applicable SRM procedure for repair.

(2) Visually examine the internal support structure behind or adjacent to the 12 in. (305 mm) external area around the impact location.

(a) If you find signs of damage, refer to the applicable SRM procedure for repair.

SUBTASK 05-51-59-960-001

(3) If a part has departed the airplane, do the steps that follow.

(a) Replace the departed part.

(b) Visually examine the airplane structure aft of the location of the departed part.

NOTE: Look for multiple impact locations along the structure.

1) If you find signs of damage, refer to the applicable SRM procedure for repair.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIRFRAME VIBRATION CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure has this task:

(1) A conditional inspection of the airframe vibration.

TASK 05-51-67-280-801

2. Airframe Vibration - Conditional Inspection

A. General

(1) This task includes the steps to check the cause of in-flight airframe vibration and eliminate or reduce this vibration.

(2) You can possibly feel a vibration similar to Mach buffet in the forward cabin when these conditions occur:

(a) At high speed, the trailing edge flaps can cause the vibration.

(b) At low speeds on climb and in cruise, these parts can cause the vibration:

1) ground spoiler float

2) too much play in the outboard aileron and tab

3) loose doors and fairings.

B. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
21-51-23-400-801	Ram Air Inlet Modulation Panels Installation (P/B 401)
27-09-91-200-804	Horizontal Stabilizer - Inspection (P/B 601)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-32 P/B 601	ELEVATOR AND ELEVATOR TAB FREEPLAY - INSPECTION/CHECK
27-31-34 P/B 601	ELEVATOR TAB CONTROL MECHANISM - INSPECTION/CHECK
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-12-200-801	Inboard Aft Flap Inspection (P/B 601)
27-51-13-200-803	Inboard Flap Inboard Carriage Inspection (P/B 601)
27-51-21-820-801	Outboard Trailing Edge Flap Adjustment (P/B 501)
27-51-28-820-801	Outboard Flap Outboard Support Fairing Adjustment (P/B 501)
27-81-11 P/B 401	LEADING EDGE FLAPS - REMOVAL/INSTALLATION
32-13-11 P/B 501	MAIN LANDING GEAR SHOCK STRUT DOOR - ADJUSTMENT/TEST
32-13-21 P/B 501	MAIN LANDING GEAR WING DOOR - ADJUSTMENT/TEST
32-13-21-000-802	Main Landing Gear Wing Door Adjustment (P/B 501)
32-22-11-000-802	Nose Landing Gear Wheel Well Doors Adjustment (P/B 501)
49-11-00-200-801	APU Power Plant Inspection (P/B 601)
49-15-00-710-801	APU Air Inlet Door Test (P/B 501)
49-15-11-200-801	Air Inlet Seal Inspection (P/B 601)
53-51-21 P/B 401	WING-TO-BODY FAIRING PANELS - REMOVAL/INSTALLATION

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(Continued)

Reference	Title
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
72-22-01-200-801-G00	Front Spinner Cone Inspection (P/B 601)
73-21-05-970-803-G00	Electronic Engine Control (EEC) Data Download (On-Wing)(Selection) (P/B 201)

C. Airframe Vibration Conditional Inspection

SUBTASK 05-51-67-210-001

(1) For vibration symptoms and maintenance actions, see Table 201.

SUBTASK 05-51-67-280-001

(2) If you feel vibration mostly in the forward and aft fuselage, with the trailing edge flaps in the up position, do these steps:

(a) Examine the elevator.

1) Make sure that there is not too much play in these parts for the elevator trim tab:

a) the tab pushrod

b) the tab lockout mechanism

c) the tab hinges.

2) Make sure that the hinge fitting bolts on the elevator are not loose.

3) Do a check for the hinge bolts on the elevator balance panel .

4) Make sure that these parts on the elevator output torque tubes are not loose or worn:

a) the bolts that attach the inner or outer torque tubes

b) the crank

c) the control

d) the pushrod

e) the elevator mast fitting.

5) Make sure that the elevator hinge seals and nose seals are installed correctly without damage.

(b) Examine the doors, panels, and fairings.

1) Make sure that all the doors, panels, and fairings are correctly aligned and attached.

2) Examine all wing-to-body fairing panels with particular attention to the 149A7654 and 149A7555 fairing panels for loose attach fasteners.

a) Do a visual check of the panels from inside the wheel well for signs of failure or wear grooves.

b) Apply axial pressure to the fastener heads from outside in an upward direction to check for loose fasteners.

<1> If you find loose fasteners, retighten and check the self locking feature of nutplates (TASK 20-50-11-910-801).

(c) Examine the landing gear doors.

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1) Make sure that the nose and main gear doors are aligned and all the seals are installed without damage.

2) Make sure that the nose gear doors are adjusted correctly.

(d) Examine the horizontal stabilizer.

1) Make sure that the hinge bearing for the horizontal stabilizer does not have too much freeplay (TASK 27-09-91-200-804).

(e) Check the elevator control cable tension (TASK 27-31-00-820-801).

(f) Examine the Auxiliary Power Unit (APU).

1) Inspect the APU, do this task: APU Power Plant Inspection, TASK 49-11-00-200-801.

2) Make sure that the compartments for the APU inlet door are not loose or worn (TASK 49-15-11-200-801).

3) Make sure that the APU inlet door is in the closed position (TASK 49-15-00-710-801).

SUBTASK 05-51-67-280-002

(3) If you feel vibration mostly in the wing root area, with the trailing edge flaps in the up position, do these steps:

(a) Examine the trailing edge flaps.

1) Make sure that the trailing edge panels on the top of the wings are adjusted correctly (TASK 27-51-21-820-801).

2) Make sure that the flap track fairing on the outboard trailing edge is not loose and does not have damage (TASK 27-51-28-820-801).

3) Make sure that the aft flap on the trailing edge is not delaminated and is in good condition (SRM PROCEDURE 51-40-06-0G-0).

(b) Examine the ram air system.

1) Make sure that the seals on the ram air inlet modulation panel are attached (TASK 21-51-23-400-801).

(c) Examine the main landing gear door.

1) Make sure that the wing door for the main landing gear is adjusted correctly (TASK 32-13-21-000-802).

SUBTASK 05-51-67-280-003

(4) If you feel vibration in the wing root area, with the trailing flaps extended between 2 and 5 units, do these steps:

(a) Examine the flaps.

1) Do a check for vertical play in the flap assembly on the trailing edge (TASK 27-51-11-820-801).

2) Make sure that the sequence carriage bearing for the inboard flap is not worn (TASK 27-51-13-200-803).

3) Do a check for play in the aft segment of the inboard flap on the trailing edge.

a) Make sure that the cam roller bearing of the actuating mechanism on the aft segment is not worn and is in good condition.

4) Make sure that the panels on the trailing edge on the top of the wing are adjusted correctly (TASK 27-51-21-820-801).

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5) Do a check for the interface blade seals.

6) Make sure that the access panel to the slat actuator for the leading edge is not loose.

7) Make sure that the trailing edge aft flaps are in good condition.

8) Make sure that the inboard mid flap seal plate is in good condition (TASK 27-51-12-200-801).

NOTE: A seal plate that is not adjusted correctly can cause a humming noise that can be heard at the seats just aft of the wing.

SUBTASK 05-51-67-280-004

(5) If you feel vibration with noise, do these steps:

(a) Make sure that there is sealant at the wing-to-body fairing.

(b) Do a check of the ground service check swing valve.

(c) Make sure that the air distribution ducts are not loose and do not vibrate.

(d) Make sure that the nose landing gear door is adjusted correctly (TASK 32-22-11-000-802).

SUBTASK 05-51-67-280-005

(6) If you feel vibration and possibly noise mostly over the wing, do these steps:

(a) Make sure that the seal door for the krueger flap on the wing leading edge is adjusted correctly.

(b) Do a check for low frequency engine vibrations.

(c) Make sure that the fan rotor is balanced (TASK 72-22-01-200-801-G00).

D. Vibration Symptom and Maintenance Action

SUBTASK 05-51-67-970-001

(1) The table below shows the vibration symptoms and maintenance actions in-flight airframe vibration.

Table 201/05-51-67-993-801 Vibration Symptom and Maintenance Action

CHECK NUMBER	CHECKS AND SYMPTOMS	INSPECTIONS/TEST/MAINTENANCE
1	N2 Tone (HPC Mode) • Clear, audible harmonic tone/ vibration during descent above 15,000 ft. at idle power, or on ground at idle power. The tone can be detected by flight crew or passengers (highest nose in mid cabin). • Tactile vibration and/or AVM gauge movement from a normal position of approximately 0.20 scalar units (no tone) to approximately 1.0 units (with tone).	Do the following: • Other than an optional slight throttle increase by the flight crew to minimize the noise tone level, no crew or maintenance action is required. • AVM download can confirm HPC and/or HPT peak vibration levels, near 68% N2. Otherwise refer to Engine Operation Limits, TASK 71-00-00-800-802-G00.

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Table 201/05-51-67-993-801 Vibration Symptom and Maintenance Action (Continued)

CHECK NUMBER	CHECKS AND SYMPTOMS	INSPECTIONS/TEST/MAINTENANCE
2	Buzzsaw Noise (Fan Blade) • Audible response during takeoff and low altitude climb described as a buzzsaw noise. No increase in vibration levels. Most perceptible in the forward cabin.	Do the following: • No action required. This is a characteristic of the engine.
3	N1 Related Vibration (Fan Unbalance) • Typically characterized as an audible "rumble" or "growl". • Increased noise level at cruise, top of climb and/on hold power. • Flight deck and cabin noise are roughly similar. • Tactile vibration may exist.	Do the following: • Download the electronic engine control data to determine if trim balance is required (TASK 73-21-05-970-803-G00). • Note: Some operators have adopted the practice of performing an engine trim balance per Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00, when engine vibration is between 1.5 and 2.0 units. These operators have found this to be an effective means for managing engine vibration.
4	Start Rumble • Very loud growl that occurs at low power such as during starts and near minimum idle. • No observable change in engine vibration level. Ground phenomena only.	Do the following: • No action required. This is a characteristic of the engine.

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Table 201/05-51-67-993-801 Vibration Symptom and Maintenance Action (Continued)

CHECK NUMBER	CHECKS AND SYMPTOMS	INSPECTIONS/TEST/MAINTENANCE
5	• High frequency vibration usually manifested as a sound but in some cases might be felt. • The vibration can be constant during all phases of flight and may vary with increased airspeed. • Vibration might be heard/felt in a specified location or may be transmitted to a broad area of the airframe.	Do the following: • Adjust nose landing gear doors (TASK 32-22-11-000-802). Check all fasteners for security of attachment. • Adjust main landing gear shock strut and wing doors (PAGEBLOCK 32-13-11/501 and PAGEBLOCK 32-13-21/501). Check all fasteners for security of attachment. • Check for missing sealant at the wing-to-body fairing (PAGEBLOCK 53-51-21/401). • Check lower fuselage doors, access panels and hatches located in the vicinity of the crew compartment. • Examine all wing-to-body fairing panels for loose attach fasteners (PAGEBLOCK 53-51-21/401). • Check seals common to leading edge flap (PAGEBLOCK 27-81-11/401).
6	• Vibration in rudder pedals • Vibration may be felt or heard in aft galley. • Vibration may disappear with rudder or rudder trim input.	Do the following: • Confirm vibration is unrelated to an engine or engines. • Check nose landing gear (TASK 32-22-11-000-802), main landing gear (PAGEBLOCK 32-13-11/501 and PAGEBLOCK 32-13-21/501), door rigging, and seal installation.
7	Elevator/Elevator Tab Related Vibration • Vibration felt in the flight deck with higher levels felt in the aft cabin that can occur during any phase of flight. • Vibration levels diminish with reduction in airspeed. • Vibration may be triggered by deployment of speedbrakes.	Do the following: • Do the elevator control tab mechanism inspection (PAGEBLOCK 27-31-34/601). • Do the elevator inspections (PAGEBLOCK 27-31-32/601).

END OF TASK

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MAIN LANDING GEAR SHIMMY/VIBRATION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This inspection applies to airplanes during landing and takeoff that have experienced a main landing gear (MLG) shimmy or vibration.

NOTE: Shimmy is defined as a rapid and violent torsional oscillation/vibration of the MLG caused by asymmetric wheel spin on one main landing gear.

B. The Inspection

(1) This inspection is divided into two phases (Phase I and Phase II).

(2) The Phase I inspection is applicable when a MLG shimmy/vibration occurs during landing and takeoff.

(a) If the inspection during Phase I does not show that damage has occurred, no more inspections are necessary.

(b) If the Phase I inspection shows that damage has occurred, the Phase II inspection must be done.

(3) The pilot must make the decision if this structural examination is necessary.

(a) If a structural examination is necessary, do the procedure "Phase I Inspection" in this section.

C. When the conditional inspection tells you to "examine" a component, look for these conditions (replace or repair components, if it is necessary).

(1) Cracks

(2) Pulled apart structure

(3) Loose paint (paint flakes)

(4) Twisted parts (distortion)

(5) Bent components

(6) Fasteners holes that become larger or longer

(7) Loose fasteners

(8) Fasteners that have pulled out or are gone

(9) Delaminations

(10) Misalignment

(11) Interference

(12) Other signs of damage.

TASK 05-51-68-210-801

2. Phase I Inspection

A. References

Reference	Title
05-41-05-210-802	EXTERNAL - ZONAL (GVI): Winglet - Left Wing (P/B 201)
05-41-06-210-802	EXTERNAL - ZONAL (GVI): Winglet - Right Wing. (P/B 201)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-210-801	Trailing Edge Flap Drive System Inspection (P/B 601)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)

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Reference	Title
27-51-00-710-801	Trailing Edge Flap System Operational Test (P/B 501)
27-61-00-710-801	Spoiler Control System Operational Test (P/B 501)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
32-11-00-200-801	Main Landing Gear Inspection (P/B 601)
32-11-51-000-801	Main Landing Gear Upper Torsion Link Removal (P/B 401)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-11-51-200-801	Main Landing Gear Torsion Links Inspection (P/B 601)
32-11-51-400-801	Main Landing Gear Upper Torsion Link Installation (P/B 401)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)
49-13-00-200-802	APU Mounts Inspection (APU Installed) (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

C. Main Landing Gear Areas

SUBTASK 05-51-68-210-001

(1) Examine the Main Landing Gear (MLG) areas as follows:

(a) The tires

(b) The wheels

(c) The shock strut of the main gear for fluid leakage

(d) The doors and linkage of the main gear strut

(e) The wiring

(f) The brackets, sensors, and hydraulic tubing (TASK 32-11-00-200-801)

(g) The actuators

(h) The top end of the shock strut for cracks and bolt distortion

(i) The landing gear beam

(j) The trunnion

(k) The stabilizer beams

(l) The inboard attachment of main landing gear beam

(m) Outboard attachment of main landing gear beam

(n) Examine the MLG torsion links and attachments, do this task: Main Landing Gear Torsion Links Inspection, TASK 32-11-51-200-801.

NOTE: Look for damage and deformation.

1) Remove the upper torsion link pin, do this task: Main Landing Gear Upper Torsion Link Removal, TASK 32-11-51-000-801.

a) Examine the pin for damage, cracks, missing chrome, or a bent shaft.

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b) Examine the bushings in the outer cylinder lugs and the upper torsion link lugs for damage, cracks, and deformation.

c) If damage, cracks, or deformation is found on the bushings or the pin, further inspection of the MLG must be performed per the Phase II inspection.

d) If no damage is found, install the upper torsion link pin, do this task: Main Landing Gear Upper Torsion Link Installation, TASK 32-11-51-400-801.

2) Disconnect the apex joint of the torsion links, do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

a) Examine damper shaft and attachment hardware for damage, wear, and deformation.

b) Verify that straight and tapered sleeves are installed in the correct positions.

c) If damage, cracks, or deformation is found on the damper shaft, further inspection of the MLG must be performed per the Phase II inspection.

d) If no damage is found, connect the apex joint of the torsion links, do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

D. Fuselage Areas

SUBTASK 05-51-68-210-002

(1) Examine the fuselage areas that follow:

(a) Examine the upper and lower fuselage skin panels forward and aft of the wing.

NOTE: Permanent wrinkles frequently occur on the lower side of the fuselage skins aft of STA 727.

(b) If you found wrinkles on the skin panels and you did not know they were there before, make an internal inspection.

NOTE: If the wrinkles were there before, make a careful inspection of the wrinkles for cracks.

(c) Examine the keel beam chords, stiffeners, webs, and splices at station 663 thru 727A.

SUBTASK 05-51-68-200-001

(2) Examine the external fuselage skin at BS 1016 and the aft side of the pressure bulkhead for signs of buckling, deformation or damage.

(a) Do a detailed visual inspection of the exterior fuselage skin 10 in. (25.4 cm) forward and aft of the BS 1016 skin butt joint between stringers S-10L and S-10R.

NOTE: Gain access to the exterior of the fuselage skin at BS 1016 under the vertical stabilizer fairing, 323AL and 323AR.

(b) Do a detailed visual inspection of the entire aft side of the BS 1016 aft pressure bulkhead. Give particular attention to the upper bulkhead structure between skin stringers S-10L and S-10R.

NOTE: Access to the interior unpressurized aft pressure bulkhead area is through the aft access hatch, 311BL.

E. Electrical Equipment Rack Inspection

SUBTASK 05-51-68-210-010

(1) Do the visual inspection of the electrical equipment racks for damage:

(a) Visually inspect the electrical equipment racks attachments to structure, shelves, and trays.

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(b) If damage is found do the detailed electrical equipment rack inspection (TASK 05-51-68-210-802).

F. APU Structure Inspection

SUBTASK 05-51-68-200-003

(1) Inspect the Auxiliary Power Unit (APU) Mounts (APU Installed), do this task: APU Mounts Inspection (APU Installed), TASK 49-13-00-200-802.

G. Wing Areas

SUBTASK 05-51-68-210-003

(1) Examine the wing areas that follow:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Deactivate the leading edge flaps and slats.

1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Deactivate the trailing edge flaps.

1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(c) The strut-to-wing fairing panels on the upper and lower nacelle.

1) Make sure that the fasteners are installed in the correct position.

(d) Examine the wing leading edge fairing for displacement and other signs of damage.

(e) Do the following inspections with attention to find possible damage to:

1) Carriage number 2, both inboard and outboard side of the carriage.
2) Track number 1 and track number 2 of the inboard and outboard lower flanges of the track.
3) Carriage number 1 and number 2 forward and aft roller bolts.
4) Bellcrank number 1 and number 2.
5) Aft flap program tracks, Outboard flap.
6) Trailing edge flaps.
   a) Examine the trailing edge flaps, do this task: Trailing Edge Flap Drive System Inspection, TASK 27-51-00-210-801.
      <1> If damage is found, refer to the Phase II for more detailed examinations of the flaps (TASK 05-51-68-210-802).
7) Perform the following operational checks:
   a) Do this task: Trailing Edge Flap System Operational Test, TASK 27-51-00-710-801.
   b) Do this task: Spoiler Control System Operational Test, TASK 27-61-00-710-801.
   c) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

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(f) Examine winglets for damage, cracks, and deformation (TASK 05-41-05-210-802, TASK 05-41-06-210-802)

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-68-410-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-68-440-001

(2) Do the activation procedure for the trailing edge flap system.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-68-440-002

(3) Do the activation procedure for the leading edge flap and slat system.

(a) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

I. Cabin Inspections

SUBTASK 05-51-68-210-004

(1) Do the following cabin inspections that follow to find possible damage:

(a) Do a visual inspection of all ceiling panels for dislodging.

(b) Do a visual inspection of all ceiling overhead bins for evidence of looseness.

(c) Do a detailed visual inspection of all ceiling panels equipped with video monitors.

1) Inspect latches, stops, potting inserts, and all fasteners.

(d) Continue the Cabin Inspection in the Phase II inspection if any cabin damage was found in the above Cabin Inspections including the ceiling panels, central overhead bins, or the video monitors. If no damage was found, no further cabin inspection is required and it is not necessary to continue the cabin inspection in Phase II.

END OF TASK

TASK 05-51-68-210-802

3. Phase II Inspection

A. References

Reference	Title
05-51-01 P/B 201	HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
05-51-10 P/B 201	DRAGGED ENGINE NACELLE/FAN BLADE OUT/ENGINE SEIZURE/THRUST REVERSER AND ATTACH POINTS DAMAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
20-10-07-000-801	E/E Box Removal (P/B 401)
25-23-61-000-801	Passenger Service Unit - Removal (P/B 401)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-12-200-801	Inboard Aft Flap Inspection (P/B 601)
27-51-22-200-801	Outboard Aft Flap Inspection (P/B 601)
27-51-23-200-802	Outboard Flap Outboard Carriage Inspection (P/B 601)

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Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
32-11-00-200-801	Main Landing Gear Inspection (P/B 601)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-34-00-730-801	Main Gear Manual Extension System Test (Airplane on Jacks) (P/B 501)
57-16-06-000-801	Main Landing Gear Support Beam Removal (P/B 401)
57-16-06-200-801	Landing Gear Support Beam Inspection (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

C. Landing gear

SUBTASK 05-51-68-710-001

(1) Do the landing gear checks that follow to make sure that the landing gear operates correctly.

NOTE: Continue the following inspection if any landing gear damage was found in the above Phase I inspection.

(a) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

(b) Retract and then extend the Main Landing Gear (MLG) with the normal system to make sure that they operate correctly.

1) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

(c) Retract and then extend the Nose Landing Gear (NLG) with the normal system to make sure that it operates correctly.

1) Do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

(d) Retract and then extend the MLG with the manual system to make sure that they operate correctly.

1) Do this task: Main Gear Manual Extension System Test (Airplane on Jacks), TASK 32-34-00-730-801.

(e) Retract and then extend the NLG with the manual system to make sure that it operates correctly.

1) Do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

D. Main Landing Gear Areas

SUBTASK 05-51-68-210-005

(1) Do the steps that follow to examine the MLG:

NOTE: Continue the following inspection if any landing gear area damage was found in the above Phase I inspection.

(a) Do these steps if you remove a wheel because of a blown tire:

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1) Examine the brake assemblies for damage.
2) Examine the wheel bearings for roughness.
3) Examine the axle for damage.

(b) Examine the forward and aft trunnion area of the outer cylinder.

(c) Examine the sliding surface of the inner cylinder.

(d) Remove the upper and lower torsion links and examine the following:

1) Visually inspect the bushings and lugs.
2) Examine and inspect the dimensions of the attachment lugs and pins.

NOTE: If deformation is found, remove the bushings and check the dimensions of the lug bores and faces.

(e) Examine the axle.

(f) Examine and check the dimensions of the outer and inner cylinder torsion link lugs.

NOTE: If deformation is found, remove the bushings and check the dimensions of the lug bores and faces.

(g) Examine the side strut linkage.

(h) Examine the actuator walking beam and linkage fittings.

(i) Examine the fuse pins at the inboard attachment and outboard attachment of the main landing gear beam.

(j) Examine the shock strut of the main landing gear for damage or signs of excessive leakage, do this task: Main Landing Gear Inspection, TASK 32-11-00-200-801.

1) If excessive leakage is found, do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

If the shock strut servicing was not correct, do the following steps:

a) Remove the inner cylinder and examine it for distortion or cracks.
b) Check the dimensions of the barrel of the inner cylinder and axles for distortion, bending and cracking.
c) Examine the interior components of the shock strut for damage.

(k) Examine the wheel well of the main landing gear for fuel or other fluid leaks.

(l) If the results of any of the above checks and inspections requires removal of the MLG, inspect the fuse pins at the inboard attachment and outboard attachment of the MLG beam.

1) Remove the inboard fuse pin of the landing gear support beam: do this task: Main Landing Gear Support Beam Removal, TASK 57-16-06-000-801.

NOTE: Complete removal of the landing gear support beam is not required.

NOTE: External support of the landing gear support beam may be necessary to remove load on the beam.

2) Inspect the fuse pin, bolts, and bushings: do this task: Landing Gear Support Beam Inspection, TASK 57-16-06-200-801.

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3) Remove the outboard fuse pin of the landing gear support beam: do this task: Main Landing Gear Support Beam Removal, TASK 57-16-06-000-801.

NOTE: Complete removal of the landing gear support beam is not required.

NOTE: External support of the landing gear support beam may be necessary to remove load on the beam.

4) Inspect the fuse pin, bolts, and bushings: do this task: Landing Gear Support Beam Inspection, TASK 57-16-06-200-801.

E. Fuselage Areas

SUBTASK 05-51-68-040-001

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-68-040-002

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-51-68-210-006

(3) Examine the fuselage areas that follow:

NOTE: Continue the following inspection if any fuselage damage was found in the above Phase I inspection.

(a) Examine the lower fuselage structure.

NOTE: Make sure that you examine the area below the body crease and from station 727 to 100 in. (254 cm) aft.

(b) Examine the wing-to-fuselage joints at stations 540 and 664.

(c) Examine the upper fuselage structure at station 540, 664, and 727.

(d) Examine the bulkheads at body stations 294.5 and 360 and the fuselage structure immediately outboard of the nose wheel well.

NOTE: You can examine the forward side if you use the access holes in the sidewalls of the nose wheel well. You can examine the aft side of the bulkhead at station 294.5 from the electronics compartment.

(e) Examine the keel beam chords, stiffeners, webs and splices at station 540 thru 727A.

(f) Examine the external surface of the fuselage for fuel or other fluid leaks.

(g) Remove the upper wing-to-body fairing forward of the aft fairing.

(h) Examine the trunnion support fitting at body station 706 for the main landing gear for cracks and bolt distortion.

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(i) If you found wrinkles on the skin panels and you did not know they were there before, do an internal inspection.

NOTE: Permanent wrinkles frequently occur on the lowerside of the fuselage skin aft of STA 727. If the wrinkles were there before, do a careful inspection of the wrinkles for cracks.

(j) Examine at BS 1016 the upper center web installation on the forward side of the pressure bulkhead for any damage.

NOTE: Gain access to inspect the upper center web installation by removing the galley and lavatories as necessary.

F. Electrical Equipment Rack Inspection

SUBTASK 05-51-68-200-004

(1) Inspect for damage in the following areas (TASK 20-10-07-000-801):

NOTE: Repair all damage found per the standard Structural Repair Manual (SRM) procedure.

(a) E1 electrical equipment rack stanchions and its attachments to structure, shelves, and trays.

(b) E2 electrical equipment rack stanchions and its attachments to structure, shelves, and trays.

(c) E3 electrical equipment rack stanchions and its attachments to structure, shelves, and trays.

(d) E4 electrical equipment rack stanchions and its attachments to structure, shelves, and trays.

(e) E5 electrical equipment rack stanchions and its attachments to structure, shelves, and trays.

(f) E8 electrical equipment rack stanchions and its attachments to structure, shelves, and trays, if installed.

G. Wing Areas

SUBTASK 05-51-68-040-003



WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Make sure that the trailing edge flap system has been deactivated. Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-51-68-040-004



WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Make sure that the leading edge flap and slat system has been deactivated. Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-51-68-210-007

(3) Do an inspection of the wing areas that follow:

NOTE: Continue the following inspection if any wing or flap damage was found in the above Phase I inspection.

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(a) Examine the wings, nacelles, and upper and lower nacelle strut-to-wing fairing for fuel or other fluid leaks.

(b) Examine the wing ribs along the aft side of the rear spar, WBL 92.5 and 114.0 for cracks.

(c) Examine the upper and lower trailing edge panels and the airplane structure for damage.

NOTE: Make sure that you examine the area near the main landing gear beam carefully.

(d) Examine the flap areas that follow for sheared rivets and structure damage:

1) The inboard and outboard trailing edge flaps, do these tasks:

a) Inboard Aft Flap Inspection, TASK 27-51-12-200-801

b) Outboard Aft Flap Inspection, TASK 27-51-22-200-801

c) Outboard Flap Outboard Carriage Inspection, TASK 27-51-23-200-802.

2) The flap tracks

3) The drive screws

4) The linkages and fairings.

(e) Examine the inboard and outboard spoilers for sheared rivets and structure damage.

(f) Examine the flap track bolts on the outboard trailing edge flap for damage that can occur if the flaps touched the ground.

(g) Examine the flap track bolts on the inboard trailing edge flap for damage that can occur if the flaps touched the ground.

(h) Make sure that the flight controls move freely and the cable tension is satisfactory.

H. Cabin Inspections

SUBTASK 05-51-68-210-008

(1) Continue the following inspection if any cabin damage was found in the above Cabin Inspection including the ceiling panels, central overhead bins, or the video monitors. If no damage was found, no further cabin inspection is required and it is not necessary to continue this Cabin Inspection.

(a) If ceiling panels without video monitors are found dislodged or damaged, do a check for proper latching and installation. Repair as necessary.

(b) If central overhead stowage bins are found with impact damage or are loose, or if ceiling panels equipped with video monitors are found with broken latches; repair as necessary.

(c) Do an inspection of all the tie rods of affected zones for evidence of buckling or rupture. If any defect is found, replace as necessary.

(d) Examine lavatory tie rod attachments for damage consisting of breakage, cracks, and deformation. Do an inspection of the tie rods for evidence of buckling or rupture.

(e) Examine the lavatory floor fittings for damage consisting of breakage, cracks, and deformation.

(f) Examine galley tie rod attachments for damage consisting of breakage, cracks, and deformation. Inspect the tie rods for evidence of buckling or rupture.

(g) Examine the galley floor fittings for damage consisting of breakage, cracks, and deformation.

(h) Examine, closet tie rod attachments for damage consisting of breakage, cracks, and deformation.

(i) Examine the Passenger Service Unit (PSU), do this task: Passenger Service Unit - Removal, TASK 25-23-61-000-801.

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(j) Examine the closet floor fittings for damage consisting of breakage, cracks, and deformation.

I. Engine Areas

SUBTASK 05-51-68-210-009

(1) Do the steps that follow to examine the engine areas:

(a) Do an engine inspection after a hard landing.

1) Do this task: HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

(b) Examine the engine nacelle if it touched the ground.

1) Do this task: DRAGGED ENGINE NACELLE/FAN BLADE OUT/ENGINE SEIZURE/THRUST REVERSER AND ATTACH POINTS DAMAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-10/201.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-68-410-002

(1) Install the components you removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-68-440-003

(2) Do the activation procedure for the trailing edge flap system.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 05-51-68-440-004

(3) Do the activation procedure for the leading edge flap and slat system.

(a) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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CONDITIONED AIR PACK OUTLET DUCT SYSTEM FAILURE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to examine the applicable air check valves after an upstream duct failure.

(1) The conditioned air check valves can be damaged when there is an upstream duct failure of the pack outlet.

(a) When there is a duct failure, the applicable conditioned air check valve must be removed and inspected.

(b) There is one conditioned air check valve found downstream of each air cooling pack, near the forward bulkhead of the ECS bay.

(2) The trim air check valves can be damaged when there is an upstream duct failure of the pack outlet.

(a) When there is a duct failure, the applicable trim air check valve must be removed and inspected.

(b) There is one trim air check valve found in each of the air conditioning bays.

TASK 05-51-80-210-801

2. Conditioned Air Pack Outlet Conditional Inspection

A. References

Reference	Title
21-51-07-000-801	Conditioned Air Check Valve Removal (P/B 401)
21-51-07-400-801	Conditioned Air Check Valve Installation (P/B 401)
21-61-09-200-801	Trim Air Check Valve Inspection (P/B 601)

B. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540

C. Procedure

SUBTASK 05-51-80-000-001

(1) Examine the conditioned air check valve.

(a) Remove the applicable conditioned air check valve for inspection, do this task: Conditioned Air Check Valve Removal, TASK 21-51-07-000-801.

(b) Do a visual examination of the conditioned air check valve for the following conditions:

1) missing parts

2) cracks

3) corrosion

4) flapper deformation

5) restricted flapper movement.

(c) Install the conditioned air check valve, do this task: Conditioned Air Check Valve Installation, TASK 21-51-07-400-801.

SUBTASK 05-51-80-210-001

(2) Examine the trim air check valve.

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(a) Do this task: Trim Air Check Valve Inspection, TASK 21-61-09-200-801.

D. Put the Airplane Back to It's Usual Condition

SUBTASK 05-51-80-410-001

(1) Install the components you removed if they are serviceable, or install replacement parts.

——— END OF TASK ———

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CABIN DEPRESSURIZATION CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure contains the task to inspect the airplane after cabin depressurization condition.

TASK 05-51-81-210-801

2. Cabin Depressurization Conditional - Inspection

A. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
21-32-05-000-801	Ceiling Blowout Panel Removal (P/B 401)
21-32-05-400-801	Ceiling Blowout Panel Installation (P/B 401)
50-10-00-200-801	Forward Cargo Compartment Liner Inspection (P/B 601)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
190	Subzone - Wing-to-Body Fairing

C. Cabin Depressurization Conditional Inspection

SUBTASK 05-51-81-210-001

(1) If cabin depressurization condition occurs, do these steps:

(a) Before opening the forward cargo door, examine the door seals to see if they have flipped outward.

(b) Examine the decompression panels for movement.

1) Do this task: Ceiling Blowout Panel Removal, TASK 21-32-05-000-801.

2) Do this task: Ceiling Blowout Panel Installation, TASK 21-32-05-400-801.

(c) Examine the cargo compartment for the conditions that follow (TASK 50-10-00-200-801):

- sidewall panels that have moved
- insulation that has moved
- lining that has moved.

(d) Examine the aft outflow valve, do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

1) Examine the area around the aft outflow valve for damage.

SUBTASK 05-51-81-210-002

(2) Examine the equipments that follow for damage.

(a) All emergency equipment:

1) All lifevests (passenger, crew, infant, etc)

2) Survival packs

3) Halon extinguishers

4) Water extinguishers

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5) First aid kits
6) Eyewash kit
7) Megaphone Transmitter with battery
8) Defibrillators.

CAUTION DO NOT PUT THE OXYGEN MASK HOSE NEAR THE ADJACENT STRUCTURE TO PREVENT RUBS OR KINKS. RUBS OR KINKS CAN CAUSE DAMAGE.

(b) All pressure sensitive equipment for Oxygen System:
1) Crew Oxygen Cylinders
2) Crew Oxygen Regulator
3) Flight Deck Oxygen Masks
4) Disposable oxygen masks with pouch (leave connected to oxygen bottles)
5) Smoke Hoods (Protective Breathing Equipment)
6) All Portable Oxygen Bottles
7) All Chemical Oxygen Generators (located in PSUs throughout main cabin; including crew rests, attendant stations, and galleys).

(c) ADM Systems:
1) All pitot and static lines.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-81-410-001

(1) Install the components you removed if they are serviceable or install replacement parts.

——— END OF TASK ———

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EXCESSIVE CABIN PRESSURE LEAKAGE - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure gives steps to do a leakage test to the airplane fuselage.

NOTE: Any pressure over 4.5 psid (31.0 kPa) differential needs to be coordinated with Boeing before the test is conducted.

TASK 05-51-91-790-801

2. Cabin Pressure Leak Test

(Figure 201, Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test uses the airplane equipment to pressurize the airplane and monitor the procedure.

(2) This leakage test is used to make sure the fuselage pressurized areas are tight.

(3) Many small leakage areas can add to create an excessive fuselage leakage rate.

(a) It is recommended that all sources of leakage be repaired.

(4) If leakage is found during this inspection, go to the related maintenance manual section for the repair.

WARNING

MAKE SURE YOU OBEY THE PROPER PROCEDURES FOR COMPRESSION AND DECOMPRESSION WHEN YOU USE PERSONS IN A PRESSURIZED AREA. PRESSURE CHANGES THAT CAUSE PAIN MUST NOT BE DONE. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURY TO PERSONS CAN OCCUR.

(5) When persons are in the airplane during this test, they must be in a good physical condition. When a person feels pain during a pressure change, you must lower the rate of change of pressure, or make the absolute pressure stable immediately. Do this to make sure the person can make the pressure equal in their ears. Also, do this to remove the person from the airplane.

(6) Pressurize the airplane with one of the steps that follow:

NOTE: A method to calculate absolute pressure is as follows:

Determine the field atmospheric pressure (in inches of mercury). Divide field atmospheric pressure by 2.036 and add the result to the gauge pressure or differential pressure inside the cabin to give the absolute pressure (PSIA). For example: If the field atmospheric pressure is 29.86 in/Hg (101.12 kPa) and the cabin differential pressure is 4.0 psid (27.6 kPa), divide 29.86 in/Hg (101.12 kPa) by 2.036 which equals 14.67 psi (101.15 kPa). Add the cabin differential pressure (in psi) to the field pressure (4.0 psid (27.6 kPa) + 14.67 psi (101.15 kPa) = 18.67 psia (128.73 kPa)) to obtain an absolute pressure of 18.67 psi (128.73 kPa).

(a) The primary air pressure source is the Auxiliary Power Unit (APU).

(b) You can use the engine bleed air for the pressure source.

(c) You can use an external ground source.

NOTE: The external ground source must supply 2000 cfm (56.6 M^3/minute) of air at 10 psig (69 kPa).

(d) The cabin differential pressure and the rates of pressurization and bleed-down are monitored in the control cabin.

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1) They are monitored on the differential pressure indicator.

(e) Figure 201 shows how to get a correction factor used in connection with some test data.

(f) Figure 202 shows the straight line curves for the permitted airplane leakage rates.

(7) Boeing has determined that all of the following listed equipment may be susceptible to damage anytime the cabin pressure exceeds 20 psia (138 kPa). This list is not meant to be all inclusive since there may be additional equipment sensitive to 20 psia (138 kPa).

(a) All emergency equipment:

1) All lifevests (passenger, crew, infant, etc)
2) Survival packs
3) Halon extinguishers
4) Water extinguishers
5) First aid kits
6) Eyewash kit
7) Megaphone Transmitter with battery
8) Defibrillators.

(b) All pressure sensitive equipment for Oxygen System:

1) Crew Oxygen Cylinders
2) Crew Oxygen Regulator
3) Flight Deck Oxygen Masks
4) Disposable oxygen masks
5) Smoke Hoods (Protective Breathing Equipment)
6) All Portable Oxygen Bottles
7) All Chemical Oxygen Generators (located in PSUs throughout main cabin; including crew rests, attendant stations, and galleys).

(c) ADM Systems:

1) All pitot and static lines.

B. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
35-12-85-000-802	Oxygen Mask/Regulator Removal (P/B 401)
35-12-85-400-802	Oxygen Mask/Regulator Installation (P/B 401)
49-11-00-860-801	APU Starting and Operation (P/B 201)
49-11-00-860-802	APU Usual Shutdown (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-7626	Cup - Air Flow, Lo/Hi Blow Test Part #: G21007-1 Supplier: 81205

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(Continued)

Reference	Description
STD-836	Stopwatch

D. Prepare for the Cabin Pressure Leakage Test

SUBTASK 05-51-91-780-001

(1) Do the steps that follow to prepare the airplane for the leakage test:

(a) Put the L and R RECIRC FAN switches on the Air Conditioning Panel, P5 to the AUTO position.

1) Make sure to limit the operation of fans at high cabin pressure to less than 30 min.

NOTE: Equipment cooling fans work harder at cabin pressures more than 15.7 psia (108.2 kPa).

(b) If the absolute pressure is at or below 20 psi (138 kPa) or if the differential pressure is at or below 7 psi (48 kPa), then remove the oxygen mask regulators from the oxygen system in the crew cabin. Do this task: Oxygen Mask/Regulator Removal, TASK 35-12-85-000-802.

NOTE: A method to calculate absolute pressure is as follows:

Determine the field atmospheric pressure (in inches of mercury). Divide field atmospheric pressure by 2.036 and add the result to the gauge pressure or differential pressure inside the cabin to give the absolute pressure (PSIA). For example: If the field atmospheric pressure is 29.86 in/Hg (101.12 kPa) and the cabin differential pressure is 4.0 psid (27.6 kPa), divide 29.86 in/Hg (101.12 kPa) by 2.036 which equals 14.67 psi (101.15 kPa). Add the cabin differential pressure (in psi) to the field pressure (4.0 psid (27.6 kPa) + 14.67 psi (101.15 kPa) = 18.67 psia (128.73 kPa)) to obtain an absolute pressure of 18.67 psi (128.73 kPa).

(c) Make sure that the pitot-static system is fully installed and operational.

(d) Start the APU and use it as a pressure source (TASK 49-11-00-860-801).

1) As an alternative, you can use a ground pneumatic cart as an air source.

NOTE: If you use a ground cart to supply pneumatic power, it must supply 2000 cfm (56.6 M^3/minute) of air at 10 psig (69 kPa).

(e) Operate an air cycle cooling pack, do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

NOTE: You can use the left or the right air cycle cooling pack.

1) Make sure that the pressurization mode selector on the P5 forward overhead panel is in AUTO position.

2) Operate the pack in AUTO at approximately 70°F (21°C).

3) Make sure that the aft outflow valve position indicator show the valve is fully open.

E. Cabin Pressure Leakage Test

SUBTASK 05-51-91-790-001

(1) Do the cabin pressure leakage test:

(a) Make sure that all of the airplane doors are properly closed and sealed.

(b) Put the pressurization mode selector on the P5 forward overhead panel to MAN position to turn off the auto control for the valve.

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(c) Record the cabin temperature, external ambient pressure, and external ambient temperature.

(d) Start to close the aft outflow valve when the air condition system has become stable.

WARNING DO NOT ALLOW THE INDICATION ON THE CABIN CLIMB GAUGE TO SHOW MORE THAN 1000 FEET PER MINUTE. WHEN YOU ALLOW A RATE OF CABIN CLIMB OF MORE THAN 1000 FEET PER MINUTE, DAMAGE TO THE AIRPLANE STRUCTURE OR INJURY TO PERSONNEL CAN OCCUR.

(e) Slowly increase the cabin pressure with the manual control toggle, at a rate of approximately 300 ft (91 m) a minute.

NOTE: The rate of pressure can be more than 300 ft (91 m) a minute but not more than 1000 ft (305 m) a minute.

1) Adjust the outflow valve position to allow for a comfortable rate of change in cabin pressure.

NOTE: The increase in pressure can cause the lavatory or galley smoke detectors to sound an alarm when there is no smoke.

WARNING DO NOT LET THE CABIN DIFFERENTIAL INDICATOR INCREASE TO MORE THAN 4.0 PSI DURING A USUAL TEST. IF YOU LET THE PRESSURE INCREASE TO MORE THAN 4.0 PSI, DAMAGE TO THE AIRPLANE STRUCTURE OR INJURY TO PERSONNEL CAN OCCUR.

(f) Increase the cabin pressure until the cabin differential indicator shows a differential pressure of 4.0 psid (27.6 kPa).

NOTE: Equipment cooling fans work harder at cabin pressures more than 15.7 psia (108.2 kPa). Limit the operation of fans at high cabin pressure to less than 30 minutes.

NOTE: If a fan overheats and stops during high cabin pressure operation, the equipment cooling OFF lights will illuminate. If this occurs, select ALTERNATE to restore the equipment cooling air flow. After a cool-down period, the internal thermostats in the overheat fan will automatically reset. The fan will become operational again.

(g) When you have a differential pressure of 4.0 psid (27.6 kPa), do the steps that follow:

1) Fully close the outflow valve.

a) Make sure that the valve position indicator shows that the valve is in the fully closed position.

2) Turn the air conditioning pack that is in operation to OFF, do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

(h) Shutdown the APU (or other pressurization source) if it is necessary (TASK 49-11-00-860-802).

(i) In less than five minutes after you turn off the air conditioning packs, make a record of the data.

NOTE: If the aft outflow valve did not stay fully closed during the leakage test, the pressure/time data is not correct.

(j) Make a record of the pressure differential and the time during the cabin bleed down.

1) Start the test at 4.0 psid (27.6 kPa) (time zero).

2) Stop the test at 2.5 psid (17.2 kPa).

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(k) Make five to ten data sets of each of the items that follow:

1) The time on the stopwatch, STD-836.

2) The cabin differential pressure.

3) The cabin temperature

4) The external ambient pressure.

(l) If you will do the test again, make sure you do a cool down time of not less than 20 minutes.

1) Do the cool down time with the equipment cooling fans operating at zero differential pressure.

NOTE: This will make sure the fans will not overheat.

F. Leakage Rate Analysis

SUBTASK 05-51-91-720-001

(1) Do the leakage rate analysis.

(a) Get the correction factor from Figure 201 to correct the time data for each data point.

NOTE: You want to use the correction factors with the bleed down time data when the ambient pressure does not equal 14.7 psia (101.4 kPa) and/or the cabin temperature does not equal 70°F (21°C).

(b) Make a graph from the points that follow:

1) The differential pressure data on the vertical axis of Figure 202.

2) The time on the horizontal axis of Figure 202.

(c) Plot each time and pressure data point onto the chart shown in Figure 202.

(d) Make a straight line through the two axis points Figure 202.

1) If the plotted line is in the upper zone in Figure 202, the pressure leakage rate is satisfactory.

G. Areas of Possible Leakage

SUBTASK 05-51-91-790-002

(1) Do a check of the areas that follow for excessive leakage:

NOTE: Do this while the airplane is pressurized between 3 psid (21 kPa) and 4 psid (28 kPa) and the test shows possible leaks.

NOTE: You can use an air flow blow cup, SPL-7626, or an equivalent substitute, to check for leakage.

NOTE: Listen for air leaks through the fuselage, especially from the pressure bulkheads and structure hidden by fairings.

(a) Make sure that the electronic cooling system automatic flow control valve is closed correctly.

NOTE: The valve should be fully closed at a differential pressure of 1.1 psid (8 kPa).

(b) Make sure that the bilge drains which are found along the bottom center line of the airplane are correctly closed.

NOTE: The bilge drain valves close at approximately 2.0 psid (14 kPa).

(c) Make a check of all door and hatch seals for leakage.

(d) Make a check of all flight deck windows for leakage.

(e) Make a check of the aft outflow valve for leakage.

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(f) Make a check of the aft outflow valve seal for leakage.

(g) Make a check of the cabin pressure safety relief valves for leakage.

(h) Make a check of the cabin pressure negative relief valve seal for leakage.

(i) Make a check of the water service panel seals for leakage.

(j) Make a check of all air conditioning and APU duct seals at the pressure bulkhead penetrations for leakage.

NOTE: These are found in the A/C bay, aft pressure dome, right wheel well, and forward pack bays.

(k) Make a check of the control cable seals in the pressure bulkhead penetrations for excessive leakage.

(l) Do a check of the areas which are difficult to access because they can also have leaks.

1) Structure hidden by the wing-to-body fairing and the APU bulkhead area should be checked for potential structure and seal leaks.

(m) Make a check of the ball valve drain located in the main landing gear wheel well.

(n) Make a check of the forward and aft nose wheel well access panels for leakage.

1) Check the side and top access panels for leakage.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-91-860-004

(1) Make sure that the cabin pressure and altitude decreases to normal ground conditions.

SUBTASK 05-51-91-410-001

(2) Install the components that were removed if they are serviceable, or install replacement parts.

SUBTASK 05-51-91-860-001

(3) If the equipment cooling supply exhaust fans were selected from NORMAL to ALTERNATE, the switches must be returned to the NORMAL position.

SUBTASK 05-51-91-860-002

(4) Make sure that the L and R RECIRC FAN switches on the Air Conditioning Panel, P5 is in the AUTO position.

SUBTASK 05-51-91-360-001

(5) If you found leaks during the test, repair the leaks.

SUBTASK 05-51-91-860-003

(6) Move the pressurization mode selector on the P5 forward overhead panel to AUTO.

SUBTASK 05-51-91-420-001

(7) If the crew oxygen mask regulator was removed, then install and test the mask demand regulator, do this task: Oxygen Mask/Regulator Installation, TASK 35-12-85-400-802.

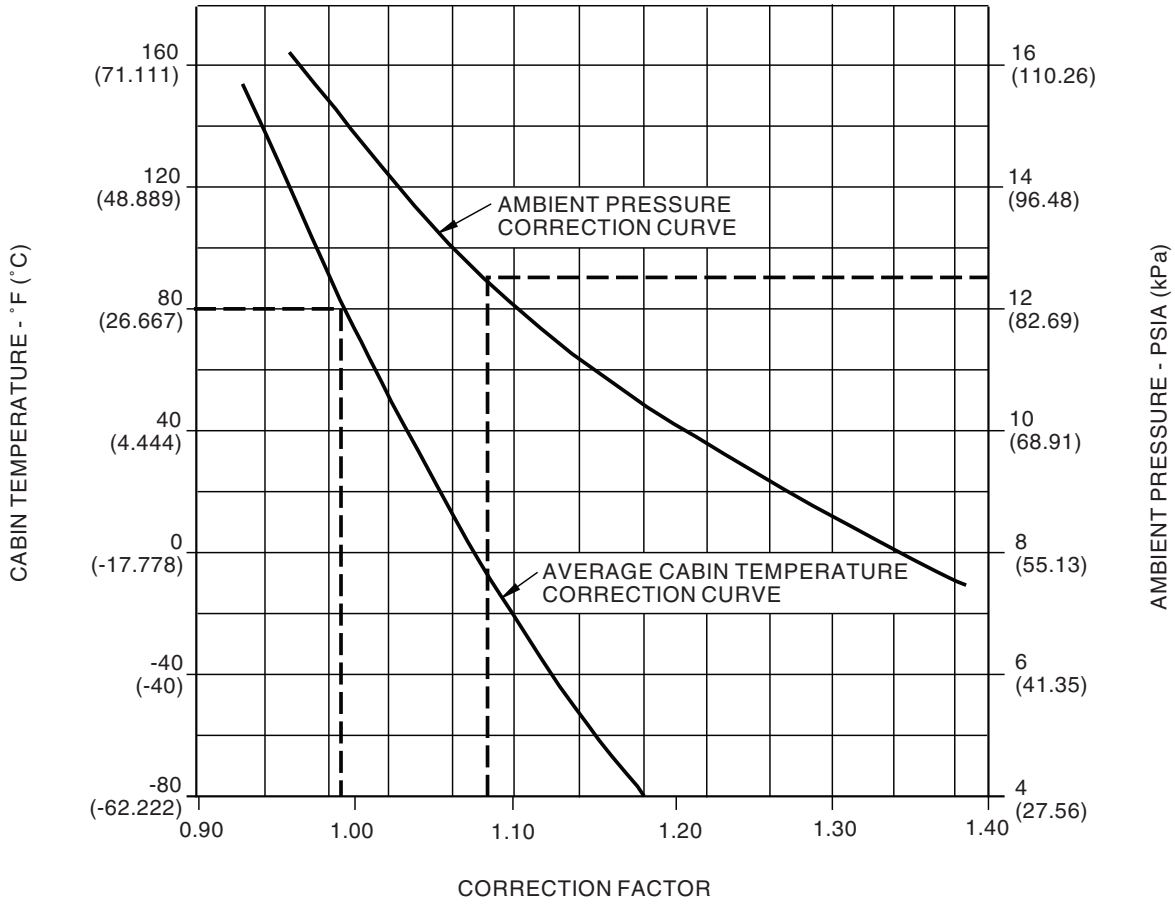
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HOW TO USE CURVE:

EXAMPLE: ASSUME A DIFFERENTIAL PRESSURE OF 3.3 PSIG (22.75 kPa) IS MEASURED AFTER 50 SECONDS THE CABIN TEMPERATURE IS 80˚F (26.669˚C) AND THE AMBIENT PRESSURE IS 12.50 PSIA (86.18 kPa).

THE CORRECTED TIME, TO BE PLOTTED ON THE FIGURE 202 CURVE, IS EQUAL TO THE MEASURED TIME DIVIDED BY BOTH THE TEMPERATURE AND PRESSURE CORRECTION FACTORS. FROM CURVES BELOW: TEMPERATURE CORRECTION FACTOR IS 0.99 AND PRESSURE CORRECTION FACTOR IS 1.08. THE PRODUCT OF THE TWO FACTORS IS 1.07. THUS, TIME CORRECTED TO AN AMBIENT PRESSURE OF 14.7 PSIA (101.35 kPa) AND CABIN TEMPERATURE OF 70˚F (21.111˚C) IS EQUAL TO 50 DIVIDED BY 1.07=47 SECONDS.



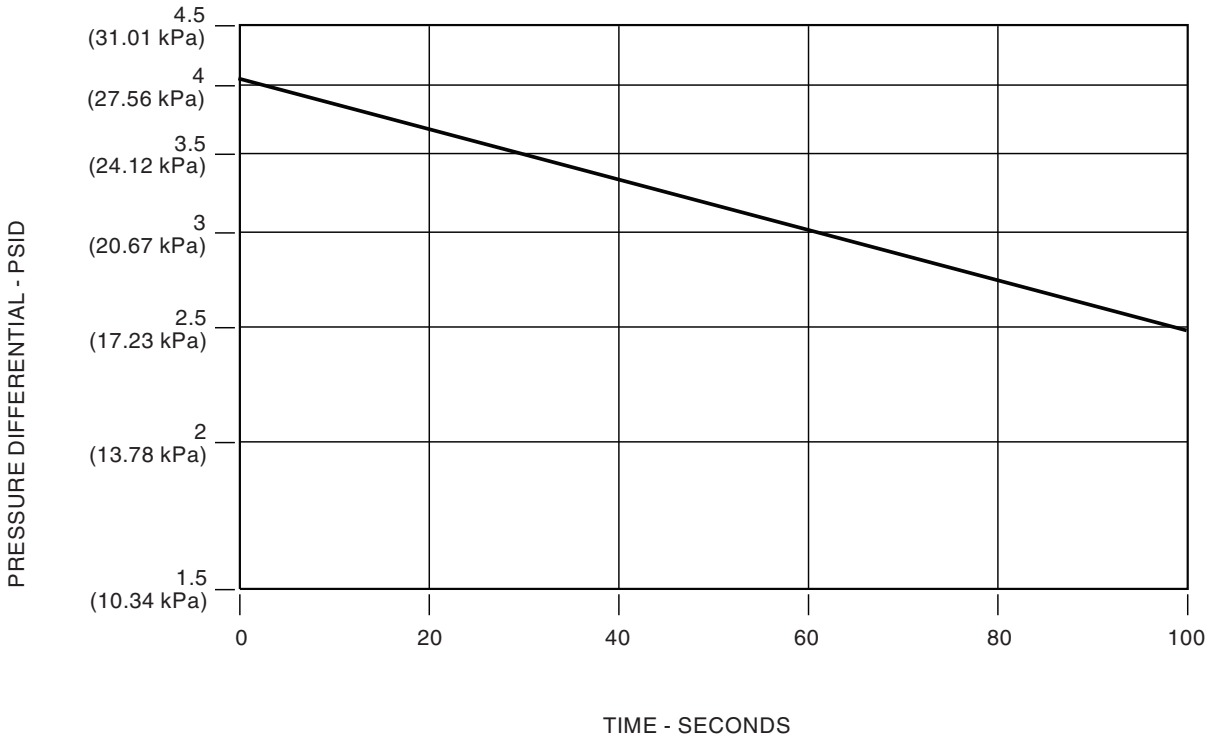
NOTE:

CORRECTION FACTOR TO BE APPLIED TO BLEED DOWN TIME WHEN AMBIENT PRESSURE DOES NOT EQUAL 14.7 PSIA (101.35 kPa) AND/OR CABIN TEMPERATURE DOES NOT EQUAL 70˚F (21.111˚C).

2410419 S00061525615_V1

Pressure Leakage Test Correction Factor
Figure 201/05-51-91-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

2.5 PSIG (17.23 kPa) IS THE RECOMMENDED MINIMUM TEST POINT.

2410420 S00061525616_V1

Pressure Leakage Rate Check Chart
Figure 202/05-51-91-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COLD-SOAKED FUEL FROST - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Cold-soaked fuel frost maintenance:

NOTE: For non FAA operators, Boeing advises operators to contact their respective regulatory authority to obtain approval to use the Cold-Soaked Fuel Frost (CSFF) procedure.

(a) Conditional inspection for cold-soaked fuel frost on the upper wing surfaces when the fuel temperature as given on the flight deck is below 32°F (0°C).

1) If the fuel temperature indicator or probe does not work, begin the inspection for cold-soaked fuel frost.

(b) Removal of ice or frost from the upper wing surfaces if certain conditions are not met.

NOTE: Light coatings of frost due to cold-soaked fuel are allowed on the lower wing surfaces per Cold Weather Maintenance Procedure, TASK 12-33-01-600-801. Cold-soaked fuel frost is also allowed on the upper surface if it is identified by meeting the conditions given in this procedure.

TASK 05-51-92-660-801

2. Cold-Soaked Fuel Frost Maintenance

(Figure 201 or Figure 202)

A. References

Reference	Title
12-33-01-600-801	Cold Weather Maintenance Procedure (P/B 301)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

C. Location Zones

Zone	Area
500	Left Wing
600	Right Wing

D. Inspection for Cold-Soaked Fuel Frost

SUBTASK 05-51-92-210-001

(1) If the temperature of the fuel in the main fuel tanks is greater than 32°F (0°C), then no additional inspection for cold-soaked fuel frost is necessary.

(a) If the fuel temperature indicator or probe does not work, begin the inspection for cold-soaked fuel frost.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-51-92-210-002

(2) If the temperature of the fuel in the main fuel tanks is less than or equal to 32°F (0°C), then do these steps:

(a) Inspect the wing lower surfaces for ice or frost. If ice or frost is present but is only inboard of the fuel measuring stick No. 4, no additional inspection of the upper surfaces is required (see warning below).

(b) If ice or frost is present on the wing lower surface outboard of fuel measuring stick No. 4, use a work platform, COM-2480, or equivalent hoist to inspect the upper wing surfaces for the presence of ice or frost.

NOTE: The distance that the frost extends outboard of fuel measuring stick No. 4 can be used as an indication of the extent of the frost on the upper wing surface.

(c) If no ice or frost is present on the wing upper surfaces, no additional inspection is required.

E. Procedure

SUBTASK 05-51-92-660-001

(1) When ice or frost is present on the top wing surface, do these steps:

(a) Determine if these conditions are met. If all of these conditions are met, the frost can be identified as permissible cold-soaked fuel frost and no further action is necessary. The airplane may be released for flight.

1) The outside air temperature is at or above 39.2°F (4°C).

2) The temperature of the fuel in the main fuel tanks is at or above 3°F (-16°C).

3) There is no precipitation or visible moisture (rain, snow, drizzle, fog with less than 1 mile visibility, etc.).

4) The frost is on or between the main black lines marking the wing fuel tanks.

5) There is no ice or frost anywhere on the upper surface of the wing, including the leading edges and the control surfaces, beyond the outer edge of the black lines defining the cold soaked fuel frost area.

(b) If any of the above conditions is not met, do this step:

1) Apply deicing/anti-icing fluids to the wings with the appropriate deicing/anti-icing equipment to remove all of the ice or frost present (TASK 12-33-01-600-801).

SUBTASK 05-51-92-660-002

WARNING REMOVE ALL ICE FROM ALL LEADING EDGE DEVICES, ALL CONTROL SURFACES, TAB SURFACES, AND BALANCE PANEL CAVITIES BEFORE TAKEOFF. ICE CAN CAUSE AN IN-FLIGHT SAFETY PROBLEM.

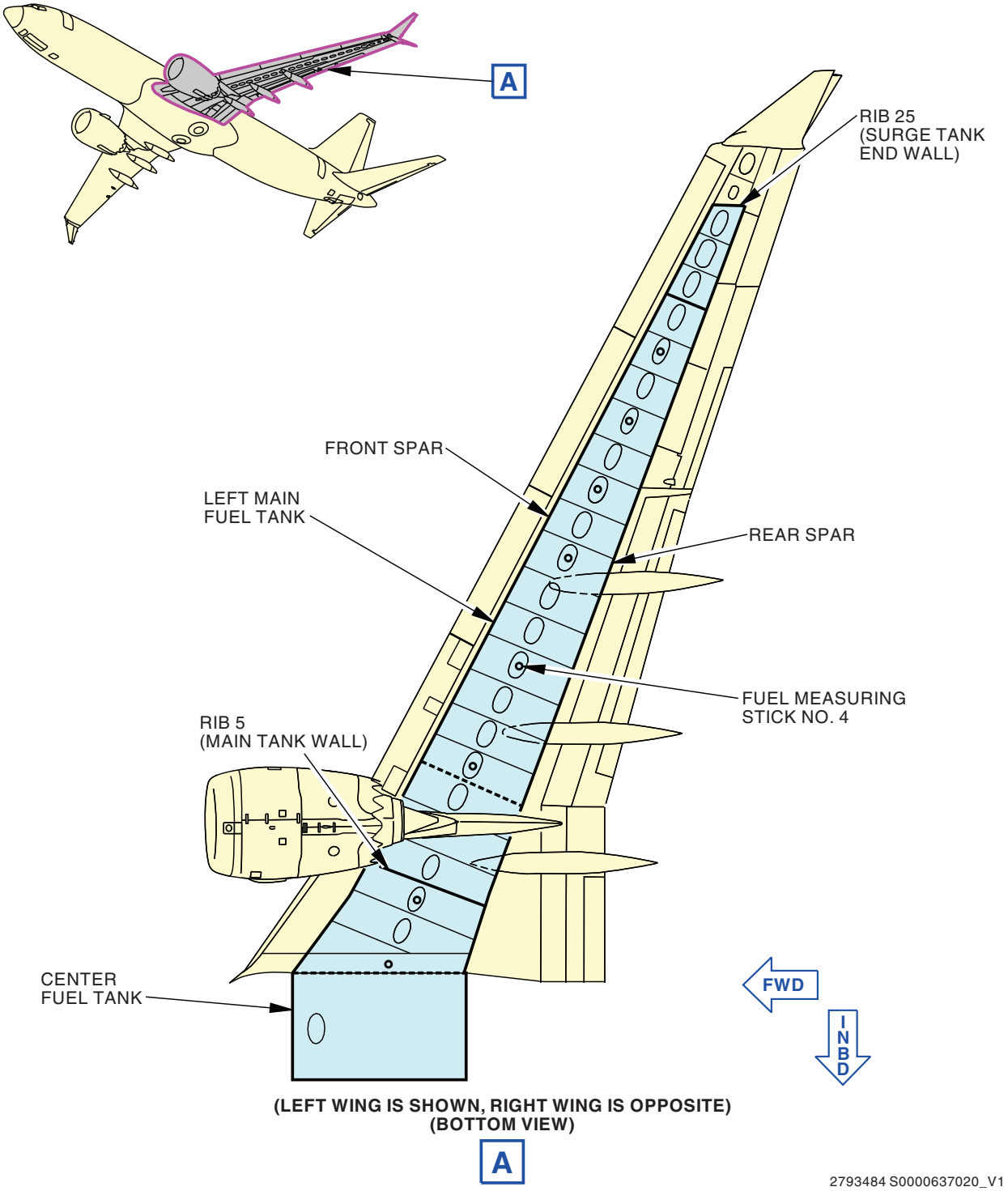
(2) Make sure that all ice and frost is removed from leading edge devices, control surfaces, tab surfaces, winglets (if installed), and balance panel cavities prior to takeoff.

SUBTASK 05-51-92-660-003

(3) If ice or frost re-forms on the wings prior to takeoff, do the step (1) of the above procedure to determine if the frost is acceptable for flight or to remove ice and frost from the wings.

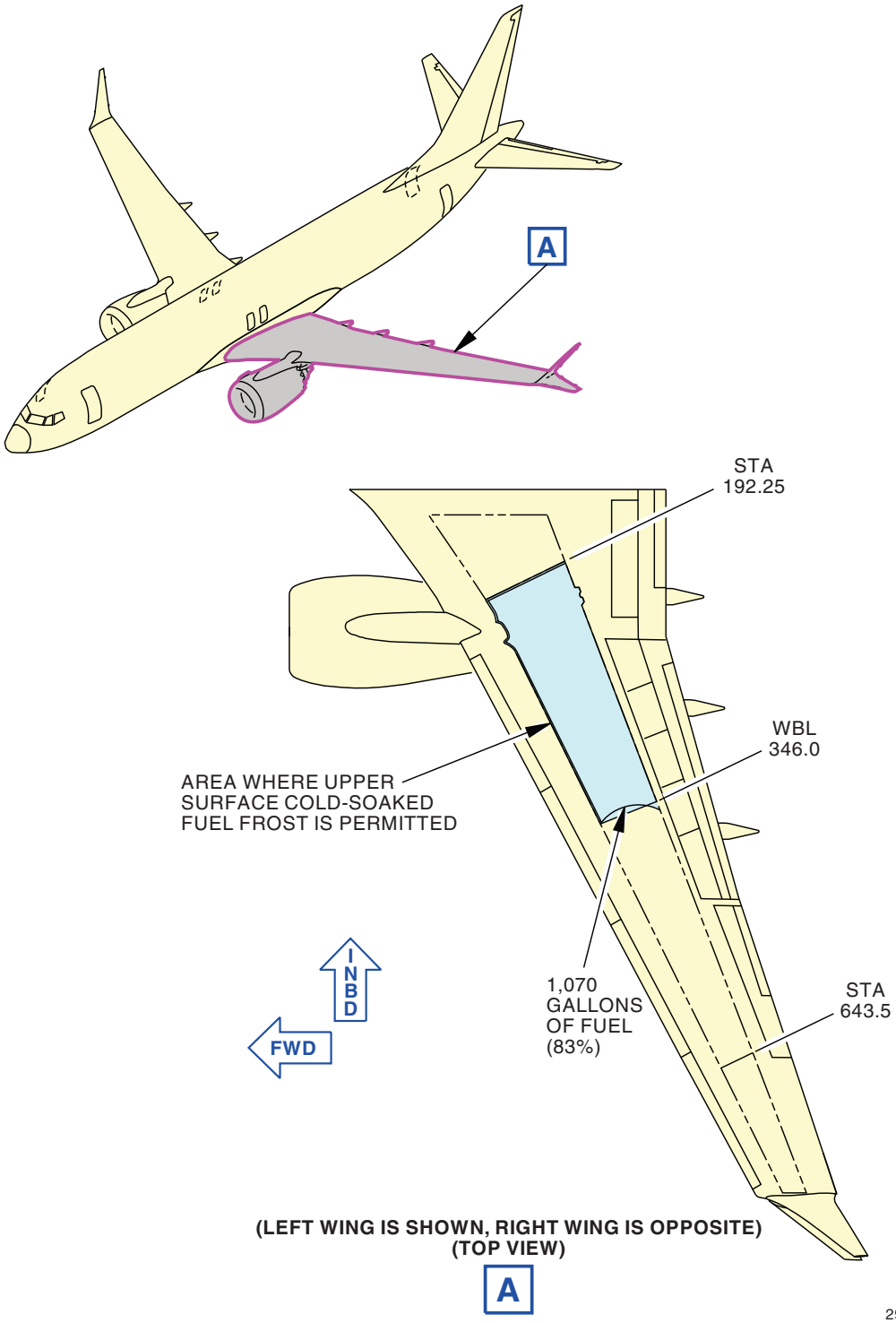
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ice/Frost Examination Areas
Figure 201/05-51-92-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Allowable Cold-Soaked Fuel Frost - Upper Wing Surface
Figure 202/05-51-92-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUNWAY/TAXIWAY EXCURSIONS - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. Runway/Taxiway Excursions

(1) A runway/taxiway excursion occurs if the airplane takes off or makes a landing with one or more of the conditions that follow:

(a) The airplane skids off or overruns the specified runway/taxiway onto an unprepared surface.

(b) The airplane lands short of the specified runway.

(c) The airplane rejects take off and departs from the specified runway.

(d) The airplane uses a runway/taxiway not specified for taking off or landing.

(e) The airplane loses directional control and skids on the runway such that damage is suspected.

NOTE: If the only inspection trigger was for two or more flat tires, then this does not count as “damage” and a Phase II inspection is not necessary. Also, if the Phase I inspection does not show more than flat tires and damaged wheels or brakes, then the aircraft can be returned to service after maintenance. In this condition it is not necessary to complete a Phase II inspection.

B. The Inspections

(1) This inspection is in two phases (Phase I and Phase II)

(a) The Phase I inspection is applicable when a runway/taxiway excursion occurs.

(b) The Phase II inspection must be done if Phase I shows that damage has occurred.

C. Inspections, Repairs and Replacements

(1) When the inspection procedure tells you to "examine" a component, look for the conditions that follow:

(a) Cracks

(b) Structures that are pulled out of their position

(c) Loose paint (paint flakes)

(d) Twisted parts (distortion)

(e) Bent parts

(f) Wrinkles or buckles in the airplane structure

(g) Fasteners that are loose or pulled out

(h) Fastener holes that became larger or longer.

(i) Delamination

(j) Discoloration (heat damage)

(k) Parts that do not align correctly

(l) Signs of impact damage

(2) Replace or repair the components that have one or more of the conditions given above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-51-94-210-801

2. Phase I Inspection

A. General

(1) This task includes the procedure necessary to complete the Phase I inspection after a runway/taxiway excursion.

B. References

Reference	Title
05-51-01 P/B 201	HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

D. Runway/Taxiway Excursions Phase I Inspection

SUBTASK 05-51-94-210-002

(1) Do this task:HARD LANDING OR HIGH DRAG/SIDE LOAD LANDING - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION), PAGEBLOCK 05-51-01/201.

——— END OF TASK ———

TASK 05-51-94-210-802

3. Phase II Inspection

A. General

(1) This task includes the procedure necessary to complete the Phase II inspection after a runway/taxiway excursion.

B. References

Reference	Title
05-51-01-210-806	Phase II Inspection (P/B 201)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage
500	Left Wing
600	Right Wing
700	Landing Gear and Landing Gear Doors

D. Runway/Taxiway Excursions Phase II Inspection

SUBTASK 05-51-94-210-003

(1) Do this task:Phase II Inspection, TASK 05-51-01-210-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DRY CHEMICAL FIRE EXTINGUISHER USED - MAINTENANCE PRACTICES

1. General

A. This procedure contains one task to clean and inspect the aircraft after a dry chemical fire extinguisher has been used. This task is necessary any time a dry chemical fire extinguisher is discharged.

(1) If a dry chemical fire extinguisher was used to extinguish a landing gear brake fire, refer to High Energy Stop, TASK 05-51-07-210-801, to inspect the landing gear.

B. Dry chemical extinguishers can cause extensive corrosion damage to airplane structure, electrical systems, carbon brakes, and electronic equipment. Dry chemical extinguisher residue is difficult to clean up. Dry chemical fire extinguishers should only be used for airplane fire-fighting if there are no other extinguishers available and there is imminent danger to property or personnel.

C. Disassembly of aluminum airplane structure may be necessary for cleanup and corrosion prevention after use of a dry chemical fire extinguisher. Electrical and electronic equipment may need to be scrapped after being contaminated with dry chemical extinguishing agent.

TASK 05-51-95-100-801

2. Cleaning After Dry Chemical Fire Extinguisher Used

A. General

(1) This task includes the steps to clean and inspect the airplane after a dry chemical fire extinguisher has been used.

(2) If a dry chemical fire extinguisher was used to extinguish a landing gear brake fire, refer to AMM 05-51 High Energy Stop to inspect the landing gear. All carbon brake assemblies that have visible dry chemical fire extinguisher residue need to be removed from the airplane, and dispositioned per the applicable instructions in the carbon brake supplier’s Component Maintenance Manual (CMM).

(3) If a dry chemical fire extinguisher was used to extinguish an engine fire, refer to AMM 71-00 Power Plant for the cleaning of external areas of the engine (engine cowls open or closed). All engine parts that cannot be cleaned when attached will need to be removed and cleaned per the applicable instructions in the engine supplier’s Component Maintenance Manual (CMM).

(4) If necessary, review the manufacturer's Material Safety Data Sheets (MSDS) for the particular fire extinguishing agents that were used for additional recommendations.

(5) If a dry chemical fire extinguisher was used to extinguish the engines fire, refer to AMM 71-00 or AMM 72-00 Inspection After an Engine Fire.

(6) If a dry chemical fire extinguisher was used to extinguish the APU fire, refer to AMM 49-11-00-6 Inspection After an APU Fire.

B. References

Reference	Title
SWPM 20-60-01	Cleaning of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-2618	Cleaner - Vacuum Part #: 106542 Supplier: 3LFF2 Part #: AVIATION VAC 106542 Supplier: 03FQ4 Part #: AVQ250-2 Supplier: U1761 Part #: AVQ350-2 Supplier: 03FQ4 Part #: RSAV130 Supplier: $1291 Part #: RSV150 Supplier: U1761 Opt Part #: 02146A Supplier: 0A5X2 Opt Part #: 44SPEC Supplier: 0Y8U0 Opt Part #: 655406-7M Supplier: 0Y8U0 Opt Part #: BP80 Supplier: $0373 Opt Part #: C-39485-41 Supplier: 16893 Opt Part #: C-39485-42 Supplier: 16893 Opt Part #: MORCLEAN 300/15D Supplier: $1316 Opt Part #: R80 Supplier: $0373 Opt Part #: RSV130 Supplier: $1291 Opt Part #: WD80 Supplier: $0373
STD-739	Respirator - Half-Mask, Air Purifying (with filters/cartridges)
STD-6087	Brush - Acid Resistant
STD-6935	Air Source - Regulated, Dry Filtered, Compressed 30 PSIG Maximum Pressure

D. Consumable Materials

Reference	Description	Specification
B00095	Compound - Sodium Bicarbonate	ASTM D928
G50138	Cloth - Soft Cotton	
G50305	Sheeting - Plastic, Polyethylene	
G50412	Paper - Litmus (Used to find the acidity or alkalinity of a liquid solution)	
G50933	Cloth - Cleaning, Low-Lint Cloth (General Use)	A-A-59323 Type II (Supersedes MIL-C-85043)

E. Prepare to Clean

CAUTION

DO NOT USE WATER ON DRY CHEMICAL FIRE EXTINGUISHING AGENTS. THESE AGENTS CAN MIX WITH WATER AND MAKE A FLUID COMPOUND THAT CAUSES CORROSION. THIS CAN CAUSE DAMAGE TO THE EQUIPMENT.

SUBTASK 05-51-95-110-001

(1) Use a half-mask respirator, STD-739, or equivalent respiratory protection.

SUBTASK 05-51-95-110-002

(2) Install plastic sheeting, G50305, as necessary to prevent the airborne dry chemical agent from further contaminating the surrounding area.

SUBTASK 05-51-95-010-001

(3) Get access to all areas where smoke and the fire extinguishing agent could have possibly contaminated to facilitate the cleaning process.

(a) Joints that cannot be cleaned in the assembled state must be disassembled for cleaning.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-51-95-110-009

(4) Prepare the neutralizing solution according to the type of fire extinguishing agent.

(a) Use a 2% vinegar and 98% hot water solution to clean a potassium bicarbonate (purple in color) or sodium bicarbonate (light blue in color) dry chemical extinguishing agent.

(b) Use a solution with compound, B00095 (baking soda) and hot water to clean monoammonium phosphate (yellow in color) dry chemical extinguishing agent.

NOTE: The extinguishing agents are potentially acidic when exposed to moisture.

F. Procedure

SUBTASK 05-51-95-140-001

(1) Do these steps to remove as much of the dry, loose agent as possible:

(a) Use a vacuum cleaner, COM-2618 with a HEPA filter to remove the unmelted, dry chemical from all accessible surfaces, including structure, tubing, wiring, and systems.

(b) Wipe down all surfaces with a soft cotton cloth, G50138 and continue to use the vacuum to clean the surfaces after they are wiped down.

(c) Any high strength steel parts exposed to the dry chemical agent must be cleaned first.

(d) It is acceptable to use a regulated air source, STD-6935 at 30 psi (207 kPa) or less to loosen the material.

SUBTASK 05-51-95-110-004

(2) Apply the solution and clean with an acid resistant brush, STD-6087, low-lint cleaning cloth, G50933 or sprayer nozzle.

(a) Make sure that the solution penetrates the unsealed faying surface joints.

(b) If there are unsealed faying surfaces, it is recommended to disassemble them and clean the individual components.

SUBTASK 05-51-95-110-005

(3) Apply the solution until the bubbles stop.

NOTE: When the bubbles stop, the acid has become neutral.

SUBTASK 05-51-95-110-006

(4) Allow the solution to remain for an additional 5 minutes after the bubbles stop.

SUBTASK 05-51-95-170-001

(5) Do these steps to remove the neutralized solution:

(a) Rinse the area with fresh water.

(b) Wipe up residual material with a low-lint cleaning cloth, G50933, or equivalent.

(c) Air dry the area immediately using a regulated air source, STD-6935.

SUBTASK 05-51-95-110-007

(6) Repeat the wash and rinse steps as many times as is necessary to pass litmus paper pH test below.

SUBTASK 05-51-95-110-008

(7) Do a test of the area with litmus paper, G50412. The pH factor must be between about 7 and 9.

(a) If the pH is less than 7 (acidic), repeat the cleaning procedure.

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SUBTASK 05-51-95-420-001

(8) Replace the parts in areas where the fire has caused the dry chemical extinguishing agent to melt.

NOTE: Melted dry chemical extinguishing agent will deeply penetrate joints and cause corrosion issues. It can also compromise the heat treatment of parts.

(a) If you suspect that the structure has heat damage, review SRM 51-20–03 for the appropriate actions.

SUBTASK 05-51-95-020-001

(9) If there is evidence of dry chemical agent on electrical equipment, remove the equipment to clean, inspect, and test it.

(a) After you reinstall the equipment, perform the appropriate functional test.

SUBTASK 05-51-95-020-002

(10) Remove any equipment that has fire damage.

SUBTASK 05-51-95-140-002

(11) If control cables are affected, clean using only a dry cloth.

(a) Lubricate the cable as necessary.

SUBTASK 05-51-95-140-003

(12) Do these steps to clean affected electrical wire bundles and external connector surfaces:

WARNING

OBEY THE MANUFACTURER'S PROCEDURES WHEN YOU DO EZAP MAINTENANCE, OR MAINTENANCE THAT HAS AN EFFECT ON EWIS. IF YOU DO NOT OBEY THE PROCEDURES, IT CAN INCREASE THE RISK OF AN IGNITION SOURCE FOR FIRES. INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Follow the instructions in AMM 20-60 for EWIS Protection and Cleaning.

(b) Clean external surfaces prior to disconnecting connectors.

(c) Place protective covers over the ends of connectors immediately after disconnecting them.

(d) If necessary, remove bundle ties to vacuum up agent that may have entered in between the wiring.

SUBTASK 05-51-95-140-004

(13) Do these steps to clean open connectors:

(a) Use a vacuum to remove dry chemical agent from the area.

(b) Wipe down all surfaces.

(c) Follow the steps in SWPM 20-60-01.

SUBTASK 05-51-95-140-005

(14) Do these steps to clean aluminum hydraulic tubing:

(a) Remove the tubing clamps.

(b) Use the vacuuming and wiping method to remove the extinguishing agent from the tubing.

(c) Use the cleaning steps above to neutralize any remaining surface residue and perform the pH check.

(d) Remove the tubing clamp block components and apply same cleaning process as above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Replace any contaminated cushioned-type loop or saddle clamps with new clamps.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-95-210-001

(1) It is recommended to review AMM 05-51 Volcanic Ash Conditional Inspection as a guide for inspection, if any other areas of the datum airplane such as landing gear or air conditioning equipment were exposed.

SUBTASK 05-51-95-620-001

(2) Reapply any corrosion prevention material that was removed during the cleaning process.

SUBTASK 05-51-95-210-002

(3) It is recommended to perform a one-time general visual inspection at 7 to 10 days for signs of corrosion or evidence of dry chemical extinguishing agent.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN WHEEL WELL CORROSION PROTECTION OF ELECTRICAL CONNECTORS

1. General

A. This procedure contains a task to inspect the electrical connectors from corrosion in the Main Wheel Well.

B. Corrosion is an indicator that the electrical connectors may have been exposed to potassium formate and/or potassium acetate. Electrical shorting of the electrical connectors in the wheel well can cause incorrect function of critical systems necessary for safe flight.

24-CMR-01

TASK 05-51-96-211-801

2. Electrical Connectors in the Main Wheel Well Detail Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-51-00-740-803	Trailing Edge Flap Load Relief System BITE Test (P/B 501)
27-51-02-720-801	Trailing Edge Flap Bypass Valve Functional Test (P/B 501)
27-61-12-400-803	Spoiler Shutoff Valve Test (P/B 501)
27-81-91-720-801	Leading Edge Standby Shutoff Valve Test (P/B 201)
29-11-81-710-802	EDP Supply Shutoff Valve Operational Test (P/B 501)
29-33-21-400-801	Standby Reservoir Low Quantity Switch Installation (P/B 401)
32-09-01-710-801	Main Landing Gear Air/Ground Sensor Operational Test (P/B 501)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)
32-61-21-710-801	Main Landing Gear Up-and-Locked Sensor Operational Test (P/B 501)
32-61-31-710-801	Main Landing Gear Down-and-Locked Sensor Operational Test (P/B 501)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-25-00	SHIELDED CABLES
SWPM 20-60-02	Replacement of Contacts
SWPM 20-60-03	Special Protection of Electrical Connectors
SWPM 20-60-09	Standard Wiring Practices Manual
SWPM 20-61-11	MIL-C-26500 Front Release Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-6402	Swab - Cotton

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G50170	Compound - Corrosion Inhibiting Compound, Soft Film, Exterior Use - Ardrox AV25	
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Electrical Connectors Detail Inspection

SUBTASK 05-51-96-860-002

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 05-51-96-010-001

(2) Get access to the wheel well of the main landing gear.

SUBTASK 05-51-96-211-001

(3) Measure the resistance of the D4850J/P connectors [11] as follows:

(a) Visually check the connectors [11].

1) Make sure that the connectors [11] are not loose or damaged.

2) Make sure that the plug backshells [4] are not loose or damaged.

3) Make sure that the strain relief clamp [12] is not loose.

(b) Measure the DC resistance between the connector receptacle backshell [1] and airplane structure [3] or bracket disconnect (SWPM 20-20-00).

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 10.0 milliohms.

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(c) Measure the DC resistance between the connector plug backshell [4] and connector receptacle [2] (SWPM 20-20-00).

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 15.0 milliohms.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(d) Measure the DC resistance between the connector plug backshell [4] and connector plug [5] (SWPM 20-20-00).

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 5.0 milliohms.

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

SUBTASK 05-51-96-211-002

(4) Examine the connectors [11] (D762, D4850J/P, D46038J/P, D764, D1582, D1014, and D46020J/P) and plug backshells [4] for exterior corrosion as follows:

(a) Examine the outer surface of the connectors [11] and plug backshells [4] for corrosion material deposits.

1) If corrosion is visible, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(b) Examine the connectors [11] for signs of corrosion.

1) Look for the presence of moisture, corrosion pits and white-colored material buildup.

NOTE: Accumulation of white-colored material is an indication of corrosion.

(c) Examine the backshell for signs of corrosion.

1) Look for the presence of corrosion pits and white-colored material buildup.

NOTE: Accumulation of white-colored material is an indication of corrosion.

2) If there are signs of corrosion on the backshell, estimate the area of corrosion.

a) If the cumulative area of corrosion is less than 10% of the total backshell surface area, clean the backshell with a cotton swab, STD-6402, moistened with alcohol, B00130.

b) If the cumulative area of corrosion is greater than 10% of the total backshell surface area, replace the connector and backshell (SWPM 20-60-03 and SWPM 20-60-09).

SUBTASK 05-51-96-211-003

(5) Examine the connectors [11] (D762, D4850J/P, D46038J/P, D764, D1582, D1014, and D46020J/P), backshells and contacts for interior corrosion or fluid contamination as follows:

(a) Disconnect the connectors [11] from the LRU connectors [9].

(b) Examine the LRU connectors [9] for signs of corrosion.

1) Look for the presence of moisture, corrosion pits, white-colored material buildup.

NOTE: Accumulation of white-colored material is an indication of corrosion.

(c) Examine the LRU connector contacts [8] for signs of corrosion.

1) Look for exposure of the copper base metal or blue-green deposits that indicate the presence of copper corrosion.

2) Look for a black or reddish discoloration of the gold-plated contacts.

(d) If there are no signs of corrosion, clean the LRU connectors [9] and LRU connector contacts [8] with a cotton swab, STD-6402, moistened with alcohol, B00130.

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(e) If there are signs of corrosion, replace the Line Replaceable Unit (LRU).

(f) Apply D5026NS or ZC-026 compound, G50171, to the pin contacts of the receptacle of LRU connector [9].

1) Spray with a 2-4 second burst 4 in. (102 mm) - 8 in. (203 mm) away from the LRU connector [9] (SWPM 20-60-03).

(g) Examine the connector contacts [10] for signs of corrosion.

1) Look for exposure of the copper base metal or blue-green deposits that indicate the presence of copper corrosion.

2) If there are no signs of corrosion, clean the connector contacts [10] with a cotton swab, STD-6402, moistened with alcohol, B00130.

3) If there are signs of corrosion, replace all connector contacts [10] in the connector [11] per SWPM 20-60-02.

(h) Apply D5026NS or ZC-026 compound, G50171, to the connectors [11] (SWPM 20-60-03).

1) Spray with a 2-4 second burst 4 in. (102 mm) - 8 in. (203 mm) away from the connector [11].

(i) Re-connect the connectors [11] to the LRU connectors [9].

(j) Measure the DC resistance between the D4850J/P connector receptacle backshell [1] and airplane structure [3] or bracket disconnect (SWPM 20-20-00).

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 10.0 milliohms.

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(k) Measure the DC resistance between the D4850J/P connector plug backshell [4] and connector receptacle [2] (SWPM 20-20-00).

1) Use intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 15.0 milliohms.

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(l) Measure the DC resistance between the D4850J/P connector plug backshell [4] and connector plug [5] (SWPM 20-20-00).

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the maximum resistance is less than 5.0 milliohms.

3) If the resistance measurements are greater than the specified milliohms, refer to SWPM 20-20-00, SWPM 20-25-00, SWPM 20-60-09 and SWPM 20-61-11 for repair instructions.

(m) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

(n) Do the following applicable operational tests:

1) For connector D762, do this task: EDP Supply Shutoff Valve Operational Test, TASK 29-11-81-710-802.

2) For connector D4850J/P, do this task: Spoiler Shutoff Valve Test, TASK 27-61-12-400-803.

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3) For connector D46038J/P, do the following tasks:
   a) Main Landing Gear Air/Ground Sensor Operational Test, TASK 32-09-01-710-801.
   b) Main Landing Gear Up-and-Locked Sensor Operational Test, TASK 32-61-21-710-801.
   c) Main Landing Gear Down-and-Locked Sensor Operational Test, TASK 32-61-31-710-801.
   d) Antiskid Valve Operational Test, TASK 32-42-00-720-803.
   e) Transducer Operational Test, TASK 32-42-00-710-803.
   f) Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.
   g) Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.
   h) Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.
4) For connector D764, do this task: EDP Supply Shutoff Valve Operational Test, TASK 29-11-81-710-802.
5) For connector D1582, do this task: Trailing Edge Flap Load Relief System BITE Test, TASK 27-51-00-740-803.
6) For connector D1014, do this task: Trailing Edge Flap Bypass Valve Functional Test, TASK 27-51-02-720-801.
7) For connector D46020J/P do the following tasks:
   a) Leading Edge Standby Shutoff Valve Test, TASK 27-81-91-720-801.
   b) Installation Test in Standby Reservoir Low Quantity Switch Installation, TASK 29-33-21-400-801.
      <1> As an alternative, do these steps:

      NOTE: This alternate test method eliminate the requirement to drain and refill the standby hydraulic system reservoir.

      <a> Remove the plug D678 from the S49 Standby Hydraulic System Low Quantity Indication Switch.
      <b> Apply a ground to pin 2 of the D678 plug.
      <c> Make sure that the Standby Hydraulic System Low Oil Quantity light on the flight control panel (P5-3) comes on.
      <d> Remove the ground from pin 2 of the D678 plug.
      <e> Reconnect the plug to the S49 Standby Hydraulic System Low Quantity Indication Switch.
      <f> Make sure that the Low Quantity light on the flight control panel (P5-3) goes out.

(o) Apply Ardrox AV25 compound, G50170, to the mated connector assembly including the grommets and backshells.
   1) Spray the connector assembly completely until the Ardrox AV25 compound, G50170, drips from it.

      NOTE: Excess Ardrox AV25 compound, G50170, should not be removed.

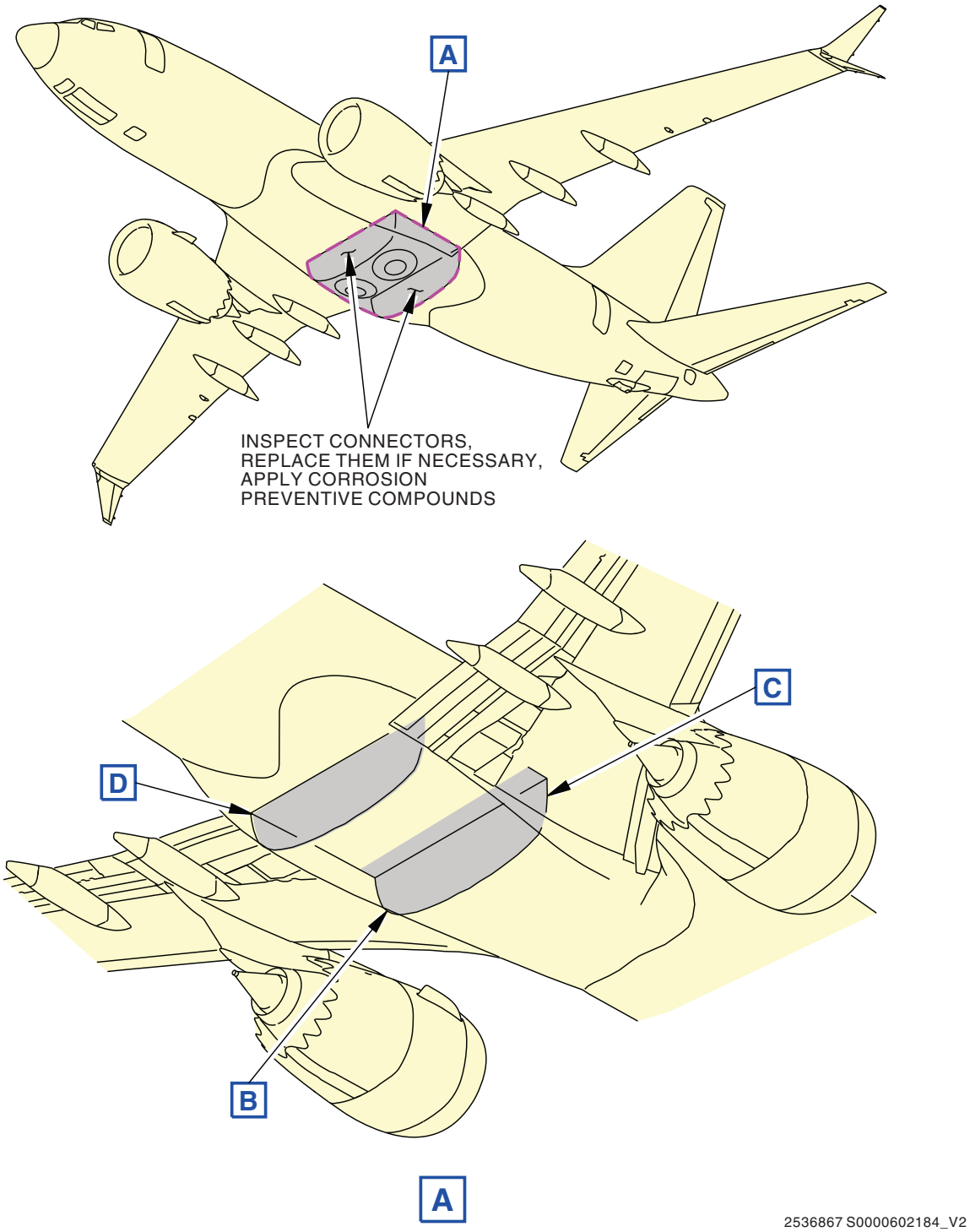
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-51-96-860-001

(6) Put the airplane back to its serviceable condition.

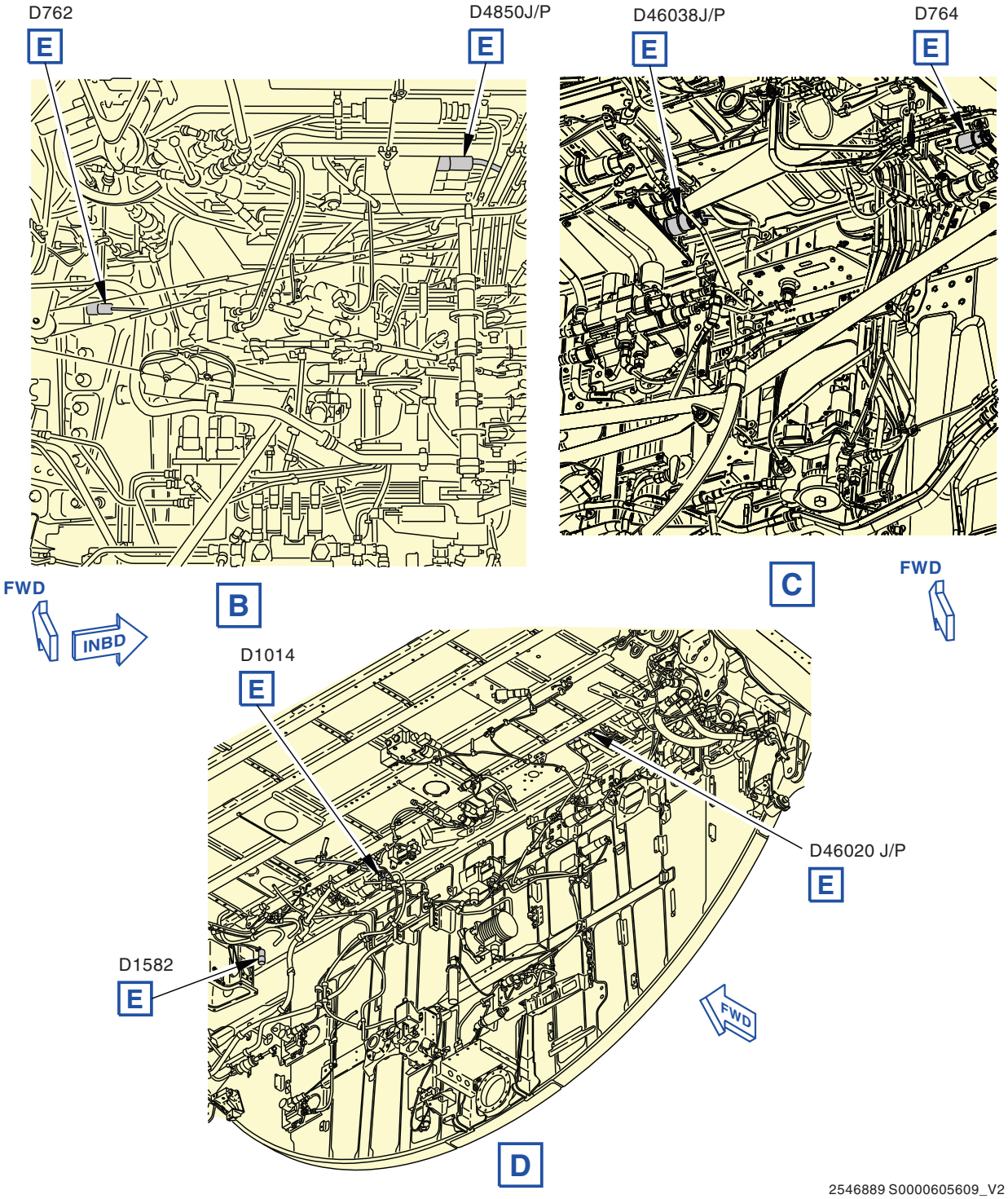
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



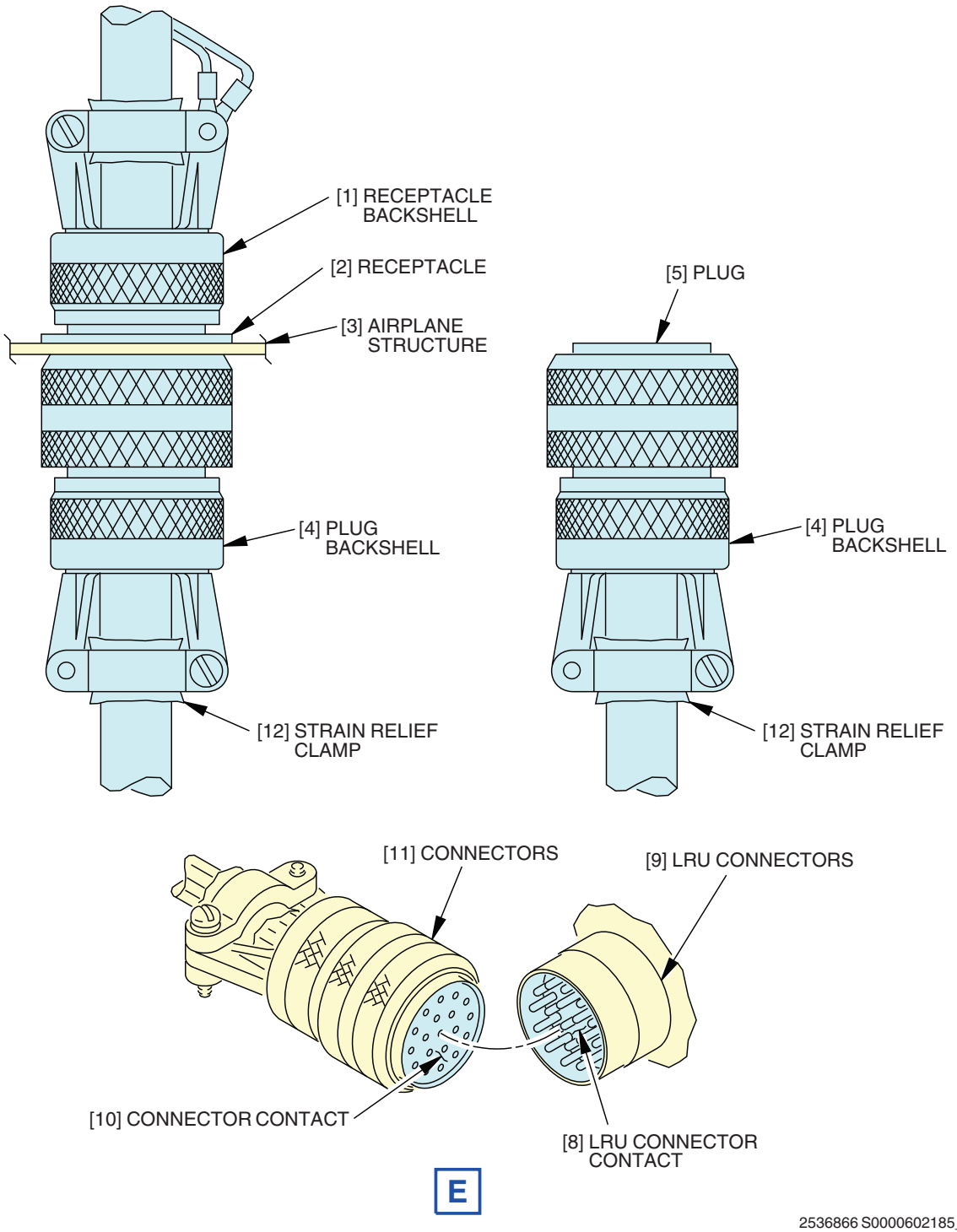
Electrical Connectors in the Main Wheel Well - Detail Inspection
Figure 601/05-51-96-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electrical Connectors in the Main Wheel Well - Detail Inspection
Figure 601/05-51-96-990-801 (Sheet 2 of 3)

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Electrical Connectors in the Main Wheel Well - Detail Inspection
Figure 601/05-51-96-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

External Doors Bonding Jumper

1. General

A. This procedure provides guidance information for an aircraft with missing or degraded bonding jumpers.

TASK 05-51-97-200-801

2. Bonding Jumper - Inspection/Check

A. References

Reference	Title
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

B. Tools/Equipment

Reference	Description
STD-1231	Multimeter - Standard

C. Location Zones

Zone	Area
110	Subzone - Body Station 130 to Station 396
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
140	Subzone - Body Station 727.00 to Body Station 1016.00
191	Lower Wing-To-Body Fairing - Forward of Wing Box
621	Right Wing - Leading Edge to Front Spar
711	Nose Landing Gear Door - Left
712	Nose Landing Gear Door - Right
820	Doors - Lower Half of Fuselage - Right Side

D. Access Panels

Number	Name/Location
114AR	External Power Receptacle Door
145AL	Waste Service Door
146AR	Water Service Door
191E	Low Pressure ECS Panel - Forward
621GB	Refuel Valve Door
711AL	Forward Nose Wheel Door
712AR	Forward Nose Wheel Door

E. Degraded Bond Jumpers Damage Limits

SUBTASK 05-51-97-210-001

(1) The following instructions pertain to these exterior doors.

External Doors

Number	Name/Location
114AR	External Power Receptacle Door
145AL	Waste Service Door
146AR	Water Service Door
191E	Low Pressure ECS Panel - Forward
621GB	Refuel Valve Door
711AL	Forward Nose Wheel Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
712AR	Forward Nose Wheel Door

SUBTASK 05-51-97-210-002

(2) Do this check when the P-Static bonding jumpers are found to be broken, missing, or damaged:

(a) Inspect the door for multiple P-Static bonding jumpers.

1) When one or more bonding jumpers are found to be in good condition, it is acceptable to defer the repair of the broken, missing, or damaged bonding jumpers to the next maintenance interval.

(b) When there are no bonding jumpers in good condition, measure the resistance of the doors in the closed position using a multimeter, STD-1231.

NOTE: The Nose Landing Gear Doors measurements will be made in the open position (measure resistance from moveable hinge to primary structure).

1) If the door to structure resistance is less than 1 Mega Ohm, then it is acceptable to defer the repair or replacement of the bonding jumpers to the next maintenance interval.

2) If the door to structure resistance is more than 1 Mega Ohm, repair or replace the broken, missing, or damaged bonding jumpers (Refer to SWPM 20-20-10 for repair).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NITROGEN GENERATION SYSTEM (NGS) OXYGEN ENRICHED AIR (OEA) EXHAUST PORT FUEL LEAK CONDITION - MAINTENANCE PRACTICES (CONDITIONAL INSPECTION)

1. General

A. This procedure has this task:

(1) A fuel leak conditional inspection of the nitrogen generation system.

TASK 05-51-98-210-801

2. Nitrogen Generation System (NGS) Fuel Leak Conditional Inspection

(Figure 201, Figure 202, and Figure 203)

A. General

(1) This task includes the steps to inspect the airplane after fuel is found leaking from the Nitrogen Generation System (NGS) Air Separation Module (ASM) Oxygen Enriched Air (OEA) exhaust port.

B. References

Reference	Title
28-11-31-400-801	Center Tank Access Door - Installation (P/B 401)
47-00-00-790-801	Leak Check of the Nitrogen Generation System (P/B 601)
47-00-00-800-801	Nitrogen Generation System Ground Operation (P/B 201)
47-11-01-000-801	Air Separation Module (ASM) - Removal (P/B 401)
47-11-01-420-801	Air Separation Module (ASM) Installation (P/B 401)
47-11-02-000-801	High Flow Valve Removal (P/B 401)
47-11-02-420-801	High Flow Valve Installation (P/B 401)
47-11-04-000-801	Overtemperature Shutoff Valve Removal (P/B 401)
47-11-04-400-801	Overtemperature Shutoff Valve Installation (P/B 401)
47-21-04-000-801	Primary Backflow Prevention Check Valve Removal (P/B 401)
47-21-04-000-802	Secondary Backflow Prevention Check Valve Removal (P/B 401)
47-21-04-400-801	Primary Backflow Prevention Check Valve Installation (P/B 401)
47-21-04-400-802	Secondary Backflow Prevention Check Valve Installation (P/B 401)
47-32-01-000-801	NGS Shutoff Valve Removal (P/B 401)
47-32-01-400-801	NGS Shutoff Valve Installation (P/B 401)
47-32-02-000-801	Ozone Converter Removal (P/B 401)
47-32-02-400-801	Ozone Converter Installation (P/B 401)
47-32-03-000-801	Heat Exchanger Removal (P/B 401)
47-32-03-400-801	Heat Exchanger Installation (P/B 401)
47-32-04-000-801	NGS Filter Assembly - Removal (P/B 401)
47-32-04-000-802	NGS Filter Element - Removal (P/B 401)
47-32-04-400-801	NGS Filter Assembly - Installation (P/B 401)
47-32-04-400-802	NGS Filter Element - Installation (P/B 401)
47-42-02-020-801	High Flow Valve Differential Pressure Sensor Removal (P/B 401)
47-42-02-400-801	High Flow Valve Differential Pressure Sensor Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
47-42-03-020-801	NGS Oxygen Sensor - Removal (P/B 401)
47-42-03-420-801	NGS Oxygen Sensor - Installation (P/B 401)
47-43-01-000-801	Filter Differential Pressure Switch Removal (P/B 401)
47-43-01-400-801	Filter Differential Pressure Switch Installation (P/B 401)
47-43-02-000-801	Thermal Switch Removal (P/B 401)
47-43-02-400-801	Thermal Switch Installation (P/B 401)
47-43-03-000-801	Temperature Sensor Removal (P/B 401)
47-43-03-400-801	Temperature Sensor Installation (P/B 401)
47-43-04-000-801	NGS Pressure Sensor Removal (P/B 401)
47-43-04-400-801	NGS Pressure Sensor Installation (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SOPM 20-30-03	General Cleaning Procedures

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
531AB	Center Tank Access Door - Wing Station 168

E. Nitrogen Generation System (NGS) Fuel Leak Conditional Inspection

SUBTASK 05-51-98-860-002

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	17	C01657	NITROGEN GENERATION CONTROL
E	15	C01680	NGS ALT PWR

SUBTASK 05-51-98-210-001

(2) If there is fuel leakage at the NGS ASM OEA exhaust port, do these steps:

NOTE: It is not necessary to do the installation/operational test after each component installation, as this will be done at the end of this task.

(a) Examine all NGS ducting and tubing downstream of the ASM [1] for fuel contamination (smell or visual).

1) Clean and dry all contaminated ducting and tubing downstream of the ASM [1] (with insulation blankets removed) per SOPM 20-30-03, Immersion Cleaning.

(b) Remove and install a new primary backflow prevention primmary check valve [11] (TASK 47-21-04-000-801 and TASK 47-21-04-400-801).

(c) Remove and install a new secondary backflow prevention secondary check valve [10] (TASK 47-21-04-000-802 and TASK 47-21-04-400-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Examine these Line Replaceable Unit (LRU)s for fuel contamination (smell or visual).

1) If there is contamination, do these steps:

a) Remove and install a new high flow valve differential pressure sensor [2] (TASK 47-42-02-020-801 and TASK 47-42-02-400-801).

b) Remove and install a new NGS oxygen sensor [3] (TASK 47-42-03-020-801 and TASK 47-42-03-420-801).

NOTE: It is not necessary to do drain cap fuel leak check, when you install the new NGS oxygen sensor.

(e) Remove and install a new ASM [1] (TASK 47-11-01-000-801 and TASK 47-11-01-420-801).

(f) Remove and install a new high flow valve (TASK 47-11-02-000-801 and TASK 47-11-02-420-801).

SUBTASK 05-51-98-210-002

(3) If ducting/tubing/LRUs upstream of the ASM [1] are contaminated, do these steps:

(a) Clean and dry all contaminated ducting and tubing between the heat exchanger and ASM [1] (with insulation blankets removed) per SOPM 20-30-03, Immersion Cleaning.

(b) Examine these LRUs for fuel contamination (smell or visual).

1) If there is contamination, do these steps:

NOTE: It is not necessary to do the installation/operational test after each component installation, as this will be done at the end of this task.

a) Remove and install a new overtemperature shutoff valve [4] (TASK 47-11-04-000-801 and TASK 47-11-04-400-801).

b) Remove and install a new thermal switch [5] (TASK 47-43-02-000-801 and TASK 47-43-02-400-801).

c) Remove and install a new temperature sensor [6] (TASK 47-43-03-000-801 and TASK 47-43-03-400-801).

d) Remove and install a new filter differenial pressure switch [7] (TASK 47-43-01-000-801 and TASK 47-43-01-400-801).

e) Do the following steps for the NGS filter assembly [8]:

<1> Do this task: NGS Filter Assembly - Removal, TASK 47-32-04-000-801.

<2> Do this task: NGS Filter Element - Removal, TASK 47-32-04-000-802.

<3> Clean and dry the filter housing: SOPM 20-30-03, Immersion Cleaning.

<4> Install a new NGS filter element (TASK 47-32-04-400-802).

<5> Do this task: NGS Filter Assembly - Installation, TASK 47-32-04-400-801.

f) Do the following steps for the heat exchanger [9]:

<1> Do this task: Heat Exchanger Removal, TASK 47-32-03-000-801.

<2> Clean the heat exchanger [9]: SOPM 20-30-03, Immersion Cleaning.

<3> Examine the upstream of the heat exchanger [9] for fuel contamination (smell or visual).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<a> If there is contamination in the upstream of the heat exchanger [9], do this task: Heat Exchanger Installation, TASK 47-32-03-400-801 and continue to the steps below related to contamination upstream of the heat exchanger [9].

<b> If there is no contamination in the upstream of the heat exchanger [9], do this task: Heat Exchanger Installation, TASK 47-32-03-400-801 and continue to the step below that closes the Circuit Breakers.

SUBTASK 05-51-98-210-003

(4) If ducting/tubing/LRUs upstream of the heat exchanger [9] are contaminated, do these steps:

(a) Clean and dry all contaminated ducting and tubing upstream of the heat exchanger [9] (with insulation blankets removed): SOPM 20-30-03, Immersion Cleaning.

(b) Examine these LRUs for fuel contamination (smell or visual).

1) If there is contamination, do these steps:

NOTE: It is not necessary to do the installation/operational test after each component installation, as this will be done at the end of this task.

a) Remove and install a new ozone converter [12] (TASK 47-32-02-000-801 and TASK 47-32-02-400-801).

b) Remove and install a new NGS shutoff valve [13] (TASK 47-32-01-000-801 and TASK 47-32-01-400-801).

c) Remove and install a new NGS pressure sensor [14] (TASK 47-43-04-000-801 and TASK 47-43-04-400-801).

SUBTASK 05-51-98-860-001

(5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	17	C01657	NITROGEN GENERATION CONTROL
E	15	C01680	NGS ALT PWR

SUBTASK 05-51-98-790-001

(6) Do this task: Leak Check of the Nitrogen Generation System, TASK 47-00-00-790-801.

(a) Repair the leaks that you find.

(b) Make sure that there are no BITE messages.

SUBTASK 05-51-98-710-001

(7) Do this task: Nitrogen Generation System Ground Operation, TASK 47-00-00-800-801.

(a) Make sure that the electrical test is good.

(b) Make sure that the oxygen sensor operates correctly.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 05-51-98-410-001

(1) Install this access panel, if applicable:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-51-98-410-002

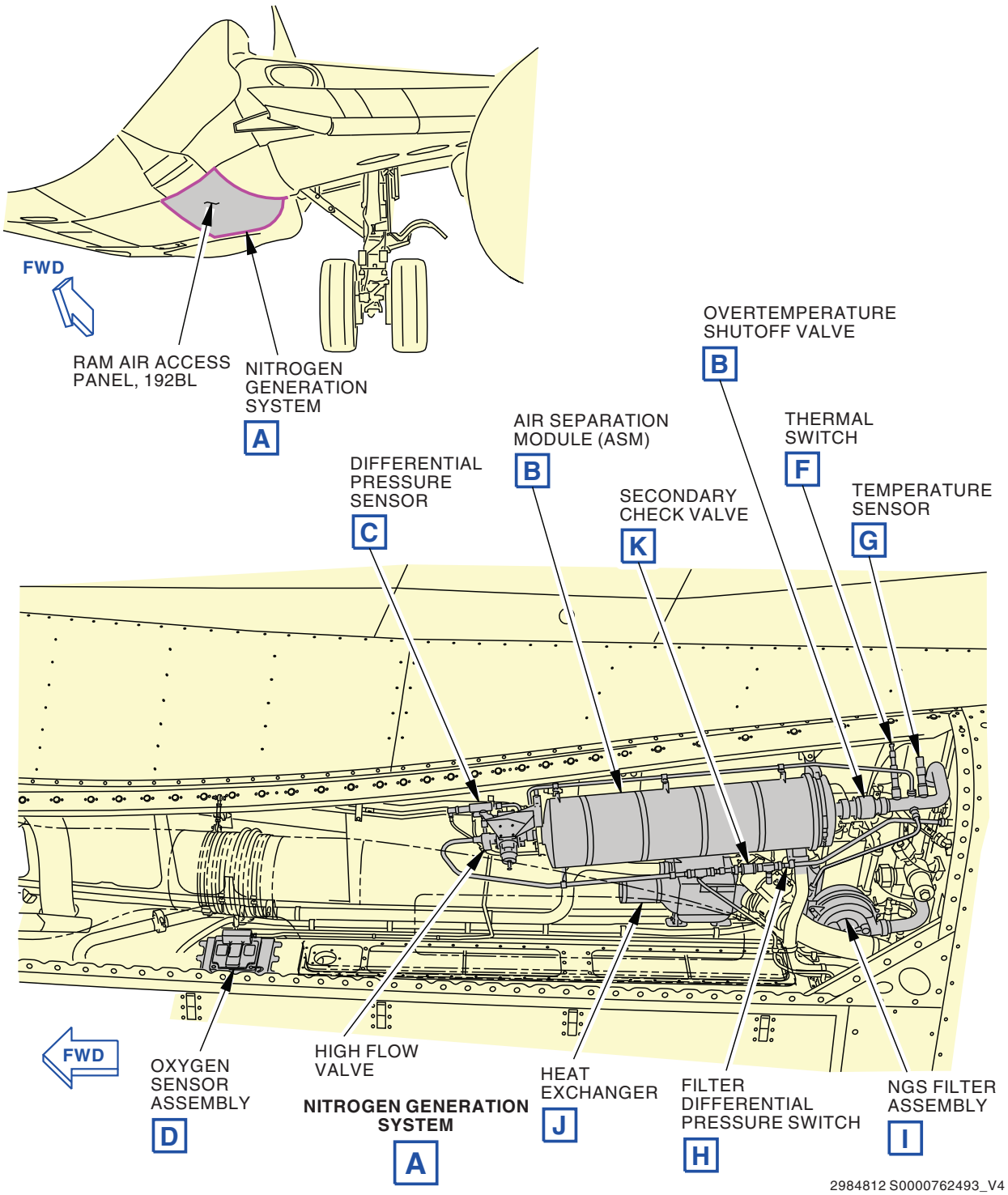
(2) Close the center tank access door, if applicable:
(TASK 28-11-31-400-801)

(a) Close this access panel:

Number	Name/Location
531AB	Center Tank Access Door - Wing Station 168

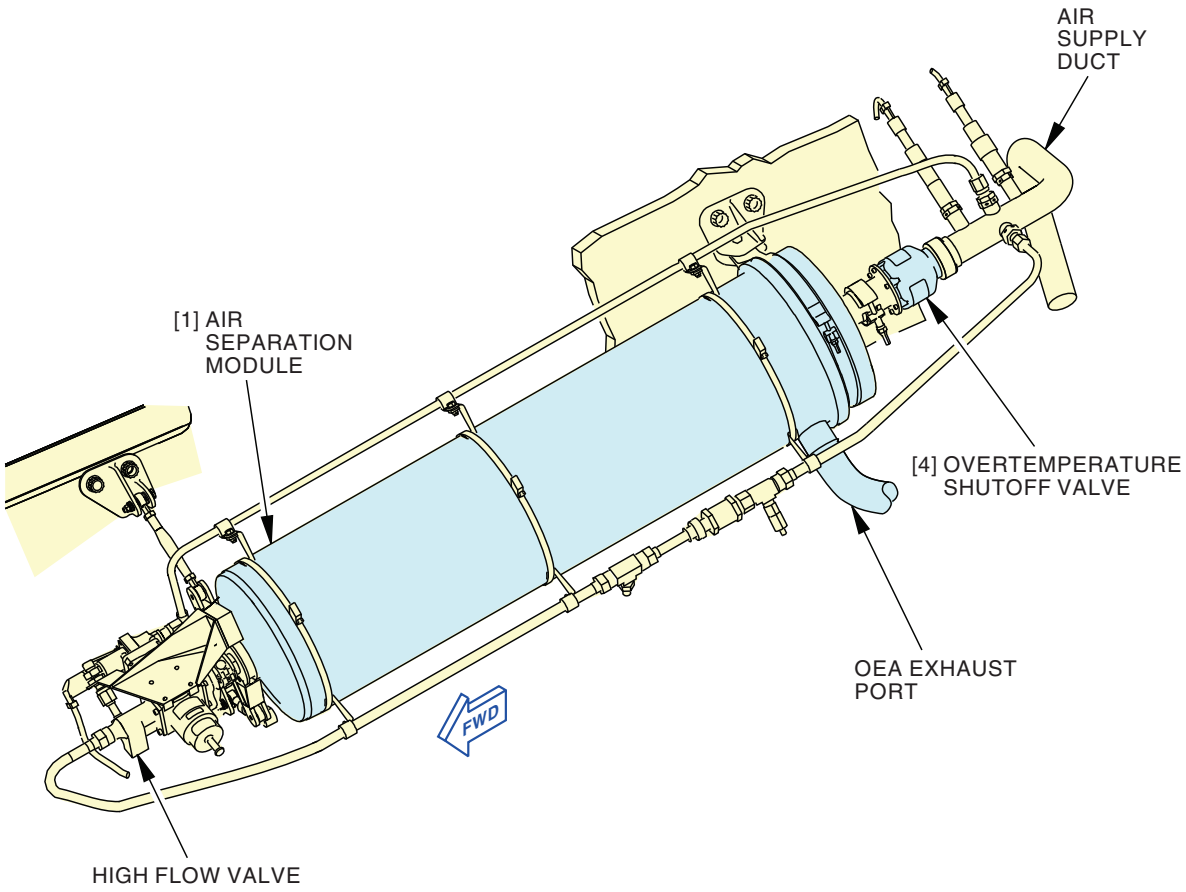
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 1 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ASM AND OVERTEMPERATURE SHUTOFF VALVE (EXAMPLE)

B

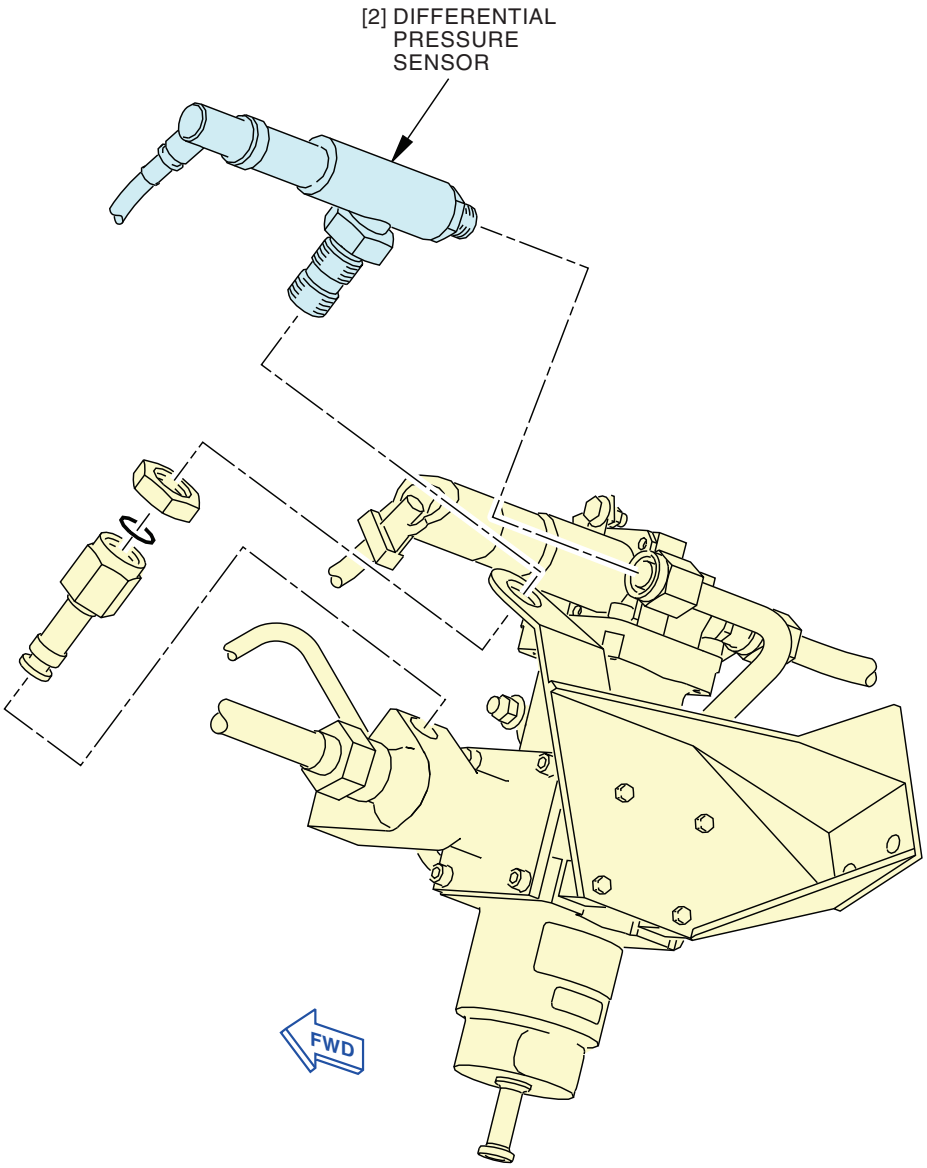
NOTE:

THE AIR SEPARATION MODULE IS SHOWN WITHOUT THE INSULATION.

2984816 S0000762799_V4

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 2 of 10)

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**DIFFERENTIAL PRESSURE SENSOR
(EXAMPLE)**

C

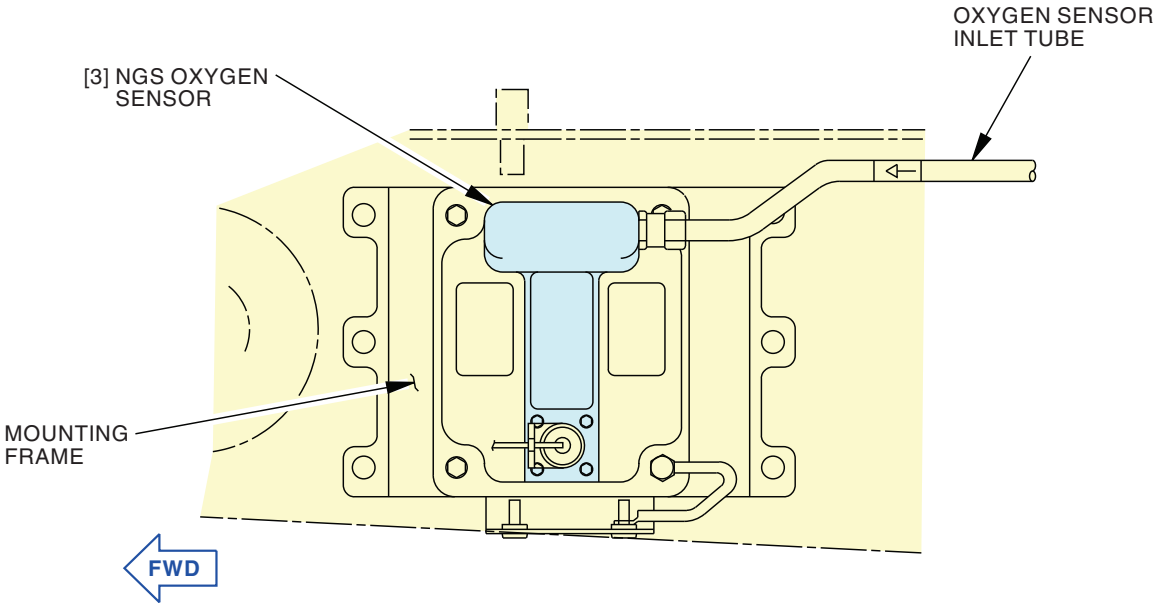
NOTE:

INSULATION NOT SHOWN FOR CLARITY

3001516 S0000777466_V1

**Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 3 of 10)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



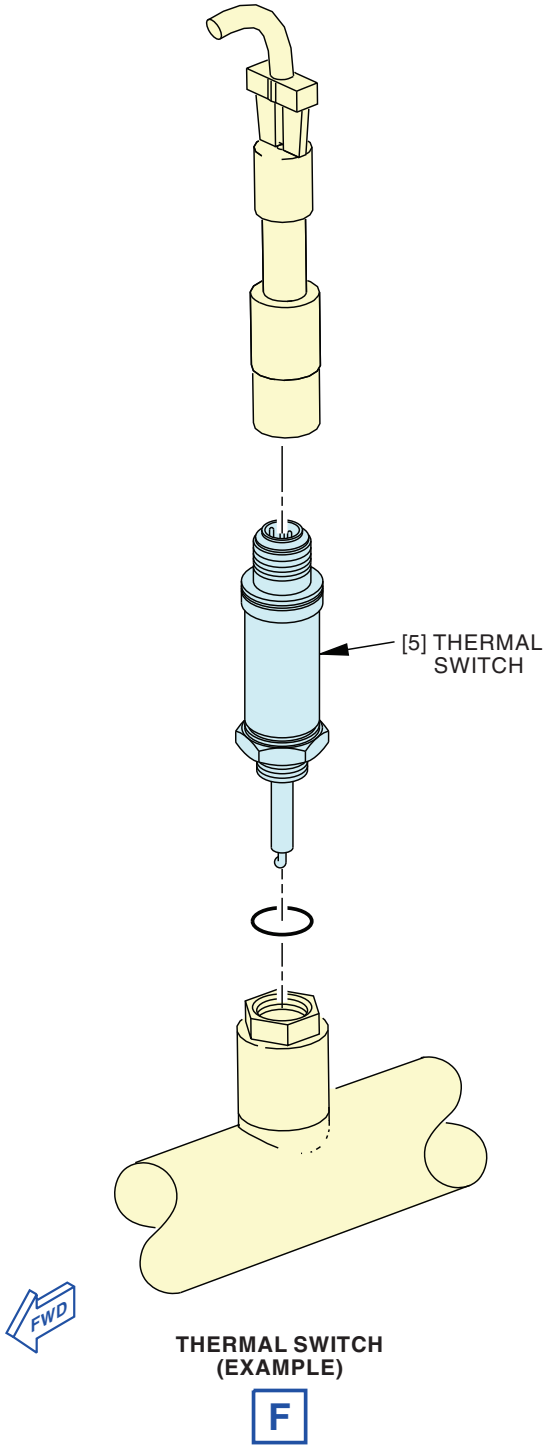
**OXYGEN SENSOR ASSEMBLY
(EXAMPLE)
(VIEW IN THE INBOARD DIRECTION)**

D

3001517 S0000777468_V1

**Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 4 of 10)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



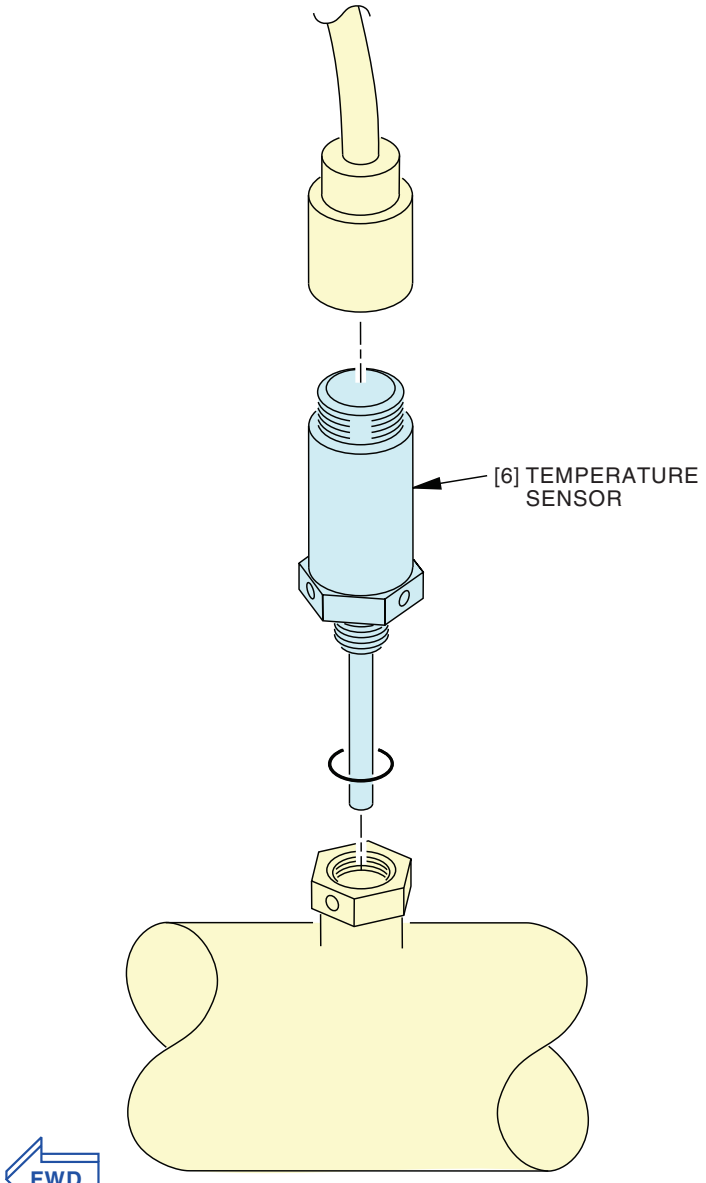
THERMAL SWITCH
(EXAMPLE)

F

3001520 S0000777469_V1

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 5 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



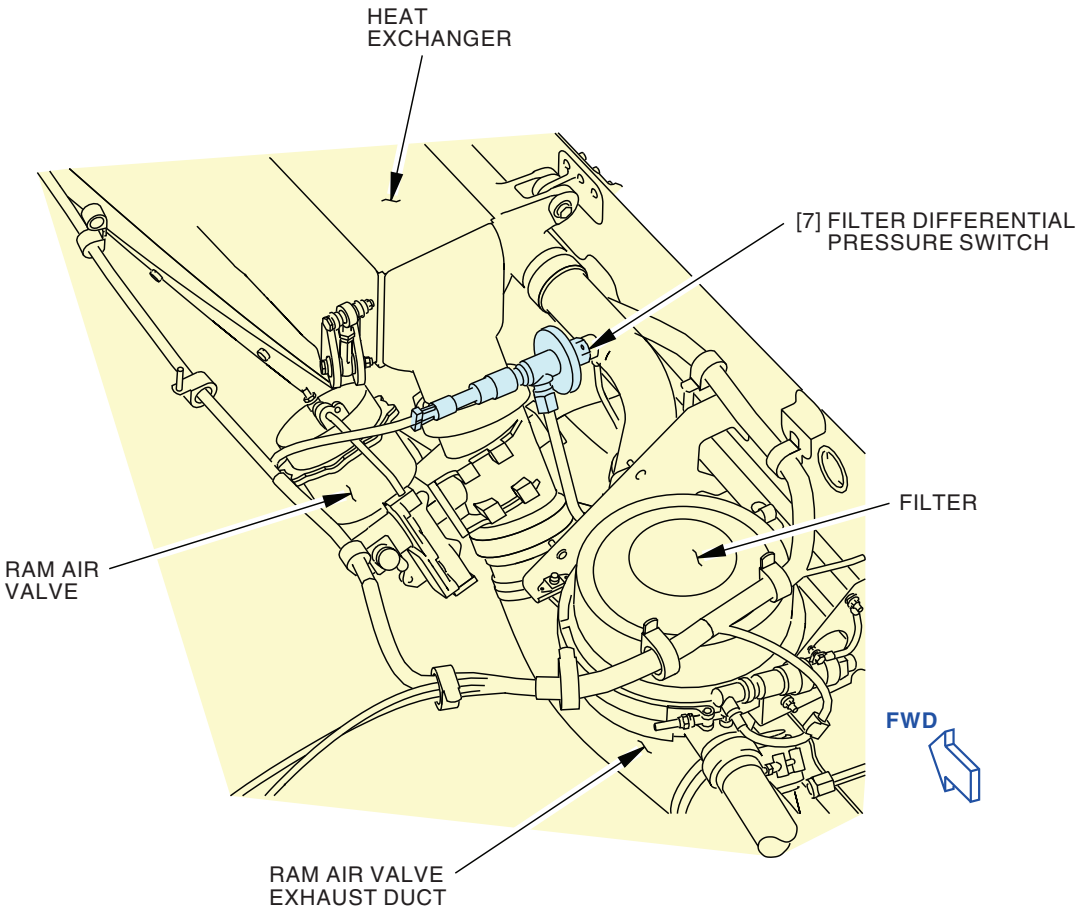
TEMPERATURE SENSOR (EXAMPLE)

G

3001521 S0000777470_V1

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



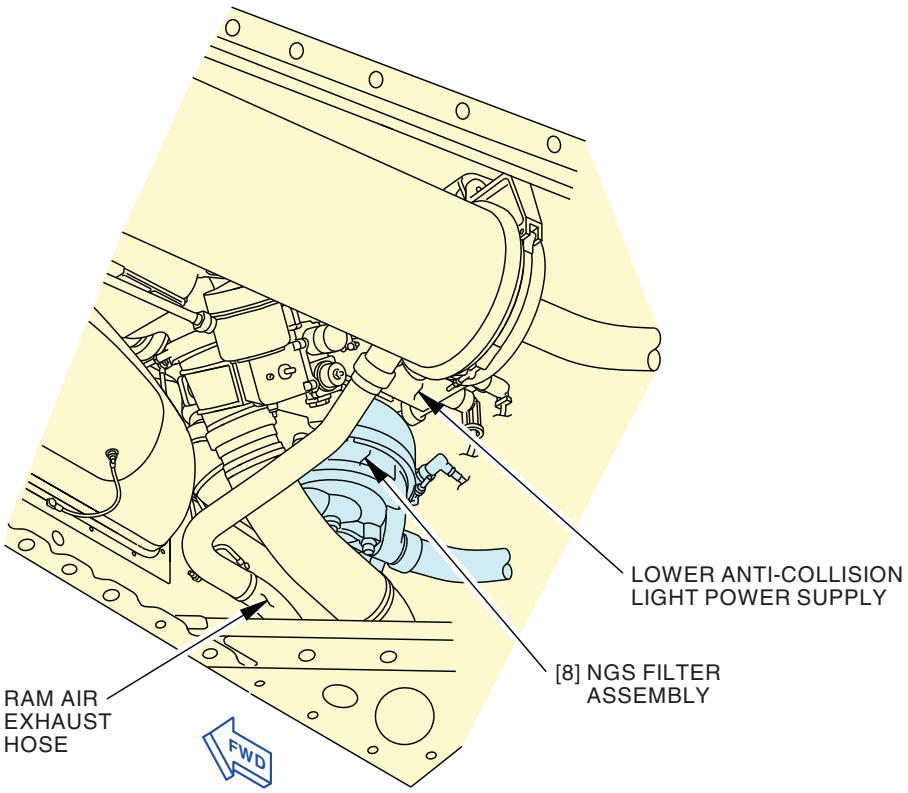
FILTER DIFFERENTIAL PRESSURE SWITCH (EXAMPLE)

H

3001522 S0000777471_V1

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



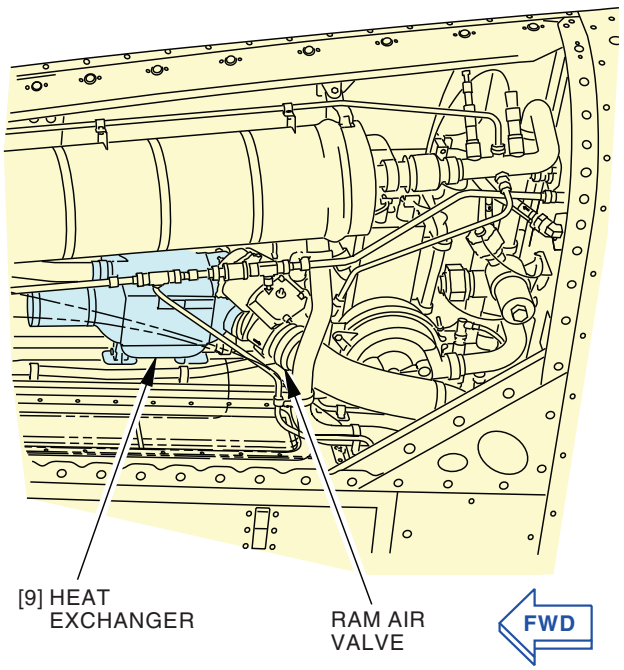
**NGS FILTER ASSEMBLY
(EXAMPLE)**

I

3001523 S0000777472_V1

**Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 8 of 10)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



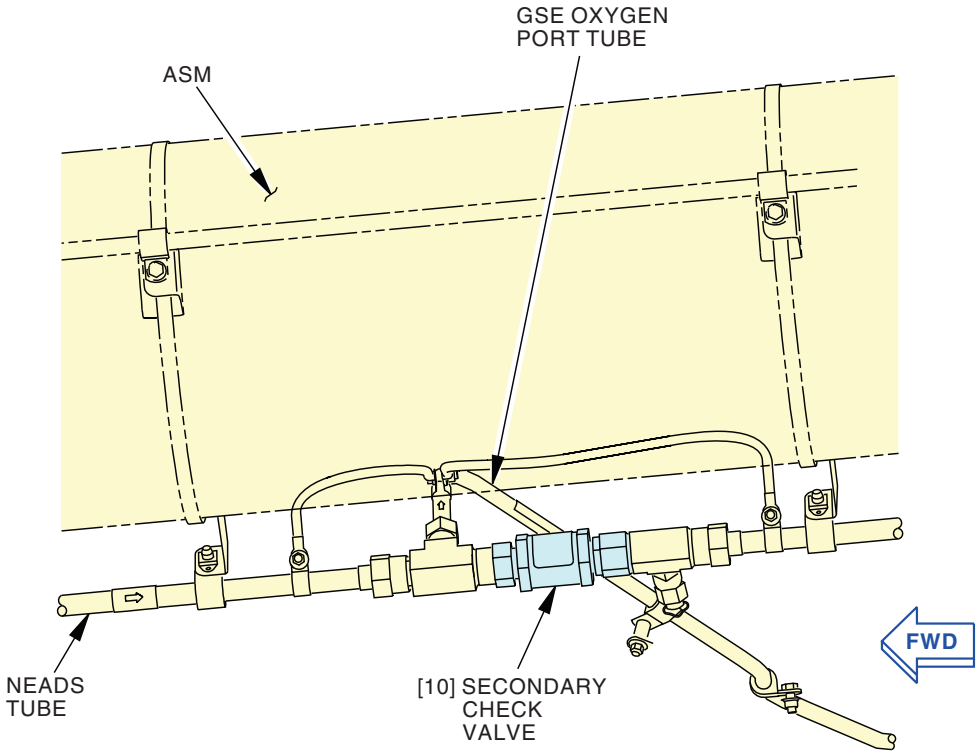
HEAT EXCHANGER
(EXAMPLE)
(VIEW IN THE INBOARD DIRECTION)

J

3001524 S0000777473_V1

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 9 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



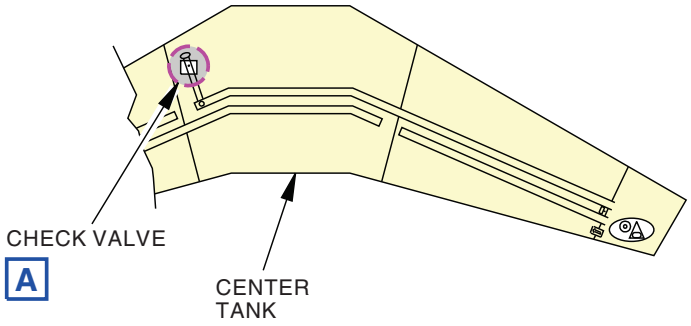
SECONDARY CHECK VALVE

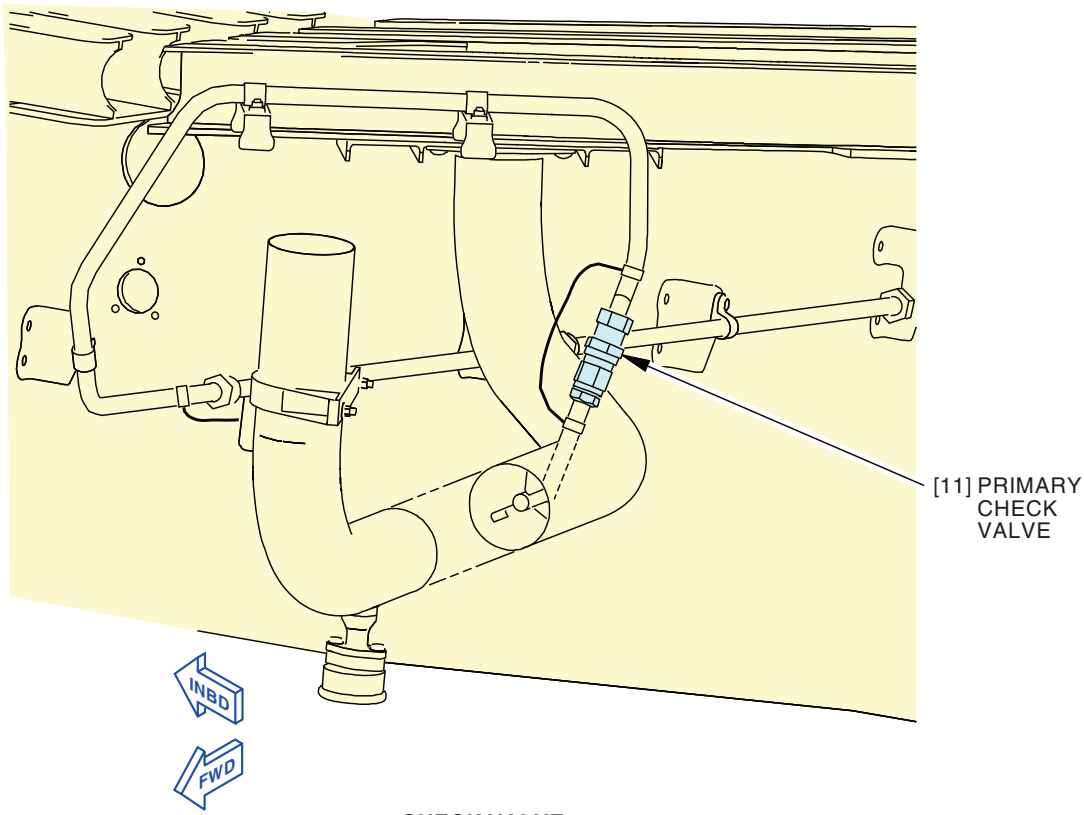
K

3001526 S0000777474_V1

Nitrogen Generation System Fuel Leak Conditional Inspection
Figure 201/05-51-98-990-801 (Sheet 10 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





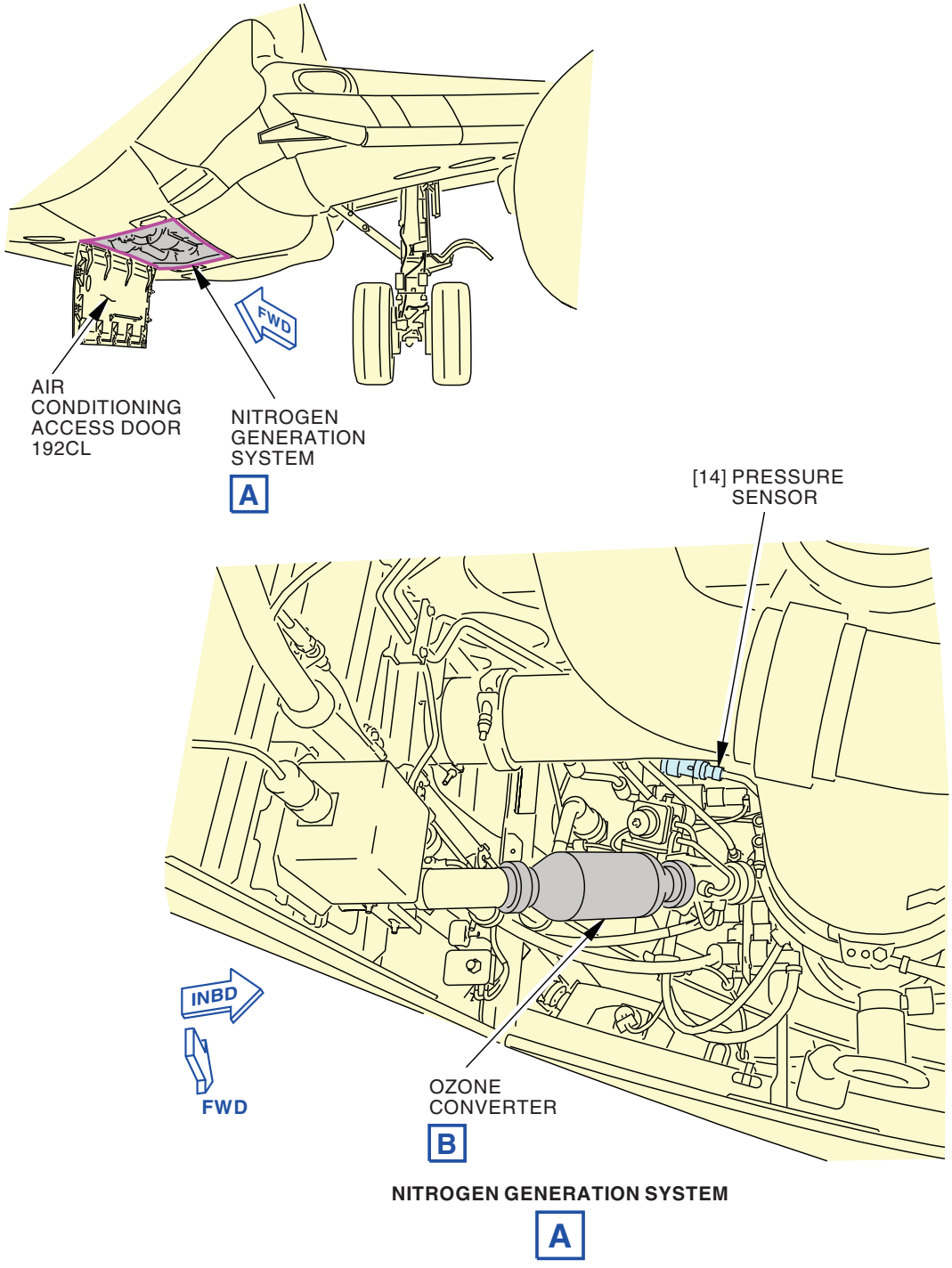
CHECK VALVE
(EXAMPLE)

A

3001531 S0000777475_V1

Nitrogen Generation System Fuel Leak Conditional Inspection - Primary Check Valve
Figure 202/05-51-98-990-802

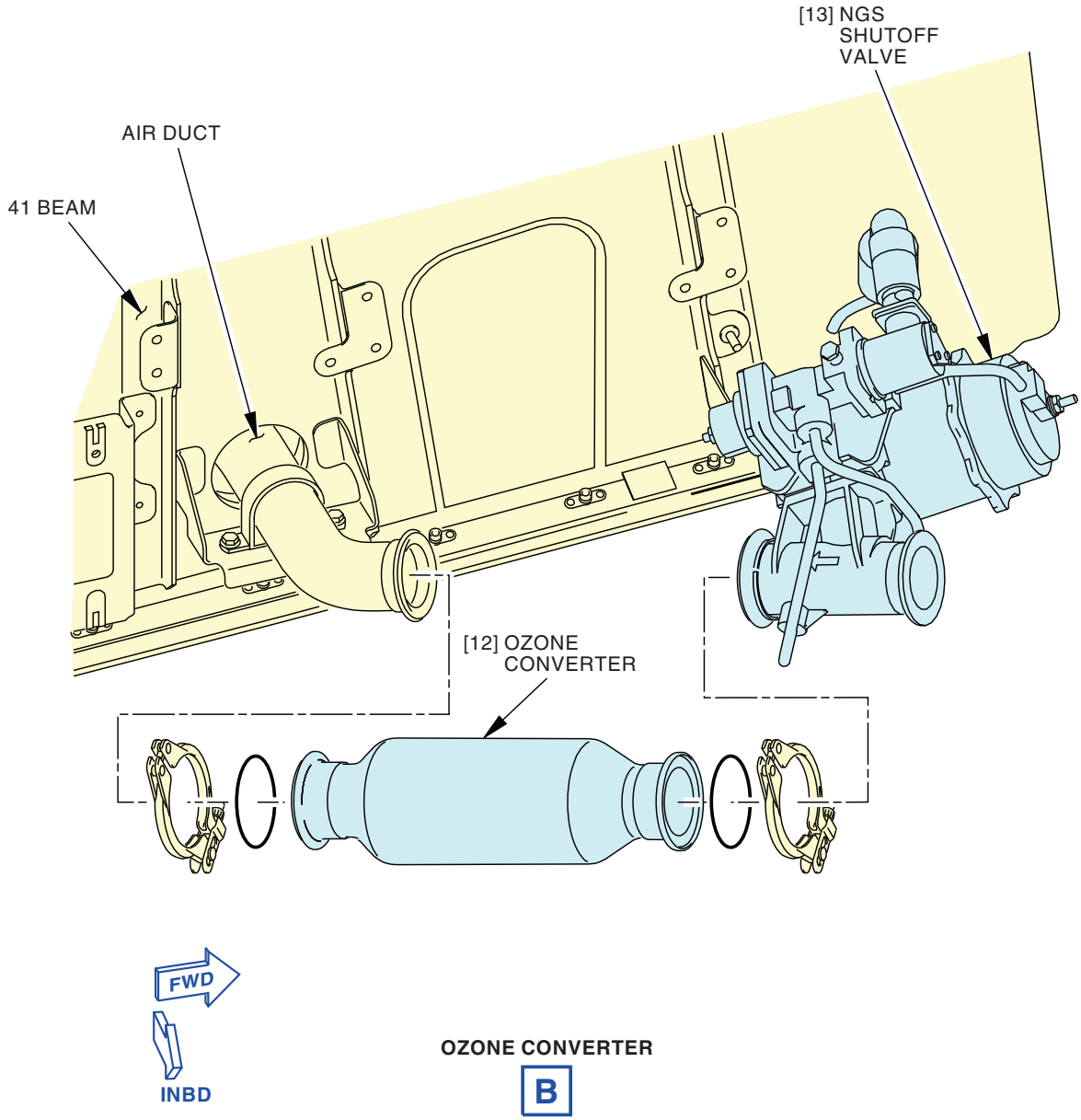
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3001533 S0000777476_V1

Nitrogen Generation System Fuel Leak Conditional Inspection - Ozone Converter and Pressure Sensor
Figure 203/05-51-98-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3001535 S0000777477_V1

Nitrogen Generation System Fuel Leak Conditional Inspection - Ozone Converter and Pressure Sensor
Figure 203/05-51-98-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

IN-FLIGHT WINDMILLING - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Inspection of the engine after in-flight windmilling.

TASK 05-51-99-200-801

2. Inspection of the Engine After In-Flight Windmilling

A. General

(1) This task gives the inspection of the engine after an in-flight windmilling condition.

B. References

Reference	Title
71-00-00-200-807-G00	In-Flight Windmilling Inspection (P/B 601)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Inspection of the Engine After In-Flight Windmilling

SUBTASK 05-51-99-200-001

(1) Do this task: In-Flight Windmilling Inspection, TASK 71-00-00-200-807-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FOIL-LINED PANELS ON LEFT SIDE OF THE AIRPLANE - INSPECTION/CHECK

1. General

A. This procedure contains tasks that perform inspections of foil-lined panels on the left side of the airplane.

(1) Aluminum Adhesive Foil Laminate or Aluminum Foil Tape is applied on the inner most layer (IML) of composite panels and used to protect underlying systems from the indirect effects of lightning.

TASK 05-55-21-210-801

2. General Visual Inspection - Left Wing Leading Edge (MRB 20-155-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar

B. Access Panels

Number	Name/Location
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

C. Prepare for the procedure

SUBTASK 05-55-21-010-001

(1) Open these access panels:

Number	Name/Location
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

D. Procedure

SUBTASK 05-55-21-210-001

(1) Perform a general visual inspection of the foil-lined panels shown in Figure 601.

SUBTASK 05-55-21-300-001

(2) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-21-410-001

(1) Close these access panels:

Number	Name/Location
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel

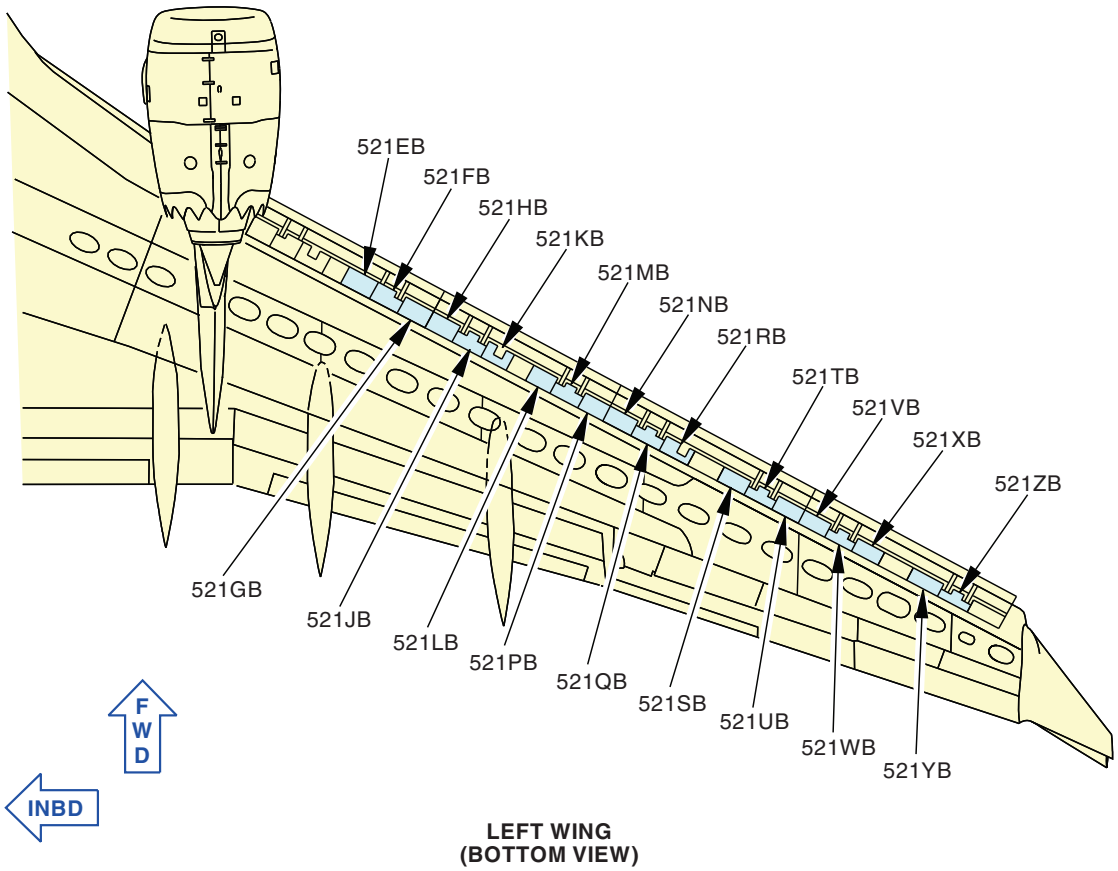
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(BOTTOM VIEW)

2532132 S0000598980_V2

LH Wing Lower Surface (Foil-Lined Panels)
Figure 601/05-55-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-21-211-802

3. Detailed Inspection - Left Wing Leading Edge (MRB 20-169-00)

(Figure 602, Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar

B. Access Panels

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel

C. Prepare for the procedure

SUBTASK 05-55-21-010-002

(1) Open these access panels:

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel

D. Procedure

SUBTASK 05-55-21-211-001

(1) Perform a detailed inspection of the foil-lined panels shown in Figure 602 and Figure 603.

SUBTASK 05-55-21-300-002

(2) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

E. Put the Airplane Back to Its Usual Condition

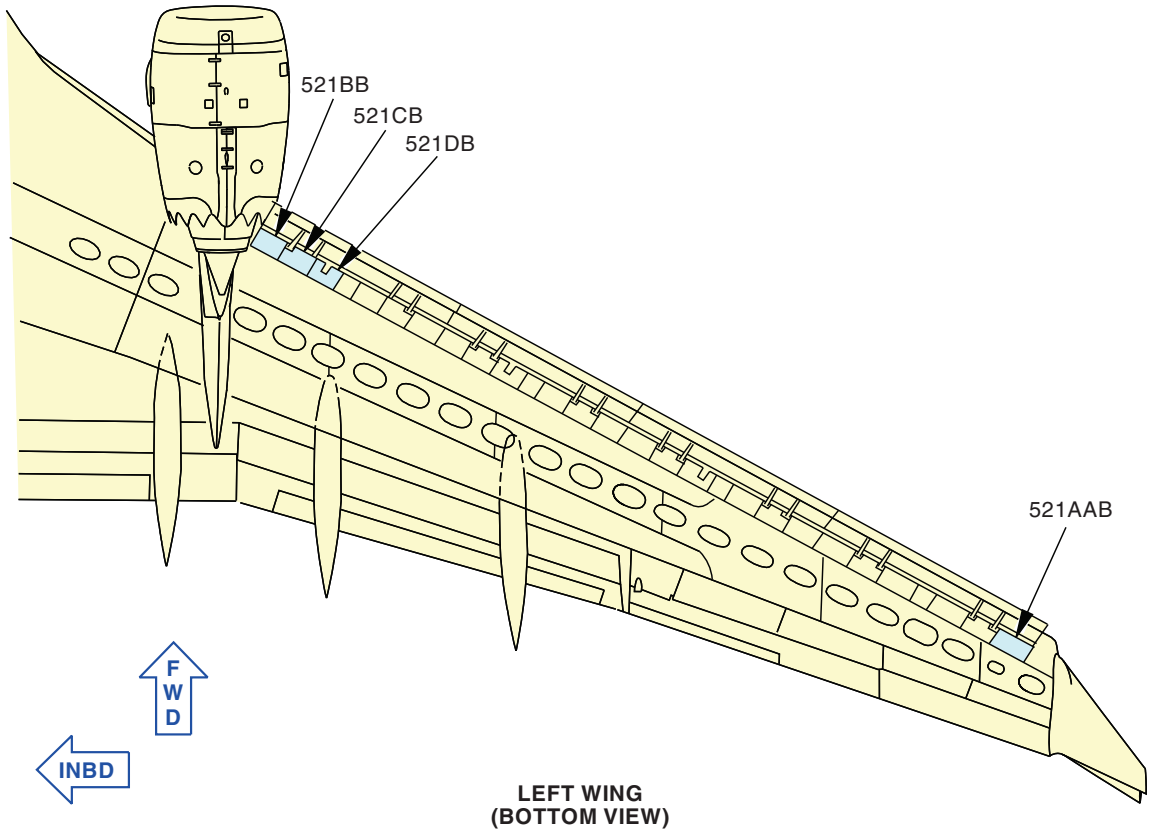
SUBTASK 05-55-21-410-002

(1) Close these access panels:

Number	Name/Location
511AB	Inboard Leading Edge, Lower Removable Panel
511AT	Inboard Leading Edge, Strakelet Upper Panel
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

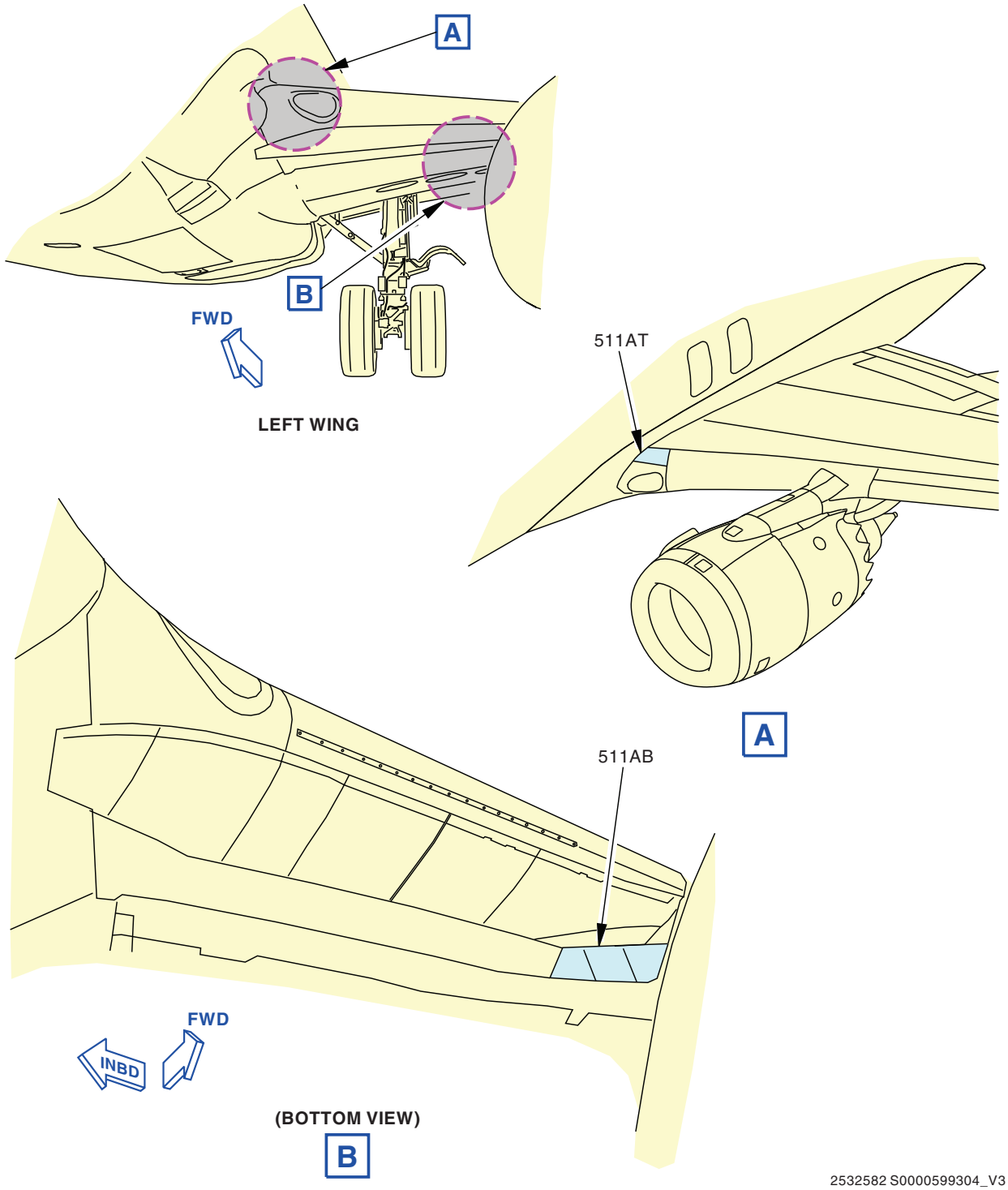


LEFT WING
(BOTTOM VIEW)

2532163 S0000599299_V3

LH Wing Lower Surface (Foil-Lined Panels)
Figure 602/05-55-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LH Wing Fixed Leading Edge Removable Panels
Figure 603/05-55-21-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-21-211-804

4. Detailed Inspection - Left Strut Fairings (MRB 20-171-00)

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
54-52-01-000-801	Forward Fairings Inspection (P/B 601)

B. Location Zones

Zone	Area
431	Engine 1 - Forward Strut Fairing

C. Access Panels

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431BT	Forward Strut Fairing, Pressure Relief Door, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

D. Prepare for the Inspection

SUBTASK 05-55-21-010-004

(1) Open these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431BT	Forward Strut Fairing, Pressure Relief Door, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

E. Left Strut Fairings Inspection

SUBTASK 05-55-21-211-003

(1) Perform a detailed inspection of the foil-lined panels shown in Figure 604.

SUBTASK 05-55-21-211-006

(2) Perform a detailed inspection of the foil-lined structure under panels 431BL, 431BR, 431AT, 431CL, and 431CR.

SUBTASK 05-55-21-300-004

(3) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

SUBTASK 05-55-21-350-001

(4) If the foil tape is damaged, replace the foil tape (TASK 54-52-01-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

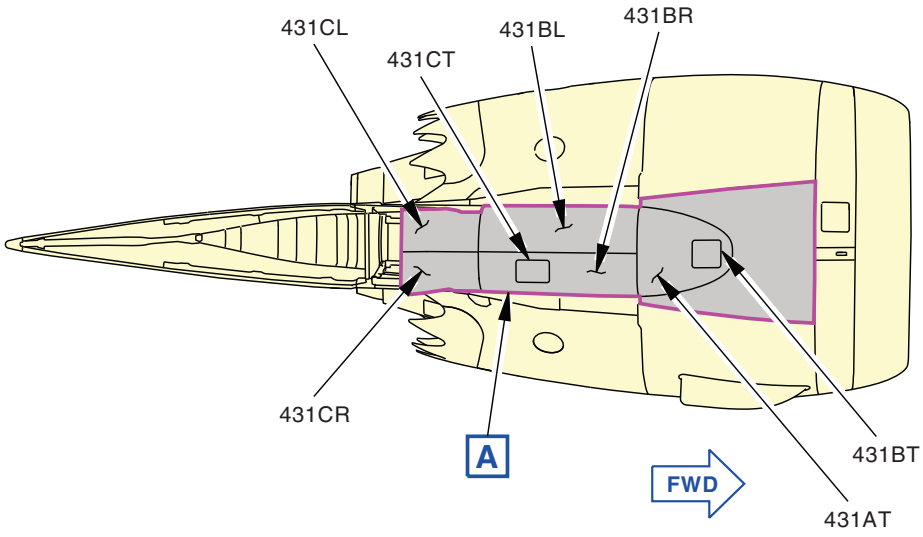
SUBTASK 05-55-21-410-004

(1) Close these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 1
431BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 1
431BT	Forward Strut Fairing, Pressure Relief Door, Strut 1
431CL	Forward Strut Fairing, Left Overwing Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
431CT	Forward Strut Fairing, Right Mid-Strut Fairing Pressure Relief Door, Strut 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

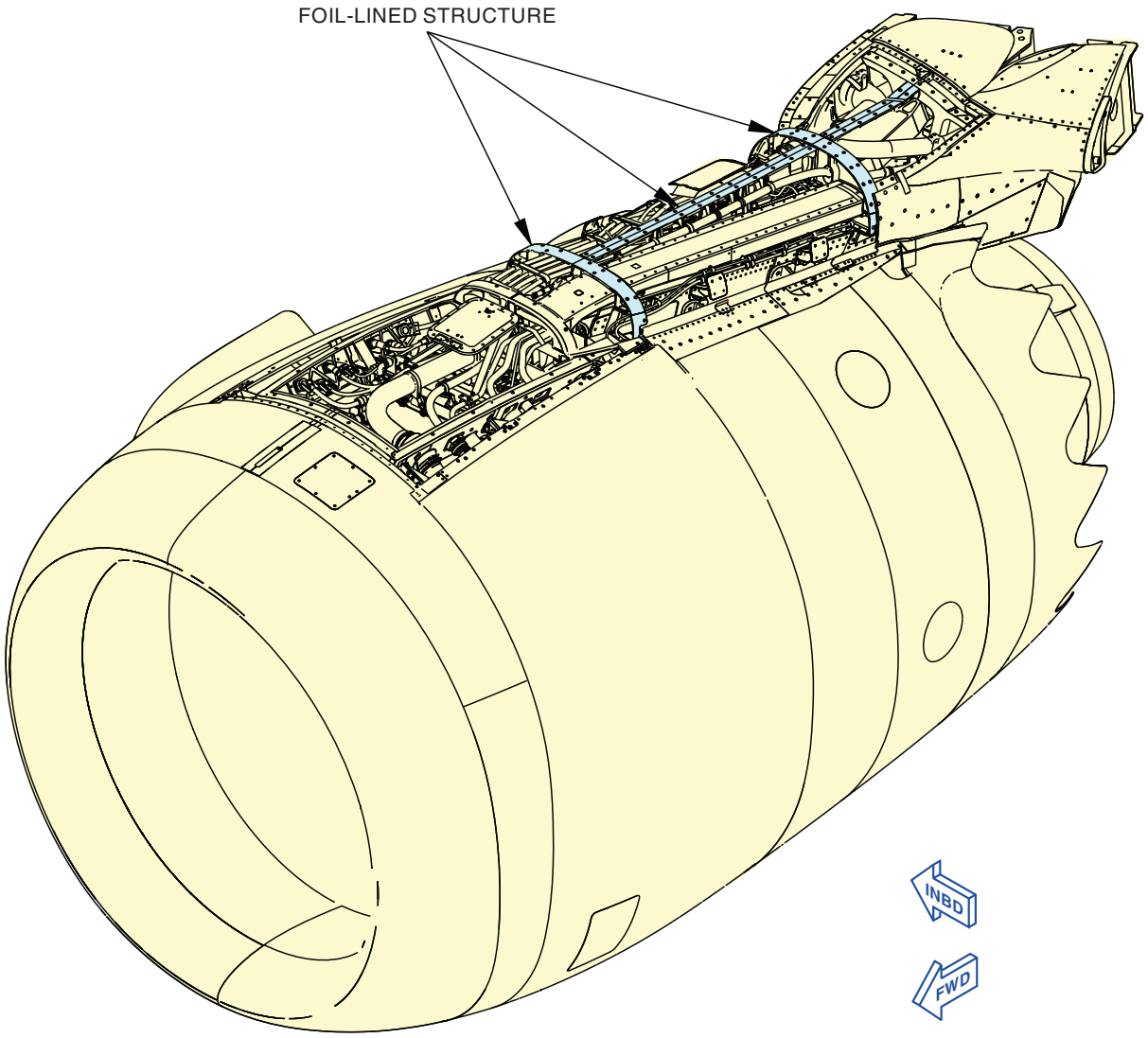


**LEFT ENGINE
(TOP VIEW)**

2532584 S0000599376_V3

**Left Mid Strut Fairing, Strut 1
Figure 604/05-55-21-990-805 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW ON THE LEFT SIDE OF ENGINE POSITION 1)

A

3059783 S0000822894_V1

Left Mid Strut Fairing, Strut 1
Figure 604/05-55-21-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FOIL-LINED PANELS ON RIGHT SIDE OF THE AIRPLANE - INSPECTION/CHECK

1. General

A. This procedure contains tasks that perform inspections of foil-lined panels on the right side of the airplane.

(1) Aluminum Adhesive Foil Laminate or Aluminum Foil Tape is applied on the inner most layer (IML) of composite panels and used to protect underlying systems from the indirect effects of lightning.

TASK 05-55-22-210-801

2. General Visual Inspection - Right Wing Leading Edge (MRB 20-155-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

B. Access Panels

Number	Name/Location
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

C. Prepare for the procedure

SUBTASK 05-55-22-010-001

(1) Open these access panels:

Number	Name/Location
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

D. Procedure

SUBTASK 05-55-22-210-001

(1) Perform a general visual inspection of the foil-lined panels (Figure 601).

SUBTASK 05-55-22-300-001

(2) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

E. Put the Airplane Back to Its Usual Condition

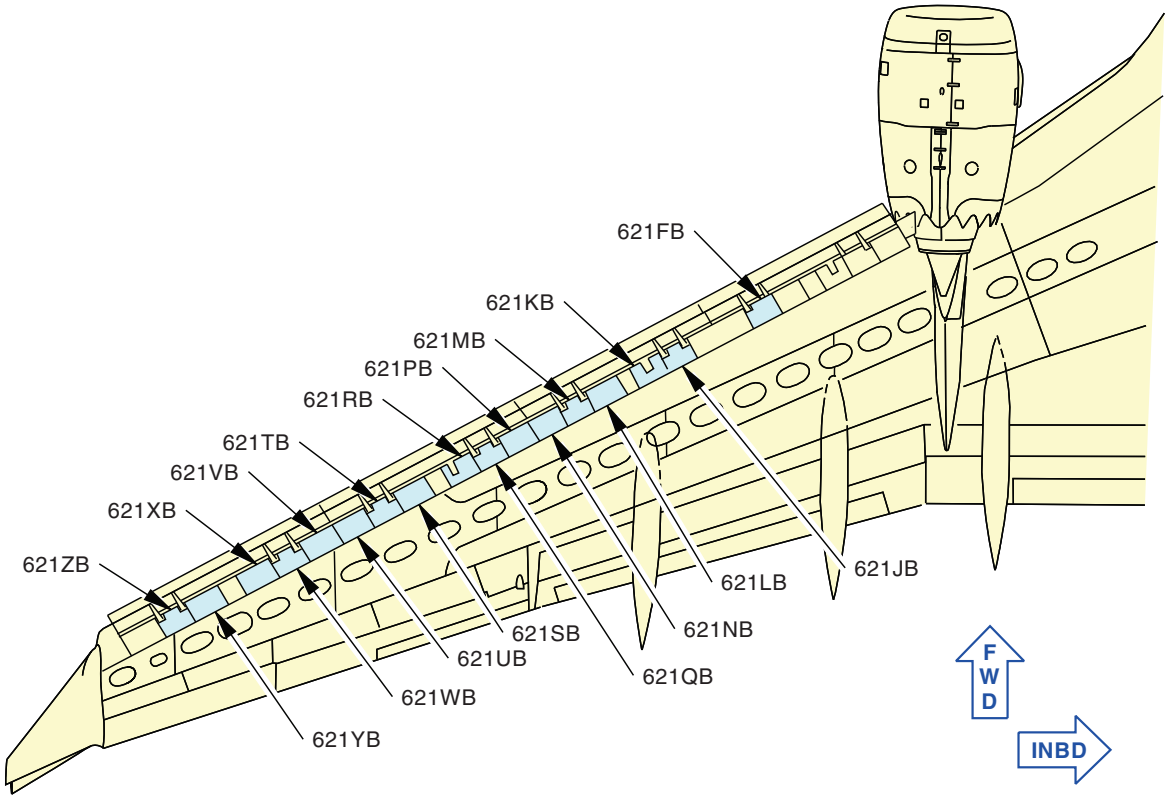
SUBTASK 05-55-22-410-001

(1) Close these access panels:

Number	Name/Location
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**RIGHT WING
(BOTTOM VIEW)**

2533069 S0000599117_V3

**RH Wing Lower Surface (Foil-Lined Panels)
Figure 601/05-55-22-990-801**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-22-211-801

3. Detailed Inspection - Right Wing Leading Edge (MRB 20-168-00)

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

B. Access Panels

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door

C. Prepare for the procedure

SUBTASK 05-55-22-010-002

(1) Open these access panels:

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door

D. Procedure

SUBTASK 05-55-22-211-001

(1) Perform a detailed inspection of the foil-lined panels shown in Figure 602.

SUBTASK 05-55-22-300-002

(2) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

E. Put the Airplane Back to Its Usual Condition

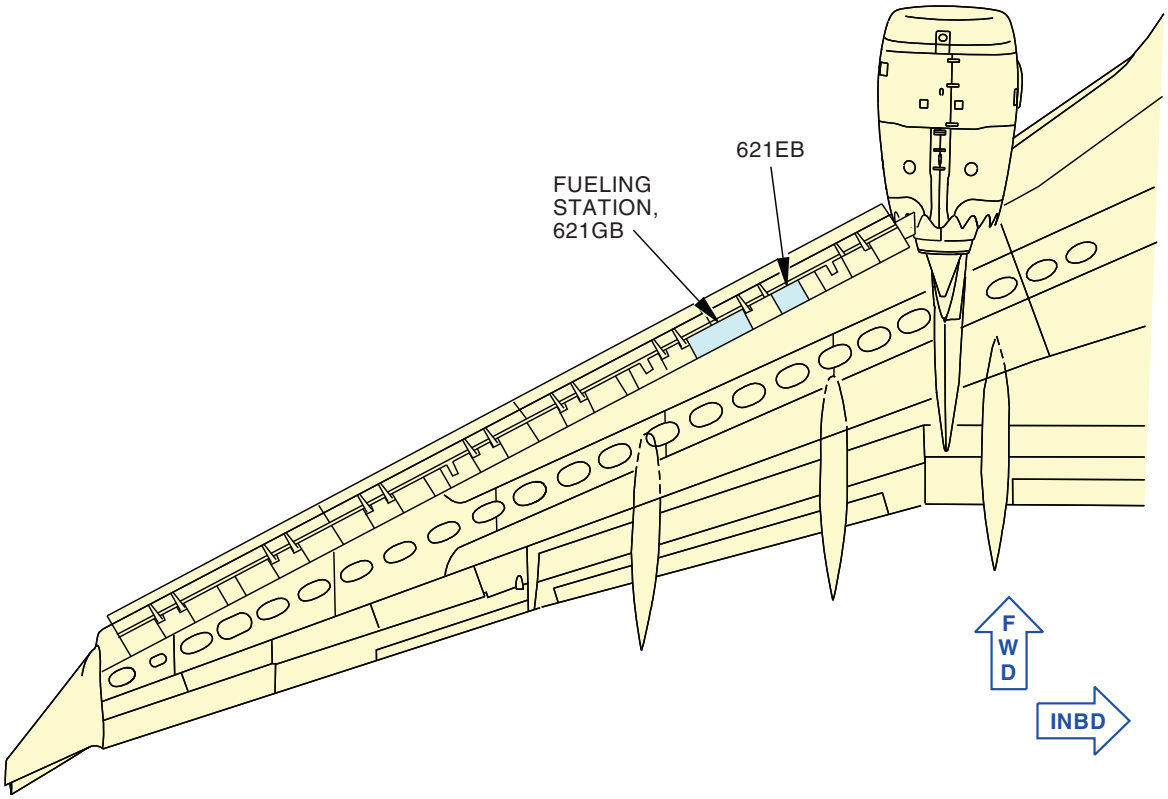
SUBTASK 05-55-22-410-002

(1) Close these access panels:

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**RIGHT WING
(BOTTOM VIEW)**

2533070 S0000599298_V2

**RH Wing Lower Surface (Foil-Lined Panels at Defuel/Refuel Stations)
Figure 602/05-55-22-990-802**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-22-211-802

4. Detailed Inspection - Right Wing Leading Edge (MRB 20-169-00)

(Figure 603, Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

B. Access Panels

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel

C. Prepare for the procedure

SUBTASK 05-55-22-010-003

(1) Open these access panels:

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel

D. Procedure

SUBTASK 05-55-22-211-002

(1) Perform a detailed inspection of the foil-lined panels (Figure 603 and Figure 604).

SUBTASK 05-55-22-300-003

(2) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

E. Put the Airplane Back to Its Usual Condition

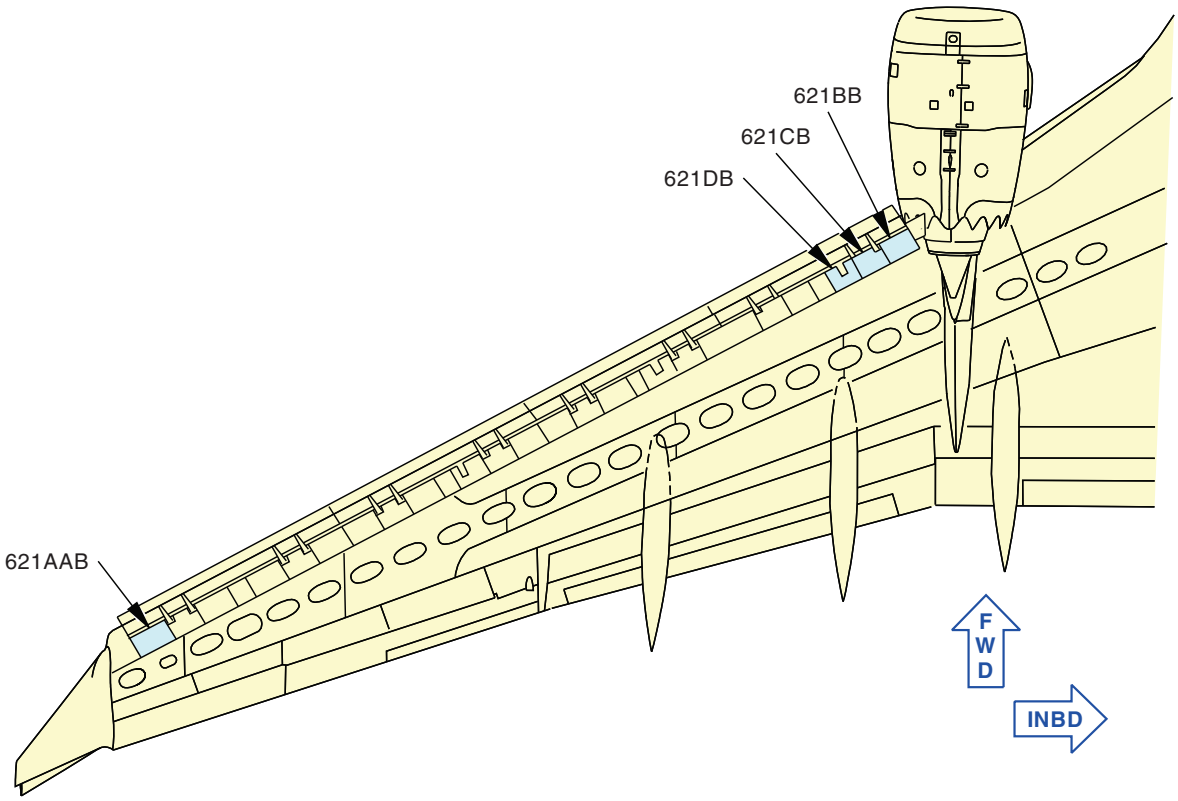
SUBTASK 05-55-22-410-003

(1) Close these access panels:

Number	Name/Location
611AB	Inboard Leading Edge, Lower Removable Access Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

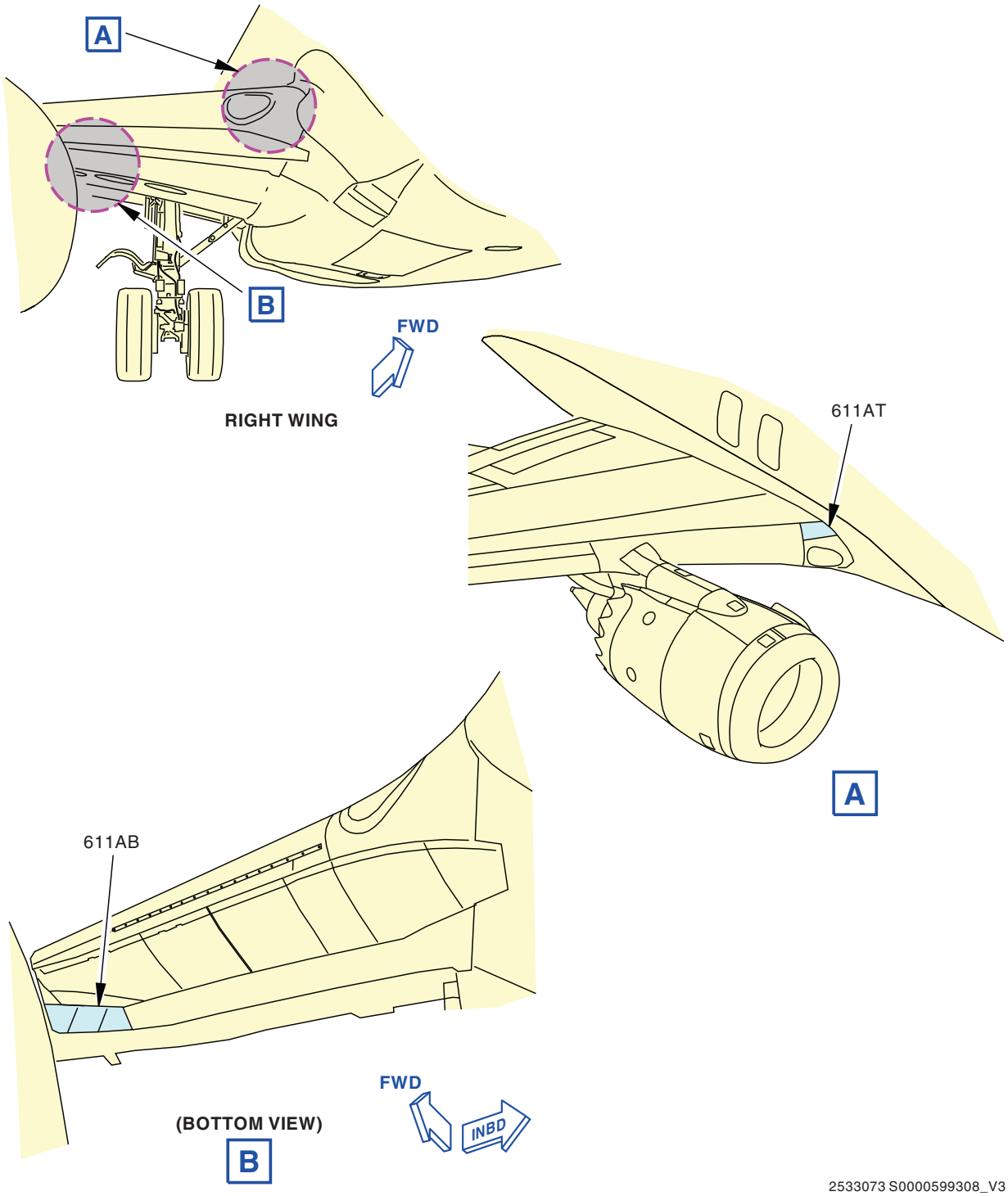


**RIGHT WING
(BOTTOM VIEW)**

2533071 S0000599300_V4

**RH Wing Lower Surface (Foil-Lined Panels)
Figure 603/05-55-22-990-803**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RH Wing Fixed Leading Edge Removable Panels
Figure 604/05-55-22-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-22-211-804

5. Detailed Inspection - Right Strut Fairings (MRB 20-171-00)

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
54-52-01-000-801	Forward Fairings Inspection (P/B 601)

B. Location Zones

Zone	Area
441	Engine 2 - Forward Strut Fairing

C. Access Panels

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441BT	Forward Strut Fairing, Pressure Relief Door, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

D. Prepare for the Inspection

SUBTASK 05-55-22-010-005

(1) Open these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441BT	Forward Strut Fairing, Pressure Relief Door, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

E. Right Strut Fairings Inspection

SUBTASK 05-55-22-211-004

(1) Do a detailed inspection of the foil-lined panel (Figure 605).

SUBTASK 05-55-22-211-007

(2) Do a detailed inspection of the foil-lined structure under panels 441BL, 441BR, 441AT, 441CL, and 441CR.

SUBTASK 05-55-22-300-005

(3) If damage is found, refer to Structures With Aluminum Coatings and Foils/SRM PROCEDURE 51-70-14-0G-0.

SUBTASK 05-55-22-350-001

(4) (4) If the foil tape is damaged, replace the foil tape (TASK 54-52-01-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

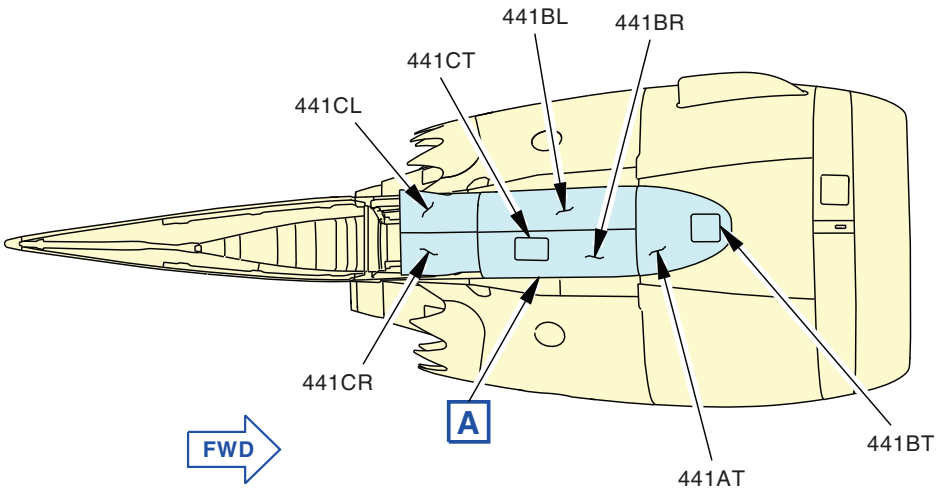
SUBTASK 05-55-22-410-005

(1) Close these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441BL	Forward Strut Fairing, Left Mid Strut Fairing, Strut 2
441BR	Forward Strut Fairing, Right Mid Strut Fairing, Strut 2
441BT	Forward Strut Fairing, Pressure Relief Door, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
441CR	Forward Strut Fairing, Right Overwing Fairing, Strut 2
441CT	Forward Strut Fairing, Forward Fuel/Hydraulic Tube Shroud Access, Strut 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

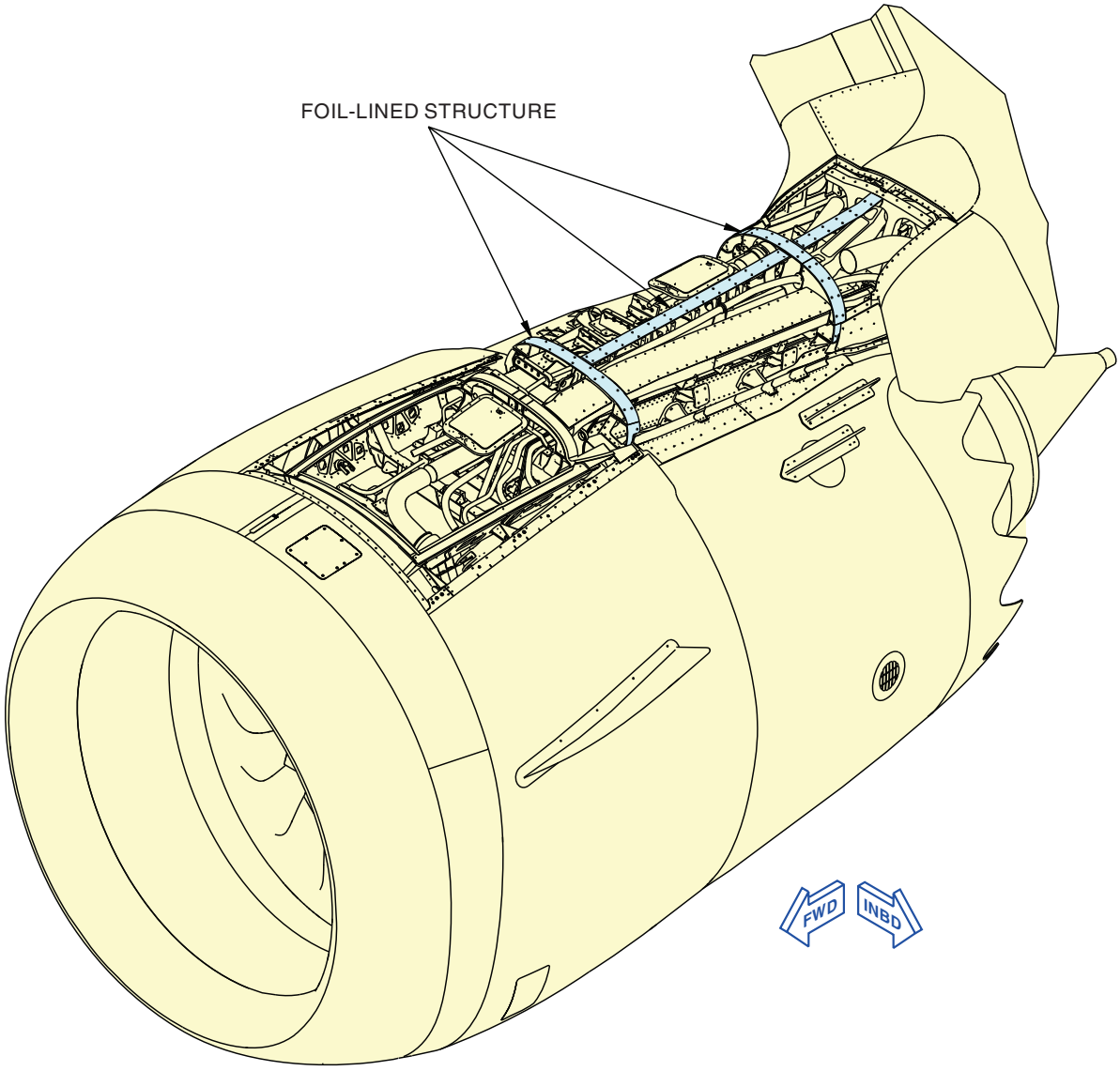


**RIGHT ENGINE
(TOP VIEW)**

2533076 S0000599383_V4

**Right Mid Strut Faring, Strut 2
Figure 605/05-55-22-990-806 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW ON THE LEFT SIDE OF ENGINE POSITION 2)

A

3056628 S0000820387_V1

Right Mid Strut Faring, Strut 2
Figure 605/05-55-22-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WIRE BUNDLES ON LEFT SIDE OF THE AIRPLANE - INSPECTION/CHECK

1. General

A. This procedure contains tasks that perform a loop resistance test of wire bundles on left side of the airplane.

TASK 05-55-25-280-801

2. Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-150-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-003

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-860-002

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-015

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 601.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 601).

Table 601/05-55-25-993-842 Wire Bundle Loop Resistance Values - Left MLG Wheel Well (MRB 20-150-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1022	D42102P		52/123		
W7230	D46028P		15/36		
W7522	D46030P		13/30		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 601).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 601 were tested and passed.

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SUBTASK 05-55-25-810-008

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-001

(3) The connectors listed in Table 602 are used for the steps above.

Table 602/05-55-25-993-814 Left MLG Wheel Well Connectors (MRB 20–150–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1022	D42102P	22–11–11, 22–12–31	AL0720A

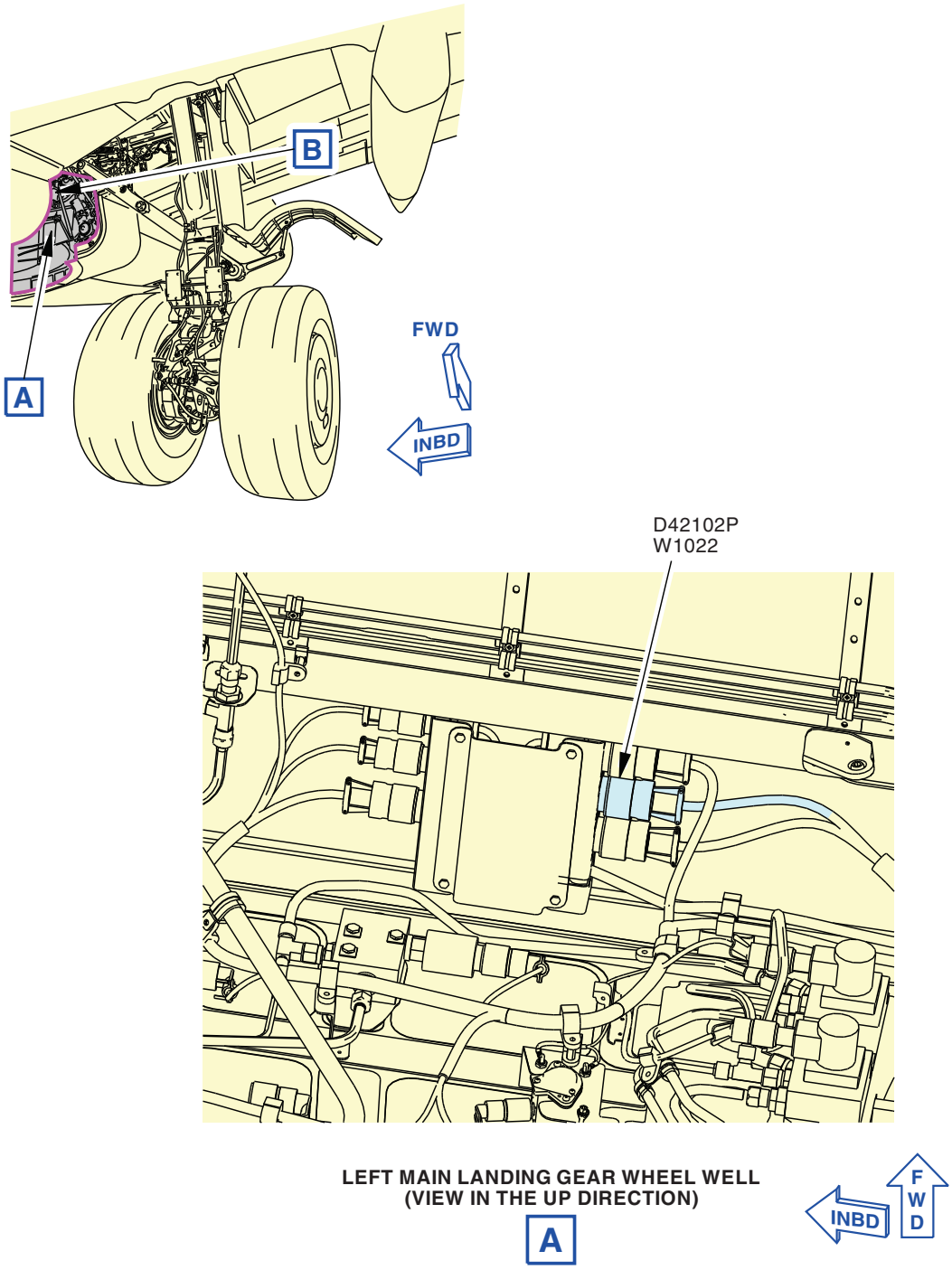
ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 602/05-55-25-993-814 Left MLG Wheel Well Connectors (MRB 20–150–00) (Continued)

Wire Bundle	Connector	WDM	Mate With Equipment
W7230	D46028P	27–61–49, 28-42-11, 29–32–11, 32–31–11, 32–41–11, 32–64–11	AL0675B
W7522	D46030P	29–11–11, 31–65–14, 78–34–11	AL0675B

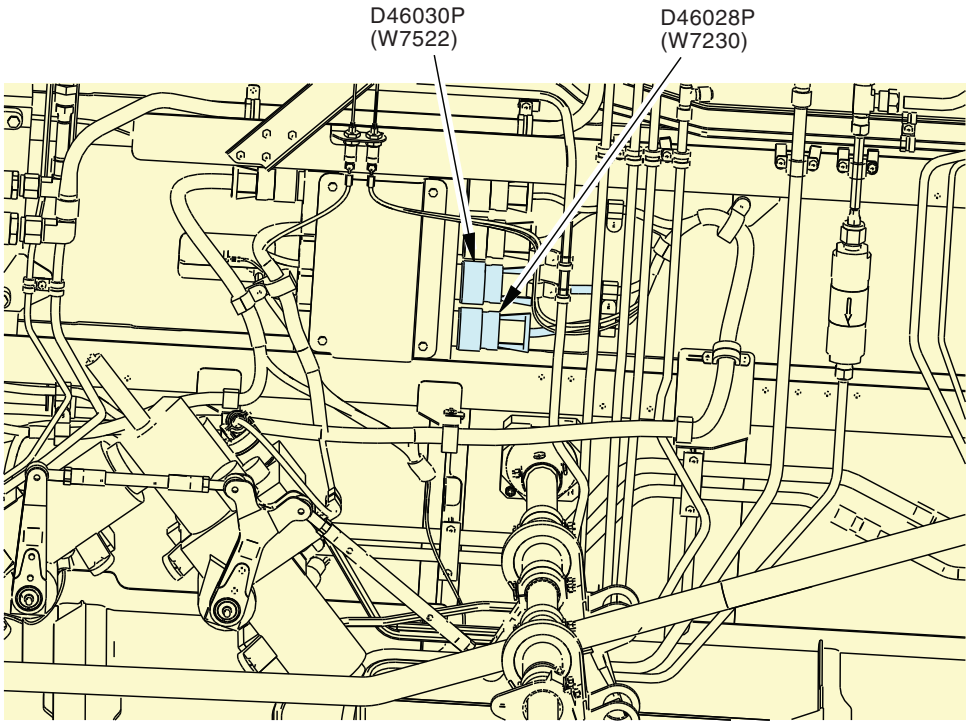
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-150-00) - Connector Location
Figure 601/05-55-25-990-819 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT MAIN LANDING GEAR WHEEL WELL
(VIEW IN THE UP DIRECTION)

B

FWD

INBD

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Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-150-00) - Connector Location
Figure 601/05-55-25-990-819 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-803

3. Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-152-00)

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-005

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-860-003

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-016

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 603.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 603).

Table 603/05-55-25-993-843 Wire Bundle Loop Resistance Values - Left MLG Wheel Well (MRB 20-152-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1136	GD03908		30/70		
W5375	GD01505		3.5/10		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 603).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 603 were tested and passed.

SUBTASK 05-55-25-810-009

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

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2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-002

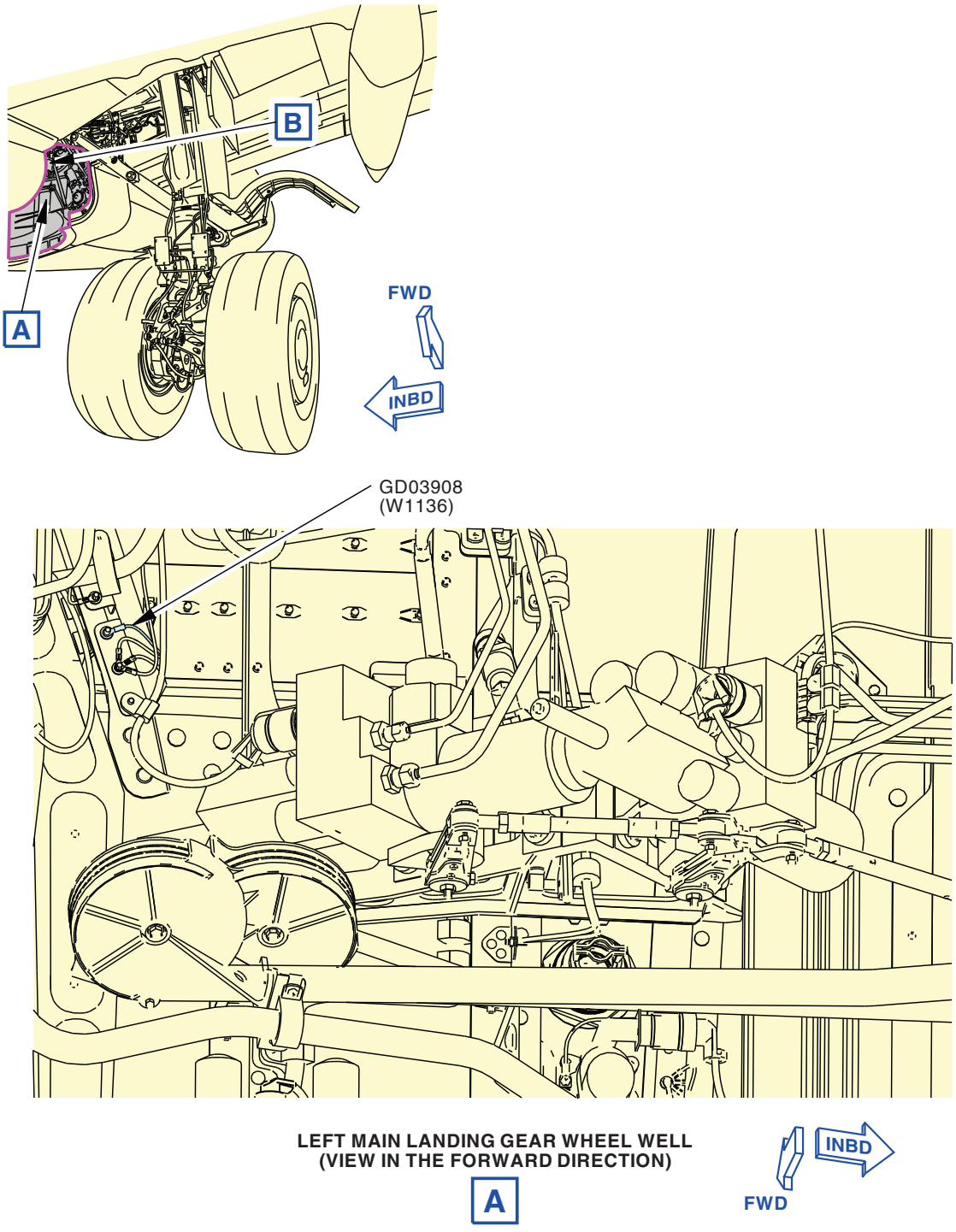
(3) The connectors/grounds listed in Table 604 is used for the steps above.

Table 604/05-55-25-993-815 Left MLG Wheel Well Connectors (MRB 20–152–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1136	GD03908	28-23-11	AB0500DR
W5375	GD01505	22-11-31	N/A

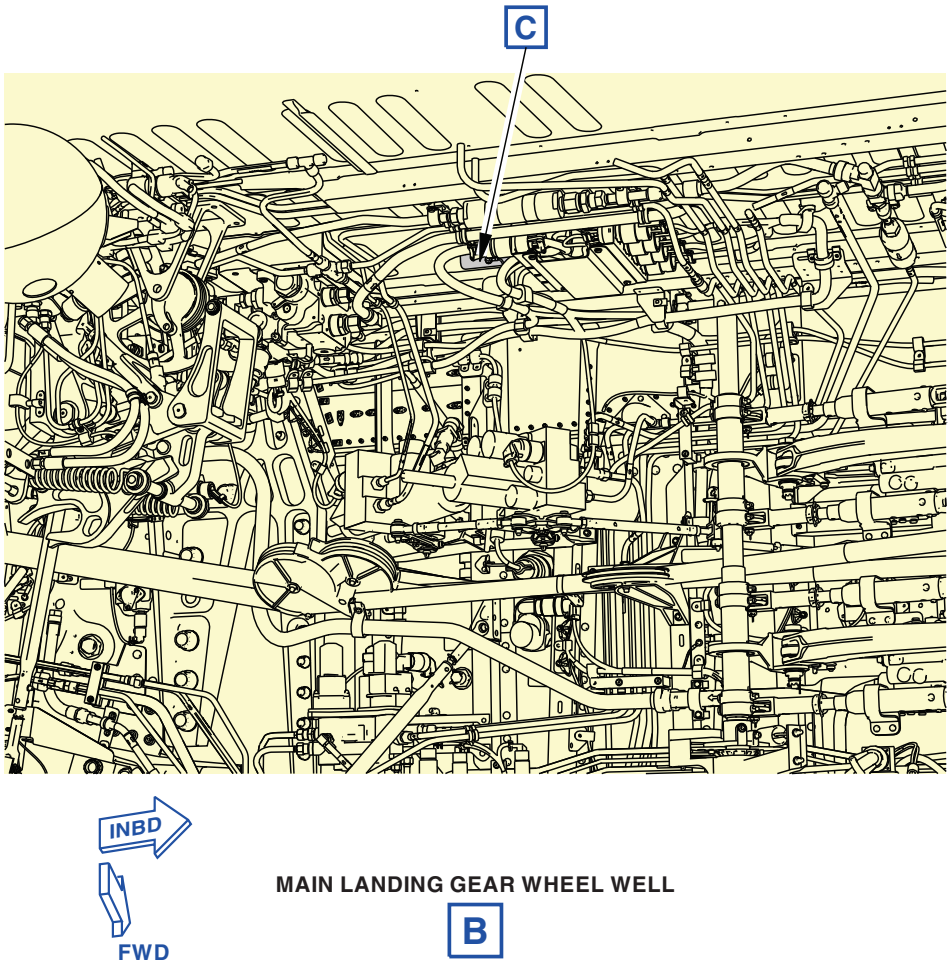
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundles - Left MLG Wheel Well (MRB 20-152-00) - Connector Locations
Figure 602/05-55-25-990-848 (Sheet 1 of 3)

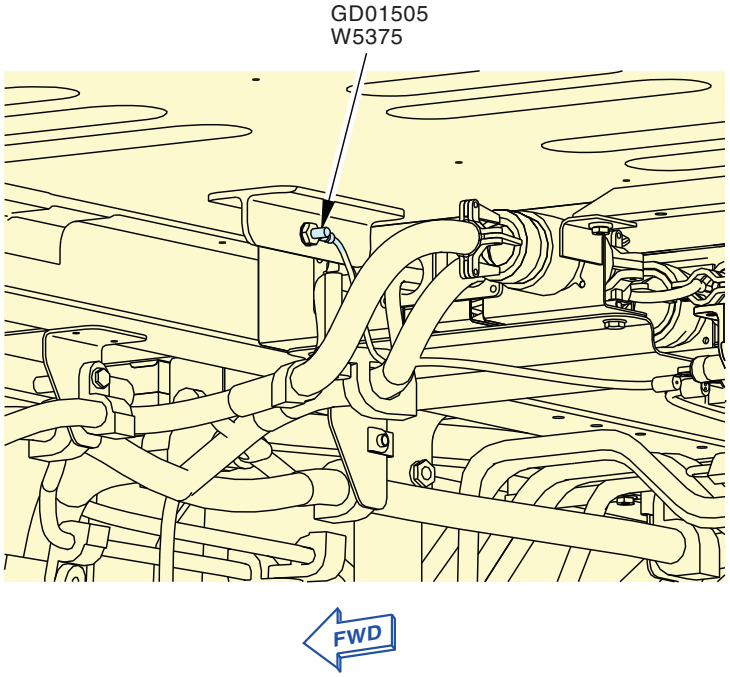
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2973960 S0000752417_V1

Loop Resistance Test of Wire Bundles - Left MLG Wheel Well (MRB 20-152-00) - Connector Locations
Figure 602/05-55-25-990-848 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2973968 S0000752418_V1

Loop Resistance Test of Wire Bundles - Left MLG Wheel Well (MRB 20-152-00) - Connector Locations
Figure 602/05-55-25-990-848 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-804

4. Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-176-00)

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-007

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-860-004

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-017

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 605.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 605).

Table 605/05-55-25-993-844 Wire Bundle Loop Resistance Values - Left MLG Wheel Well (MRB 20-176-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W7580	D46202		7/22		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 605).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 605 were tested and passed.

SUBTASK 05-55-25-810-010

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

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2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-003

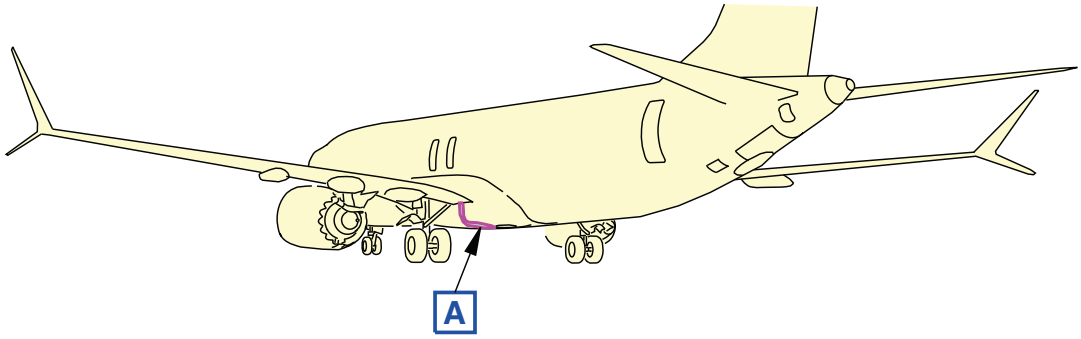
(3) The connector listed in Table 606 is used for the steps above.

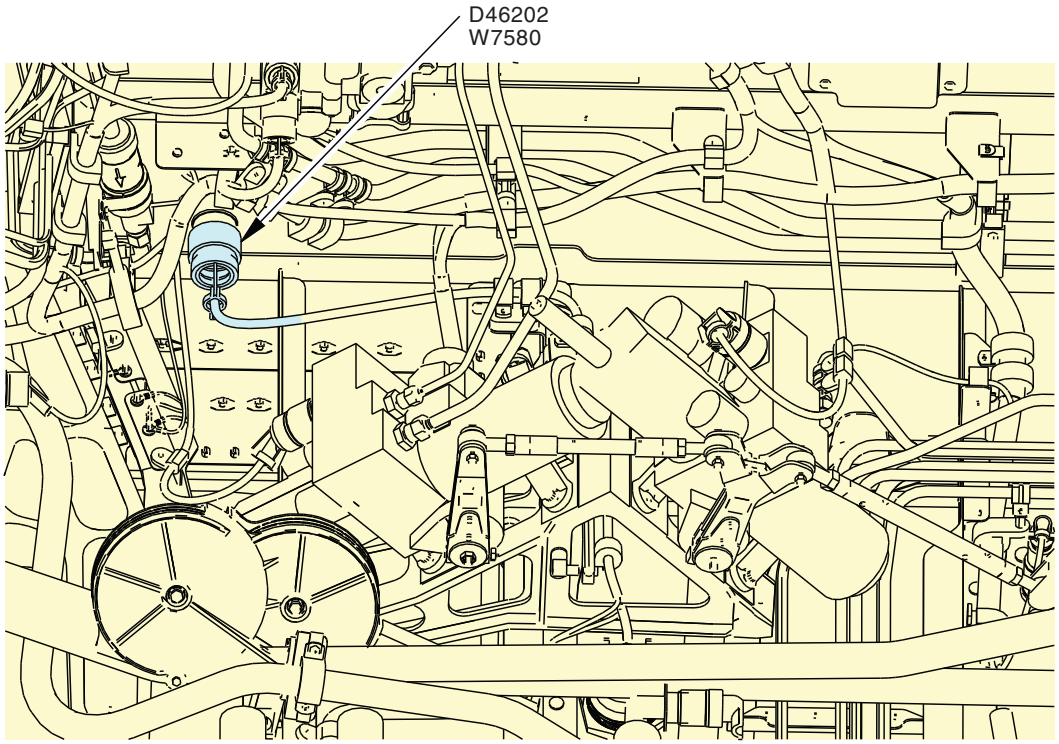
Table 606/05-55-25-993-816 Left MLG Wheel Well Connectors (MRB 20–176–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W7580	D46202	28–41–11	DM0210

——— END OF TASK ———

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Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-176-00) - Connector Location
Figure 603/05-55-25-990-823

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-805

5. Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-160-00)

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box

E. Access Panels

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-009

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-040-011

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-25-010-012

(3) Open this access panel:

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-018

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 607.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 607).

Table 607/05-55-25-993-845 Wire Bundle Loop Resistance Values - Left Wing to Body Fairing (MRB 20-160-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1178	D39927		4.3/15		
W1182	D39931		6/15		
W1184	D39935		11/26		
W1190	D39973		7/17		
W1188	D39903		16/38		
W1662	D39957		6/14		
W1140	D39965		11/36		

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Table 607/05-55-25-993-845 Wire Bundle Loop Resistance Values - Left Wing to Body Fairing (MRB 20-160-00) (Continued)

W1192	D39985		4/10		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 607).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 607 were tested and passed.

SUBTASK 05-55-25-810-011

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

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(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-004

(3) The connectors listed in Table 608 are used for the steps above.

Table 608/05-55-25-993-817 Left Wing to Body Fairing Connectors (MRB 20-160-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1178	D39927	26–11–11, 28–21–11, 36–11–12, 36–11–13, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–11, 78–35–11, 79–31–11	DM0055
W1182	D39931	26–11–11, 26–12–11, 30–21–31, 36–11–13, 73–21–12, 73–22–11, 73–22–41, 74–11–11, 76–21–11, 77–12–21	DM0015
W1184	D39935	24–11–11, 24–21–11, 24–24–11, 29–11–11	DM0017
W1190	D39973	46–13–11, 73–21–11, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0011
W1188	D39903	26–12–11, 27–81–12, 30–11–11	DM0001
W1662	D39957	28–44–11, 30–11–11, 33–42–21, 33–43–11, 33–44–12	DM0052
W1140	D39965	28–41–11	DM0053
W1192	D39985	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0013

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-008

(1) Close this access panel:

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

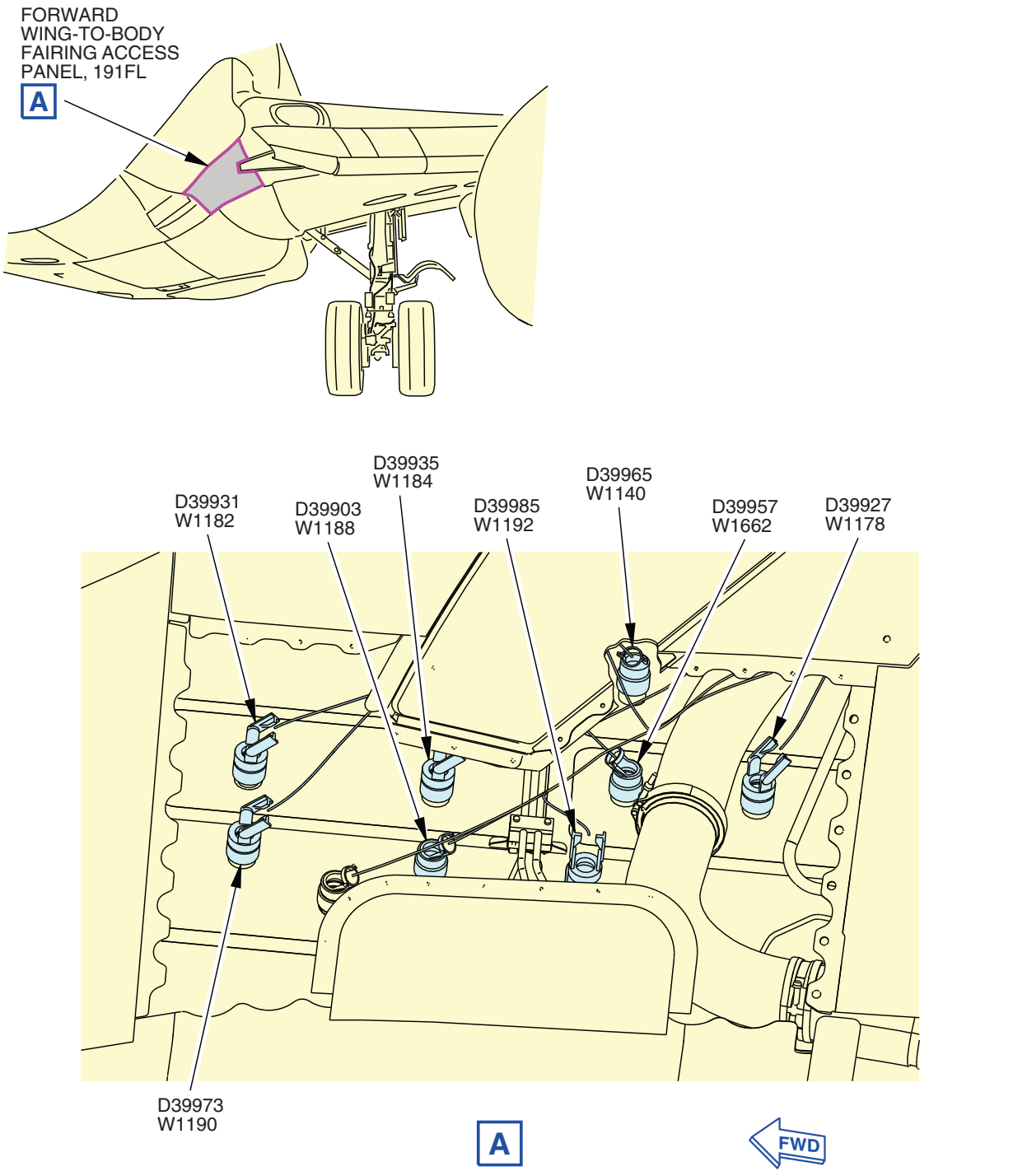
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-440-010

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-160-00) - Connector Location
Figure 604/05-55-25-990-825

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-806

6. Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-161-00)

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CL	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-011

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-840-002

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-25-010-013

(3) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-019

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 609.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 609).

Table 609/05-55-25-993-846 Wire Bundle Loop Resistance Values - Left Wing to Body Fairing (MRB 20-161-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W5526	D04580P		500/750		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 609).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 609 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-810-012

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-005

(3) The connector listed in Table 610 is used for the steps above.

Table 610/05-55-25-993-818 Left Wing to Body Fairing Connectors (MRB 20-161-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5526	D04580P	21-51-14, 36–21–11	AB0536L

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-009

(1) Close this access panel:

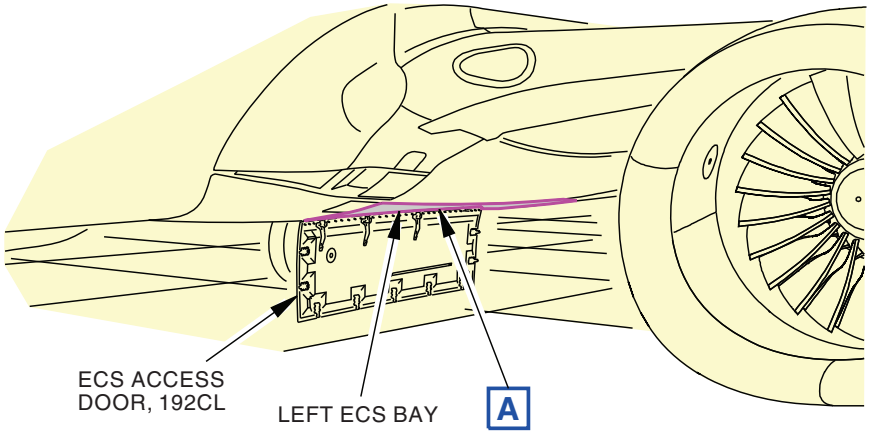
Number	Name/Location
192CL	ECS Access Door

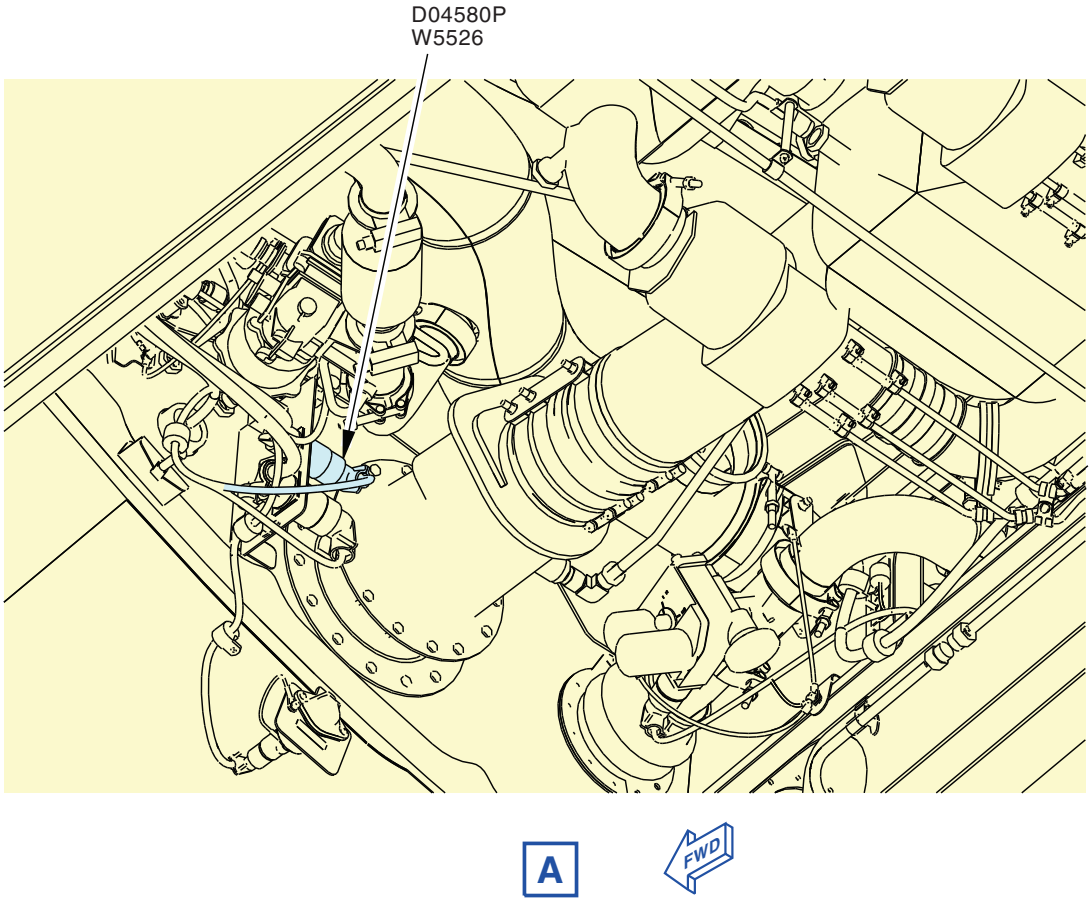
SUBTASK 05-55-25-840-003

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546262 S0000605057_V1

Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-161-00) - Connector Location
Figure 605/05-55-25-990-827

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-809

7. Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-164-00)

(Figure 606)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CL	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-001

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-840-001

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-25-010-011

(3) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-020

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 611.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 611).

Table 611/05-55-25-993-847 Wire Bundle Loop Resistance Values - Left Wing to Body Fairing (MRB 20-164-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W5502	D40844P		16/38		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 611).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 611 were tested and passed.

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SUBTASK 05-55-25-810-007

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-007

(3) The connector listed in Table 612 is used for the steps above.

Table 612/05-55-25-993-820 Left Wing to Body Fairing Connectors (MRB 20-164-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5502	D40844P	21-42-11, 21–51–11	D40844J

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-011

(1) Close this access panel:

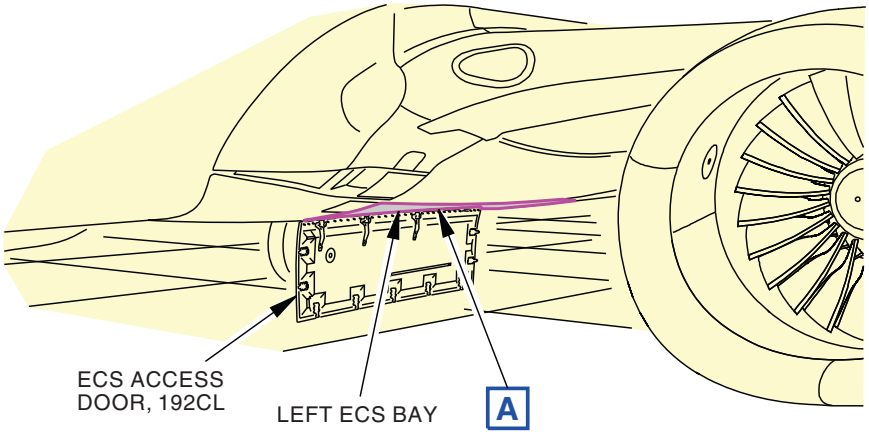
Number	Name/Location
192CL	ECS Access Door

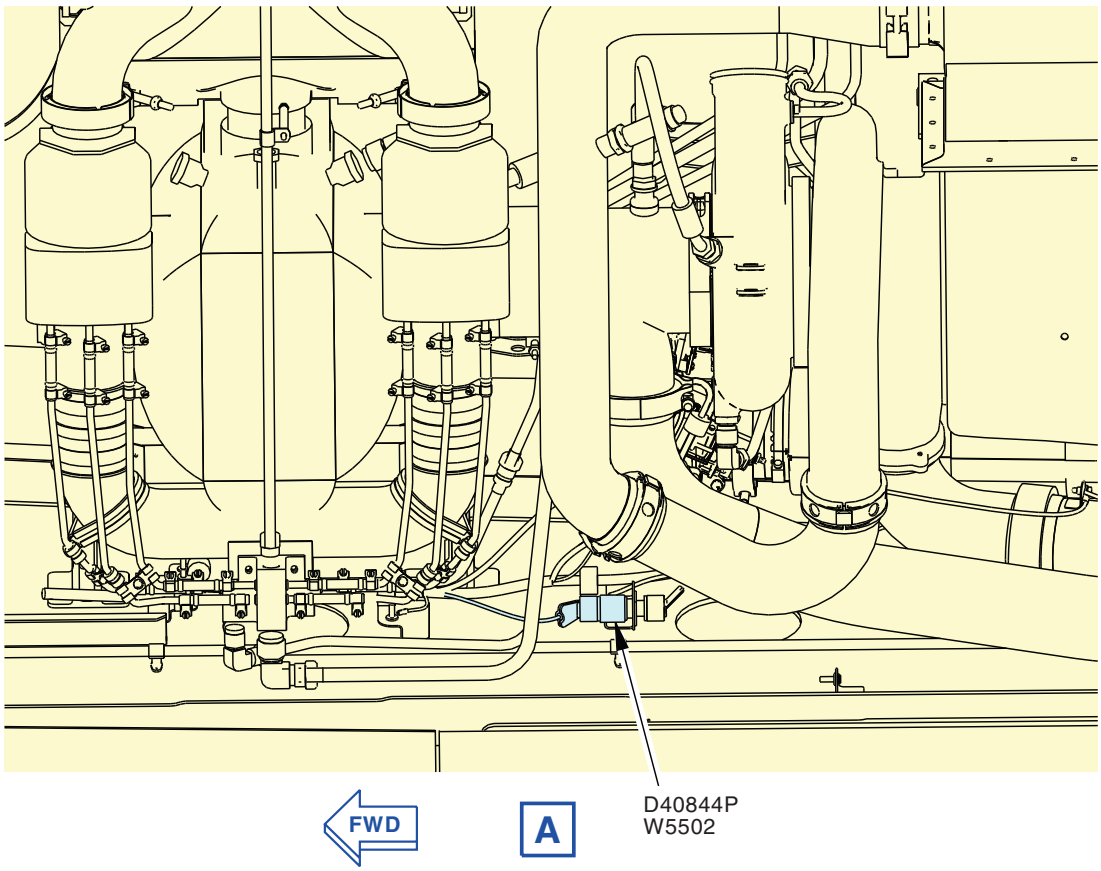
SUBTASK 05-55-25-840-004

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546378 S0000605061_V1

Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-164-00) - Connector Location
Figure 606/05-55-25-990-831

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-810

8. Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-178-00)

(Figure 607)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CL	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-016

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-840-005

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-25-010-015

(3) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-021

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 613.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 613).

Table 613/05-55-25-993-848 Wire Bundle Loop Resistance Values - Left Wing to Body Fairing (MRB 20-178-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W5502	D41500		3/8		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 613).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 613 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-810-014

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-008

(3) The connector listed in Table 614 is used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 614/05-55-25-993-821 Left Wing to Body Fairing Connectors (MRB 20-178-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5502	D41500	21–42–11, 21–51–11, 21–51–12, 21–61–11, 26–12–11, 33–33–11, 36–11–31	DM0050

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-012

(1) Close this access panel:

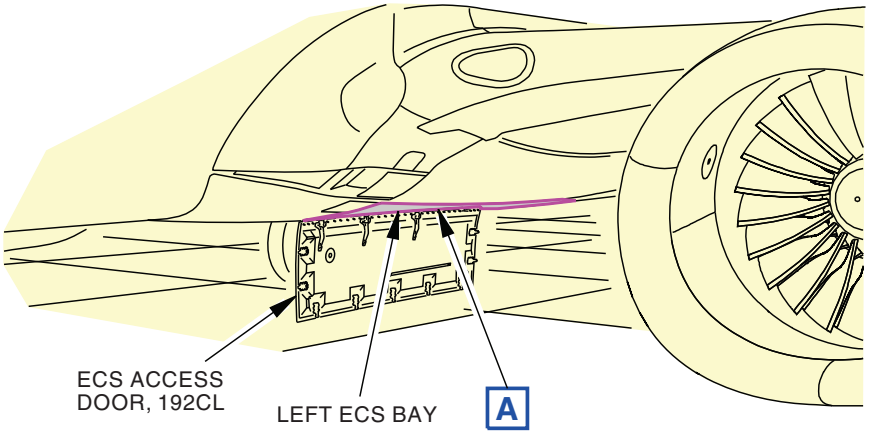
Number	Name/Location
192CL	ECS Access Door

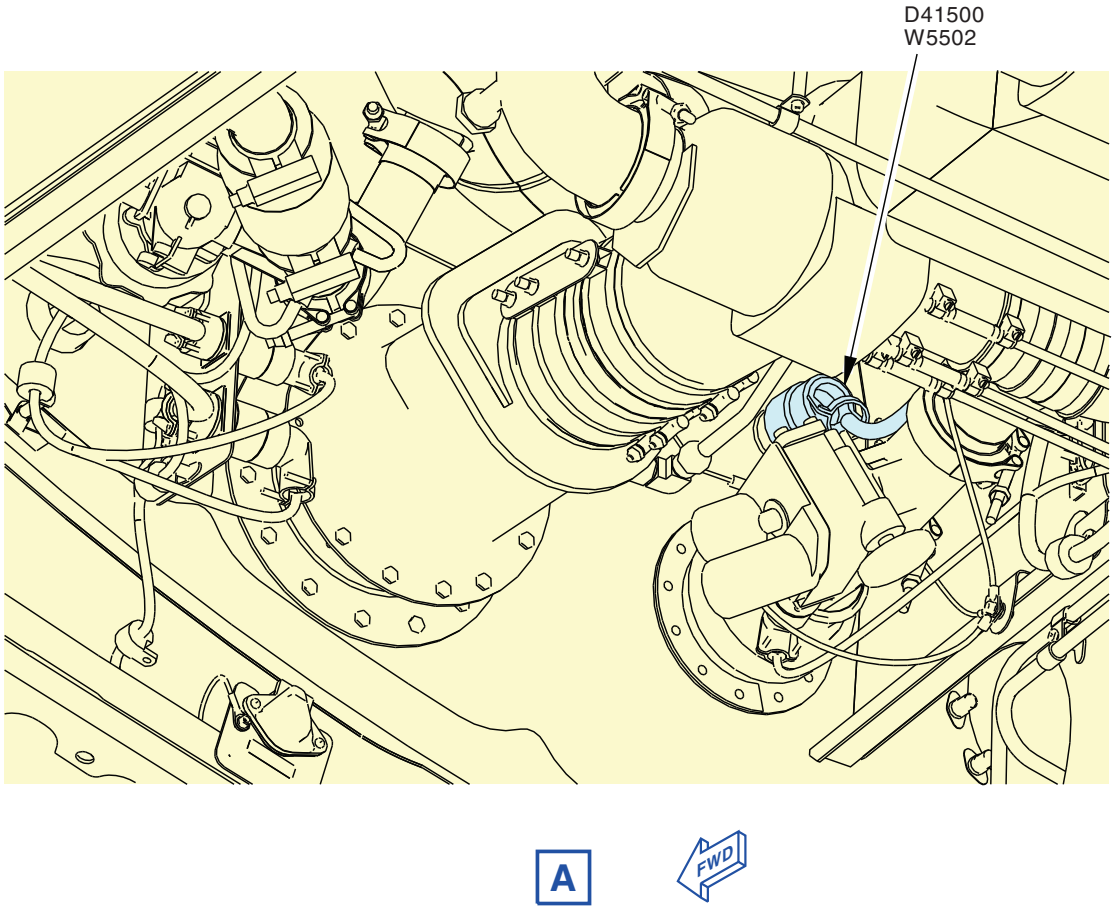
SUBTASK 05-55-25-840-006

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546331 S0000605064_V1

Loop Resistance Test of Wire Bundle - Left Wing to Body Fairing (MRB 20-178-00) - Connector Location
Figure 607/05-55-25-990-833

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-811

9. Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-156-00)

(Figure 608)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam

E. Access Panels

Number	Name/Location
551GB	Lower inboard fixed trailing edge

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-018

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-010-023

(2) Do one of the two optional steps that follow to get access to the wire bundle:

(a) To get access by the Trailing Edge Flaps: (Figure 609).

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Extend the Trailing Edge Flaps to 40 degrees (Extend the Trailing Edge Flaps, TASK 27-51-00-860-803).

2) Do this task: Trailing Edge Flaps System Deactivation Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(b) To get access by the removal of panel 551GB: (Figure 608).

1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

2) Remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
551GB	Lower inboard fixed trailing edge

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-022

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 615.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 615).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 615/05-55-25-993-854 Wire Bundle Loop Resistance Values - Left Wing Trailing Edge (MRB 20-156-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1024	D40024J		12/28		
W1022	D40024P		8/20		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 615).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 615 were tested and passed.

SUBTASK 05-55-25-810-015

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-009

(3) The connectors listed in Table 616 are used for the steps above.

Table 616/05-55-25-993-822 Left Wing Trailing Edge Connectors (MRB 20-156-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1024	D40024J	22–11–11, 22–12–31, 27–18–11, 27–32–21, 27–52–11, 27–53–21, 28–21–11	AW0254L
W1022	D40024P	22–11–11, 22–12–31, 27–18–11, 27–32–21, 27–52–11, 27–53–21, 28–21–11	AW0254L

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-860-012

(1) If necessary, retract the trailing edge flaps:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 05-55-25-410-019

(2) If removed, install this access panel:

(TASK 57-50-02-400-801)

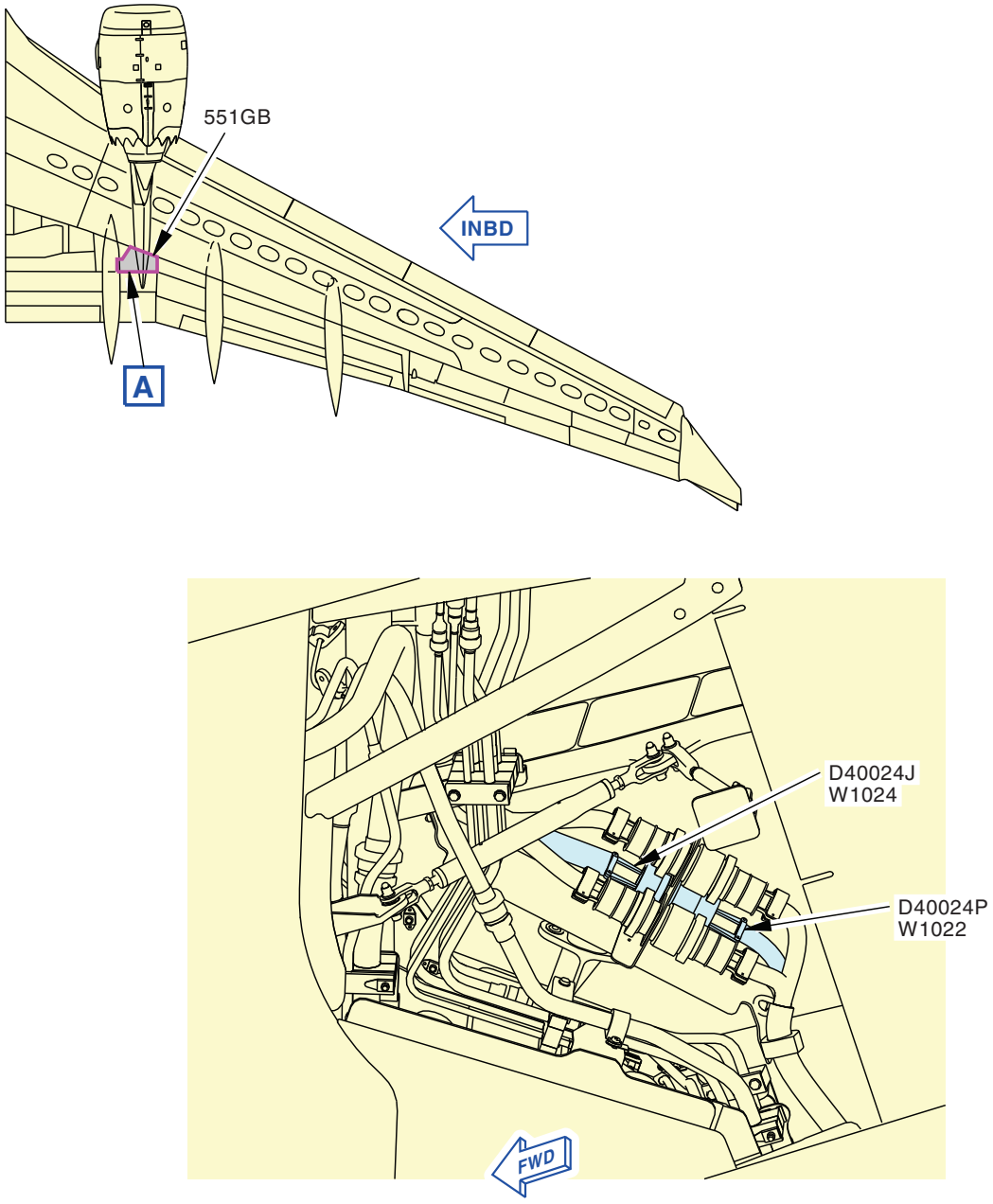
Number	Name/Location
551GB	Lower inboard fixed trailing edge

SUBTASK 05-55-25-440-019

(3) If necessary, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

END OF TASK

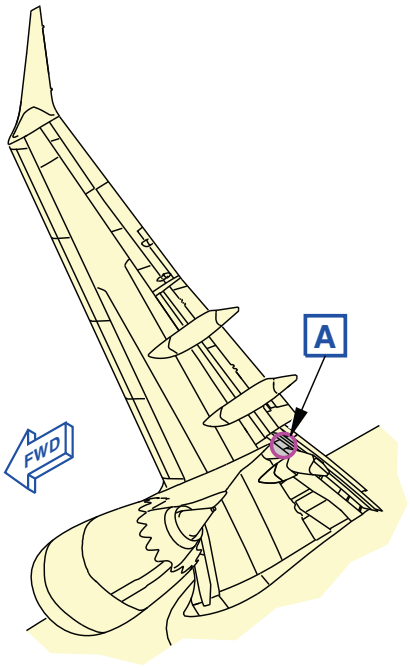
ECCN 9E991 BOEING PROPRIETARY - See title page for details

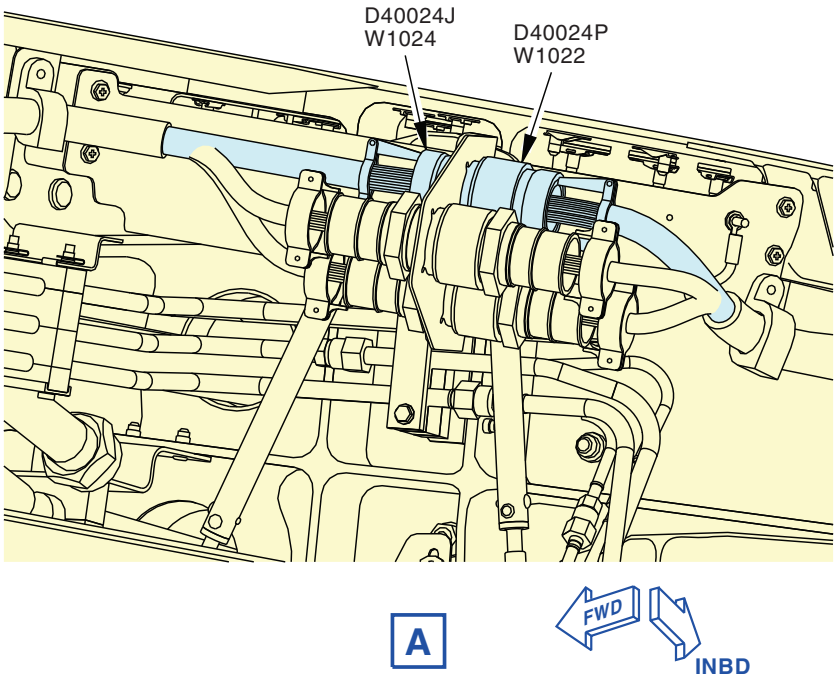


2546330 S0000605066_V2

Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-156-00) - Connector Location - Access Panel
Figure 608/05-55-25-990-843

ECCN 9E991 BOEING PROPRIETARY - See title page for details





3051788 S0000817064_V1

Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-156-00) - Connector Location - Trailing Edge Flap (40 degrees)
Figure 609/05-55-25-990-856

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-812

10. Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-159-00)

(Figure 610)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam

E. Access Panels

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-020

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-040-014

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-55-25-010-017

(3) Do one of the two steps that follow to get access to the wire bundle:

(a) To get access by the Main Landing Gear (MLG) (Figure 611).

(b) To get access by the removal of panel 551BB (Figure 610).

1) Remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-023

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) For each wire bundle listed in the Table 617, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 617).

Table 617/05-55-25-993-849 Wire Bundle Loop Resistance Values - Left Wing Trailing Edge (MRB 20-159-00)

Plane:					
Date:					
Technician:					
Bundle Number	Ground Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1136	GD04428		30/70		

3) Make sure that that the measured value is within the MIN/MAX value range listed in the data sheet (Table 617).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 617 were tested and passed.

SUBTASK 05-55-25-810-016

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-010

(3) The connector listed in the Table 618 is used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 618/05-55-25-993-823 Left Wing Trailing Edge Connectors (MRB 20-159-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1136	D16336	28-23-11	M03046

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-014

(1) If removed, install this access panel:

(TASK 57-50-02-400-801)

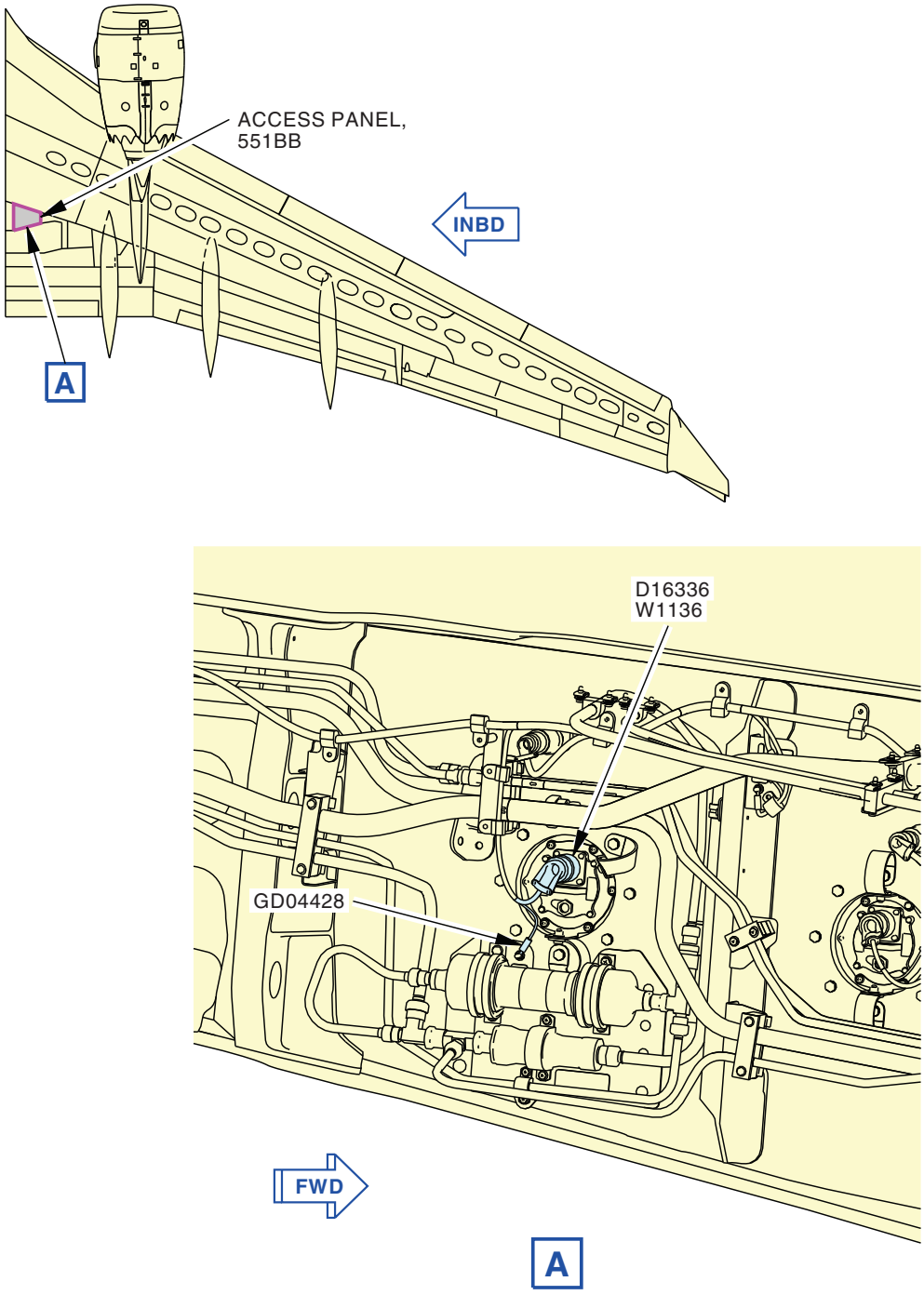
Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

SUBTASK 05-55-25-440-013

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

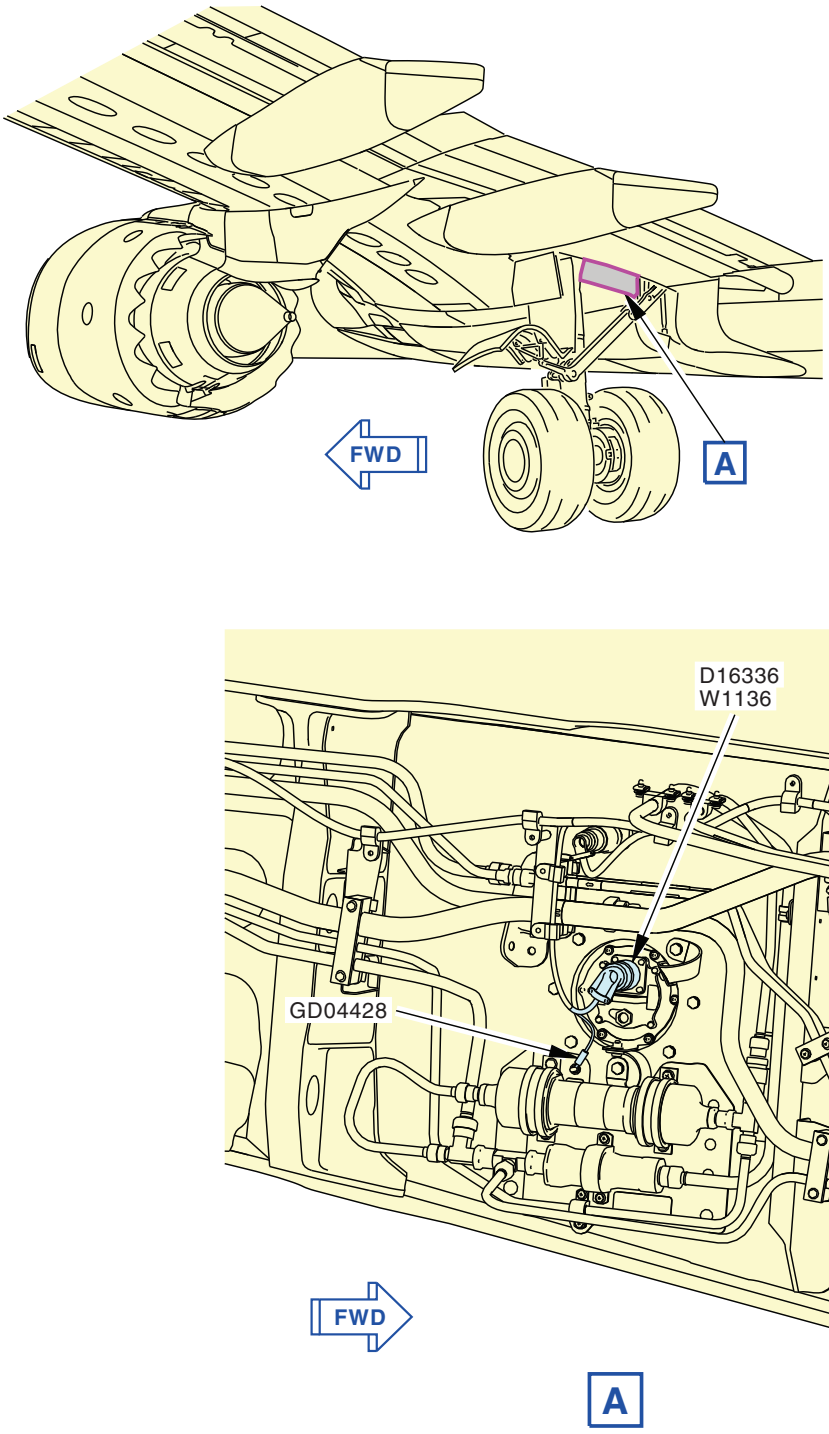
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546312 S0000605069_V1

Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-159-00) - Connector Location
Figure 610/05-55-25-990-835

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3056643 S0000820552_V1

Loop Resistance Test of Wire Bundle - Left Wing Trailing Edge (MRB 20-159-00) - Connector Location - Main Landing Gear
Figure 611/05-55-25-990-857

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-813

11. Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-154-00)

(Figure 612)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar

E. Access Panels

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-022

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-040-015

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-25-010-018

(3) Open this access panel:

Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-024

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 619.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 619).

Table 619/05-55-25-993-850 Wire Bundle Loop Resistance Values - Left Wing Leading Edge (MRB 20-154-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1188	D00738		78/184		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 619).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 619 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-810-017

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-011

(3) The connector listed in Table 620 is used for the steps above.

Table 620/05-55-25-993-824 Left Wing Leading Edge Connectors (MRB 20-154-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1188	D00738	30–11–11	S00118

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-015

(1) Close this access panel:

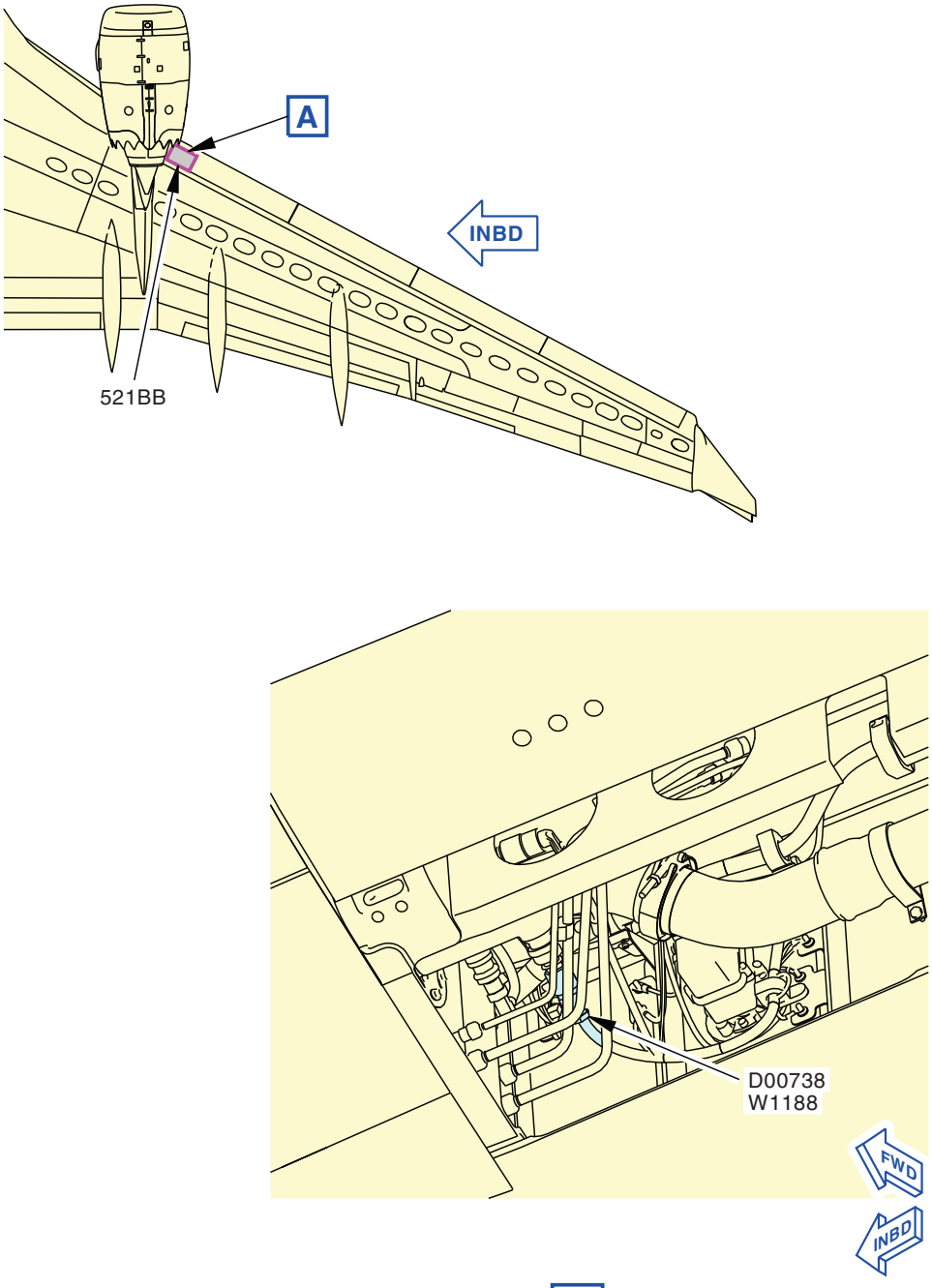
Number	Name/Location
521BB	Outboard Lower Fixed Leading Edge Access Panel

SUBTASK 05-55-25-440-014

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546201 S0000605072_V2

Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-154-00) - Connector Location
Figure 612/05-55-25-990-837

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-815

12. Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-158-00)

(Figure 613)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
431	Engine 1 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar

E. Access Panels

Number	Name/Location
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
511AB	Inboard Leading Edge, Lower Removable Panel

F. Prepare for the procedure

SUBTASK 05-55-25-970-024

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-040-016

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-25-010-019

(3) Open these access panels:

Number	Name/Location
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
511AB	Inboard Leading Edge, Lower Removable Panel

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-025

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 621.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 621).

Table 621/05-55-25-993-851 Wire Bundle Loop Resistance Values - Left Wing Leading Edge (MRB 20-158-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1184	D30014		10/23		
W1164	D30016		10/24		
W1172	D30042		8/19		
W1178	D30048		6/13		
W1170	D30052		5/12		
W1182	D30062		6/15		
W1166	D30064		5/12		

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Table 621/05-55-25-993-851 Wire Bundle Loop Resistance Values - Left Wing Leading Edge (MRB 20-158-00) (Continued)

W1192	D30068		7/15		
W1168	D30084		6/15		
W1190	D30092		7/17		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 621).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 621 were tested and passed.

SUBTASK 05-55-25-810-018

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-012

(3) The connectors listed in Table 622 are used for the steps above.

Table 622/05-55-25-993-825 Left Wing Leading Edge Connectors (MRB 20-158-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1184	D30014	24–11–11, 24–21–11, 24–24–11, 29–11–11	DM0023
W1164	D30016	24–11–11, 24–21–11, 24–24–11, 29–11–11	DM0023
W1172	D30042	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0025
W1178	D30048	26–11–11, 36–11–12, 36–11–13, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–11, 78–35–11, 79–31–11	DM0037
W1170	D30052	26–11–11, 36–11–12, 36–11–13, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–11, 78–35–11, 79–31–11	DM0037

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 622/05-55-25-993-825 Left Wing Leading Edge Connectors (MRB 20-158-00) (Continued)

Wire Bundle	Connector	WDM	Mate With Equipment
W1182	D30062	26–11–11, 26–12–11, 30–21–31, 36–11–13, 73–21–12, 73–22–11, 73–22–41, 74–11–11, 76–21–11, 77–12–21	DM0019
W1166	D30064	26–11–11, 26–12–11, 30–21–31, 36–11–13, 73–21–12, 73–22–11, 73–22–41, 74–11–11, 76–21–11, 77–12–21	DM0019
W1192	D30068	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0025
W1168	D30084	46–13–11, 73–21–11, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0027
W1190	D30092	46–13–11, 73–21–11, 73–21–21, 73–24–11, 73–24–12, 78–32–51	DM0027

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-410-016

(1) Close these access panels:

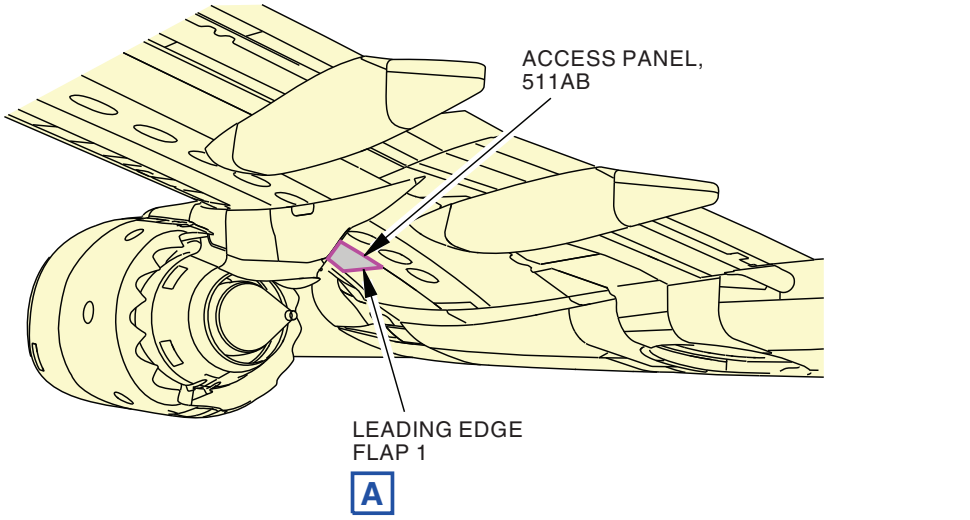
Number	Name/Location
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1
511AB	Inboard Leading Edge, Lower Removable Panel

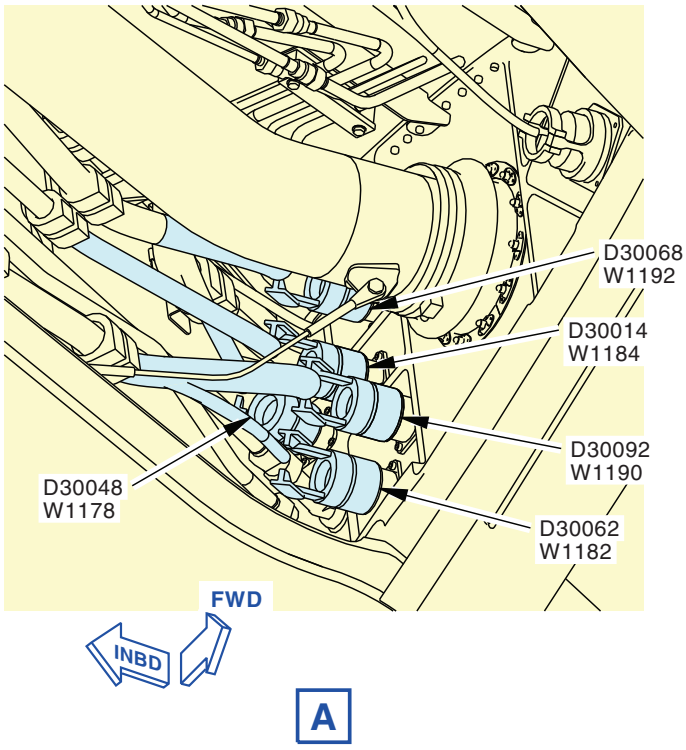
SUBTASK 05-55-25-440-015

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

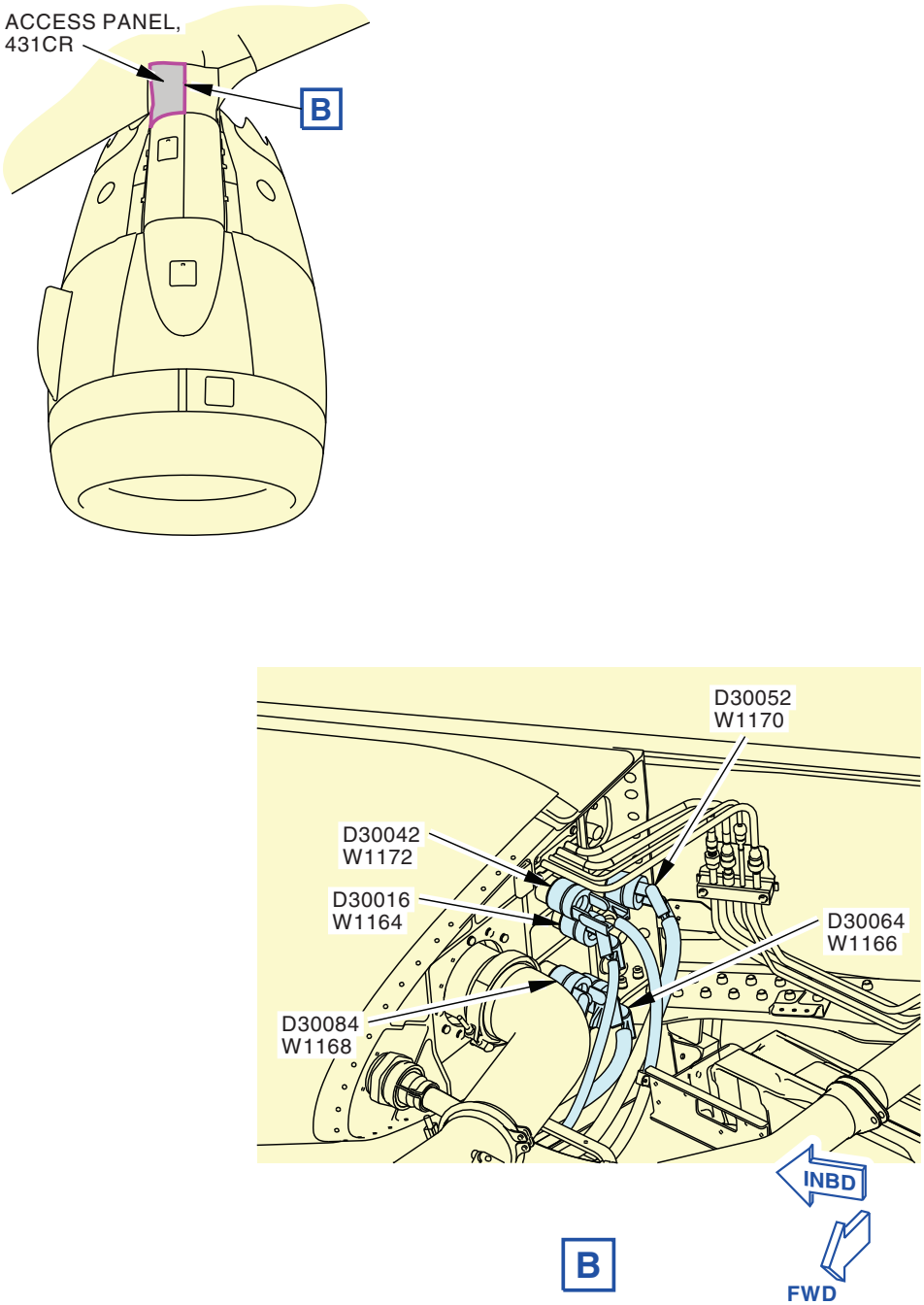




2545994 S0000605225_V1

Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-158-00) - Connector Location
Figure 613/05-55-25-990-839 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546002 S0000605226_V1

Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-158-00) - Connector Location
Figure 613/05-55-25-990-839 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-25-280-816

13. Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-177-00)

(Figure 614)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
510	Subzone - Left Wing: Leading Edge, Fwd of Front Spar, Inbd of Strut and Nacelle Gap Cover Area

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-026

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-840-007

(2) Do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 05-55-25-040-017

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-026

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 623.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 623).

Table 623/05-55-25-993-852 Wire Bundle Loop Resistance Values - Left Wing Leading Edge (MRB 20-177-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1140	GD00191		11/36		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 623).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 623 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-25-810-019

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-013

(3) The connector listed in Table 624 is used for the steps above.

Table 624/05-55-25-993-826 Left Wing Leading Edge Connectors (MRB 20-177-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1140	GD00191	28–41–11	M02945

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-25-440-016

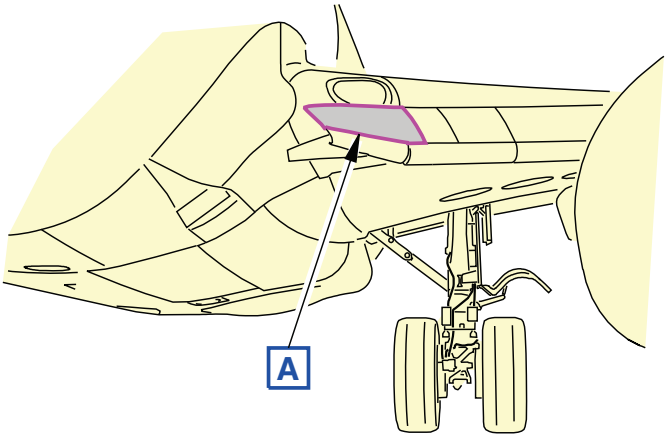
(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

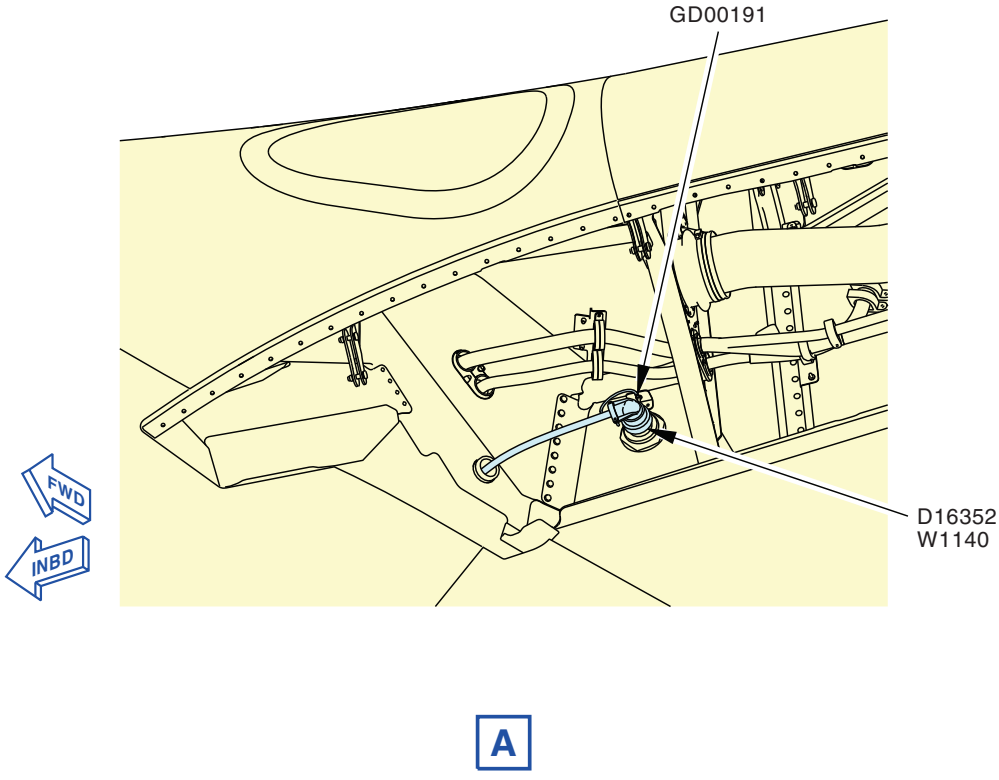
SUBTASK 05-55-25-840-008

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Loop Resistance Test of Wire Bundle - Left Wing Leading Edge (MRB 20-177-00) - Connector Location
Figure 614/05-55-25-990-841

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233

TASK 05-55-25-280-817

14. Loop Resistance Test of Wire Bundle - Left Engine (MRB 20-182-00)

(Figure 615 or Figure 616)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
431	Engine 1 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar

E. Access Panels

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-028

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233 (Continued)

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-010-020

(2) Open these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-027

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 625.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 625).

Table 625/05-55-25-993-853 Wire Bundle Loop Resistance Values - Left Engine (MRB 20-182-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1062	D30038		6/15		
W1162	D08094P		4/12		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 625).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 625 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233 (Continued)

SUBTASK 05-55-25-610-001

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-014

(3) The connectors listed in Table 626 are used in the steps above.

Table 626/05-55-25-993-827 Left Engine Connectors (MRB 20-182-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1062	D30038	24-21-11	AS0004L

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233 (Continued)

Table 626/05-55-25-993-827 Left Engine Connectors (MRB 20-182-00) (Continued)

Wire Bundle	Connector	WDM	Mate With Equipment
W1162	D08094P	24-21-11	AW0258L

H. Put the Airplane Back to Its Usual Condition

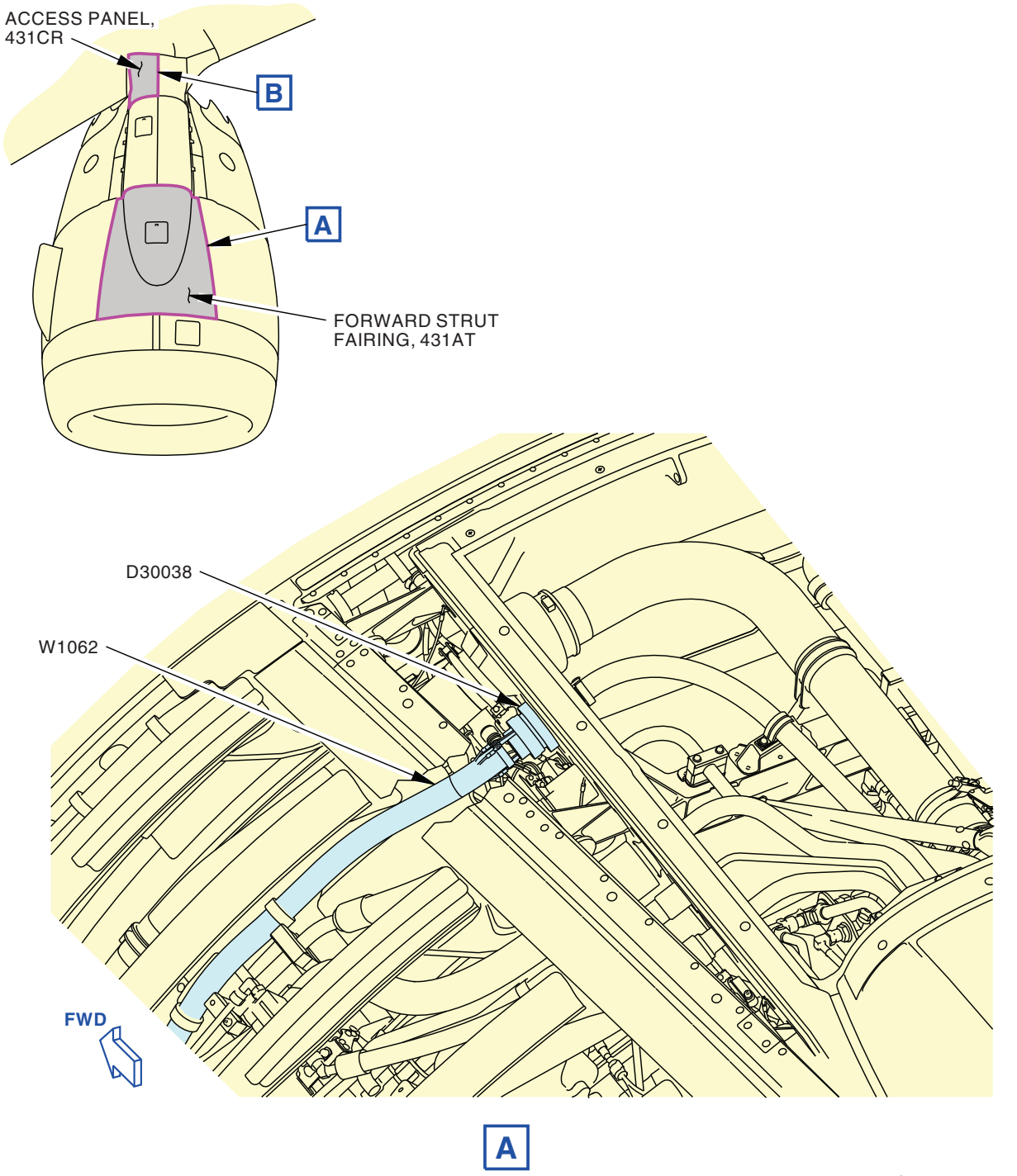
SUBTASK 05-55-25-410-017

(1) Close these access panels:

Number	Name/Location
431AT	Forward Strut Fairing, Thumbnail Fairing, Strut 1
431CR	Forward Strut Fairing, Right Overwing Fairing, Strut 1

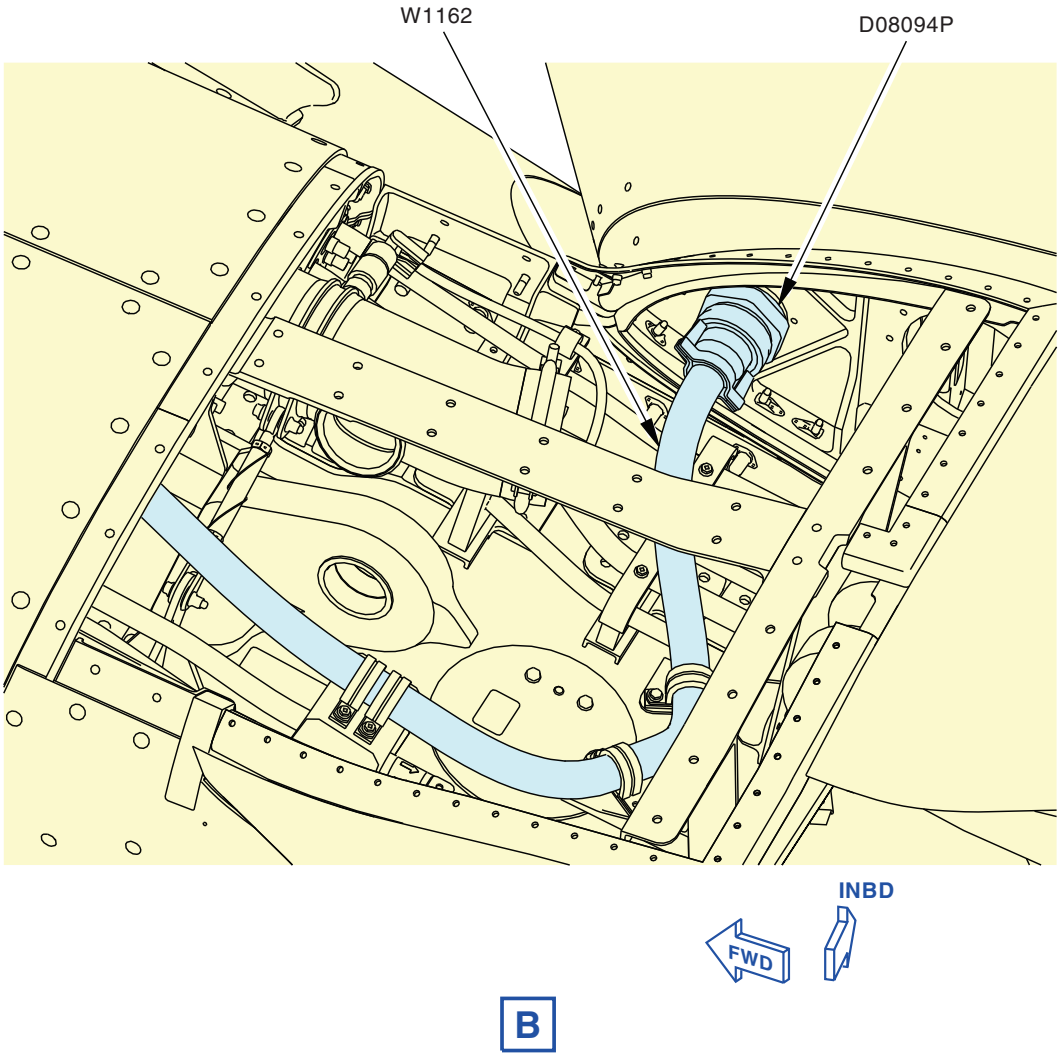
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Left Engine (MRB 20-182-00)
Figure 615/05-55-25-990-845

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundle - Left Engine (MRB 20-182-00)
Figure 616/05-55-25-990-846

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 05-55-25-280-821

15. Loop Resistance Test of Wire Bundle - Left MLG Wheel Well (MRB 20-188-00)

(Figure 617)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-25-970-032

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-25-860-008

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-25-720-031

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 627.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 627).

Table 627/05-55-25-993-862 Wire Bundle Loop Resistance Values - Left MLG Wheel Well (MRB 20-188-00)

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W5375	GD00947		3.5/10		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 627).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in Table 627 the were tested and passed.

SUBTASK 05-55-25-810-023

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-25-800-018

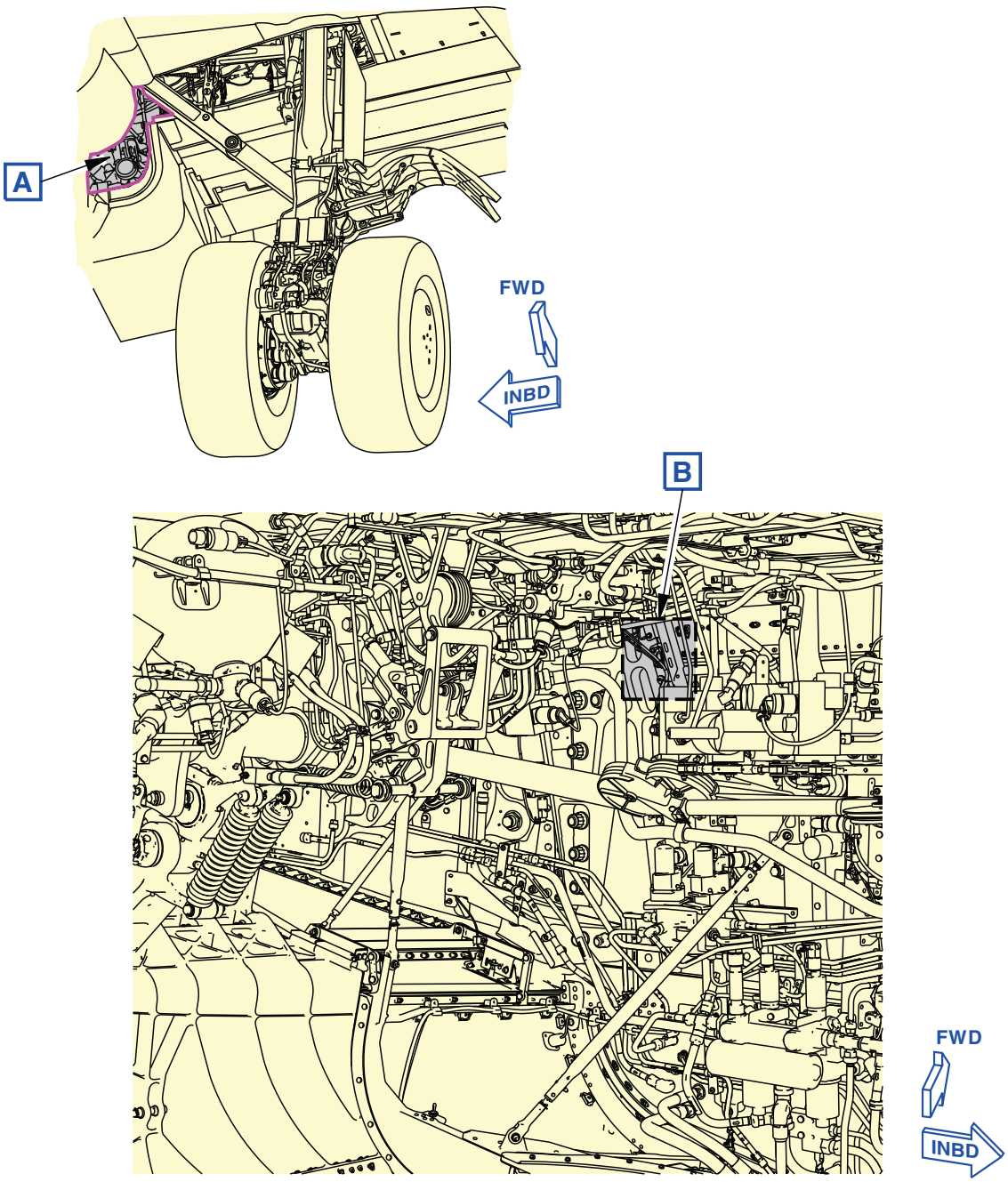
(3) The connectors/grounds listed in Table 628 is used for the steps above.

Table 628/05-55-25-993-861 Left MLG Wheel Well Connectors (MRB 20-188-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5375	GD00947	22-11-31	N/A

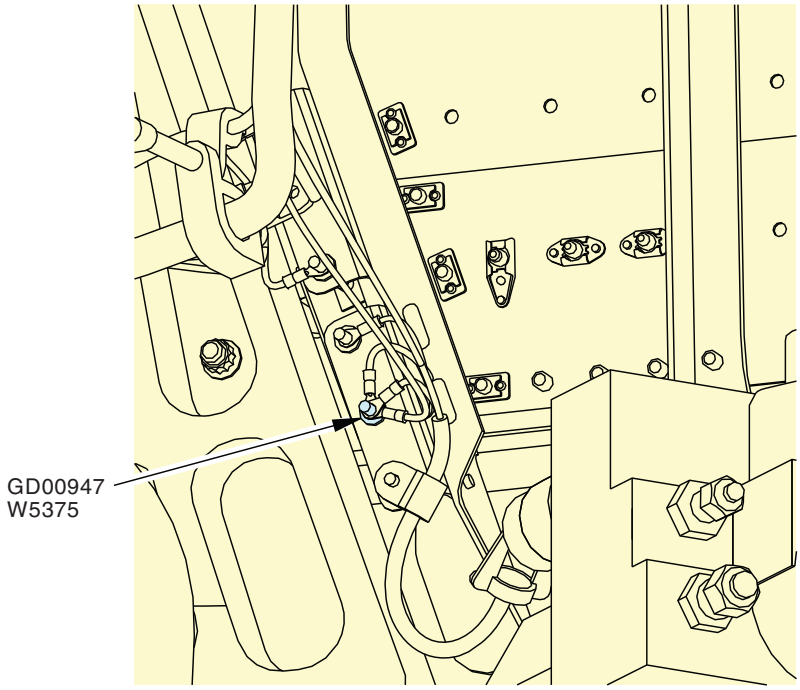
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundles - Left MLG Wheel Well (MRB 20-188-00) - Connector Locations
Figure 617/05-55-25-990-849 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundles - Left MLG Wheel Well (MRB 20-188-00) - Connector Locations
Figure 617/05-55-25-990-849 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WIRE BUNDLES ON RIGHT SIDE OF THE AIRPLANE - INSPECTION/CHECK

1. General

A. This procedure contains tasks that perform a loop resistance test of wire bundles on right side of the airplane.

TASK 05-55-26-280-801

2. Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-150-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-002

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-860-001

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-030

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 601.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 601).

Table 601/05-55-26-993-846 Wire Bundle Loop Resistance Values - Right MLG Wheel Well (MRB 20-150-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1032	D43102P		52/123		
W8122	D46046P		11/26		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 601).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 601 were tested and passed.

SUBTASK 05-55-26-810-007

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-001

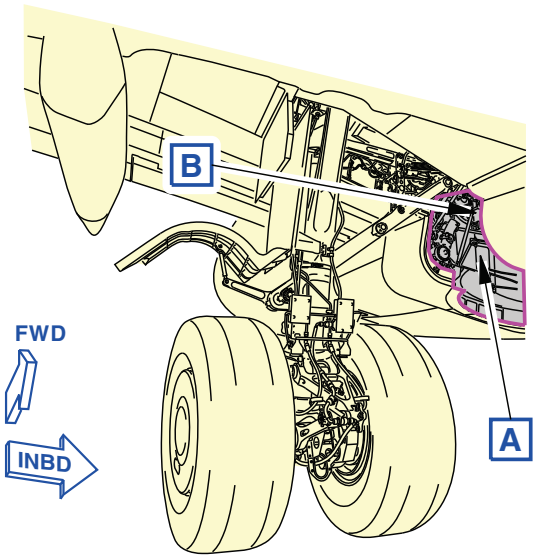
(3) The connectors listed in Table 602 are used for the steps above.

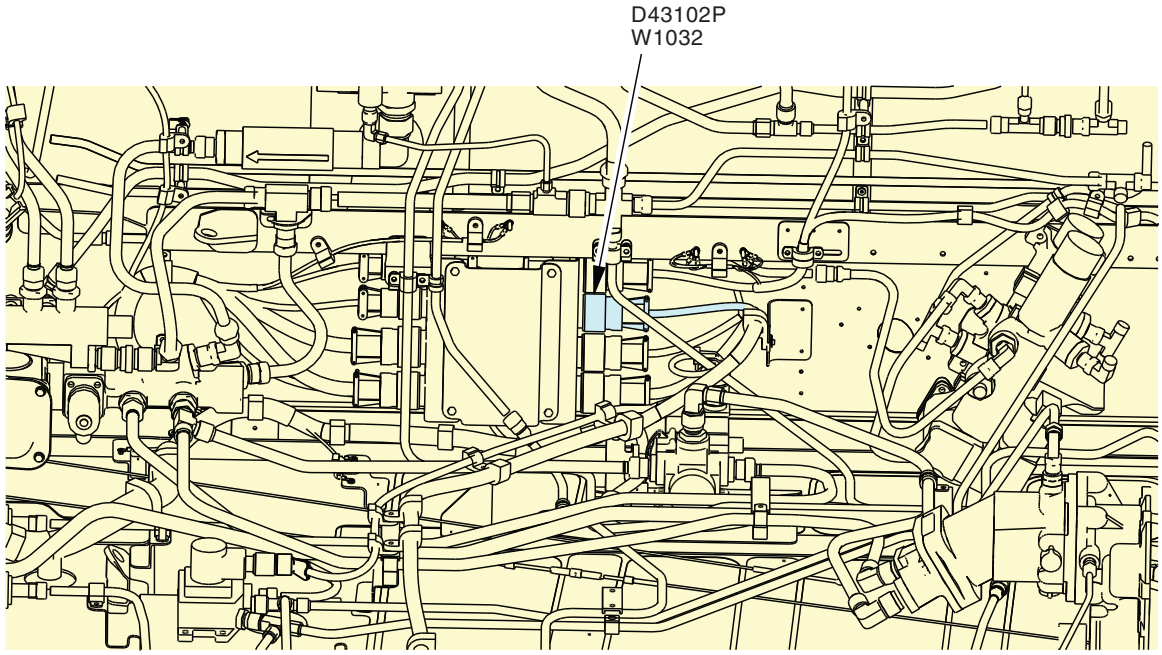
Table 602/05-55-26-993-814 Right MLG Wheel Well Connectors (MRB 20–150–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1032	D43102P	22–11–11, 22–12–41	AL0720D
W8122	D46046P	27–61–49, 27–83–11, 29–23–11, 29–25–11	AL0675C

——— END OF TASK ———

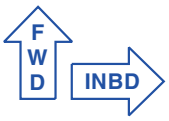
ECCN 9E991 BOEING PROPRIETARY - See title page for details





RIGHT MAIN LANDING GEAR WHEEL
WELL (VIEW IN THE UP DIRECTION)

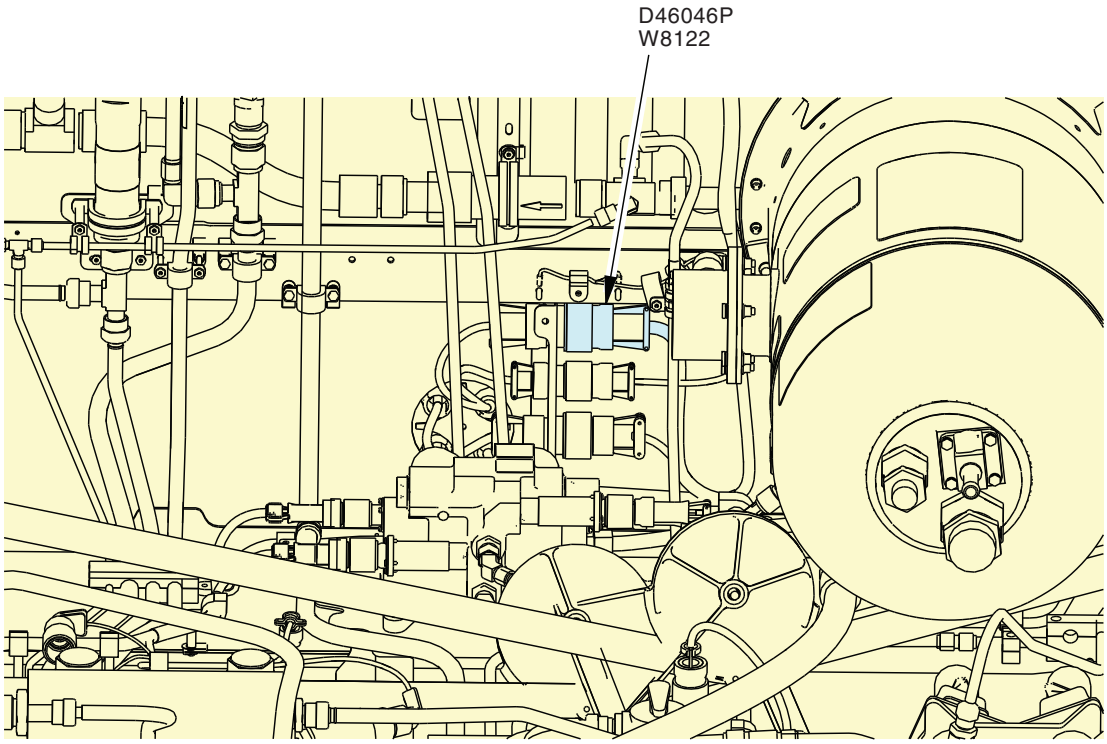
A



2546390 S0000605107_V1

Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-150-00) - Connector Location
Figure 601/05-55-26-990-819 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546794 S0000605108_V1

Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-150-00) - Connector Location
Figure 601/05-55-26-990-819 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-802

3. Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-151-00)

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-004

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-860-002

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-017

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 603.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 603).

Table 603/05-55-26-993-845 Wire Bundle Loop Resistance Values - Right MLG Wheel Well (MRB 20-151-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W8122	D16424		37/86		
W8122	D16428		39/92		
W7230	D16426		255/600		
W7230	D16496		255/600		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 603).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 603 were tested and passed.

SUBTASK 05-55-26-810-022

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-002

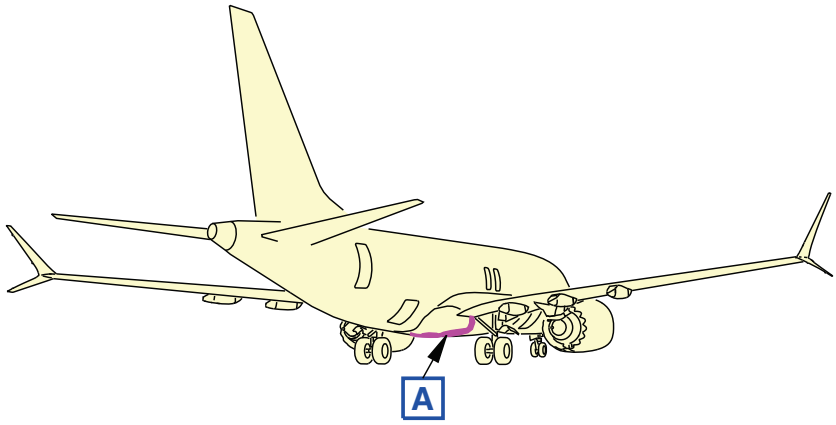
(3) The connectors listed in Table 604 are used for the steps above.

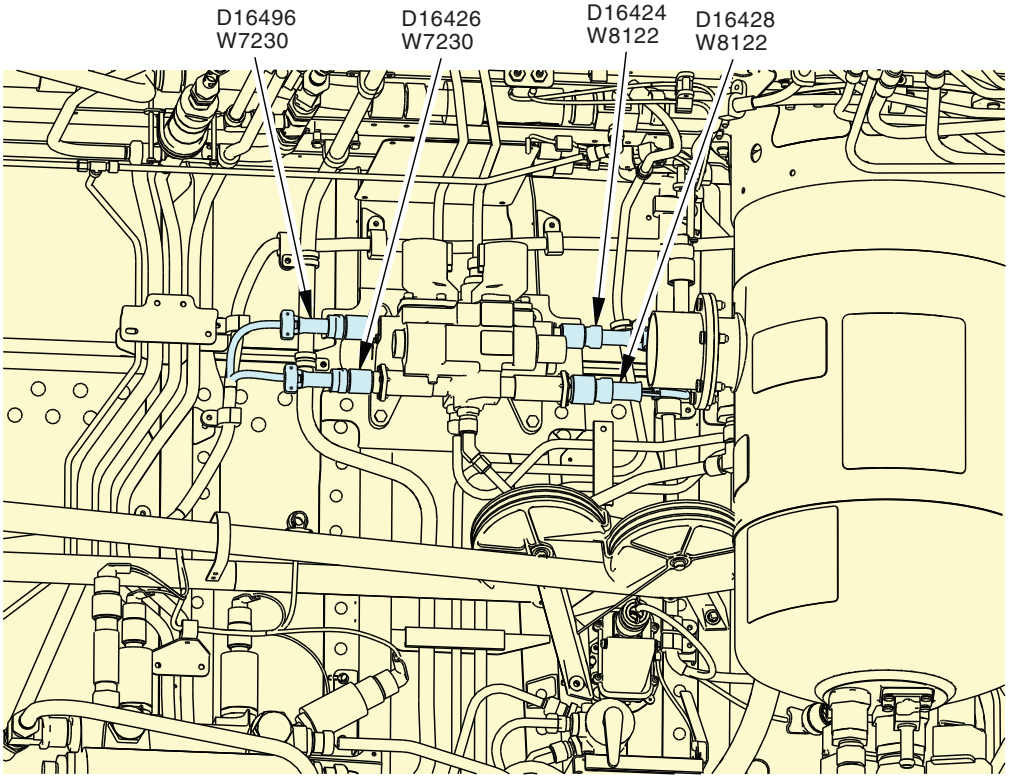
Table 604/05-55-26-993-815 Right MLG Wheel Well Connectors (MRB 20–151–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W8122	D16424	27–61–49	M02988
	D16428	27–61–49	M02988
W7230	D16426	27–61–49	M02988
	D16496	27–61–49	M02988

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-151-00) - Connector Location
Figure 602/05-55-26-990-821

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-805

4. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-160-00)

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box

E. Access Panels

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-006

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-011

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-26-010-011

(3) Open this access panel:

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-018

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 605.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 605).

Table 605/05-55-26-993-847 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing (MRB 20-160-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1278	D39928		6/15		
W1282	D39932		6/15		
W1290	D39974		8/20		
W1292	D39986		7/15		
W1288	D39904		12/28		
W1664	D39943		6/13		
W1138	D39944		11/36		

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2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 605).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 605 were tested and passed.

SUBTASK 05-55-26-810-009

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-003

(3) The connectors listed in Table 606 are used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 606/05-55-26-993-816 Right Wing to Body Fairing Connectors (MRB 20-160-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1278	D39928	26–11–21, 28–21–11, 36–11–22, 36–11–23, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–21, 78–35–21, 79–31–11	DM0056
W1292	D39986	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–12, 73–24–21, 78–32–61	DM0014
W1282	D39932	26–11–21, 26–12–11, 30–21–32, 36–11–23, 73–21–12, 73–22–11, 73–22–41, 74–11–11, 76–21–21, 77–12–21	DM0016
W1290	D39974	46–13–11, 73–21–11, 73–21–21, 73–24–12, 73–24–21, 78–32–61	DM0012
W1288	D39904	26–12–11, 27–81–22, 30–11–11	DM0002
W1664	D39943	23–41–11, 28–41–11, 28–44–11, 30–11–11, 33–42–21, 33–43–11, 33–44–12	DM0051
W1138	D39944	28–41–11	DM0054

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-008

(1) Close this access panel:

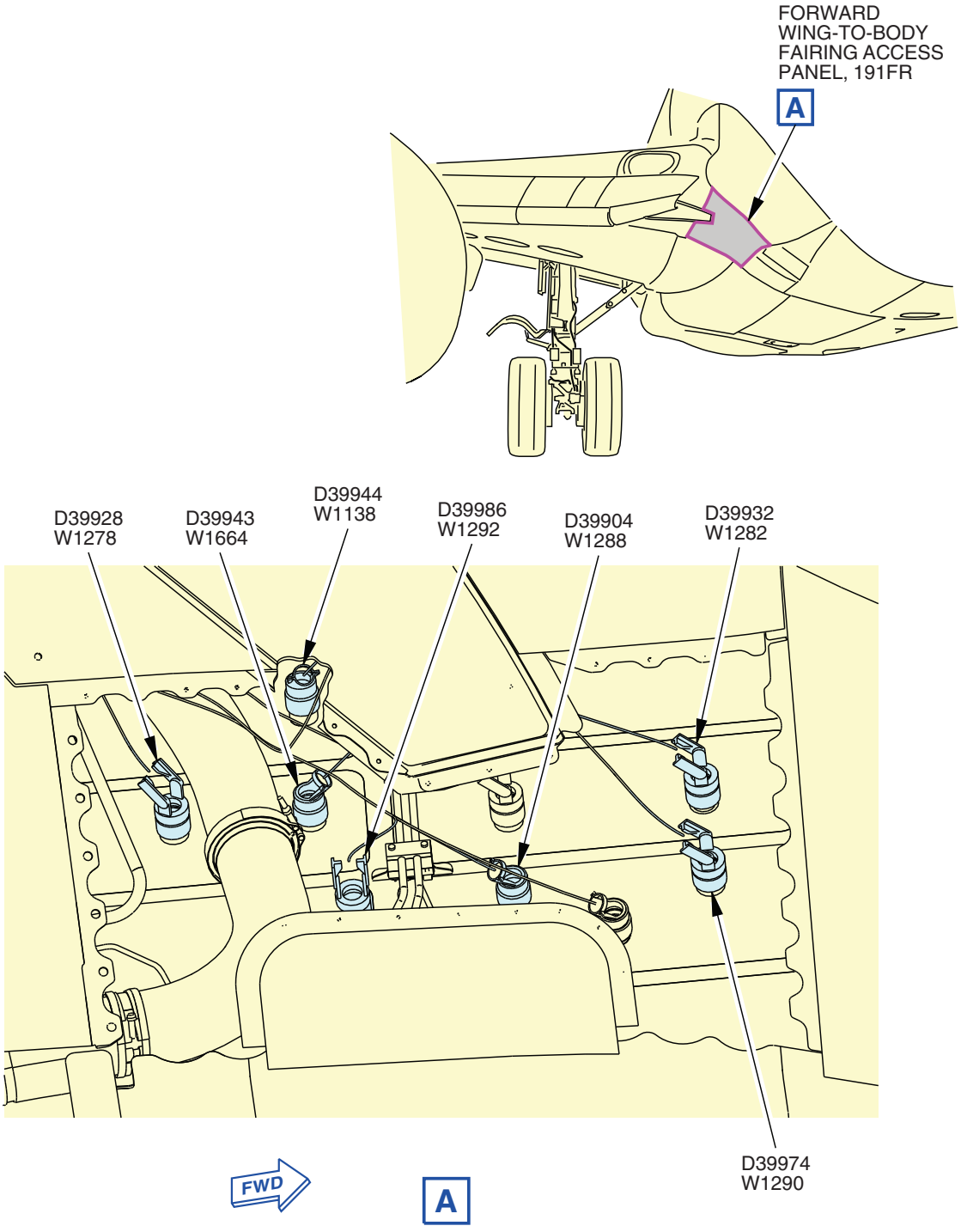
Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

SUBTASK 05-55-26-440-012

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-160-00) - Connector Location
Figure 603/05-55-26-990-823

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-806

5. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-161-00)

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CR	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-008

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-840-001

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-26-010-012

(3) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-019

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 607.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 607).

Table 607/05-55-26-993-848 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing (MRB 20-161-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W6230	D40058P		156/368		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 607).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 607 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-810-010

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-004

(3) The connector listed in Table 608 is used for the steps above.

Table 608/05-55-26-993-817 Right Wing to Body Fairing Connectors (MRB 20-161-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W6230	D40058P	21–51–24, 36–21–11	AB0536R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-009

(1) Close this access panel:

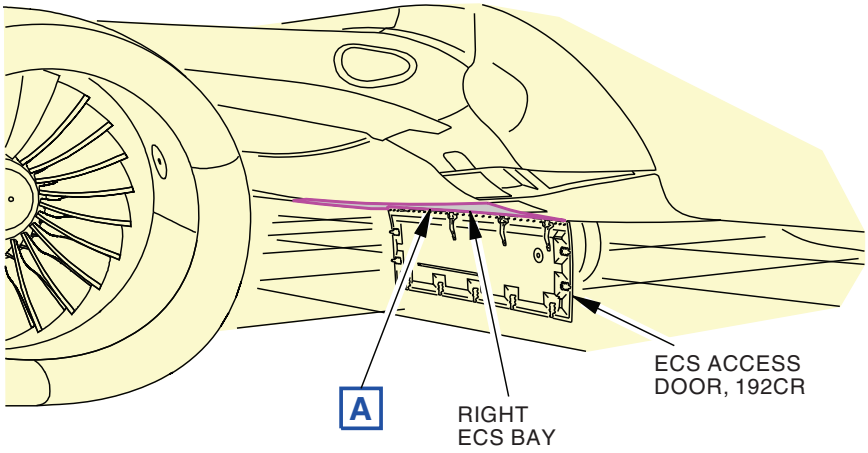
Number	Name/Location
192CR	ECS Access Door

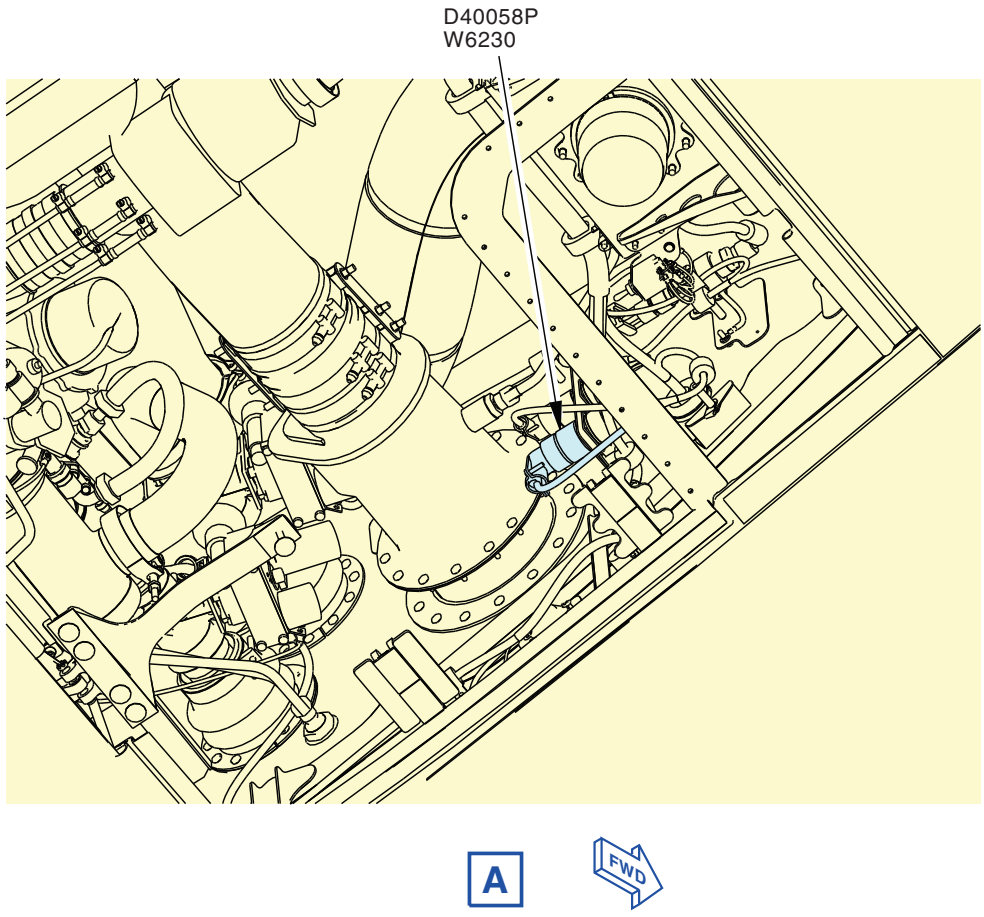
SUBTASK 05-55-26-840-002

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546510 S0000605115_V1

Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-161-00) - Connector Location
Figure 604/05-55-26-990-825

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-807

6. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-162-00)

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box

E. Access Panels

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-010

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-012

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-26-010-013

(3) Open this access panel:

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-020

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 609.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 609).

Table 609/05-55-26-993-849 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing (MRB 20-162-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1284	D39936		11/25		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 609).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 609 were tested and passed.

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SUBTASK 05-55-26-810-011

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-005

(3) The connector listed in Table 610 is used for the steps above.

Table 610/05-55-26-993-818 Right Wing to Body Fairing Connectors (MRB 20-162-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1284	D39936	24–11–21, 24–21–21, 24–24–21, 29–11–11	DM0018

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-010

(1) Close this access panel:

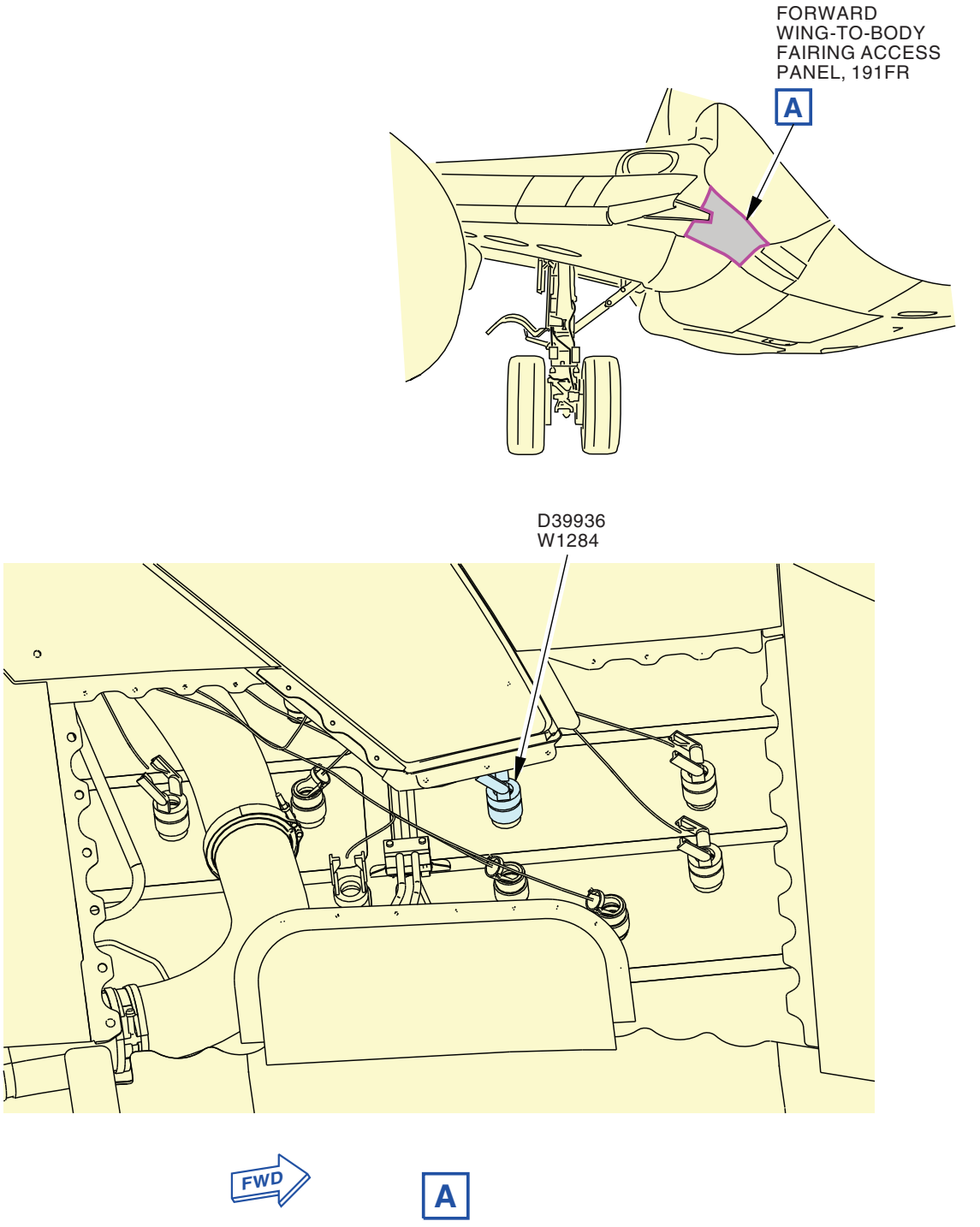
Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet

SUBTASK 05-55-26-440-013

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-162-00) - Connector Location
Figure 605/05-55-26-990-827

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-809

7. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-164-00)

(Figure 606)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CR	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-014

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-840-003

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-26-010-015

(3) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-021

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 611.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties, to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 611).

Table 611/05-55-26-993-850 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing (MRB 20-164-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/ Comments**
W6504	D40846J		7/16		
W6502	D40846P		8/18		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 611).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 611 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-810-013

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-007

(3) The connectors listed in Table 612 are used for the steps above.

Table 612/05-55-26-993-820 Right Wing to Body Fairing Connectors (MRB 20-164-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W6504	D40846J	21–42–11, 21–51–21	D40846P
W6502	D40846P	21–42–11, 21–51–21	D40846J

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-012

(1) Close this access panel:

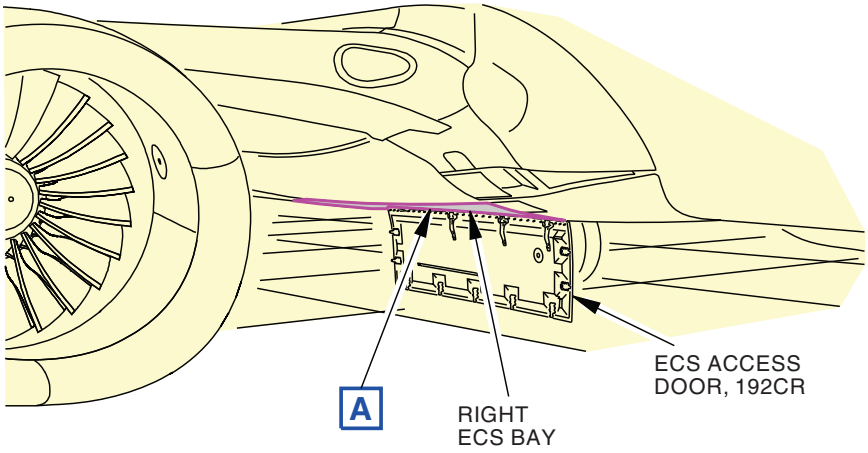
Number	Name/Location
192CR	ECS Access Door

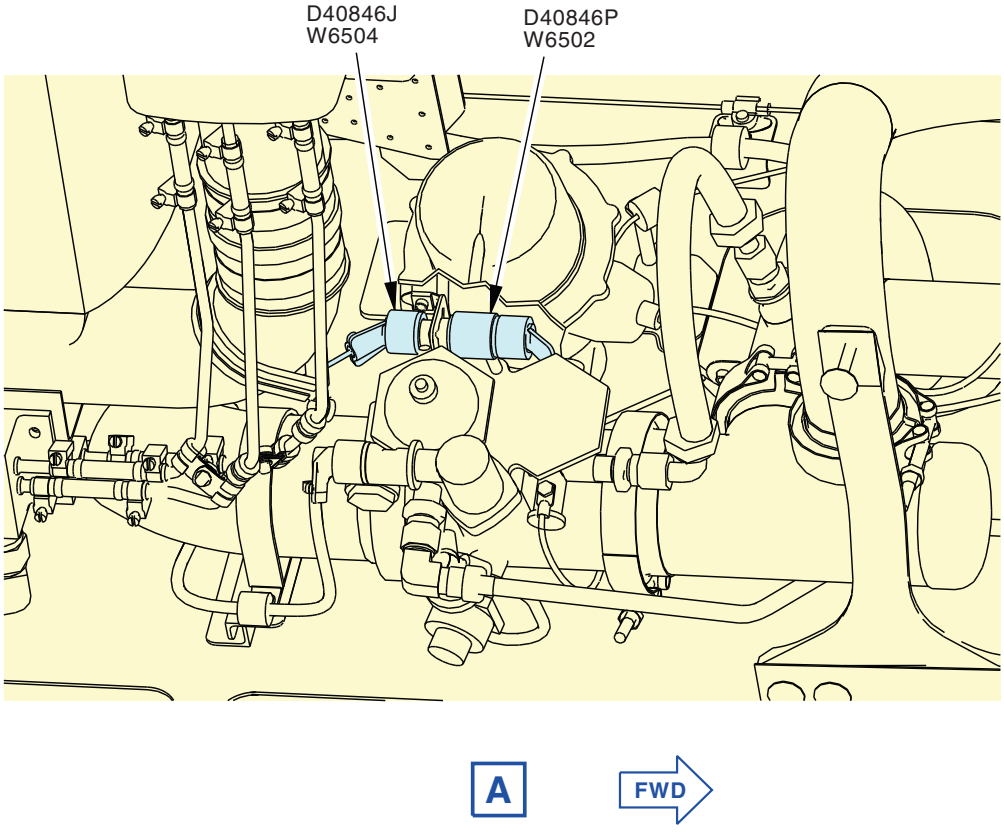
SUBTASK 05-55-26-840-004

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-164-00) - Connector Location
Figure 606/05-55-26-990-831

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-810

8. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-178-00)

(Figure 607)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
21-51-00-800-801	Air Cycle Air Conditioning and Temperature Control System - Deactivation (P/B 201)
21-51-00-800-802	Air Cycle Air Conditioning and Temperature Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CR	ECS Access Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-016

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-840-005

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Deactivation, TASK 21-51-00-800-801.

SUBTASK 05-55-26-010-016

(3) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-022

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 613.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 613).

Table 613/05-55-26-993-851 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing Connectors (MRB 20-178-00)

Plane:					
Data:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Vale/ Comments**
W6502	D41600		3/8		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 613).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 613 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-810-014

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-008

(3) The connector listed in Table 614 is used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 614/05-55-26-993-821 Right Wing to Body Fairing Connectors (MRB 20-178-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W6502	D41600	21–42–11, 21–51–21, 21–51–22, 21–61–15, 26–12–11, 33–33–11	DM0060

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-013

(1) Close this access panel:

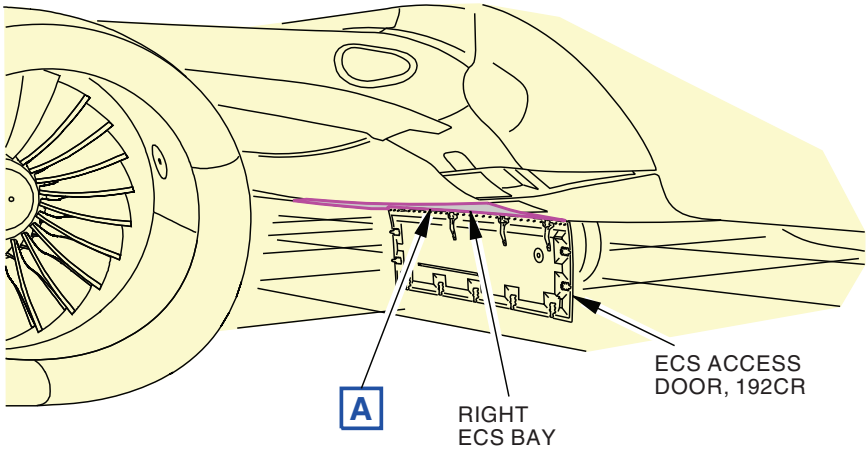
Number	Name/Location
192CR	ECS Access Door

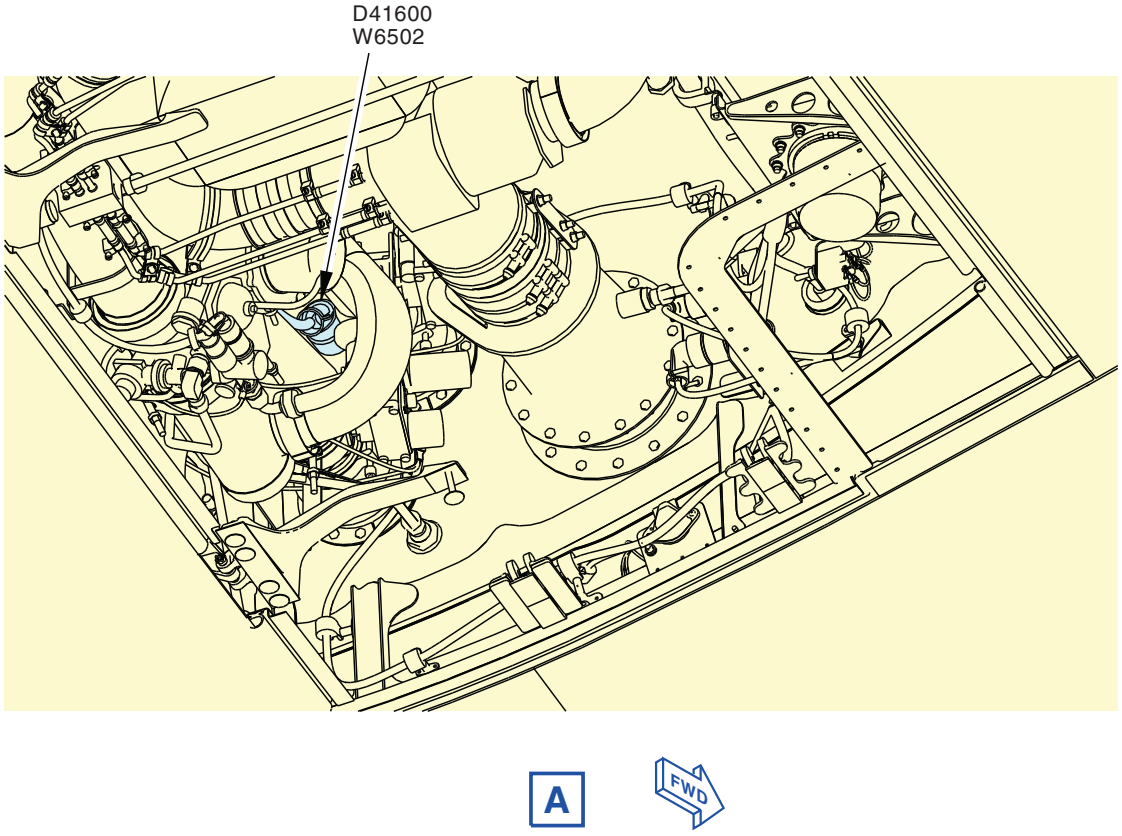
SUBTASK 05-55-26-840-006

(2) Do this task: Air Cycle Air Conditioning and Temperature Control System - Activation, TASK 21-51-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-178-00) - Connector Location
Figure 607/05-55-26-990-833

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-811

9. Loop Resistance Test of Wire Bundle - Right Wing Trailing Edge (MRB 20-156-00)

(Figure 608)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Access Panels

Number	Name/Location
651GB	Lower inboard fixed trailing edge

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-018

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-010-017

(2) Do one of the following step to get access to the wire bundle:

(a) Option I:

1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

2) Remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
651GB	Lower inboard fixed trailing edge

(b) Option II:

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

1) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-023

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) For each wire bundle listed in the Table 615, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This could require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 615).

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Table 615/05-55-26-993-852 Wire Bundle Loop Resistance Values - Right Wing Trailing Edge (MRB 20-156-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1034	D40034J		8/20		
W1032	D40034P		7/16		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 615).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 615 were tested and passed.

SUBTASK 05-55-26-810-015

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

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1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-009

(3) The connectors listed in Table 616 are used for the steps above.

Table 616/05-55-26-993-822 Right Wing Trailing Edge Connectors (MRB 20-156-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1034	D40034J	22–11–11, 22–12–41, 27–18–11, 27–32–11, 27–52–11, 27–53–21, 28–21–11	AW0254R
W1032	D40034P	22–11–11, 22–12–41, 27–18–11, 27–32–11, 27–52–11, 27–53–21, 28–21–11	AW0254R

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-014

(1) Do one of the following step:

(a) Option I:

1) Install this access panel:

(TASK 57-50-02-400-801)

Number	Name/Location
651GB	Lower inboard fixed trailing edge

2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Option II:

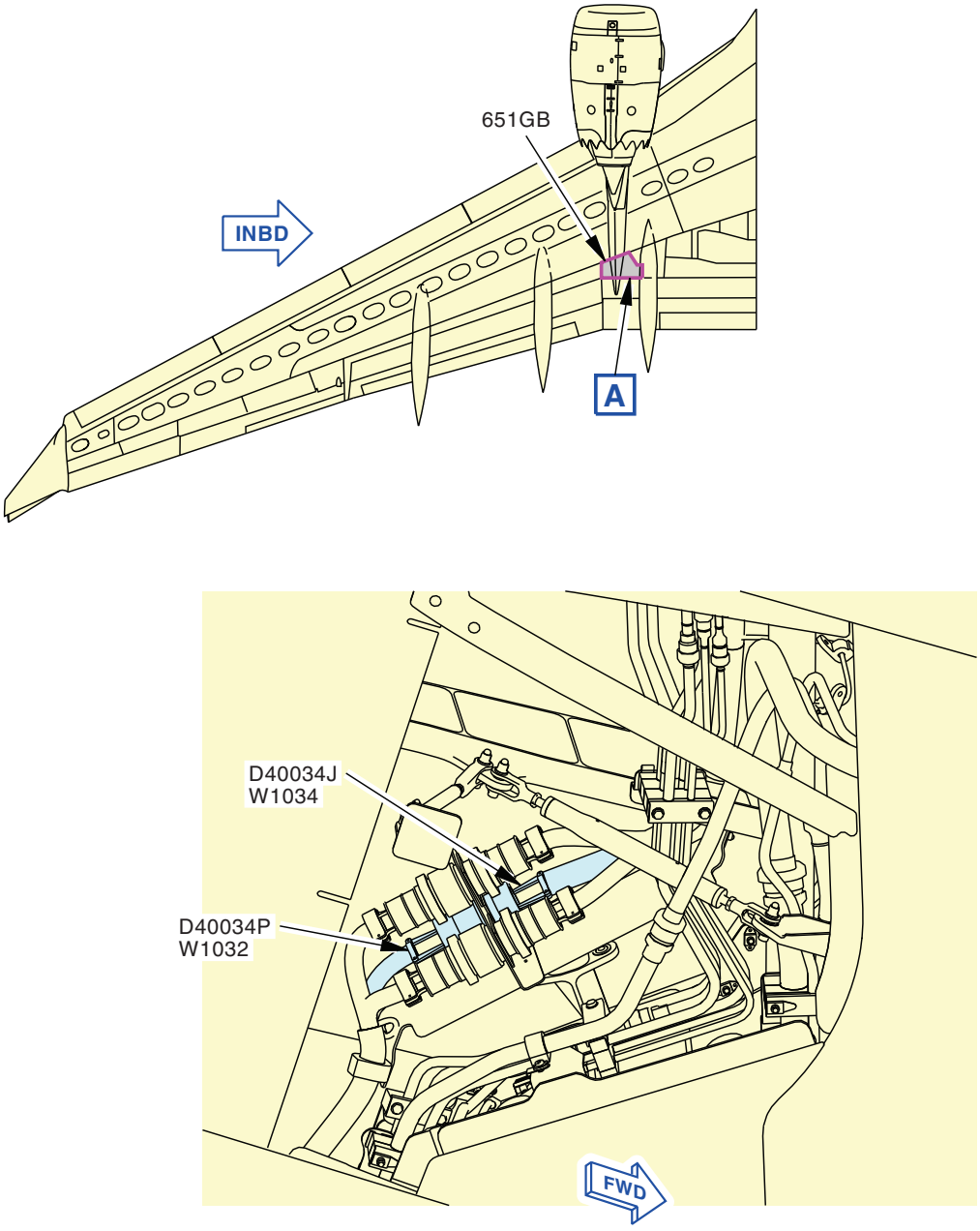
1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

2) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546534 S0000605127_V2

Loop Resistance Test of Wire Bundle - Right Wing Trailing Edge (MRB 20-156-00) - Connector Location
Figure 608/05-55-26-990-835

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-812

10. Loop Resistance Test of Wire Bundle - Right Wing Trailing Edge (MRB 20-159-00)

(Figure 609)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Access Panels

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-020

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-022

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 05-55-26-010-026

(3) Do one of the two steps that follow to get access to the wire bundle:

(a) To get access by the Main Landing Gear (MLG) (Figure 610).

(b) To get access by the removal of panel 651BB: (Figure 609).

1) Remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-024

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 617.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 617).

Table 617/05-55-26-993-853 Wire Bundle Loop Resistance Values - Right Wing Trailing Edge (MRB 20-159-00)

Plane:					
Data:					
Technician:					
Bundle Number	**Ground Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1134	GD04426		30/70		

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 617).

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4) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 617 were tested and passed.

SUBTASK 05-55-26-720-011

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-010

(3) The connector listed in the Table 618 is used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 618/05-55-26-993-823 Right Wing Trailing Edge Connectors (MRB 20-159-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1134	D16340	28–23–11	M03048

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-015

(1) If removed, install this access panel:

(TASK 57-50-02-400-801)

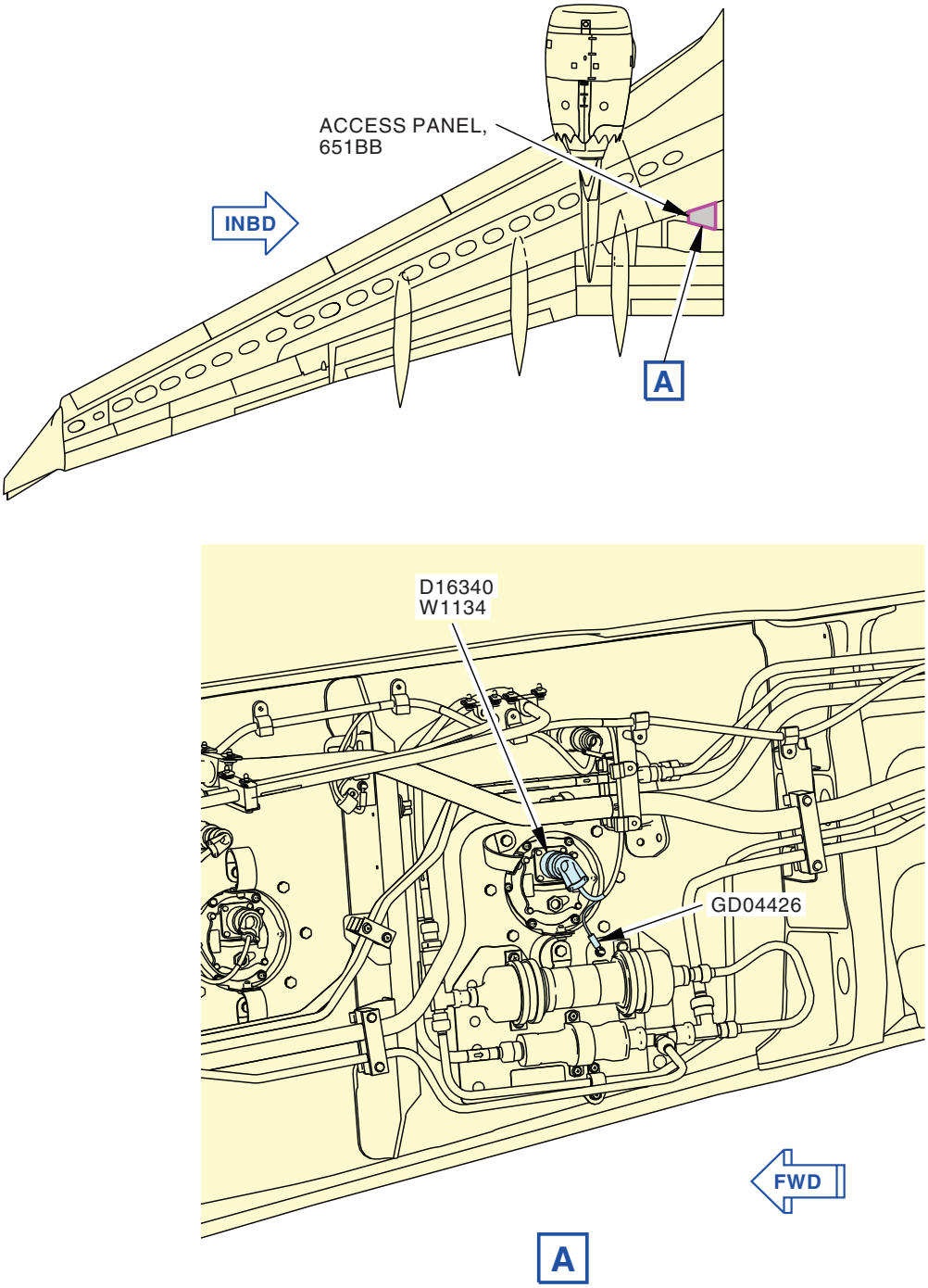
Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door

SUBTASK 05-55-26-440-016

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

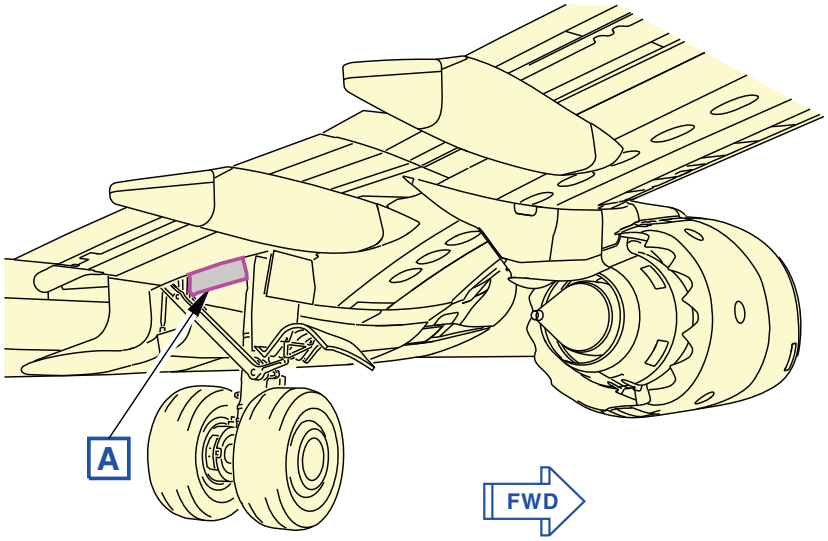
ECCN 9E991 BOEING PROPRIETARY - See title page for details

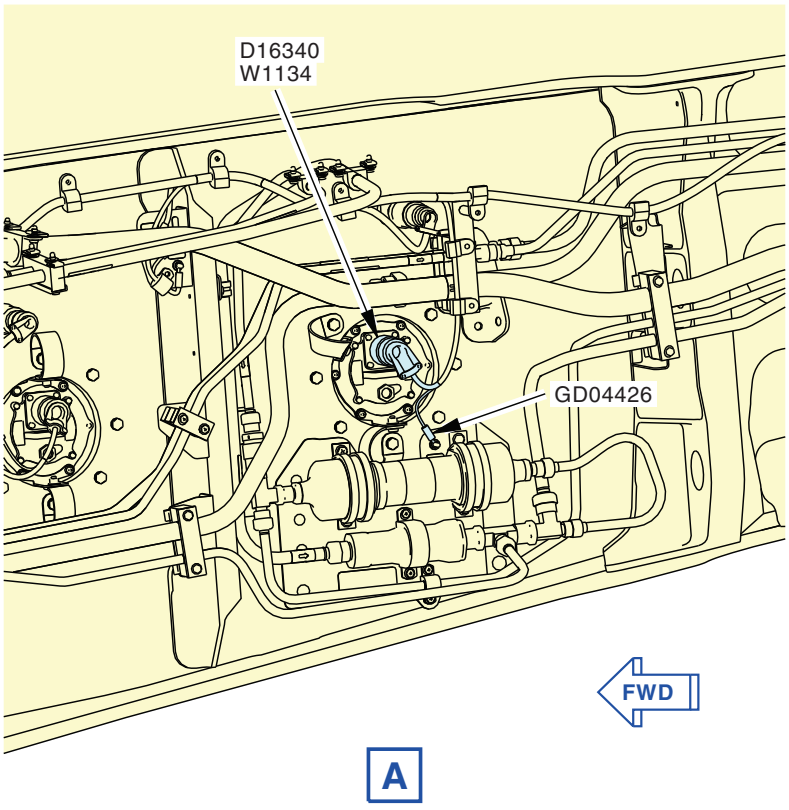


2546548 S0000605130_V1

Loop Resistance Test of Wire Bundle - Right Wing Trailing Edge (MRB 20-159-00) - Connector Location - Access Panel
Figure 609/05-55-26-990-837

ECCN 9E991 BOEING PROPRIETARY - See title page for details





3051801 S0000816986_V1

Loop Resistance Test of Wire Bundle - Right Wing Trailing Edge (MRB 20-159-00) - Connector Location - Main Landing Gear
Figure 610/05-55-26-990-858

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-813

11. Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-154-00)

(Figure 611)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
621	Right Wing - Leading Edge to Front Spar

E. Access Panels

Number	Name/Location
621BB	Outboard Lower Fixed Leading Edge Access Panel

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-022

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-016

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-26-010-019

(3) Open this access panel:

Number	Name/Location
621BB	Outboard Lower Fixed Leading Edge Access Panel

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-025

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 619.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 619).

Table 619/05-55-26-993-854 Wire Bundle Loop Resistance Values - Right Wing Leading Edge (MRB 20-154-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1288	D00736		77/182		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 619).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 619 were tested and passed.

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SUBTASK 05-55-26-810-016

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-011

(3) The connector listed in Table 620 is used for the steps above.

Table 620/05-55-26-993-824 Right Wing Leading Edge Connectors (MRB 20-154-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1288	D00736	30–11–11	S00117

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-016

(1) Close this access panel:

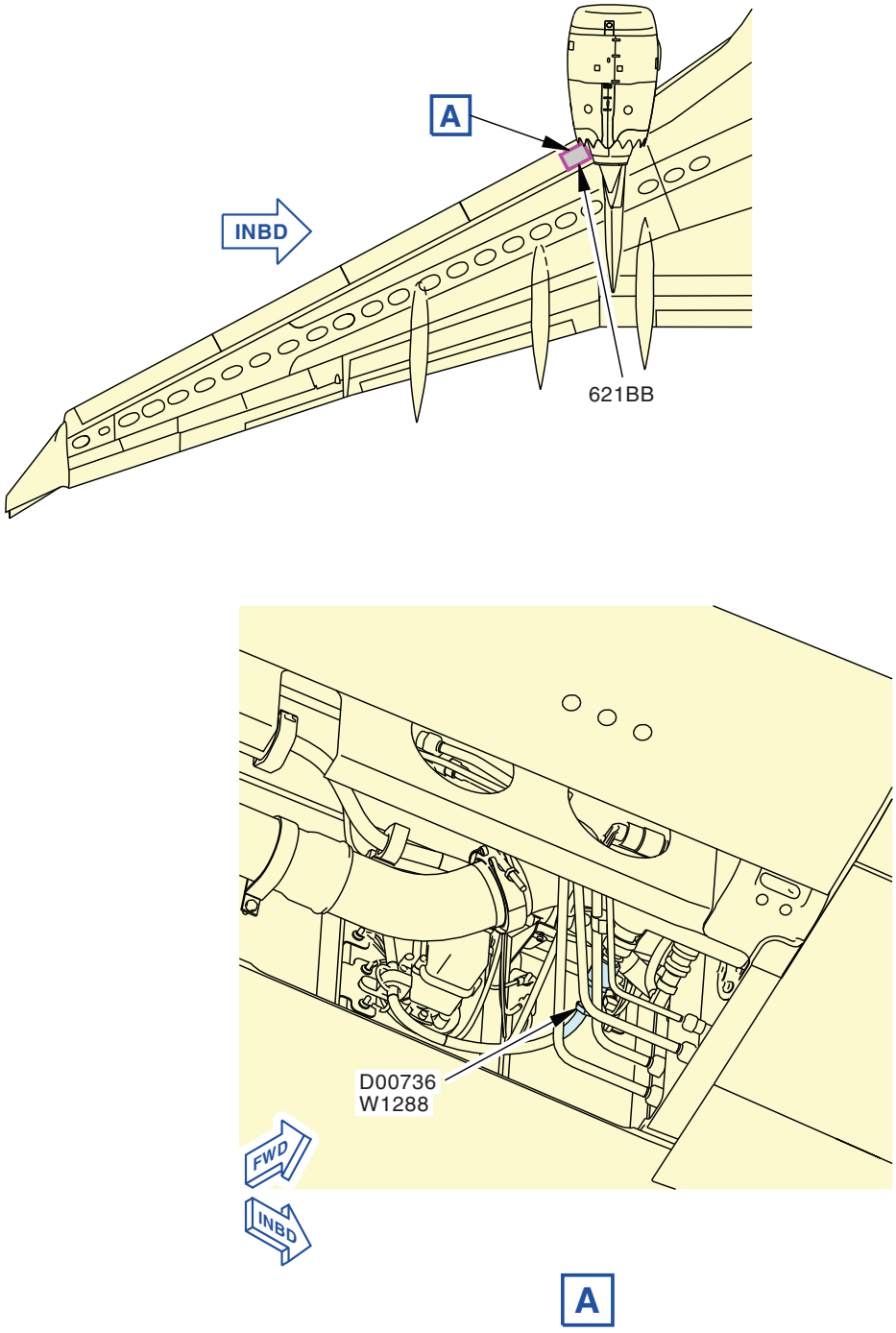
Number	Name/Location
621BB	Outboard Lower Fixed Leading Edge Access Panel

SUBTASK 05-55-26-440-017

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546557 S0000605133_V2

Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-154-00) - Connector Location
Figure 611/05-55-26-990-839

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-814

12. Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-157-00)

(Figure 612)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
441	Engine 2 - Forward Strut Fairing

E. Access Panels

Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-024

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-017

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-26-010-020

(3) Open this access panel:

Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-026

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 621.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 621).

Table 621/05-55-26-993-855 Wire Bundle Loop Resistance Values - Right Wing Leading Edge (MRB 20-157-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1266	D30164		9/22		
W1268	D30184		9/22		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 621).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 621 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-810-017

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-012

(3) The connectors listed in Table 622 are used for the steps above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 622/05-55-26-993-825 Right Wing Leading Edge Connectors (MRB 20-157-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1266	D30164	26–11–21, 26-12-11, 30-21-32, 36-11-23, 73-21-12, 73-22-11, 73-22-41, 74-11-11, 76-21-21, 77-12-21	DM0020
W1268	D30184	46-13-11, 73-21-11, 73-21-21, 73-24-12, 73-21-21, 78-32-61	DM0028

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-017

(1) Close this access panel:

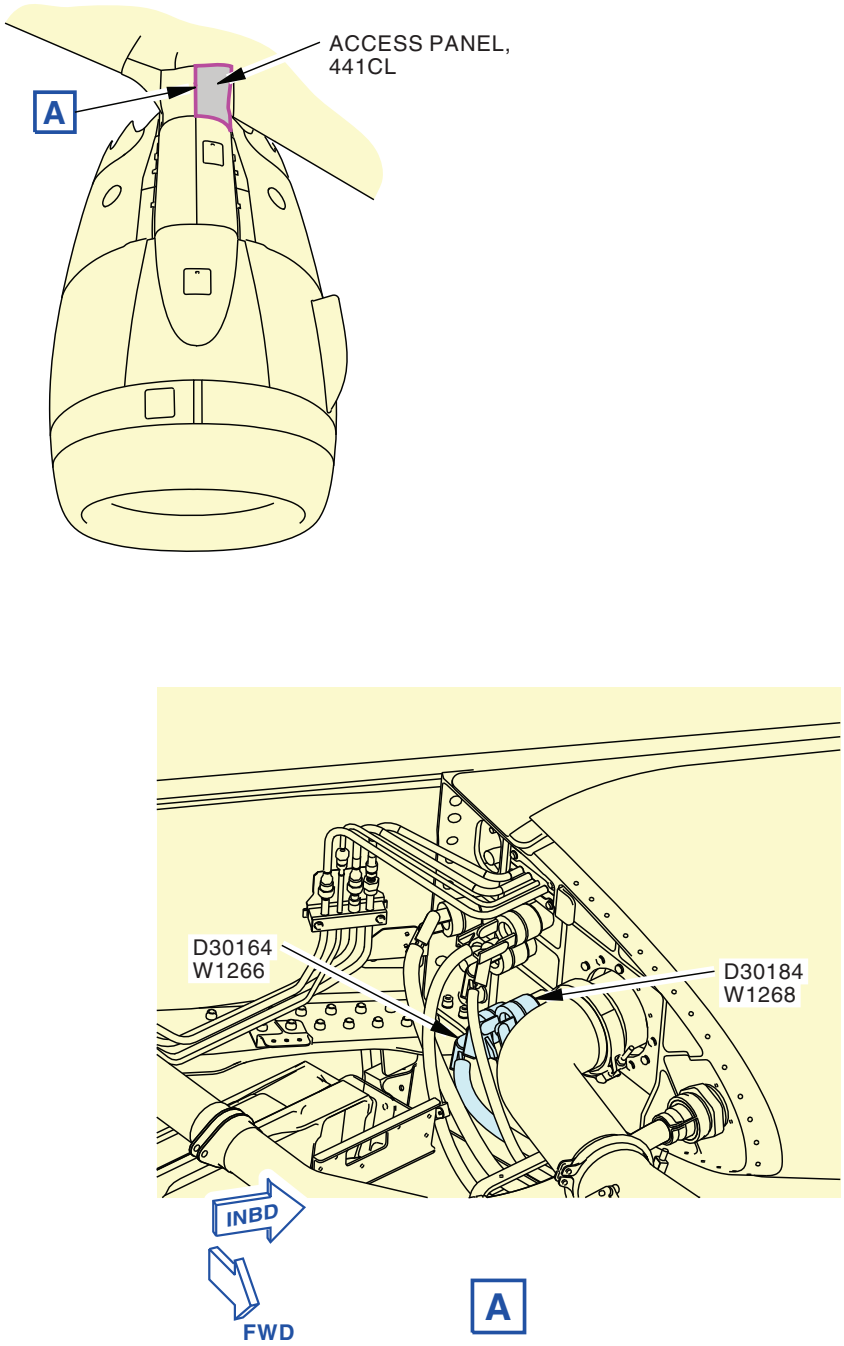
Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

SUBTASK 05-55-26-440-018

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-157-00) - Connector Location
Figure 612/05-55-26-990-841

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-815

13. Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-158-00)

(Figure 613)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
441	Engine 2 - Forward Strut Fairing
611	Right Wing - Leading Edge to Front Spar

E. Access Panels

Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
611AB	Inboard Leading Edge, Lower Removable Access Panel

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-026

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-040-018

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 05-55-26-010-021

(3) Open these access panels:

Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
611AB	Inboard Leading Edge, Lower Removable Access Panel

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-027

(1) Do the functional check of the wire bundle shielding:

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 623:

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 623).

Table 623/05-55-26-993-856 Wire Bundle Loop Resistance Values - Right Wing Leading Edge (MRB 20-158-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1284	D30114		10/23		
W1264	D30116		10/24		
W1272	D30142		8/19		
W1278	D30148		6/13		
W1270	D30152		5/12		
W1282	D30162		6/15		
W1292	D30168		7/15		

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 623/05-55-26-993-856 Wire Bundle Loop Resistance Values - Right Wing Leading Edge (MRB 20-158-00) (Continued)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1290	D30192		7/17		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 623).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 623 were tested and passed.

SUBTASK 05-55-26-810-018

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

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a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-013

(3) The connectors listed in Table 624 are used for the steps above.

Table 624/05-55-26-993-826 Right Wing Leading Edge Connectors (MRB 20-158-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1284	D30114	24–11–21, 24–21–21, 24–24–21, 29–11–11	DM0024
W1264	D30116	24–11–21, 24–21–21, 24–24–21, 29–11–11	DM0024
W1272	D30142	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–12, 73–24–21, 78–32–61	DM0026
W1278	D30148	26–11–21, 36–11–22, 36–11–23, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–21, 78–35–21, 79–31–11	DM0038
W1270	D30152	26–11–21, 36–11–22, 36–11–23, 73–21–12, 73–22–11, 74–11–11, 77–12–11, 78–34–21, 78–35–21, 79–31–11	DM0038
W1282	D30162	26–11–21, 26–12–11, 30–21–32, 36–11–23, 73–21–12, 73–22–11, 73–22–41, 74–11–11, 76–21–21, 77–12–21	DM0020
W1292	D30168	46–13–11, 73–21–11, 73–21–12, 73–21–21, 73–24–12, 73–24–21, 78–32–61	DM0026
W1290	D30192	46–13–11, 73–21–11, 73–21–21, 73–24–12, 73–24–21, 78–32–61	DM0028

H. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-410-018

(1) Close these access panels:

Number	Name/Location
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2
611AB	Inboard Leading Edge, Lower Removable Access Panel

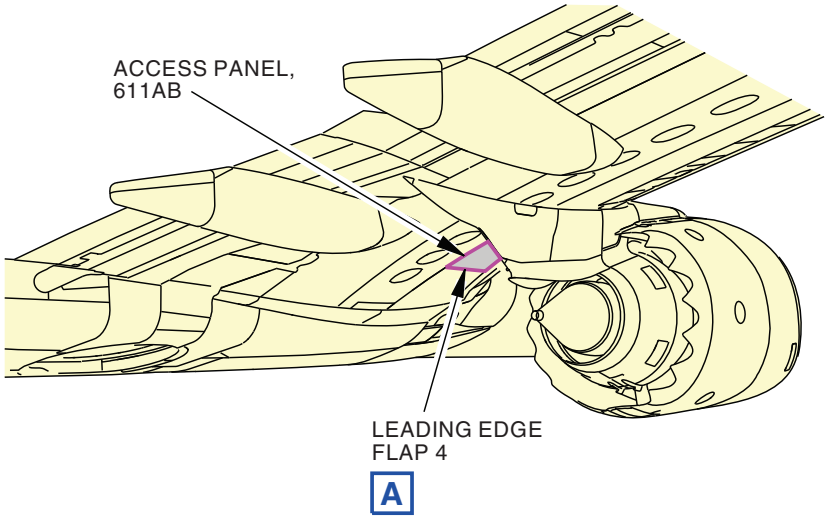
ECCN 9E991 BOEING PROPRIETARY - See title page for details

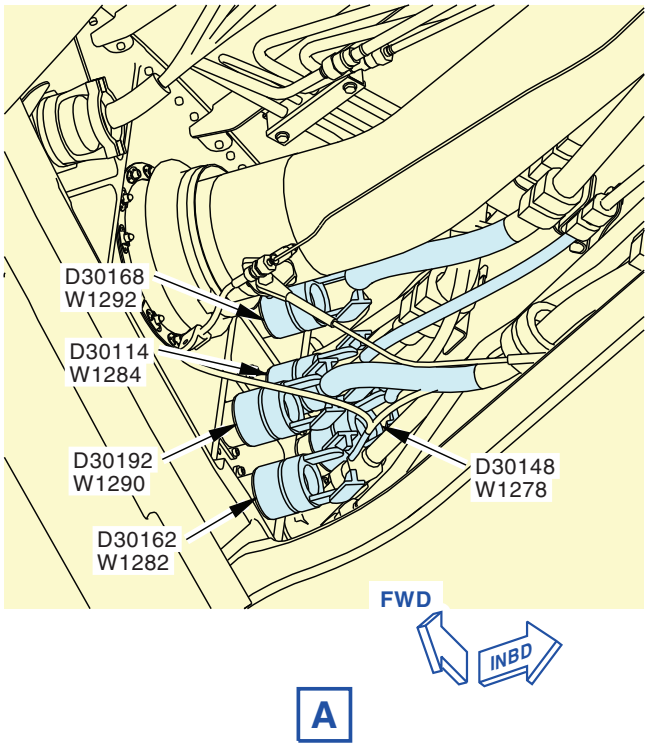
SUBTASK 05-55-26-440-019

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

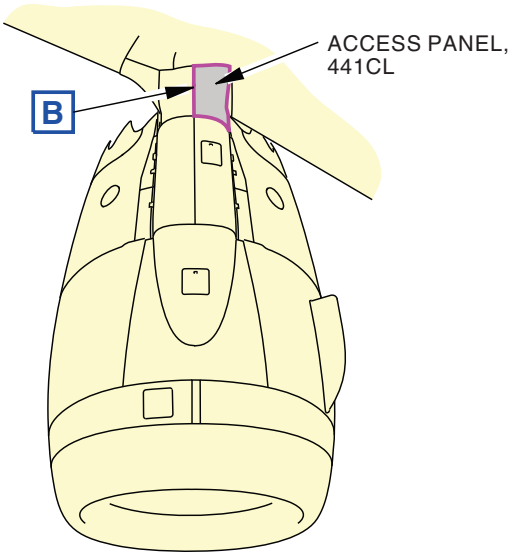


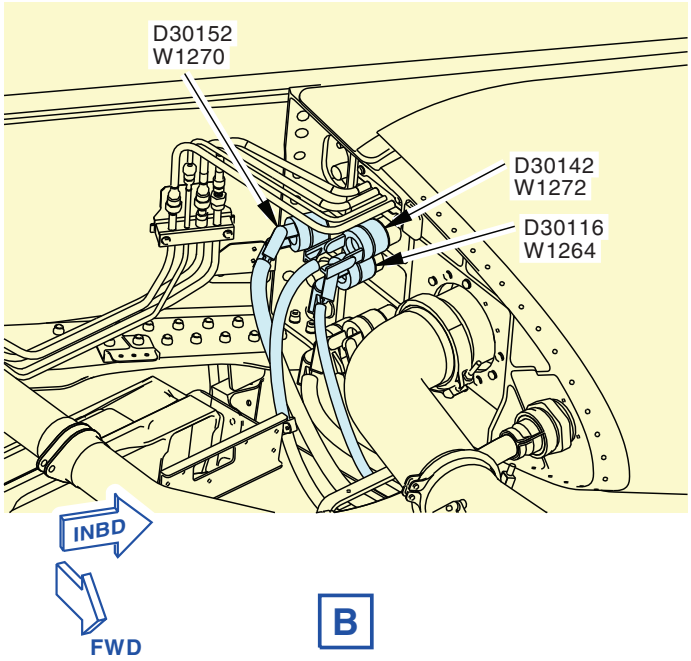


2546522 S0000605231_V1

Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-158-00) - Connector Location
Figure 613/05-55-26-990-843 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546527 S0000605232_V1

Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-158-00) - Connector Location
Figure 613/05-55-26-990-843 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-816

14. Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-177-00)

(Figure 614)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
610	Subzone - Right Wing: Leading Edge, Forward of Front Spar, Inboard of Nacelle Strut, Including Gap Cover Area

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-028

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-840-007

(2) Do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 05-55-26-040-019

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-028

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 625.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 625).

Table 625/05-55-26-993-857 Wire Bundle Loop Resistance Values - Right Wing Leading Edge (MRB 20-177-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1138	GD00193		11/36		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 625).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 625 were tested and passed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-26-810-019

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-014

(3) The connector listed in Table 626 is used for the steps above.

Table 626/05-55-26-993-827 Right Wing Leading Edge Connectors (MRB 20-177-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W1138	GD00193	28–41–11	M02946

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-26-440-020

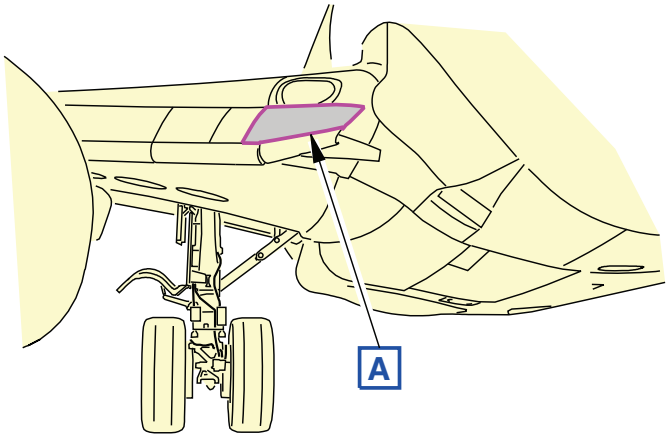
(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

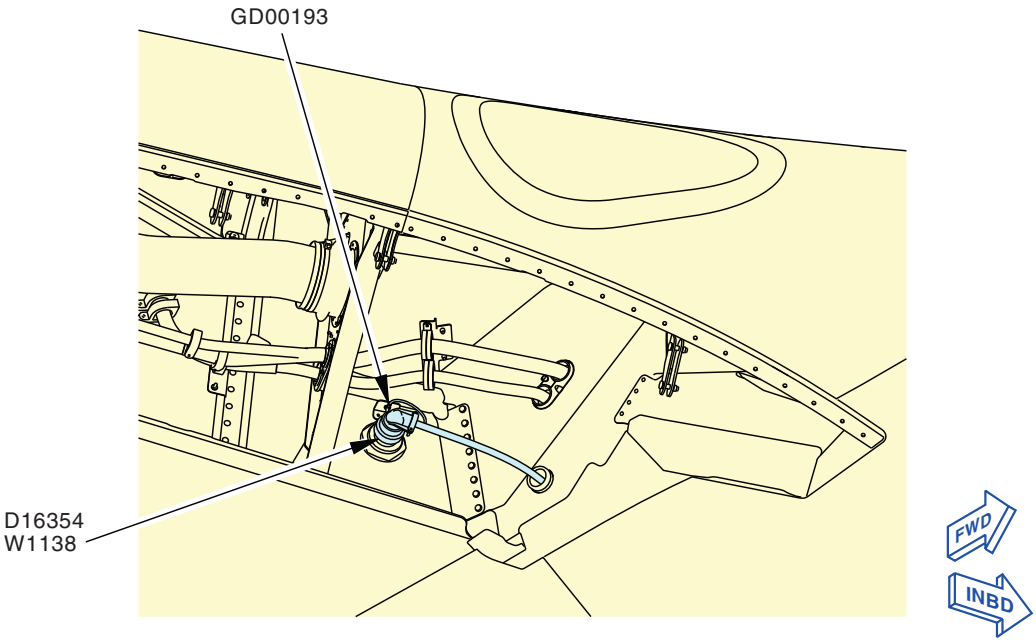
SUBTASK 05-55-26-840-008

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2546477 S0000605234_V2

Loop Resistance Test of Wire Bundle - Right Wing Leading Edge (MRB 20-177-00) - Connector Location
Figure 614/05-55-26-990-845

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233

TASK 05-55-26-280-817

15. Loop Resistance Test of Wire Bundle - Right Engine (MRB 20-182-00)

(Figure 615 or Figure 616)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
441	Engine 2 - Forward Strut Fairing
611	Right Wing - Leading Edge to Front Spar

E. Access Panels

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-030

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233 (Continued)

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-010-022

(2) Open these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-029

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 627.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This can require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 627).

Table 627/05-55-26-993-858 Wire Bundle Loop Resistance Values - Right Engine (MRB 20-182-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W1062	D30038		6/15		
W1262	D08194P		4/12		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 627).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 627 were tested and passed.

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PTF ALL PRE SB 737-24-1233 (Continued)

SUBTASK 05-55-26-810-020

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-015

(3) The connectors listed in Table 628 are used in the steps above.

Table 628/05-55-26-993-828 Right Engine Connectors (MRB 20-182-00)

Wire Bundle	Connector	WDM	Mate With Equipment
1062	D30038	24-21-11	AS0004R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-24-1233 (Continued)

Table 628/05-55-26-993-828 Right Engine Connectors (MRB 20-182-00) (Continued)

Wire Bundle	Connector	WDM	Mate With Equipment
1262	D08194P	24-21-11	AW0258R

H. Put the Airplane Back to Its Usual Condition

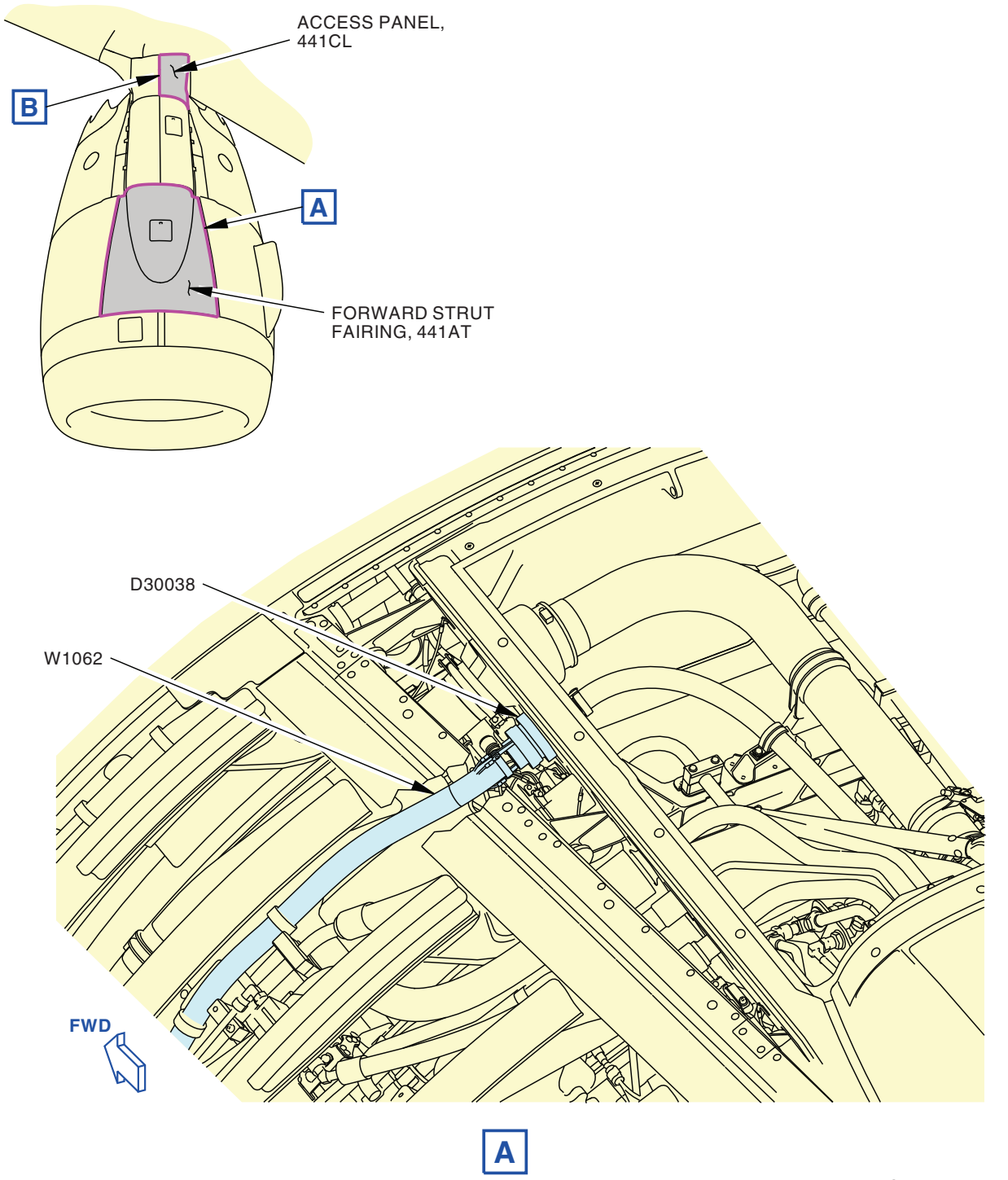
SUBTASK 05-55-26-410-019

(1) Close these access panels:

Number	Name/Location
441AT	Forward Strut Fairing, Thumbnail Fairing, Strut 2
441CL	Forward Strut Fairing, Left Overwing Fairing, Strut 2

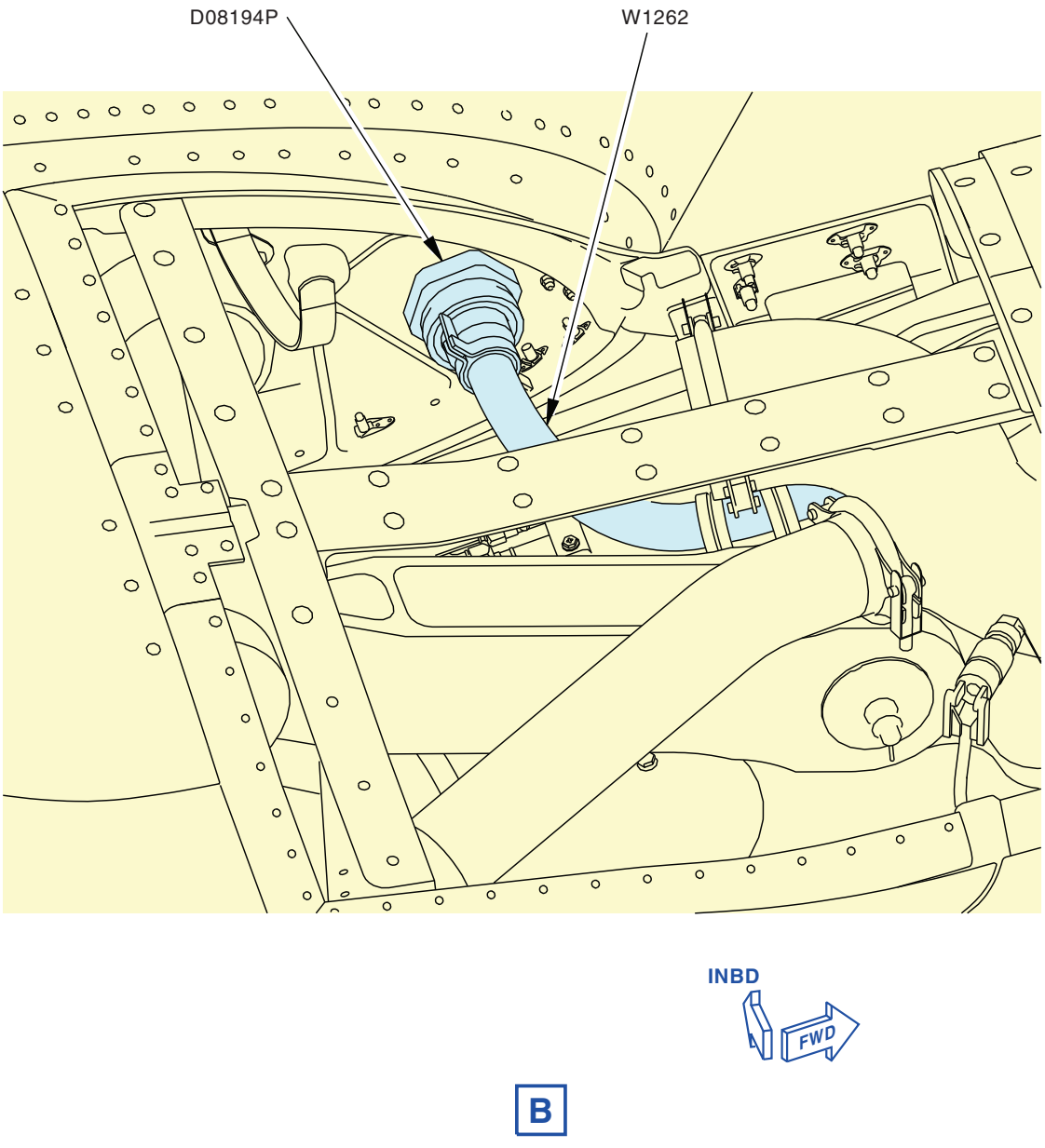
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Right Engine (MRB 20-182-00)
Figure 615/05-55-26-990-847

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2832833 S0000658039_V1

Loop Resistance Test of Wire Bundle - Right Engine (MRB 20-182-00)
Figure 616/05-55-26-990-848

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 05-55-26-280-818

16. Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-183-00)

(Figure 617)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-031

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-860-003

(2) Make sure that the landing gear downlock pins are installed (TASK 32-00-01-480-801).

F. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-031

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 629.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 629).

Table 629/05-55-26-993-859 Wire Bundle Loop Resistance Values - Right MLG Wheel Well (MRB 20-183-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W5204	D13838		23/55		
W63	D14112		3.7/9		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 629).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 629 were tested and passed.

SUBTASK 05-55-26-810-023

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-016

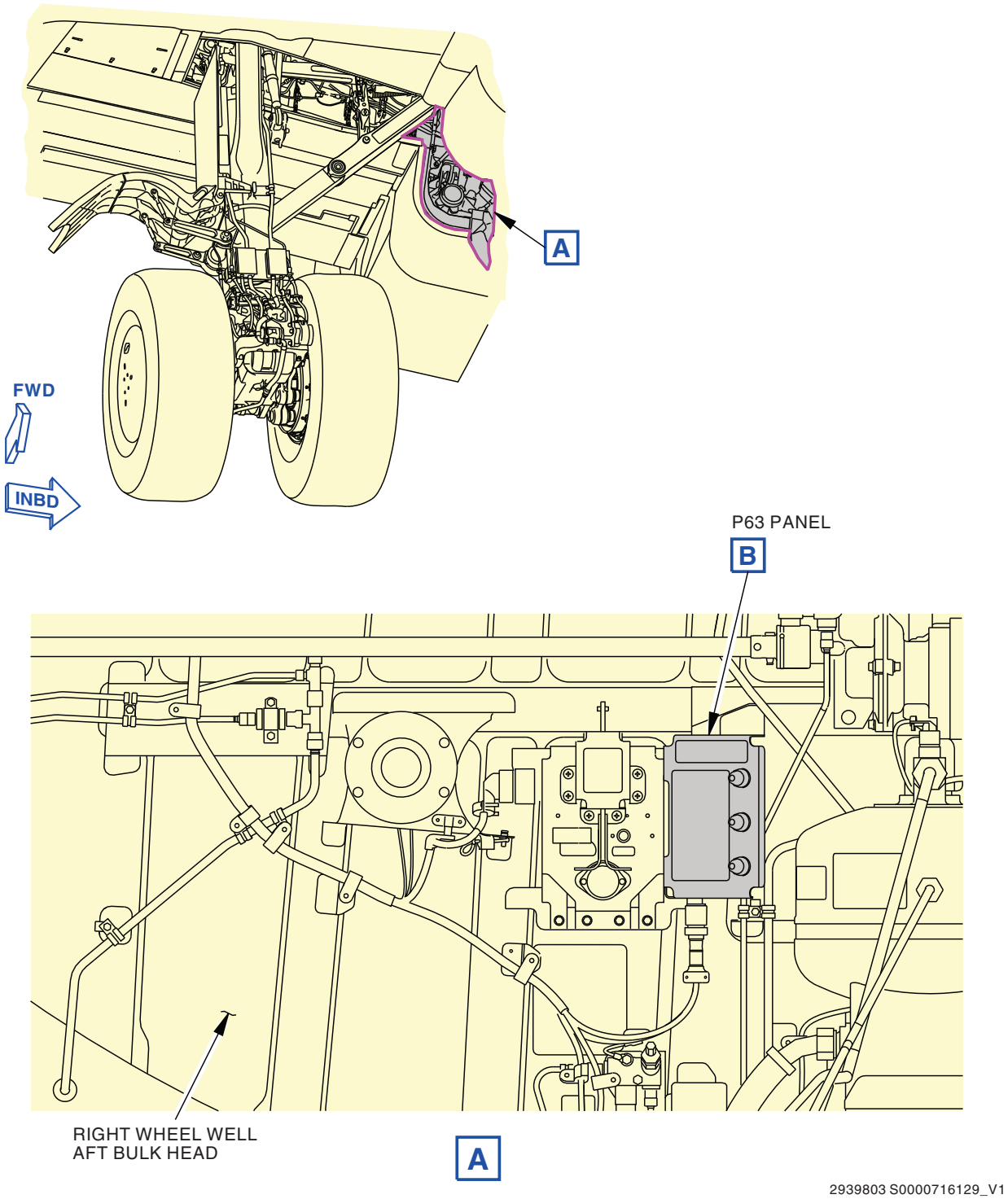
(3) The connectors listed in Table 630 are used for the steps above.

Table 630/05-55-26-993-860 Right MLG Wheel Well Connectors (MRB 20–183–00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5204	D13838	47-30-11	P63
W63	D14112	47-30-11	P63

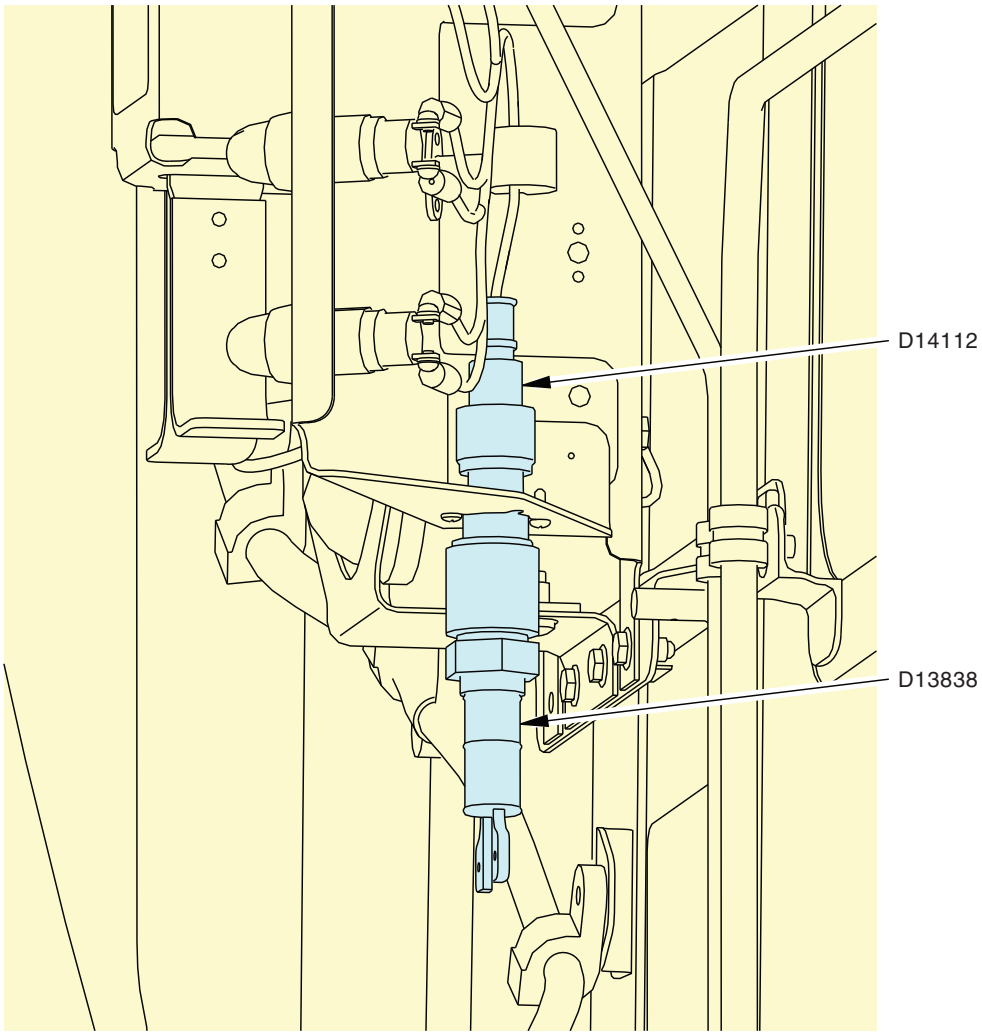
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-183-00)
Figure 617/05-55-26-990-850 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundle - Right MLG Wheel Well (MRB 20-183-00)
Figure 617/05-55-26-990-850 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-26-280-822

17. Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-189-00)

(Figure 618)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-26-970-035

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-26-010-024

(2) Open this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-26-720-036

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the (Table 631).

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 631).

Table 631/05-55-26-993-867 Wire Bundle Loop Resistance Values - Right Wing to Body Fairing Connectors (MRB 20-189-00)

Plane:					
Data:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Vale/ Comments**
W8122	D10533		6/16		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 631).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 631 were tested and passed.

SUBTASK 05-55-26-810-027

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

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1) Check the test setup.
   a) Make sure that the couplers are only around the desired wire bundle and closed properly.
2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.
3) If no problem is found, check the entire length of the wire bundle.
   a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

      NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

   b) If a problem is found, repair or replace (SWPM 20).
4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:
1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).
   a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).
   b) If the measured resistance value stays more than the given maximum value for the joint, do this step:
      <1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-26-800-020

(3) The connectors/grounds listed in Table 632 is used for the steps above.

Table 632/05-55-26-993-868 Right Wing to Body Fairing Connectors (MRB 20-189-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W8122	D10533	27-81-51	V00162

H. Put the Airplane Back to Its Usual Condition

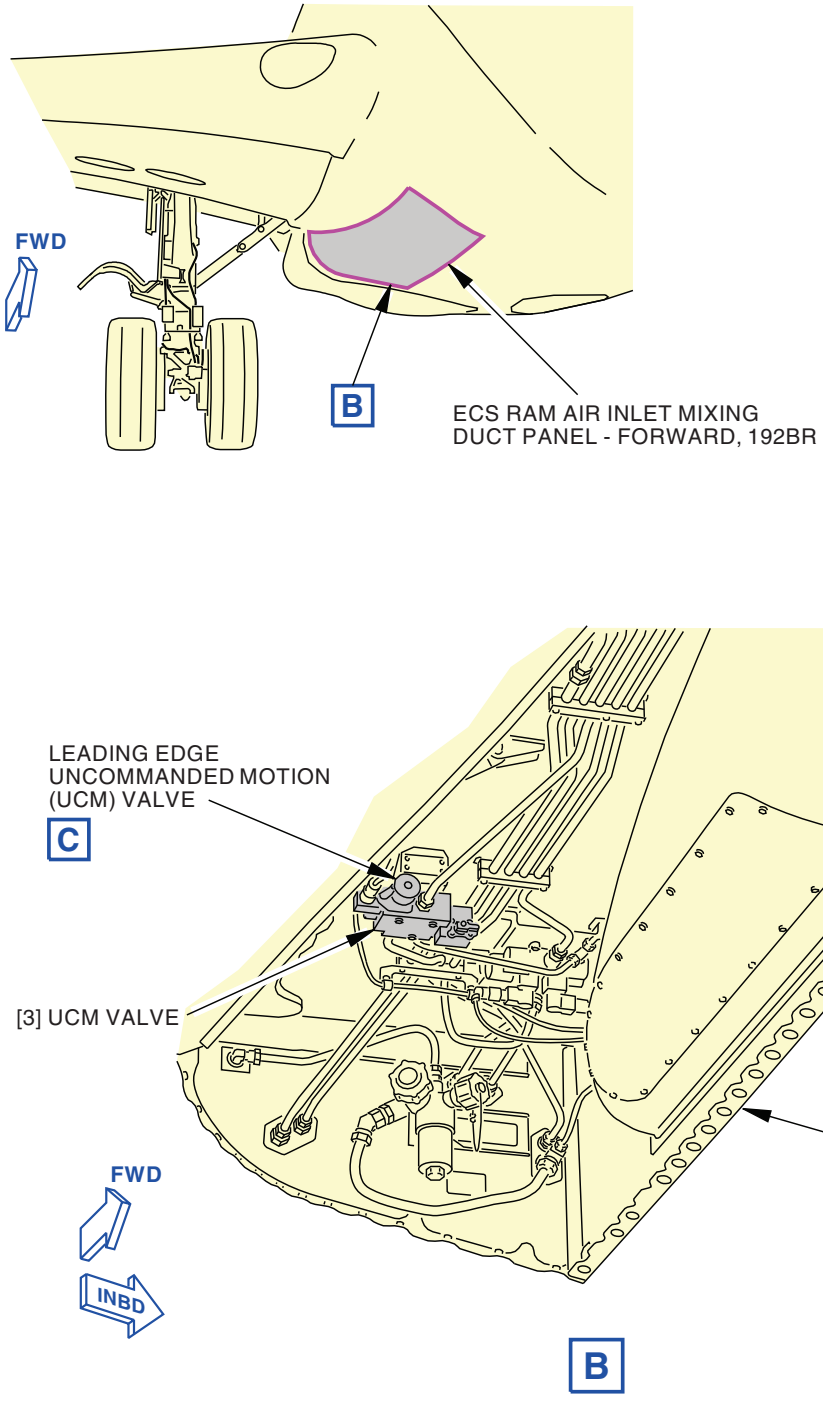
SUBTASK 05-55-26-410-021

(1) Close this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

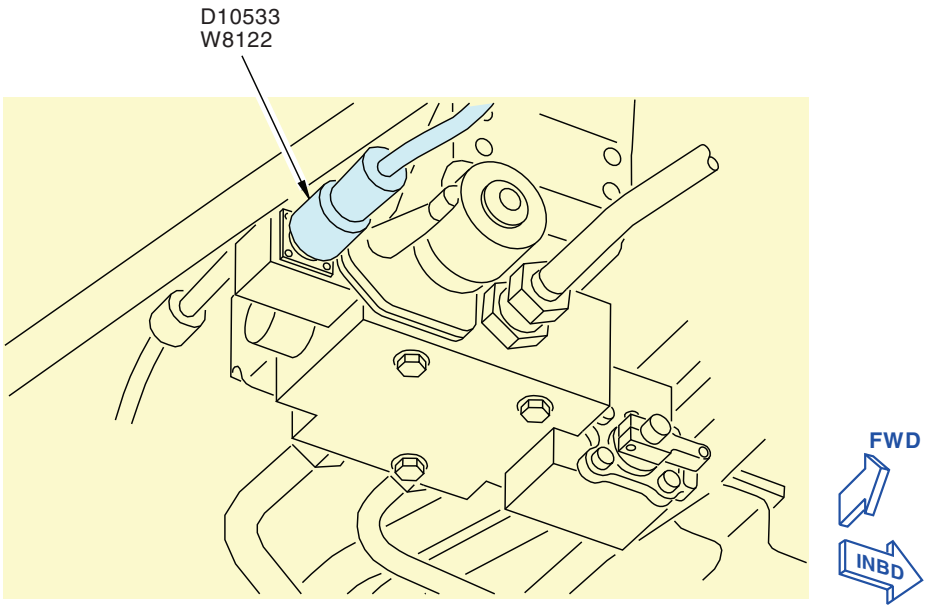
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-189-00) - Connector Location
Figure 618/05-55-26-990-851 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test of Wire Bundle - Right Wing to Body Fairing (MRB 20-189-00) - Connector Location
Figure 618/05-55-26-990-851 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMPENNAGE WIRE BUNDLES - INSPECTION/CHECK

1. General

A. This procedure contains tasks that perform a loop resistance test of wire bundles in the empennage area of the airplane.

TASK 05-55-46-280-801

2. Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-165-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-002

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-001

(2) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-008

(1) Do the Functional Check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for each wire bundle listed in the Table 601, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 601).

Table 601/05-55-46-993-812 Wire Bundle Loop Resistance Values - Tailcone Access Compartment (MRB 20-165-00)

Plane:						
Date:						
Technician:						
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min Value (mΩ)**	**Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comment**
W5726	D40790P		29	40		
	D40854P		32	40		
W3377	D43397P		65	154		
W3379	D43399P		55	70		
W6134*[1]	D40842P		500	800		

*[1] A minimum of one inch of coupler separation is required.

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 601).

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4) If the measured value is not within the MIN/MAX value range, do the Troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 601 were tested and passed.

SUBTASK 05-55-46-810-010

(2) Do the Troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

(b) Record all the measurement data to the data sheet (Table 601).

NOTE: This should include both initial and final values.

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the LRT.

a) Check the LRT operation (TASK 05-56-02-200-801 or TASK 05-56-02-200-802).

3) If no problem is found, check the entire length of the wire bundle.

4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.

5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:

1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-001

(3) The connectors listed in Table 602 are used for the steps above.

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Table 602/05-55-46-993-801 Tailcone Access Compartment Connectors (MRB 20-165-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W5726	D40790P	24–24–31, 49–52–31, 49–62–12, 49–71–21, 49–94–21	AB1050R
	D40854P	49–41–11, 49–52–31, 49–62–12	AB1050R
W3377	D43397P	22–12–31, 22–23–11, 22–23–12, 27–32–31	AB1050L
W3379	D43399P	22–12–41, 27–24–11, 27–32–31	AB1050R
W6134	D40842P	27–28–11	AB1050L

H. Put the Airplane Back to Its Usual Condition

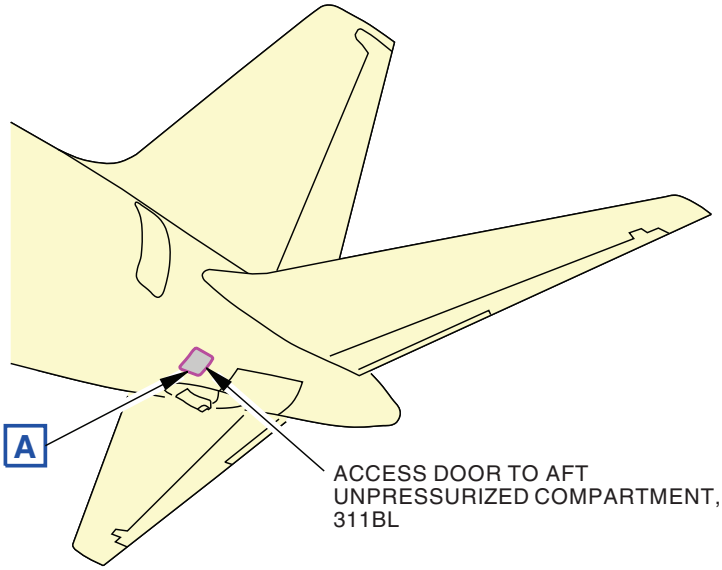
SUBTASK 05-55-46-410-001

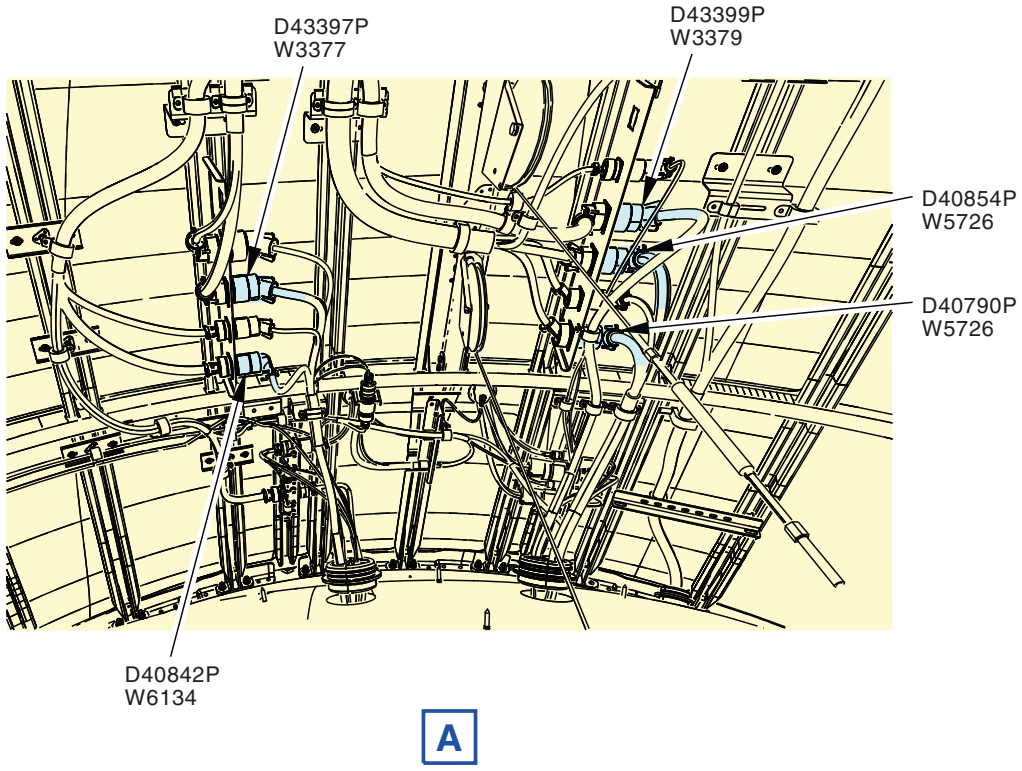
(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-165-00) - Connector Location
Figure 601/05-55-46-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-46-280-802

3. Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-166-00)

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-8965	LRT Jumper Cable Kit (12-ft cable IA2401-1 & 42-ft cable IA2402-1) Part #: 906-10281-1 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-004

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-002

(2) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-010

(1) Do the Functional Check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for each wire bundle listed in the Table 603, do this task:Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 603).

Table 603/05-55-46-993-814 Wire Bundle Loop Resistance Values - Tailcone Access Compartment (MRB 20-166-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W3391	D00381		25/58		
W4222	D00700		44/104		
W3399	D01857		52/122		
W7030	D10531		81/190		
W4724	D15980		32/76		
W7030	D40236P		183/430		

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3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 603).

4) If the measured value is not within the MIN/MAX value range, do the Troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 603 were tested and passed.

(b) Do this task to test W3399-D1857:

1) Connect one end of the 12 foot jumper, (jumper cable assembly, LRT (loop resistance tester), SPL-8965) to the housing of switch S773 (NOT the switch connector itself).

2) Connect the other end of the 12 foot jumper, (jumper cable assembly, LRT (loop resistance tester), SPL-8965) to ground terminal GD4811-ST.

3) Attach the driver coupler of the loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, to the 12-foot jumper cable.

4) Attach the sense coupler of the loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, to the W3399 wire bundle.

5) Do the Modified Loop Resistance Measurement (Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801) for this setup loop. Make sure the loop resistance value is between 10 - 75 milliohms.

NOTE: The loop resistance value is for reference only (do not record).

SUBTASK 05-55-46-810-008

(2) Do the Troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

(b) Record all the measurement data to the data sheet (Table 603).

NOTE: This should include both initial and final values.

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565 (TASK 05-56-02-200-801 or TASK 05-56-02-200-802).

3) If no problem is found, check the entire length of the wire bundle.

4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

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a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.

5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:

1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-002

(3) The connectors listed in Table 604 are used for the steps above.

Table 604/05-55-46-993-802 Tailcone Access Compartment Connectors (MRB 20-166-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W3391	D00381	22–13–11, 27–41–12	M01659
W4222	D00700	22–13–11, 27–41–12	M01659
W3399	D01857	22–11–12	S00773
W7030	D10531	27–31–11	V00160
W7030	D40236P	27–31–11	D40236J
W4724	D15980	33–43–31	T03033

H. Put the Airplane Back to Its Usual Condition

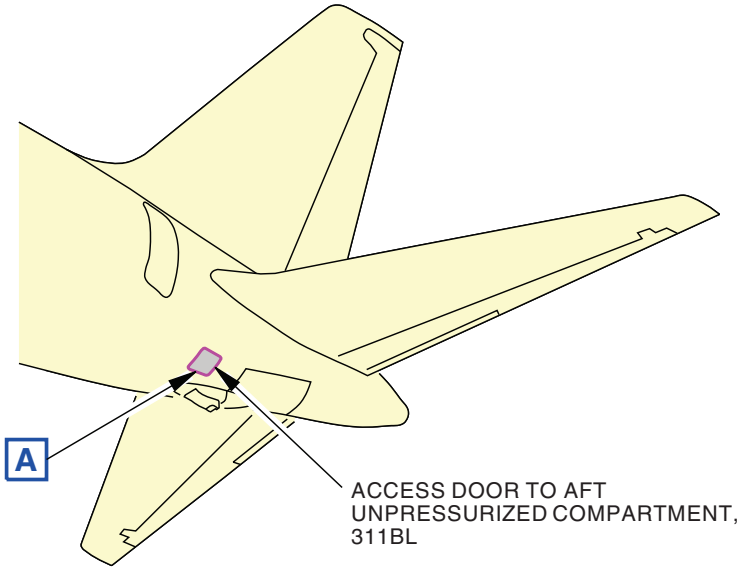
SUBTASK 05-55-46-410-002

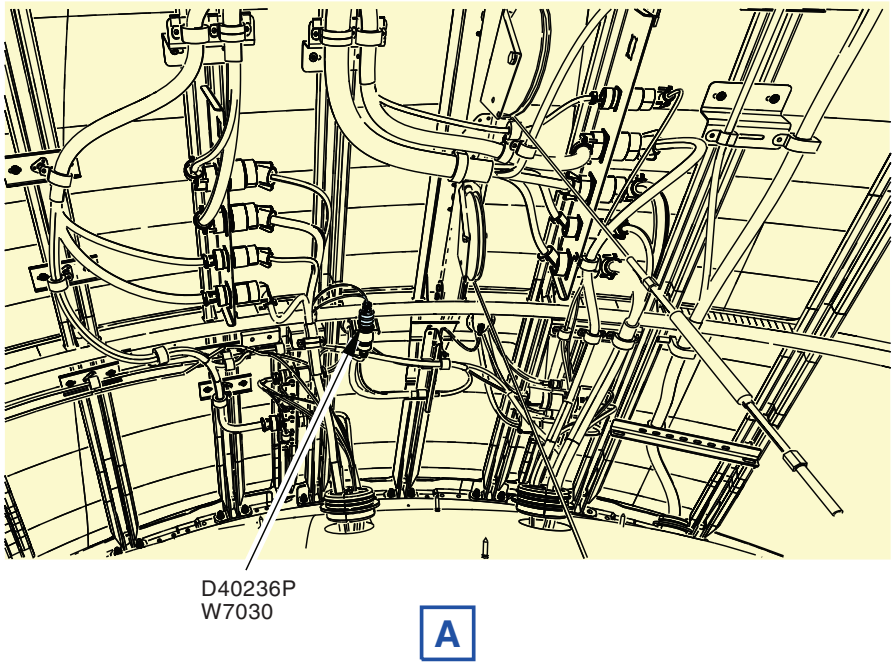
(1) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

——— END OF TASK ———

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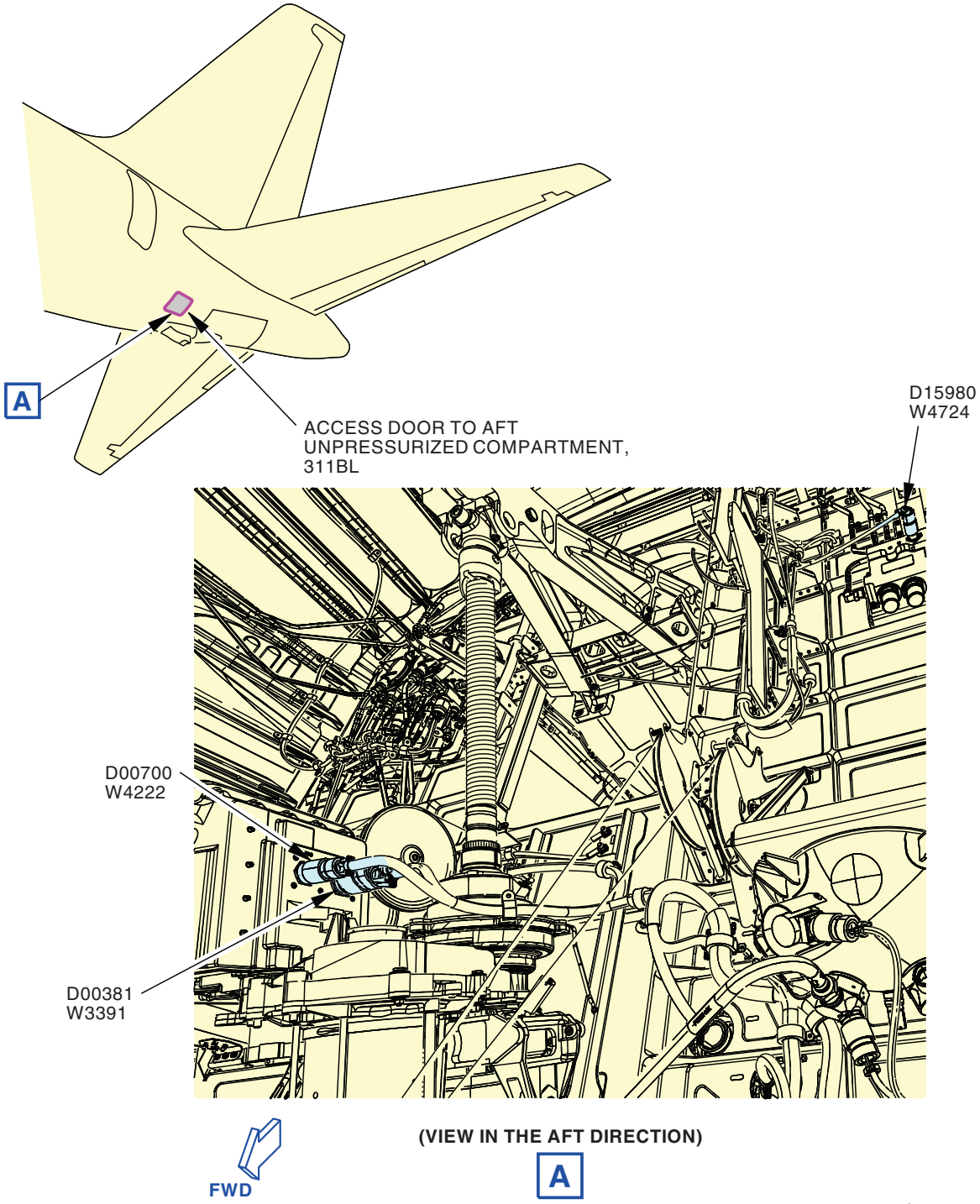




2546728 S0000605242_V1

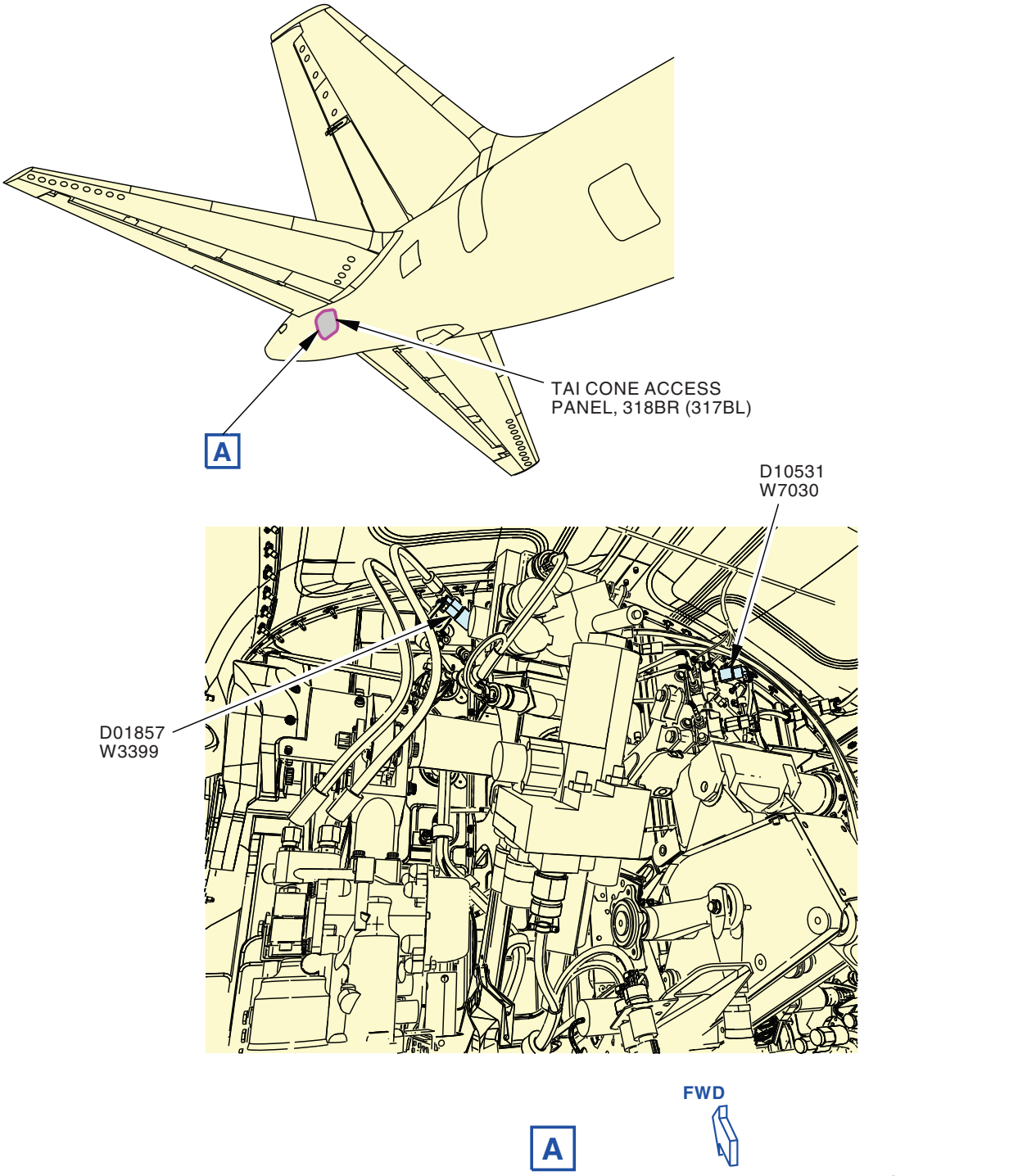
Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-166-00) - Connector Location
Figure 602/05-55-46-990-804 (Sheet 1 of 3)

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Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-166-00) - Connector Location
Figure 602/05-55-46-990-804 (Sheet 2 of 3)

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Loop Resistance Test of Wire Bundle - Tailcone Access Compartment (MRB 20-166-00) - Connector Location
Figure 602/05-55-46-990-804 (Sheet 3 of 3)

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TASK 05-55-46-280-803

4. Loop Resistance Test of Wire Bundle - Right Tailcone Access Compartment (MRB 20-167-00)

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-006

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-003

(2) Open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-011

(1) Do the Functional Check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for each wire bundle listed in the Table 605, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 605).

Table 605/05-55-46-993-815 Wire Bundle Loop Resistance Values - Right Tailcone Access Compartment (MRB 20-167-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W3399	D01677		48/114		
W3399	D01679		48/108		
W3397	D01711		46/108		
W3397	D01829		47/110		

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 605).

4) If the measured value is not within the MIN/MAX value range, do the Troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 605 were tested and passed.

SUBTASK 05-55-46-810-007

(2) Do the Troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

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(b) Record all the measurement data to the data sheet (Table 605).

NOTE: This should include both initial and final values.

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the LRT.

a) Check the LRT operation (LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802).

3) If no problem is found, check the entire length of the wire bundle.

4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.

5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:

1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-003

(3) The connectors listed in Table 606 are used for the steps above.

Table 606/05-55-46-993-803 Right Tailcone Access Compartment Connectors (MRB 20-167-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W3399	D01677	22-12-41	M00958
	D01679	22-12-41	M00929

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Table 606/05-55-46-993-803 Right Tailcone Access Compartment Connectors (MRB 20-167-00) (Continued)

Wire Bundle	Connector	WDM	Mate With Equipment
W3397	D01711	22-12-31	M00929
	D01829	22-12-31	M00958

H. Put the Airplane Back to Its Usual Condition

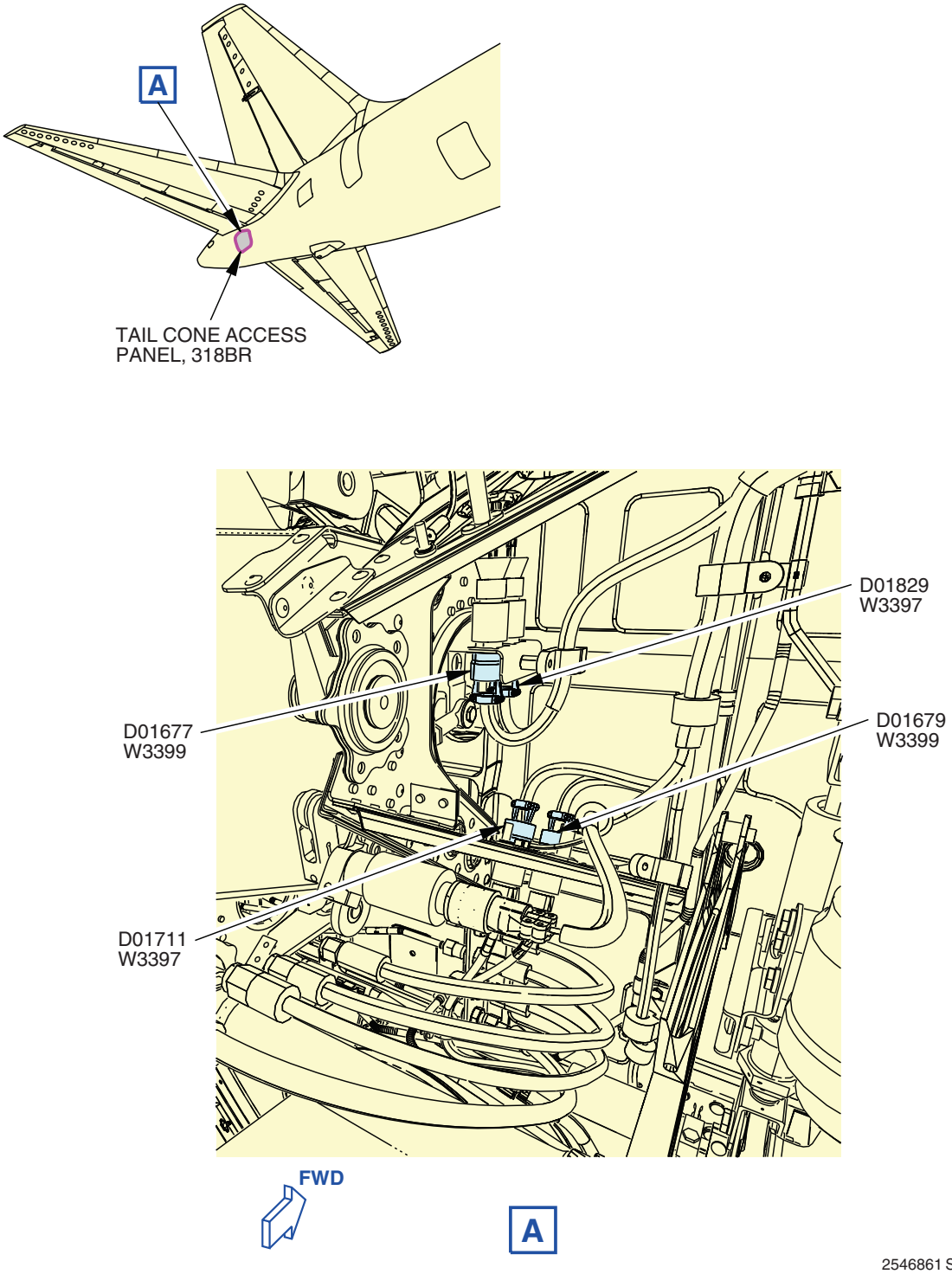
SUBTASK 05-55-46-410-003

(1) Close this access panel:

Number	Name/Location
318BR	Tailcone Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Right Tailcone Access Compartment (MRB 20-167-00) - Connector Location
Figure 603/05-55-46-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-46-280-810

5. Loop Resistance Test of Wire Bundle - Left Tailcone Access Compartment (MRB 20-167-00)

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left

E. Access Panels

Number	Name/Location
317BL	Tailcone Access Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-016

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

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3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-010

(2) Open this access panel:

Number	Name/Location
317BL	Tailcone Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-016

(1) Do the Functional Check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for the wire bundle listed in the Table 607, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 607).

Table 607/05-55-46-993-821 Wire Bundle Loop Resistance Values - Left Tailcone Access Compartment (MRB 20-167-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W3397	D02205		43/101		

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 607).

4) If the measured value is not within the MIN/MAX value range, do the Troubleshooting step below for the wire bundle shielding failure.

SUBTASK 05-55-46-810-013

(2) Do the Troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

(b) Record all the measurement data to the data sheet (Table 607).

NOTE: This should include both initial and final values.

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

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1) Check the test setup.
   a) Make sure that the couplers are only around the wire bundle and closed properly.
2) If the test setup is OK, check the operation of the LRT.
   a) Check the LRT operation (LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802).
3) If no problem is found, check the entire length of the wire bundle.
4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

   NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

   a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.
5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:
1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).
   a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.
   b) If the measured resistance value stays more than the given maximum value for the joint, do this step:
      <1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again.

SUBTASK 05-55-46-800-009

(3) The connector listed in Table 608 is used for the steps above.

Table 608/05-55-46-993-822 Left Tailcone Access Compartment Connector (MRB 20-167-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W3397	D02205	22-12-31	M01192

H. Put the Airplane Back to Its Usual Condition

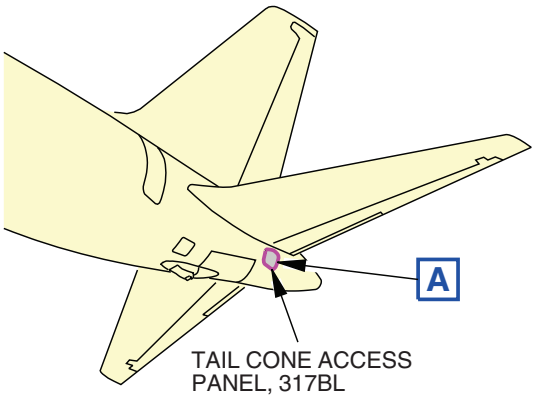
SUBTASK 05-55-46-410-010

(1) Close this access panel:

Number	Name/Location
317BL	Tailcone Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





3132140 S0000862056_V1

Loop Resistance Test of Wire Bundle - Left Tailcone Access Compartment (MRB 20-167-00) - Connector Location
Figure 604/05-55-46-990-817

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-46-280-804

6. Loop Resistance Test of Wire Bundle - APU Compartment (MRB 20-174-00)

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right

E. Access Panels

Number	Name/Location
315A	APU Cowl Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-008

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-004

(2) Open this access panel:

Number	Name/Location
315A	APU Cowl Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-012

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for each wire bundle listed in the Table 609, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 609).

Table 609/05-55-46-993-816 Wire Bundle Loop Resistance Values - APU Compartment (MRB 20-174-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
3888448–2	P5		6/24		
	P6		6/24		

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 609).

4) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the Table 609 were tested and passed.

SUBTASK 05-55-46-810-009

(2) Do the troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the functional check of the wire bundle shielding is not within the MIN/MAX value range.

(b) Record all the measurement data to the data sheet (Table 609).

NOTE: This should include both initial and final values.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the LRT.

a) Check the LRT operation (TASK 05-56-02-200-801 or TASK 05-56-02-200-802).

3) If no problem is found, check the entire length of the wire bundle.

4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.

5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:

1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the functional check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-004

(3) The connectors listed in Table 610 are used for the steps above.

Table 610/05-55-46-993-804 APU Compartment Connectors (MRB 20-174-00)

Wire Bundle	Connector	WDM	Mate With Equipment
3888448-2	P5	3888448–2, 49–41–11	M83723/83G1415N
	P6	3888448–2, 49–41–11	M83723/83G1808N

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

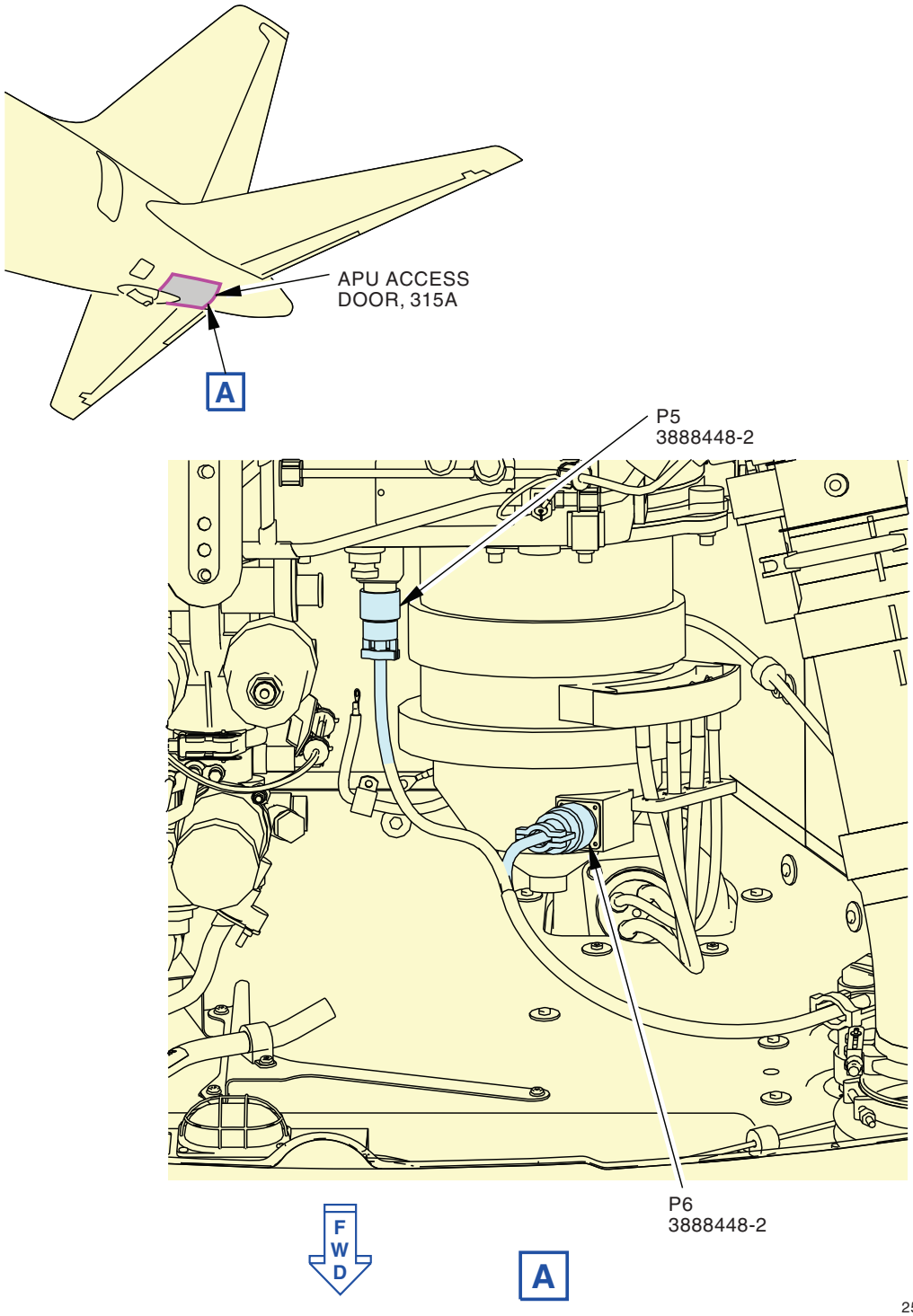
SUBTASK 05-55-46-410-004

(1) Close this access panel:

Number	Name/Location
315A	APU Cowl Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - APU Compartment (MRB 20-174-00) - Connector Location
Figure 605/05-55-46-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-46-280-808

7. Loop Resistance Test Wire Bundle - Vertical Stabilizer (MRB 20-184-00)

(Figure 606)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636 or the MK loop resistance tester, SPL-18565 is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
324DL	Trailing Edge Access

F. Prepare for the procedure

SUBTASK 05-55-46-720-013

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-008

(2) Open this access panel:

Number	Name/Location
324DL	Trailing Edge Access

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-014

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801 for each wire bundle listed in the Table 611.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 611).

Table 611/05-55-46-993-818 Wire Bundle Loop Resistance Values - Vertical Stabilizer (MRB 20-184-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W6136	D10197		24/57		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 611).

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 611 were tested and passed.

SUBTASK 05-55-46-810-011

(2) Do the troubleshooting step for the wire bundle shielding.

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

NOTE: Record all the measurement data to the data sheet. This should include both initial and final values. Also record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(a) If the wire bundle shield resistance is less than the MIN value, do these steps:

1) Check the test setup.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

3) If no problem is found, check the entire length of the wire bundle.

a) Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example could be an area where bundle insulation has degraded and portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another could be some metallic object penetrating the insulation and making contact between the shield and structure.

b) If a problem is found, repair or replace (SWPM 20).

4) If no defects are found with the wire bundle and loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(b) If the wire bundle shield loop resistance is more than the MAX value, do this step:

1) Measure the joint resistance across the connector from the ground terminal to primary structure for each wire bundle listed in the table below for applicable connectors (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair the connector (SWPM 20).

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(c) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-007

(3) The connectors listed in Table 612 are used for the steps above.

Table 612/05-55-46-993-817 Power Control Unit Connectors (MRB 20-184-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W6136	D10197	27-23-11, 27-25-11	M175

H. Put the Airplane Back to Its Usual Condition

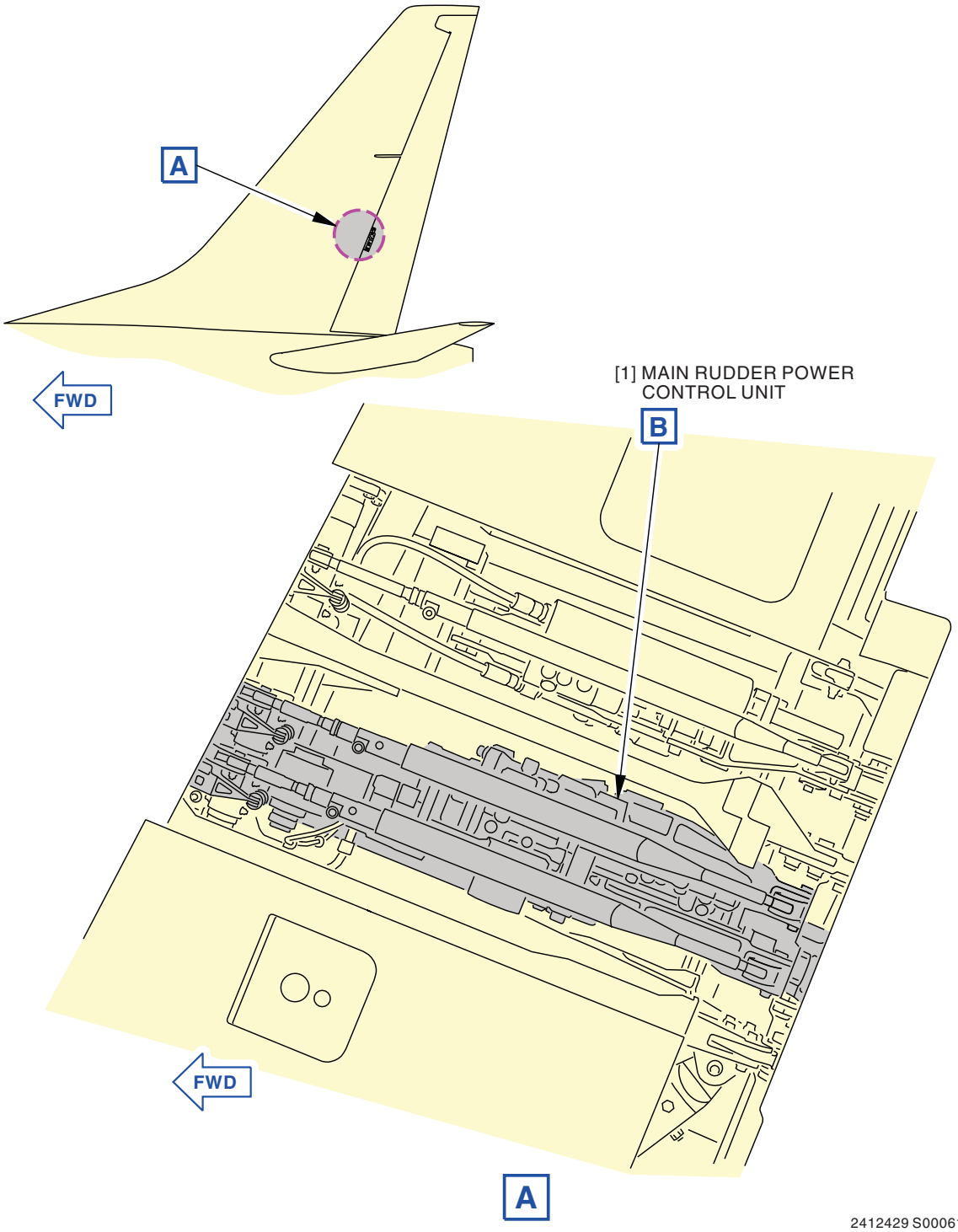
SUBTASK 05-55-46-410-008

(1) Close this access panel:

Number	Name/Location
324DL	Trailing Edge Access

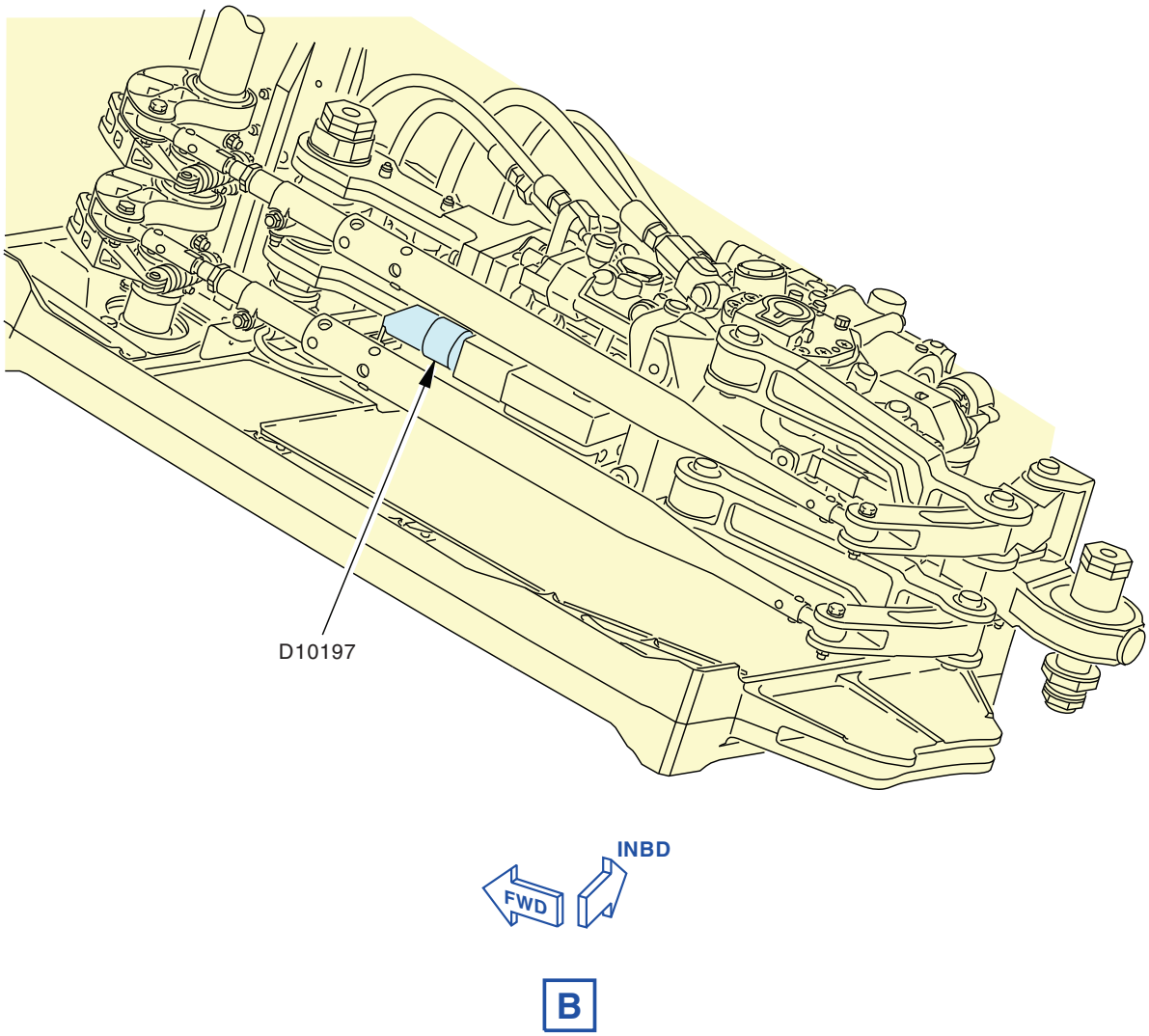
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test Wire Bundle - Vertical Stabilizer (MRB 20-184-00)
Figure 606/05-55-46-990-815 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Loop Resistance Test Wire Bundle - Vertical Stabilizer (MRB 20-184-00)
Figure 606/05-55-46-990-815 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-46-280-809

8. Loop Resistance Test of Wire Bundle - Aft Pressure Bulkhead (MRB 20-190-00)

(Figure 607)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The loop resistance tester, SPL-1636, or the MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-970-015

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-46-010-009

(2) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-46-720-015

(1) Do the Functional Check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) Measure the loop resistance for each wire bundle listed in theTable 613, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

1) In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle.

NOTE: This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

2) Record the measured value to the data sheet (Table 613).

Table 613/05-55-46-993-819 Wire Bundle Loop Resistance Values - Aft Pressure Bulkhead (MRB 20-190-00)

Plane:					
Date:					
Technician:					
Bundle Number	**Connector Number**	**Measured Loop Resistance Value (mΩ)**	**Min/Max Value (mΩ)**	**Pass Fail**	**Retest Measured Value/Comments**
W6136	D40268P		10/15		

3) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet (Table 613).

4) If the measured value is not within the MIN/MAX value range, do the Troubleshooting step below for the wire bundle shielding failure.

5) This step is complete when all wire bundles listed in the(Table 613) were tested and passed.

SUBTASK 05-55-46-810-012

(2) Do the Troubleshooting step for the wire bundle shielding.

(a) Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

(b) Record all the measurement data to the data sheet (Table 613).

NOTE: This should include both initial and final values.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(d) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the LRT.

a) Check the LRT operation (TASK 05-56-02-200-801 or TASK 05-56-02-200-802).

3) If no problem is found, check the entire length of the wire bundle.

4) Look for signs of chafing or damage to the wire bundle, that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

a) If a problem is found, repair or replace in accordance with the SWPM 20-10-13.

5) If no defects are found with the wire bundle and the loop resistance value stays less than the MIN value, record this value and contact Boeing for support.

(e) If the wire bundle shield loop resistance is more than the MAX value, do the following:

1) Do the joint resistance measurement across the connector from the ground terminal to primary structure (TASK 05-56-04-200-801 or TASK 05-56-05-200-801).

a) If the joint value measured is more than the maximum joint resistance value in the data table listed in TASK 05-56-04-200-801 or TASK 05-56-05-200-801, repair in accordance with the SWPM 20-10-13.

b) If the measured resistance value stays more than the given maximum value for the joint, do this step:

<1> Record the resistance value and contact Boeing for support.

(f) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

SUBTASK 05-55-46-800-008

(3) The connectors listed in Table 614 are used for the steps above.

Table 614/05-55-46-993-820 Tailcone Access Compartment Connectors (MRB 20-166-00)

Wire Bundle	Connector	WDM	Mate With Equipment
W6136	D40268P	27-28-11	AB1080

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

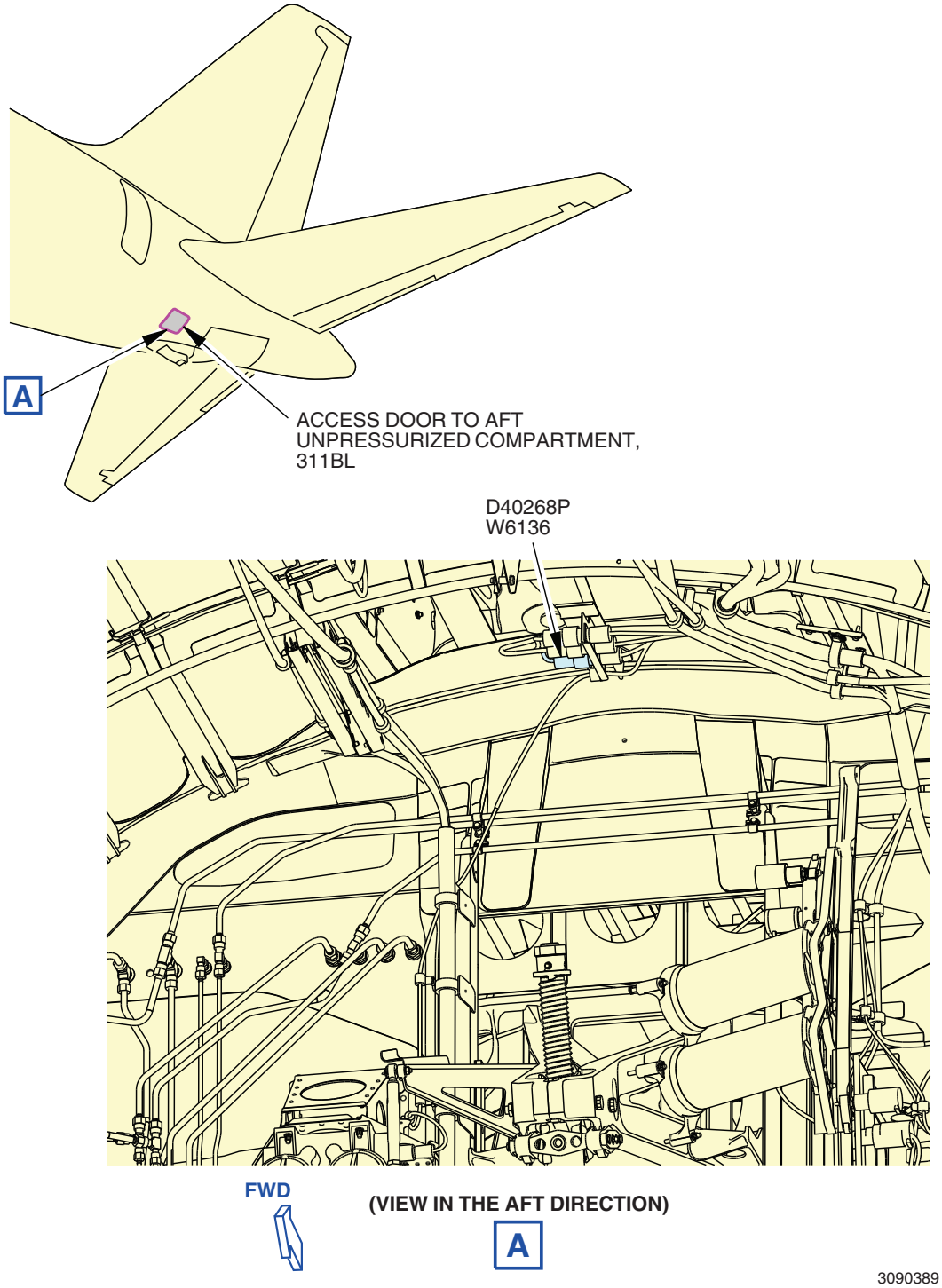
SUBTASK 05-55-46-410-009

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - Aft Pressure Bulkhead (MRB 20–190–00) - Connector Location
Figure 607/05-55-46-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE VESSEL WIRE BUNDLES - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Functionally check the Lightning/HIRF protection components in the LRRA Antenna installation.

TASK 05-55-47-280-801

2. Loop Resistance Test of Wire Bundle - LRRA (MRB 20-153-00)

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is a Functional check of the Lightning/HIRF protection components in the LRRA antenna for electrical bond degradation using the Loop Resistance Tester (LRT).

(2) For ease of access, perform the functional check of the LRRA wire bundles at the EE bay area.

(3) The loop resistance tester, SPL-1636, or MK loop resistance tester, SPL-18565, is required for this task.

B. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
34-33-11 P/B 401	LOW RANGE RADIO ALTIMETER ANTENNA - REMOVAL/INSTALLATION
SWPM Chapter 20	Standard Wiring Practices Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
117AW	Equipment Access Door Cover

F. Prepare for the Loop Resistance Test

SUBTASK 05-55-47-970-004

(1) Boeing requests that all data recorded while doing this procedure be submitted to The Boeing Company for engineering analysis.

(a) Send the data to one of the following addresses:

1) Electronically: MTCProgLHIRFTaskCardData@boeing.com

2) By Mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 110-SC15, BLDG 80, 2201 Seal Beach Blvd., Seal Beach, CA USA 90740, USA".

3) By Fax: 206-655-4991.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

SUBTASK 05-55-47-010-002

(2) Open these access panels only when the troubleshooting task is required:

Number	Name/Location
117A	Electronic Equipment Access Door
117AW	Equipment Access Door Cover

G. Loop Resistance Test of Wire Bundle

SUBTASK 05-55-47-720-002

(1) Do the functional check of the wire bundle shielding.

NOTE: This step checks the wire bundle shielding for the scheduled maintenance check. This step also checks the wire bundle shielding after any repair is done with the wire bundle or connector.

(a) For each wire bundle listed in the Table 601, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure that the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

1) Record the measured value to the data sheet (Table 601).

Table 601/05-55-47-993-804 Wire Bundle Loop Resistance Values - LRRA

Plane:		
Date:		
Technician:		

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Table 601/05-55-47-993-804 Wire Bundle Loop Resistance Values - LRRA (Continued)

Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/Comments
W5295	D13993A		19-51		
W5293	D13993B		19-51		
W5299	D13995A		19-51		
W5297	D13995B		19-51		

2) Make sure that the measured value is within the MIN/MAX value range listed in the data sheet.

3) If the measured value is not within the MIN/MAX value range, do the troubleshooting step below for the wire bundle shielding failure.

4) This step is complete when all wire bundles listed in the Table 601 were tested and passed.

SUBTASK 05-55-47-800-001

(2) Connectors listed in Table 602 are used for the steps above.

Table 602/05-55-47-993-801 LRRA Connectors (MRB 20-153-00)

WIRE BUNDLE	CONNECTOR	WDM	PNL OR MODULE
W5293	D13993B	34-33-11	M1735
W5295	D13993A	34-33-11	M1735
W5297	D13995B	34-33-21	M1736
W5299	D13995A	34-33-21	M1736

SUBTASK 05-55-47-810-001

(3) Procedure to do the troubleshooting step for the wire bundle shielding:

NOTE: Perform this step only if the measured value of the Functional Check of the wire bundle shielding is not within the MIN/MAX value range.

(a) Record all the measurement data to the data sheet.

NOTE: This should include both initial and final values.

1) Record the troubleshooting steps taken to the RETEST MEASURED VALUE/COMMENTS column.

(b) If the wire bundle shield resistance is less than the MIN value, do the following:

1) Check the test setup.

a) Make sure that the couplers are only around the desired wire bundle and closed properly.

2) If the test setup is OK, check the operation of the LRT (TASK 05-56-02-200-801 or TASK 05-56-02-200-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If no problem is found, check the entire length of the wire bundle. Look for signs of chafing or damage to the wire bundle that can cause the shield to short to structure.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

a) If a problem is found, repair or replace in accordance with the SWPM Chapter 20.

4) If no defects are found with the wire bundle and the loop resistance value is still less than the MIN value, record this value and contact Boeing for further support.

(c) If the wire bundle shield loop resistance is greater than the MAX value, do the troubleshooting at the antenna end first (Table 603).

NOTE: When the loop resistance is high, the failure likely is at the antenna end.

1) On the coax connector from plug to airplane structure, do this task: Joint Resistance Measurement, TASK 05-56-04-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

a) Make sure that the joint resistance value is not greater than 0.0085 ohm (8.5 milliohms).

b) If the joint resistance value is greater than 0.0085 ohm (8.5 milliohms), do the following:

<1> On the coax connector from plug to receptacle, do this task: Joint Resistance Measurement, TASK 05-56-04-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

<2> Make sure that the joint resistance value from plug to receptacle is not greater than 0.005 ohm (5 milliohms).

<3> If the joint resistance value is greater than 0.005 ohm (5 milliohms), repair in accordance with the SWPM Chapter 20.

<4> On the coax connector from receptacle to airplane structure, if the joint resistance value is not greater than 0.005 ohm (5 milliohms), do this task: Joint Resistance Measurement, TASK 05-56-04-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801.

<5> Make sure that the joint resistance value is not greater than 0.0035 ohm (3.5 milliohms).

<a> If the joint resistance value is greater than 0.0035 ohm (3.5 milliohms), replace the antenna (PAGEBLOCK 34-33-11/401).

Table 603/05-55-47-993-805 LRRA Connector Troubleshooting (MRB 20-153-00)

WIRE BUNDLE	CONNECTOR AT LRRA END	CONNECTOR AT ANTENNA END	WDM
W5293	D13993B	D3671	34-33-11
W5295	D13993A	D3675	34-33-11
W5297	D13995B	D3673	34-33-21
W5299	D13995A	D3677	34-33-21

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c) If the joint resistance value is not greater than 0.0085 ohm (8.5 milliohms), move to connector at the LRRA transceiver end of same wire bundle.

(d) After the wire bundle or connector is repaired, do the Functional Check of the wire bundle shielding again on this wire bundle.

H. Put the Airplane Back to Its Usual Condition

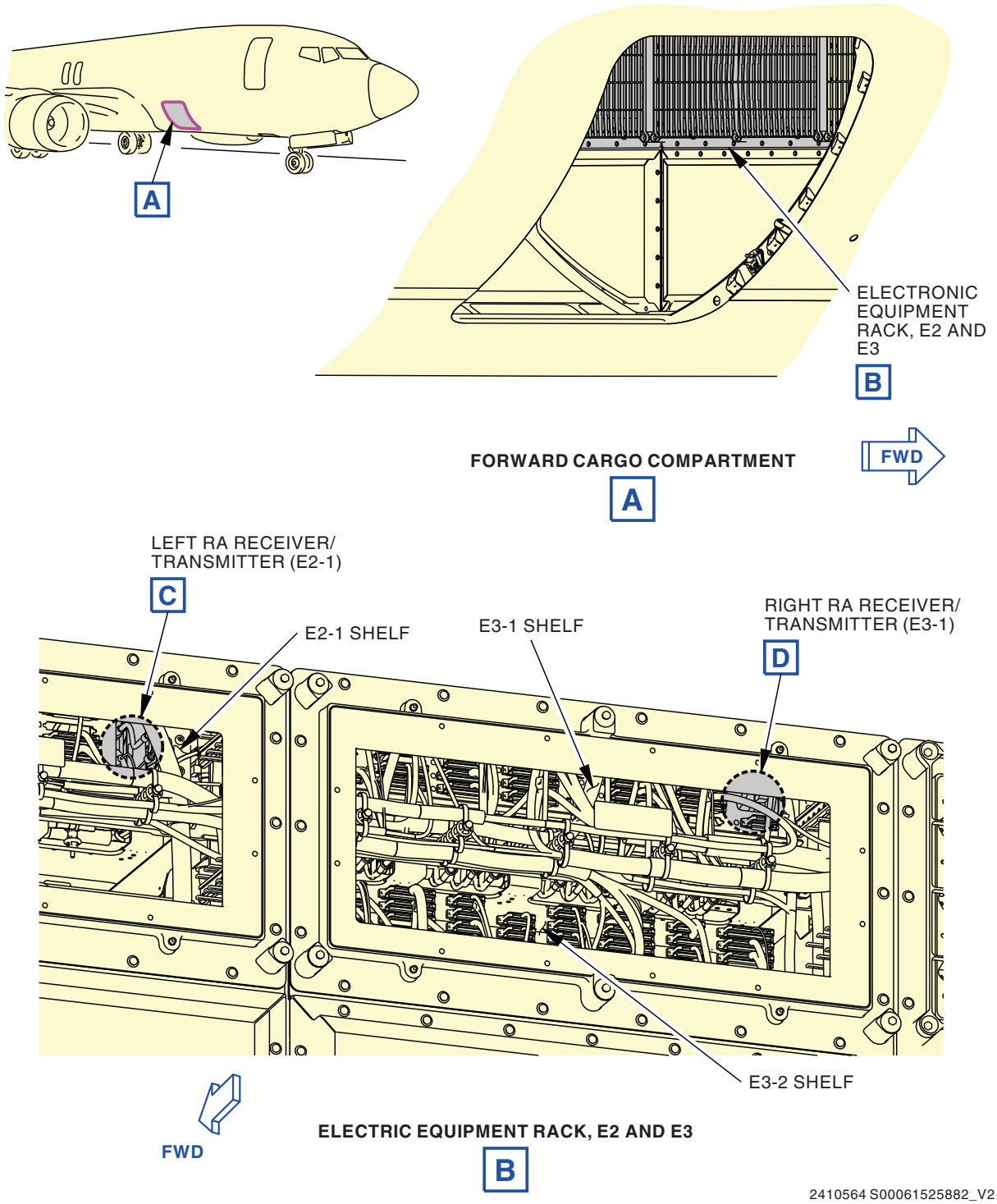
SUBTASK 05-55-47-410-002

(1) Make sure that these access panels are closed:

Number	Name/Location
117A	Electronic Equipment Access Door
117AW	Equipment Access Door Cover

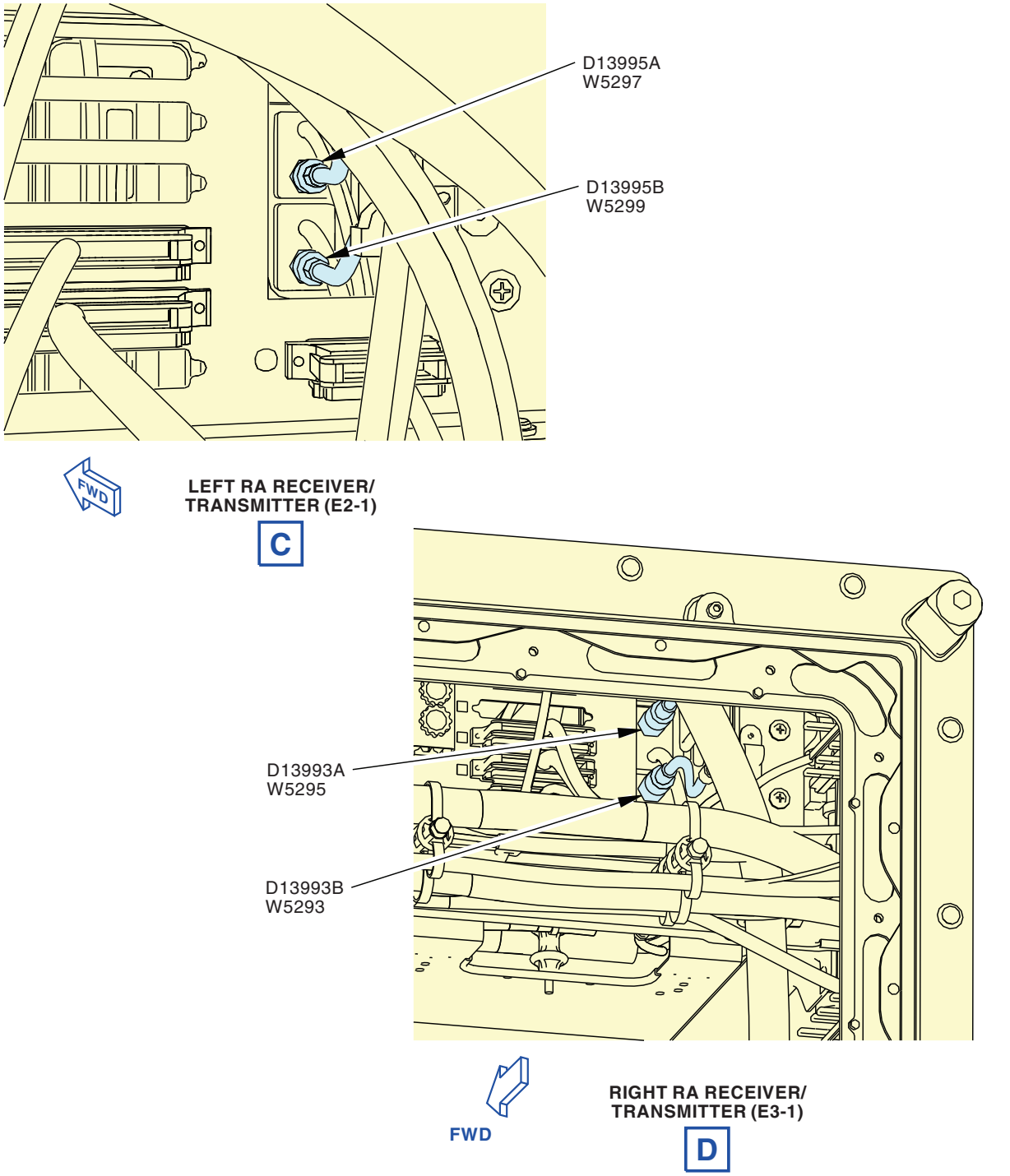
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - LRRA - Connector Location
Figure 601/05-55-47-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Test of Wire Bundle - LRRA - Connector Location
Figure 601/05-55-47-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIRF/LIGHTNING - FQIS WIRING AND BONDING - INSPECTION/CHECK

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure contains tasks that inspect the bonding of the Fuel Quantity Indicating System (FQIS) wiring connectors at the fuel tank receptacles and inboard disconnect panels.

C. Tasks contained in this procedure are:

(1) FQIS Wiring And Bonding - Inspection

NOTE: This task checks the shielding and connector bonding of the Fuel Quantity Indicating System out-tank wire bundles at the spar penetrations.

(2) FQIS - Out Tank Connectors - Electrical Bonding Resistance Check

(3) FQIS Wire Bundles - Fault isolation

NOTE: This task is only used when the loop resistance value is out-of-range. During the test, record the Loop Resistance and Joint Resistance values to Data sheets.

TASK 05-55-54-200-801

2. FQIS Wiring And Bonding - Inspection

(Figure 601, Figure 602, Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has one or more steps which are a means to satisfy Airworthiness Limitation Instruction (ALI) requirements. An ALI note will follow the step to which it applies. Any step or sub-step that precedes or follows an ALI identified step is not subject to the ALI requirement.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-03.

(2) To inspect the FQIS connectors at the fuel tank spar penetrations, a ladder and safety equipment may be required.

(3) You will do Loop Resistance measurements on the wire bundles at the connectors listed in (Table 602) at the tank end of the bundles.

NOTE: If the Loop Resistance measurement is not within limits, fault isolation on the tested wire bundle is required.

(4) Boeing requests that all data recorded while doing this procedure be submitted to the Boeing Company for engineering analysis. Send the data to one of the following addresses:

(a) Electronically: MTCProgLHIRFTaskCardData@boeing.com.

(b) By mail : "The Boeing Company, Attention: Manager, Lightning / HIRF, Mail Code: MC 2J-51, BLDG 11-14N, 2925 S. 112th St., Tukwila, WA 98108, USA".

(c) By Fax: 206-662-3672.

NOTE: Data sent will help Boeing understand the reliability of the components and will be used to improve the maintenance program.

B. References

Reference	Title
05-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
28-41-00-710-801	Fuel Quantity Indicating System - Operational Test (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM Chapter 20	Standard Wiring Practices Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
510	Subzone - Left Wing: Leading Edge, Fwd of Front Spar, Inbd of Strut and Nacelle Gap Cover Area
610	Subzone - Right Wing: Leading Edge, Forward of Front Spar, Inboard of Nacelle Strut, Including Gap Cover Area

E. Prepare for the Inspection Procedure

SUBTASK 05-55-54-410-002

(1) Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 05-55-54-010-002

(2) Do this step to get access to the bussing plug for No. 1 tank or No. 2 tank located on the front spar:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Operate the LE flaps to the Full Extend position (TASK 27-81-00-860-803).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT YOU INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION. THE LEADING EDGE FLAP AND SLAT CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 05-55-54-200-004

28-AWL-03: ALI

(3) To prepare the loop resistance tester, SPL-1636, (LRT) (BAE Systems Dallas Service Center (3X2T2), P/N 906-10246-2 or 906-10246-3 only), for use, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-03.

F. FQIS Connector Wiring and Bonding Inspection

SUBTASK 05-55-54-760-005

28-AWL-03: ALI

(1) Do a Loop Resistance Measurement on every wire bundle listed in Table 602.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-03.

NOTE: Typical LRT connections for the FQIS wire bundles are shown on: Figure 601.

28-AWL-03: ALI

(a) Put the LRT Drive and Sense couplers on the wire bundle before the point where the out-tank wire bundle shields breakout from the bundle, and do this task: Loop Resistance Measurement, TASK 05-56-03-200-801.

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-03.

(b) Record the value to the data sheet (Table 601).

Table 601/05-55-54-993-807 Wire Bundle Loop Resistance Values

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Resistance Value (mΩ)	Min/Max Value (mΩ)	Pass Fail	Retest Measured Value/ Comments
W1140	D16352		11/36		
W1138	D16354		11/36		
W7580	D11316		7/22		

ECCN 9E991 BOEING PROPRIETARY - See title page for details

28-AWL-03: ALI

(c) Make sure the measured resistance value of the shield to ground termination is within the MIN/MAX values listed in the data sheet (Table 601).

NOTE: ALI - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for important Information on airworthiness limitation instructions (ALIs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-03.

(d) If the measured loop resistance value is not within the MIN/MAX values listed in the data sheet, do this task: FQIS Wiring And Bonding - Fault Isolation, TASK 05-55-54-810-801.

(e) If the measured loop resistance value is within the MIN/MAX values, continue to test on the next wire bundle in the Table.

Table 602/05-55-54-993-804

WIRE BUNDLE NUMBER	CONNECTOR	CONN NUMBER	GROUND	WDM
W1140	LEFT MAIN TANK W/O DENSITOMETER	D16352	GD191-ST	28-41-11
W1138	RIGHT MAIN TANK W/O DENSITOMETER	D16354	GD193-ST	28-41-11
W7580	CENTER MAIN TANK W/O DENSITOMETER	D11316	GD3774-ST	28-41-11

SUBTASK 05-55-54-710-003

(2) After doing the loop resistance measurement, perform the visual check on the ground wire terminations, connectors, and backshells associated with the wire bundle and make sure they are hand tight, and that there is no evidence of gross degradation or accidental damage. Also, for connectors with a ground wire which goes from pigtail-to-backshell and ground stud, visually/physically inspect the pigtail-to-backshell terminations and make sure the pigtail is securely fastened to the connector backshell.

(a) If a loose connector or backshell is found, record that condition to the data sheet (Table 601). Repair in accordance with the (SWPM Chapter 20).

(b) If a loose connector or backshell is found and tightened, retest the wire bundle (FQIS Wiring And Bonding - Inspection, TASK 05-55-54-200-801).

(c) Make a note of that condition for future reference.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-54-710-004

(1) If any FQIS wire bundles were disturbed during the accomplishment of this procedure, then do this task: Fuel Quantity Indicating System - Operational Test, TASK 28-41-00-710-801.

SUBTASK 05-55-54-080-001

WARNING

REMOVE THE LOCK FROM LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(2) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-55-54-710-005

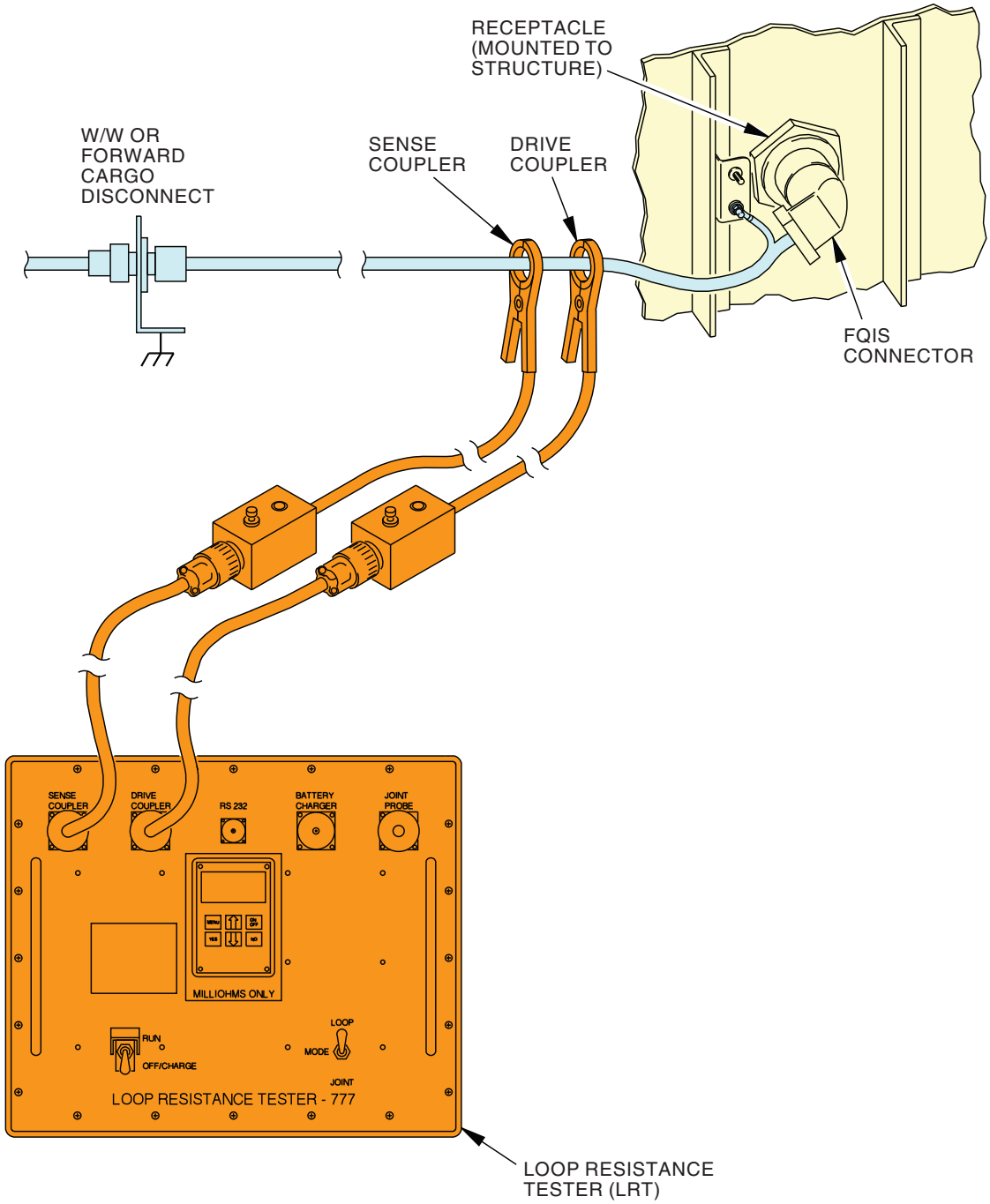
WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Operate the LE flaps to the 0 detent and stowed (LE Flaps retracted) position (TASK 27-81-00-860-804).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

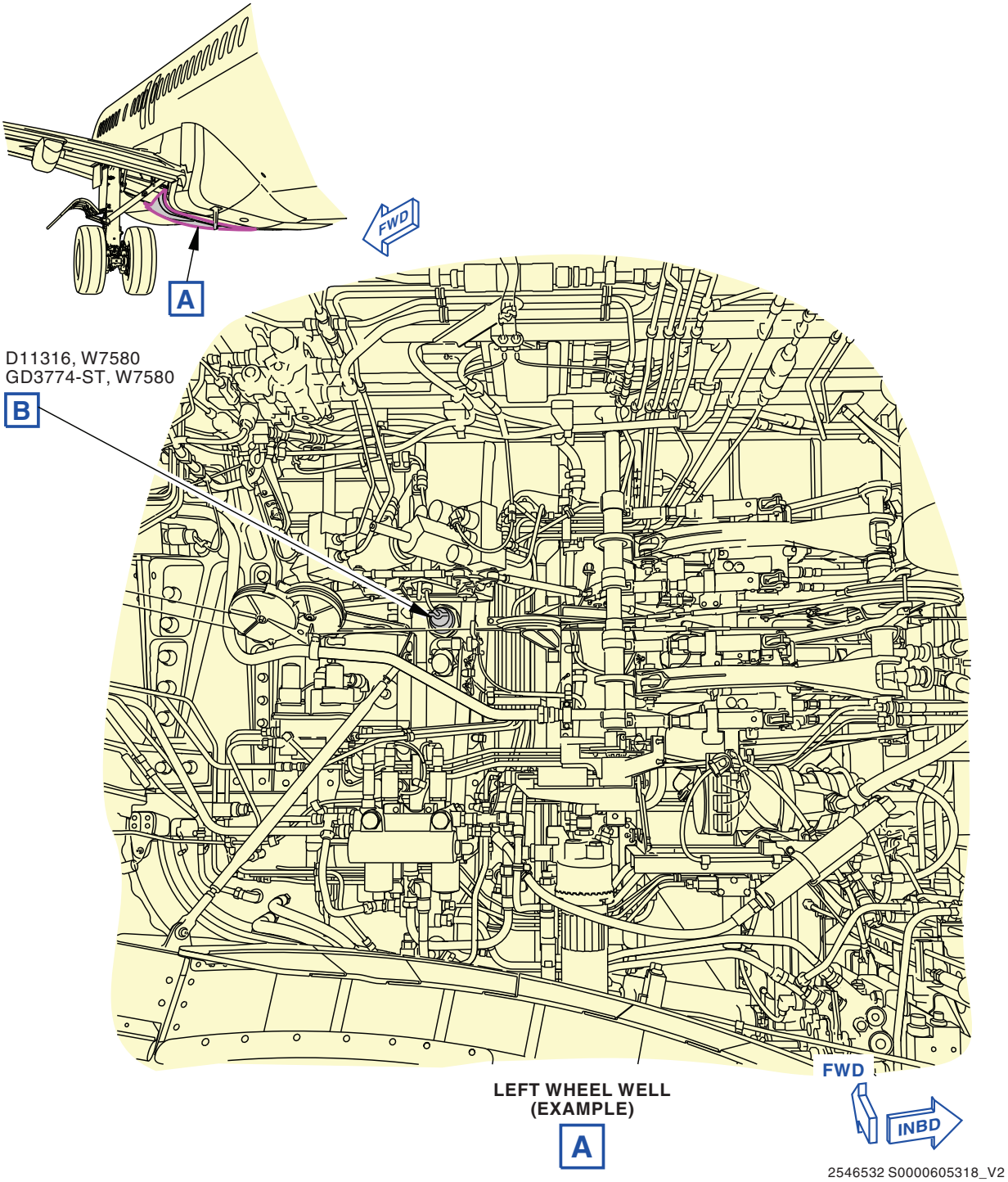


LOOP RESISTANCE TESTER CONNECTIONS - LOOP TEST

2410567 S00061525888_V1

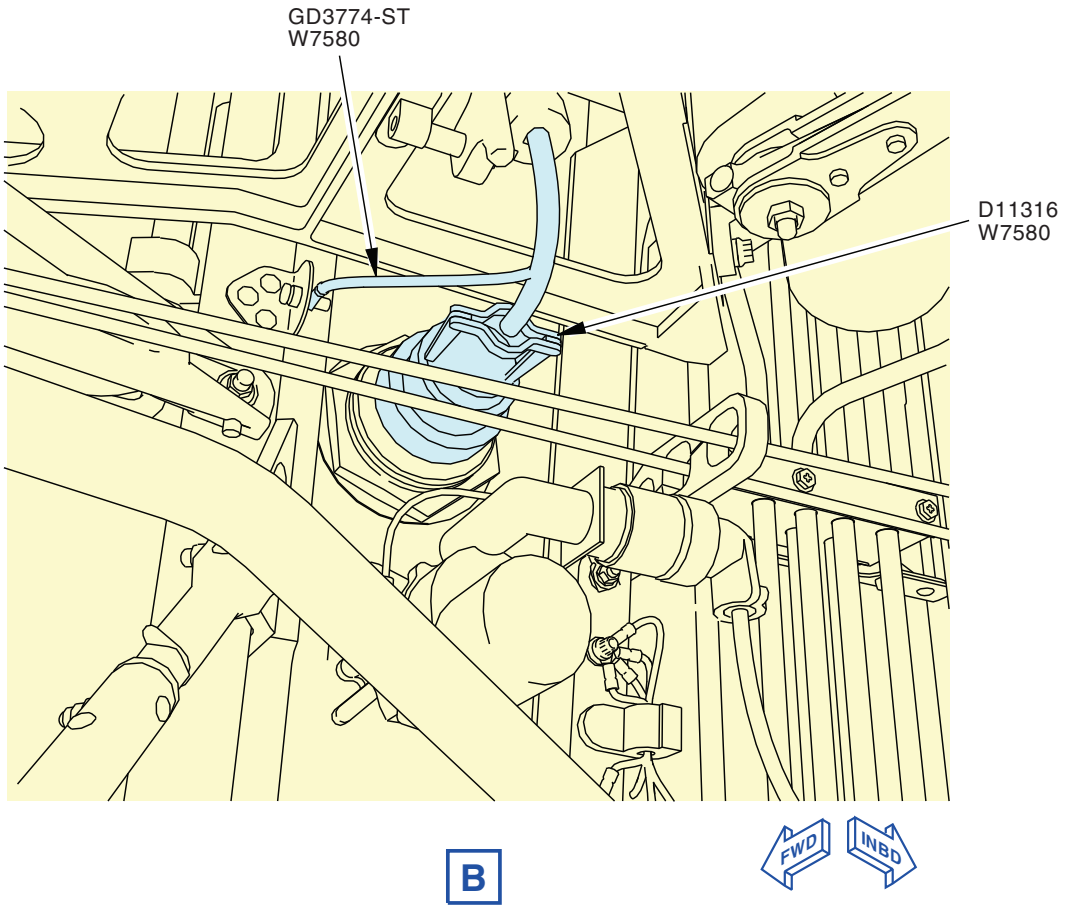
FQIS Wiring And Bonding - Inspection - Loop Resistance Test Connections
Figure 601/05-55-54-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS Wiring And Bonding - Inspection - Connector Locations
Figure 602/05-55-54-990-817 (Sheet 1 of 2)

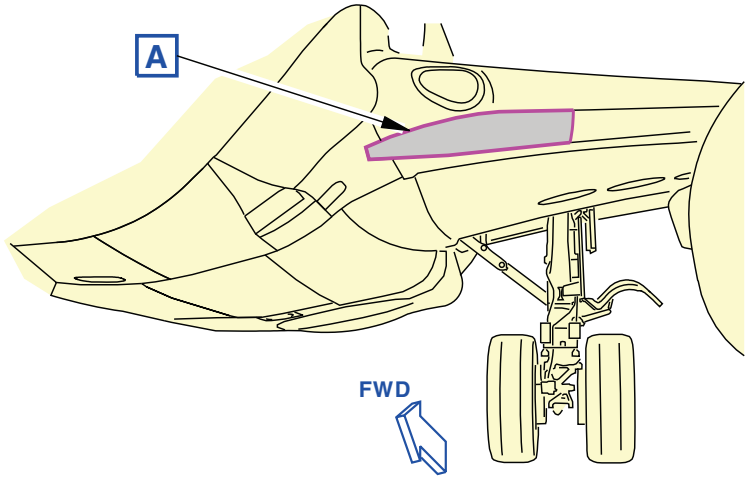
ECCN 9E991 BOEING PROPRIETARY - See title page for details



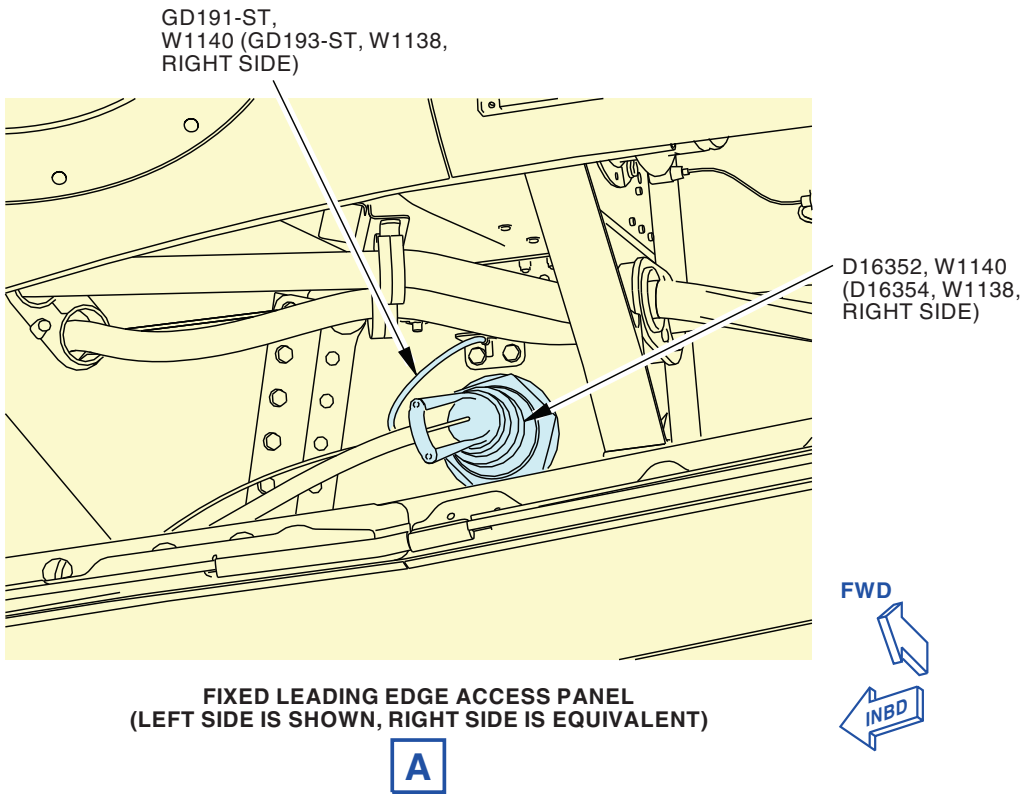
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FQIS Wiring And Bonding - Inspection - Connector Locations
Figure 602/05-55-54-990-817 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



FIXED LEADING EDGE ACCESS PANEL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2546759 S0000605323_V2

FQIS Wiring and Bonding Inspection - Connector Locations Engine Valve Shutoff
Figure 603/05-55-54-990-818

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-54-200-802

3. FQIS - Out Tank Connectors - Electrical Bonding Resistance Check

(Figure 604)

A. General

(1) This procedure is an unscheduled maintenance task.

(2) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

B. References

Reference	Title
05-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM Chapter 20	Standard Wiring Practices Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. FQIS - Out Tank Connectors - Electrical Bonding Resistance Check

SUBTASK 05-55-54-200-005

28-AWL-04: CDCCL

(1) Measure the electrical bonding resistance from the shield terminal lug on the connector backshell to airplane structure for each connector listed in Table 603 (SWPM 20-20-00):

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) To measure the joint resistance, do these steps:

1) Connect one probe of the intrinsically safe approved bonding meter, COM-1550, to the airplane structure within 1 in. (25 mm) of the joint being measured.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Connect the second joint probe to the shield terminal lug on the connector backshell.

3) Record the resistance value.

28-AWL-04: CDCCL

(c) Make sure the electrical bonding resistance from the shield terminal lug on the connector backshell to airplane structure is the “MAX JOINT VALUE” or less listed in Table 603.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

(d) If the electrical bonding resistance from the shield terminal lug on the connector backshell to airplane structure is greater than the “MAX JOINT VALUE” listed in Table 603, repair the wiring (SWPM Chapter 20).

Table 603/05-55-54-993-803

WIRE BUNDLE NUMBER	FUNCTION	CONNECTOR	WDM	MAX JOINT VALUE (mOhm)
W1140	LEFT MAIN TANK	D39965	28-41-11	3.0
W1138	RIGHT MAIN TANK	D39944	28-41-11	3.0
W7580	CENTER MAIN TANK	D46202	28-41-11	3.0

E. Put the Airplane Back to Its Usual Condition

SUBTASK 05-55-54-200-006

28-AWL-04: CDCCL

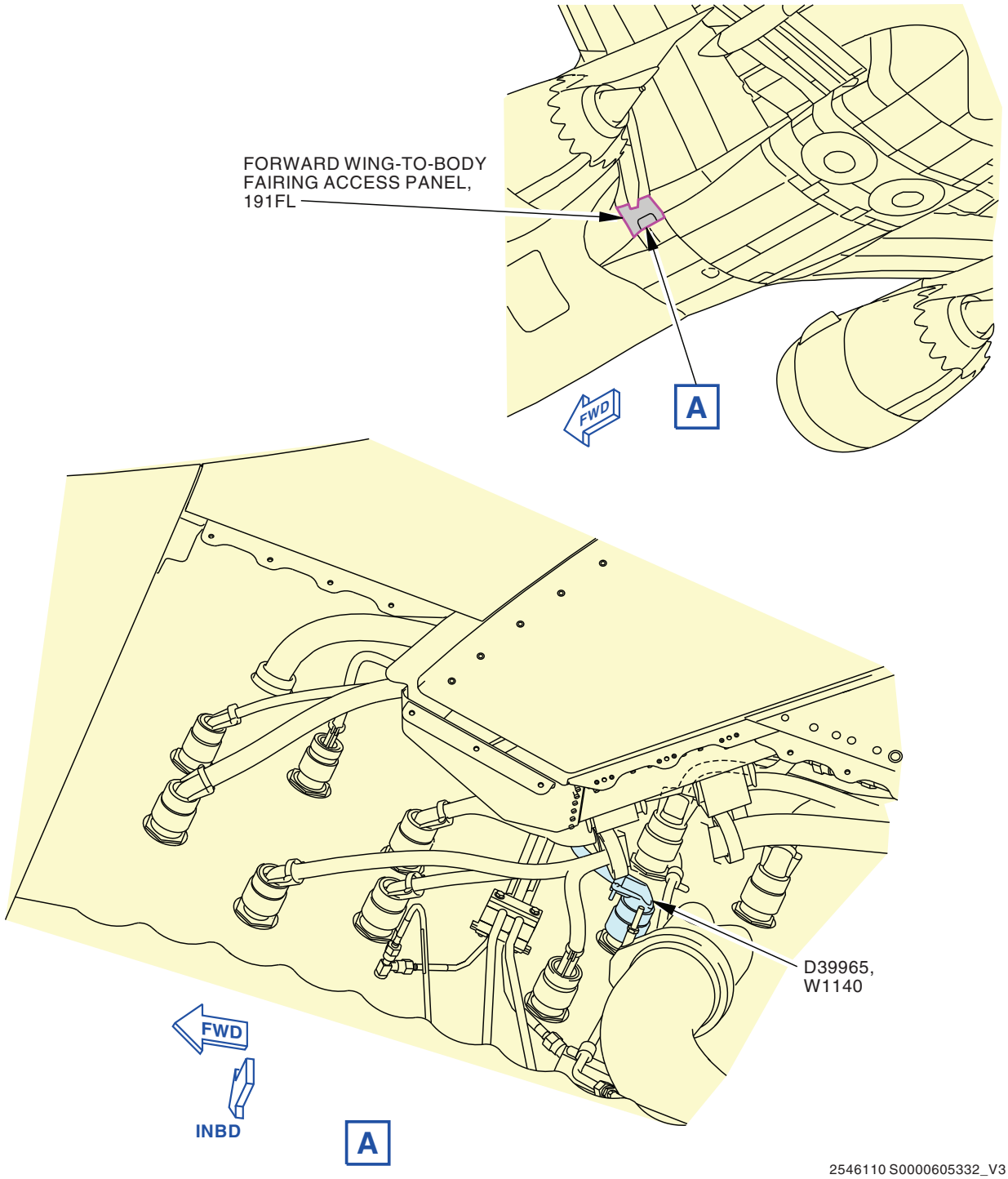
(1) If any FQIS out-tank wire bundle is replaced or the FQIS wire bundle's shield is spliced, repaired, or re-connected, do this task: FQIS Wiring And Bonding - Inspection, TASK 05-55-54-200-801.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 05-00-00-910-801, for information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 28-AWL-04.

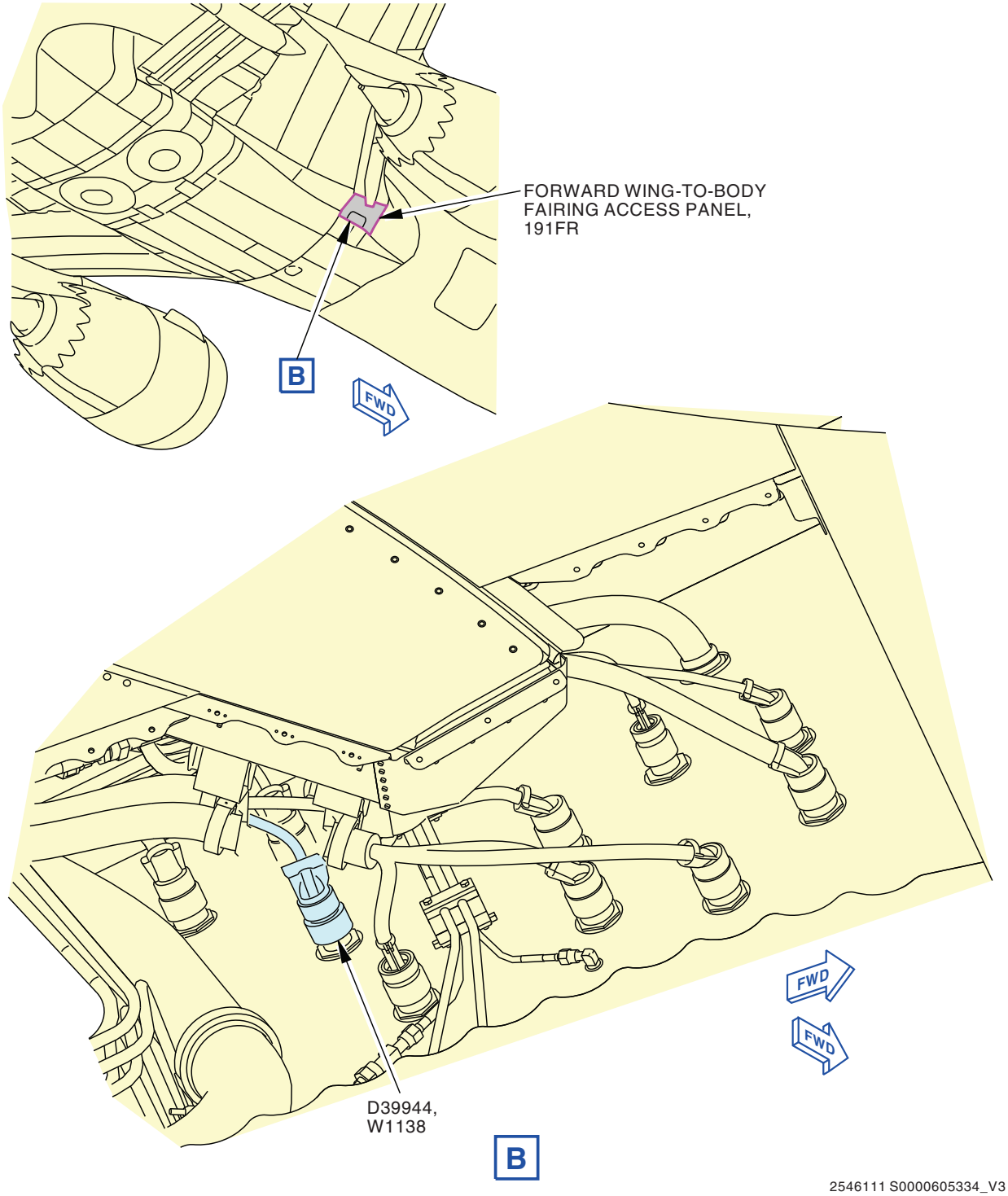
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



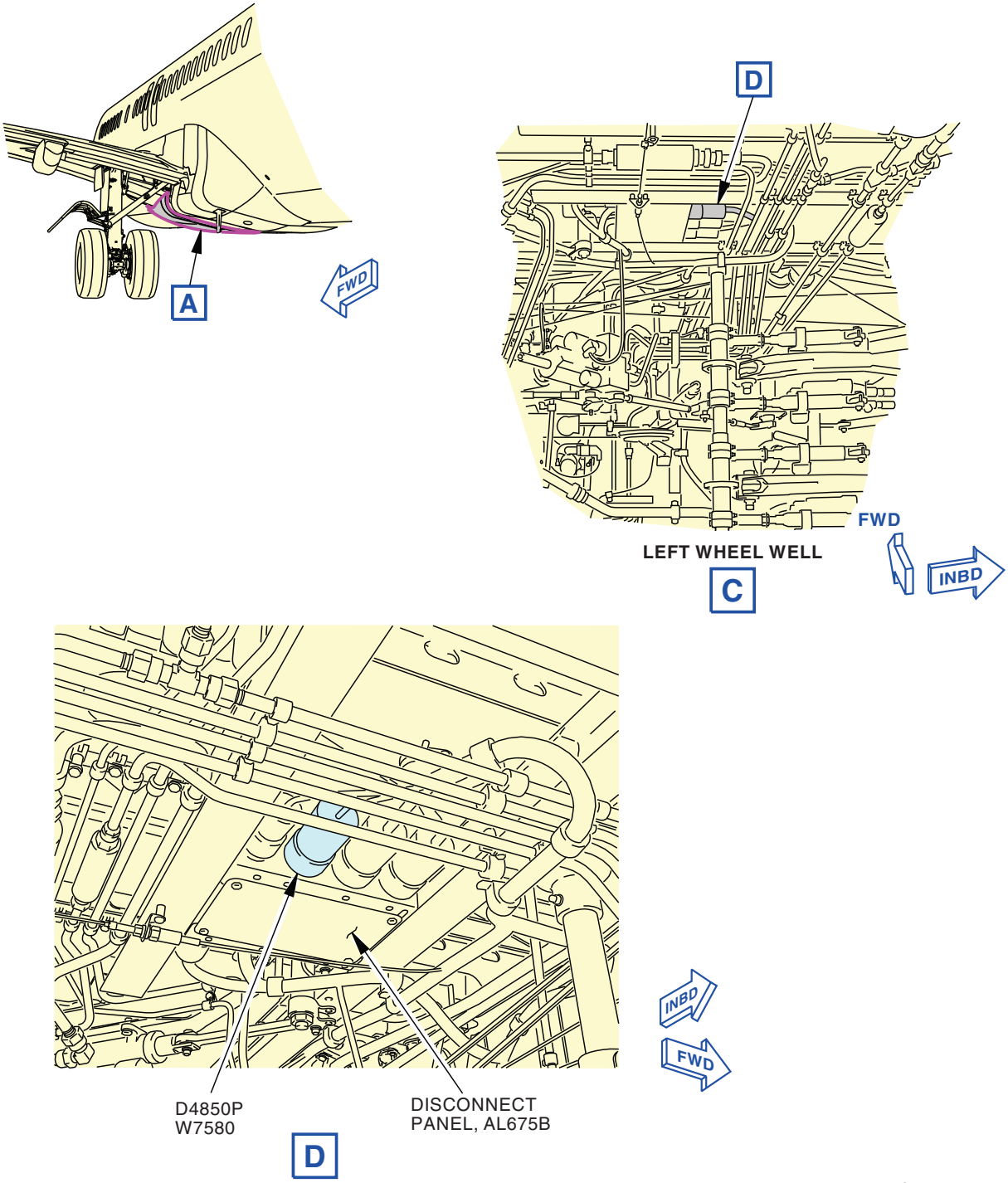
FQIS - Out Tank Connectors
Figure 604/05-55-54-990-810 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS - Out Tank Connectors
Figure 604/05-55-54-990-810 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546112 S0000605335_V2

FQIS - Out Tank Connectors
Figure 604/05-55-54-990-810 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-55-54-810-801

4. FQIS Wiring And Bonding - Fault Isolation

(Figure 605 or Figure 606 or Figure 607)

A. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
05-56-05-200-801	Loop Resistance and Joint Resistance Measurement (P/B 201)
SWPM Chapter 20	Standard Wiring Practices Manual

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
510	Subzone - Left Wing: Leading Edge, Fwd of Front Spar, Inbd of Strut and Nacelle Gap Cover Area
610	Subzone - Right Wing: Leading Edge, Forward of Front Spar, Inboard of Nacelle Strut, Including Gap Cover Area

C. Access Panels

Number	Name/Location
511AT	Inboard Leading Edge, Strakelet Upper Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel

D. Procedure

SUBTASK 05-55-54-940-005

(1) These panels must be opened to inspect/check the left and right wing FQIS wire bundle installations.

Number	Name/Location
511AT	Inboard Leading Edge, Strakelet Upper Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel

SUBTASK 05-55-54-760-006

(2) To isolate a fault associated with the Loop Resistance Measurement of the FQIS system wire bundles, do the following:

SUBTASK 05-55-54-760-007

(3) If the wire bundle shield loop resistance is GREATER than the MAX VALUE in the Table 601, do these tasks:

NOTE: The main current paths making up the bond at the fuel tank end of the bundle consist of three main parts: the bundle connector to the fuel tank receptacle; the receptacle to structure attachment itself; and the bondstrap from receptacle to structure.

(a) Do this task: Joint Resistance Measurement, TASK 05-56-04-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801, from the ground terminal to primary structure.

1) Record the measured resistance values in the Table 604.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 604/05-55-54-993-806 Wire Bundle Loop Resistance Values

Plane:					
Date:					
Technician:					
Bundle Number	Connector Number	Measured Loop Value (mΩ)	Max Value (mΩ)	Pass Fail	Retest Measured Value/Comments
Joint Resistance, Ground Terminal to Bracket:					
W1140	GD-191-ST		1.0		
W1138	GD-193–ST		1.0		
W7580	GD-3774–ST		1.0		
Joint Resistance, Bracket to Spar:					
W1140	GD-191–ST		1.0		
W1138	GD-193–ST		1.0		
W7580	GD-3774–ST		1.0		
Joint Resistance, Backshell to Receptacle:					
W7580	D4850P		5.0		
Joint Resistance, Backshell to Plug:					
W1140	D16352		2.5		
W1138	D39944		2.5		
W7580	D4850P		5.0		
Joint Resistance, Plug to Receptacle;					
W1140	D16352		5.0		
W1138	D39944		5.0		
Joint Resistance, Receptacle to Mating Surface:					
W1140	D16352		2.5		
W1138	D39944		2.5		

2) If the joint value measured is GREATER than the MAX VALUE in the Table 604, repair in accordance with the (SWPM Chapter 20).

3) If the joint value measured is LESS than the MAX VALUE in the Table 604, move to the inboard end of that wire bundle and perform the step below:

(b) Do this task: Joint Resistance Measurement, TASK 05-56-04-200-801 or Loop Resistance and Joint Resistance Measurement, TASK 05-56-05-200-801, from the backshell to primary structure at the inboard end of that wire bundle.

1) Record the measurement value in the Table 604.

2) If the joint value measured is GREATER than the MAX VALUE in the Table 604, repair in accordance with the (SWPM Chapter 20).

3) If the joint value measured is still LESS than the MAX VALUE in the Table 604, repair the wire bundle shield termination in accordance with the (SWPM Chapter 20).

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a) If the shield termination is good and does not require rework, the wire bundle itself has failure. Repair or replace the wire bundle in accordance with the (SWPM Chapter 20).

(c) After the shield termination or wire bundle is repaired or replaced, do this task: FQIS Wiring And Bonding - Inspection, TASK 05-55-54-200-801 on this wire bundle.

SUBTASK 05-55-54-760-008

(4) If the wire bundle shield resistance is LESS than the MIN VALUE in Table 601, do the following:

(a) Check the operation of the LRT. To check the LRT operation, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801 or ExLRT Standard Measurement, TASK 05-56-02-200-802.

(b) If no problem is found with the LRT, check the test setup. Make sure the couplers only be clamped around the wire bundle being measured and must be closed properly.

(c) If no problem is found, do a detailed visual inspection along the entire length of the wire bundle being measured. Look for signs of chafing or damage to the wire bundle that would result in the shielding shorting to structure somewhere along its length.

NOTE: An example might be an area where bundle insulation has degraded and a portion of the shield is exposed and making contact with structure at an intermediate point in the bundle length. Another might be some metallic object penetrating the insulation and making contact between the shield and structure.

1) If a problem is found, repair or replace in accordance with the (SWPM Chapter 20).

2) If a problem was found and repaired, retest the wire bundle (FQIS Wiring And Bonding - Inspection, TASK 05-55-54-200-801).

(d) If no defects are found with the wire bundle and the loop resistance value is still LESS than the MIN VALUE, record this value and report this condition to Boeing for investigation and disposition.

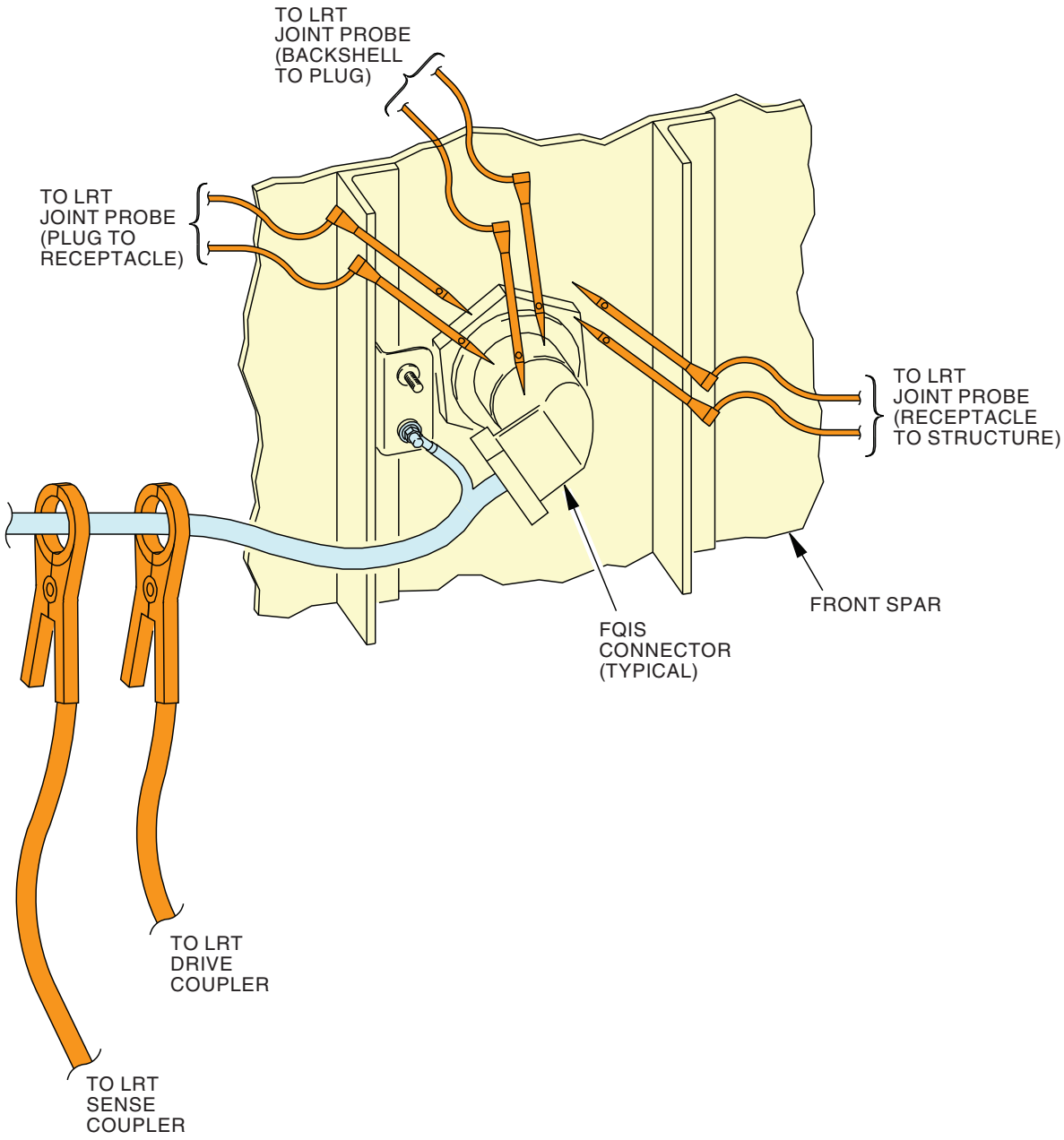
SUBTASK 05-55-54-940-006

(5) Close these access panels:

Number	Name/Location
511AT	Inboard Leading Edge, Strakelet Upper Panel
611AT	Inboard Leading Edge, Strakelet Upper Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

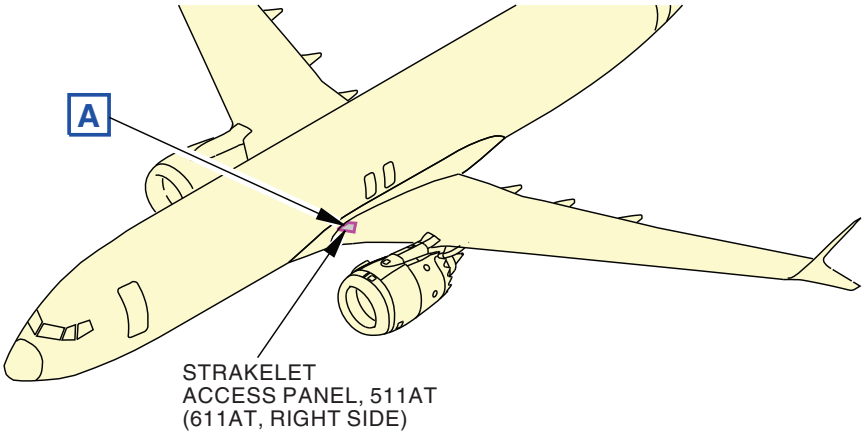


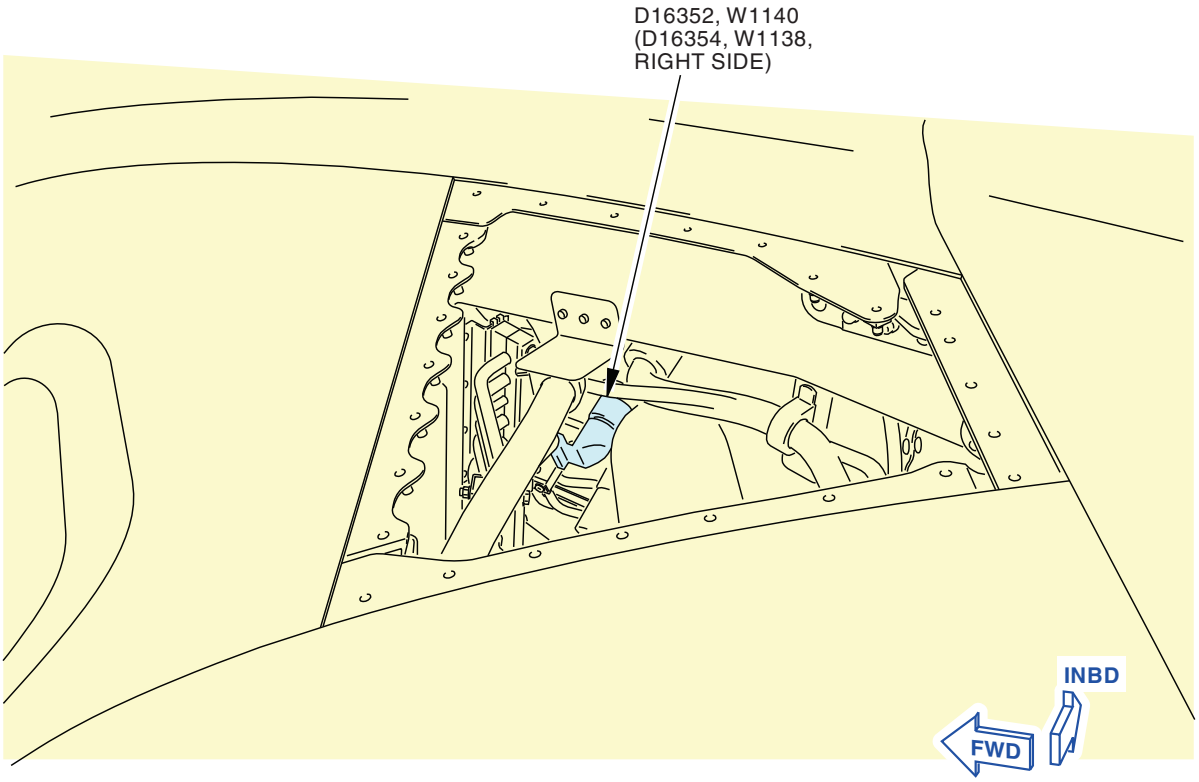
LOOP RESISTANCE TESTER CONNECTIONS - JOINT TEST

2546114 S0000605346_V1

FQIS Wiring And Bonding - Inspection - Joint Resistance Test Connections
Figure 605/05-55-54-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details





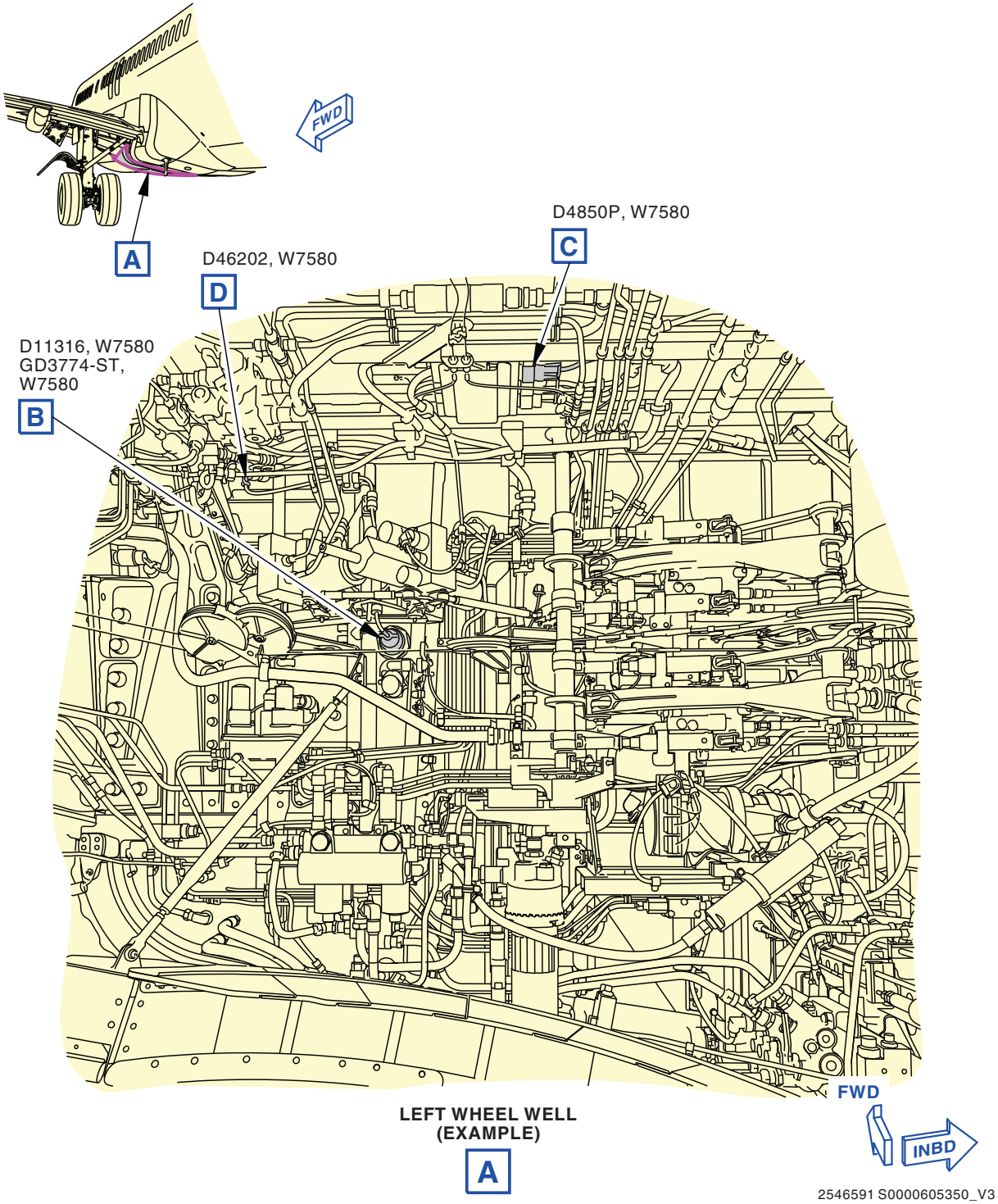
STRAKELET ACCESS PANEL REMOVED
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2546116 S0000605347_V1

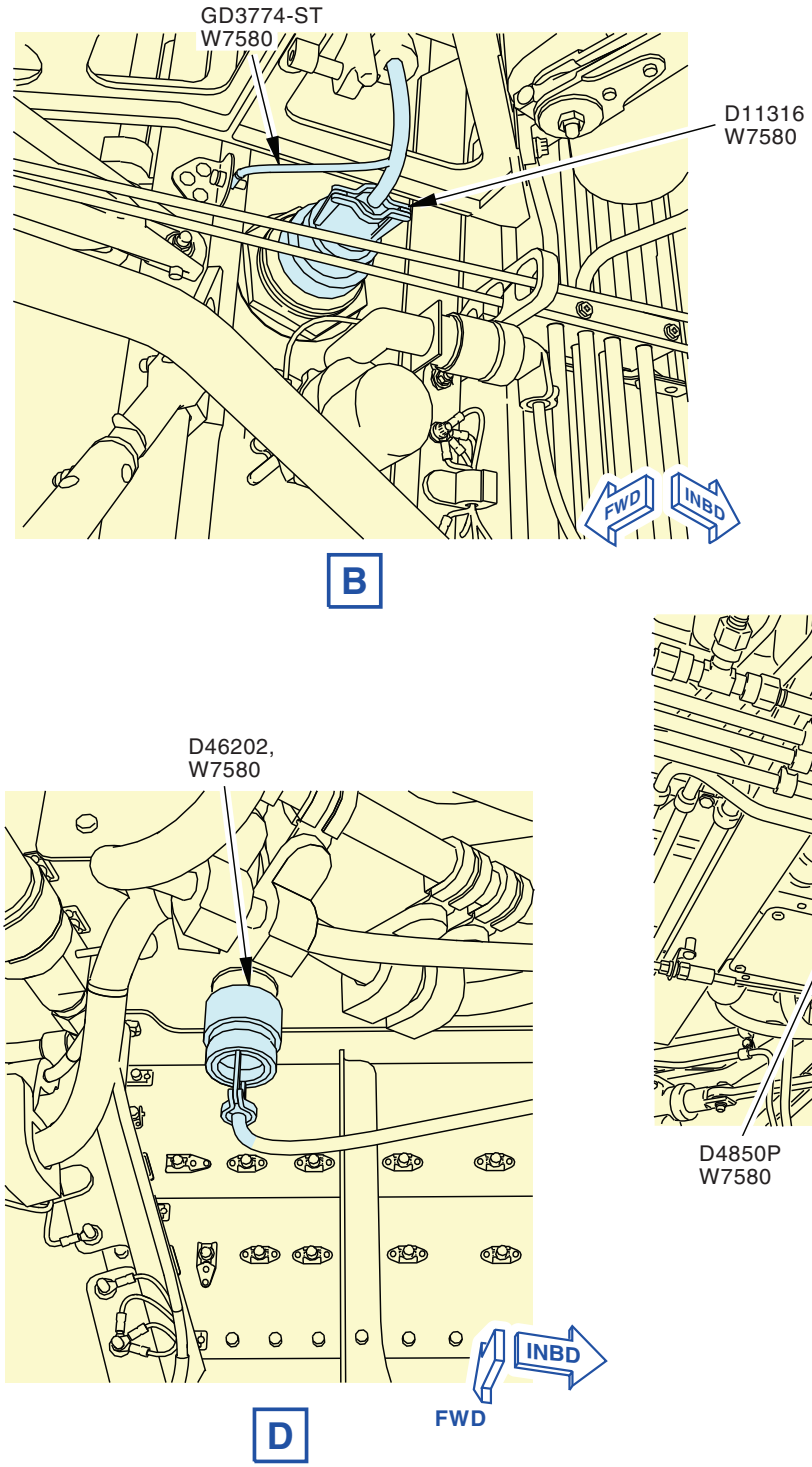
FQIS Wiring And Bonding - Connectors - Strakelet Access
Figure 606/05-55-54-990-813

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FQIS Wiring And Bonding - Inspection - Connector Locations
Figure 607/05-55-54-990-814 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546593 S0000605351_V2

FQIS Wiring And Bonding - Inspection - Connector Locations
Figure 607/05-55-54-990-814 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BOND RESISTANCE MEASUREMENT - MAINTENANCE PRACTICES

1. General

A. This task contains instructions on how to use the Avtron Bond Resistance Tester intrinsically safe approved bonding meter, COM-1550 (Model T477W) or equivalent, to measure and evaluate the resistance across a built-up joint or individual electrical connections/joints.

TASK 05-56-01-760-801

2. Bond Resistance Measurement

(Figure 201)

A. References

Reference	Title
SWPM 20-14-11	Standard Wiring Practices Manual
SWPM Chapter 20	Standard Wiring Practices Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Procedure

SUBTASK 05-56-01-760-001

(1) Connect one intrinsically safe approved bonding meter, COM-1550 probe to one side of a complex joint build-up (structure, LRU, receptacle, bracket, or connector backshell for example) within 1 in. (25 mm) of the joint being measured.

NOTE: A joint may have several electrical connections. For example, a connector mounted on a bracket which is mounted to structure can provide five connections between the bundle shield and "ground": The shield is (1) connected to the connector backshell which is (2) connected to the connector itself which (3) connects to the receptacle which (4) is mounted on the bracket which is (5) connected to structure.

(a) Several kinds of possible measurement connections are shown in Figure 201.

SUBTASK 05-56-01-760-002

(2) Connect the second joint probe to the other side of the joint being measured, within 1 in. (25 mm) of the joint/connector.

NOTE: It may be necessary to wiggle the probes in order to make electrical contact through surface coatings.

SUBTASK 05-56-01-760-003

(3) Record the resistance value of this joint from the Bond Meter display.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The resistance values shown in Table 201 are maximum joint resistance values for common electrical connections/joints.

Table 201/05-56-01-993-801

TYPE OF JOINT	MAXIMUM RESISTANCE (ALUMINUM CONNECTORS) (MILLIOHMS)	MAXIMUM RESISTANCE (STAINLESS CONNECTORS) (MILLIOHMS)
BACKSHELL-TO-RECEPTACLE	2.5	5.0
BACKSHELL-TO-PLUG BODY	2.5	5.0
PLUG-TO-RECEPTACLE	5.0	12.5
FEEDTHROUGH-TO-STRUCTURE/SKIN	10.0	N/A
RECEPTACLE-TO-BRACKET	2.5	5.0
BRACKET-TO-BOND STRAP	1.0	1.0
BOND STRAP-TO-STRUCTURE	1.0	1.0
BRACKET-TO-STRUCTURE	1.0	1.0
RIVETED BRACKET-TO-STRUCTURE	1.0	1.0
OTHER FAY SURFACE BONDS	1.0	1.0

(b) To calculate the maximum allowable resistance for a built-up joint, do the following:

1) Determine what type of electrical connections/joints and how many of each are contained within the built-up joint.
2) Find the maximum allowable resistance for each type of electrical connection/joint contained within the built-up joint.
3) Sum the individual maximum allowable resistances for each electrical connection/joint contained within the built-up joint. This sum is the calculated maximum allowable resistance for the built-up joint.

NOTE: For example, the maximum resistance from a backshell to structure could be the sum of the following values: Backshell-to-Receptacle + Receptacle-to-Bracket + Bracket-to-Structure. Adding up the values in Table 201 (for an aluminum connector) of 2.5 + 2.5 + 1 milliohms each, totals 6 milliohms as a maximum value.

(c) If the measured built-up joint resistance is higher than the calculated maximum allowable resistance, then do a joint resistance measurement across each individual connection/joint to isolate the specific high resistance connection/joint.

1) If an individual connection/joint resistance exceeds the maximum allowable resistance for that type of connection/joint, then repair the faulty connection/joint.

NOTE: Loose connectors and backshells or pigtails can cause an out of tolerance resistance reading.

a) If the resistance of the BACKSHELL-TO-RECEPTACLE is not within limits, refer to: SWPM 20-14-11 for connector replacement or repair due to corrosion or fretting between connectors.

b) If the resistance of the RECEPTACLE-TO-BRACKET is not within limits, refer to: SWPM Chapter 20 for procedures to repair corrosion between surfaces.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) If the resistance of the BRACKET-TO-STRUCTURE is not within limits, refer to: SWPM Chapter 20 for procedures to repair corrosion between surfaces of a critical bond.

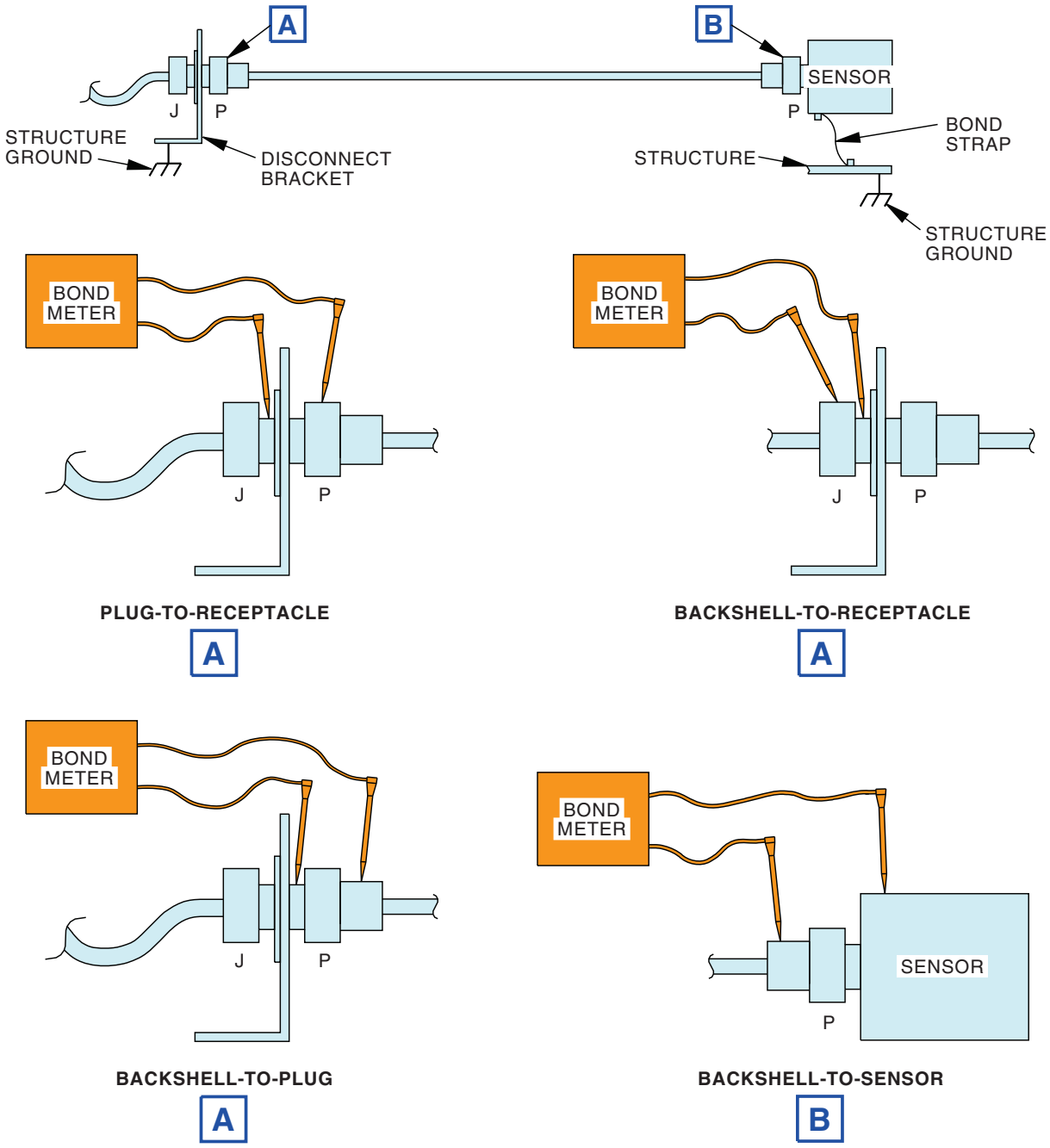
2) After you complete the joint repair, repeat the Joint Buildup Resistance Measurement to verify the repair.

(d) If the measured joint resistance is less than the calculated maximum allowable resistance for the built-up joint, then do the Joint Buildup Resistance Measurement on the next joint, if any, to be tested.

(e) If all of the built-up joints/connections for a particular wire bundle have been measured and joint resistances are less than the calculated maximum allowable resistance for the built-up joints, then the Joint Buildup Resistance Measurement Task is complete.

——— END OF TASK ———

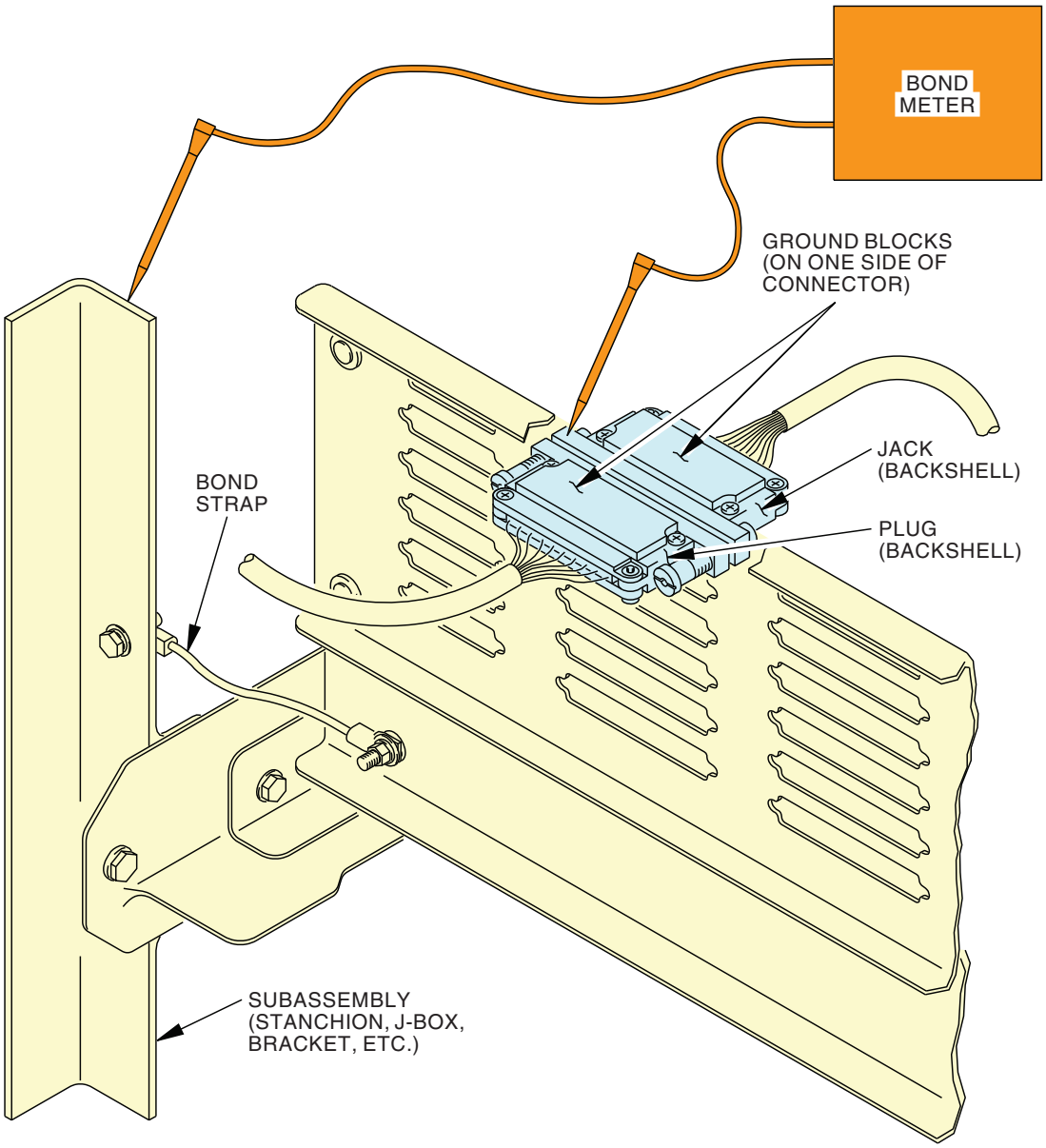
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410579 S00061525906_V1

Bond Meter General Connections for Joint Build-up Test High Intensity Radiated Field (HIRF) Inspection
Figure 201/05-56-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RECTANGULAR CONNECTORS

2410580 S00061525907_V1

Bond Meter General Connections for Joint Build-up Test High Intensity Radiated Field (HIRF) Inspection
Figure 201/05-56-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOOP RESISTANCE TESTER (LRT)- MAINTENANCE PRACTICES

1. General

A. This Procedure has these tasks:

(1) A Lid Standard Measurement of the Loop Resistance Tester (LRT).

(2) A Standard Measurement of the MK Loop Resistance Tester (ExLRT).

B. The loop resistance tester, SPL-1636 (LRT) and MK loop resistance tester, SPL-18565 (ExLRT) are specialized pieces of test equipment which are used to make non-intrusive wire bundle and coax cable shield resistance measurements. The LRT and the ExLRT can also be used to make resistance measurements across a built-up joint or individual electrical connections/joints.

TASK 05-56-02-200-801

2. LRT Lid Standard Measurement

A. General

(1) This task contains the procedure for performing a Lid Standard Measurement with the LRT. The Lid Standard Measurement is a check of the (LRT) that should be performed whenever the LRT does not appear to be operating correctly. Measurement of the standard resistances on the LRT storage lid provides a quick indication of proper LRT operation. The Lid Standard Measurement consists of the following two parts:

(a) Loop Mode, which makes a check of the loop resistance measurement function.

(b) Joint Mode, which makes a check of the joint resistance measurement function.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2

C. Procedure

SUBTASK 05-56-02-820-001

(1) Loop Mode

(a) The various components of the loop resistance tester, SPL-1636 (LRT) are illustrated in Figure 201.

(b) Open the LRT assembly and slide the lid aside to separate it from the base.

(c) Open the lid and remove the Sense/Drive couplers and the Joint Probe assembly.

(d) Connect the blue-coded Sense coupler cable to the blue-coded Sense Coupler connector on the LRT.

(e) Connect the red-coded Drive coupler cable to the red-coded Drive Coupler connector on the LRT.

(f) Connect the Joint Probe assembly to the Joint Probe connector on the LRT.

(g) Lift the safety cover and set the RUN-OFF/CHARGE switch to the RUN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Set the MODE switch to the LOOP position.

(i) Push and hold the ON/OFF push-button switch on the BITE module until "Testing Hardware" shows on the BITE display, then release the ON/OFF switch. A four character symbol will move around the BITE display during the hardware self-test.

NOTE: To conserve the battery, the LRT will automatically power down after 10 minutes of inactivity. To re-apply power, press the “ON” button.

(j) When the "Press Start" indication shows on the BITE display, make sure the LED on each Sense/Drive coupler control box is red.

(k) Connect the Sense/Drive couplers to the cutout in the Reference Standard, built into the storage lid of the LRT.

1) Position the couplers at each end of the cutout, near the Sense/Drive coupler alignment marks (View C, Figure 201).

(l) Begin the Lid Standard Test by pushing the START push-button switch on either of the Sense/Drive coupler control boxes.

(m) Make sure the LED on each of the Sense/Drive coupler control boxes is green.

1) If either LED is red and a "DRV Coup Is Open" or "SNS Coup Is Open" message shows on the BITE display, then the couplers should be checked for proper connection.

a) Gently squeeze and release the coupler handles to improve the connection of the couplers to the Lid Standard. If adjusting the couplers does not change the LEDs to green, replace the LRT.

b) If "Press Start" shows on the BITE display, then push the START push-button on either of the Sense/Drive coupler control boxes and begin the Lid Standard Test again.

2) If either LED is red and a "DRV Coup Is Open" or "SNS Coup Is Open" message does not show on the BITE display, replace the LRT.

3) If the LED on one or both couplers flashes red, then re-connect the couplers to the lid after the "Press Start" message shows on the BITE display.

a) Push the START push-button on either of the Sense/Drive coupler control boxes and begin the Lid Standard Test again.

4) If the message "UNSTABLE" shows on the BITE display and does not clear, set the MODE switch to the JOINT position.

a) When the message "CONNECT PROBES" shows on the BITE display, set the MODE switch back to the LOOP position.

b) Push the START push-button on either of the Sense/Drive coupler control boxes and begin the Lid Standard Test again.

c) If the "UNSTABLE" message continues to show, replace the LRT.

(n) When the Sense/Drive coupler control box LEDs flash green, note the measured resistance value which shows on the BITE display.

1) Make sure the displayed resistance value is 0.5 - 1.0 milliohm.

2) If the displayed resistance value is not 0.5 - 1.0 milliohm, replace the LRT.

SUBTASK 05-56-02-820-002

(2) Joint Mode

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) With the Sense/Drive couplers still connected to the Lid Reference Standard, change the MODE switch to the JOINT position.

(b) Make sure the Joint Probe LEDs are red (View B, Figure 201).

(c) Place the tip of one Joint Probe on the first Joint Probe reference mark on the lid Reference Standard (View C, Figure 201).

(d) Place the tip of the other Joint Probe on the second Joint Probe reference mark.

NOTE: The LRT will start the Joint Test when the LEDs on both Joint Probes are green.

(e) When the LED on either Joint Probe flashes green, the measurement shows, in milliohms, on the bottom line of the BITE display.

NOTE: The loop resistance from the previous loop measurement is displayed on the top line of the BITE display during a joint mode measurement.

(f) Note the joint resistance value.

(g) Keep the first Joint Probe in place on the first Joint Probe reference mark and move the second Joint Probe to the next Joint Probe reference mark.

(h) Repeat the above steps until a resistance value has been measured for each Joint Probe reference mark, 2 through 6.

(i) Compare the measured resistance values with the acceptable resistance values listed below:

Table 201/05-56-02-993-801

Measurement Points (Referenced to First Joint Probe Mark)	Resistance Measurement Acceptable Values
Second Joint Probe Reference Mark	0.01 - 0.06 milliohms
Third Joint Probe Reference Mark	0.04 - 0.11 milliohms
Fourth Joint Probe Reference Mark	0.07 - 0.16 milliohms
Fifth Joint Probe Reference Mark	0.10 - 0.21 milliohms
Sixth Joint Probe Reference Mark	0.12 - 0.26 milliohms

1) If all five resistance measurements taken above are within the range of Acceptable Values from Table 201, then this task is complete and the LRT is operational.

2) If any of the five resistance measurements taken above are outside the range of Acceptable Values from the table, then replace the LRT.

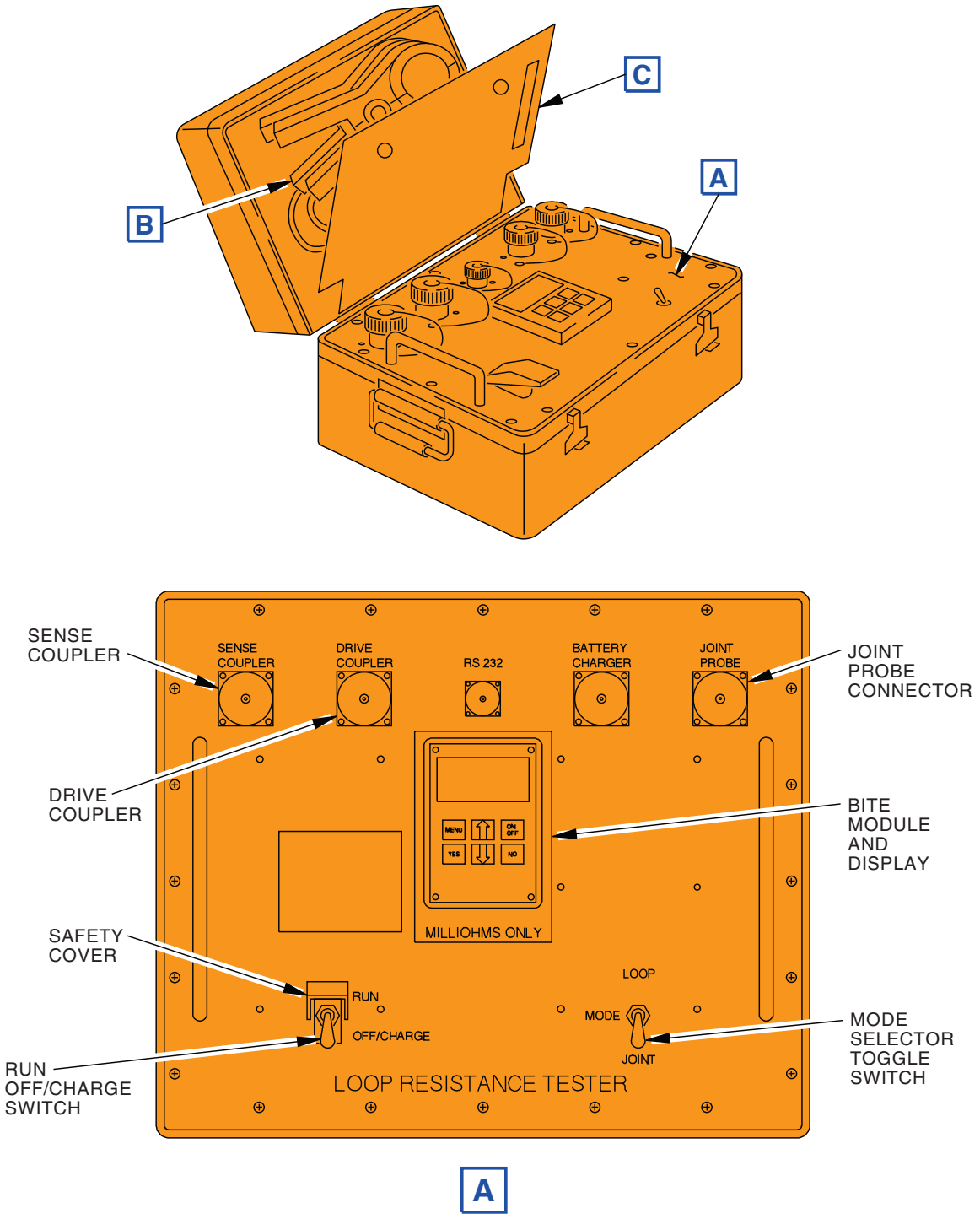
(j) Change the MODE switch to LOOP.

(k) Disconnect the Sense/Drive couplers from the Lid Reference Standard.

(l) This completes the LRT Lid Standard Measurement.

——— END OF TASK ———

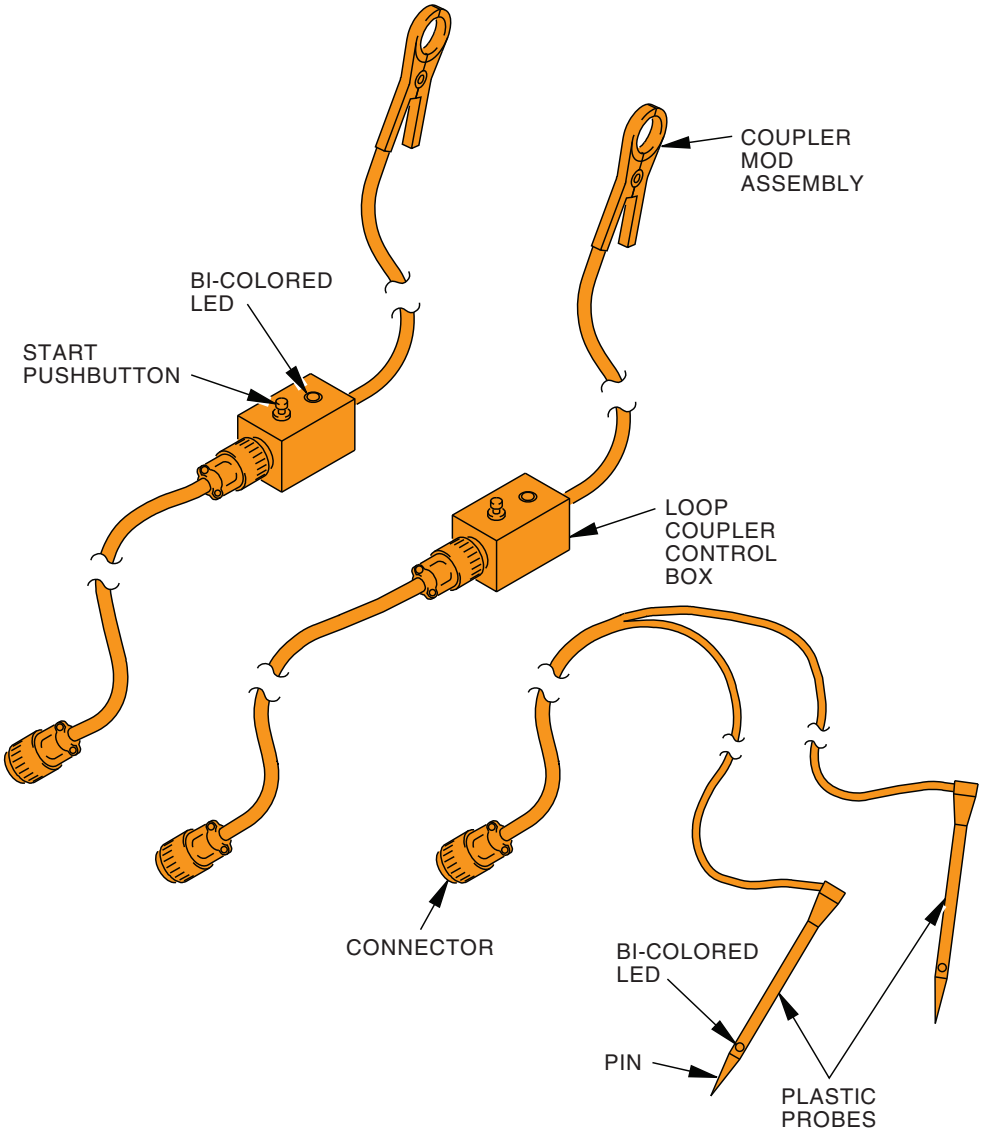
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410581 S00061525911_V1

High Intensity Radiated Fields (HIRF) Inspection - Loop Resistance Tester (LRT) - Maintenance Practices
Figure 201/05-56-02-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

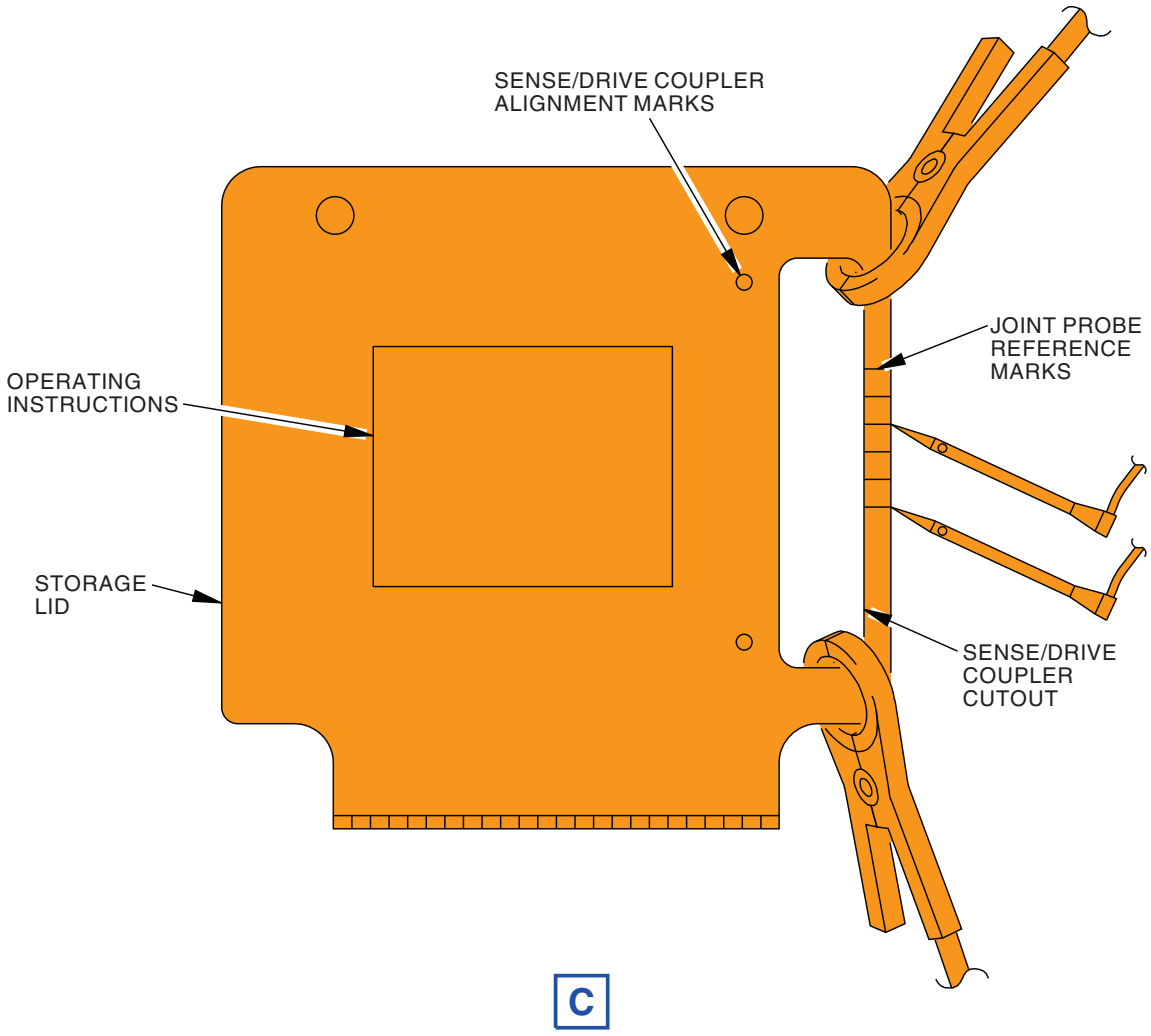


B

2410582 S00061525912_V1

High Intensity Radiated Fields (HIRF) Inspection - Loop Resistance Tester (LRT) - Maintenance Practices
Figure 201/05-56-02-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410583 S00061525913_V1

High Intensity Radiated Fields (HIRF) Inspection - Loop Resistance Tester (LRT) - Maintenance Practices
Figure 201/05-56-02-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 05-56-02-200-802

3. ExLRT Standard Measurement

Figure 202

A. General

(1) This task contains the procedure for performing a standard handle measurement with the ExLRT. The standard handle Measurement is a check of the Loop Resistance Tester that should be performed whenever the ExLRT does not appear to be operating correctly. Measurement of the standard resistances on the LRT Handle provides a quick indication of proper ExLRT operation. The Standard Handle Measurement consists of the following two parts:

(a) Loop Measurement, which makes a check of the loop resistance measurement function.

(b) Joint Measurement, which makes a check of the joint resistance measurement function.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

C. Prepare for the Measurement

SUBTASK 05-56-02-820-003

(1) Prepare the MK loop resistance tester, SPL-18565, for the measurement (Figure 202).

(a) Remove the ExLRT assembly from it's box.

(b) Connect the Loop cable [4] to the Loop connector [2].

(c) Connect the Joint cable [5] to the Joint connector [3].

(d) To power on, push the Power-on button [1] on the rear of the front panel.

NOTE: The LCD display will show the unit information.

1) If the ExLRT does not turn on, do these steps:

a) Open the battery compartment.

b) Remove the battery.

c) Put the charged battery into the unit.

d) Close the battery compartment.

2) If necessary, refer to the ExLRT user’s guide and contact the device manufacturer.

SUBTASK 05-56-02-820-004

(2) Configure the ExLRT, as follows:

(a) Make sure that LCD display shows 'Manual (continuous)'.

1) If it is not displayed, push the Select button [9].

2) Use the Scroll up button [7] and Scroll down button [8] to select 'Continuous (Default)'.

3) Push the Select button [9].

4) Push the Stop measurement button [10] to go back to Main menu.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Select “Manual (Continuous)”.

6) Push the Select button [9].

7) Select “Continuous (Default)”.

8) Push the Select button [9].

D. ExLRT Standard Measurement

SUBTASK 05-56-02-820-005

(1) Do the self-test of the ExLRT, as follows:

(a) Clamp the couplers of the Loop cable [4] to the self-test handle of ExLRT.

(b) Push the Start measurement button [6].

(c) Make sure that the loop status LED indicator is amber.

(d) Record the value of the Loop resistance.

NOTE: The loop measurements are displayed as a number with two decimal places on left top of LCD screen.

1) If the loop status LED indicator is red and either “Drive coupler is open” or “Sense Coupler is Open” message shows on the LCD display, do these steps:

a) Gently squeeze and release the coupler handles to improve the connection of the coupler around the wire bundle being measured.

b) If adjusting the couplers does not change the loop status indicator to amber, use a different set of calibrated loop couplers.

c) If neither of these actions address the issue, contact the manufacturer.

d) If error message disappears and loop status LED indicator is flashing green, push the Start measurement button [6]. to begin Loop Resistance measurement.

2) If the loop status LED indicator is red and “Loop Unstable” message shows on the LCD display, contact the manufacturer.

(e) Attach one pin of the Joint cable [5] on the joint probe reference mark 1 on self-test handle of ExLRT.

(f) Attach the second pin of the Joint cable [5] on the joint probe reference mark 4.

(g) Record the value of the Joint 1 resistance.

NOTE: The joint measurement will be displayed as a number with two decimal places on right top of LCD screen.

1) If joint status LED indicator is red, check the LCD display for an error message.

2) If necessary, refer to the ExLRT user’s guide for corrective action based on display error message.

(h) Attach the one pin of the Joint cable [5] on the joint probe reference mark 2.

(i) Attach the second pin of the Joint cable [5] on the joint probe reference mark 3.

(j) Record the value of the Joint 2 resistance.

NOTE: The joint measurement will be displayed as a number with two decimal places on right top of LCD screen.

1) If joint status LED indicator is red, check the LCD display for an error message.

2) If necessary, refer to the ExLRT user’s guide for corrective action based on display error message.

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SUBTASK 05-56-02-820-006

(2) If unable to complete the joint resistance measurement, do the steps that follows:

(a) If the joint status indicator is red, check the LCD display for an error message.

1) If the message on the LCD displays "Joint Intermittent", do the following:

a) Push the Stop measurement button [10] and remove the joint probes from the connector to stop the measurement.

b) Push the Start measurement button [6] and re-position probes to correct locations to start measurement.

c) If this fails to clear the "Joint Intermittent" message, contact the manufacturer.

2) If the message on the LCD displays a “Probes not connected to UUT", do the following:

a) Push the Stop measurement button [10] and remove the joint probes from the connector to stop the measurement.

b) Push the Start measurement button [6] and re-position probes to correct locations to start measurement.

3) If this fails to clear the message, then use a different set of calibrated probes.

4) If another set of calibrated probes isn't available or doesn't resolve the error message, contact the manufacturer.

SUBTASK 05-56-02-820-007

(3) Compare the measured resistance values with the acceptable resistance values (Table 202).

Table 202/05-56-02-993-802

Measurement Types	Resistant Measurement Acceptable value
Loop Resistance	9.0–10.0 Milioms
Joint 1 Resistance	7.5–8.5 Milioms
Joint 2 Resistance	0.1–0.3 Milioms

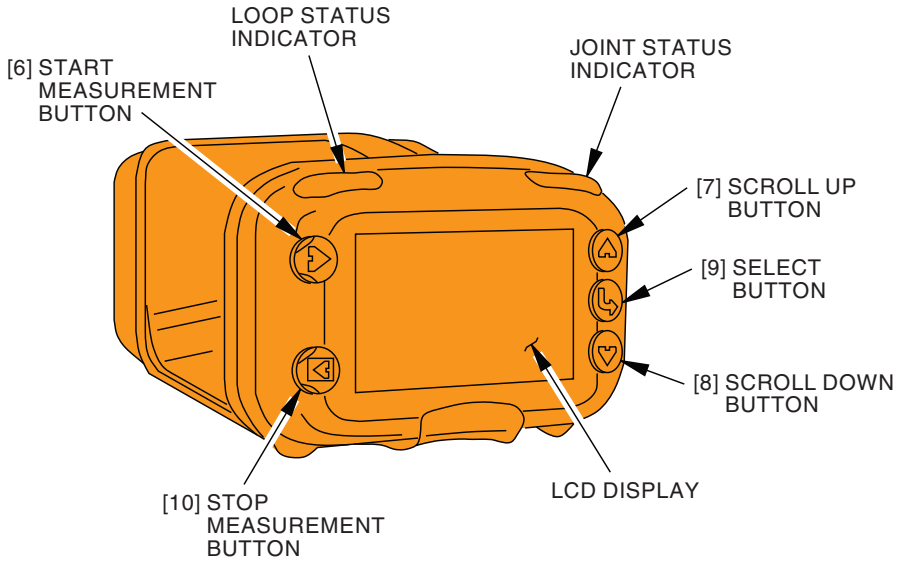
(a) If either the loop or joint resistance values are not in the acceptable range, contact the manufacturer.

(b) If all resistance measurements are in the acceptable range, then this task is completed and the ExLRT is operational.

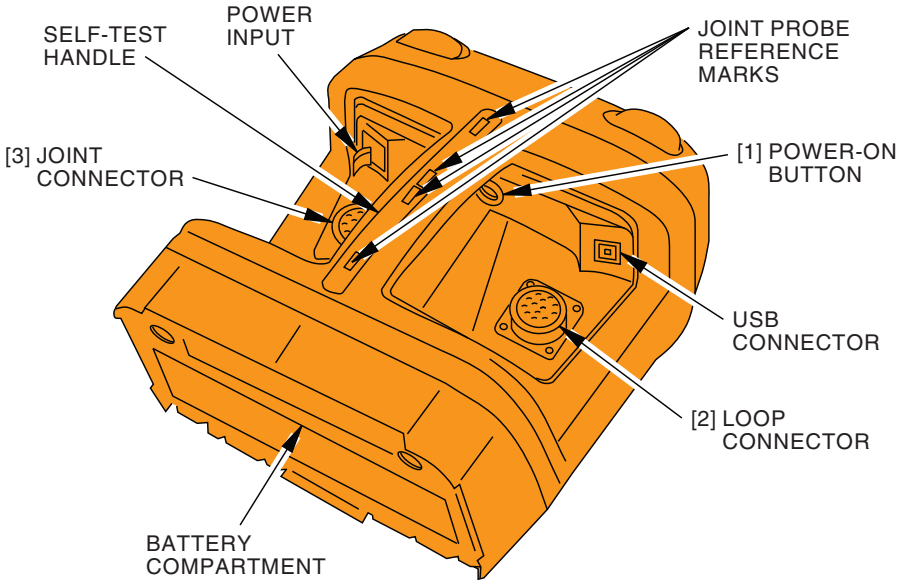
(c) Remove the couplers from the self-test handle of the ExLRT.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**MK LOOP RESISTANCE TESTER (ExLRT)
(FRONT VIEW)**

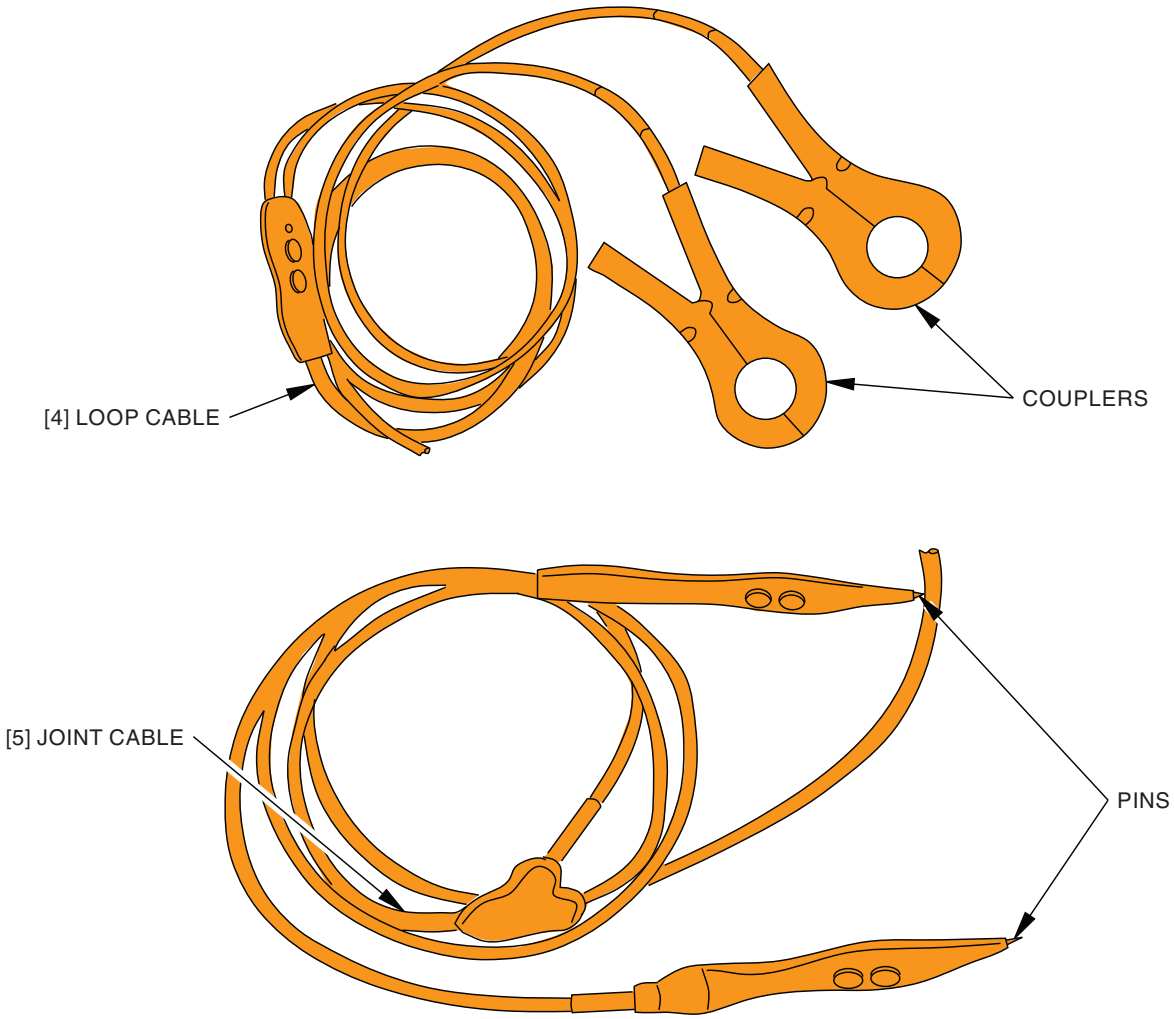


**MK LOOP RESISTANCE TESTER (ExLRT)
(TOP VIEW)**

2929035 S0000708126_V1

**Loop Resistance Tester
Figure 202/05-56-02-990-802 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Resistance Tester
Figure 202/05-56-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIRF/LIGHTNING LOOP RESISTANCE MEASUREMENT - MAINTENANCE PRACTICES

1. General

A. The comments and instructions contained in this procedure apply to loop resistance tester, SPL-1636 (LRT) when so noted.

B. The LRT is used to make non-intrusive wire bundle and coax cable shield resistance measurements by inductively coupling a low voltage into the shield and measuring the current flowing through the shield. The LRT can also be used to make resistance measurements across a built-up joint or individual electrical connections/joints after an electrical loop is first established.

C. This procedure does the Loop Resistance Measurement task.

(1) The Loop Resistance Measurement task contains instructions on how to use the LRT to measure the shield resistance of a wire bundle or coaxial cable. The shield must be electrically connected to airplane structure at both ends in order to form an electrical loop for current to flow. In certain cases, a ground return jumper cable must be used to provide a ground return path because the wire bundle is not connected to airplane structure.

TASK 05-56-03-200-801

2. Loop Resistance Measurement

(Figure 201, Figure 202)

A. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-04-200-801	Joint Resistance Measurement (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2

C. Prepare for the Loop Resistance Measurement

SUBTASK 05-56-03-860-001

WARNING REMOVE ELECTRICAL POWER. THIS IS A HIGH-VOLTAGE AND HIGH-CURRENT CONNECTION. ELECTRICAL SHOCK CAN CAUSE INJURIES AND KILL YOU.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 05-56-03-940-001

(2) Prepare the loop resistance tester, SPL-1636 (LRT) for use (Figure 201).

(a) Open the LRT assembly and slide the lid aside to remove it from the base.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open the lid and remove the Sense/Drive couplers and the joint probe assembly.

(c) Connect the blue-coded Sense coupler cable to the blue-coded Sense Coupler connector on the LRT.

(d) Connect the red-coded Drive coupler cable to the red-coded Drive Coupler connector on the LRT.

(e) Connect the Joint Probe assembly to the Joint Probe connector on the LRT.

(f) Lift the safety cover and set the RUN-OFF/CHARGE switch to the RUN position.

(g) Set the LRT MODE switch to the LOOP position.

(h) Push and hold the ON/OFF push-button switch on the BITE module until "Testing Hardware" shows on the BITE display, then release the ON/OFF switch. A four character symbol will move around the BITE display during the hardware self-test.

NOTE: To conserve the battery, the LRT will automatically power down after 10 minutes of inactivity. To re-apply power, do the above steps again.

(i) When the "Press Start" indication shows on the BITE display, make sure the LED on each Sense/Drive coupler control box is red.

D. Loop Resistance Measurement

SUBTASK 05-56-03-760-001

(1) Connect the couplers.

(a) For a normal Loop Resistance Measurement (Figure 202):

1) Clamp both of the LRT couplers around the wire bundle to be measured.

a) If there are ground wires, such as shield pigtail connections, at the point of measurement on the measured bundle, make sure that such wiring is NOT included within either coupler.

b) Make sure that there is at least 4 in. (102 mm) separation between the two couplers. If such separation is impossible to maintain, use the Mu-Metal shields (P/N 906-10249-9) provided with the LRT to obtain the necessary isolation.

NOTE: In the case of multiple wire bundles going to/from a connector, make sure the coupler is only around the desired bundle. This may require removing (and replacing) wire ties to separate the bundles enough for coupler connection.

SUBTASK 05-56-03-760-002

(2) Initiate a Loop Resistance Measurement by pushing the START push-button on either of the Sense/Drive Coupler Control Boxes, causing the LEDs on both boxes to turn green.

(a) When the LEDs flash green, note the loop resistance value shown on the LRT BITE display.

(b) This completes the Loop Resistance measurement.

SUBTASK 05-56-03-810-001

(3) If you are unable to complete the Loop Resistance Measurement, do the steps that follow:

(a) If either LED is red and a "DRV Coup Is Open" or "SNS Coup Is Open" message shows on the BITE display, then the couplers should be checked for proper connection.

1) Gently squeeze and release the coupler handles to improve the connection of the couplers around the wire bundle being measured.

a) If adjusting the couplers does not change the LEDs to green, replace the LRT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If "Press Start" shows on the BITE display, then push the START push-button on either of the Sense/Drive coupler control boxes and begin the Loop Resistance Measurement again.

(b) If either LED is red and a "DRV Coup Is Open" or "SNS Coup Is Open" message does not show on the BITE display, replace the LRT.

(c) If the LED on one or both couplers flashes red, then re-connect the couplers to the wire bundle being measured after the "Press Start" message shows on the BITE display.

1) Push the START push-button on either coupler control box and begin the Loop Resistance Measurement again.

(d) If the message "UNSTABLE" is displayed and does not clear, set the MODE switch to the JOINT position.

1) When the message "Connect Probes" shows on the BITE display, set the MODE switch back to the LOOP position.

2) When the "Press Start" message appears, push the Start push-button on either of the coupler control boxes and begin the Loop Resistance Measurement again.

3) If this fails to clear the "UNSTABLE" message, then there is a problem with the wire bundle being measured or with the LRT.

NOTE: The "UNSTABLE" message is caused by the LRT detecting a varying resistance. This varying resistance can be caused by a loose connection, a corroded or intermittent electrical joint, a faulty Transorb adapter, or a faulty LRT.

a) Make sure that the LRT is operating correctly, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801.

b) If the LRT is operating correctly, then the wire bundle or a connection/joint is faulty and must be repaired or replaced before continuing the measurement.

<1> Check each electrical connection in the circuit being measured to isolate the faulty connection/joint, do this task: Joint Resistance Measurement, TASK 05-56-04-200-801.

(e) If the BITE display shows "Joint V TooBig" (Dash-2 LRT only), "Loop I TooBig" or "Loop V TooBig", then stray currents may be circulating through the shield of the wire bundle being measured.

1) If power is applied to any of the equipment attached to this bundle, remove power from the equipment.

2) Any on-board electrical equipment that does not need to be ON should be turned OFF.

3) Make sure that no other sources of current or voltage are connected on or near the bundle being tested.

4) In extreme cases, all power may need to be removed from the airplane.

SUBTASK 05-56-03-810-002

(4) For a Dash-2 or -3 LRT, the following BITE display messages may also appear:

(a) Joint TooSmall:

1) This indication is displayed when the Joint Value is too small compared to the Loop Value to give an accurate Joint reading. This condition exceeds the operating capabilities of the LRT.

(b) LNoise V TooBig:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) This indication is displayed when excessive noise voltage on the wire bundle shield interferes with the LRTs ability to read accurately during Loop Mode. Stray signals on the wire shield may be produced by motors, electronic equipment, or adjacent electronic circuits. This condition can be alleviated by de-energizing equipment in the vicinity of the circuit being tested.

(c) LNoise I TooBig:

1) This indication is displayed when excessive noise current on the wire bundle shield interferes with the LRTs ability to read accurately during Loop Mode. Stray signals on the wire shield may be produced by motors, electronic equipment, or adjacent electronic circuits. This condition can be alleviated by de-energizing equipment in the vicinity of the circuit being tested.

(d) JNoise V TooBig:

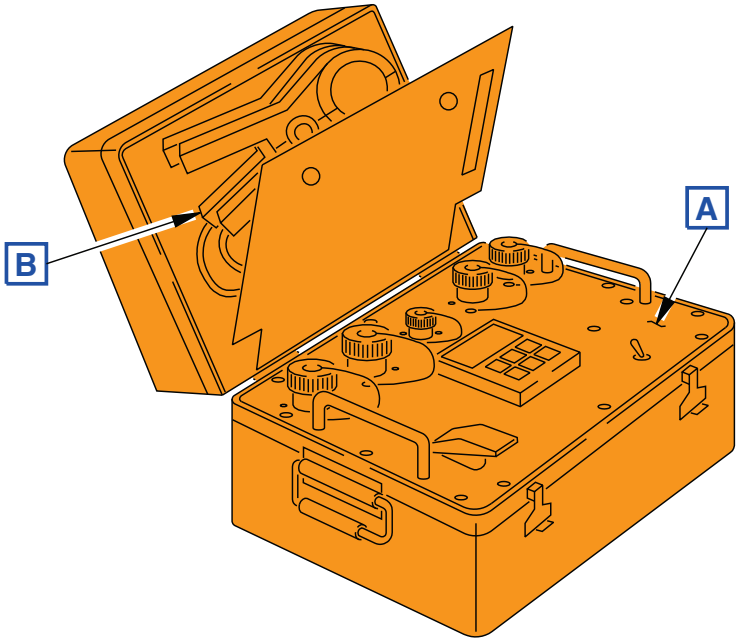
1) This indication is displayed when excessive noise voltage on the wire bundle shield interferes with the LRTs ability to read accurately during Joint Mode. Stray signals on the wire shield may be produced by motors, electronic equipment, or adjacent electronic circuits. This condition can be alleviated by de-energizing equipment in the vicinity of the circuit being tested.

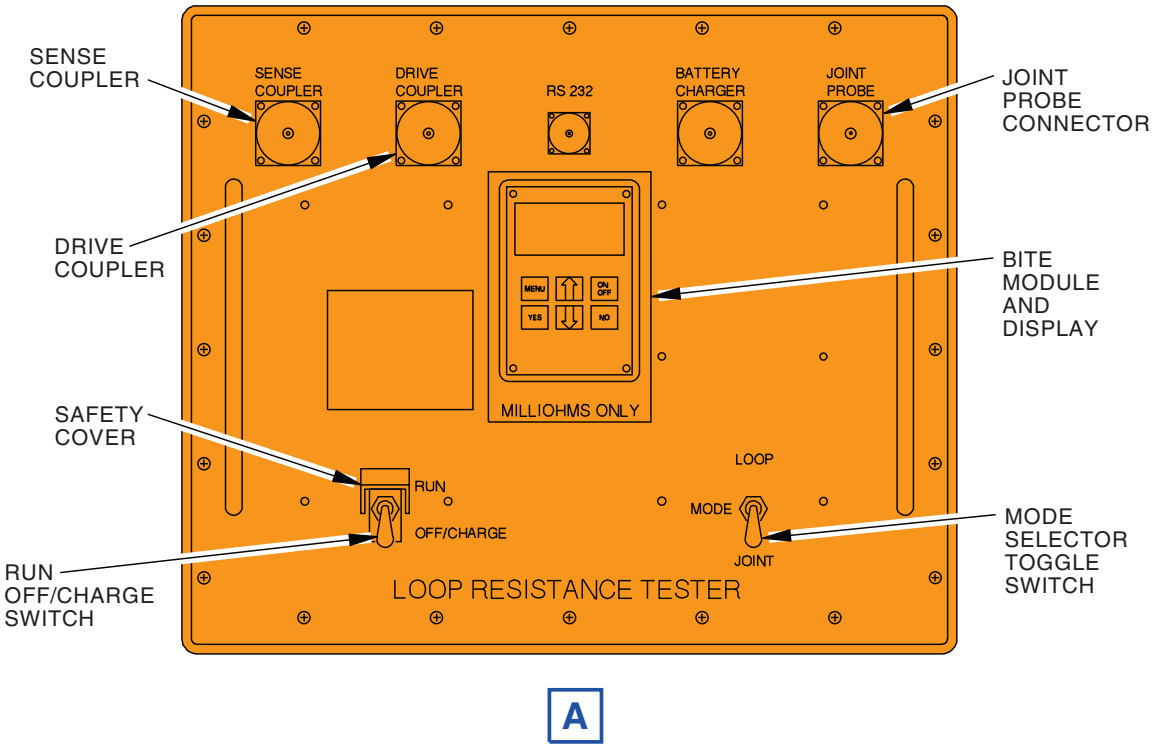
(e) JNoise I TooBig:

1) This indication is displayed when excessive noise current on the wire bundle shield interferes with the LRTs ability to read accurately during Joint Mode. Stray signals on the wire shield may be produced by motors, electronic equipment, or adjacent electronic circuits. This condition can be alleviated by de-energizing equipment in the vicinity of the circuit being tested.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

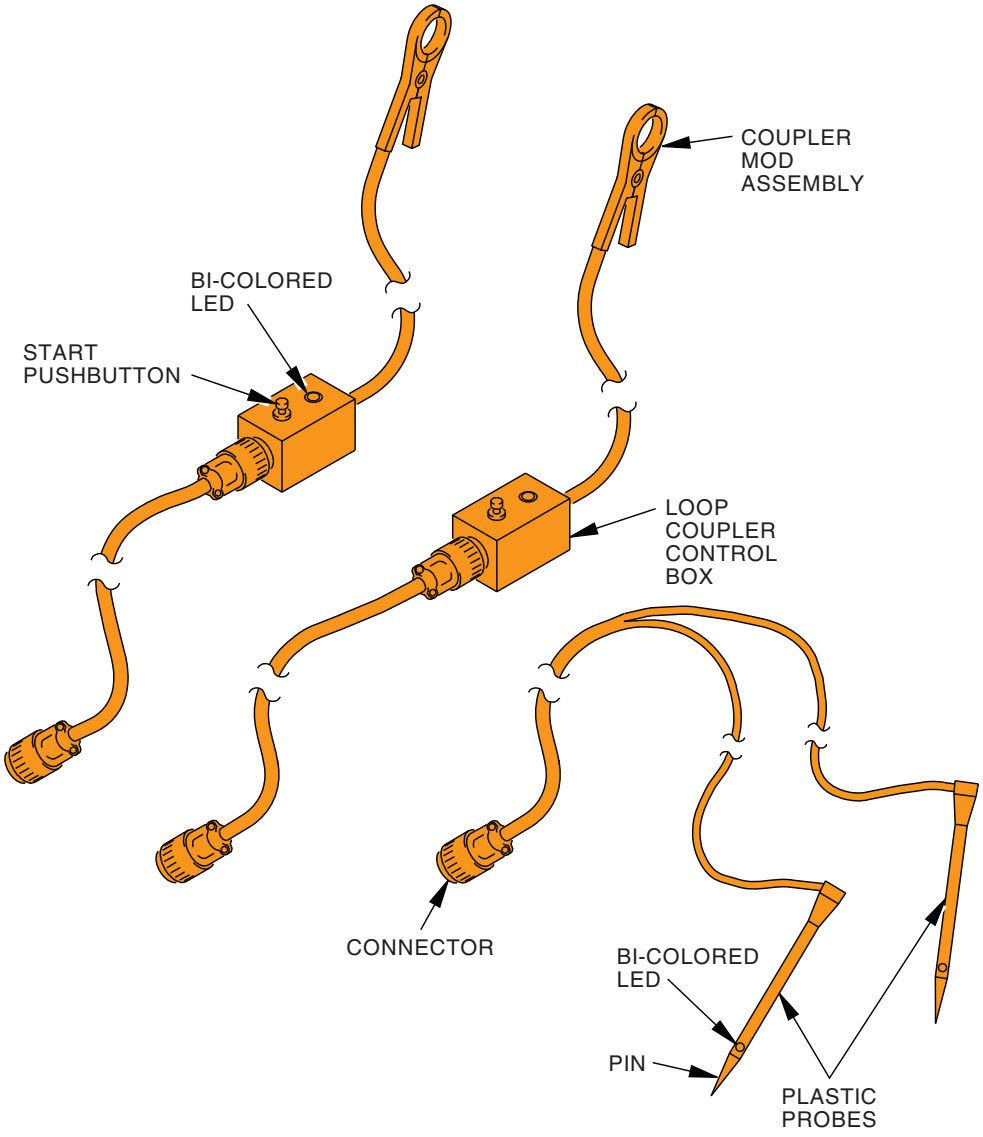




2410584 S00061525917_V1

High Intensity Radiated Fields (HIRF) Inspection Loop Resistance Tester Maintenance Practices
Figure 201/05-56-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

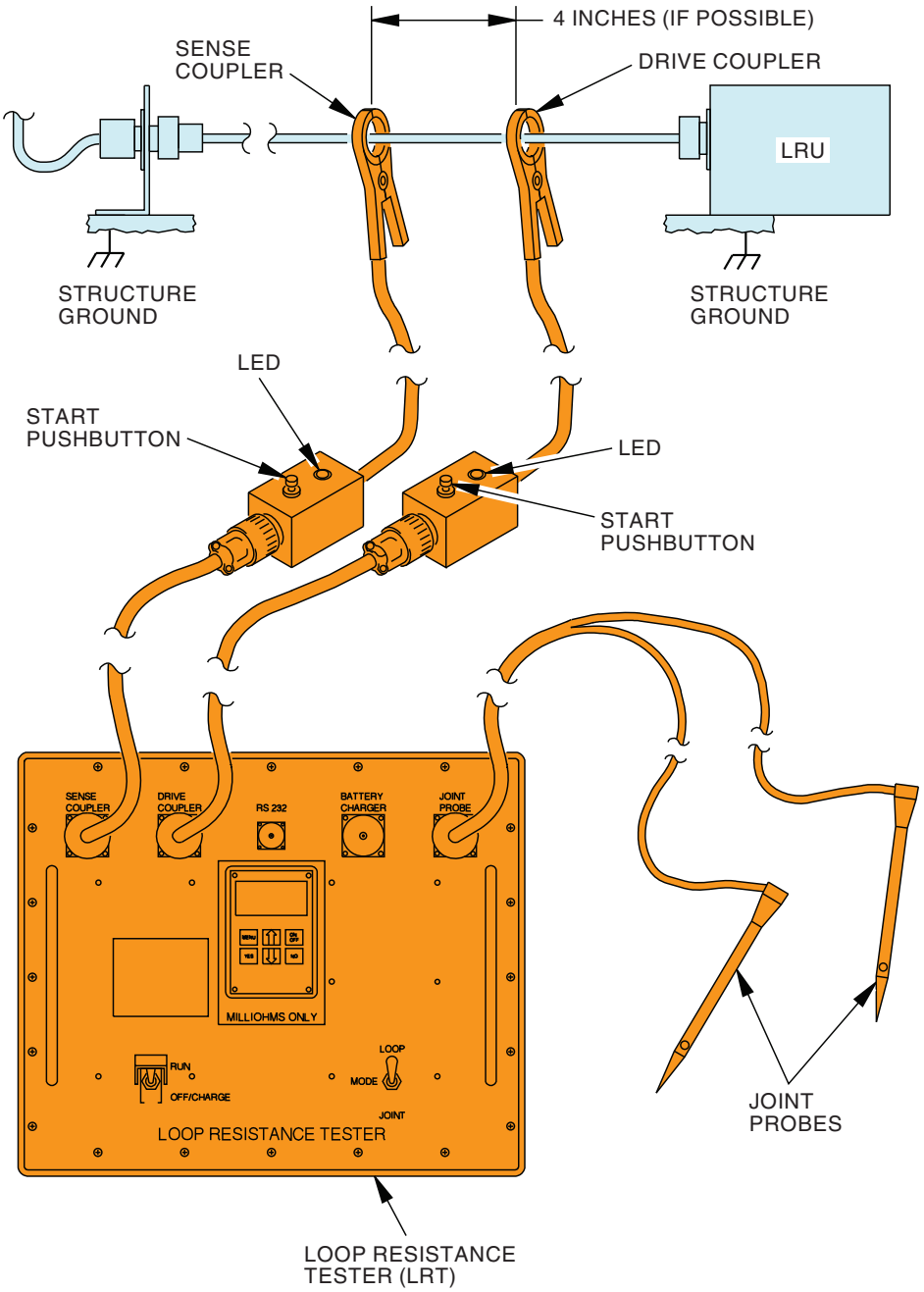


B

2410585 S00061525918_V1

High Intensity Radiated Fields (HIRF) Inspection Loop Resistance Tester Maintenance Practices
Figure 201/05-56-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410586 S00061525919_V1

Typical Loop Resistance Tester (LRT) Hookup for Loop Resistance Measurement
Figure 202/05-56-03-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIRF/LIGHTNING - JOINT RESISTANCE MEASUREMENT - MAINTENANCE PRACTICES

1. General

A. This task contains instructions on how to use the loop resistance tester, SPL-1636 (LRT) to measure the resistance across a built-up joint or individual electrical connections/joints.

(1) The direct contact voltage and current measurement capabilities of the LRT Joint Probes are used to perform a Joint Resistance Measurement.

(2) The connections/joints to be measured must be contained within the shielding system being measured in the Loop Resistance Mode (the loop current must flow through these joints/ connections).

B. In some cases, the current path to structure may include a subassembly. An example would be one of the connectors mounted to an FSEU Gearbox. In such a case, the bonding of the subassembly-to-structure may also need to be measured. If an out-of-spec joint reading is found, refer to the AMM chapter associated with that particular subsystem for component bonding requirements.

TASK 05-56-04-200-801

2. Joint Resistance Measurement

(Figure 201)

A. References

Reference	Title
05-56-02-200-801	LRT Lid Standard Measurement (P/B 201)
05-56-03-200-801	Loop Resistance Measurement (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-25-11	Overbraid Shields on Wire Bundles

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1636	Tester - Loop Resistance (LRT "-1" for checks up to & including 4 years ONLY, LRT "-2 & -3" applicable for all HIRF/FQIS check/inspection intervals) Part #: 906-10246-3 Supplier: 3X2T2 Opt Part #: 906-10246-1 Supplier: 3X2T2 Opt Part #: 906-10246-2 Supplier: 3X2T2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

SUBTASK 05-56-04-760-001

(1) To establish an electrical current flow in the wire bundle, joints/connections and bonding straps being measured, do this task: Loop Resistance Measurement, TASK 05-56-03-200-801 before starting the Joint Resistance Measurement.

NOTE: Before a Joint Resistance measurement can be performed, a Loop measurement must have been completed on the wire bundle associated with the joint being measured. The loop resistance tester, SPL-1636 (LRT) continues to monitor the loop resistance of the wire bundle while the joint measurement is being made. If the loop resistance changes by too large an amount, the LEDs on both joint probes will blink red and the message "REDO LOOP" will appear on the BITE display. This message cannot be erased without switching the LRT back to the Loop mode and doing a loop resistance measurement again.

SUBTASK 05-56-04-760-002

(2) Set the Mode Selector switch on the loop resistance tester, SPL-1636 (LRT) to the Joint position.

SUBTASK 05-56-04-760-003

(3) Make sure the LEDs on both Joint Probes are red.

SUBTASK 05-56-04-760-004

(4) Connect one joint probe to one side of a complex joint build-up (structure, LRU, receptacle, bracket, or connector backshell for example) within 1 in. (25 mm) of the joint being measured.

NOTE: A joint may have several electrical connections. For example, a connector mounted on a bracket which is mounted to structure can provide five connections between the bundle shield and "ground": The shield is (1) connected to the connector backshell which is (2) connected to the connector itself which (3) connects to the receptacle which (4) is mounted on the bracket which is (5) connected to structure.

SUBTASK 05-56-04-760-005

(5) Connect the second joint probe to the other side of the joint being measured, within 1 in. (25 mm) of the joint/connector.

NOTE: The LRT will start the Joint Resistance Measurement when the LEDs on both Joint Probes are green. The LEDs change from red to green when electrical continuity is established between the probes. It may be necessary to wiggle the probes in order to make electrical contact through surface coatings.

(a) If the Joint Probes appear to be making good electrical contact and the Joint Probe LEDs do not change to green and no error messages are displayed on the LRT, then, do this task: LRT Lid Standard Measurement, TASK 05-56-02-200-801.

(b) If the BITE display shows "Joint V Too Big", then stray currents may be circulating through the shield of the wire bundle being measured.

1) If power is applied to any of the equipment attached to this bundle, remove power from the equipment.

2) Any on-board electrical equipment that does not need to be ON should be turned OFF.

3) Make sure no other sources of current or voltage are connected on or near the bundle being tested.

4) In extreme cases, all power may need to be removed from the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 05-56-04-760-006

(6) When the LEDs on the probes flash green, record the resistance value of this joint from the bottom line of the BITE display.

(a) The resistance values shown in Table 201 are maximum joint resistance values for common electrical connections/joints.

Table 201/05-56-04-993-801

TYPE OF JOINT	MAXIMUM ALLOWABLE RESISTANCE (ALUMINUM CONNECTORS) IN MILLIOHMS	MAXIMUM ALLOWABLE RESISTANCE (STAINLESS/ STEEL) IN MILLIOHMS
BACKSHELL-TO-RECEPTACLE	2.5	5.0
BACKSHELL-TO-PLUG BODY	2.5	5.0
PLUG-TO-RECEPTACLE	5.0	12.5
RECEPTACLE-TO-BRACKET	2.5	5.0
BRACKET-TO-BOND STRAP	1.0	1.0
BOND STRAP-TO-STRUCTURE	1.0	1.0
BRACKET-TO-STRUCTURE	1.0	1.0
RIVETED BRACKET-TO-STRUCTURE	1.0	1.0
OTHER FAY SURFACE BONDS	1.0	1.0

(b) To calculate the maximum allowable resistance for a built-up joint, do the following:

1) Determine what type of electrical connections/joints and how many of each are contained within the built-up joint.

2) Find the maximum allowable resistance for each type of electrical connection/joint contained within the built-up joint.

3) Sum the individual maximum allowable resistances for each electrical connection/joint contained within the built-up joint. This sum is the calculated maximum allowable resistance for the built-up joint.

NOTE: For example, the maximum resistance from a backshell to structure could be the sum of the following values: Backshell-to-Receptacle + Receptacle-to-Bracket + Bracket-to-Structure. Adding up the values in Table 201 (for an aluminum connector) of 2.5 + 2.5 + 1 milliohms each, totals 6 milliohms as a maximum value.

(c) If the measured built-up joint resistance is higher than the calculated maximum allowable resistance, then do a joint resistance measurement across each individual connection/joint to isolate the specific high resistance connection/joint.

1) If an individual connection/joint resistance exceeds the maximum allowable resistance for that type of connection/joint, then repair the faulty connection/joint.

NOTE: Loose connectors and backshells or pigtails can cause an out of tolerance resistance reading.

a) Do the following task: SWPM 20-20-00, or SWPM 20-25-11.

2) After you complete the joint repair, repeat the Joint Buildup Resistance Measurement to verify the repair.

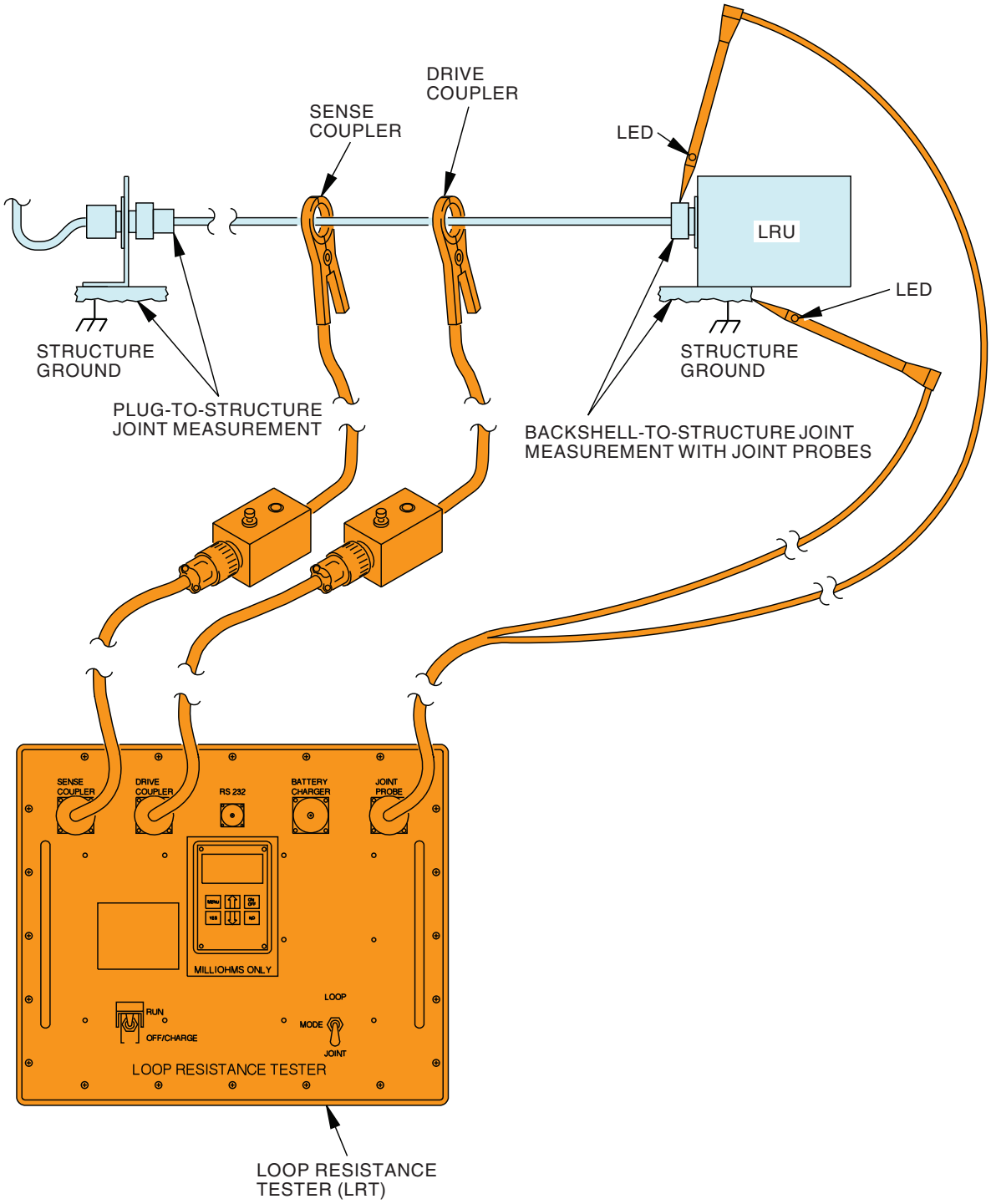
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If the measured joint resistance is less than the calculated maximum allowable resistance for the built-up joint, then do the Joint Buildup Resistance Measurement on the next joint in the wire bundle.

(e) If all of the built-up joints/connections for a particular wire bundle have been measured and joint resistances are less than the calculated maximum allowable resistance for the built-up joints, then the Joint Buildup Resistance Measurement Task is complete.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410588 S00061525924_V1

Typical Loop Resistance Tester (LRT) Hookup for Joint Buildup Test
Figure 201/05-56-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOOP RESISTANCE MEASUREMENT AND JOINT RESISTANCE MEASUREMENT - MAINTENANCE PRACTICES

1. General

A. The MK loop resistance tester, SPL-18565 (ExLRT), is a specialized piece of test equipment that is used to make non-intrusive wire bundle and coax cable shield resistance measurements. The ExLRT can also be used to make resistance measurements across a built-up joint or individual electrical connections/joints.

B. This task contains instructions on how to use ExLRT to measure the loop resistance of a wire bundle or coax cable shield and how to measure resistance across a built-up joint or individual electrical connections/joints.

(1) The connections/joints to be measured must be contained within the shielding system being measured in loop resistance mode (the loop current must flow through these joints/ connections).

(2) The direct contact voltage and current measurement capabilities of the ExLRT joint probes used to perform a joint resistance measurement.

C. In some cases, the current path to structure may include a subassembly. An example would be of the connectors mounted to an FSEU gearbox. In such a case, the bonding of the subassembly to structure may also need to be measured. If an out of spec joint reading is found, refer to AMM chapter associated with that particular subsystem for component bonding requirements.

TASK 05-56-05-200-801

2. Loop Resistance and Joint Resistance Measurement

Figure 201Figure 202 or Figure 203Figure 203

A. References

Reference	Title
05-56-02-200-802	ExLRT Standard Measurement (P/B 201)
05-56-02-990-802	Figure: Loop Resistance Tester (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-25-11	Overbraid Shields on Wire Bundles

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-18565	Tester - Loop Resistance, MK Part #: XLR-9703-01 Supplier: KC432 Part #: XLR-9703-02 Supplier: KC432

C. Prepare for the Measurement

SUBTASK 05-56-05-940-001

(1) Prepare the MK loop resistance tester, SPL-18565 (ExLRT), for measurement (Figure 05-56-02-990-802).

(a) Remove the ExLRT assembly from its box.

(b) Connect the Loop cable [4] to the Loop connector [2].

(c) Connect theJoint cable [5] to the Joint connector [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Push the Power-on button [1] on the rear of the front panel.

NOTE: The LCD display will show the unit information.

1) If ExLRT will not turn on, do these steps:

a) Change the battery in a safe zone:

<1> Open the battery compartment.

<2> Remove the battery.

<3> Put a charged battery into the unit.

<4> Close the battery compartment.

b) If necessary, refer to ExLRT user's guide and contact the device manufacturer.

D. Loop Resistance Measurement

SUBTASK 05-56-05-760-001

(1) For a normal loop resistance measurement (Figure 202 or Figure 203):

(a) Clamp two couplers around the wire bundle.

1) Make sure that ground wires, such as shield pigtail connections, at the point of measurement on the measured bundle are not included within either coupler.

2) Make sure that there is minimum 4 in. (102 mm) separation between the two couplers.

a) If necessary, skew the couplers 45 degrees.

3) Make sure that the coupler is only around the desired bundle.

a) If necessary, remove (and replace) wire ties to separate the bundles enough for coupler connection.

4) Push the Start measurement button [6] or green button on the loop controller to start a loop resistance measurement.

NOTE: The loop status indicator will flash green when the unit is ready to take a loop resistance measurement.

5) When the loop status indicator is amber, the loop resistance measurement is displayed as a number with two decimal places on top left of LCD display.

SUBTASK 05-56-05-820-002

(2) If either the loop status indicator is red and "Drive coupler is open" or "Sense Coupler is Open" message shows on the LCD display, do these steps:

(a) Gently squeeze and release the coupler handles to improve the connection of the coupler around the wire bundle being measured.

(b) If adjusting the couplers does not change the loop status indicator to amber, do this task: ExLRT Standard Measurement, TASK 05-56-02-200-802.

(c) If error message disappears and loop status indicator is flashing green, push the Start measurement button [6] to begin loop resistance measurement again.

SUBTASK 05-56-05-820-003

(3) If the loop status indicator is red, do these steps:

(a) Push the Stop measurement button [10] to stop the measurement.

(b) Re-connect the couplers to the wire bundle.

(c) Push the Start measurement button [6] to start new measurement.

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SUBTASK 05-56-05-820-005

(4) If this fails to clear the “Loop Unstable” message, then there is a problem with the wire bundle being measured or with the ExLRT.

NOTE: The "Loop Unstable" message is caused by the ExLRT detecting a varying resistance. This varying resistance can be caused by a loose connection, a corroded or intermittent electrical joint, a faulty transorb adapter, or a faulty ExLRT.

(a) Make sure that the ExLRT is operating correctly (ExLRT Standard Measurement, TASK 05-56-02-200-802).

(b) If the ExLRT is operating correctly, then the wire bundle or a connection/joint is faulty and must be repaired or replaced before continuing the measurement.

1) Check each electrical connection in the circuit being measured to isolate the faulty connection/joint.

SUBTASK 05-56-05-820-006

(5) If the loop status indicator is red and “Loop Resistance Low” message displays, do these steps:

(a) Make sure that ground wires, such as shield pigtail connections, at the point of measurement on the measured bundle are not included within either coupler.

(b) Make sure that there is minimum 4 in. (102 mm) separation between the two couplers.

1) If necessary, skew the couplers 45 degrees.

(c) Make sure that the coupler is only around the desired bundle.

1) If necessary, remove (and replace) wire ties to separate the bundles enough for coupler connection.

(d) Make sure that the ExLRT is operating correctly (ExLRT Standard Measurement, TASK 05-56-02-200-802).

SUBTASK 05-56-05-820-007

(6) If the loop status indicator is red and a "Loop Resistance High" message displays, do these steps:

(a) Make sure that the placement of couplers are correct on bundles under test.

(b) Make sure that the ExLRT is operating correctly (ExLRT Standard Measurement, TASK 05-56-02-200-802).

SUBTASK 05-56-05-820-008

(7) If the ExLRT is operating correctly, then the wire bundle or a connection/joint is faulty.

(a) Repair or replace the wire bundle or a connection/joint before continuing the measurement.

(b) Check each electrical connection in the circuit being measured to isolate the faulty connection/joint.

E. Joint Resistance Measurement

SUBTASK 05-56-05-760-003

(1) Complete the loop measurement on the wire bundle associated with the joint being measured.

NOTE: The ExLRT continues to monitor the loop resistance of the wire bundle while the joint measurement is being made.

NOTE: Keep couplers in the same configuration.

SUBTASK 05-56-05-760-004

(2) Make sure that the joint status indicator is flashing green.

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SUBTASK 05-56-05-760-005

(3) Connect one joint probe to one side of a complex joint build-up (structure, Line Replaceable Unit (LRU), receptacle, bracket, or connector backshell for example) within 1 in. (25 mm) of the joint being measured (Figure 203.

NOTE: A joint may have several electrical connections. For example, a connector mounted on a bracket that is mounted to structure can provide five connections between the bundle shield and "ground": The shield is (1) connected to the connector backshell that is (2) connected to the connector itself that (3) connects to the receptacle that (4) is mounted on the bracket that is (5) connected to structure.

SUBTASK 05-56-05-760-006

(4) Connect the second joint probe to the other side of the joint being measured, within 1 in. (25 mm) of the joint/connector.

(a) Make sure that the joint status indicator turn amber.

(b) Make sure that the joint measurement is displayed in the top right of the LCD display.

(c) If necessary, shake the probes to make electrical contact through surface coatings.

(d) If the joint probes appear to be making good electrical contact and the joint probe LED indicator does not change to amber and no error messages are displayed, do this task: ExLRT Standard Measurement, TASK 05-56-02-200-802.

(e) If the joint probe LED indicator is red and the LCD display shows "Joint Low vs Loop", do these steps:

1) Attach one end of the ground return test cable to the shield termination point of the electrical connector under test.

2) Thread the ground return test cable and wire bundle under test through both loop couplers.

3) Connect the other end of the ground return test cable to structure.

4) Push the Start measurement button [6] to start a new measurement, the loop resistance should be much lower than it was previously.

a) If the loop resistance is not lower, check the connection of the ground return test cable.

5) Connect one joint probe to one side of a complex joint build-up (structure, LRU, receptacle, bracket, or connector backshell for example) within 1 in. (25 mm) of the joint being measured (Figure 203).

6) Connect the second joint probe to the other side of the joint being measured, within 1 in. (25 mm) of the joint/connector.

a) Make sure that the joint status indicator turn amber.

b) Make sure that the joint measurement is displayed in the top right of the LCD display.

SUBTASK 05-56-05-760-007

(5) If the joint probe LED indicator is red and the LCD display shows “Joint Intermittent”, do these steps:

(a) Push the Stop measurement button [10] and remove joint probes from the connector to stop the measurement.

(b) Push the Start measurement button [6] and re-position probes to correct locations to start measurement.

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(c) If this fails to clear the “Joint Intermittent” message, then there is a problem with the wire bundle being measured or with the ExLRT.

(d) Make sure that the ExLRT is operating correctly, do this task: ExLRT Standard Measurement, TASK 05-56-02-200-802.

(e) If the ExLRT is operating correctly, then there is a problem with the connection/joint and it must be repaired or replaced before continuing the measurement.

SUBTASK 05-56-05-760-008

(6) When the LED is amber, record the resistance value of the joint shown in the top right of the LCD display.

(a) The following resistance values (Table 201) are maximum joint resistance values for common electrical connections/joints.

Table 201/05-56-05-993-801

TYPE OF JOINT	MAXIMUM ALLOWABLE RESISTANCE (ALUMINUM CONNECTORS) IN MILLIOHMS	MAXIMUM ALLOWABLE RESISTANCE (STAINLESS/ STEEL) IN MILLIOHMS
BACKSHELL-TO-RECEPTACLE	2.5	5.0
BACKSHELL-TO-PLUG BODY	2.5	5.0
PLUG-TO-RECEPTACLE	5.0	12.5
RECEPTACLE-TO-BRACKET	2.5	5.0
BRACKET-TO-BOND STRAP	1.0	1.0
BOND STRAP-TO-STRUCTURE	1.0	1.0
BRACKET-TO-STRUCTURE	1.0	1.0
RIVETED BRACKET-TO-STRUCTURE	1.0	1.0
OTHER FAY SURFACE BONDS	1.0	1.0

(b) Calculate the maximum allowable resistance for a built-up joint.

1) Determine what type of electrical connections/joints and how many of each are contained within the built-up joint.

2) Find the maximum allowable resistance for each type of electrical connection/joint (Table 201) contained within the built-up joint.

3) Sum the individual maximum allowable resistances for each electrical connection/joint contained within the built-up joint.

NOTE: This sum is the calculated maximum allowable resistance for the built-up joint.

(c) If the measured built-up joint resistance is higher than the calculated maximum allowable resistance, then do a joint resistance measurement across each individual connection/joint to isolate the specific high resistance connection/joint.

1) If an individual connection/joint resistance exceeds the maximum allowable resistance (Table 201) for that type of connection/joint, then repair the faulty connection/joint.

NOTE: Loose connectors and backshells or pigtails can cause an out of tolerance resistance reading.

a) Do the following task: (SWPM 20-20-00) or (SWPM 20-25-11).

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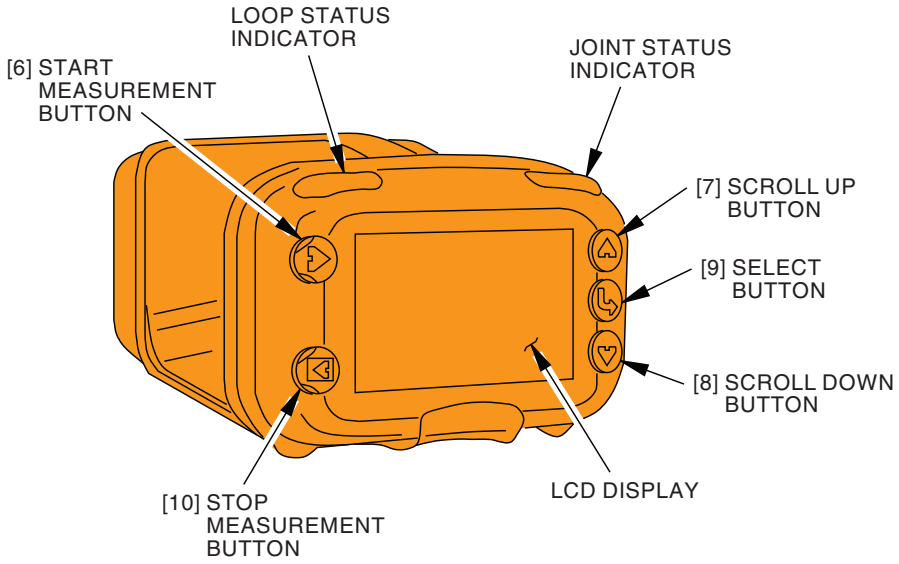
2) After you complete the joint repair, repeat the joint buildup resistance measurement to verify the repair.

(d) If the measured joint resistance is less than the calculated maximum allowable resistance for the built-up joint, then do the joint buildup resistance measurement on the next joint in the wire bundle.

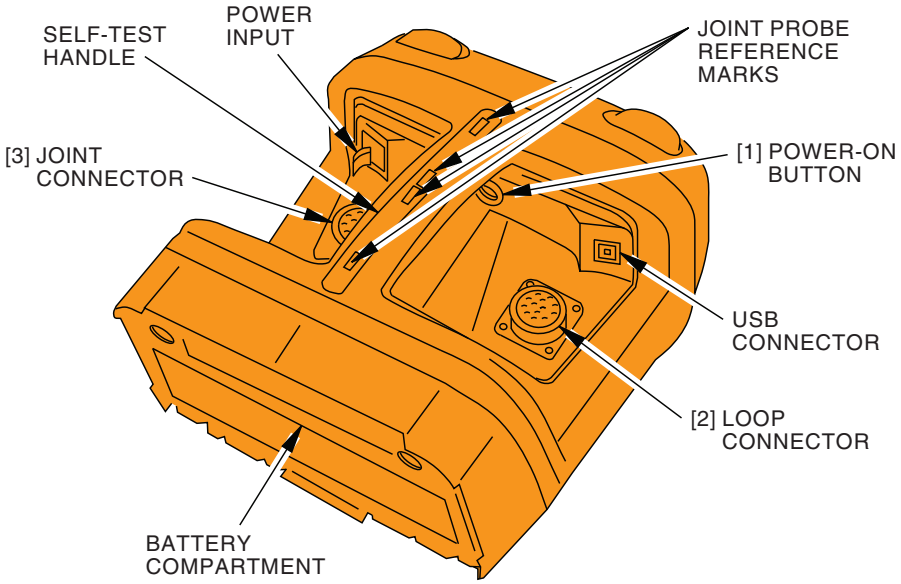
(e) If all of the built-up joints/connections for a particular wire bundle have been measured and joint resistances are less than the calculated maximum allowable resistance for the built-up joints, then the joint buildup resistance measurement task is complete.

——— END OF TASK ———

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**MK LOOP RESISTANCE TESTER (ExLRT)
(FRONT VIEW)**

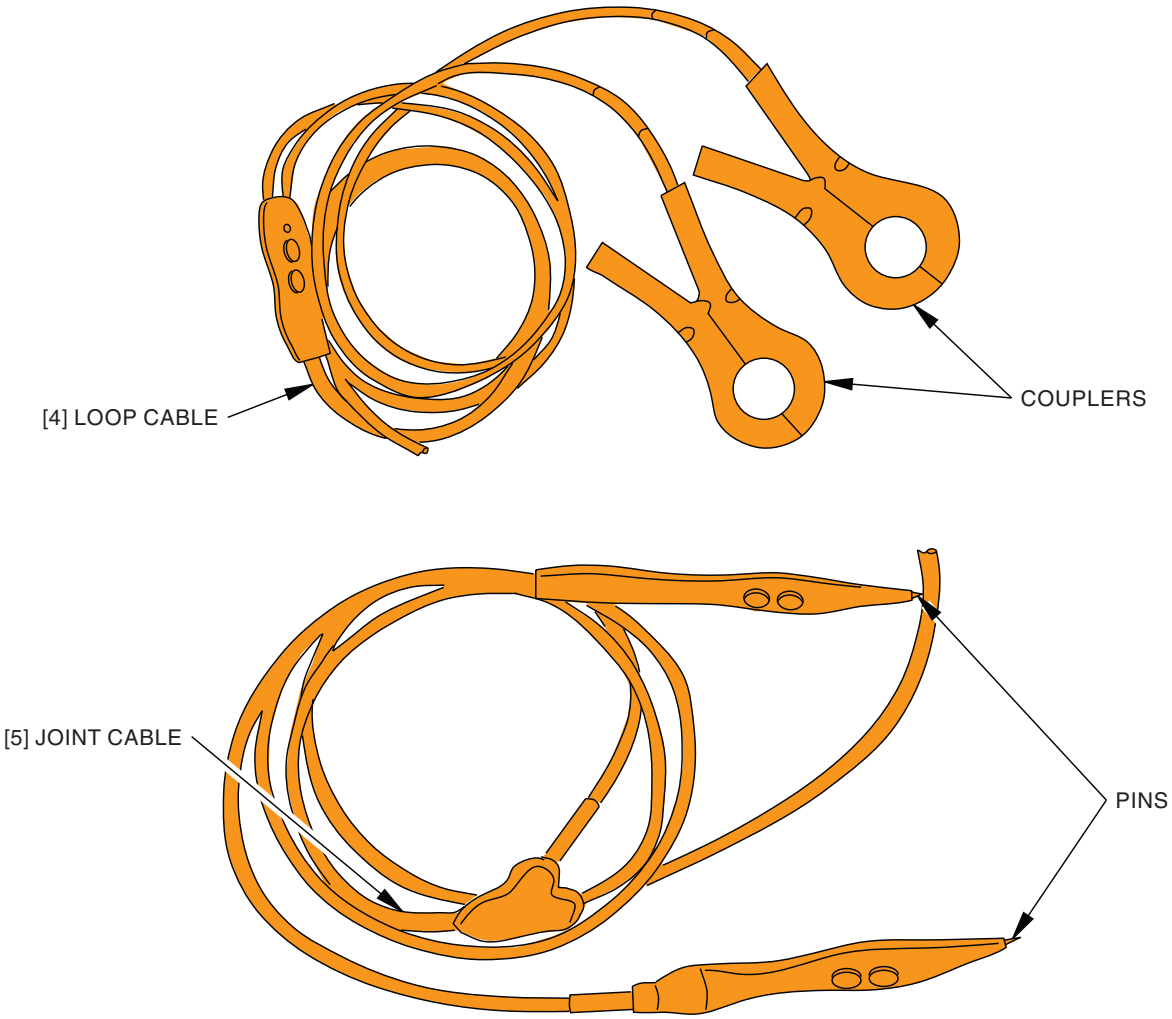


**MK LOOP RESISTANCE TESTER (ExLRT)
(TOP VIEW)**

2929035 S0000708126_V1

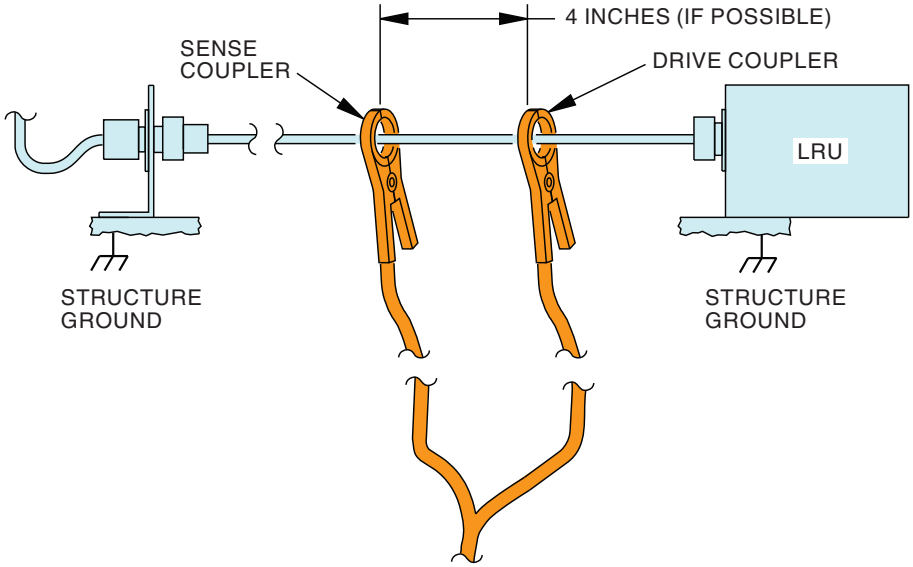
**Loop Resistance Tester
Figure 201/05-56-05-990-801 (Sheet 1 of 2)**

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Loop Resistance Tester
Figure 201/05-56-05-990-801 (Sheet 2 of 2)

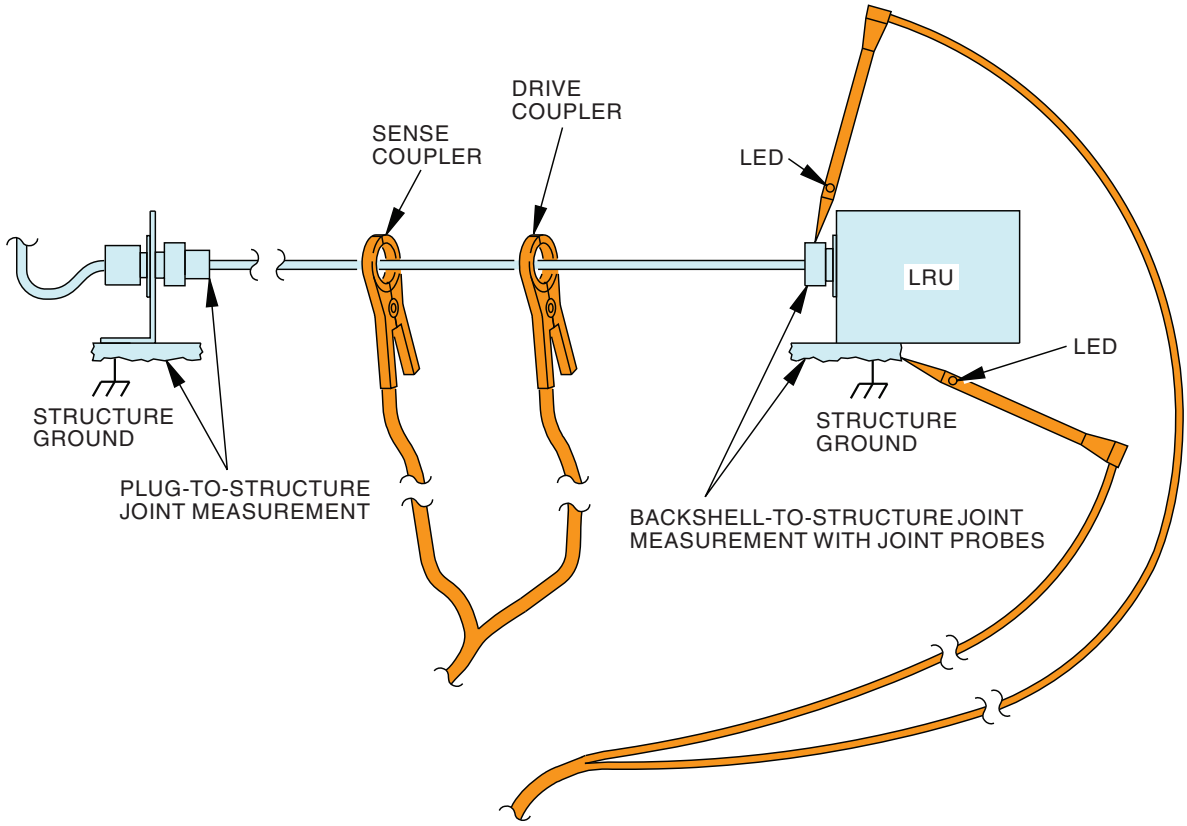
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2929038 S0000708130_V1

Typical Loop Resistance Tester (ExLRT) Hookup for Loop Resistance Measurement
Figure 202/05-56-05-990-802

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2929040 S0000708132_V1

Typical Loop Resistance Tester (ExLRT) Hookup for Joint Buildup Test
Figure 203/05-56-05-990-804

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